HL7 FHIR R4 — Chapter 11: Medications Module (sections 11 – 11.31.3)

Source: HL7 FHIR R4 (v4.0.1) specification, https://hl7.org/fhir/R4/ — text-only rendition for AI ingestion; page order follows toc.html.

11 Medications Module

URL: https://hl7.org/fhir/R4/medications-module.html

There are other resources that are of particular interest in the medication domain.

Work Group Pharmacy & Public Health and Emergency Response | Standards Status: Informative

11.0 Medications Module

11.0.1 Introduction

This module is concerned with resources and functionality in 3 main domains:

• The ordering, dispensing, administration of medications and recording statements of medication use.

• Recording of Immunizations given (or not given), evaluation of given immunizations and recommendations for an individual patient at a point in time.

• The creation or querying for medications as part of drug information or drug knowledge.

11.0.2 Index

MedicationRequest MedicationDispense MedicationAdministration | MedicationStatement Medication MedicationKnowledge | Immunization ImmunizationEvaluation ImmunizationRecommendation

11.0.2.1 Medications

Name	Description
MedicationRequest	Represents an instruction for the administration of medication to a patient - both in the inpatient (hospital) and community setting. It can also include instructions for the dispensing, the reasons why the administration should occur and other data. It is called an 'Request' to be consistent with other FHIR resources and the workflow pattern, but a common alias for this resource is a 'Prescription' or an 'Order'. The Order itself represents the content of the instruction and is not, by itself, actionable. The workflow process around 'fulfilling' the order is part of the generic FHIR workflow (see below), with the MedicationRequest representing the contents.
MedicationDispense	The provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).
MedicationAdministration	A record of a patient actually consuming a medicine, or if it has otherwise been administered to them
MedicationStatement	This is a record indicating that a patient may be taking a medication now, has taken the medication in the past, or will be taking the medication in the future. The source for this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. A medication statement is not a part of the prescribe->dispense->administer sequence, but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication. It may be used to construct a patients 'Current Medications' list.
Medication	The medication resource represents an actual medication that can be given to a patient, and referenced by the other medication resources. In many cases, this resource is not needed and the drug is indicated by a reference to the appropriate terminology and so can be represented using a codeable concept. In other cases, however, it may be desired to indicate more details than the simple drug (such as the packaging, whether it is a generic medication or the active and inactive ingredients) and so the Medication resource can be used for this.
MedicationKnowledge	The MedicationKnowledge resource is draft and is included for comment purposes. This resource represents information about a medication, for example, details about the medication including interactions, contraindications, cost, regulatory status, administration guidelines, etc.

11.0.2.2 Immunizations

Name	Description
Immunization	The Immunization resource is intended to cover the recording of current and historical administration of vaccines to patients across all healthcare disciplines in all care settings and all regions. This includes immunization of both humans and animals, but does not include the administration of non-vaccine agents, even those that may have or claim to have immunological effects.
ImmunizationRecommendation	A patient's point-in-time immunization and recommendation (i.e. forecasting a patient's immunization eligibility according to a published schedule) with optional supporting justification
ImmunizationEvaluation	The ImmunizationEvaluation resource is intended to cover communicating the results of an evaluation of a vaccine administration event (documented using the Immunization resource) against a set of published recommendations (protocols).

11.0.3 Common Use Cases

• Placing a Medication Request (aka Prescription or Order). The MedicationRequest resource represents the details of what medication the prescriber intends the patient to receive and other details such as the dose, timing and route. However, while it represents that order, the actual workflow around the supply (dispensing) and administration of that medication is managed by the common workflow functionality, resulting in the creation of the other medication resources (MedicationDispense, MedicationAdministration) as it executes.

• Listing a patient's current medications. This is represented using the List resource, which allows an author to construct a 'curated' list. This has features beyond a simple collection of resources - such as the ability to state that the patient is not currently taking any medications at the time that the list is constructed, or changes that are made at a point in time (sometimes called medication reconciliation). FHIR defines a number of specific functional lists for this purpose, and a specific operation to retrieve it. Note that FHIR does not describe how the list should be maintained by the individual system, as that will vary widely between implementations.

• Retrieving a list of Immunizations. This can be achieved using a simple query against the Immunization resource type. Note that an Immunization resource can specifically state that it was not given, as well as any reactions that occurred. Note that if the reaction is considered to be an indication of an allergy or intolerance, then a separate AllergyIntolerance resource should be created.

• Retrieving recommendations for a specific patient at a point in time. The recommendation for immunizations to give would be represented by an ImmunizationRecommendation resource, but the determination of that recommendation is a part of Decision Support, with the actual immunization schedule being represented by a generic Plan Definition, individual Care Plan or some other record.

• Dispensing a medication based on a Medication Request (aka Prescription or Order). The MedicationDispense resource represents the details of the dispensing event including the actual product provided to the patient and the dosage to be administered or consumed.

• Retrieving the planned medication therapy for specific patient at a point in time. The intended medication therapy may be represented by an individual Care Plan.

• Retrieving information about a medication either as a stand alone request or as while reviewing a patients' medication list

11.0.4 Other resources

• SupplyRequest and SupplyDelivery which, like MedicationRequest, are 'detail' resources used as part of a workflow. They are concerned with the request of supplies used in the healthcare process. This includes supplies specifically used in the treatment of patients as well as supply movement within an institution (transport a set of supplies from materials management to a service unit (nurse station).

• This specification also includes a draft set of resources that are used for registration of medications for regulatory purposes: MedicinalProduct MedicinalProductAuthorization MedicinalProductContraindication MedicinalProductIndication MedicinalProductIngredient MedicinalProductInteraction MedicinalProductManufactured MedicinalProductPackaged MedicinalProductPharmaceutical MedicinalProductUndesirableEffect SubstancePolymer SubstanceReferenceInformation SubstanceSpecification These resources are under development in association with several medication regulation authorities.

11.0.5 Security and Privacy

As with all clinical data, Medications (in particular) can be sensitive information as specific medications can indicate the presence of private information such as mental health disorders or HIV. However, withholding information about what medications a person is taking can lead to catastrophic results, and so needs to be considered very carefully. At the least, a clinician should be made aware that there is information available that they have not been given when making clinical decisions.

For more general considerations, see the Security and Privacy module.

11.0.6 Developmental Roadmap

The Pharmacy workgroup has plans to improve all existing resources e.g. adding in features that support detailing our conditional orders in a structured way; evaluating requirements for supporting drug formularies and medication knowledge. This work is expected to include the development and approval of a new resource and may involve updates to the Medication Resource.

11.1 Resource MedicationRequest - Content

URL: https://hl7.org/fhir/R4/medicationrequest.html
Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

An order or request for both supply of the medication and the instructions for administration of the medication to a patient. The resource is called "MedicationRequest" rather than "MedicationPrescription" or "MedicationOrder" to generalize the use across inpatient and outpatient settings, including care plans, etc., and to harmonize with workflow patterns.

11.1.1 Scope and Usage

This resource covers all type of orders for medications for a patient. This includes inpatient medication orders as well as community orders (whether filled by the prescriber or by a pharmacy). It also includes orders for over-the-counter medications (e.g. Aspirin), total parenteral nutrition and diet/ vitamin supplements. It may be used to support the order of medication-related devices. It is not intended for use in prescribing particular

diets, or for ordering non-medication-related items (eyeglasses, supplies, etc.). In addition, the MedicationRequest may be used to report orders/request from external systems that have been reported for informational purposes and are not authoritative and are not expected to be acted upon (e.g. dispensed or administered).

The MedicationRequest resource is a "request" resource from a FHIR workflow perspective - see Workflow Request.

The MedicationRequest resource allows requesting only a single medication. If a workflow requires requesting multiple items simultaneously, this is done using multiple instances of this resource. These instances can be linked in different ways, depending on the needs of the workflow. For guidance, refer to the Request pattern

11.1.2 Boundaries and Relationships

The MedicationRequest resource is used to request or order medication for a subject. It may also be used to report a medication request or order from one organization or source to another. When requesting supplies or devices when there is a patient focus or instructions regarding their use, SupplyRequest or DeviceRequest should be used instead. When reporting on the usage of a medication by a patient, the MedicationStatementt resource should be used.

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.
MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).
MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them
MedicationStatement | This is a record of medication being taken by a patient, or that the medication has been given to a patient where the record is the result of a report from the patient, or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence but is a report that such a sequence (or at least a part of it) did take place resulting in a belief that the patient has received a particular medication.

This resource is referenced by CarePlan, Claim, DiagnosticReport, ExplanationOfBenefit, MedicationAdministration, MedicationDispense, itself, MedicationStatement, Observation and ServiceRequest

11.1.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationRequest | TU | | DomainResource | Ordering of medication for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External ids for this request
status | ?!Σ | 1..1 | code | active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown Medicationrequest status (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status Medication request status reason codes (Example)
intent | ?!Σ | 1..1 | code | proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option Medication request intent (Required)
category | | 0..* | CodeableConcept | Type of medication usage Medication request category codes (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if request is prohibiting action
reported[x] | Σ | 0..1 | | Reported rather than primary record
reportedBoolean | | | boolean
reportedReference | | | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)
medication[x] | Σ | 1..1 | | Medication to be taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 1..1 | Reference(Patient | Group) | Who or group medication request is for
encounter | | 0..1 | Reference(Encounter) | Encounter created as part of encounter/admission/stay
supportingInformation | | 0..* | Reference(Any) | Information to support ordering of the medication
authoredOn | Σ | 0..1 | dateTime | When request was initially authored
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/What requested the Request
performer | | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson | CareTeam) | Intended performer of administration
performerType | Σ | 0..1 | CodeableConcept | Desired kind of performer of the medication administration Procedure Performer Role Codes (Example)

recorder | | 0..1 | Reference(Practitioner | PractitionerRole) | Person who entered the request
reasonCode | | 0..* | CodeableConcept | Reason or indication for ordering or not ordering the medication Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or observation that supports why the prescription is being written
instantiatesCanonical | Σ | 0..* | canonical() | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan | MedicationRequest | ServiceRequest | ImmunizationRecommendation) | What request fulfills
groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of
courseOfTherapyType | | 0..1 | CodeableConcept | Overall pattern of medication administration Medication request course of therapy codes (Example)
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
note | | 0..* | Annotation | Information about the prescription
dosageInstruction | | 0..* | Dosage | How the medication should be taken
dispenseRequest | | 0..1 | BackboneElement | Medication supply authorization
initialFill | | 0..1 | BackboneElement | First fill details
quantity | | 0..1 | SimpleQuantity | First fill quantity
duration | | 0..1 | Duration | First fill duration
dispenseInterval | | 0..1 | Duration | Minimum period of time between dispenses
validityPeriod | | 0..1 | Period | Time period supply is authorized for
numberOfRepeatsAllowed | | 0..1 | unsignedInt | Number of refills authorized
quantity | | 0..1 | SimpleQuantity | Amount of medication to supply per dispense
expectedSupplyDuration | | 0..1 | Duration | Number of days supply per dispense
performer | | 0..1 | Reference(Organization) | Intended dispenser
substitution | | 0..1 | BackboneElement | Any restrictions on medication substitution
allowed[x] | | 1..1 | | Whether substitution is allowed or not V3 Value SetActSubstanceAdminSubstitutionCode (Example)
allowedBoolean | | | boolean
allowedCodeableConcept | | | CodeableConcept
reason | | 0..1 | CodeableConcept | Why should (not) substitution be made V3 Value SetSubstanceAdminSubstitutionReason (Example)
priorPrescription | | 0..1 | Reference(MedicationRequest) | An order/prescription that is being replaced
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical Issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this request --></identifier>
 <status value="[code]"/><!-- 1..1 active | on-hold | cancelled | completed | entered-in-error | stopped | draft |
    unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | original-order | reflex-order | filler-order | instance-
    order | option -->
 <category><!-- 0..* CodeableConcept Type of medication usage --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if request is prohibiting action -->
 <reported[x]><!-- 0..1 boolean|Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Reported rather than primary record --></reported[x]>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) Medication to be taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who or group medication request is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of encounter/admission/stay --></encounter>
 <supportingInformation><!-- 0..* Reference(Any) Information to support ordering of the medication
    --></supportingInformation>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request was initially authored -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/What requested the Request --></requester>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|Device|RelatedPerson|CareTeam) Intended performer of administration --></performer>
 <performerType><!-- 0..1 CodeableConcept Desired kind of performer of the medication administration --></performerType>
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole) Person who entered the request --></recorder>
 <reasonCode><!-- 0..* CodeableConcept Reason or indication for ordering or not ordering the medication --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or observation that supports why the prescription
     is being written --></reasonReference>
 <instantiatesCanonical value="[canonical]"/><!-- 0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|MedicationRequest|ServiceRequest|
   ImmunizationRecommendation) What request fulfills --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <courseOfTherapyType><!-- 0..1 CodeableConcept Overall pattern of medication administration --></courseOfTherapyType>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <note><!-- 0..* Annotation Information about the prescription --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication should be taken --></dosageInstruction>
 <dispenseRequest>  <!-- 0..1 Medication supply authorization -->
  <initialFill>  <!-- 0..1 First fill details -->
   <quantity><!-- 0..1 Quantity(SimpleQuantity) First fill quantity --></quantity>
   <duration><!-- 0..1 Duration First fill duration --></duration>
  </initialFill>
```

```
  <dispenseInterval><!-- 0..1 Duration Minimum period of time between dispenses --></dispenseInterval>
  <validityPeriod><!-- 0..1 Period Time period supply is authorized for --></validityPeriod>
  <numberOfRepeatsAllowed value="[unsignedInt]"/><!-- 0..1 Number of refills authorized -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of medication to supply per dispense --></quantity>
  <expectedSupplyDuration><!-- 0..1 Duration Number of days supply per dispense --></expectedSupplyDuration>
  <performer><!-- 0..1 Reference(Organization) Intended dispenser --></performer>
 </dispenseRequest>
 <substitution>  <!-- 0..1 Any restrictions on medication substitution -->
  <allowed[x]><!-- 1..1 boolean|CodeableConcept Whether substitution is allowed or not --></allowed[x]>
  <reason><!-- 0..1 CodeableConcept Why should (not) substitution be made --></reason>
 </substitution>
 <priorPrescription><!-- 0..1 Reference(MedicationRequest) An order/prescription that is being replaced
    --></priorPrescription>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical Issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationRequest>
```

JSON Template

```
{
  "resourceType" : "MedicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this request
  "status" : "<code>", // R!  active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "intent" : "<code>", // R!  proposal | plan | order | original-order | reflex-order | filler-order | instance-order |
    option
  "category" : [{ CodeableConcept }], // Type of medication usage
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if request is prohibiting action
  // reported[x]: Reported rather than primary record. One of these 2:
  "reportedBoolean" : <boolean>,
  "reportedReference" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) },
  // medication[x]: Medication to be taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who or group medication request is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of encounter/admission/stay
  "supportingInformation" : [{ Reference(Any) }], // Information to support ordering of the medication
  "authoredOn" : "<dateTime>", // When request was initially authored
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/What requested the Request
  "performer" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|Device|RelatedPerson|CareTeam) }, // Intended performer of administration
  "performerType" : { CodeableConcept }, // Desired kind of performer of the medication administration
  "recorder" : { Reference(Practitioner|PractitionerRole) }, // Person who entered the request
  "reasonCode" : [{ CodeableConcept }], // Reason or indication for ordering or not ordering the medication
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or observation that supports why the
    prescription is being written
  "instantiatesCanonical" : ["<canonical>"], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|MedicationRequest|ServiceRequest|
   ImmunizationRecommendation) }], // What request fulfills
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "courseOfTherapyType" : { CodeableConcept }, // Overall pattern of medication administration
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "note" : [{ Annotation }], // Information about the prescription
  "dosageInstruction" : [{ Dosage }], // How the medication should be taken
  "dispenseRequest" : { // Medication supply authorization
    "initialFill" : { // First fill details
      "quantity" : { Quantity(SimpleQuantity) }, // First fill quantity
      "duration" : { Duration } // First fill duration
    },
    "dispenseInterval" : { Duration }, // Minimum period of time between dispenses
    "validityPeriod" : { Period }, // Time period supply is authorized for
    "numberOfRepeatsAllowed" : "<unsignedInt>", // Number of refills authorized
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of medication to supply per dispense
    "expectedSupplyDuration" : { Duration }, // Number of days supply per dispense
    "performer" : { Reference(Organization) } // Intended dispenser
  },
  "substitution" : { // Any restrictions on medication substitution
    // allowed[x]: Whether substitution is allowed or not. One of these 2:
    "allowedBoolean" : <boolean>,
    "allowedCodeableConcept" : { CodeableConcept },
    "reason" : { CodeableConcept } // Why should (not) substitution be made
  },
  "priorPrescription" : { Reference(MedicationRequest) }, // An order/prescription that is being replaced
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical Issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationRequest.identifier [ Identifier ], ... ; # 0..* External ids for this request
  fhir:MedicationRequest.status [ code ]; # 1..1 active | on-hold | cancelled | completed | entered-in-error | stopped |
     draft | unknown
  fhir:MedicationRequest.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:MedicationRequest.intent [ code ]; # 1..1 proposal | plan | order | original-order | reflex-order | filler-order
    | instance-order | option
  fhir:MedicationRequest.category [ CodeableConcept ], ... ; # 0..* Type of medication usage
  fhir:MedicationRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:MedicationRequest.doNotPerform [ boolean ]; # 0..1 True if request is prohibiting action
  # MedicationRequest.reported[x] : 0..1 Reported rather than primary record. One of these 2
    fhir:MedicationRequest.reportedBoolean [ boolean ]
    fhir:MedicationRequest.reportedReference [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ]
  # MedicationRequest.medication[x] : 1..1 Medication to be taken. One of these 2
    fhir:MedicationRequest.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.medicationReference [ Reference(Medication) ]
  fhir:MedicationRequest.subject [ Reference(Patient|Group) ]; # 1..1 Who or group medication request is for
  fhir:MedicationRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
    encounter/admission/stay
  fhir:MedicationRequest.supportingInformation [ Reference(Any) ], ... ; # 0..* Information to support ordering of the
    medication
  fhir:MedicationRequest.authoredOn [ dateTime ]; # 0..1 When request was initially authored
  fhir:MedicationRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device)
    ]; # 0..1 Who/What requested the Request
  fhir:MedicationRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson|CareTeam) ]; # 0..1 Intended
    performer of administration
  fhir:MedicationRequest.performerType [ CodeableConcept ]; # 0..1 Desired kind of performer of the medication
    administration
  fhir:MedicationRequest.recorder [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Person who entered the request
  fhir:MedicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason or indication for ordering or not ordering
    the medication
  fhir:MedicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or observation
    that supports why the prescription is being written
  fhir:MedicationRequest.instantiatesCanonical [ canonical ], ... ; # 0..* Instantiates FHIR protocol or definition
  fhir:MedicationRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:MedicationRequest.basedOn [ Reference(CarePlan|MedicationRequest|ServiceRequest|ImmunizationRecommendation) ],
    ... ; # 0..* What request fulfills
  fhir:MedicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:MedicationRequest.courseOfTherapyType [ CodeableConcept ]; # 0..1 Overall pattern of medication administration
  fhir:MedicationRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:MedicationRequest.note [ Annotation ], ... ; # 0..* Information about the prescription
  fhir:MedicationRequest.dosageInstruction [ Dosage ], ... ; # 0..* How the medication should be taken
  fhir:MedicationRequest.dispenseRequest [ # 0..1 Medication supply authorization
    fhir:MedicationRequest.dispenseRequest.initialFill [ # 0..1 First fill details
      fhir:MedicationRequest.dispenseRequest.initialFill.quantity [ Quantity(SimpleQuantity) ]; # 0..1 First fill
    quantity
      fhir:MedicationRequest.dispenseRequest.initialFill.duration [ Duration ]; # 0..1 First fill duration
    ];
    fhir:MedicationRequest.dispenseRequest.dispenseInterval [ Duration ]; # 0..1 Minimum period of time between
    dispenses
    fhir:MedicationRequest.dispenseRequest.validityPeriod [ Period ]; # 0..1 Time period supply is authorized for
    fhir:MedicationRequest.dispenseRequest.numberOfRepeatsAllowed [ unsignedInt ]; # 0..1 Number of refills authorized
    fhir:MedicationRequest.dispenseRequest.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication to supply
    per dispense
    fhir:MedicationRequest.dispenseRequest.expectedSupplyDuration [ Duration ]; # 0..1 Number of days supply per
    dispense
    fhir:MedicationRequest.dispenseRequest.performer [ Reference(Organization) ]; # 0..1 Intended dispenser
  ];
  fhir:MedicationRequest.substitution [ # 0..1 Any restrictions on medication substitution
    # MedicationRequest.substitution.allowed[x] : 1..1 Whether substitution is allowed or not. One of these 2
      fhir:MedicationRequest.substitution.allowedBoolean [ boolean ]
      fhir:MedicationRequest.substitution.allowedCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.substitution.reason [ CodeableConcept ]; # 0..1 Why should (not) substitution be made
  ];
  fhir:MedicationRequest.priorPrescription [ Reference(MedicationRequest) ]; # 0..1 An order/prescription that is being
    replaced
  fhir:MedicationRequest.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical Issue with action
  fhir:MedicationRequest.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
    lifecycle
]
```

Changes since R3

MedicationRequest

MedicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-request-status to http://hl7.org/fhir/ValueSet/medicationrequest-status|4.0.1

MedicationRequest.statusReason | Added Element

MedicationRequest.intent | Change value set from http://hl7.org/fhir/ValueSet/medication-request-intent to http://hl7.org/fhir/ValueSet/medicationrequest-intent|4.0.1

MedicationRequest.category | Max Cardinality changed from 1 to *

MedicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/medication-request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

MedicationRequest.doNotPerform | Added Element

MedicationRequest.reported[x] | Added Element

MedicationRequest.encounter | Added Element

MedicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
MedicationRequest.performer | Added Element
MedicationRequest.performerType | Added Element
MedicationRequest.recorder | Type Reference: Added Target Type PractitionerRole
MedicationRequest.instantiatesCanonical | Added Element
MedicationRequest.instantiatesUri | Added Element
MedicationRequest.basedOn | Type Reference: Added Target Types ServiceRequest, ImmunizationRecommendation Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
MedicationRequest.courseOfTherapyType | Added Element
MedicationRequest.insurance | Added Element
MedicationRequest.dispenseRequest.initialFill | Added Element
MedicationRequest.dispenseRequest.initialFill.quantity | Added Element
MedicationRequest.dispenseRequest.initialFill.duration | Added Element
MedicationRequest.dispenseRequest.dispenseInterval | Added Element
MedicationRequest.dispenseRequest.numberOfRepeatsAllowed | Type changed from positiveInt to unsignedInt
MedicationRequest.substitution.allowed[x] | Renamed from allowed to allowed[x] Add Type CodeableConcept No longer marked as Modifier
MedicationRequest.definition | deleted
MedicationRequest.context | deleted
MedicationRequest.requester.agent | deleted
MedicationRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 36 tests that all execute ok. 8 fail round-trip testing and 36 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationRequest | TU | | DomainResource | Ordering of medication for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External ids for this request
status | ?!Σ | 1..1 | code | active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown Medicationrequest status (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status Medication request status reason codes (Example)
intent | ?!Σ | 1..1 | code | proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option Medication request intent (Required)
category | | 0..* | CodeableConcept | Type of medication usage Medication request category codes (Example)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
doNotPerform | ?!Σ | 0..1 | boolean | True if request is prohibiting action
reported[x] | Σ | 0..1 | | Reported rather than primary record
reportedBoolean | | | boolean
reportedReference | | | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)
medication[x] | Σ | 1..1 | | Medication to be taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 1..1 | Reference(Patient | Group) | Who or group medication request is for
encounter | | 0..1 | Reference(Encounter) | Encounter created as part of encounter/admission/stay
supportingInformation | | 0..* | Reference(Any) | Information to support ordering of the medication
authoredOn | Σ | 0..1 | dateTime | When request was initially authored
requester | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Who/What requested the Request
performer | | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson | CareTeam) | Intended performer of administration
performerType | Σ | 0..1 | CodeableConcept | Desired kind of performer of the medication administration Procedure Performer Role Codes (Example)
recorder | | 0..1 | Reference(Practitioner | PractitionerRole) | Person who entered the request
reasonCode | | 0..* | CodeableConcept | Reason or indication for ordering or not ordering the medication Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation) | Condition or observation that supports why the prescription is being written
instantiatesCanonical | Σ | 0..* | canonical() | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan | MedicationRequest | ServiceRequest | ImmunizationRecommendation) | What request fulfills
groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of
courseOfTherapyType | | 0..1 | CodeableConcept | Overall pattern of medication administration Medication request course of therapy codes (Example)
insurance | | 0..* | Reference(Coverage | ClaimResponse) | Associated insurance coverage
note | | 0..* | Annotation | Information about the prescription
dosageInstruction | | 0..* | Dosage | How the medication should be taken
dispenseRequest | | 0..1 | BackboneElement | Medication supply authorization

initialFill | | 0..1 | BackboneElement | First fill details
quantity | | 0..1 | SimpleQuantity | First fill quantity
duration | | 0..1 | Duration | First fill duration
dispenseInterval | | 0..1 | Duration | Minimum period of time between dispenses
validityPeriod | | 0..1 | Period | Time period supply is authorized for
numberOfRepeatsAllowed | | 0..1 | unsignedInt | Number of refills authorized
quantity | | 0..1 | SimpleQuantity | Amount of medication to supply per dispense
expectedSupplyDuration | | 0..1 | Duration | Number of days supply per dispense
performer | | 0..1 | Reference(Organization) | Intended dispenser
substitution | | 0..1 | BackboneElement | Any restrictions on medication substitution
allowed[x] | | 1..1 | | Whether substitution is allowed or not V3 Value SetActSubstanceAdminSubstitutionCode (Example)
allowedBoolean | | | boolean
allowedCodeableConcept | | | CodeableConcept
reason | | 0..1 | CodeableConcept | Why should (not) substitution be made V3 Value SetSubstanceAdminSubstitutionReason (Example)
priorPrescription | | 0..1 | Reference(MedicationRequest) | An order/prescription that is being replaced
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical Issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this request --></identifier>
 <status value="[code]"/><!-- 1..1 active | on-hold | cancelled | completed | entered-in-error | stopped | draft |
    unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | original-order | reflex-order | filler-order | instance-
    order | option -->
 <category><!-- 0..* CodeableConcept Type of medication usage --></category>
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <doNotPerform value="[boolean]"/><!-- 0..1 True if request is prohibiting action -->
 <reported[x]><!-- 0..1 boolean|Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Reported rather than primary record --></reported[x]>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) Medication to be taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who or group medication request is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of encounter/admission/stay --></encounter>
 <supportingInformation><!-- 0..* Reference(Any) Information to support ordering of the medication
    --></supportingInformation>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request was initially authored -->
 <requester><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) Who/What requested the Request --></requester>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Organization|
   Patient|Device|RelatedPerson|CareTeam) Intended performer of administration --></performer>
 <performerType><!-- 0..1 CodeableConcept Desired kind of performer of the medication administration --></performerType>
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole) Person who entered the request --></recorder>
 <reasonCode><!-- 0..* CodeableConcept Reason or indication for ordering or not ordering the medication --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Condition or observation that supports why the prescription
     is being written --></reasonReference>
 <instantiatesCanonical value="[canonical]"/><!-- 0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|MedicationRequest|ServiceRequest|
   ImmunizationRecommendation) What request fulfills --></basedOn>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <courseOfTherapyType><!-- 0..1 CodeableConcept Overall pattern of medication administration --></courseOfTherapyType>
 <insurance><!-- 0..* Reference(Coverage|ClaimResponse) Associated insurance coverage --></insurance>
 <note><!-- 0..* Annotation Information about the prescription --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication should be taken --></dosageInstruction>
 <dispenseRequest>  <!-- 0..1 Medication supply authorization -->
  <initialFill>  <!-- 0..1 First fill details -->
   <quantity><!-- 0..1 Quantity(SimpleQuantity) First fill quantity --></quantity>
   <duration><!-- 0..1 Duration First fill duration --></duration>
  </initialFill>
  <dispenseInterval><!-- 0..1 Duration Minimum period of time between dispenses --></dispenseInterval>
  <validityPeriod><!-- 0..1 Period Time period supply is authorized for --></validityPeriod>
  <numberOfRepeatsAllowed value="[unsignedInt]"/><!-- 0..1 Number of refills authorized -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of medication to supply per dispense --></quantity>
  <expectedSupplyDuration><!-- 0..1 Duration Number of days supply per dispense --></expectedSupplyDuration>
  <performer><!-- 0..1 Reference(Organization) Intended dispenser --></performer>
 </dispenseRequest>
 <substitution>  <!-- 0..1 Any restrictions on medication substitution -->
  <allowed[x]><!-- 1..1 boolean|CodeableConcept Whether substitution is allowed or not --></allowed[x]>
  <reason><!-- 0..1 CodeableConcept Why should (not) substitution be made --></reason>
 </substitution>
 <priorPrescription><!-- 0..1 Reference(MedicationRequest) An order/prescription that is being replaced
    --></priorPrescription>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical Issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationRequest>
```

JSON Template

```
{
```

```
  "resourceType" : "MedicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this request
  "status" : "<code>", // R!  active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "intent" : "<code>", // R!  proposal | plan | order | original-order | reflex-order | filler-order | instance-order |
    option
  "category" : [{ CodeableConcept }], // Type of medication usage
  "priority" : "<code>", // routine | urgent | asap | stat
  "doNotPerform" : <boolean>, // True if request is prohibiting action
  // reported[x]: Reported rather than primary record. One of these 2:
  "reportedBoolean" : <boolean>,
  "reportedReference" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) },
  // medication[x]: Medication to be taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who or group medication request is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of encounter/admission/stay
  "supportingInformation" : [{ Reference(Any) }], // Information to support ordering of the medication
  "authoredOn" : "<dateTime>", // When request was initially authored
  "requester" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|RelatedPerson|Device) }, // Who/What requested the Request
  "performer" : { Reference(Practitioner|PractitionerRole|Organization|
   Patient|Device|RelatedPerson|CareTeam) }, // Intended performer of administration
  "performerType" : { CodeableConcept }, // Desired kind of performer of the medication administration
  "recorder" : { Reference(Practitioner|PractitionerRole) }, // Person who entered the request
  "reasonCode" : [{ CodeableConcept }], // Reason or indication for ordering or not ordering the medication
  "reasonReference" : [{ Reference(Condition|Observation) }], // Condition or observation that supports why the
    prescription is being written
  "instantiatesCanonical" : ["<canonical>"], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|MedicationRequest|ServiceRequest|
   ImmunizationRecommendation) }], // What request fulfills
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "courseOfTherapyType" : { CodeableConcept }, // Overall pattern of medication administration
  "insurance" : [{ Reference(Coverage|ClaimResponse) }], // Associated insurance coverage
  "note" : [{ Annotation }], // Information about the prescription
  "dosageInstruction" : [{ Dosage }], // How the medication should be taken
  "dispenseRequest" : { // Medication supply authorization
    "initialFill" : { // First fill details
      "quantity" : { Quantity(SimpleQuantity) }, // First fill quantity
      "duration" : { Duration } // First fill duration
    },
    "dispenseInterval" : { Duration }, // Minimum period of time between dispenses
    "validityPeriod" : { Period }, // Time period supply is authorized for
    "numberOfRepeatsAllowed" : "<unsignedInt>", // Number of refills authorized
    "quantity" : { Quantity(SimpleQuantity) }, // Amount of medication to supply per dispense
    "expectedSupplyDuration" : { Duration }, // Number of days supply per dispense
    "performer" : { Reference(Organization) } // Intended dispenser
  },
  "substitution" : { // Any restrictions on medication substitution
    // allowed[x]: Whether substitution is allowed or not. One of these 2:
    "allowedBoolean" : <boolean>,
    "allowedCodeableConcept" : { CodeableConcept },
    "reason" : { CodeableConcept } // Why should (not) substitution be made
  },
  "priorPrescription" : { Reference(MedicationRequest) }, // An order/prescription that is being replaced
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical Issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationRequest.identifier [ Identifier ], ... ; # 0..* External ids for this request
  fhir:MedicationRequest.status [ code ]; # 1..1 active | on-hold | cancelled | completed | entered-in-error | stopped |
     draft | unknown
  fhir:MedicationRequest.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:MedicationRequest.intent [ code ]; # 1..1 proposal | plan | order | original-order | reflex-order | filler-order
    | instance-order | option
  fhir:MedicationRequest.category [ CodeableConcept ], ... ; # 0..* Type of medication usage
  fhir:MedicationRequest.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:MedicationRequest.doNotPerform [ boolean ]; # 0..1 True if request is prohibiting action
  # MedicationRequest.reported[x] : 0..1 Reported rather than primary record. One of these 2
    fhir:MedicationRequest.reportedBoolean [ boolean ]
    fhir:MedicationRequest.reportedReference [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ]
  # MedicationRequest.medication[x] : 1..1 Medication to be taken. One of these 2
    fhir:MedicationRequest.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.medicationReference [ Reference(Medication) ]
  fhir:MedicationRequest.subject [ Reference(Patient|Group) ]; # 1..1 Who or group medication request is for
```

```
  fhir:MedicationRequest.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
    encounter/admission/stay
  fhir:MedicationRequest.supportingInformation [ Reference(Any) ], ... ; # 0..* Information to support ordering of the
    medication
  fhir:MedicationRequest.authoredOn [ dateTime ]; # 0..1 When request was initially authored
  fhir:MedicationRequest.requester [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device)
    ]; # 0..1 Who/What requested the Request
  fhir:MedicationRequest.performer [
    Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson|CareTeam) ]; # 0..1 Intended
    performer of administration
  fhir:MedicationRequest.performerType [ CodeableConcept ]; # 0..1 Desired kind of performer of the medication
    administration
  fhir:MedicationRequest.recorder [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Person who entered the request
  fhir:MedicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Reason or indication for ordering or not ordering
    the medication
  fhir:MedicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Condition or observation
    that supports why the prescription is being written
  fhir:MedicationRequest.instantiatesCanonical [ canonical ], ... ; # 0..* Instantiates FHIR protocol or definition
  fhir:MedicationRequest.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:MedicationRequest.basedOn [ Reference(CarePlan|MedicationRequest|ServiceRequest|ImmunizationRecommendation) ],
    ... ; # 0..* What request fulfills
  fhir:MedicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:MedicationRequest.courseOfTherapyType [ CodeableConcept ]; # 0..1 Overall pattern of medication administration
  fhir:MedicationRequest.insurance [ Reference(Coverage|ClaimResponse) ], ... ; # 0..* Associated insurance coverage
  fhir:MedicationRequest.note [ Annotation ], ... ; # 0..* Information about the prescription
  fhir:MedicationRequest.dosageInstruction [ Dosage ], ... ; # 0..* How the medication should be taken
  fhir:MedicationRequest.dispenseRequest [ # 0..1 Medication supply authorization
    fhir:MedicationRequest.dispenseRequest.initialFill [ # 0..1 First fill details
      fhir:MedicationRequest.dispenseRequest.initialFill.quantity [ Quantity(SimpleQuantity) ]; # 0..1 First fill
    quantity
      fhir:MedicationRequest.dispenseRequest.initialFill.duration [ Duration ]; # 0..1 First fill duration
    ];
    fhir:MedicationRequest.dispenseRequest.dispenseInterval [ Duration ]; # 0..1 Minimum period of time between
    dispenses
    fhir:MedicationRequest.dispenseRequest.validityPeriod [ Period ]; # 0..1 Time period supply is authorized for
    fhir:MedicationRequest.dispenseRequest.numberOfRepeatsAllowed [ unsignedInt ]; # 0..1 Number of refills authorized
    fhir:MedicationRequest.dispenseRequest.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication to supply
    per dispense
    fhir:MedicationRequest.dispenseRequest.expectedSupplyDuration [ Duration ]; # 0..1 Number of days supply per
    dispense
    fhir:MedicationRequest.dispenseRequest.performer [ Reference(Organization) ]; # 0..1 Intended dispenser
  ];
  fhir:MedicationRequest.substitution [ # 0..1 Any restrictions on medication substitution
    # MedicationRequest.substitution.allowed[x] : 1..1 Whether substitution is allowed or not. One of these 2
      fhir:MedicationRequest.substitution.allowedBoolean [ boolean ]
      fhir:MedicationRequest.substitution.allowedCodeableConcept [ CodeableConcept ]
    fhir:MedicationRequest.substitution.reason [ CodeableConcept ]; # 0..1 Why should (not) substitution be made
  ];
  fhir:MedicationRequest.priorPrescription [ Reference(MedicationRequest) ]; # 0..1 An order/prescription that is being
    replaced
  fhir:MedicationRequest.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical Issue with action
  fhir:MedicationRequest.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
    lifecycle
]
```

Changes since Release 3

MedicationRequest

MedicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-request-status to http://hl7.org/fhir/ValueSet/medicationrequest-status|4.0.1

MedicationRequest.statusReason | Added Element

MedicationRequest.intent | Change value set from http://hl7.org/fhir/ValueSet/medication-request-intent to http://hl7.org/fhir/ValueSet/medicationrequest-intent|4.0.1

MedicationRequest.category | Max Cardinality changed from 1 to *

MedicationRequest.priority | Change value set from http://hl7.org/fhir/ValueSet/medication-request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

MedicationRequest.doNotPerform | Added Element

MedicationRequest.reported[x] | Added Element

MedicationRequest.encounter | Added Element

MedicationRequest.requester | Type changed from BackboneElement to Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)

MedicationRequest.performer | Added Element

MedicationRequest.performerType | Added Element

MedicationRequest.recorder | Type Reference: Added Target Type PractitionerRole

MedicationRequest.instantiatesCanonical | Added Element

MedicationRequest.instantiatesUri | Added Element

MedicationRequest.basedOn | Type Reference: Added Target Types ServiceRequest, ImmunizationRecommendation Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

MedicationRequest.courseOfTherapyType | Added Element

MedicationRequest.insurance | Added Element

MedicationRequest.dispenseRequest.initialFill | Added Element

MedicationRequest.dispenseRequest.initialFill.quantity | Added Element

MedicationRequest.dispenseRequest.initialFill.duration | Added Element

MedicationRequest.dispenseRequest.dispenseInterval | Added Element

MedicationRequest.dispenseRequest.numberOfRepeatsAllowed | Type changed from positiveInt to unsignedInt
MedicationRequest.substitution.allowed[x] | Renamed from allowed to allowed[x] Add Type CodeableConcept No longer marked as Modifier
MedicationRequest.definition | deleted
MedicationRequest.context | deleted
MedicationRequest.requester.agent | deleted
MedicationRequest.requester.onBehalfOf | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 36 tests that all execute ok. 8 fail round-trip testing and 36 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.1.3.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationRequest.status | A coded concept specifying the state of the prescribing event. Describes the lifecycle of the prescription. | Required | medicationrequest Status
MedicationRequest.statusReason | Identifies the reasons for a given status. | Example | medicationRequest Status Reason Codes
MedicationRequest.intent | The kind of medication order. | Required | medicationRequest Intent
MedicationRequest.category | A coded concept identifying the category of medication request. For example, where the medication is to be consumed or administered, or the type of medication treatment. | Example | medicationRequest Category Codes
MedicationRequest.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
MedicationRequest.medication[x] | A coded concept identifying substance or product that can be ordered. | Example | SNOMEDCTMedicationCodes
MedicationRequest.performerType | Identifies the type of individual that is desired to administer the medication. | Example | ProcedurePerformerRoleCodes
MedicationRequest.reasonCode | A coded concept indicating why the medication was ordered. | Example | Condition/Problem/DiagnosisCodes
MedicationRequest.courseOfTherapyType | Identifies the overall pattern of medication administratio. | Example | medicationRequest Course of Therapy Codes
MedicationRequest.substitution.allowed[x] | Identifies the type of substitution allowed. | Example | v3.ActSubstanceAdminSubstitutionCode
MedicationRequest.substitution.reason | A coded concept describing the reason that a different medication should (or should not) be substituted from what was prescribed. | Example | v3.SubstanceAdminSubstitutionReason

11.1.3.2 Dosage Instructions

Free text dosage instructions can be used for cases where the instructions are too complex to code. The content of this attribute does not include the name or description of the medication. When coded instructions are present, the free text instructions may still be present for display to humans taking or administering the medication. It is expected that the text instructions will always be populated. If the dosage.timing attribute is also populated, then the dosage.text should reflect the same information as the timing.

11.1.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authoredon | date | Return prescriptions written on this date | MedicationRequest.authoredOn
category | token | Returns prescriptions with different categories | MedicationRequest.category
code | token | Return prescriptions of this medication code | (MedicationRequest.medication as CodeableConcept) | 13 Resources
date | date | Returns medication request to be administered on a specific date | MedicationRequest.dosageInstruction.timing.event | 3 Resources
encounter | reference | Return prescriptions with this encounter identifier | MedicationRequest.encounter (Encounter) | 1 Resources
identifier | token | Return prescriptions with this external identifier | MedicationRequest.identifier | 30 Resources
intended-dispenser | reference | Returns prescriptions intended to be dispensed by this Organization | MedicationRequest.dispenseRequest.performer (Organization)
intended-performer | reference | Returns the intended performer of the administration of the medication request | MedicationRequest.performer (Practitioner, Organization, CareTeam, Device, Patient, PractitionerRole, RelatedPerson)
intended-performertype | token | Returns requests for a specific type of performer | MedicationRequest.performerType
intent | token | Returns prescriptions with different intents | MedicationRequest.intent
medication | reference | Return prescriptions for this medication reference | (MedicationRequest.medication as Reference) (Medication) | 3 Resources
patient | reference | Returns prescriptions for a specific patient | MedicationRequest.subject.where(resolve() is Patient) (Patient) | 33 Resources
priority | token | Returns prescriptions with different priorities | MedicationRequest.priority
requester | reference | Returns prescriptions prescribed by this prescriber | MedicationRequest.requester (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
status | token | Status of the prescription | MedicationRequest.status | 3 Resources
subject | reference | The identity of a patient to list orders for | MedicationRequest.subject (Group, Patient)

11.1.5 Resource MedicationRequest - Examples

URL: https://hl7.org/fhir/R4/medicationrequest-examples.html

11.1.5 Resource MedicationRequest - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

ORAL DOSAGE FORMS

The following examples are MedicationRequest resources for Oral Dosage Forms that reference Medication Resources for generic formulations:

• Active Request with link to encounter, reasonCode, note multiple dosage lines, dispenseRequest and substitution - Azithromycin. Also includes non-clinical supportingInformation

• Request with as needed (PRN) dosage with pre-condition - Oxycodone - active with reasonCode, note, dispenseRequest and substitution

• Request for Tapering Dose - Prednisone - active with note and multiple DosageInstructions with boundsPeriod

• Request for Oral Drops - Nystatin Suspension - completed - with Dispense Request

• Request with Dosage with a Pre-Condition - Alprazolam - dosageInstructions includes boundsPeriod with periodMax and asNeededCodeableConcept

• Request for a Dosage that needs to be dispensed as different strength tablets - Warfarin - active

The following examples are MedicationRequest resources for Oral Dosage Forms that reference Medication Resources for manufactured products:

• Oral Chemotherapy Request - Myleran - completed - with reasonCode

• Request with as needed (PRN) dosage - Percocet - completed with link to encounter, dispenseRequest and substitution

• Request for as needed (PRN) dosage - Vicodin - completed with dosageInstruction with additionalInstruction (coded) and as needed with reason, dispenseRequest and substitution

• Oral Chemotherapy Request - Capecitabine - active medication with dosing by Body Surface Area (BSA) (mg/m2)

• Request for Over the Counter Medication - Tylenol PM - with dosage range as needed with pre-condition for use

• Request for a medication that includes the dosage of a prescription in text

• Fully populated example of a MedicationRequest - Chlorthalidone - active - with link to encounter, reasonReference, note, dispenseRequest, substitution and eventHistory

The following examples are MedicationRequest resources for Oral Dosage Forms that include a Medication Code

• Liquid Medication - Phenytoin with link to prior prescription

• Request with a single dosage line - Azithromycin - completed - with link to prior prescription and reason code

• Request - Levothyroxine - completed with reasonCode, dosageInstructions and dispenseRequest

INJECTABLE DOSAGE FORMS

The following examples of MedicationRequest resources for Injectable Dosage Forms that reference a Medication Resource for a generic formulation

• Request for an injectable medication - Lorazepam with Dosage Rate (as rateRange) - active - with link to encounter, reasonCode and dosageInstruction

The following examples of MedicationRequest resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

• Request for Intravenous Chemotherapy (inpatient Request) - Brentuximab Vedotin - completed

• Request for Intravenous Chemotherapy (inpatient Request) - Alemtuzumab - completed with multiple dosageInstruction lines

• Request for Intravenous Antibiotic - Vancomycin - active - with link to encounter, reasonCode, and note

• Request for Intravenous Antibiotic with a Dosage Rate (as RateQuantity) - Piperacillin - completed with reason code

• Request for Subcutaneous injection with multiple dosage lines - Insulin - Novolog - active - includes reasonCode and note. Dosage includes additionalInstructions as text

The following examples of MedicationRequest resources for Injectable Dosage Forms that include a Medication Code:

• Request for Subcutaneous injection - Insulin - Lantus - completed, with reason code and dispense request

The following examples of MedicationRequest resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

• Request for Intravenous Admixture with Dosage Rate (as rateRatio) - Potassium Chloride in D5W - completed with reason code

• Request for TPN Solution with Dosage Rate (as rateRatio) - completed

MISCELLAENOUS DOSAGE FORMS

The following examples of MedicationRequest resources including miscellaneous dosage forms:

• Request using Medication Code for Rectal Suppository - Acetaminophen - completed with reasonCode, dispenseRequest and substitution

• Request using Medication Code for Topical Ointment - Nystatin - on hold and with dispense request

• Request using Medication Code for Inhaler - Proventil - on hold with dispenseRequest and substitution

• Request using Medication Code for Patch - Fentanyl - active, with dispense request

• Medication Resource Reference - Nasal Spray - Nasonex

• Request referencing Medication Resource for Compounded Product - HC 1%, Salicylic Acid 5% in Glaxal Base - on hold with dispenseRequest

• Request referencing Medication Resource for Eye Drops - Timoptic - active - with link to encounter, dispenseRequest and substitution flag

• Request referencing Medication Resource for a Kit that contains 2 different dosage forms - Vagistat

MISCELLAENOUS DOSAGE INSTRUCTIONS

The following examples of MedicationRequest resources including miscellaneous dosage instructions:

• Request that includes a dosage instruction for a medication that is intended to be given once only as single dose

• Request for a medication that is intended to be given in the morning

• Request that includes a dosage instruction for a medication that is given on a specific day of the week

• Request that includes a dosage instruction for a medication that is given at a specific time of the day

• Request with as needed (PRN) dosage with pre-condition - Oxycodone - active with reasonCode, note, dispenseRequest and substitution

• Request for Tapering Dose - Prednisone - active with note and multiple DosageInstructions with boundsPeriod

• Request with Dosage with a Pre-Condition - Alprazolam - dosageInstructions includes boundsPeriod with periodMax and asNeededCodeableConcept

• Request for an injectable medication - Lorazepam with Dosage Rate (as rateRange) - active - with link to encounter, reasonCode and dosageInstruction

• Request for Intravenous Admixture with Dosage Rate (as rateRatio) - Potassium Chloride in D5W - completed with reason code

• Request for Intravenous Dopamine - Weight based calculation with rateQuantity

• Request for Intravenous Admixture Dopamine - with rateQuantity

• Request for Intravenous Admixture Dopamine - with doseQuantity and rateQuantity

• Request for Subcutaneous injection with multiple dosage lines - Insulin - Novolog - active - includes reasonCode and note. Dosage includes additionalInstructions as text

• Request for Intravenous Antibiotic with a Dosage Rate (as RateQuantity) - Piperacillin - completed with reason code

• Oral Chemotherapy Request - Capecitabine - active medication with dosing by Body Surface Area (BSA) (mg/m2)

• Request for as needed (PRN) dosage - Vicodin - completed with dosageInstruction with additionalInstruction (coded) and as needed with reason, dispenseRequest and substitution

Example Name | id | Format
Order for a medication that includes the dosage of a prescription in text | medrx002 | XML | JSON | Turtle
Active Order with link to encounter, reasonCode, note multiple dosage lines, dispenseRequest and substitution - Azithromycin | medrx0302 | XML | JSON | Turtle
Order with as needed (PRN) dosage with pre-condition - Oxycodone - active with reasonCode, note, dispenseRequest and substitution | medrx0301 | XML | JSON | Turtle
Order for Tapering Dose - Prednisone - active with note and multiple DosageInstructions with boundsPeriod | medrx0303 | XML | JSON | Turtle
Order for Oral Drops - Nystatin Suspension - completed - with Dispense Request | medrx0304 | XML | JSON | Turtle
Order with Dosage with a Pre-Condition - Alprazolam - dosageInstructions includes boundsPeriod with periodMax and asNeededCodeableConcept | medrx0305 | XML | JSON | Turtle
Oral Chemotherapy Order - Myleran - completed - with reasonCode | medrx0306 | XML | JSON | Turtle

Order with as needed (PRN) dosage - Percocet - completed with link to encounter, dispenseRequest and substitution | medrx0307 | XML | JSON | Turtle
Order for as needed (PRN) dosage - Vicodin - completed with dosageInstruction with additionalInstruction (coded) and as needed with reason, dispenseRequest and substitution | medrx0308 | XML | JSON | Turtle
Oral Chemotherapy Order - Capecitabine - active medication with dosing by Body Surface Area (BSA) (mg/m2) | medrx0309 | XML | JSON | Turtle
Request for Over the Counter Medication - Tylenol PM - with dosage range as needed with pre-condition for use | medrx0310 | XML | JSON | Turtle
Fully populated example of a MedicationRequest - Chlorthalidone - active - with link to encounter, reasonReference, note, dispenseRequest, substitution and eventHistory | medrx0311 | XML | JSON | Turtle
Liquid Medication - Phenytoin with link to prior prescription | medrx0312 | XML | JSON | Turtle
Request with a single dosage line - Azithromycin - completed - with link to prior prescription and reason code | medrx0313 | XML | JSON | Turtle
Request - Levothyroxine - completed with reasonCode, dosageInstructions and dispenseRequest | medrx0314 | XML | JSON | Turtle
Request for an injectable medication - Lorazepam - active - with link to encounter, reasonCode and dosageInstruction | medrx0315 | XML | JSON | Turtle
Request for Intravenous Chemotherapy (inpatient order) - Brentuximab Vedotin - completed | medrx0316 | XML | JSON | Turtle
Request for Intravenous Chemotherapy (inpatient order) - Alemtuzumab - completed with multiple dosageInstruction lines | medrx0317 | XML | JSON | Turtle
Request for Intravenous Antibiotic - Vancomycin - active - with link to encounter, reasonCode, and note | medrx0318 | XML | JSON | Turtle
Request for Intravenous Antibiotic with a Dosage Rate - Piperacillin - completed with reason code | medrx0319 | XML | JSON | Turtle
Request for Subcutaneous injection - Insulin - Lantus - completed, with reason code and dispense request | medrx0320 | XML | JSON | Turtle
Request for Subcutaneous injection with multiple dosage lines - Insulin - Novolog - active - includes reasonCode and note. Dosage includes additionalInstructions as text | medrx0321 | XML | JSON | Turtle
Request for Intravenous Admixture - Potassium Chloride in D5W - completed with reason code | medrx0322 | XML | JSON | Turtle
Request for TPN Solution - completed | medrx0323 | XML | JSON | Turtle
Request using Medication Code for Rectal Suppository - Acetaminophen - completed with reasonCode, dispenseRequest and substitution | medrx0324 | XML | JSON | Turtle
Request using Medication Code for Topical Ointment - Nystatin - on hold and with dispense request | medrx0325 | XML | JSON | Turtle
Request using Medication Code for Inhaler - Proventil - on hold with dispenseRequest and substitution | medrx0326 | XML | JSON | Turtle
Request using Medication Code for Patch - Fentanyl - active, with dispense request | medrx0327 | XML | JSON | Turtle
Request referencing Medication Resource for Nasal Spray - Nasonex | medrx0328 | XML | JSON | Turtle
Request referencing Medication Resource for Compounded Product - HC 1%, Salicylic Acid 5% in Glaxal Base - on hold with dispenseRequest | medrx0329 | XML | JSON | Turtle
Request referencing Medication Resource for Eye Drops - Timoptic - active - with link to encounter, dispenseRequest and substitution flag | medrx0330 | XML | JSON | Turtle
Request for a Dosage that needs to be dispensed as different strength tablets - Warfarin - active | medrx0331 | XML | JSON | Turtle
Request for a medication that is intended to be given as a single, once only dose | medrx0332 | XML | JSON | Turtle
Request for a medication that is intended to be given in the morning | medrx0333 | XML | JSON | Turtle
Request for a medication that is intended to be given on a specific day | medrx0334 | XML | JSON | Turtle
Request for a medication that is intended to be given at a specific time | medrx0335 | XML | JSON | Turtle
Request for Intravenous Dopamine - Weight based calculation with rateQuantity | medrx0336 | XML | JSON | Turtle
Request for Intravenous Admixture Dopamine - with rateQuantity | medrx0337 | XML | JSON | Turtle
Request for Intravenous Admixture Dopamine - with doseQuantity and rateQuantity | medrx0338 | XML | JSON | Turtle
Request for a product that contains multiple dosage forms | medrx0339 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.1.6 Resource MedicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicationrequest-definitions.html

11.1.6 Resource MedicationRequest - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the MedicationRequest resource.

MedicationRequest
Element Id | MedicationRequest
Definition | An order or request for both supply of the medication and the instructions for administration of the medication to a patient. The resource is called "MedicationRequest" rather than "MedicationPrescription" or "MedicationOrder" to generalize the use across inpatient and outpatient settings, including care plans, etc., and to harmonize with workflow patterns.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Prescription; Order
MedicationRequest.identifier
Element Id | MedicationRequest.identifier
Definition | Identifiers associated with this medication request that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business identifiers assigned to this resource by the

performer or other systems and remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | This is a business identifier, not a resource identifier.
MedicationRequest.status
Element Id | MedicationRequest.status
Definition | A code specifying the current state of the order. Generally, this will be active or completed state.
Cardinality | 1..1
Terminology Binding | Medicationrequest status (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationRequest.statusReason
Element Id | MedicationRequest.statusReason
Definition | Captures the reason for the current state of the MedicationRequest.
Cardinality | 0..1
Terminology Binding | Medication request status reason codes (Example)
Type | CodeableConcept
Comments | This is generally only used for "exception" statuses such as "suspended" or "cancelled". The reason why the MedicationRequest was created at all is captured in reasonCode, not here.
MedicationRequest.intent
Element Id | MedicationRequest.intent
Definition | Whether the request is a proposal, plan, or an original order.
Cardinality | 1..1
Terminology Binding | Medication request intent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request)
Summary | true
Comments | It is expected that the type of requester will be restricted for different stages of a MedicationRequest. For example, Proposals can be created by a patient, relatedPerson, Practitioner or Device. Plans can be created by Practitioners, Patients, RelatedPersons and Devices. Original orders can be created by a Practitioner only. An instance-order is an instantiation of a request or order and may be used to populate Medication Administration Record. This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
MedicationRequest.category
Element Id | MedicationRequest.category
Definition | Indicates the type of medication request (for example, where the medication is expected to be consumed or administered (i.e. inpatient or outpatient)).
Cardinality | 0..*
Terminology Binding | Medication request category codes (Example)
Type | CodeableConcept
Comments | The category can be used to include where the medication is expected to be consumed or other types of requests.
MedicationRequest.priority
Element Id | MedicationRequest.priority
Definition | Indicates how quickly the Medication Request should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Summary | true
MedicationRequest.doNotPerform
Element Id | MedicationRequest.doNotPerform
Definition | If true indicates that the provider is asking for the medication request not to occur.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: This element is labeled as a modifier because this element negates the request to occur (ie, this is a request for the medication not to be ordered or prescribed, etc))
Summary | true
Comments | If do not perform is not specified, the request is a positive request e.g. "do perform".
MedicationRequest.reported[x]
Element Id | MedicationRequest.reported[x]
Definition | Indicates if this record was captured as a secondary 'reported' record rather than as an original primary source-of-truth record. It may also indicate the source of the report.
Cardinality | 0..1
Type | boolean|Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicationRequest.medication[x]
Element Id | MedicationRequest.medication[x]

Definition | Identifies the medication being requested. This is a link to a resource that represents the medication which may be the details of the medication or simply an attribute carrying a code that identifies the medication from a known list of medications.
Cardinality | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the Medication resource is recommended. For example, if you require form or lot number or if the medication is compounded or extemporaneously prepared, then you must reference the Medication resource.
MedicationRequest.subject
Element Id | MedicationRequest.subject
Definition | A link to a resource representing the person or set of individuals to whom the medication will be given.
Cardinality | 1..1
Type | Reference(Patient | Group)
Summary | true
Comments | The subject on a medication request is mandatory. For the secondary use case where the actual subject is not provided, there still must be an anonymized subject specified.
MedicationRequest.encounter
Element Id | MedicationRequest.encounter
Definition | The Encounter during which this [x] was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter." If there is a need to link to episodes of care they will be handled with an extension.
MedicationRequest.supportingInformation
Element Id | MedicationRequest.supportingInformation
Definition | Include additional information (for example, patient height and weight) that supports the ordering of the medication.
Cardinality | 0..*
Type | Reference(Any)
MedicationRequest.authoredOn
Element Id | MedicationRequest.authoredOn
Definition | The date (and perhaps time) when the prescription was initially written or authored on.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicationRequest.requester
Element Id | MedicationRequest.requester
Definition | The individual, organization, or device that initiated the request and has responsibility for its activation.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
Summary | true
MedicationRequest.performer
Element Id | MedicationRequest.performer
Definition | The specified desired performer of the medication treatment (e.g. the performer of the medication administration).
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson | CareTeam)
MedicationRequest.performerType
Element Id | MedicationRequest.performerType
Definition | Indicates the type of performer of the administration of the medication.
Cardinality | 0..1
Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Summary | true
Comments | If specified without indicating a performer, this indicates that the performer must be of the specified type. If specified with a performer then it indicates the requirements of the performer if the designated performer is not available.
MedicationRequest.recorder
Element Id | MedicationRequest.recorder
Definition | The person who entered the order on behalf of another individual for example in the case of a verbal or a telephone order.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
MedicationRequest.reasonCode
Element Id | MedicationRequest.reasonCode
Definition | The reason or the indication for ordering or not ordering the medication.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonReference.
MedicationRequest.reasonReference
Element Id | MedicationRequest.reasonReference
Definition | Condition or observation that supports why the medication was ordered.

Cardinality | 0..*
Type | Reference(Condition | Observation)
Comments | This is a reference to a condition or observation that is the reason for the medication order. If only a code exists, use reasonCode.
MedicationRequest.instantiatesCanonical
Element Id | MedicationRequest.instantiatesCanonical
Definition | The URL pointing to a protocol, guideline, orderset, or other definition that is adhered to in whole or in part by this MedicationRequest.
Cardinality | 0..*
Type | canonical
Summary | true
MedicationRequest.instantiatesUri
Element Id | MedicationRequest.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this MedicationRequest.
Cardinality | 0..*
Type | uri
Summary | true
MedicationRequest.basedOn
Element Id | MedicationRequest.basedOn
Definition | A plan or request that is fulfilled in whole or in part by this medication request.
Cardinality | 0..*
Type | Reference(CarePlan | MedicationRequest | ServiceRequest | ImmunizationRecommendation)
Summary | true
MedicationRequest.groupIdentifier
Element Id | MedicationRequest.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition or prescription.
Cardinality | 0..1
Type | Identifier
Requirements | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
Summary | true
MedicationRequest.courseOfTherapyType
Element Id | MedicationRequest.courseOfTherapyType
Definition | The description of the overall patte3rn of the administration of the medication to the patient.
Cardinality | 0..1
Terminology Binding | Medication request course of therapy codes (Example)
Type | CodeableConcept
Comments | This attribute should not be confused with the protocol of the medication.
MedicationRequest.insurance
Element Id | MedicationRequest.insurance
Definition | Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be required for delivering the requested service.
Cardinality | 0..*
Type | Reference(Coverage | ClaimResponse)
MedicationRequest.note
Element Id | MedicationRequest.note
Definition | Extra information about the prescription that could not be conveyed by the other attributes.
Cardinality | 0..*
Type | Annotation
MedicationRequest.dosageInstruction
Element Id | MedicationRequest.dosageInstruction
Definition | Indicates how the medication is to be used by the patient.
Cardinality | 0..*
Type | Dosage
Comments | There are examples where a medication request may include the option of an oral dose or an Intravenous or Intramuscular dose. For example, "Ondansetron 8mg orally or IV twice a day as needed for nausea" or "Compazine® (prochlorperazine) 5-10mg PO or 25mg PR bid prn nausea or vomiting". In these cases, two medication requests would be created that could be grouped together. The decision on which dose and route of administration to use is based on the patient's condition at the time the dose is needed.
MedicationRequest.dispenseRequest
Element Id | MedicationRequest.dispenseRequest
Definition | Indicates the specific details for the dispense or medication supply part of a medication request (also known as a Medication Prescription or Medication Order). Note that this information is not always sent with the order. There may be in some settings (e.g. hospitals) institutional or system support for completing the dispense details in the pharmacy department.
Cardinality | 0..1
MedicationRequest.dispenseRequest.initialFill
Element Id | MedicationRequest.dispenseRequest.initialFill
Definition | Indicates the quantity or duration for the first dispense of the medication.
Cardinality | 0..1
Comments | If populating this element, either the quantity or the duration must be included.

MedicationRequest.dispenseRequest.initialFill.quantity
Element Id | MedicationRequest.dispenseRequest.initialFill.quantity
Definition | The amount or quantity to provide as part of the first dispense.
Cardinality | 0..1
Type | SimpleQuantity
MedicationRequest.dispenseRequest.initialFill.duration
Element Id | MedicationRequest.dispenseRequest.initialFill.duration
Definition | The length of time that the first dispense is expected to last.
Cardinality | 0..1
Type | Duration
MedicationRequest.dispenseRequest.dispenseInterval
Element Id | MedicationRequest.dispenseRequest.dispenseInterval
Definition | The minimum period of time that must occur between dispenses of the medication.
Cardinality | 0..1
Type | Duration
MedicationRequest.dispenseRequest.validityPeriod
Element Id | MedicationRequest.dispenseRequest.validityPeriod
Definition | This indicates the validity period of a prescription (stale dating the Prescription).
Cardinality | 0..1
Type | Period
Requirements | Indicates when the Prescription becomes valid, and when it ceases to be a dispensable Prescription.
Comments | It reflects the prescribers' perspective for the validity of the prescription. Dispenses must not be made against the prescription outside of this period. The lower-bound of the Dispensing Window signifies the earliest date that the prescription can be filled for the first time. If an upper-bound is not specified then the Prescription is open-ended or will default to a stale-date based on regulations.
MedicationRequest.dispenseRequest.numberOfRepeatsAllowed
Element Id | MedicationRequest.dispenseRequest.numberOfRepeatsAllowed
Definition | An integer indicating the number of times, in addition to the original dispense, (aka refills or repeats) that the patient can receive the prescribed medication. Usage Notes: This integer does not include the original order dispense. This means that if an order indicates dispense 30 tablets plus "3 repeats", then the order can be dispensed a total of 4 times and the patient can receive a total of 120 tablets. A prescriber may explicitly say that zero refills are permitted after the initial dispense.
Cardinality | 0..1
Type | unsignedInt
Comments | If displaying "number of authorized fills", add 1 to this number.
MedicationRequest.dispenseRequest.quantity
Element Id | MedicationRequest.dispenseRequest.quantity
Definition | The amount that is to be dispensed for one fill.
Cardinality | 0..1
Type | SimpleQuantity
MedicationRequest.dispenseRequest.expectedSupplyDuration
Element Id | MedicationRequest.dispenseRequest.expectedSupplyDuration
Definition | Identifies the period time over which the supplied product is expected to be used, or the length of time the dispense is expected to last.
Cardinality | 0..1
Type | Duration
Comments | In some situations, this attribute may be used instead of quantity to identify the amount supplied by how long it is expected to last, rather than the physical quantity issued, e.g. 90 days supply of medication (based on an ordered dosage). When possible, it is always better to specify quantity, as this tends to be more precise. expectedSupplyDuration will always be an estimate that can be influenced by external factors.
MedicationRequest.dispenseRequest.performer
Element Id | MedicationRequest.dispenseRequest.performer
Definition | Indicates the intended dispensing Organization specified by the prescriber.
Cardinality | 0..1
Type | Reference(Organization)
MedicationRequest.substitution
Element Id | MedicationRequest.substitution
Definition | Indicates whether or not substitution can or should be part of the dispense. In some cases, substitution must happen, in other cases substitution must not happen. This block explains the prescriber's intent. If nothing is specified substitution may be done.
Cardinality | 0..1
MedicationRequest.substitution.allowed[x]
Element Id | MedicationRequest.substitution.allowed[x]
Definition | True if the prescriber allows a different drug to be dispensed from what was prescribed.
Cardinality | 1..1
Terminology Binding | V3 Value SetActSubstanceAdminSubstitutionCode (Example)
Type | boolean|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | This element is labeled as a modifier because whether substitution is allow or not, it cannot be ignored.
MedicationRequest.substitution.reason
Element Id | MedicationRequest.substitution.reason
Definition | Indicates the reason for the substitution, or why substitution must or must not be performed.
Cardinality | 0..1
Terminology Binding | V3 Value SetSubstanceAdminSubstitutionReason (Example)

Type | CodeableConcept
MedicationRequest.priorPrescription
Element Id | MedicationRequest.priorPrescription
Definition | A link to a resource representing an earlier order related order or prescription.
Cardinality | 0..1
Type | Reference(MedicationRequest)
MedicationRequest.detectedIssue
Element Id | MedicationRequest.detectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, duplicate therapy, dosage alert etc.
Cardinality | 0..*
Type | Reference(DetectedIssue)
Alternate Names | Contraindication; Drug Utilization Review (DUR); Alert
Comments | This element can include a detected issue that has been identified either by a decision support system or by a clinician and may include information on the steps that were taken to address the issue.
MedicationRequest.eventHistory
Element Id | MedicationRequest.eventHistory
Definition | Links to Provenance records for past versions of this resource or fulfilling request or event resources that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the resource.
Cardinality | 0..*
Type | Reference(Provenance)
Comments | This might not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.).

11.1.7 Resource MedicationRequest - Mappings

URL: https://hl7.org/fhir/R4/medicationrequest-mappings.html

11.1.7 Resource MedicationRequest - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mapping to NCPDP SCRIPT 10.6

Mappings for the medicationrequest resource (see Mappings to Other Standards for further information & status).

11.1.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MedicationRequest
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
category | FiveWs.class
priority | FiveWs.grade
medication[x] | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
supportingInformation | FiveWs.context
authoredOn | FiveWs.recorded
requester | FiveWs.author
performer | FiveWs.actor
recorder | FiveWs.who
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]
performer | FiveWs.who

11.1.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

MedicationRequest | Request
identifier | Request.identifier
status | Request.status
statusReason | Request.statusReason
intent | Request.intent
priority | Request.priority
medication[x] | Request.code

subject | Request.subject
encounter | Request.context
supportingInformation | Request.supportingInfo
authoredOn | Request.authoredOn
requester | Request.requester
performer | Request.performer
performerType | Request.performerType
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
instantiatesCanonical | Request.instantiates
basedOn | Request.basedOn
groupIdentifier | Request.groupIdentifier
insurance | Request.insurance
note | Request.note
dosageInstruction | Request.occurrence[x]
priorPrescription | Request.replaces
eventHistory | Request.relevantHistory

11.1.7.3 HL7 v2 Mapping (http://hl7.org/v2)

MedicationRequest
identifier | ORC-2-Placer Order Number / ORC-3-Filler Order Number
status
statusReason
intent
category
priority
doNotPerform
reported[x]
medication[x] | RXE-2-Give Code / RXO-1-Requested Give Code / RXC-2-Component Code
subject | PID-3-Patient ID List
encounter | PV1-19-Visit Number
supportingInformation
authoredOn | RXE-32-Original Order Date/Time / ORC-9-Date/Time of Transaction
requester
performer
performerType
recorder
reasonCode | ORC-16-Order Control Code Reason /RXE-27-Give Indication/RXO-20-Indication / RXD-21-Indication / RXG-22-Indication / RXA-19-Indication
reasonReference
instantiatesCanonical
instantiatesUri
basedOn
groupIdentifier
courseOfTherapyType
insurance
note
dosageInstruction
dispenseRequest
initialFill
quantity
duration
dispenseInterval
validityPeriod
numberOfRepeatsAllowed | RXE-12-Number of Refills
quantity | RXD-4-Actual Dispense Amount / RXD-5.1-Actual Dispense Units.code / RXD-5.3-Actual Dispense Units.name of coding system
expectedSupplyDuration
performer
substitution
allowed[x] | RXO-9-Allow Substitutions / RXE-9-Substitution Status
reason | RXE-9 Substition status
priorPrescription
detectedIssue
eventHistory

11.1.7.4 RIM Mapping (http://hl7.org/v3)

MedicationRequest | CombinedMedicationRequest
identifier | .id
status | .statusCode
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN].reasonCOde
intent | .moodCode (nuances beyond PRP/PLAN/RQO would need to be elsewhere)

category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
priority | .priorityCode
doNotPerform | SubstanceAdministration.actionNegationInd
reported[x] | .participation[typeCode=INF].role
medication[x] | consumable.administrableMedication
subject | .participation[typeCode=AUT].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
authoredOn | author.time
requester | .participation[typeCode=AUT].role
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
performerType | .participation[typeCode=PRF].role[scoper.determinerCode=KIND].code
recorder | .participation[typeCode=TRANS].role[classCode=ASSIGNED].code (HealthcareProviderType)
reasonCode | reason.observation.reasonCode
reasonReference | reason.observation[code=ASSERTION].value
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=SBADM or PROC or PCPR or OBS, moodCode=RQO orPLAN or PRP]
groupIdentifier | .outboundRelationship(typeCode=COMP].target[classCode=SBADM, moodCode=INT].id
courseOfTherapyType | Act.code where classCode = LIST and moodCode = EVN
insurance | .outboundRelationship[typeCode=COVBY].target
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosageInstruction | see dosageInstruction mapping
dispenseRequest | component.supplyEvent
initialFill | SubstanceAdministration -> ActRelationship[sequenceNumber = '1'] -> Supply
quantity | Supply.quantity[moodCode=RQO]
duration | Supply.effectivetime[moodCode=RQO]
dispenseInterval | Supply.effectivetime[moodCode=RQO]
validityPeriod | effectiveTime
numberOfRepeatsAllowed | repeatNumber
quantity | quantity
expectedSupplyDuration | expectedUseTime
performer | .outboundRelationship[typeCode=COMP].target[classCode=SPLY, moodCode=RQO] .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
substitution | subjectOf.substitutionPersmission
allowed[x] | code
reason | reasonCode
priorPrescription | .outboundRelationship[typeCode=?RPLC or ?SUCC]/target[classCode=SBADM,moodCode=RQO]
detectedIssue | .inboundRelationship[typeCode=SUBJ]/source[classCode=ALRT,moodCode=EVN].value
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.1.7.5 Mapping to NCPDP SCRIPT 10.6 (http://ncpdp.org/SCRIPT10_6)

MedicationRequest | Message/Body/NewRx
identifier | Message/Header/PrescriberOrderNumber
status | no mapping
statusReason
intent
category | Message/Body/NewRx/MedicationPrescribed/Directions or Message/Body/NewRx/MedicationPrescribed/StructuredSIG
priority
doNotPerform
reported[x]
medication[x] | Message/Body/NewRx/MedicationPrescribed Medication.code.coding.code = Message/Body/NewRx/MedicationPrescribed/DrugCoded/ProductCode Medication.code.coding.system = Message/Body/NewRx/MedicationPrescribed/DrugCoded/ProductCodeQualifier Medication.code.coding.display = Message/Body/NewRx/MedicationPrescribed/DrugDescription
subject | Message/Body/NewRx/Patient (need detail to link to specific patient ... Patient.Identification in SCRIPT)
encounter | no mapping
supportingInformation
authoredOn | Message/Body/NewRx/MedicationPrescribed/WrittenDate
requester
performer
performerType
recorder
reasonCode | Message/Body/NewRx/MedicationPrescribed/Diagnosis/Primary/Value
reasonReference | no mapping
instantiatesCanonical
instantiatesUri
basedOn

groupIdentifier
courseOfTherapyType
insurance
note | Message/Body/NewRx/MedicationPrescribed/Note
dosageInstruction
dispenseRequest | Message/Body/NewRx/MedicationPrescribed/ExpirationDate
initialFill
quantity
duration
dispenseInterval
validityPeriod | Message/Body/NewRx/MedicationPrescribed/Refills
numberOfRepeatsAllowed | Message/Body/NewRx/MedicationPrescribed/Quantity
quantity | Message/Body/NewRx/MedicationPrescribed/DaysSupply
expectedSupplyDuration | Message/Body/NewRx/MedicationPrescribed/Substitutions
performer
substitution | specific values within Message/Body/NewRx/MedicationPrescribed/Substitutions
allowed[x] | specific values within Message/Body/NewRx/MedicationPrescribed/Substitutions
reason | not mapped
priorPrescription | not mapped
detectedIssue
eventHistory

11.1.8 Resource MedicationRequest - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicationrequest-profiles.html

11.1.8 Resource MedicationRequest - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

This table lists profiles and extensions for the MedicationRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.1.8.1 Profiles

No Profiles defined for this resource

11.1.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.1.8.3 Search Extensions

No Search Extensions defined for this resource

11.1.9 Resource MedicationRequest - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/medicationrequest-version-maps.html

No validation errors - all conversions are clean

11.1.9 Resource MedicationRequest - R3/R4 Conversions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

R3 : R4 Conversion maps for MedicationRequest.

Functional status for this map: 36 tests that all execute ok. 8 fail round-trip testing and 36 r3 resources are invalid (0 errors). (see documentation)

11.1.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MedicationRequest3to4" = "R3 to R4 Conversions for MedicationRequest"

uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationRequest" alias MedicationRequestR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationRequest" alias MedicationRequest as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MedicationRequest(source src : MedicationRequestR3, target tgt : MedicationRequest) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiatesCanonical;
  src.basedOn -> tgt.basedOn;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.supportingInformation -> tgt.supportingInformation;
  src.authoredOn -> tgt.authoredOn;
  src.requester as vs0 then requester(vs0, tgt);
  src.recorder -> tgt.recorder;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.dosageInstruction -> tgt.dosageInstruction;
  src.dispenseRequest as vs0 -> tgt.dispenseRequest as vt0 then dispense(vs0, vt0);
  src.substitution as vs0 -> tgt.substitution as vt0 then subst(vs0, vt0);
  src.priorPrescription -> tgt.priorPrescription;
  src.detectedIssue -> tgt.detectedIssue;
  src.eventHistory -> tgt.eventHistory;
}

group requester(source src, target tgt) {
  src.agent -> tgt.requester;
  src.onBehalfOf as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationRequest.requester.onBehalfOf',  vt.value = v;
}

group dispense(source src, target tgt) {
  src.validityPeriod -> tgt.validityPeriod;
  src.numberOfRepeatsAllowed -> tgt.numberOfRepeatsAllowed;
  src.quantity -> tgt.quantity;
  src.expectedSupplyDuration -> tgt.expectedSupplyDuration;
  src.performer -> tgt.performer;
}

group subst(source src, target tgt) {
  src.allowed as vs -> tgt.allowed = create('boolean') as vt then boolean(vs, vt);
  src.reason -> tgt.reason;
}
```

11.1.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationRequest4to3" = "R4 to R3 Conversion for MedicationRequest"

uses "http://hl7.org/fhir/StructureDefinition/MedicationRequest" alias MedicationRequest as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationRequest" alias MedicationRequestR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MedicationRequest(source src : MedicationRequest, target tgt : MedicationRequestR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.definition;
  src.basedOn -> tgt.basedOn;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.supportingInformation -> tgt.supportingInformation;
  src.authoredOn -> tgt.authoredOn;
  src as vs0 where requester.exists() or extension.where(url = '').exists() -> tgt.requester as vt then requester(src,
    vt) "requester";
  src.recorder -> tgt.recorder;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.dosageInstruction -> tgt.dosageInstruction;
  src.dispenseRequest as vs0 -> tgt.dispenseRequest as vt0 then dispense(vs0, vt0);
```

```
  src.substitution as vs0 -> tgt.substitution as vt0 then subst(vs0, vt0);
  src.priorPrescription -> tgt.priorPrescription;
  src.detectedIssue -> tgt.detectedIssue;
  src.eventHistory -> tgt.eventHistory;
}

group requester(source src, target tgt) {
  src.requester -> tgt.agent;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationRequest.requester.onBehalfOf' then {
    e.value -> tgt.onBehalfOf;
  };
}

group dispense(source src, target tgt) {
  src.validityPeriod -> tgt.validityPeriod;
  src.numberOfRepeatsAllowed -> tgt.numberOfRepeatsAllowed;
  src.quantity -> tgt.quantity;
  src.expectedSupplyDuration -> tgt.expectedSupplyDuration;
  src.performer -> tgt.performer;
}

group subst(source src, target tgt) {
  src.allowed : boolean as vs -> tgt.allowed as vt then boolean(vs, vt);
  src.reason -> tgt.reason;
}
```

11.1.9.3 R4 Validation Errors

11.1.10 MedicationRequest HL7 Extensions

URL: https://hl7.org/fhir/R4/medicationrequest-extensions.html

Defines common extensions used with or related to the MedicationRequest resource

11.1.10 MedicationRequest HL7 Extensions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

11.1.10.1 Content

11.1.11 JSON Schema for MedicationRequest

URL: https://hl7.org/fhir/R4/medicationrequest.schema.json.html

11.1.11 JSON Schema for MedicationRequest

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is a representation of the json schema for MedicationRequest, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicationRequest",
  "$ref": "#/definitions/MedicationRequest",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicationRequest": {
      "description": "An order or request for both supply of the medication and the instructions for administration of
    the medication to a patient. The resource is called \"MedicationRequest\" rather than \"MedicationPrescription\" or
    \"MedicationOrder\" to generalize the use across inpatient and outpatient settings, including care plans, etc., and
    to harmonize with workflow patterns.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicationRequest resource",
          "const": "MedicationRequest"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers associated with this medication request that are defined by business processes
and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business
identifiers assigned to this resource by the performer or other systems and remain constant as the resource is
updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "A code specifying the current state of the order.  Generally, this will be active or completed
 state.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current state of the MedicationRequest.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "intent": {
      "description": "Whether the request is a proposal, plan, or an original order.",
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Indicates the type of medication request (for example, where the medication is expected to be
consumed or administered (i.e. inpatient or outpatient)).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "priority": {
      "description": "Indicates how quickly the Medication Request should be addressed with respect to other
```

```
requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doNotPerform": {
      "description": "If true indicates that the provider is asking for the medication request not to occur.",
      "$ref": "#/definitions/boolean"
    },
    "_doNotPerform": {
      "description": "Extensions for doNotPerform",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reportedBoolean": {
      "description": "Indicates if this record was captured as a secondary \u0027reported\u0027 record rather than
as an original primary source-of-truth record.  It may also indicate the source of the report.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_reportedBoolean": {
      "description": "Extensions for reportedBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reportedReference": {
      "description": "Indicates if this record was captured as a secondary \u0027reported\u0027 record rather than
as an original primary source-of-truth record.  It may also indicate the source of the report.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "medicationCodeableConcept": {
      "description": "Identifies the medication being requested. This is a link to a resource that represents the
medication which may be the details of the medication or simply an attribute carrying a code that identifies the
medication from a known list of medications.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationReference": {
      "description": "Identifies the medication being requested. This is a link to a resource that represents the
medication which may be the details of the medication or simply an attribute carrying a code that identifies the
medication from a known list of medications.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "subject": {
      "description": "A link to a resource representing the person or set of individuals to whom the medication will
 be given.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this [x] was created or to which the creation of this record is
tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "supportingInformation": {
      "description": "Include additional information (for example, patient height and weight) that supports the
ordering of the medication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "authoredOn": {
      "description": "The date (and perhaps time) when the prescription was initially written or authored on.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requester": {
      "description": "The individual, organization, or device that initiated the request and has responsibility for
its activation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performer": {
      "description": "The specified desired performer of the medication treatment (e.g. the performer of the
medication administration).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performerType": {
      "description": "Indicates the type of performer of the administration of the medication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "recorder": {
      "description": "The person who entered the order on behalf of another individual for example in the case of a
verbal or a telephone order.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "The reason or the indication for ordering or not ordering the medication.",
```

```
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Condition or observation that supports why the medication was ordered.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a protocol, guideline, orderset, or other definition that is adhered to in
 whole or in part by this MedicationRequest.",
      "items": {
        "$ref": "#/definitions/canonical"
      },
      "type": "array"
    },
    "_instantiatesCanonical": {
      "description": "Extensions for instantiatesCanonical",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition
 that is adhered to in whole or in part by this MedicationRequest.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A plan or request that is fulfilled in whole or in part by this medication request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "groupIdentifier": {
      "description": "A shared identifier common to all requests that were authorized more or less simultaneously by
 a single author, representing the identifier of the requisition or prescription.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "courseOfTherapyType": {
      "description": "The description of the overall patte3rn of the administration of the medication to the
patient.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "insurance": {
      "description": "Insurance plans, coverage extensions, pre-authorizations and/or pre-determinations that may be
 required for delivering the requested service.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Extra information about the prescription that could not be conveyed by the other attributes.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "dosageInstruction": {
      "description": "Indicates how the medication is to be used by the patient.",
      "items": {
        "$ref": "Dosage.schema.json#/definitions/Dosage"
      },
      "type": "array"
    },
    "dispenseRequest": {
      "description": "Indicates the specific details for the dispense or medication supply part of a medication
request (also known as a Medication Prescription or Medication Order).  Note that this information is not always
sent with the order.  There may be in some settings (e.g. hospitals) institutional or system support for completing
the dispense details in the pharmacy department.",
      "$ref": "#/definitions/MedicationRequest_DispenseRequest"
    },
    "substitution": {
```

```
      "description": "Indicates whether or not substitution can or should be part of the dispense. In some cases, substitution must happen, in other cases substitution must not happen. This block explains the prescriber\u0027s intent. If nothing is specified substitution may be done.",
      "$ref": "#/definitions/MedicationRequest_Substitution"
    },
    "priorPrescription": {
      "description": "A link to a resource representing an earlier order related order or prescription.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "detectedIssue": {
      "description": "Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, duplicate therapy, dosage alert etc.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "eventHistory": {
      "description": "Links to Provenance records for past versions of this resource or fulfilling request or event resources that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the resource.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"MedicationRequest_DispenseRequest": {
  "description": "An order or request for both supply of the medication and the instructions for administration of the medication to a patient. The resource is called \"MedicationRequest\" rather than \"MedicationPrescription\" or \"MedicationOrder\" to generalize the use across inpatient and outpatient settings, including care plans, etc., and to harmonize with workflow patterns.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "initialFill": {
      "description": "Indicates the quantity or duration for the first dispense of the medication.",
      "$ref": "#/definitions/MedicationRequest_InitialFill"
    },
    "dispenseInterval": {
      "description": "The minimum period of time that must occur between dispenses of the medication.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "validityPeriod": {
      "description": "This indicates the validity period of a prescription (stale dating the Prescription).",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "numberOfRepeatsAllowed": {
      "description": "An integer indicating the number of times, in addition to the original dispense, (aka refills or repeats) that the patient can receive the prescribed medication. Usage Notes: This integer does not include the original order dispense. This means that if an order indicates dispense 30 tablets plus \"3 repeats\", then the order can be dispensed a total of 4 times and the patient can receive a total of 120 tablets.  A prescriber may explicitly say that zero refills are permitted after the initial dispense.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_numberOfRepeatsAllowed": {
```

```
      "description": "Extensions for numberOfRepeatsAllowed",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "quantity": {
      "description": "The amount that is to be dispensed for one fill.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "expectedSupplyDuration": {
      "description": "Identifies the period time over which the supplied product is expected to be used, or the
length of time the dispense is expected to last.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "performer": {
      "description": "Indicates the intended dispensing Organization specified by the prescriber.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"MedicationRequest_InitialFill": {
  "description": "An order or request for both supply of the medication and the instructions for administration of
the medication to a patient. The resource is called \"MedicationRequest\" rather than \"MedicationPrescription\" or
\"MedicationOrder\" to generalize the use across inpatient and outpatient settings, including care plans, etc., and
to harmonize with workflow patterns.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "quantity": {
      "description": "The amount or quantity to provide as part of the first dispense.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "duration": {
      "description": "The length of time that the first dispense is expected to last.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    }
  },
  "additionalProperties": false
},
"MedicationRequest_Substitution": {
  "description": "An order or request for both supply of the medication and the instructions for administration of
the medication to a patient. The resource is called \"MedicationRequest\" rather than \"MedicationPrescription\" or
\"MedicationOrder\" to generalize the use across inpatient and outpatient settings, including care plans, etc., and
to harmonize with workflow patterns.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
```

```
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "allowedBoolean": {
          "description": "True if the prescriber allows a different drug to be dispensed from what was prescribed.",
          "pattern": "^true|false$",
          "type": "boolean"
        },
        "_allowedBoolean": {
          "description": "Extensions for allowedBoolean",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "allowedCodeableConcept": {
          "description": "True if the prescriber allows a different drug to be dispensed from what was prescribed.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "reason": {
          "description": "Indicates the reason for the substitution, or why substitution must or must not be
    performed.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.1.12 Resource MedicationRequest - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicationrequest-dependencies.html

11.1.12 Resource MedicationRequest - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner

Dependency Graph for MedicationRequest FMM level 3

Name | Card. | Type | Dependency Analysis
MedicationRequest | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
intent | 1..1 | code
category | 0..* | CodeableConcept
priority | 0..1 | code
doNotPerform | 0..1 | boolean
reported[x] | 0..1
reportedBoolean | | boolean
reportedReference | | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(Medication
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
supportingInformation | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
authoredOn | 0..1 | dateTime
requester | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
performer | 0..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson | CareTeam) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (CareTeam = FMM2-Trial Use vs. Element = FMM3-Trial Use)
performerType | 0..1 | CodeableConcept
recorder | 0..1 | Reference(Practitioner | PractitionerRole) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation)
instantiatesCanonical | 0..* | canonical()
instantiatesUri | 0..* | uri

basedOn | 0..* | Reference(CarePlan | MedicationRequest | ServiceRequest | ImmunizationRecommendation) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ImmunizationRecommendation = FMM1-Trial Use vs. Element = FMM3-Trial Use)
groupIdentifier | 0..1 | Identifier
courseOfTherapyType | 0..1 | CodeableConcept
insurance | 0..* | Reference(Coverage | ClaimResponse) | Type Warning: (Coverage = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ClaimResponse = FMM2-Trial Use vs. Element = FMM3-Trial Use)
note | 0..* | Annotation
dosageInstruction | 0..* | Dosage
dispenseRequest | 0..1 | BackboneElement
initialFill | 0..1 | BackboneElement
quantity | 0..1 | SimpleQuantity
duration | 0..1 | Duration
dispenseInterval | 0..1 | Duration
validityPeriod | 0..1 | Period
numberOfRepeatsAllowed | 0..1 | unsignedInt
quantity | 0..1 | SimpleQuantity
expectedSupplyDuration | 0..1 | Duration
performer | 0..1 | Reference(Organization)
substitution | 0..1 | BackboneElement
allowed[x] | 1..1
allowedBoolean | | boolean
allowedCodeableConcept | | CodeableConcept
reason | 0..1 | CodeableConcept
priorPrescription | 0..1 | Reference(MedicationRequest)
detectedIssue | 0..* | Reference(DetectedIssue) | Type Warning: (DetectedIssue = FMM1-Trial Use vs. Element = FMM3-Trial Use)
eventHistory | 0..* | Reference(Provenance)

11.1.13 Value Set Sc-valueset-medicationrequest-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medicationrequest-status.html

11.1.13 Value Set Sc-valueset-medicationrequest-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.1.14 medicationrequest StatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medicationrequest-status)

Mapping from http://hl7.org/fhir/ValueSet/medicationrequest-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "MedicationRequest Status Codes"

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
draft | equivalent | draft
active | equivalent | active
on-hold | equivalent | suspended
stopped | equivalent | failed
completed | equivalent | complete
cancelled | equivalent | abandoned
unknown | equivalent | unknown

11.2 Resource MedicationAdministration - Content

URL: https://hl7.org/fhir/R4/medicationadministration.html
Pharmacy Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Describes the event of a patient consuming or otherwise being administered a medication. This may be as simple as swallowing a tablet or it may be a long running infusion. Related resources tie this event to the authorizing prescription, and the specific encounter between patient and health care practitioner.

11.2.1 Scope and Usage

This resource covers the administration of all medications and vaccines. Please refer to the Immunization Resource/Profile for the treatment of vaccines. It will principally be used within care settings (including

inpatient) to record the capture of medication administrations, including self-administrations of oral medications, injections, intra-venous adjustments, etc. It can also be used in outpatient settings to record allergy shots and other non-immunization administrations. In some cases, it might be used for home-health reporting, such as recording self-administered or even device-administered insulin.

MedicationAdministration is an event resource from a FHIR workflow perspective - see Workflow Event

11.2.2 Boundaries and Relationships

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.

MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them

MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

MedicationAdministration is intended for tracking the administration of non-vaccine medications. Administration of vaccines is intended to be handled using the Immunization resource. Some systems treat immunizations in the same way as any other medication administration. Such systems SHOULD use an immunization resource to represent these. If systems need to use a MedicationAdministration resource to capture vaccinations for workflow or other reasons, they SHOULD also create and expose an equivalent Immunization instance.

This resource is referenced by AdverseEvent, ChargeItem, itself, MedicationStatement, Observation and Procedure

11.2.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicationAdministration | TU | | DomainResource | Administration of medication to a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | External identifier

instantiates | Σ | 0..* | uri | Instantiates protocol or definition

partOf | Σ | 0..* | Reference(MedicationAdministration | Procedure) | Part of referenced event

status | ?!Σ | 1..1 | code | in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown Medication administration status codes (Required)

statusReason | | 0..* | CodeableConcept | Reason administration not performed SNOMED CT Reason Medication Not Given Codes (Example)

category | | 0..1 | CodeableConcept | Type of medication usage Medication administration category codes (Preferred)

medication[x] | Σ | 1..1 | | What was administered SNOMED CT Medication Codes (Example)

medicationCodeableConcept | | | CodeableConcept

medicationReference | | | Reference(Medication)

subject | Σ | 1..1 | Reference(Patient | Group) | Who received medication

context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode of Care administered as part of

supportingInformation | | 0..* | Reference(Any) | Additional information to support administration

effective[x] | Σ | 1..1 | | Start and end time of administration

effectiveDateTime | | | dateTime

effectivePeriod | | | Period

performer | Σ | 0..* | BackboneElement | Who performed the medication administration and what they did

function | | 0..1 | CodeableConcept | Type of performance Medication administration performer function codes (Example)

actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Device) | Who performed the medication administration

reasonCode | | 0..* | CodeableConcept | Reason administration performed Reason Medication Given Codes (Example)

reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Condition or observation that supports why the medication was administered

request | | 0..1 | Reference(MedicationRequest) | Request administration performed against

device | | 0..* | Reference(Device) | Device used to administer

note | | 0..* | Annotation | Information about the administration

dosage | I | 0..1 | BackboneElement | Details of how medication was taken + Rule: SHALL have at least one of dosage.dose or dosage.rate[x]

text | | 0..1 | string | Free text dosage instructions e.g. SIG
site | | 0..1 | CodeableConcept | Body site administered to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | | 0..1 | CodeableConcept | Path of substance into body SNOMED CT Route Codes (Example)
method | | 0..1 | CodeableConcept | How drug was administered SNOMED CT Administration Method Codes (Example)
dose | | 0..1 | SimpleQuantity | Amount of medication per dose
rate[x] | | 0..1 | | Dose quantity per unit of time
rateRatio | | | Ratio
rateQuantity | | | SimpleQuantity
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationAdministration xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <instantiates value="[uri]"/><!-- 0..* Instantiates protocol or definition -->
 <partOf><!-- 0..* Reference(MedicationAdministration|Procedure) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown
    -->
 <statusReason><!-- 0..* CodeableConcept Reason administration not performed --></statusReason>
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What was administered --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who received medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode of Care administered as part of
    --></context>
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support administration
    --></supportingInformation>
 <effective[x]><!-- 1..1 dateTime|Period Start and end time of administration --></effective[x]>
 <performer>  <!-- 0..* Who performed the medication administration and what they did -->
  <function><!-- 0..1 CodeableConcept Type of performance --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
    Device) Who performed the medication administration --></actor>
 </performer>
 <reasonCode><!-- 0..* CodeableConcept Reason administration performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Condition or observation that supports why
     the medication was administered --></reasonReference>
 <request><!-- 0..1 Reference(MedicationRequest) Request administration performed against --></request>
 <device><!-- 0..* Reference(Device) Device used to administer --></device>
 <note><!-- 0..* Annotation Information about the administration --></note>
 <dosage>  <!-- 0..1 Details of how medication was taken -->
  <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
  <site><!-- 0..1 CodeableConcept Body site administered to --></site>
  <route><!-- 0..1 CodeableConcept Path of substance into body --></route>
  <method><!-- 0..1 CodeableConcept How drug was administered --></method>
  <dose><!-- 0..1 Quantity(SimpleQuantity) Amount of medication per dose --></dose>
  <rate[x]><!-- 0..1 Ratio|Quantity(SimpleQuantity) Dose quantity per unit of time --></rate[x]>
 </dosage>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationAdministration>
```

JSON Template

```
{
  "resourceType" : "MedicationAdministration",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "instantiates" : ["<uri>"], // Instantiates protocol or definition
  "partOf" : [{ Reference(MedicationAdministration|Procedure) }], // Part of referenced event
  "status" : "<code>", // R!  in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown
  "statusReason" : [{ CodeableConcept }], // Reason administration not performed
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What was administered. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who received medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode of Care administered as part of
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support administration
  // effective[x]: Start and end time of administration. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "performer" : [{ // Who performed the medication administration and what they did
    "function" : { CodeableConcept }, // Type of performance
    "actor" : { Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
    Device) } // R!  Who performed the medication administration
  }],
  "reasonCode" : [{ CodeableConcept }], // Reason administration performed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Condition or observation that supports
     why the medication was administered
  "request" : { Reference(MedicationRequest) }, // Request administration performed against
  "device" : [{ Reference(Device) }], // Device used to administer
  "note" : [{ Annotation }], // Information about the administration
  "dosage" : { // Details of how medication was taken
    "text" : "<string>", // Free text dosage instructions e.g. SIG
```

```
    "site" : { CodeableConcept }, // Body site administered to
    "route" : { CodeableConcept }, // Path of substance into body
    "method" : { CodeableConcept }, // How drug was administered
    "dose" : { Quantity(SimpleQuantity) }, // Amount of medication per dose
    // rate[x]: Dose quantity per unit of time. One of these 2:
    "rateRatio" : { Ratio }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  },
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationAdministration;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationAdministration.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationAdministration.instantiates [ uri ], ... ; # 0..* Instantiates protocol or definition
  fhir:MedicationAdministration.partOf [ Reference(MedicationAdministration|Procedure) ], ... ; # 0..* Part of
    referenced event
  fhir:MedicationAdministration.status [ code ]; # 1..1 in-progress | not-done | on-hold | completed | entered-in-error
    | stopped | unknown
  fhir:MedicationAdministration.statusReason [ CodeableConcept ], ... ; # 0..* Reason administration not performed
  fhir:MedicationAdministration.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationAdministration.medication[x] : 1..1 What was administered. One of these 2
    fhir:MedicationAdministration.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationAdministration.medicationReference [ Reference(Medication) ]
  fhir:MedicationAdministration.subject [ Reference(Patient|Group) ]; # 1..1 Who received medication
  fhir:MedicationAdministration.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode of Care
    administered as part of
  fhir:MedicationAdministration.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support
     administration
  # MedicationAdministration.effective[x] : 1..1 Start and end time of administration. One of these 2
    fhir:MedicationAdministration.effectiveDateTime [ dateTime ]
    fhir:MedicationAdministration.effectivePeriod [ Period ]
  fhir:MedicationAdministration.performer [ # 0..* Who performed the medication administration and what they did
    fhir:MedicationAdministration.performer.function [ CodeableConcept ]; # 0..1 Type of performance
    fhir:MedicationAdministration.performer.actor [
    Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|Device) ]; # 1..1 Who performed the medication
    administration
  ], ...;
  fhir:MedicationAdministration.reasonCode [ CodeableConcept ], ... ; # 0..* Reason administration performed
  fhir:MedicationAdministration.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..*
    Condition or observation that supports why the medication was administered
  fhir:MedicationAdministration.request [ Reference(MedicationRequest) ]; # 0..1 Request administration performed
    against
  fhir:MedicationAdministration.device [ Reference(Device) ], ... ; # 0..* Device used to administer
  fhir:MedicationAdministration.note [ Annotation ], ... ; # 0..* Information about the administration
  fhir:MedicationAdministration.dosage [ # 0..1 Details of how medication was taken
    fhir:MedicationAdministration.dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
    fhir:MedicationAdministration.dosage.site [ CodeableConcept ]; # 0..1 Body site administered to
    fhir:MedicationAdministration.dosage.route [ CodeableConcept ]; # 0..1 Path of substance into body
    fhir:MedicationAdministration.dosage.method [ CodeableConcept ]; # 0..1 How drug was administered
    fhir:MedicationAdministration.dosage.dose [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication per dose
    # MedicationAdministration.dosage.rate[x] : 0..1 Dose quantity per unit of time. One of these 2
      fhir:MedicationAdministration.dosage.rateRatio [ Ratio ]
      fhir:MedicationAdministration.dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ];
  fhir:MedicationAdministration.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
     lifecycle
]
```

Changes since R3

MedicationAdministration
MedicationAdministration.instantiates | Added Element
MedicationAdministration.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/medication-admin-status|4.0.1
MedicationAdministration.statusReason | Added Element
MedicationAdministration.performer.function | Added Element
MedicationAdministration.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationAdministration.reasonReference | Type Reference: Added Target Type DiagnosticReport
MedicationAdministration.request | Added Element
MedicationAdministration.dosage.rate[x] | Add Type Quantity() Remove Type Quantity()
MedicationAdministration.definition | deleted
MedicationAdministration.performer.onBehalfOf | deleted
MedicationAdministration.notGiven | deleted
MedicationAdministration.reasonNotGiven | deleted
MedicationAdministration.prescription | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. 4 fail round-trip testing and 14 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationAdministration | TU | | DomainResource | Administration of medication to a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
instantiates | Σ | 0..* | uri | Instantiates protocol or definition
partOf | Σ | 0..* | Reference(MedicationAdministration | Procedure) | Part of referenced event
status | ?!Σ | 1..1 | code | in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown Medication administration status codes (Required)
statusReason | | 0..* | CodeableConcept | Reason administration not performed SNOMED CT Reason Medication Not Given Codes (Example)
category | | 0..1 | CodeableConcept | Type of medication usage Medication administration category codes (Preferred)
medication[x] | Σ | 1..1 | | What was administered SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 1..1 | Reference(Patient | Group) | Who received medication
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode of Care administered as part of
supportingInformation | | 0..* | Reference(Any) | Additional information to support administration
effective[x] | Σ | 1..1 | | Start and end time of administration
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
performer | Σ | 0..* | BackboneElement | Who performed the medication administration and what they did
function | | 0..1 | CodeableConcept | Type of performance Medication administration performer function codes (Example)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Device) | Who performed the medication administration
reasonCode | | 0..* | CodeableConcept | Reason administration performed Reason Medication Given Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Condition or observation that supports why the medication was administered
request | | 0..1 | Reference(MedicationRequest) | Request administration performed against
device | | 0..* | Reference(Device) | Device used to administer
note | | 0..* | Annotation | Information about the administration
dosage | I | 0..1 | BackboneElement | Details of how medication was taken + Rule: SHALL have at least one of dosage.dose or dosage.rate[x]
text | | 0..1 | string | Free text dosage instructions e.g. SIG
site | | 0..1 | CodeableConcept | Body site administered to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | | 0..1 | CodeableConcept | Path of substance into body SNOMED CT Route Codes (Example)
method | | 0..1 | CodeableConcept | How drug was administered SNOMED CT Administration Method Codes (Example)
dose | | 0..1 | SimpleQuantity | Amount of medication per dose
rate[x] | | 0..1 | | Dose quantity per unit of time
rateRatio | | | Ratio
rateQuantity | | | SimpleQuantity
eventHistory | | 0..* | Reference(Provenance) | A list of events of interest in the lifecycle
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationAdministration xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <instantiates value="[uri]"/><!-- 0..* Instantiates protocol or definition -->
 <partOf><!-- 0..* Reference(MedicationAdministration|Procedure) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown
    -->
 <statusReason><!-- 0..* CodeableConcept Reason administration not performed --></statusReason>
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What was administered --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who received medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode of Care administered as part of
    --></context>
 <supportingInformation><!-- 0..* Reference(Any) Additional information to support administration
    --></supportingInformation>
 <effective[x]><!-- 1..1 dateTime|Period Start and end time of administration --></effective[x]>
 <performer>  <!-- 0..* Who performed the medication administration and what they did -->
  <function><!-- 0..1 CodeableConcept Type of performance --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
    Device) Who performed the medication administration --></actor>
 </performer>
 <reasonCode><!-- 0..* CodeableConcept Reason administration performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Condition or observation that supports why
     the medication was administered --></reasonReference>
 <request><!-- 0..1 Reference(MedicationRequest) Request administration performed against --></request>
 <device><!-- 0..* Reference(Device) Device used to administer --></device>
 <note><!-- 0..* Annotation Information about the administration --></note>
```

```
 <dosage>  <!-- 0..1 Details of how medication was taken -->
  <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
  <site><!-- 0..1 CodeableConcept Body site administered to --></site>
  <route><!-- 0..1 CodeableConcept Path of substance into body --></route>
  <method><!-- 0..1 CodeableConcept How drug was administered --></method>
  <dose><!-- 0..1 Quantity(SimpleQuantity) Amount of medication per dose --></dose>
  <rate[x]><!-- 0..1 Ratio|Quantity(SimpleQuantity) Dose quantity per unit of time --></rate[x]>
 </dosage>
 <eventHistory><!-- 0..* Reference(Provenance) A list of events of interest in the lifecycle --></eventHistory>
</MedicationAdministration>
```

JSON Template

```
{
  "resourceType" : "MedicationAdministration",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "instantiates" : ["<uri>"], // Instantiates protocol or definition
  "partOf" : [{ Reference(MedicationAdministration|Procedure) }], // Part of referenced event
  "status" : "<code>", // R!  in-progress | not-done | on-hold | completed | entered-in-error | stopped | unknown
  "statusReason" : [{ CodeableConcept }], // Reason administration not performed
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What was administered. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who received medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode of Care administered as part of
  "supportingInformation" : [{ Reference(Any) }], // Additional information to support administration
  // effective[x]: Start and end time of administration. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "performer" : [{ // Who performed the medication administration and what they did
    "function" : { CodeableConcept }, // Type of performance
    "actor" : { Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|
    Device) } // R!  Who performed the medication administration
  }],
  "reasonCode" : [{ CodeableConcept }], // Reason administration performed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Condition or observation that supports
     why the medication was administered
  "request" : { Reference(MedicationRequest) }, // Request administration performed against
  "device" : [{ Reference(Device) }], // Device used to administer
  "note" : [{ Annotation }], // Information about the administration
  "dosage" : { // Details of how medication was taken
    "text" : "<string>", // Free text dosage instructions e.g. SIG
    "site" : { CodeableConcept }, // Body site administered to
    "route" : { CodeableConcept }, // Path of substance into body
    "method" : { CodeableConcept }, // How drug was administered
    "dose" : { Quantity(SimpleQuantity) }, // Amount of medication per dose
    // rate[x]: Dose quantity per unit of time. One of these 2:
    "rateRatio" : { Ratio }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  },
  "eventHistory" : [{ Reference(Provenance) }] // A list of events of interest in the lifecycle
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationAdministration;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationAdministration.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationAdministration.instantiates [ uri ], ... ; # 0..* Instantiates protocol or definition
  fhir:MedicationAdministration.partOf [ Reference(MedicationAdministration|Procedure) ], ... ; # 0..* Part of
    referenced event
  fhir:MedicationAdministration.status [ code ]; # 1..1 in-progress | not-done | on-hold | completed | entered-in-error
    | stopped | unknown
  fhir:MedicationAdministration.statusReason [ CodeableConcept ], ... ; # 0..* Reason administration not performed
  fhir:MedicationAdministration.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationAdministration.medication[x] : 1..1 What was administered. One of these 2
    fhir:MedicationAdministration.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationAdministration.medicationReference [ Reference(Medication) ]
  fhir:MedicationAdministration.subject [ Reference(Patient|Group) ]; # 1..1 Who received medication
  fhir:MedicationAdministration.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode of Care
    administered as part of
  fhir:MedicationAdministration.supportingInformation [ Reference(Any) ], ... ; # 0..* Additional information to support
     administration
  # MedicationAdministration.effective[x] : 1..1 Start and end time of administration. One of these 2
    fhir:MedicationAdministration.effectiveDateTime [ dateTime ]
    fhir:MedicationAdministration.effectivePeriod [ Period ]
  fhir:MedicationAdministration.performer [ # 0..* Who performed the medication administration and what they did
    fhir:MedicationAdministration.performer.function [ CodeableConcept ]; # 0..1 Type of performance
    fhir:MedicationAdministration.performer.actor [
    Reference(Practitioner|PractitionerRole|Patient|RelatedPerson|Device) ]; # 1..1 Who performed the medication
    administration
  ], ...;
```

```
  fhir:MedicationAdministration.reasonCode [ CodeableConcept ], ... ; # 0..* Reason administration performed
  fhir:MedicationAdministration.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..*
    Condition or observation that supports why the medication was administered
  fhir:MedicationAdministration.request [ Reference(MedicationRequest) ]; # 0..1 Request administration performed
    against
  fhir:MedicationAdministration.device [ Reference(Device) ], ... ; # 0..* Device used to administer
  fhir:MedicationAdministration.note [ Annotation ], ... ; # 0..* Information about the administration
  fhir:MedicationAdministration.dosage [ # 0..1 Details of how medication was taken
    fhir:MedicationAdministration.dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
    fhir:MedicationAdministration.dosage.site [ CodeableConcept ]; # 0..1 Body site administered to
    fhir:MedicationAdministration.dosage.route [ CodeableConcept ]; # 0..1 Path of substance into body
    fhir:MedicationAdministration.dosage.method [ CodeableConcept ]; # 0..1 How drug was administered
    fhir:MedicationAdministration.dosage.dose [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication per dose
    # MedicationAdministration.dosage.rate[x] : 0..1 Dose quantity per unit of time. One of these 2
      fhir:MedicationAdministration.dosage.rateRatio [ Ratio ]
      fhir:MedicationAdministration.dosage.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ];
  fhir:MedicationAdministration.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of events of interest in the
     lifecycle
]
```

Changes since Release 3

MedicationAdministration
MedicationAdministration.instantiates | Added Element
MedicationAdministration.status | Change value set from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/medication-admin-status|4.0.1
MedicationAdministration.statusReason | Added Element
MedicationAdministration.performer.function | Added Element
MedicationAdministration.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationAdministration.reasonReference | Type Reference: Added Target Type DiagnosticReport
MedicationAdministration.request | Added Element
MedicationAdministration.dosage.rate[x] | Add Type Quantity() Remove Type Quantity()
MedicationAdministration.definition | deleted
MedicationAdministration.performer.onBehalfOf | deleted
MedicationAdministration.notGiven | deleted
MedicationAdministration.reasonNotGiven | deleted
MedicationAdministration.prescription | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 14 tests that all execute ok. 4 fail round-trip testing and 14 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationAdministration.status | A set of codes indicating the current status of a MedicationAdministration. | Required | MedicationAdministration Status Codes
MedicationAdministration.statusReason | A set of codes indicating the reason why the MedicationAdministration is negated. | Example | SNOMEDCTReasonMedicationNotGivenCodes
MedicationAdministration.category | A coded concept describing where the medication administered is expected to occur. | Preferred | MedicationAdministration Category Codes
MedicationAdministration.medication[x] | Codes identifying substance or product that can be administered. | Example | SNOMEDCTMedicationCodes
MedicationAdministration.performer.function | A code describing the role an individual played in administering the medication. | Example | MedicationAdministration Performer Function Codes
MedicationAdministration.reasonCode | A set of codes indicating the reason why the MedicationAdministration was made. | Example | ReasonMedicationGivenCodes
MedicationAdministration.dosage.site | A coded concept describing the site location the medicine enters into or onto the body. | Example | SNOMEDCTAnatomicalStructureForAdministrationSiteCodes
MedicationAdministration.dosage.route | A coded concept describing the route or physiological path of administration of a therapeutic agent into or onto the body of a subject. | Example | SNOMEDCTRouteCodes
MedicationAdministration.dosage.method | A coded concept describing the technique by which the medicine is administered. | Example | SNOMEDCTAdministrationMethodCodes

11.2.3.2 Constraints

id | Level | Location | Description | Expression
mad-1 | Rule | MedicationAdministration.dosage | SHALL have at least one of dosage.dose or dosage.rate[x] | dose.exists() or rate.exists()

11.2.4 Known Issues

Issue | Comments
Medication Resource | A medication will typically be referred to by means of a code drawn from a suitable medication terminology. However, on occasion a product will be required for which the "recipe" must be specified. This implies a requirement to deal with a choice of either a code or a much more complete resource. Currently that resource has not been created.

Contrast Media | Is this resource adequate for administering contrast media to a patient?
Author (accountability) | Authorship (and any other accountability) is assumed to be dealt with by the standard FHIR methods.

11.2.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Return administrations of this medication code | (MedicationAdministration.medication as CodeableConcept) | 13 Resources
context | reference | Return administrations that share this encounter or episode of care | MedicationAdministration.context (EpisodeOfCare, Encounter)
device | reference | Return administrations with this administration device identity | MedicationAdministration.device (Device)
effective-time | date | Date administration happened (or did not happen) | MedicationAdministration.effective
identifier | token | Return administrations with this external identifier | MedicationAdministration.identifier | 30 Resources
medication | reference | Return administrations of this medication resource | (MedicationAdministration.medication as Reference) (Medication) | 3 Resources
patient | reference | The identity of a patient to list administrations for | MedicationAdministration.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | The identity of the individual who administered the medication | MedicationAdministration.performer.actor (Practitioner, Device, Patient, PractitionerRole, RelatedPerson)
reason-given | token | Reasons for administering the medication | MedicationAdministration.reasonCode
reason-not-given | token | Reasons for not administering the medication | MedicationAdministration.statusReason
request | reference | The identity of a request to list administrations from | MedicationAdministration.request (MedicationRequest)
status | token | MedicationAdministration event status (for example one of active/paused/completed/nullified) | MedicationAdministration.status | 3 Resources
subject | reference | The identity of the individual or group to list administrations for | MedicationAdministration.subject (Group, Patient)

11.2.6 Resource MedicationAdministration - Examples

URL: https://hl7.org/fhir/R4/medicationadministration-examples.html

11.2.6 Resource MedicationAdministration - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

The following are examples of MedicationAdministration resources:

INJECTABLE DOSAGE FORMS

The following examples of MedicationAdministration resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

• Intravenous Antibiotic - Vancomycin

• Intravenous Antibiotic with a Dosage Rate - Piperacillin

• Intravenous Chemotherapy (Inpatient order) - Alemtuzumab - completed

• Intravenous Chemotherapy (Inpatient order) - Brentuximab Vedotin

• Example of an IV administration not given with a note - Alemtuzumab

The following examples of MedicationAdministration resources for Injectable Dosage Forms that reference a Medication Resource for a generic formulation:

• Oral Antibiotic - Azithromycin

• Lorazepam with a Device

The following examples of MedicationAdministration resources for Injectable Dosage Forms that include a Medication Code:

• Subcutaneous Injection - Insulin - Lantus

The following examples of MedicationAdministration resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

• Intravenous Admixture - Potassium Chloride in D5W

• TPN Solution with a Device

ORAL DOSAGE FORMS

The following examples are MedicationAdministration resources for Oral Dosage Forms that reference Medication Resources for manufactured products:

• Oral Chemotherapy - Myleran

• Dosage with a Pre-Condition - Alprazolam

MISCELLANEOUS DOSAGE FORMS

The following examples of MedicationAdministration resources including miscellaneous dosage forms:

- Medication Code - Patch - Fentanyl
- Medication Code - Rectal Suppository - Acetaminophen

Example Name | id | Format
Example of an IV administration not given with a note - Alemtuzumab | medadminexample03 | XML | JSON | Turtle
Intravenous Antibiotic - Vancomycin | medadmin0301 | XML | JSON | Turtle
Intravenous Admixture - Potassium Chloride in D5W | medadmin0302 | XML | JSON | Turtle
Intravenous Antibiotic with a Dosage Rate - Piperacillin | medadmin0303 | XML | JSON | Turtle
Intravenous Chemotherapy (Inpatient order) - Alemtuzumab - completed | medadmin0304 | XML | JSON | Turtle
Intravenous Chemotherapy (Inpatient order) - Brentuximab Vedotin | medadmin0305 | XML | JSON | Turtle
Oral Antibiotic - Azithromycin | medadmin0306 | XML | JSON | Turtle
Lorazepam with a Device | medadmin0307 | XML | JSON | Turtle
Subcutaneous Injection - Insulin - Lantus | medadmin0308 | XML | JSON | Turtle
TPN Solution with a Device | medadmin0309 | XML | JSON | Turtle
Oral Chemotherapy - Myleran | medadmin0310 | XML | JSON | Turtle
Dosage with a Pre-Condition - Alprazolam | medadmin0311 | XML | JSON | Turtle
Medication Code - Patch - Fentanyl | medadmin0312 | XML | JSON | Turtle
Medication Code - Rectal Suppository - Acetaminophen | medadmin0313 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.2.7 Resource MedicationAdministration - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicationadministration-definitions.html

11.2.7 Resource MedicationAdministration - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the MedicationAdministration resource.

MedicationAdministration
Element Id | MedicationAdministration
Definition | Describes the event of a patient consuming or otherwise being administered a medication. This may be as simple as swallowing a tablet or it may be a long running infusion. Related resources tie this event to the authorizing prescription, and the specific encounter between patient and health care practitioner.
Cardinality | 0..*
Type | DomainResource
MedicationAdministration.identifier
Element Id | MedicationAdministration.identifier
Definition | Identifiers associated with this Medication Administration that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business identifiers assigned to this resource by the performer or other systems and remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | This is a business identifier, not a resource identifier.
MedicationAdministration.instantiates
Element Id | MedicationAdministration.instantiates
Definition | A protocol, guideline, orderset, or other definition that was adhered to in whole or in part by this event.
Cardinality | 0..*
Type | uri
Summary | true
MedicationAdministration.partOf
Element Id | MedicationAdministration.partOf
Definition | A larger event of which this particular event is a component or step.
Cardinality | 0..*
Type | Reference(MedicationAdministration | Procedure)
Summary | true
MedicationAdministration.status
Element Id | MedicationAdministration.status
Definition | Will generally be set to show that the administration has been completed. For some long running administrations such as infusions, it is possible for an administration to be started but not completed or it may be paused while some other process is under way.
Cardinality | 1..1
Terminology Binding | Medication administration status codes (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationAdministration.statusReason

Element Id | MedicationAdministration.statusReason
Definition | A code indicating why the administration was not performed.
Cardinality | 0..*
Terminology Binding | SNOMED CT Reason Medication Not Given Codes (Example)
Type | CodeableConcept
MedicationAdministration.category
Element Id | MedicationAdministration.category
Definition | Indicates where the medication is expected to be consumed or administered.
Cardinality | 0..1
Terminology Binding | Medication administration category codes (Preferred)
Type | CodeableConcept
MedicationAdministration.medication[x]
Element Id | MedicationAdministration.medication[x]
Definition | Identifies the medication that was administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Cardinality | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example, if you require form or lot number, then you must reference the Medication resource.
MedicationAdministration.subject
Element Id | MedicationAdministration.subject
Definition | The person or animal or group receiving the medication.
Cardinality | 1..1
Type | Reference(Patient | Group)
Summary | true
MedicationAdministration.context
Element Id | MedicationAdministration.context
Definition | The visit, admission, or other contact between patient and health care provider during which the medication administration was performed.
Cardinality | 0..1
Type | Reference(Encounter | EpisodeOfCare)
MedicationAdministration.supportingInformation
Element Id | MedicationAdministration.supportingInformation
Definition | Additional information (for example, patient height and weight) that supports the administration of the medication.
Cardinality | 0..*
Type | Reference(Any)
MedicationAdministration.effective[x]
Element Id | MedicationAdministration.effective[x]
Definition | A specific date/time or interval of time during which the administration took place (or did not take place, when the 'notGiven' attribute is true). For many administrations, such as swallowing a tablet the use of dateTime is more appropriate.
Cardinality | 1..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicationAdministration.performer
Element Id | MedicationAdministration.performer
Definition | Indicates who or what performed the medication administration and how they were involved.
Cardinality | 0..*
Summary | true
MedicationAdministration.performer.function
Element Id | MedicationAdministration.performer.function
Definition | Distinguishes the type of involvement of the performer in the medication administration.
Cardinality | 0..1
Terminology Binding | Medication administration performer function codes (Example)
Type | CodeableConcept
MedicationAdministration.performer.actor
Element Id | MedicationAdministration.performer.actor
Definition | Indicates who or what performed the medication administration.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Device)
Summary | true
MedicationAdministration.reasonCode
Element Id | MedicationAdministration.reasonCode
Definition | A code indicating why the medication was given.
Cardinality | 0..*
Terminology Binding | Reason Medication Given Codes (Example)
Type | CodeableConcept
MedicationAdministration.reasonReference
Element Id | MedicationAdministration.reasonReference

Definition | Condition or observation that supports why the medication was administered.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport)
Comments | This is a reference to a condition that is the reason for the medication request. If only a code exists, use reasonCode.
MedicationAdministration.request
Element Id | MedicationAdministration.request
Definition | The original request, instruction or authority to perform the administration.
Cardinality | 0..1
Type | Reference(MedicationRequest)
Comments | This is a reference to the MedicationRequest where the intent is either order or instance-order. It should not reference MedicationRequests where the intent is any other value.
MedicationAdministration.device
Element Id | MedicationAdministration.device
Definition | The device used in administering the medication to the patient. For example, a particular infusion pump.
Cardinality | 0..*
Type | Reference(Device)
MedicationAdministration.note
Element Id | MedicationAdministration.note
Definition | Extra information about the medication administration that is not conveyed by the other attributes.
Cardinality | 0..*
Type | Annotation
MedicationAdministration.dosage
Element Id | MedicationAdministration.dosage
Definition | Describes the medication dosage information details e.g. dose, rate, site, route, etc.
Cardinality | 0..1
Invariants | Defined on this element mad-1 Rule SHALL have at least one of dosage.dose or dosage.rate[x] dose.exists() or rate.exists()
Defined on this element
mad-1 | Rule | SHALL have at least one of dosage.dose or dosage.rate[x] | dose.exists() or rate.exists()
MedicationAdministration.dosage.text
Element Id | MedicationAdministration.dosage.text
Definition | Free text dosage can be used for cases where the dosage administered is too complex to code. When coded dosage is present, the free text dosage may still be present for display to humans. The dosage instructions should reflect the dosage of the medication that was administered.
Cardinality | 0..1
Type | string
MedicationAdministration.dosage.site
Element Id | MedicationAdministration.dosage.site
Definition | A coded specification of the anatomic site where the medication first entered the body. For example, "left arm".
Cardinality | 0..1
Terminology Binding | SNOMED CT Anatomical Structure for Administration Site Codes (Example)
Type | CodeableConcept
Comments | If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension bodySite. May be a summary code, or a reference to a very precise definition of the location, or both.
MedicationAdministration.dosage.route
Element Id | MedicationAdministration.dosage.route
Definition | A code specifying the route or physiological path of administration of a therapeutic agent into or onto the patient. For example, topical, intravenous, etc.
Cardinality | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
MedicationAdministration.dosage.method
Element Id | MedicationAdministration.dosage.method
Definition | A coded value indicating the method by which the medication is intended to be or was introduced into or on the body. This attribute will most often NOT be populated. It is most commonly used for injections. For example, Slow Push, Deep IV.
Cardinality | 0..1
Terminology Binding | SNOMED CT Administration Method Codes (Example)
Type | CodeableConcept
Comments | One of the reasons this attribute is not used often, is that the method is often pre-coordinated with the route and/or form of administration. This means the codes used in route or form may pre-coordinate the method in the route code or the form code. The implementation decision about what coding system to use for route or form code will determine how frequently the method code will be populated e.g. if route or form code pre-coordinate method code, then this attribute will not be populated often; if there is no pre-coordination then method code may be used frequently.
MedicationAdministration.dosage.dose
Element Id | MedicationAdministration.dosage.dose
Definition | The amount of the medication given at one administration event. Use this value when the administration is essentially an instantaneous event such as a swallowing a tablet or giving an injection.
Cardinality | 0..1
Type | SimpleQuantity
Comments | If the administration is not instantaneous (rate is present), this can be specified to convey the total amount administered over period of time of a single administration.
MedicationAdministration.dosage.rate[x]
Element Id | MedicationAdministration.dosage.rate[x]

Definition | Identifies the speed with which the medication was or will be introduced into the patient. Typically, the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr. May also be expressed as a rate per unit of time, e.g. 500 ml per 2 hours. Other examples: 200 mcg/min or 200 mcg/1 minute; 1 liter/8 hours.
Cardinality | 0..1
Type | Ratio|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If the rate changes over time, and you want to capture this in MedicationAdministration, then each change should be captured as a distinct MedicationAdministration, with a specific MedicationAdministration.dosage.rate, and the date time when the rate change occurred. Typically, the MedicationAdministration.dosage.rate element is not used to convey an average rate.
MedicationAdministration.eventHistory
Element Id | MedicationAdministration.eventHistory
Definition | A summary of the events of interest that have occurred, such as when the administration was verified.
Cardinality | 0..*
Type | Reference(Provenance)
Comments | This might not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.

11.2.8 Resource MedicationAdministration - Mappings

URL: https://hl7.org/fhir/R4/medicationadministration-mappings.html

11.2.8 Resource MedicationAdministration - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

W3C PROV

Mappings for the medicationadministration resource (see Mappings to Other Standards for further information & status).

11.2.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MedicationAdministration
identifier | FiveWs.identifier
status | FiveWs.status
medication[x] | FiveWs.what[x]
subject | FiveWs.subject[x]
context | FiveWs.context
supportingInformation | FiveWs.context
effective[x] | FiveWs.done[x]
performer | FiveWs.actor
reasonReference | FiveWs.why[x]

11.2.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

MedicationAdministration | Event
identifier | Event.identifier
instantiates | Event.instantiates
partOf | Event.partOf
status | Event.status
statusReason | Event.statusReason
medication[x] | Event.code
subject | Event.subject
context | Event.context
effective[x] | Event.occurrence[x]
performer | Event.performer
function | Event.performer.function
actor | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
request | Event.basedOn
note | Event.note

11.2.8.3 HL7 v2 Mapping (http://hl7.org/v2)

MedicationAdministration
identifier

instantiates
partOf
status | RXA-20-Completion Status
statusReason | RXA-9-Administration Notes / RXA-18 Substance/Treatment Refusal Reason
category
medication[x] | RXA-5-Administered Code
subject | PID-3-Patient ID List
context | PV1-19-Visit Number
supportingInformation
effective[x] | RXA-3-Date/Time Start of Administration / RXA-4-Date/Time End of Administration
performer | RXA-10-Administering Provider / PRT-5-Participation Person: PRT-4-Participation='AP' (RXA-10 is deprecated)
function
actor
reasonCode | RXE-27 Give Indication
reasonReference
request | ORC-2-Placer Order Number / ORC-3-Filler Order Number
device | PRT-10 Participation Device / RXR-3 Administrative Device
note
dosage
text
site | RXR-2 Administration Site / RXR-6 Administration Site Modifier
route | RXR-1-Route
method | RXR-4-Administration Method
dose | RXA-6 Administered Amount / RXA-7 Administered Units
rate[x] | RXA-12-Administered Per (Time Unit)
eventHistory

11.2.8.4 RIM Mapping (http://hl7.org/v3)

MedicationAdministration | SubstanceAdministration
identifier | .id
instantiates | .outboundRelationship[typeCode=DEFN].target[classCode=unspecified]
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=SBADM or PROC,moodCode=EVN]
status | .statusCode
statusReason | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason not given"].value
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
medication[x] | .participation[typeCode=CSM].role[classCode=ADMM]
subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
effective[x] | .effectiveTime
performer | .participation[typeCode=PRF]
function | participation[typeCode=PRF].functionCode
actor | .role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason for use ASSERTION"].value
request | InFullfillmentOf->SubstanceAdministration
device | device->Access OR device->AssignedDevice
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosage | component->SubstanceAdministrationEvent
text | .text
site | .approachSiteCode
route | .routeCode
method | .methodCode
dose | .doseQuantity
rate[x] | .rateQuantity
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.2.8.5 W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

MedicationAdministration
identifier
instantiates
partOf

status
statusReason
category
medication[x]
subject
context
supportingInformation
effective[x]
performer
function
actor
reasonCode
reasonReference | no mapping
request
device
note
dosage
text
site
route
method
dose
rate[x]
eventHistory

11.2.9 Resource MedicationAdministration - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicationadministration-profiles.html

11.2.9 Resource MedicationAdministration - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the MedicationAdministration resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.2.9.1 Profiles

No Profiles defined for this resource

11.2.9.2 Extensions

workflow-researchStudy | MedicationAdministration | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
openEHR-administration | administration | for openEHR Archetype Profile
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.2.9.3 Search Extensions

No Search Extensions defined for this resource

11.2.10 Resource MedicationAdministration - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/medicationadministration-version-maps.html

No validation errors - all conversions are clean

11.2.10 Resource MedicationAdministration - R3/R4 Conversions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for MedicationAdministration.

Functional status for this map: 14 tests that all execute ok. 4 fail round-trip testing and 14 r3 resources are invalid (0 errors). (see documentation)

11.2.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MedicationAdministration3to4" = "R3 to R4 Conversions for
    MedicationAdministration"

uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationAdministration" alias MedicationAdministrationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationAdministration" alias MedicationAdministration as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MedicationAdministration(source src : MedicationAdministrationR3, target tgt : MedicationAdministration) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiates;
  src.partOf -> tgt.partOf;
  src.status as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.status',  vt.value = v;
  src.notGiven as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.notGiven',  vt.value = v;
  src.status where src.notGiven.not() -> tgt.status;
  src.notGiven -> tgt.status = 'not-done';
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.subject -> tgt.subject;
  src.context -> tgt.context;
  src.supportingInformation -> tgt.supportingInformation;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.performer as vs0 -> tgt.performer as vt0 then actor(vs0, vt0);
  src.statusReason -> tgt.statusReason;
  src.reasonNotGiven -> tgt.reasonCode;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.prescription -> tgt.request;
  src.device -> tgt.device;
  src.note -> tgt.note;
  src.dosage as s -> tgt.dosage as t then dosage(s, t);
  src.eventHistory -> tgt.eventHistory;
}

group actor(source src, target tgt) extends BackboneElement {
  src.actor -> tgt.actor;
  src.onBehalfOf as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.performer.onBehalfOf',  vt.value = v;
}

group dosage(source src, target tgt) extends BackboneElement {
  src.text -> tgt.text;
  src.site -> tgt.site;
  src.route -> tgt.route;
  src.method -> tgt.method;
  src.dose -> tgt.dose;
  src.rate : Ratio as vs -> tgt.rate = create('Ratio') as vt then Ratio(vs, vt);
  src.rate : Quantity as vs -> tgt.rate = create('Quantity') as vt then Quantity(vs, vt);
}
```

11.2.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationAdministration4to3" = "R4 to R3 Conversion for MedicationAdministration"

uses "http://hl7.org/fhir/StructureDefinition/MedicationAdministration" alias MedicationAdministration as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationAdministration" alias MedicationAdministrationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MedicationAdministration(source src : MedicationAdministrationR3, target tgt : MedicationAdministration) extends
    DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiates -> tgt.definition;
  src.partOf -> tgt.partOf;
  src where src.extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.status').exists() then statusR3(src, tgt) "status1";
  src where src.extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.status').empty() then statusNR3(src, tgt) "status2";
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
```

```
  src.subject -> tgt.subject;
  src.context -> tgt.context;
  src.supportingInformation -> tgt.supportingInformation;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.performer as vs0 -> tgt.performer as vt0 then actor(vs0, vt0);
  src.statusReason -> tgt.statusReason;
  src.reasonCode where value = 'not-done' -> tgt.reasonNotGiven;
  src.reasonCode where value != 'not-done' -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.request -> tgt.prescription;
  src.device -> tgt.device;
  src.note -> tgt.note;
  src.dosage as s -> tgt.dosage as t then dosage(s, t);
  src.eventHistory -> tgt.eventHistory;
}

group actor(source src, target tgt) extends BackboneElement {
  src.actor -> tgt.actor;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.performer.onBehalfOf' then {
    e.value -> tgt.onBehalfOf;
  };
}

group dosage(source src, target tgt) extends BackboneElement {
  src.text -> tgt.text;
  src.site -> tgt.site;
  src.route -> tgt.route;
  src.method -> tgt.method;
  src.dose -> tgt.dose;
  src.rate : Ratio as vs -> tgt.rate = create('Ratio') as vt then Ratio(vs, vt);
  src.rate : Quantity as vs -> tgt.rate = create('Quantity') as vt then Quantity(vs, vt);
}

group statusR3(source src, target tgt) {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationAdministration.notGiven' then {
    e.value -> tgt.notGiven;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationAdministration.status'
     then {
    e.value -> tgt.status;
  };
}

group statusNR3(source src, target tgt) {
  src.status where value != 'not-done' -> tgt.status;
  src.status where value = 'not-done' -> tgt.notGiven = false;
}
```

11.2.10.3 R4 Validation Errors

11.2.11 MedicationAdministration HL7 Extensions

URL: https://hl7.org/fhir/R4/medicationadministration-extensions.html

Defines common extensions used with or related to the MedicationAdministration resource

11.2.11 MedicationAdministration HL7 Extensions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

11.2.11.1 Content

11.2.12 JSON Schema for MedicationAdministration

URL: https://hl7.org/fhir/R4/medicationadministration.schema.json.html

11.2.12 JSON Schema for MedicationAdministration

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for MedicationAdministration, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicationAdministration",
  "$ref": "#/definitions/MedicationAdministration",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicationAdministration": {
      "description": "Describes the event of a patient consuming or otherwise being administered a medication.  This may
     be as simple as swallowing a tablet or it may be a long running infusion.  Related resources tie this event to the
    authorizing prescription, and the specific encounter between patient and health care practitioner.",
```

```
  "properties": {
    "resourceType": {
      "description": "This is a MedicationAdministration resource",
      "const": "MedicationAdministration"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers associated with this Medication Administration that are defined by business
processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are
 business identifiers assigned to this resource by the performer or other systems and remain constant as the
resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiates": {
      "description": "A protocol, guideline, orderset, or other definition that was adhered to in whole or in part
by this event.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
```

```
    },
    "_instantiates": {
      "description": "Extensions for instantiates",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular event is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "Will generally be set to show that the administration has been completed.  For some long
running administrations such as infusions, it is possible for an administration to be started but not completed or
it may be paused while some other process is under way.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "A code indicating why the administration was not performed.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "category": {
      "description": "Indicates where the medication is expected to be consumed or administered.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationCodeableConcept": {
      "description": "Identifies the medication that was administered. This is either a link to a resource
representing the details of the medication or a simple attribute carrying a code that identifies the medication from
 a known list of medications.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationReference": {
      "description": "Identifies the medication that was administered. This is either a link to a resource
representing the details of the medication or a simple attribute carrying a code that identifies the medication from
 a known list of medications.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "subject": {
      "description": "The person or animal or group receiving the medication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "context": {
      "description": "The visit, admission, or other contact between patient and health care provider during which
the medication administration was performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "supportingInformation": {
      "description": "Additional information (for example, patient height and weight) that supports the
administration of the medication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "effectiveDateTime": {
      "description": "A specific date/time or interval of time during which the administration took place (or did
not take place, when the \u0027notGiven\u0027 attribute is true). For many administrations, such as swallowing a
tablet the use of dateTime is more appropriate.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_effectiveDateTime": {
      "description": "Extensions for effectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "A specific date/time or interval of time during which the administration took place (or did
not take place, when the \u0027notGiven\u0027 attribute is true). For many administrations, such as swallowing a
tablet the use of dateTime is more appropriate.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "performer": {
      "description": "Indicates who or what performed the medication administration and how they were involved.",
      "items": {
        "$ref": "#/definitions/MedicationAdministration_Performer"
      },
      "type": "array"
```

```
    },
    "reasonCode": {
      "description": "A code indicating why the medication was given.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Condition or observation that supports why the medication was administered.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "request": {
      "description": "The original request, instruction or authority to perform the administration.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "device": {
      "description": "The device used in administering the medication to the patient.  For example, a particular
infusion pump.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Extra information about the medication administration that is not conveyed by the other
attributes.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "dosage": {
      "description": "Describes the medication dosage information details e.g. dose, rate, site, route, etc.",
      "$ref": "#/definitions/MedicationAdministration_Dosage"
    },
    "eventHistory": {
      "description": "A summary of the events of interest that have occurred, such as when the administration was
verified.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"MedicationAdministration_Performer": {
  "description": "Describes the event of a patient consuming or otherwise being administered a medication.  This may
 be as simple as swallowing a tablet or it may be a long running infusion.  Related resources tie this event to the
authorizing prescription, and the specific encounter between patient and health care practitioner.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "function": {
```

```
      "description": "Distinguishes the type of involvement of the performer in the medication administration.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "actor": {
      "description": "Indicates who or what performed the medication administration.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "actor"
  ]
},
"MedicationAdministration_Dosage": {
  "description": "Describes the event of a patient consuming or otherwise being administered a medication.  This may
 be as simple as swallowing a tablet or it may be a long running infusion.  Related resources tie this event to the
authorizing prescription, and the specific encounter between patient and health care practitioner.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "text": {
      "description": "Free text dosage can be used for cases where the dosage administered is too complex to code.
When coded dosage is present, the free text dosage may still be present for display to humans.\r\rThe dosage
instructions should reflect the dosage of the medication that was administered.",
      "$ref": "#/definitions/string"
    },
    "_text": {
      "description": "Extensions for text",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "site": {
      "description": "A coded specification of the anatomic site where the medication first entered the body.  For
example, \"left arm\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "route": {
      "description": "A code specifying the route or physiological path of administration of a therapeutic agent
into or onto the patient.  For example, topical, intravenous, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "method": {
      "description": "A coded value indicating the method by which the medication is intended to be or was
introduced into or on the body.  This attribute will most often NOT be populated.  It is most commonly used for
injections.  For example, Slow Push, Deep IV.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "dose": {
      "description": "The amount of the medication given at one administration event.   Use this value when the
administration is essentially an instantaneous event such as a swallowing a tablet or giving an injection.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "rateRatio": {
      "description": "Identifies the speed with which the medication was or will be introduced into the patient.
Typically, the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr.  May also be expressed as a rate per unit
of time, e.g. 500 ml per 2 hours.  Other examples:  200 mcg/min or 200 mcg/1 minute; 1 liter/8 hours.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"

    },
    "rateQuantity": {
      "description": "Identifies the speed with which the medication was or will be introduced into the patient.
Typically, the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr.  May also be expressed as a rate per unit
of time, e.g. 500 ml per 2 hours.  Other examples:  200 mcg/min or 200 mcg/1 minute; 1 liter/8 hours.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
```

```
        }
      },
      "additionalProperties": false
    }
  }
}
```

11.2.13 Resource MedicationAdministration - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicationadministration-dependencies.html

11.2.13 Resource MedicationAdministration - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for MedicationAdministration FMM level 2

Name | Card. | Type | Dependency Analysis
MedicationAdministration | | DomainResource
identifier | 0..* | Identifier
instantiates | 0..* | uri
partOf | 0..* | Reference(MedicationAdministration | Procedure)
status | 1..1 | code
statusReason | 0..* | CodeableConcept
category | 0..1 | CodeableConcept
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(Medication
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
context | 0..1 | Reference(Encounter | EpisodeOfCare)
supportingInformation | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
effective[x] | 1..1
effectiveDateTime | | dateTime
effectivePeriod | | Period
performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson | Device)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport)
request | 0..1 | Reference(MedicationRequest)
device | 0..* | Reference(Device)
note | 0..* | Annotation
dosage | 0..1 | BackboneElement
text | 0..1 | string
site | 0..1 | CodeableConcept
route | 0..1 | CodeableConcept
method | 0..1 | CodeableConcept
dose | 0..1 | SimpleQuantity
rate[x] | 0..1
rateRatio | | Ratio
rateQuantity | | SimpleQuantity
eventHistory | 0..* | Reference(Provenance)

11.2.14 Value Set Sc-valueset-medication-admin-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medication-admin-status.html

11.2.14 Value Set Sc-valueset-medication-admin-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.2.15 MedicationAdministration Status CodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medication-admin-status)

Mapping from http://hl7.org/fhir/ValueSet/medication-admin-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "MedicationAdministration Status Codes"

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
in-progress | equivalent | active
on-hold | equivalent | suspended
stopped | equivalent | failed
completed | equivalent | complete
unknown | equivalent | unknown
not-done | equivalent | not-done

11.3 Resource MedicationDispense - Content

URL: https://hl7.org/fhir/R4/medicationdispense.html
Pharmacy Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Indicates that a medication product is to be or has been dispensed for a named person/patient. This includes a description of the medication product (supply) provided and the instructions for administering the medication. The medication dispense is the result of a pharmacy system responding to a medication order.

11.3.1 Scope and Usage

This resource covers the supply of medications to a patient. Examples include dispensing and pick-up from an outpatient or community pharmacy, dispensing patient-specific medications from inpatient pharmacy to ward, as well as issuing a single dose from ward stock to a patient for consumption. The medication dispense is the result of a pharmacy system responding to a medication order.

MedicationDispense is an event resource from a FHIR workflow perspective - see Workflow Event

11.3.2 Boundaries and Relationships

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.
MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).
MedicationAdministration | When a patient actually consumes a medicine, or the medication is otherwise administered to them
MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician. A medication statement is not a part of the prescribe->dispense->administer sequence but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

This resource does not deal with the supply or transfer of non-medication related items to a patient.

11.3.3 Background and Context

The supply and the associated administration instructions might not exactly follow the original order (prescription), either because some details were left for completion at this point in the process or because the dispenser exercised their clinical judgment to make some appropriate modification.

This resource is referenced by ChargeItem, MedicationStatement and Observation

11.3.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationDispense | ITU | | DomainResource | Dispensing a medication to a named patient + Rule: whenHandedOver cannot be before whenPrepared Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
partOf | | 0..* | Reference(Procedure) | Event that dispense is part of
status | ?!Σ | 1..1 | code | preparation | in-progress | cancelled | on-hold | completed | entered-in-error | stopped | declined | unknown Medication dispense status codes (Required)
statusReason[x] | | 0..1 | | Why a dispense was not performed Medication dispense status reason codes (Example)
statusReasonCodeableConcept | | | CodeableConcept
statusReasonReference | | | Reference(DetectedIssue)
category | | 0..1 | CodeableConcept | Type of medication dispense Medication dispense category codes (Preferred)
medication[x] | Σ | 1..1 | | What medication was supplied SNOMED CT Medication Codes (Example)

medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 0..1 | Reference(Patient | Group) | Who the dispense is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
supportingInformation | | 0..* | Reference(Any) | Information that supports the dispensing of the medication
performer | | 0..* | BackboneElement | Who performed event
function | | 0..1 | CodeableConcept | Who performed the dispense and what they did Medication dispense performer function codes (Example)
actor | | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson) | Individual who was performing
location | | 0..1 | Reference(Location) | Where the dispense occurred
authorizingPrescription | | 0..* | Reference(MedicationRequest) | Medication order that authorizes the dispense
type | | 0..1 | CodeableConcept | Trial fill, partial fill, emergency fill, etc. V3 Value SetActPharmacySupplyType (Example)
quantity | | 0..1 | SimpleQuantity | Amount dispensed
daysSupply | | 0..1 | SimpleQuantity | Amount of medication expressed as a timing amount
whenPrepared | Σ | 0..1 | dateTime | When product was packaged and reviewed
whenHandedOver | | 0..1 | dateTime | When product was given out
destination | | 0..1 | Reference(Location) | Where the medication was sent
receiver | | 0..* | Reference(Patient | Practitioner) | Who collected the medication
note | | 0..* | Annotation | Information about the dispense
dosageInstruction | | 0..* | Dosage | How the medication is to be used by the patient or administered by the caregiver
substitution | | 0..1 | BackboneElement | Whether a substitution was performed on the dispense
wasSubstituted | | 1..1 | boolean | Whether a substitution was or was not performed on the dispense
type | | 0..1 | CodeableConcept | Code signifying whether a different drug was dispensed from what was prescribed V3 Value SetActSubstanceAdminSubstitutionCode (Example)
reason | | 0..* | CodeableConcept | Why was substitution made V3 Value SetSubstanceAdminSubstitutionReason (Example)
responsibleParty | | 0..* | Reference(Practitioner | PractitionerRole) | Who is responsible for the substitution
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of relevant lifecycle events
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationDispense xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <partOf><!-- 0..* Reference(Procedure) Event that dispense is part of --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | cancelled | on-hold | completed | entered-in-error |
    stopped | declined | unknown -->
 <statusReason[x]><!-- 0..1 CodeableConcept|Reference(DetectedIssue) Why a dispense was not performed
    --></statusReason[x]>
 <category><!-- 0..1 CodeableConcept Type of medication dispense --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was supplied --></medication[x]>
 <subject><!-- 0..1 Reference(Patient|Group) Who the dispense is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information that supports the dispensing of the medication
    --></supportingInformation>
 <performer>  <!-- 0..* Who performed event -->
  <function><!-- 0..1 CodeableConcept Who performed the dispense and what they did --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
    Device|RelatedPerson) Individual who was performing --></actor>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the dispense occurred --></location>
 <authorizingPrescription><!-- 0..* Reference(MedicationRequest) Medication order that authorizes the dispense
    --></authorizingPrescription>
 <type><!-- 0..1 CodeableConcept Trial fill, partial fill, emergency fill, etc. --></type>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
 <daysSupply><!-- 0..1 Quantity(SimpleQuantity) Amount of medication expressed as a timing amount --></daysSupply>
 <whenPrepared value="[dateTime]"/><!-- 0..1 When product was packaged and reviewed -->
 <whenHandedOver value="[dateTime]"/><!-- 0..1 When product was given out -->
 <destination><!-- 0..1 Reference(Location) Where the medication was sent --></destination>
 <receiver><!-- 0..* Reference(Patient|Practitioner) Who collected the medication --></receiver>
 <note><!-- 0..* Annotation Information about the dispense --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication is to be used by the patient or administered by the caregiver
    --></dosageInstruction>
 <substitution>  <!-- 0..1 Whether a substitution was performed on the dispense -->
  <wasSubstituted value="[boolean]"/><!-- 1..1 Whether a substitution was or was not performed on the dispense -->
  <type><!-- 0..1 CodeableConcept Code signifying whether a different drug was dispensed from what was prescribed
    --></type>
  <reason><!-- 0..* CodeableConcept Why was substitution made --></reason>
  <responsibleParty><!-- 0..* Reference(Practitioner|PractitionerRole) Who is responsible for the substitution
    --></responsibleParty>
 </substitution>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of relevant lifecycle events --></eventHistory>
</MedicationDispense>
```

JSON Template

```
{
  "resourceType" : "MedicationDispense",
```

```
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "partOf" : [{ Reference(Procedure) }], // Event that dispense is part of
  "status" : "<code>", // R!  preparation | in-progress | cancelled | on-hold | completed | entered-in-error | stopped |
     declined | unknown
  // statusReason[x]: Why a dispense was not performed. One of these 2:
  "statusReasonCodeableConcept" : { CodeableConcept },
  "statusReasonReference" : { Reference(DetectedIssue) },
  "category" : { CodeableConcept }, // Type of medication dispense
  // medication[x]: What medication was supplied. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // Who the dispense is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  "supportingInformation" : [{ Reference(Any) }], // Information that supports the dispensing of the medication
  "performer" : [{ // Who performed event
    "function" : { CodeableConcept }, // Who performed the dispense and what they did
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
    Device|RelatedPerson) } // R!  Individual who was performing
  }],
  "location" : { Reference(Location) }, // Where the dispense occurred
  "authorizingPrescription" : [{ Reference(MedicationRequest) }], // Medication order that authorizes the dispense
  "type" : { CodeableConcept }, // Trial fill, partial fill, emergency fill, etc.
  "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
  "daysSupply" : { Quantity(SimpleQuantity) }, // Amount of medication expressed as a timing amount
  "whenPrepared" : "<dateTime>", // When product was packaged and reviewed
  "whenHandedOver" : "<dateTime>", // When product was given out
  "destination" : { Reference(Location) }, // Where the medication was sent
  "receiver" : [{ Reference(Patient|Practitioner) }], // Who collected the medication
  "note" : [{ Annotation }], // Information about the dispense
  "dosageInstruction" : [{ Dosage }], // How the medication is to be used by the patient or administered by the
    caregiver
  "substitution" : { // Whether a substitution was performed on the dispense
    "wasSubstituted" : <boolean>, // R!  Whether a substitution was or was not performed on the dispense
    "type" : { CodeableConcept }, // Code signifying whether a different drug was dispensed from what was prescribed
    "reason" : [{ CodeableConcept }], // Why was substitution made
    "responsibleParty" : [{ Reference(Practitioner|PractitionerRole) }] // Who is responsible for the substitution
  },
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of relevant lifecycle events
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationDispense;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationDispense.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationDispense.partOf [ Reference(Procedure) ], ... ; # 0..* Event that dispense is part of
  fhir:MedicationDispense.status [ code ]; # 1..1 preparation | in-progress | cancelled | on-hold | completed | entered-
    in-error | stopped | declined | unknown
  # MedicationDispense.statusReason[x] : 0..1 Why a dispense was not performed. One of these 2
    fhir:MedicationDispense.statusReasonCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.statusReasonReference [ Reference(DetectedIssue) ]
  fhir:MedicationDispense.category [ CodeableConcept ]; # 0..1 Type of medication dispense
  # MedicationDispense.medication[x] : 1..1 What medication was supplied. One of these 2
    fhir:MedicationDispense.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.medicationReference [ Reference(Medication) ]
  fhir:MedicationDispense.subject [ Reference(Patient|Group) ]; # 0..1 Who the dispense is for
  fhir:MedicationDispense.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    event
  fhir:MedicationDispense.supportingInformation [ Reference(Any) ], ... ; # 0..* Information that supports the
    dispensing of the medication
  fhir:MedicationDispense.performer [ # 0..* Who performed event
    fhir:MedicationDispense.performer.function [ CodeableConcept ]; # 0..1 Who performed the dispense and what they did
    fhir:MedicationDispense.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson) ]; # 1..1 Individual who was
    performing
  ], ...;
  fhir:MedicationDispense.location [ Reference(Location) ]; # 0..1 Where the dispense occurred
  fhir:MedicationDispense.authorizingPrescription [ Reference(MedicationRequest) ], ... ; # 0..* Medication order that
    authorizes the dispense
  fhir:MedicationDispense.type [ CodeableConcept ]; # 0..1 Trial fill, partial fill, emergency fill, etc.
  fhir:MedicationDispense.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
  fhir:MedicationDispense.daysSupply [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication expressed as a timing
    amount
  fhir:MedicationDispense.whenPrepared [ dateTime ]; # 0..1 When product was packaged and reviewed
  fhir:MedicationDispense.whenHandedOver [ dateTime ]; # 0..1 When product was given out
  fhir:MedicationDispense.destination [ Reference(Location) ]; # 0..1 Where the medication was sent
  fhir:MedicationDispense.receiver [ Reference(Patient|Practitioner) ], ... ; # 0..* Who collected the medication
  fhir:MedicationDispense.note [ Annotation ], ... ; # 0..* Information about the dispense
  fhir:MedicationDispense.dosageInstruction [ Dosage ], ... ; # 0..* How the medication is to be used by the patient or
    administered by the caregiver
  fhir:MedicationDispense.substitution [ # 0..1 Whether a substitution was performed on the dispense
```

```
    fhir:MedicationDispense.substitution.wasSubstituted [ boolean ]; # 1..1 Whether a substitution was or was not
    performed on the dispense
    fhir:MedicationDispense.substitution.type [ CodeableConcept ]; # 0..1 Code signifying whether a different drug was
    dispensed from what was prescribed
    fhir:MedicationDispense.substitution.reason [ CodeableConcept ], ... ; # 0..* Why was substitution made
    fhir:MedicationDispense.substitution.responsibleParty [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who
     is responsible for the substitution
  ];
  fhir:MedicationDispense.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical issue with action
  fhir:MedicationDispense.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of relevant lifecycle events
]
```

Changes since R3

MedicationDispense
MedicationDispense.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-dispense-status to http://hl7.org/fhir/ValueSet/medicationdispense-status|4.0.1
MedicationDispense.statusReason[x] | Added Element
MedicationDispense.performer.function | Added Element
MedicationDispense.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationDispense.location | Added Element
MedicationDispense.substitution.responsibleParty | Type Reference: Added Target Type PractitionerRole
MedicationDispense.performer.onBehalfOf | deleted
MedicationDispense.notDone | deleted
MedicationDispense.notDoneReason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 31 tests that all execute ok. 6 fail round-trip testing and 31 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationDispense | ITU | | DomainResource | Dispensing a medication to a named patient + Rule: whenHandedOver cannot be before whenPrepared Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External identifier
partOf | | 0..* | Reference(Procedure) | Event that dispense is part of
status | ?!Σ | 1..1 | code | preparation | in-progress | cancelled | on-hold | completed | entered-in-error | stopped | declined | unknown Medication dispense status codes (Required)
statusReason[x] | | 0..1 | | Why a dispense was not performed Medication dispense status reason codes (Example)
statusReasonCodeableConcept | | | CodeableConcept
statusReasonReference | | | Reference(DetectedIssue)
category | | 0..1 | CodeableConcept | Type of medication dispense Medication dispense category codes (Preferred)
medication[x] | Σ | 1..1 | | What medication was supplied SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 0..1 | Reference(Patient | Group) | Who the dispense is for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
supportingInformation | | 0..* | Reference(Any) | Information that supports the dispensing of the medication
performer | | 0..* | BackboneElement | Who performed event
function | | 0..1 | CodeableConcept | Who performed the dispense and what they did Medication dispense performer function codes (Example)
actor | | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson) | Individual who was performing
location | | 0..1 | Reference(Location) | Where the dispense occurred
authorizingPrescription | | 0..* | Reference(MedicationRequest) | Medication order that authorizes the dispense
type | | 0..1 | CodeableConcept | Trial fill, partial fill, emergency fill, etc. V3 Value SetActPharmacySupplyType (Example)
quantity | | 0..1 | SimpleQuantity | Amount dispensed
daysSupply | | 0..1 | SimpleQuantity | Amount of medication expressed as a timing amount
whenPrepared | Σ | 0..1 | dateTime | When product was packaged and reviewed
whenHandedOver | | 0..1 | dateTime | When product was given out
destination | | 0..1 | Reference(Location) | Where the medication was sent
receiver | | 0..* | Reference(Patient | Practitioner) | Who collected the medication
note | | 0..* | Annotation | Information about the dispense
dosageInstruction | | 0..* | Dosage | How the medication is to be used by the patient or administered by the caregiver
substitution | | 0..1 | BackboneElement | Whether a substitution was performed on the dispense
wasSubstituted | | 1..1 | boolean | Whether a substitution was or was not performed on the dispense
type | | 0..1 | CodeableConcept | Code signifying whether a different drug was dispensed from what was prescribed V3 Value SetActSubstanceAdminSubstitutionCode (Example)
reason | | 0..* | CodeableConcept | Why was substitution made V3 Value SetSubstanceAdminSubstitutionReason (Example)
responsibleParty | | 0..* | Reference(Practitioner | PractitionerRole) | Who is responsible for the substitution
detectedIssue | | 0..* | Reference(DetectedIssue) | Clinical issue with action
eventHistory | | 0..* | Reference(Provenance) | A list of relevant lifecycle events
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationDispense xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <partOf><!-- 0..* Reference(Procedure) Event that dispense is part of --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | cancelled | on-hold | completed | entered-in-error |
    stopped | declined | unknown -->
 <statusReason[x]><!-- 0..1 CodeableConcept|Reference(DetectedIssue) Why a dispense was not performed
    --></statusReason[x]>
 <category><!-- 0..1 CodeableConcept Type of medication dispense --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was supplied --></medication[x]>
 <subject><!-- 0..1 Reference(Patient|Group) Who the dispense is for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <supportingInformation><!-- 0..* Reference(Any) Information that supports the dispensing of the medication
    --></supportingInformation>
 <performer>  <!-- 0..* Who performed event -->
  <function><!-- 0..1 CodeableConcept Who performed the dispense and what they did --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
    Device|RelatedPerson) Individual who was performing --></actor>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the dispense occurred --></location>
 <authorizingPrescription><!-- 0..* Reference(MedicationRequest) Medication order that authorizes the dispense
    --></authorizingPrescription>
 <type><!-- 0..1 CodeableConcept Trial fill, partial fill, emergency fill, etc. --></type>
 <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount dispensed --></quantity>
 <daysSupply><!-- 0..1 Quantity(SimpleQuantity) Amount of medication expressed as a timing amount --></daysSupply>
 <whenPrepared value="[dateTime]"/><!-- 0..1 When product was packaged and reviewed -->
 <whenHandedOver value="[dateTime]"/><!-- 0..1 When product was given out -->
 <destination><!-- 0..1 Reference(Location) Where the medication was sent --></destination>
 <receiver><!-- 0..* Reference(Patient|Practitioner) Who collected the medication --></receiver>
 <note><!-- 0..* Annotation Information about the dispense --></note>
 <dosageInstruction><!-- 0..* Dosage How the medication is to be used by the patient or administered by the caregiver
    --></dosageInstruction>
 <substitution>  <!-- 0..1 Whether a substitution was performed on the dispense -->
  <wasSubstituted value="[boolean]"/><!-- 1..1 Whether a substitution was or was not performed on the dispense -->
  <type><!-- 0..1 CodeableConcept Code signifying whether a different drug was dispensed from what was prescribed
    --></type>
  <reason><!-- 0..* CodeableConcept Why was substitution made --></reason>
  <responsibleParty><!-- 0..* Reference(Practitioner|PractitionerRole) Who is responsible for the substitution
    --></responsibleParty>
 </substitution>
 <detectedIssue><!-- 0..* Reference(DetectedIssue) Clinical issue with action --></detectedIssue>
 <eventHistory><!-- 0..* Reference(Provenance) A list of relevant lifecycle events --></eventHistory>
</MedicationDispense>
```

JSON Template

```
{
  "resourceType" : "MedicationDispense",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "partOf" : [{ Reference(Procedure) }], // Event that dispense is part of
  "status" : "<code>", // R!  preparation | in-progress | cancelled | on-hold | completed | entered-in-error | stopped |
     declined | unknown
  // statusReason[x]: Why a dispense was not performed. One of these 2:
  "statusReasonCodeableConcept" : { CodeableConcept },
  "statusReasonReference" : { Reference(DetectedIssue) },
  "category" : { CodeableConcept }, // Type of medication dispense
  // medication[x]: What medication was supplied. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // Who the dispense is for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  "supportingInformation" : [{ Reference(Any) }], // Information that supports the dispensing of the medication
  "performer" : [{ // Who performed event
    "function" : { CodeableConcept }, // Who performed the dispense and what they did
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
    Device|RelatedPerson) } // R!  Individual who was performing
  }],
  "location" : { Reference(Location) }, // Where the dispense occurred
  "authorizingPrescription" : [{ Reference(MedicationRequest) }], // Medication order that authorizes the dispense
  "type" : { CodeableConcept }, // Trial fill, partial fill, emergency fill, etc.
  "quantity" : { Quantity(SimpleQuantity) }, // Amount dispensed
  "daysSupply" : { Quantity(SimpleQuantity) }, // Amount of medication expressed as a timing amount
  "whenPrepared" : "<dateTime>", // When product was packaged and reviewed
  "whenHandedOver" : "<dateTime>", // When product was given out
  "destination" : { Reference(Location) }, // Where the medication was sent
  "receiver" : [{ Reference(Patient|Practitioner) }], // Who collected the medication
  "note" : [{ Annotation }], // Information about the dispense
  "dosageInstruction" : [{ Dosage }], // How the medication is to be used by the patient or administered by the
    caregiver
  "substitution" : { // Whether a substitution was performed on the dispense
    "wasSubstituted" : <boolean>, // R!  Whether a substitution was or was not performed on the dispense
    "type" : { CodeableConcept }, // Code signifying whether a different drug was dispensed from what was prescribed
    "reason" : [{ CodeableConcept }], // Why was substitution made
    "responsibleParty" : [{ Reference(Practitioner|PractitionerRole) }] // Who is responsible for the substitution
  },
```

```
  "detectedIssue" : [{ Reference(DetectedIssue) }], // Clinical issue with action
  "eventHistory" : [{ Reference(Provenance) }] // A list of relevant lifecycle events
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationDispense;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationDispense.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationDispense.partOf [ Reference(Procedure) ], ... ; # 0..* Event that dispense is part of
  fhir:MedicationDispense.status [ code ]; # 1..1 preparation | in-progress | cancelled | on-hold | completed | entered-
    in-error | stopped | declined | unknown
  # MedicationDispense.statusReason[x] : 0..1 Why a dispense was not performed. One of these 2
    fhir:MedicationDispense.statusReasonCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.statusReasonReference [ Reference(DetectedIssue) ]
  fhir:MedicationDispense.category [ CodeableConcept ]; # 0..1 Type of medication dispense
  # MedicationDispense.medication[x] : 1..1 What medication was supplied. One of these 2
    fhir:MedicationDispense.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationDispense.medicationReference [ Reference(Medication) ]
  fhir:MedicationDispense.subject [ Reference(Patient|Group) ]; # 0..1 Who the dispense is for
  fhir:MedicationDispense.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    event
  fhir:MedicationDispense.supportingInformation [ Reference(Any) ], ... ; # 0..* Information that supports the
    dispensing of the medication
  fhir:MedicationDispense.performer [ # 0..* Who performed event
    fhir:MedicationDispense.performer.function [ CodeableConcept ]; # 0..1 Who performed the dispense and what they did
    fhir:MedicationDispense.performer.actor [
    Reference(Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson) ]; # 1..1 Individual who was
    performing
  ], ...;
  fhir:MedicationDispense.location [ Reference(Location) ]; # 0..1 Where the dispense occurred
  fhir:MedicationDispense.authorizingPrescription [ Reference(MedicationRequest) ], ... ; # 0..* Medication order that
    authorizes the dispense
  fhir:MedicationDispense.type [ CodeableConcept ]; # 0..1 Trial fill, partial fill, emergency fill, etc.
  fhir:MedicationDispense.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount dispensed
  fhir:MedicationDispense.daysSupply [ Quantity(SimpleQuantity) ]; # 0..1 Amount of medication expressed as a timing
    amount
  fhir:MedicationDispense.whenPrepared [ dateTime ]; # 0..1 When product was packaged and reviewed
  fhir:MedicationDispense.whenHandedOver [ dateTime ]; # 0..1 When product was given out
  fhir:MedicationDispense.destination [ Reference(Location) ]; # 0..1 Where the medication was sent
  fhir:MedicationDispense.receiver [ Reference(Patient|Practitioner) ], ... ; # 0..* Who collected the medication
  fhir:MedicationDispense.note [ Annotation ], ... ; # 0..* Information about the dispense
  fhir:MedicationDispense.dosageInstruction [ Dosage ], ... ; # 0..* How the medication is to be used by the patient or
    administered by the caregiver
  fhir:MedicationDispense.substitution [ # 0..1 Whether a substitution was performed on the dispense
    fhir:MedicationDispense.substitution.wasSubstituted [ boolean ]; # 1..1 Whether a substitution was or was not
    performed on the dispense
    fhir:MedicationDispense.substitution.type [ CodeableConcept ]; # 0..1 Code signifying whether a different drug was
    dispensed from what was prescribed
    fhir:MedicationDispense.substitution.reason [ CodeableConcept ], ... ; # 0..* Why was substitution made
    fhir:MedicationDispense.substitution.responsibleParty [ Reference(Practitioner|PractitionerRole) ], ... ; # 0..* Who
     is responsible for the substitution
  ];
  fhir:MedicationDispense.detectedIssue [ Reference(DetectedIssue) ], ... ; # 0..* Clinical issue with action
  fhir:MedicationDispense.eventHistory [ Reference(Provenance) ], ... ; # 0..* A list of relevant lifecycle events
]
```

Changes since Release 3

MedicationDispense
MedicationDispense.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/medication-dispense-status to http://hl7.org/fhir/ValueSet/medicationdispense-status|4.0.1
MedicationDispense.statusReason[x] | Added Element
MedicationDispense.performer.function | Added Element
MedicationDispense.performer.actor | Type Reference: Added Target Type PractitionerRole
MedicationDispense.location | Added Element
MedicationDispense.substitution.responsibleParty | Type Reference: Added Target Type PractitionerRole
MedicationDispense.performer.onBehalfOf | deleted
MedicationDispense.notDone | deleted
MedicationDispense.notDoneReason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 31 tests that all execute ok. 6 fail round-trip testing and 31 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
MedicationDispense.status | A coded concept specifying the state of the dispense event. | Required | MedicationDispense Status Codes
MedicationDispense.statusReason[x] | A code describing why a dispense was not performed. | Example | MedicationDispense Status Reason Codes
MedicationDispense.category | A code describing where the dispensed medication is expected to be consumed or administered. | Preferred | MedicationDispense Category Codes
MedicationDispense.medication[x] | A coded concept identifying which substance or product can be dispensed. | Example | SNOMEDCTMedicationCodes
MedicationDispense.performer.function | A code describing the role an individual played in dispensing a medication. | Example | MedicationDispense Performer Function Codes
MedicationDispense.type | Indicates the type of dispensing event that is performed. For example, Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc. | Example | v3.ActPharmacySupplyType
MedicationDispense.substitution.type | A coded concept describing whether a different medicinal product may be dispensed other than the product as specified exactly in the prescription. | Example | v3.ActSubstanceAdminSubstitutionCode
MedicationDispense.substitution.reason | A coded concept describing the reason that a different medication should (or should not) be substituted from what was prescribed. | Example | v3.SubstanceAdminSubstitutionReason

11.3.4.2 Constraints

id | Level | Location | Description | Expression
mdd-1 | Rule | (base) | whenHandedOver cannot be before whenPrepared | whenHandedOver.empty() or whenPrepared.empty() or whenHandedOver >= whenPrepared

11.3.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Returns dispenses of this medicine code | (MedicationDispense.medication as CodeableConcept) | 13 Resources
context | reference | Returns dispenses with a specific context (episode or episode of care) | MedicationDispense.context (EpisodeOfCare, Encounter)
destination | reference | Returns dispenses that should be sent to a specific destination | MedicationDispense.destination (Location)
identifier | token | Returns dispenses with this external identifier | MedicationDispense.identifier | 30 Resources
medication | reference | Returns dispenses of this medicine resource | (MedicationDispense.medication as Reference) (Medication) | 3 Resources
patient | reference | The identity of a patient to list dispenses for | MedicationDispense.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | Returns dispenses performed by a specific individual | MedicationDispense.performer.actor (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
prescription | reference | The identity of a prescription to list dispenses from | MedicationDispense.authorizingPrescription (MedicationRequest) | 1 Resources
receiver | reference | The identity of a receiver to list dispenses for | MedicationDispense.receiver (Practitioner, Patient)
responsibleparty | reference | Returns dispenses with the specified responsible party | MedicationDispense.substitution.responsibleParty (Practitioner, PractitionerRole)
status | token | Returns dispenses with a specified dispense status | MedicationDispense.status | 3 Resources
subject | reference | The identity of a patient for whom to list dispenses | MedicationDispense.subject (Group, Patient)
type | token | Returns dispenses of a specific type | MedicationDispense.type
whenhandedover | date | Returns dispenses handed over on this date | MedicationDispense.whenHandedOver
whenprepared | date | Returns dispenses prepared on this date | MedicationDispense.whenPrepared

11.3.6 Resource MedicationDispense - Examples

URL: https://hl7.org/fhir/R4/medicationdispense-examples.html

11.3.6 Resource MedicationDispense - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

ORAL DOSAGE FORMS

The following examples are MedicationDispense resources for Oral Dosage Forms that reference Medication Resources for a manufactured product:

• Dispense for Oral Chemotherapy - Capecitabine (RxNorm code)

• Dispense for Oxycodone (RxNorm code) - PRN dose with dose range, coded additional instructions and PRN coded pre-condition

• Dispense for Prednisone (RxNorm code) Tapering Dose - entered-in-error with multiple Dosage lines that include coded additionalInstructions and multiple EventHistory elements

• Dispense for Percocet (NDC Code) - in-progress with additionalInstructions, and as needed dosage with condition

• Dispense for Over the Counter medication - Tylenol PM (NDC code) - on hold with as needed dosage with a condition, a dose Range and an Event History

• Dispense for Alprazolam (Rx Norm code) - in progress

• Dispense for Azithromycin (NDC code) - completed with multiple dosage lines

• Dispense for Warfarin 5mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage

• Dispense for Warfarin 2mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage

The following examples are MedicationDispense resources for Oral Dosage Forms that include a Medication Code

• Dispense for Levothyroxine (NDC code) in-progress with authorizing prescription

• Dispense for Azithromycin (RxNorm Code) - in-progress with coded additionalInstructions

• Dispense for Dilantin Oral Suspension (NDC code) - completed

• Dispense for Busulfan (Myleran) - completed

• Dispense for Vicodin - in progress - PRN with pre-condition

INJECTABLE DOSAGE FORMS

The following examples of MedicationDispense resources for Injectable Dosage Forms that reference a Medication Resource for a manufactured product:

• Dispense for IV Antibiotic - in progress - Vancomycin IV (NDC Code) - in-progress

• Dispense for Subcutaneous injection with multiple dosage lines - Insulin - Novolog (NDC Code) - active - includes Destination, Receiver, note. Dosage includes additionalInstructions as text

• Dispense for Intravenous Antibiotic with a Dosage Rate - Piperacillin (NDC code) - completed with a coded site and route of administration and a rate expressed as a ratio

• Dispense for Intravenous Chemotherapy (inpatient order) - Alemtuzumab (RxNorm Code) - completed with multiple dosageInstruction lines

• Dispense for Lorazepam (Rx norm code) - completed with rate range

• ispense for Brentuximab Vedotin (NDC code) - IV Chemo - stopped

The following examples of MedicationDispense resources for Injectable Dosage Forms that include a Medication Code:

• Dispense for Insulin (Lantus) (NDC code) - in-progress with SC dosage

The following examples of MedicationDispense resources for Injectable Dosage Forms for Injectable Dosage Forms where the medication is a compounded (aka extemporaneous or magistral) product:

• Dispense for Potassium Chloride Admixture - stopped - with eventHistory and rate range for dosage

• Dispense for a TPN solution - completed

MISCELLAENOUS DOSAGE FORMS

The following examples of MedicationDispense resources including miscellaneous dosage forms:

• Dispense for Timoptic Opthalmic Drops - (NDC code) - completed

• Dispense for Nystatin drops (Snomed Code) - entered-in-error

• Dispense for Proventil (RxNorm Code) - completed - with additionalInstructions as text

• Dispense for Nystatin Topical Ointment (RxNorm code) - completed

• Dispense for Nasonex Inhaler (RxNorm Code) - in progress with additionalInstructions as text

• Dispense for Compounded ointment (HC 1%, SA 5% in White Petrolatum) - completed with additionalInstructions as text and a coded site in dosageInstructions

• Dispense for Fentanyl Patch (NDC code) - in progress

• Dispense for Paracetamol Suppositories - completed

Example Name | id | Format

Dispense for Oral Chemotherapy - Capecitabine (RxNorm code) | meddisp008 | XML | JSON | Turtle

Dispense for IV Antibiotic - in progress - Vancomycin IV (NDC Code) - in-progress | meddisp0301 | XML | JSON | Turtle

Dispense for Subcutaneous injection with multiple dosage lines - Insulin - Novolog (NDC Code) - active - includes Destination, Receiver, note. Dosage includes additionalInstructions as text | meddisp0302 | XML | JSON | Turtle

Dispense for Oxycodone (RxNorm code) - PRN dose with dose range, coded additional instructions and PRN coded pre-condition | meddisp0303 | XML | JSON | Turtle

Dispense for Intravenous Antibiotic with a Dosage Rate - Piperacillin (NDC code) - completed with a coded site and route of administration and a rate expressed as a ratio | meddisp0304 | XML | JSON | Turtle

Dispense for Prednisone (RxNorm code) Tapering Dose - entered-in-error with multiple Dosage lines that include coded additionalInstructions and multiple EventHistory elements | meddisp0305 | XML | JSON | Turtle

Dispense for Intravenous Chemotherapy (inpatient order) - Alemtuzumab (RxNorm Code) - completed with multiple dosageInstruction lines | meddisp0306 | XML | JSON | Turtle

Dispense for Busulfan (Myleran) - completed | meddisp0307 | XML | JSON | Turtle

Dispense for Timoptic Opthalmic Drops - (NDC code) - completed | meddisp0308 | XML | JSON | Turtle
Dispense for Nystatin drops (Snomed Code) - entered-in-error | meddisp0309 | XML | JSON | Turtle
Dispense for Percocet (NDC Code) - in-progress with additionalInstructions, and as needed dosage with condition | meddisp0310 | XML | JSON | Turtle
Dispense for Paracetamol Suppositories - completed | meddisp0311 | XML | JSON | Turtle
Dispense for Over the Counter medication - Tylenol PM (NDC code) - on hold with as needed dosage with a condition, a dose Range and an Event History | meddisp0312 | XML | JSON | Turtle
Dispense for Potassium Chloride Admixture - stopped - with eventHistory and rate range for dosage | meddisp0313 | XML | JSON | Turtle
Dispense for Lorazepam (Rx norm code) - completed with rate range | meddisp0314 | XML | JSON | Turtle
Dispense for Alprazolam (Rx Norm code) - in progress | meddisp0315 | XML | JSON | Turtle
Dispense for Insulin (Lantus) (NDC code) - in-progress with SC dosage | meddisp0316 | XML | JSON | Turtle
Dispense for Brentuximab Vedotin (NDC code) - IV Chemo - stopped | meddisp0317 | XML | JSON | Turtle
Dispense for Levothyroxine (NDC code) in-progress with authorizing prescription | meddisp0318 | XML | JSON | Turtle
Dispense for Azithromycin (NDC code) - completed with multiple dosage lines | meddisp0319 | XML | JSON | Turtle
Dispense for a TPN solution - completed | meddisp0320 | XML | JSON | Turtle
Dispense for Vicodin - in progress - PRN with pre-condition | meddisp0321 | XML | JSON | Turtle
Dispense for Dilantin Oral Suspension (NDC code) - completed | meddisp0322 | XML | JSON | Turtle
Dispense for Nystatin Topical Ointment (RxNorm code) - completed | meddisp0324 | XML | JSON | Turtle
Dispense for Fentanyl Patch (NDC code) - in progress | meddisp0325 | XML | JSON | Turtle
Dispense for Azithromycin (RxNorm Code) - in-progress with coded additionalInstructions | meddisp0326 | XML | JSON | Turtle
Dispense for Proventil (RxNorm Code) - completed - with additionalInstructions as text | meddisp0327 | XML | JSON | Turtle
Dispense for Nasonex Inhaler (RxNorm Code) - in progress with additionalInstructions as text | meddisp0328 | XML | JSON | Turtle
Dispense for compounded ointment - completed with additionalInstructions as text and a coded site in dosageInstructions | meddisp0329 | XML | JSON | Turtle
Dispense for Warfarin 5mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage | meddisp0330 | XML | JSON | Turtle
Dispense for Warfarin 2mg (NDC Code) where the product is dispensed as multiple strengths based on prescribe dosage | meddisp0331 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.3.7 Resource MedicationDispense - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicationdispense-definitions.html

11.3.7 Resource MedicationDispense - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the MedicationDispense resource.

MedicationDispense
Element Id | MedicationDispense
Definition | Indicates that a medication product is to be or has been dispensed for a named person/patient. This includes a description of the medication product (supply) provided and the instructions for administering the medication. The medication dispense is the result of a pharmacy system responding to a medication order.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element mdd-1 Rule whenHandedOver cannot be before whenPrepared whenHandedOver.empty() or whenPrepared.empty() or whenHandedOver >= whenPrepared
Defined on this element
mdd-1 | Rule | whenHandedOver cannot be before whenPrepared | whenHandedOver.empty() or whenPrepared.empty() or whenHandedOver >= whenPrepared
MedicationDispense.identifier
Element Id | MedicationDispense.identifier
Definition | Identifiers associated with this Medication Dispense that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business identifiers assigned to this resource by the performer or other systems and remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Comments | This is a business identifier, not a resource identifier.
MedicationDispense.partOf
Element Id | MedicationDispense.partOf
Definition | The procedure that trigger the dispense.
Cardinality | 0..*
Type | Reference(Procedure)
MedicationDispense.status
Element Id | MedicationDispense.status
Definition | A code specifying the state of the set of dispense events.
Cardinality | 1..1
Terminology Binding | Medication dispense status codes (Required)
Type | code

Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationDispense.statusReason[x]
Element Id | MedicationDispense.statusReason[x]
Definition | Indicates the reason why a dispense was not performed.
Cardinality | 0..1
Terminology Binding | Medication dispense status reason codes (Example)
Type | CodeableConcept|Reference(DetectedIssue)
[x] Note | See Choice of Data Types for further information about how to use [x]
MedicationDispense.category
Element Id | MedicationDispense.category
Definition | Indicates the type of medication dispense (for example, where the medication is expected to be consumed or administered (i.e. inpatient or outpatient)).
Cardinality | 0..1
Terminology Binding | Medication dispense category codes (Preferred)
Type | CodeableConcept
Comments | The category can be used to include where the medication is expected to be consumed or other types of dispenses. Invariants can be used to bind to different value sets when profiling to bind.
MedicationDispense.medication[x]
Element Id | MedicationDispense.medication[x]
Definition | Identifies the medication being administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Cardinality | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example, if you require form or lot number, then you must reference the Medication resource.
MedicationDispense.subject
Element Id | MedicationDispense.subject
Definition | A link to a resource representing the person or the group to whom the medication will be given.
Cardinality | 0..1
Type | Reference(Patient | Group)
Summary | true
Comments | SubstanceAdministration->subject->Patient.
MedicationDispense.context
Element Id | MedicationDispense.context
Definition | The encounter or episode of care that establishes the context for this event.
Cardinality | 0..1
Type | Reference(Encounter | EpisodeOfCare)
MedicationDispense.supportingInformation
Element Id | MedicationDispense.supportingInformation
Definition | Additional information that supports the medication being dispensed.
Cardinality | 0..*
Type | Reference(Any)
MedicationDispense.performer
Element Id | MedicationDispense.performer
Definition | Indicates who or what performed the event.
Cardinality | 0..*
MedicationDispense.performer.function
Element Id | MedicationDispense.performer.function
Definition | Distinguishes the type of performer in the dispense. For example, date enterer, packager, final checker.
Cardinality | 0..1
Terminology Binding | Medication dispense performer function codes (Example)
Type | CodeableConcept
Requirements | Allows disambiguation of the types of involvement of different performers.
MedicationDispense.performer.actor
Element Id | MedicationDispense.performer.actor
Definition | The device, practitioner, etc. who performed the action. It should be assumed that the actor is the dispenser of the medication.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson)
MedicationDispense.location
Element Id | MedicationDispense.location
Definition | The principal physical location where the dispense was performed.
Cardinality | 0..1
Type | Reference(Location)
MedicationDispense.authorizingPrescription
Element Id | MedicationDispense.authorizingPrescription

Definition | Indicates the medication order that is being dispensed against.
Cardinality | 0..*
Type | Reference(MedicationRequest)
Comments | Maps to basedOn in Event logical model.
MedicationDispense.type
Element Id | MedicationDispense.type
Definition | Indicates the type of dispensing event that is performed. For example, Trial Fill, Completion of Trial, Partial Fill, Emergency Fill, Samples, etc.
Cardinality | 0..1
Terminology Binding | V3 Value SetActPharmacySupplyType (Example)
Type | CodeableConcept
MedicationDispense.quantity
Element Id | MedicationDispense.quantity
Definition | The amount of medication that has been dispensed. Includes unit of measure.
Cardinality | 0..1
Type | SimpleQuantity
MedicationDispense.daysSupply
Element Id | MedicationDispense.daysSupply
Definition | The amount of medication expressed as a timing amount.
Cardinality | 0..1
Type | SimpleQuantity
MedicationDispense.whenPrepared
Element Id | MedicationDispense.whenPrepared
Definition | The time when the dispensed product was packaged and reviewed.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicationDispense.whenHandedOver
Element Id | MedicationDispense.whenHandedOver
Definition | The time the dispensed product was provided to the patient or their representative.
Cardinality | 0..1
Type | dateTime
MedicationDispense.destination
Element Id | MedicationDispense.destination
Definition | Identification of the facility/location where the medication was shipped to, as part of the dispense event.
Cardinality | 0..1
Type | Reference(Location)
MedicationDispense.receiver
Element Id | MedicationDispense.receiver
Definition | Identifies the person who picked up the medication. This will usually be a patient or their caregiver, but some cases exist where it can be a healthcare professional.
Cardinality | 0..*
Type | Reference(Patient | Practitioner)
MedicationDispense.note
Element Id | MedicationDispense.note
Definition | Extra information about the dispense that could not be conveyed in the other attributes.
Cardinality | 0..*
Type | Annotation
MedicationDispense.dosageInstruction
Element Id | MedicationDispense.dosageInstruction
Definition | Indicates how the medication is to be used by the patient.
Cardinality | 0..*
Type | Dosage
Comments | When the dose or rate is intended to change over the entire administration period (e.g. Tapering dose prescriptions), multiple instances of dosage instructions will need to be supplied to convey the different doses/rates. The pharmacist reviews the medication order prior to dispense and updates the dosageInstruction based on the actual product being dispensed.
MedicationDispense.substitution
Element Id | MedicationDispense.substitution
Definition | Indicates whether or not substitution was made as part of the dispense. In some cases, substitution will be expected but does not happen, in other cases substitution is not expected but does happen. This block explains what substitution did or did not happen and why. If nothing is specified, substitution was not done.
Cardinality | 0..1
MedicationDispense.substitution.wasSubstituted
Element Id | MedicationDispense.substitution.wasSubstituted
Definition | True if the dispenser dispensed a different drug or product from what was prescribed.
Cardinality | 1..1
Type | boolean
MedicationDispense.substitution.type
Element Id | MedicationDispense.substitution.type
Definition | A code signifying whether a different drug was dispensed from what was prescribed.
Cardinality | 0..1
Terminology Binding | V3 Value SetActSubstanceAdminSubstitutionCode (Example)
Type | CodeableConcept

MedicationDispense.substitution.reason
Element Id | MedicationDispense.substitution.reason
Definition | Indicates the reason for the substitution (or lack of substitution) from what was prescribed.
Cardinality | 0..*
Terminology Binding | V3 Value SetSubstanceAdminSubstitutionReason (Example)
Type | CodeableConcept
MedicationDispense.substitution.responsibleParty
Element Id | MedicationDispense.substitution.responsibleParty
Definition | The person or organization that has primary responsibility for the substitution.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole)
MedicationDispense.detectedIssue
Element Id | MedicationDispense.detectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. drug-drug interaction, duplicate therapy, dosage alert etc.
Cardinality | 0..*
Type | Reference(DetectedIssue)
Alternate Names | Contraindication; Drug Utilization Review (DUR); Alert
Comments | This element can include a detected issue that has been identified either by a decision support system or by a clinician and may include information on the steps that were taken to address the issue.
MedicationDispense.eventHistory
Element Id | MedicationDispense.eventHistory
Definition | A summary of the events of interest that have occurred, such as when the dispense was verified.
Cardinality | 0..*
Type | Reference(Provenance)
Comments | This might not include provenances for all versions of the request – only those deemed “relevant” or important. This SHALL NOT include the Provenance associated with this current version of the resource. (If that provenance is deemed to be a “relevant” change, it will need to be added as part of a later update. Until then, it can be queried directly as the Provenance that points to this version using _revinclude All Provenances should have some historical version of this Request as their subject.).

11.3.8 Resource MedicationDispense - Mappings

URL: https://hl7.org/fhir/R4/medicationdispense-mappings.html

11.3.8 Resource MedicationDispense - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

V3 Pharmacy Dispense RMIM

Mappings for the medicationdispense resource (see Mappings to Other Standards for further information & status).

11.3.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MedicationDispense
identifier | FiveWs.identifier
status | FiveWs.status
medication[x] | FiveWs.what[x]
subject | FiveWs.subject[x]
supportingInformation | FiveWs.context

11.3.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

MedicationDispense | Event
identifier | Event.identifier
partOf | Event.partOf
status | Event.status
statusReason[x] | Event.statusReason
medication[x] | Event.code
subject | Event.subject
context | Event.context
performer | Event.performer
actor | Event.performer.actor
authorizingPrescription | Event.basedOn
whenHandedOver | Event.occurrence[x]
note | Event.note

11.3.8.3 HL7 v2 Mapping (http://hl7.org/v2)

MedicationDispense
identifier
partOf
status
statusReason[x]
category
medication[x] | RXD-2-Dispense/Give Code
subject | PID-3-Patient ID List
context
supportingInformation
performer
function
actor
location
authorizingPrescription | ORC-2 Placer Order Number
type | RXD-33-Dispense Type
quantity | RXD-4-Actual Dispense Amount / RXD-5.1-Actual Dispense Units.code / RXD-5.3-Actual Dispense Units.name of coding system
daysSupply | TQ1.6 Timing/Quantity Segment Service Duration. Prior to v2.5, ORC.7.3 Common Order Segment / Quantity/Timing / Duration component. This is a formatted string, first character for the time unit (e.g., D=days), followed by the value. For example, “D14” represents “14 days supply” From v2.5 on, TQ1.6 Timing/Quantity Segment / Service Duration. This is a CQ data type (<Quantity (NM)> ^ <Units (CWE)>), thus for days supply, assuming the unit of measure is “days”, the numeric value is TQ1.6.1 (...|14^ For backwards compatibility, ORC.7 was permitted through v2.6. Both forms (field and segment) may be present in v2.5, v2.5.1, and v2.6
whenPrepared | RXD-3-Date/Time Dispensed
whenHandedOver | RXD-3-Date/Time Dispensed
destination | RXD-13 Dispense-to Location; RXD-30 Dispense to Pharmacy; RXD-31 Dispense to Pharmacy Address
receiver
note | RXD-9 Dispense Notes
dosageInstruction
substitution
wasSubstituted
type | RXO-9-Allow Substitutions / RXE-9-Substitution Status
reason
responsibleParty
detectedIssue
eventHistory

11.3.8.4 RIM Mapping (http://hl7.org/v3)

MedicationDispense | Supply[moodCode=EVN]
identifier | .id
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=PROC,moodCode=EVN]
status | .statusCode
statusReason[x] | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="not done reason"].value
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication dispense"].value
medication[x] | .participation[typeCode=PRD].role
subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
supportingInformation | .outboundRelationship[typeCode=PERT].target[A_SupportingClinicalStatement CMET minimal with many different choices of classCodes(ORG, ENC, PROC, SPLY, SBADM, OBS) and each of the act class codes draws from one or more of the following moodCodes (EVN, DEF, INT PRMS, RQO, PRP, APT, ARQ, GOL)]
performer | .participation[typeCode=PRF]
function | participation[typeCode=PRF].functionCode
actor | .role
location | .participation[typeCode=LOC].role
authorizingPrescription | .outboundRelationship[typeCode=FLFS].target[moodCode=RQO]
type | .code
quantity | .quantity
daysSupply | effectiveUseTime
whenPrepared | .effectiveTime[xmi:type=IVL_TS].low
whenHandedOver | .effectiveTime[xmi:type=IVL_TS].high
destination | .participation[typeCode=DST].role
receiver | .participation[typeCode=RCV].role
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=ACT,moodCode=EVN]
dosageInstruction | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=INT]
substitution | .outboundRelationship[typeCode=COMP].target[classCode=SUBST, moodCode=EVN]
wasSubstituted | .actionNegationInd
type | .code
reason | .reasonCode

responsibleParty | .participation[typeCode=PFM].role[classCode=ASSIGN]
detectedIssue | .inboundRelationship[typeCode=SUBJ]/source[classCode=ALRT,moodCode=EVN].value
eventHistory | .inboundRelationship(typeCode=SUBJ].source[classCode=CACT, moodCode=EVN]

11.3.8.5 V3 Pharmacy Dispense RMIM (http://www.hl7.org/v3/PORX_RM020070UV)

MedicationDispense
identifier | CombinedMedicationDispense.id
partOf
status | CombinedMedicationDispense.SupplyEvent.statusCode
statusReason[x]
category
medication[x] | CombinedMedicationRequest.component1.AdministrationRequest.consumable
subject
context
supportingInformation
performer
function
actor
location
authorizingPrescription | CombinedMedicationDispense.inFulfillmentOf
type | CombinedMedicationDispense.SupplyEvent.code
quantity | CombinedMedicationDispense.SupplyEvent.quantity
daysSupply
whenPrepared | CombinedMedicationDispense.SupplyEvent.effectiveTime
whenHandedOver | CombinedMedicationDispense.SupplyEvent.effectiveTime
destination | CombinedMedicationDispense.SupplyEvent.destination
receiver | CombinedMedicationDispense.SupplyEvent.receiver
note
dosageInstruction
substitution | CombinedMedicationDispense.substitutionMade
wasSubstituted
type | CombinedMedicationDispense.substitutionMade.code
reason | CombinedMedicationDispense.substitutionMade.reasonCode
responsibleParty | CombinedMedicationDispense.substitutionMade.ResponsibleParty
detectedIssue
eventHistory

11.3.9 Resource MedicationDispense - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicationdispense-profiles.html

11.3.9 Resource MedicationDispense - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

This table lists profiles and extensions for the MedicationDispense resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.3.9.1 Profiles

No Profiles defined for this resource

11.3.9.2 Extensions

workflow-researchStudy | MedicationDispense | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.3.9.3 Search Extensions

No Search Extensions defined for this resource

11.3.10 Resource MedicationDispense - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/medicationdispense-version-maps.html

No validation errors - all conversions are clean

11.3.10 Resource MedicationDispense - R3/R4 Conversions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

R3 : R4 Conversion maps for MedicationDispense.

Functional status for this map: 31 tests that all execute ok. 6 fail round-trip testing and 31 r3 resources are invalid (0 errors). (see documentation)

11.3.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MedicationDispense3to4" = "R3 to R4 Conversions for MedicationDispense"

uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationDispense" alias MedicationDispenseR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationDispense" alias MedicationDispense as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MedicationDispense(source src : MedicationDispenseR3, target tgt : MedicationDispense) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.subject -> tgt.subject;
  src.context -> tgt.context;
  src.supportingInformation -> tgt.supportingInformation;
  src.performer as vs0 -> tgt.performer as vt0 then actor(vs0, vt0);
  src.authorizingPrescription -> tgt.authorizingPrescription;
  src.type -> tgt.type;
  src.quantity -> tgt.quantity;
  src.daysSupply -> tgt.daysSupply;
  src.whenPrepared -> tgt.whenPrepared;
  src.whenHandedOver -> tgt.whenHandedOver;
  src.destination -> tgt.destination;
  src.receiver -> tgt.receiver;
  src.note -> tgt.note;
  src.dosageInstruction -> tgt.dosageInstruction;
  src.substitution as vs0 -> tgt.substitution as vt0 then subst(vs0, vt0);
  src.detectedIssue -> tgt.detectedIssue;
  src.eventHistory -> tgt.eventHistory;
  src.notDone as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationDispense.notDone',  vt.value = v;
  src.notDoneReason as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationDispense.notDoneReason',  vt.value = v;
}

group actor(source src, target tgt) extends BackboneElement {
  src.actor -> tgt.actor;
  src.onBehalfOf as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationDispense.performer.onBehalfOf',  vt.value = v;
}

group subst(source src, target tgt) extends BackboneElement {
  src.wasSubstituted -> tgt.wasSubstituted;
  src.type -> tgt.type;
  src.reason -> tgt.reason;
  src.responsibleParty -> tgt.responsibleParty;
}
```

11.3.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationDispense4to3" = "R4 to R3 Conversion for MedicationDispense"

uses "http://hl7.org/fhir/StructureDefinition/MedicationDispense" alias MedicationDispense as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationDispense" alias MedicationDispenseR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MedicationDispense(source src : MedicationDispense, target tgt : MedicationDispenseR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.subject -> tgt.subject;
  src.context -> tgt.context;
```

```
  src.supportingInformation -> tgt.supportingInformation;
  src.performer as vs0 -> tgt.performer as vt0 then actor(vs0, vt0);
  src.authorizingPrescription -> tgt.authorizingPrescription;
  src.type -> tgt.type;
  src.quantity -> tgt.quantity;
  src.daysSupply -> tgt.daysSupply;
  src.whenPrepared -> tgt.whenPrepared;
  src.whenHandedOver -> tgt.whenHandedOver;
  src.destination -> tgt.destination;
  src.receiver -> tgt.receiver;
  src.note -> tgt.note;
  src.dosageInstruction -> tgt.dosageInstruction;
  src.substitution as vs0 -> tgt.substitution as vt0 then subst(vs0, vt0);
  src.detectedIssue -> tgt.detectedIssue;
  src.eventHistory -> tgt.eventHistory;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationDispense.notDone' then
     {
    e.value -> tgt.notDone;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationDispense.notDoneReason' then {
    e.value -> tgt.notDoneReason;
  };
}

group actor(source src, target tgt) extends BackboneElement {
  src.actor -> tgt.actor;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationDispense.performer.onBehalfOf' then {
    e.value -> tgt.onBehalfOf;
  };
}

group subst(source src, target tgt) extends BackboneElement {
  src.wasSubstituted -> tgt.wasSubstituted;
  src.type -> tgt.type;
  src.reason -> tgt.reason;
  src.responsibleParty -> tgt.responsibleParty;
}
```

11.3.10.3 R4 Validation Errors

11.3.11 MedicationDispense HL7 Extensions

URL: https://hl7.org/fhir/R4/medicationdispense-extensions.html

Defines common extensions used with or related to the MedicationDispense resource

11.3.11 MedicationDispense HL7 Extensions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

11.3.11.1 Content

11.3.12 JSON Schema for MedicationDispense

URL: https://hl7.org/fhir/R4/medicationdispense.schema.json.html

11.3.12 JSON Schema for MedicationDispense

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is a representation of the json schema for MedicationDispense, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicationDispense",
  "$ref": "#/definitions/MedicationDispense",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicationDispense": {
      "description": "Indicates that a medication product is to be or has been dispensed for a named person/patient.
    This includes a description of the medication product (supply) provided and the instructions for administering the
    medication.  The medication dispense is the result of a pharmacy system responding to a medication order.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicationDispense resource",
          "const": "MedicationDispense"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
```

```
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers associated with this Medication Dispense that are defined by business processes
and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business
identifiers assigned to this resource by the performer or other systems and remain constant as the resource is
updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "partOf": {
      "description": "The procedure that trigger the dispense.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "A code specifying the state of the set of dispense events.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReasonCodeableConcept": {
      "description": "Indicates the reason why a dispense was not performed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
```

```
    "statusReasonReference": {
      "description": "Indicates the reason why a dispense was not performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "category": {
      "description": "Indicates the type of medication dispense (for example, where the medication is expected to be
 consumed or administered (i.e. inpatient or outpatient)).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationCodeableConcept": {
      "description": "Identifies the medication being administered. This is either a link to a resource representing
 the details of the medication or a simple attribute carrying a code that identifies the medication from a known
list of medications.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationReference": {
      "description": "Identifies the medication being administered. This is either a link to a resource representing
 the details of the medication or a simple attribute carrying a code that identifies the medication from a known
list of medications.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "subject": {
      "description": "A link to a resource representing the person or the group to whom the medication will be
given.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "context": {
      "description": "The encounter or episode of care that establishes the context for this event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "supportingInformation": {
      "description": "Additional information that supports the medication being dispensed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "performer": {
      "description": "Indicates who or what performed the event.",
      "items": {
        "$ref": "#/definitions/MedicationDispense_Performer"
      },
      "type": "array"
    },
    "location": {
      "description": "The principal physical location where the dispense was performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "authorizingPrescription": {
      "description": "Indicates the medication order that is being dispensed against.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "type": {
      "description": "Indicates the type of dispensing event that is performed. For example, Trial Fill, Completion
of Trial, Partial Fill, Emergency Fill, Samples, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "quantity": {
      "description": "The amount of medication that has been dispensed. Includes unit of measure.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "daysSupply": {
      "description": "The amount of medication expressed as a timing amount.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "whenPrepared": {
      "description": "The time when the dispensed product was packaged and reviewed.",
      "$ref": "#/definitions/dateTime"
    },
    "_whenPrepared": {
      "description": "Extensions for whenPrepared",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "whenHandedOver": {
      "description": "The time the dispensed product was provided to the patient or their representative.",
      "$ref": "#/definitions/dateTime"
    },
    "_whenHandedOver": {
      "description": "Extensions for whenHandedOver",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "destination": {
      "description": "Identification of the facility/location where the medication was shipped to, as part of the
dispense event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
```

```
    "receiver": {
      "description": "Identifies the person who picked up the medication.  This will usually be a patient or their
caregiver, but some cases exist where it can be a healthcare professional.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Extra information about the dispense that could not be conveyed in the other attributes.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "dosageInstruction": {
      "description": "Indicates how the medication is to be used by the patient.",
      "items": {
        "$ref": "Dosage.schema.json#/definitions/Dosage"
      },
      "type": "array"
    },
    "substitution": {
      "description": "Indicates whether or not substitution was made as part of the dispense.  In some cases,
substitution will be expected but does not happen, in other cases substitution is not expected but does happen.
This block explains what substitution did or did not happen and why.  If nothing is specified, substitution was not
done.",
      "$ref": "#/definitions/MedicationDispense_Substitution"
    },
    "detectedIssue": {
      "description": "Indicates an actual or potential clinical issue with or between one or more active or proposed
 clinical actions for a patient; e.g. drug-drug interaction, duplicate therapy, dosage alert etc.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "eventHistory": {
      "description": "A summary of the events of interest that have occurred, such as when the dispense was
verified.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"MedicationDispense_Performer": {
  "description": "Indicates that a medication product is to be or has been dispensed for a named person/patient.
This includes a description of the medication product (supply) provided and the instructions for administering the
medication.  The medication dispense is the result of a pharmacy system responding to a medication order.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "function": {
      "description": "Distinguishes the type of performer in the dispense.  For example, date enterer, packager,
final checker.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
        },
        "actor": {
          "description": "The device, practitioner, etc. who performed the action.  It should be assumed that the actor
    is the dispenser of the medication.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "actor"
      ]
    },
    "MedicationDispense_Substitution": {
      "description": "Indicates that a medication product is to be or has been dispensed for a named person/patient.
    This includes a description of the medication product (supply) provided and the instructions for administering the
    medication.  The medication dispense is the result of a pharmacy system responding to a medication order.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "wasSubstituted": {
          "description": "True if the dispenser dispensed a different drug or product from what was prescribed.",
          "$ref": "#/definitions/boolean"
        },
        "_wasSubstituted": {
          "description": "Extensions for wasSubstituted",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "type": {
          "description": "A code signifying whether a different drug was dispensed from what was prescribed.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "reason": {
          "description": "Indicates the reason for the substitution (or lack of substitution) from what was
    prescribed.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "responsibleParty": {
          "description": "The person or organization that has primary responsibility for the substitution.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.3.13 Resource MedicationDispense - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicationdispense-dependencies.html

11.3.13 Resource MedicationDispense - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

Dependency Graph for MedicationDispense FMM level 2

Name | Card. | Type | Dependency Analysis
MedicationDispense | | DomainResource
identifier | 0..* | Identifier
partOf | 0..* | Reference(Procedure)
status | 1..1 | code
statusReason[x] | 0..1
statusReasonCodeableConcept | | CodeableConcept
statusReasonReference | | Reference(DetectedIssue
category | 0..1 | CodeableConcept
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(Medication
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
context | 0..1 | Reference(Encounter | EpisodeOfCare)
supportingInformation | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson)
location | 0..1 | Reference(Location)
authorizingPrescription | 0..* | Reference(MedicationRequest)
type | 0..1 | CodeableConcept
quantity | 0..1 | SimpleQuantity
daysSupply | 0..1 | SimpleQuantity
whenPrepared | 0..1 | dateTime
whenHandedOver | 0..1 | dateTime
destination | 0..1 | Reference(Location)
receiver | 0..* | Reference(Patient | Practitioner)
note | 0..* | Annotation
dosageInstruction | 0..* | Dosage
substitution | 0..1 | BackboneElement
wasSubstituted | 1..1 | boolean
type | 0..1 | CodeableConcept
reason | 0..* | CodeableConcept
responsibleParty | 0..* | Reference(Practitioner | PractitionerRole)
detectedIssue | 0..* | Reference(DetectedIssue) | Type Warning: (DetectedIssue = FMM1-Trial Use vs. Element = FMM2-Trial Use)
eventHistory | 0..* | Reference(Provenance)

11.3.14 Value Set Sc-valueset-medicationdispense-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medicationdispense-status.html

11.3.14 Value Set Sc-valueset-medicationdispense-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.3.15 MedicationDispense Status CodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medicationdispense-status)

Mapping from http://hl7.org/fhir/ValueSet/medicationdispense-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "MedicationDispense Status Codes"

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
preparation | equivalent | planned
in-progress | equivalent | active
on-hold | equivalent | suspended
stopped | equivalent | failed
completed | equivalent | complete
unknown | equivalent | unknown
cancelled, declined | equivalent | not-done

11.4 Resource MedicationStatement - Content

URL: https://hl7.org/fhir/R4/medicationstatement.html
Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

A record of a medication that is being consumed by a patient. A MedicationStatement may indicate that the patient may be taking the medication now or has taken the medication in the past or will be taking the medication in the future. The source of this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. The medication information may come from sources such as the patient's memory, from a prescription bottle, or from a list of medications the patient, clinician or other party maintains.

The primary difference between a medication statement and a medication administration is that the medication administration has complete administration information and is based on actual administration information from the person who administered the medication. A medication statement is often, if not always, less specific. There is no required date/time when the medication was administered, in fact we only know that a source has reported the patient is taking this medication, where details such as time, quantity, or rate or even medication product may be incomplete or missing or less precise. As stated earlier, the medication statement information may come from the patient's memory, from a prescription bottle or from a list of medications the patient, clinician or other party maintains. Medication administration is more formal and is not missing detailed information.

11.4.1 Scope and Usage

Common usage includes:

• the recording of non-prescription and/or recreational drugs

• the recording of an intake medication list upon admission to hospital

• the summarization of a patient's "active medications" in a patient profile

A MedicationStatement may be used to record substance abuse or the use of other agents such as tobacco or alcohol. This would typically be done if these substances are intended to be included in clinical decision support checking (for example, interaction checking) and as part of an active medication list. If the intent is to populate social history and/or to include additional information (for example, desire to quit, amount per day, negative health effects), then it is better to record as an Observation that could then be used to populate Social History.

This resource does not produce a medication list, but it does produce individual medication statements that may be used in the List resource to construct various types of medication lists. Note that other medication lists can also be constructed from the other Pharmacy resources (e.g., MedicationRequest, MedicationAdministration).

A medication statement is not a part of the prescribe -> dispense -> administer sequence, but is a report by a patient, significant other or a clinician that one or more of the prescribe, dispense or administer actions has occurred, resulting is a belief that the patient is, has, or will be using a particular medication.

MedicationStatement is an event resource from a FHIR workflow perspective - see Workflow Event

11.4.2 Boundaries and Relationships

The MedicationStatement resource is used to record a medications or substances that the patient reports as being taken, not taking, have taken in the past or may take in the future. It can also be used to record medication use that is derived from other records such as a MedicationRequest. The statement is not used to request or order a medication, supply or device. When requesting medication, supplies or devices when there is a patient focus or instructions regarding their use, a MedicationRequest, SupplyRequest or DeviceRequest should be used instead

The Medication domain includes a number of related resources

MedicationRequest | An order for both supply of the medication and the instructions for administration of the medicine to a patient.

MedicationDispense | Provision of a supply of a medication with the intention that it is subsequently consumed by a patient (usually in response to a prescription).

MedicationAdministration | When a patient actually consumes a medicine, or it is otherwise administered to them

MedicationStatement | This is a record of a medication being taken by a patient or that a medication has been given to a patient, where the record is the result of a report from the patient or another clinician, or derived from supporting information (for example, Claim, Observation or MedicationRequest). A medication statement is not a part of the prescribe->dispense->administer sequence but is a report that such a sequence (or at least a part of it) did take place, resulting in a belief that the patient has received a particular medication.

This resource is distinct from MedicationRequest, MedicationDispense and MedicationAdministration. Each of those resources refers to specific events - an individual order, an individual provisioning of medication or an individual dosing. MedicationStatement is a broader assertion covering a wider timespan and is independent of specific events. The existence of resource instances of any of the preceding three types may be used to infer a medication statement. However, medication statements can also be captured on the basis of other information, including an assertion by the patient or a care-giver, the results of a lab test, etc.

This resource is referenced by AdverseEvent, Goal, itself and Observation

11.4.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationStatement | TU | | DomainResource | Record of medication being taken by a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External identifier
basedOn | Σ | 0..* | Reference(MedicationRequest | CarePlan | ServiceRequest) | Fulfils plan, proposal or order
partOf | Σ | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation) | Part of referenced event
status | ?!Σ | 1..1 | code | active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-taken Medication status codes (Required)
statusReason | | 0..* | CodeableConcept | Reason for current status SNOMED CT Drug Therapy Status codes (Example)
category | Σ | 0..1 | CodeableConcept | Type of medication usage Medication usage category codes (Preferred)
medication[x] | Σ | 1..1 | | What medication was taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 1..1 | Reference(Patient | Group) | Who is/was taking the medication
context | Σ | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with MedicationStatement
effective[x] | Σ | 0..1 | | The date/time or interval when the medication is/was/will be taken
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
dateAsserted | Σ | 0..1 | dateTime | When the statement was asserted?
informationSource | | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization) | Person or organization that provided the information about the taking of this medication
derivedFrom | | 0..* | Reference(Any) | Additional supporting information
reasonCode | | 0..* | CodeableConcept | Reason for why the medication is being/was taken Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Condition or observation that supports why the medication is being/was taken
note | | 0..* | Annotation | Further information about the statement
dosage | | 0..* | Dosage | Details of how medication is/was taken or should be taken
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(MedicationRequest|CarePlan|ServiceRequest) Fulfils plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-
    taken -->
 <statusReason><!-- 0..* CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who is/was taking  the medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with MedicationStatement
    --></context>
 <effective[x]><!-- 0..1 dateTime|Period The date/time or interval when the medication is/was/will be taken
    --></effective[x]>
 <dateAsserted value="[dateTime]"/><!-- 0..1 When the statement was asserted? -->
 <informationSource><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Person or organization that provided the information about the taking of this medication
    --></informationSource>
 <derivedFrom><!-- 0..* Reference(Any) Additional supporting information --></derivedFrom>
 <reasonCode><!-- 0..* CodeableConcept Reason for why the medication is being/was taken --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Condition or observation that supports why
     the medication is being/was taken --></reasonReference>
 <note><!-- 0..* Annotation Further information about the statement --></note>
 <dosage><!-- 0..* Dosage Details of how medication is/was taken or should be taken --></dosage>
</MedicationStatement>
```

JSON Template

```
{
```

```
  "resourceType" : "MedicationStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(MedicationRequest|CarePlan|ServiceRequest) }], // Fulfils plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) }], // Part of referenced event
  "status" : "<code>", // R!  active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-taken
  "statusReason" : [{ CodeableConcept }], // Reason for current status
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What medication was taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
  "subject" : { Reference(Patient|Group) }, // R!  Who is/was taking  the medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with MedicationStatement
  // effective[x]: The date/time or interval when the medication is/was/will be taken. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "dateAsserted" : "<dateTime>", // When the statement was asserted?
  "informationSource" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) }, // Person or organization that provided the information about the taking of this
    medication
  "derivedFrom" : [{ Reference(Any) }], // Additional supporting information
  "reasonCode" : [{ CodeableConcept }], // Reason for why the medication is being/was taken
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Condition or observation that supports
     why the medication is being/was taken
  "note" : [{ Annotation }], // Further information about the statement
  "dosage" : [{ Dosage }] // Details of how medication is/was taken or should be taken
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationStatement.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationStatement.basedOn [ Reference(MedicationRequest|CarePlan|ServiceRequest) ], ... ; # 0..* Fulfils plan,
    proposal or order
  fhir:MedicationStatement.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Observation) ], ... ; # 0..* Part of referenced event
  fhir:MedicationStatement.status [ code ]; # 1..1 active | completed | entered-in-error | intended | stopped | on-hold
    | unknown | not-taken
  fhir:MedicationStatement.statusReason [ CodeableConcept ], ... ; # 0..* Reason for current status
  fhir:MedicationStatement.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationStatement.medication[x] : 1..1 What medication was taken. One of these 2
    fhir:MedicationStatement.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationStatement.medicationReference [ Reference(Medication) ]
  fhir:MedicationStatement.subject [ Reference(Patient|Group) ]; # 1..1 Who is/was taking  the medication
  fhir:MedicationStatement.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    MedicationStatement
  # MedicationStatement.effective[x] : 0..1 The date/time or interval when the medication is/was/will be taken. One of
    these 2
    fhir:MedicationStatement.effectiveDateTime [ dateTime ]
    fhir:MedicationStatement.effectivePeriod [ Period ]
  fhir:MedicationStatement.dateAsserted [ dateTime ]; # 0..1 When the statement was asserted?
  fhir:MedicationStatement.informationSource [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ]; # 0..1 Person or organization that
    provided the information about the taking of this medication
  fhir:MedicationStatement.derivedFrom [ Reference(Any) ], ... ; # 0..* Additional supporting information
  fhir:MedicationStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for why the medication is being/was taken
  fhir:MedicationStatement.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..* Condition
     or observation that supports why the medication is being/was taken
  fhir:MedicationStatement.note [ Annotation ], ... ; # 0..* Further information about the statement
  fhir:MedicationStatement.dosage [ Dosage ], ... ; # 0..* Details of how medication is/was taken or should be taken
]
```

Changes since R3

MedicationStatement

MedicationStatement.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

MedicationStatement.status | Change value set from http://hl7.org/fhir/ValueSet/medication-statement-status to http://hl7.org/fhir/ValueSet/medication-statement-status|4.0.1

MedicationStatement.statusReason | Added Element

MedicationStatement.informationSource | Type Reference: Added Target Type PractitionerRole

MedicationStatement.reasonReference | Type Reference: Added Target Type DiagnosticReport

MedicationStatement.taken | deleted

MedicationStatement.reasonNotTaken | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 7 tests that all execute ok. 3 fail round-trip testing and 7 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationStatement | TU | | DomainResource | Record of medication being taken by a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External identifier
basedOn | Σ | 0..* | Reference(MedicationRequest | CarePlan | ServiceRequest) | Fulfils plan, proposal or order
partOf | Σ | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation) | Part of referenced event
status | ?!Σ | 1..1 | code | active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-taken Medication status codes (Required)
statusReason | | 0..* | CodeableConcept | Reason for current status SNOMED CT Drug Therapy Status codes (Example)
category | Σ | 0..1 | CodeableConcept | Type of medication usage Medication usage category codes (Preferred)
medication[x] | Σ | 1..1 | | What medication was taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(Medication)
subject | Σ | 1..1 | Reference(Patient | Group) | Who is/was taking the medication
context | Σ | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with MedicationStatement
effective[x] | Σ | 0..1 | | The date/time or interval when the medication is/was/will be taken
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
dateAsserted | Σ | 0..1 | dateTime | When the statement was asserted?
informationSource | | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization) | Person or organization that provided the information about the taking of this medication
derivedFrom | | 0..* | Reference(Any) | Additional supporting information
reasonCode | | 0..* | CodeableConcept | Reason for why the medication is being/was taken Condition/Problem/Diagnosis Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Condition or observation that supports why the medication is being/was taken
note | | 0..* | Annotation | Further information about the statement
dosage | | 0..* | Dosage | Details of how medication is/was taken or should be taken
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifier --></identifier>
 <basedOn><!-- 0..* Reference(MedicationRequest|CarePlan|ServiceRequest) Fulfils plan, proposal or order --></basedOn>
 <partOf><!-- 0..* Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-
    taken -->
 <statusReason><!-- 0..* CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..1 CodeableConcept Type of medication usage --></category>
 <medication[x]><!-- 1..1 CodeableConcept|Reference(Medication) What medication was taken --></medication[x]>
 <subject><!-- 1..1 Reference(Patient|Group) Who is/was taking  the medication --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with MedicationStatement
    --></context>
 <effective[x]><!-- 0..1 dateTime|Period The date/time or interval when the medication is/was/will be taken
    --></effective[x]>
 <dateAsserted value="[dateTime]"/><!-- 0..1 When the statement was asserted? -->
 <informationSource><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) Person or organization that provided the information about the taking of this medication
    --></informationSource>
 <derivedFrom><!-- 0..* Reference(Any) Additional supporting information --></derivedFrom>
 <reasonCode><!-- 0..* CodeableConcept Reason for why the medication is being/was taken --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Condition or observation that supports why
     the medication is being/was taken --></reasonReference>
 <note><!-- 0..* Annotation Further information about the statement --></note>
 <dosage><!-- 0..* Dosage Details of how medication is/was taken or should be taken --></dosage>
</MedicationStatement>
```

JSON Template

```
{
  "resourceType" : "MedicationStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifier
  "basedOn" : [{ Reference(MedicationRequest|CarePlan|ServiceRequest) }], // Fulfils plan, proposal or order
  "partOf" : [{ Reference(MedicationAdministration|MedicationDispense|
   MedicationStatement|Procedure|Observation) }], // Part of referenced event
  "status" : "<code>", // R!  active | completed | entered-in-error | intended | stopped | on-hold | unknown | not-taken
  "statusReason" : [{ CodeableConcept }], // Reason for current status
  "category" : { CodeableConcept }, // Type of medication usage
  // medication[x]: What medication was taken. One of these 2:
  "medicationCodeableConcept" : { CodeableConcept },
  "medicationReference" : { Reference(Medication) },
```

```
  "subject" : { Reference(Patient|Group) }, // R!  Who is/was taking  the medication
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with MedicationStatement
  // effective[x]: The date/time or interval when the medication is/was/will be taken. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "dateAsserted" : "<dateTime>", // When the statement was asserted?
  "informationSource" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson|Organization) }, // Person or organization that provided the information about the taking of this
    medication
  "derivedFrom" : [{ Reference(Any) }], // Additional supporting information
  "reasonCode" : [{ CodeableConcept }], // Reason for why the medication is being/was taken
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Condition or observation that supports
     why the medication is being/was taken
  "note" : [{ Annotation }], // Further information about the statement
  "dosage" : [{ Dosage }] // Details of how medication is/was taken or should be taken
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationStatement.identifier [ Identifier ], ... ; # 0..* External identifier
  fhir:MedicationStatement.basedOn [ Reference(MedicationRequest|CarePlan|ServiceRequest) ], ... ; # 0..* Fulfils plan,
    proposal or order
  fhir:MedicationStatement.partOf [ Reference(MedicationAdministration|MedicationDispense|MedicationStatement|Procedure|
  Observation) ], ... ; # 0..* Part of referenced event
  fhir:MedicationStatement.status [ code ]; # 1..1 active | completed | entered-in-error | intended | stopped | on-hold
    | unknown | not-taken
  fhir:MedicationStatement.statusReason [ CodeableConcept ], ... ; # 0..* Reason for current status
  fhir:MedicationStatement.category [ CodeableConcept ]; # 0..1 Type of medication usage
  # MedicationStatement.medication[x] : 1..1 What medication was taken. One of these 2
    fhir:MedicationStatement.medicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationStatement.medicationReference [ Reference(Medication) ]
  fhir:MedicationStatement.subject [ Reference(Patient|Group) ]; # 1..1 Who is/was taking  the medication
  fhir:MedicationStatement.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with
    MedicationStatement
  # MedicationStatement.effective[x] : 0..1 The date/time or interval when the medication is/was/will be taken. One of
    these 2
    fhir:MedicationStatement.effectiveDateTime [ dateTime ]
    fhir:MedicationStatement.effectivePeriod [ Period ]
  fhir:MedicationStatement.dateAsserted [ dateTime ]; # 0..1 When the statement was asserted?
  fhir:MedicationStatement.informationSource [
    Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Organization) ]; # 0..1 Person or organization that
    provided the information about the taking of this medication
  fhir:MedicationStatement.derivedFrom [ Reference(Any) ], ... ; # 0..* Additional supporting information
  fhir:MedicationStatement.reasonCode [ CodeableConcept ], ... ; # 0..* Reason for why the medication is being/was taken
  fhir:MedicationStatement.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..* Condition
     or observation that supports why the medication is being/was taken
  fhir:MedicationStatement.note [ Annotation ], ... ; # 0..* Further information about the statement
  fhir:MedicationStatement.dosage [ Dosage ], ... ; # 0..* Details of how medication is/was taken or should be taken
]
```

Changes since Release 3

MedicationStatement

MedicationStatement.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest

MedicationStatement.status | Change value set from http://hl7.org/fhir/ValueSet/medication-statement-status to http://hl7.org/fhir/ValueSet/medication-statement-status|4.0.1

MedicationStatement.statusReason | Added Element

MedicationStatement.informationSource | Type Reference: Added Target Type PractitionerRole

MedicationStatement.reasonReference | Type Reference: Added Target Type DiagnosticReport

MedicationStatement.taken | deleted

MedicationStatement.reasonNotTaken | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 7 tests that all execute ok. 3 fail round-trip testing and 7 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.4.3.1 Terminology Bindings

Path | Definition | Type | Reference

MedicationStatement.status | A coded concept indicating the current status of a MedicationStatement. | Required | Medication Status Codes

MedicationStatement.statusReason | A coded concept indicating the reason for the status of the statement. | Example | SNOMEDCTDrugTherapyStatusCodes

MedicationStatement.category | A coded concept identifying where the medication included in the MedicationStatement is expected to be consumed or administered. | Preferred | Medication usage category codes
MedicationStatement.medication[x] | A coded concept identifying the substance or product being taken. | Example | SNOMEDCTMedicationCodes
MedicationStatement.reasonCode | A coded concept identifying why the medication is being taken. | Example | Condition/Problem/DiagnosisCodes

11.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | Returns statements of this category of medicationstatement | MedicationStatement.category
code | token | Return statements of this medication code | (MedicationStatement.medication as CodeableConcept) | 13 Resources
context | reference | Returns statements for a specific context (episode or episode of Care). | MedicationStatement.context (EpisodeOfCare, Encounter)
effective | date | Date when patient was taking (or not taking) the medication | MedicationStatement.effective
identifier | token | Return statements with this external identifier | MedicationStatement.identifier | 30 Resources
medication | reference | Return statements of this medication reference | (MedicationStatement.medication as Reference) (Medication) | 3 Resources
part-of | reference | Returns statements that are part of another event. | MedicationStatement.partOf (MedicationDispense, Observation, MedicationAdministration, Procedure, MedicationStatement)
patient | reference | Returns statements for a specific patient. | MedicationStatement.subject.where(resolve() is Patient) (Patient) | 33 Resources
source | reference | Who or where the information in the statement came from | MedicationStatement.informationSource (Practitioner, Organization, Patient, PractitionerRole, RelatedPerson)
status | token | Return statements that match the given status | MedicationStatement.status | 3 Resources
subject | reference | The identity of a patient, animal or group to list statements for | MedicationStatement.subject (Group, Patient)

11.4.5 Resource MedicationStatement - Examples

URL: https://hl7.org/fhir/R4/medicationstatement-examples.html

11.4.5 Resource MedicationStatement - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

The following are examples of Medication Statement resources:

- Patient reports they are taking Tylenol PM and has the bottle
- Patient reports they are not taking Tylenol PM
- Patient reports they took Amoxycillin in the past but are not taking it now
- Patient reports they previous said they took Amoxycillin but really didn’t
- Care Giver reports that Patient took Amoxycillin
- Patient reports that they intend to use Mometasone in the future
- Patient reports they are taking the “little pink pill for water retention

Example Name | id | Format
Patient reports they are taking Tylenol PM and has the bottle | example001 | XML | JSON | Turtle
Patient reports they are not taking Tylenol PM | example002 | XML | JSON | Turtle
Patient reports they were taking Amoxycillin in the past but are not taking it now | example004 | XML | JSON | Turtle
Patient reports they previous said they took Amoxycillin but really didn't | example005 | XML | JSON | Turtle
Care Giver reports that the Patient took Amoxicillin | example006 | XML | JSON | Turtle
Patient reports they intend to use Mometasone in the future | example007 | XML | JSON | Turtle
Patient reports that they are taking the "little pink pill" | example003 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.4.6 Resource MedicationStatement - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicationstatement-definitions.html

11.4.6 Resource MedicationStatement - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the MedicationStatement resource.

MedicationStatement
Element Id | MedicationStatement
Definition | A record of a medication that is being consumed by a patient. A MedicationStatement may indicate that the patient may be taking the medication now or has taken the medication in the past or will be taking the medication in the future. The

source of this information can be the patient, significant other (such as a family member or spouse), or a clinician. A common scenario where this information is captured is during the history taking process during a patient visit or stay. The medication information may come from sources such as the patient's memory, from a prescription bottle, or from a list of medications the patient, clinician or other party maintains. The primary difference between a medication statement and a medication administration is that the medication administration has complete administration information and is based on actual administration information from the person who administered the medication. A medication statement is often, if not always, less specific. There is no required date/time when the medication was administered, in fact we only know that a source has reported the patient is taking this medication, where details such as time, quantity, or rate or even medication product may be incomplete or missing or less precise. As stated earlier, the medication statement information may come from the patient's memory, from a prescription bottle or from a list of medications the patient, clinician or other party maintains. Medication administration is more formal and is not missing detailed information.
Cardinality | 0..*
Type | DomainResource
Comments | When interpreting a medicationStatement, the value of the status and NotTaken needed to be considered: MedicationStatement.status + MedicationStatement.wasNotTaken Status=Active + NotTaken=T = Not currently taking Status=Completed + NotTaken=T = Not taken in the past Status=Intended + NotTaken=T = No intention of taking Status=Active + NotTaken=F = Taking, but not as prescribed Status=Active + NotTaken=F = Taking Status=Intended +NotTaken= F = Will be taking (not started) Status=Completed + NotTaken=F = Taken in past Status=In Error + NotTaken=N/A = In Error.
MedicationStatement.identifier
Element Id | MedicationStatement.identifier
Definition | Identifiers associated with this Medication Statement that are defined by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business identifiers assigned to this resource by the performer or other systems and remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | This is a business identifier, not a resource identifier.
MedicationStatement.basedOn
Element Id | MedicationStatement.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this event.
Cardinality | 0..*
Type | Reference(MedicationRequest | CarePlan | ServiceRequest)
Requirements | Allows tracing of authorization for the event and tracking whether proposals/recommendations were acted upon.
Summary | true
MedicationStatement.partOf
Element Id | MedicationStatement.partOf
Definition | A larger event of which this particular event is a component or step.
Cardinality | 0..*
Type | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation)
Requirements | This should not be used when indicating which resource a MedicationStatement has been derived from. If that is the use case, then MedicationStatement.derivedFrom should be used.
Summary | true
MedicationStatement.status
Element Id | MedicationStatement.status
Definition | A code representing the patient or other source's judgment about the state of the medication used that this statement is about. Generally, this will be active or completed.
Cardinality | 1..1
Terminology Binding | Medication status codes (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | MedicationStatement is a statement at a point in time. The status is only representative at the point when it was asserted. The value set for MedicationStatement.status contains codes that assert the status of the use of the medication by the patient (for example, stopped or on hold) as well as codes that assert the status of the medication statement itself (for example, entered in error). This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MedicationStatement.statusReason
Element Id | MedicationStatement.statusReason
Definition | Captures the reason for the current state of the MedicationStatement.
Cardinality | 0..*
Terminology Binding | SNOMED CT Drug Therapy Status codes (Example)
Type | CodeableConcept
Comments | This is generally only used for "exception" statuses such as "not-taken", "on-hold", "cancelled" or "entered-in-error". The reason for performing the event at all is captured in reasonCode, not here.
MedicationStatement.category
Element Id | MedicationStatement.category
Definition | Indicates where the medication is expected to be consumed or administered.
Cardinality | 0..1
Terminology Binding | Medication usage category codes (Preferred)
Type | CodeableConcept
Summary | true
MedicationStatement.medication[x]

Element Id | MedicationStatement.medication[x]
Definition | Identifies the medication being administered. This is either a link to a resource representing the details of the medication or a simple attribute carrying a code that identifies the medication from a known list of medications.
Cardinality | 1..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | If only a code is specified, then it needs to be a code for a specific product. If more information is required, then the use of the medication resource is recommended. For example, if you require form or lot number, then you must reference the Medication resource.
MedicationStatement.subject
Element Id | MedicationStatement.subject
Definition | The person, animal or group who is/was taking the medication.
Cardinality | 1..1
Type | Reference(Patient | Group)
Summary | true
MedicationStatement.context
Element Id | MedicationStatement.context
Definition | The encounter or episode of care that establishes the context for this MedicationStatement.
Cardinality | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Summary | true
MedicationStatement.effective[x]
Element Id | MedicationStatement.effective[x]
Definition | The interval of time during which it is being asserted that the patient is/was/will be taking the medication (or was not taking, when the MedicationStatement.taken element is No).
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This attribute reflects the period over which the patient consumed the medication and is expected to be populated on the majority of Medication Statements. If the medication is still being taken at the time the statement is recorded, the "end" date will be omitted. The date/time attribute supports a variety of dates - year, year/month and exact date. If something more than this is required, this should be conveyed as text.
MedicationStatement.dateAsserted
Element Id | MedicationStatement.dateAsserted
Definition | The date when the medication statement was asserted by the information source.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicationStatement.informationSource
Element Id | MedicationStatement.informationSource
Definition | The person or organization that provided the information about the taking of this medication. Note: Use derivedFrom when a MedicationStatement is derived from other resources, e.g. Claim or MedicationRequest.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)
MedicationStatement.derivedFrom
Element Id | MedicationStatement.derivedFrom
Definition | Allows linking the MedicationStatement to the underlying MedicationRequest, or to other information that supports or is used to derive the MedicationStatement.
Cardinality | 0..*
Type | Reference(Any)
Comments | Likely references would be to MedicationRequest, MedicationDispense, Claim, Observation or QuestionnaireAnswers. The most common use cases for deriving a MedicationStatement comes from creating a MedicationStatement from a MedicationRequest or from a lab observation or a claim. it should be noted that the amount of information that is available varies from the type resource that you derive the MedicationStatement from.
MedicationStatement.reasonCode
Element Id | MedicationStatement.reasonCode
Definition | A reason for why the medication is being/was taken.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonForUseReference.
MedicationStatement.reasonReference
Element Id | MedicationStatement.reasonReference
Definition | Condition or observation that supports why the medication is being/was taken.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport)
Comments | This is a reference to a condition that is the reason why the medication is being/was taken. If only a code exists, use reasonForUseCode.
MedicationStatement.note

Element Id | MedicationStatement.note
Definition | Provides extra information about the medication statement that is not conveyed by the other attributes.
Cardinality | 0..*
Type | Annotation
MedicationStatement.dosage
Element Id | MedicationStatement.dosage
Definition | Indicates how the medication is/was or should be taken by the patient.
Cardinality | 0..*
Type | Dosage
Comments | The dates included in the dosage on a Medication Statement reflect the dates for a given dose. For example, "from November 1, 2016 to November 3, 2016, take one tablet daily and from November 4, 2016 to November 7, 2016, take two tablets daily." It is expected that this specificity may only be populated where the patient brings in their labeled container or where the Medication Statement is derived from a MedicationRequest.

11.4.7 Resource MedicationStatement - Mappings

URL: https://hl7.org/fhir/R4/medicationstatement-mappings.html

11.4.7 Resource MedicationStatement - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the medicationstatement resource (see Mappings to Other Standards for further information & status).

11.4.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MedicationStatement
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
medication[x] | FiveWs.what[x]
subject | FiveWs.subject[x]
effective[x] | FiveWs.done[x]
dateAsserted | FiveWs.recorded
informationSource | FiveWs.source
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

11.4.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

MedicationStatement | Event
identifier | Event.identifier
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
statusReason | Event.statusReason
medication[x] | Event.code
subject | Event.subject
context | Event.context
effective[x] | Event.occurrence[x]
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

11.4.7.3 HL7 v2 Mapping (http://hl7.org/v2)

MedicationStatement
identifier
basedOn
partOf
status
statusReason
category
medication[x]
subject | PID-3-Patient ID List
context
effective[x]

dateAsserted
informationSource
derivedFrom
reasonCode
reasonReference
note
dosage

11.4.7.4 RIM Mapping (http://hl7.org/v3)

MedicationStatement | SubstanceAdministration
identifier | .id
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=SBADM or PROC or PCPR or OBS, moodCode=RQO orPLAN or PRP]
partOf | .outboundRelationship[typeCode=COMP]/target[classCode=SPLY or SBADM or PROC or OBS,moodCode=EVN]
status | .statusCode
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN].reasonCOde
category | .inboundRelationship[typeCode=COMP].source[classCode=OBS, moodCode=EVN, code="type of medication usage"].value
medication[x] | .participation[typeCode=CSM].role[classCode=ADMM or MANU]
subject | .participation[typeCode=SBJ].role[classCode=PAT]
context | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN, code="type of encounter or episode"]
effective[x] | .effectiveTime
dateAsserted | .participation[typeCode=AUT].time
informationSource | .participation[typeCode=INF].role[classCode=PAT, or codes for Practioner or Related Person (if PAT is the informer, then syntax for self-reported =true)
derivedFrom | .outboundRelationship[typeCode=SPRT]/target[classCode=ACT,moodCode=EVN]
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON]/target[classCode=OBS,moodCode=EVN, code="reason for use"].value
note | .inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="annotation"].value
dosage | refer dosageInstruction mapping

11.4.8 Resource MedicationStatement - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicationstatement-profiles.html

11.4.8 Resource MedicationStatement - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the MedicationStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.4.8.1 Profiles

No Profiles defined for this resource

11.4.8.2 Extensions

workflow-researchStudy | MedicationStatement | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.4.8.3 Search Extensions

No Search Extensions defined for this resource

11.4.9 Resource MedicationStatement - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/medicationstatement-version-maps.html

No validation errors - all conversions are clean

11.4.9 Resource MedicationStatement - R3/R4 Conversions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for MedicationStatement.

Functional status for this map: 7 tests that all execute ok. 3 fail round-trip testing and 7 r3 resources are invalid (0 errors). (see documentation)

11.4.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MedicationStatement3to4" = "R3 to R4 Conversions for MedicationStatement"

uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationStatement" alias MedicationStatementR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MedicationStatement" alias MedicationStatement as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MedicationStatement(source src : MedicationStatementR3, target tgt : MedicationStatement) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.context -> tgt.context;
  src.status as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationStatement.status',  vt.value = v;
  src.taken as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationStatement.taken',  vt.value = v;
  src.status where (src.taken in ('n' | 'unk')).not() -> tgt.status;
  src.taken where value = 'n' -> tgt.status = 'not-taken';
  src.taken where value = 'unk' -> tgt.status = 'unknown';
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.dateAsserted -> tgt.dateAsserted;
  src.informationSource -> tgt.informationSource;
  src.subject -> tgt.subject;
  src.derivedFrom -> tgt.derivedFrom;
  src.reasonNotTaken -> tgt.reasonCode;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.dosage -> tgt.dosage;
}
```

11.4.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MedicationStatement4to3" = "R4 to R3 Conversion for MedicationStatement"

uses "http://hl7.org/fhir/StructureDefinition/MedicationStatement" alias MedicationStatement as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MedicationStatement" alias MedicationStatementR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MedicationStatement(source src : MedicationStatement, target tgt : MedicationStatementR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.context -> tgt.context;
  src where src.extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationStatement.status').exists() then statusR3(src, tgt) "status1";
  src where src.extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationStatement.status').empty() then statusNR3(src, tgt) "status2";
  src.category -> tgt.category;
  src.medication : CodeableConcept as vs -> tgt.medication = create('CodeableConcept') as vt then CodeableConcept(vs,
    vt);
  src.medication : Reference as vs -> tgt.medication = create('Reference') as vt then Reference(vs, vt);
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.dateAsserted -> tgt.dateAsserted;
  src.informationSource -> tgt.informationSource;
  src.subject -> tgt.subject;
  src.derivedFrom -> tgt.derivedFrom;
  src.reasonCode where src.status = 'not-taken' -> tgt.reasonNotTaken;
  src.reasonCode where src.status != 'not-taken' -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.dosage -> tgt.dosage;
}

group statusR3(source src, target tgt) {
```

```
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationStatement.taken' then
    {
    e.value -> tgt.taken;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationStatement.status' then
     {
    e.value -> tgt.status;
  };
}

group statusNR3(source src, target tgt) {
  src.status where (value in ('not-taken' | 'unknown')).not() -> tgt.status;
  src.status where value = 'not-taken' ->  tgt.status = 'completed',  tgt.taken = 'n';
  src.status where value = 'unknown' ->  tgt.status = 'completed',  tgt.taken = 'unk';
}
```

11.4.9.3 R4 Validation Errors

11.4.10 MedicationStatement HL7 Extensions

URL: https://hl7.org/fhir/R4/medicationstatement-extensions.html

Defines common extensions used with or related to the MedicationStatement resource

11.4.10 MedicationStatement HL7 Extensions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

11.4.10.1 Content

11.4.11 JSON Schema for MedicationStatement

URL: https://hl7.org/fhir/R4/medicationstatement.schema.json.html

11.4.11 JSON Schema for MedicationStatement

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is a representation of the json schema for MedicationStatement, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicationStatement",
  "$ref": "#/definitions/MedicationStatement",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicationStatement": {
      "description": "A record of a medication that is being consumed by a patient.   A MedicationStatement may indicate
     that the patient may be taking the medication now or has taken the medication in the past or will be taking the
    medication in the future.  The source of this information can be the patient, significant other (such as a family
    member or spouse), or a clinician.  A common scenario where this information is captured is during the history
    taking process during a patient visit or stay.   The medication information may come from sources such as the
    patient\u0027s memory, from a prescription bottle,  or from a list of medications the patient, clinician or other
    party maintains. \n\nThe primary difference between a medication statement and a medication administration is that
    the medication administration has complete administration information and is based on actual administration
    information from the person who administered the medication.  A medication statement is often, if not always, less
    specific.  There is no required date/time when the medication was administered, in fact we only know that a source
    has reported the patient is taking this medication, where details such as time, quantity, or rate or even medication
     product may be incomplete or missing or less precise.  As stated earlier, the medication statement information may
    come from the patient\u0027s memory, from a prescription bottle or from a list of medications the patient, clinician
     or other party maintains.  Medication administration is more formal and is not missing detailed information.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicationStatement resource",
          "const": "MedicationStatement"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
```

```
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifiers associated with this Medication Statement that are defined by business processes
and/or used to refer to it when a direct URL reference to the resource itself is not appropriate. They are business
identifiers assigned to this resource by the performer or other systems and remain constant as the resource is
updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A plan, proposal or order that is fulfilled in whole or in part by this event.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular event is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "A code representing the patient or other source\u0027s judgment about the state of the
medication used that this statement is about.  Generally, this will be active or completed.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current state of the MedicationStatement.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "category": {
      "description": "Indicates where the medication is expected to be consumed or administered.",
```

```
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationCodeableConcept": {
      "description": "Identifies the medication being administered. This is either a link to a resource representing
 the details of the medication or a simple attribute carrying a code that identifies the medication from a known
list of medications.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "medicationReference": {
      "description": "Identifies the medication being administered. This is either a link to a resource representing
 the details of the medication or a simple attribute carrying a code that identifies the medication from a known
list of medications.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "subject": {
      "description": "The person, animal or group who is/was taking the medication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "context": {
      "description": "The encounter or episode of care that establishes the context for this MedicationStatement.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "effectiveDateTime": {
      "description": "The interval of time during which it is being asserted that the patient is/was/will be taking
the medication (or was not taking, when the MedicationStatement.taken element is No).",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_effectiveDateTime": {
      "description": "Extensions for effectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The interval of time during which it is being asserted that the patient is/was/will be taking
the medication (or was not taking, when the MedicationStatement.taken element is No).",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "dateAsserted": {
      "description": "The date when the medication statement was asserted by the information source.",
      "$ref": "#/definitions/dateTime"
    },
    "_dateAsserted": {
      "description": "Extensions for dateAsserted",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "informationSource": {
      "description": "The person or organization that provided the information about the taking of this medication.
Note: Use derivedFrom when a MedicationStatement is derived from other resources, e.g. Claim or MedicationRequest.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "derivedFrom": {
      "description": "Allows linking the MedicationStatement to the underlying MedicationRequest, or to other
information that supports or is used to derive the MedicationStatement.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "reasonCode": {
      "description": "A reason for why the medication is being/was taken.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Condition or observation that supports why the medication is being/was taken.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Provides extra information about the medication statement that is not conveyed by the other
attributes.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "dosage": {
      "description": "Indicates how the medication is/was or should be taken by the patient.",
      "items": {
        "$ref": "Dosage.schema.json#/definitions/Dosage"
      },
      "type": "array"
    }
  },
```

```
      "additionalProperties": false,
      "required": [
        "subject",
        "resourceType"
      ]
    }
  }
}
```

11.4.12 Resource MedicationStatement - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicationstatement-dependencies.html

11.4.12 Resource MedicationStatement - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Dependency Graph for MedicationStatement FMM level 3

Name | Card. | Type | Dependency Analysis
MedicationStatement | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..* | Reference(MedicationRequest | CarePlan | ServiceRequest) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use)
partOf | 0..* | Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Observation) | Type Warning: (MedicationAdministration = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (MedicationDispense = FMM2-Trial Use vs. Element = FMM3-Trial Use)
status | 1..1 | code
statusReason | 0..* | CodeableConcept
category | 0..1 | CodeableConcept
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(Medication
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use)
context | 0..1 | Reference(Encounter | EpisodeOfCare) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (EpisodeOfCare = FMM2-Trial Use vs. Element = FMM3-Trial Use)
effective[x] | 0..1
effectiveDateTime | | dateTime
effectivePeriod | | Period
dateAsserted | 0..1 | dateTime
informationSource | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
derivedFrom | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport)
note | 0..* | Annotation
dosage | 0..* | Dosage

11.4.13 Value Set Sc-valueset-medication-statement-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medication-statement-status.html

11.4.13 Value Set Sc-valueset-medication-statement-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.4.14 Medication Status CodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medication-statement-status)

Mapping from http://hl7.org/fhir/ValueSet/medication-statement-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "Medication Status Codes"

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
intended | equivalent | planned
active | equivalent | active
on-hold | equivalent | suspended

stopped | equivalent | failed
completed | equivalent | complete
unknown | equivalent | unknown
not-taken | equivalent | not-done

11.5 Resource Medication - Content

URL: https://hl7.org/fhir/R4/medication.html
Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

This resource is primarily used for the identification and definition of a medication for the purposes of prescribing, dispensing, and administering a medication as well as for making statements about medication use.

11.5.1 Scope and Usage

Representing medications in the majority of healthcare settings is a matter of identifying an item from a list and then conveying a reference for the item selected either into a patient-related resource or to other applications. Additional information about the medication is frequently provided for human verification, but a full representation of the details of composition and efficacy of the medicine is conveyed by referring to drug dictionaries by means of the codes they define. There are some occasions where it is necessary to identify slightly more detail, such as when dispensing a package containing a particular medication requires identification both of the medicine and the package at once. There are also some occasions (e.g. custom formulations) where the composition of a medicine must be represented. In these cases, the ingredients of the medicine have to be specified together with the amount contained, though the Medication resource does not provide full details.

The Medication resource allows for medications to be characterized by the form of the drug and the ingredient (or ingredients), as well as how it is packaged. The medication will include the ingredient(s) and their strength(s) and the package can include the amount (for example, number of tablets, volume, etc.) that is contained in a particular container (for example, 100 capsules of Amoxicillin 500mg per bottle).

The Medication resource can be used to describe a compounded (aka extemporaneous or magistral) product that is manufactured by the pharmacy at the time of dispensing. In this case there will be multiple ingredients which are typically base chemicals (for example, hydrocortisone powder) and there may be other ingredients that are manufactured products (for example, Glaxal Base).

When a medication includes a package, further details about the composition can be provided. A package has a container (vacuum packed box, jar, etc.) and a list of the products or other packages that are in the package.

This resource is referenced by ActivityDefinition, AdverseEvent, CarePlan, CatalogEntry, ChargeItem, ChargeItemDefinition, Flag, Group, itself, MedicationAdministration, MedicationDispense, MedicationKnowledge, MedicationRequest, MedicationStatement, MedicinalProductContraindication, MedicinalProductIndication, MedicinalProductInteraction, MedicinalProductUndesirableEffect, Procedure, SupplyDelivery and SupplyRequest

11.5.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Medication | TU | | DomainResource | Definition of a Medication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for this medication
code | Σ | 0..1 | CodeableConcept | Codes that identify this medication SNOMED CT Medication Codes (Example)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error Medication status codes (Required)
manufacturer | Σ | 0..1 | Reference(Organization) | Manufacturer of the item
form | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
amount | Σ | 0..1 | Ratio | Amount of drug in package
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | The actual ingredient or content
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance | Medication)

isActive | | 0..1 | boolean | Active ingredient indicator
strength | | 0..1 | Ratio | Quantity of ingredient present
batch | | 0..1 | BackboneElement | Details about packaged medications
lotNumber | | 0..1 | string | Identifier assigned to batch
expirationDate | | 0..1 | dateTime | When batch will expire
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Medication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for this medication --></identifier>
 <code><!-- 0..1 CodeableConcept Codes that identify this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <form><!-- 0..1 CodeableConcept powder | tablets | capsule + --></form>
 <amount><!-- 0..1 Ratio Amount of drug in package --></amount>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance|Medication) The actual ingredient or content --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <strength><!-- 0..1 Ratio Quantity of ingredient present --></strength>
 </ingredient>
 <batch>  <!-- 0..1 Details about packaged medications -->
  <lotNumber value="[string]"/><!-- 0..1 Identifier assigned to batch -->
  <expirationDate value="[dateTime]"/><!-- 0..1 When batch will expire -->
 </batch>
</Medication>
```

JSON Template

```
{
  "resourceType" : "Medication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for this medication
  "code" : { CodeableConcept }, // Codes that identify this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "form" : { CodeableConcept }, // powder | tablets | capsule +
  "amount" : { Ratio }, // Amount of drug in package
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: The actual ingredient or content. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance|Medication) },
    "isActive" : <boolean>, // Active ingredient indicator
    "strength" : { Ratio } // Quantity of ingredient present
  }],
  "batch" : { // Details about packaged medications
    "lotNumber" : "<string>", // Identifier assigned to batch
    "expirationDate" : "<dateTime>" // When batch will expire
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Medication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Medication.identifier [ Identifier ], ... ; # 0..* Business identifier for this medication
  fhir:Medication.code [ CodeableConcept ]; # 0..1 Codes that identify this medication
  fhir:Medication.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Medication.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:Medication.form [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:Medication.amount [ Ratio ]; # 0..1 Amount of drug in package
  fhir:Medication.ingredient [ # 0..* Active or inactive ingredient
    # Medication.ingredient.item[x] : 1..1 The actual ingredient or content. One of these 2
      fhir:Medication.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:Medication.ingredient.itemReference [ Reference(Substance|Medication) ]
    fhir:Medication.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:Medication.ingredient.strength [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:Medication.batch [ # 0..1 Details about packaged medications
    fhir:Medication.batch.lotNumber [ string ]; # 0..1 Identifier assigned to batch
    fhir:Medication.batch.expirationDate [ dateTime ]; # 0..1 When batch will expire
  ];
]
```

Changes since R3

Medication
Medication.identifier | Added Element
Medication.status | Change value set from http://hl7.org/fhir/ValueSet/medication-status to http://hl7.org/fhir/ValueSet/medication-status|4.0.1 Now marked as Modifier

Medication.amount | Added Element
Medication.ingredient.strength | Added Element
Medication.batch | Added Element
Medication.batch.lotNumber | Added Element
Medication.batch.expirationDate | Added Element
Medication.isBrand | deleted
Medication.isOverTheCounter | deleted
Medication.ingredient.amount | deleted
Medication.package | deleted
Medication.image | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 23 tests that all execute ok. 22 fail round-trip testing and 22 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Medication | TU | | DomainResource | Definition of a Medication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for this medication
code | Σ | 0..1 | CodeableConcept | Codes that identify this medication SNOMED CT Medication Codes (Example)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error Medication status codes (Required)
manufacturer | Σ | 0..1 | Reference(Organization) | Manufacturer of the item
form | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
amount | Σ | 0..1 | Ratio | Amount of drug in package
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | The actual ingredient or content
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance | Medication)
isActive | | 0..1 | boolean | Active ingredient indicator
strength | | 0..1 | Ratio | Quantity of ingredient present
batch | | 0..1 | BackboneElement | Details about packaged medications
lotNumber | | 0..1 | string | Identifier assigned to batch
expirationDate | | 0..1 | dateTime | When batch will expire
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Medication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for this medication --></identifier>
 <code><!-- 0..1 CodeableConcept Codes that identify this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <form><!-- 0..1 CodeableConcept powder | tablets | capsule + --></form>
 <amount><!-- 0..1 Ratio Amount of drug in package --></amount>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance|Medication) The actual ingredient or content --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <strength><!-- 0..1 Ratio Quantity of ingredient present --></strength>
 </ingredient>
 <batch>  <!-- 0..1 Details about packaged medications -->
  <lotNumber value="[string]"/><!-- 0..1 Identifier assigned to batch -->
  <expirationDate value="[dateTime]"/><!-- 0..1 When batch will expire -->
 </batch>
</Medication>
```

JSON Template

```
{
  "resourceType" : "Medication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for this medication
  "code" : { CodeableConcept }, // Codes that identify this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "form" : { CodeableConcept }, // powder | tablets | capsule +
  "amount" : { Ratio }, // Amount of drug in package
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: The actual ingredient or content. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance|Medication) },
    "isActive" : <boolean>, // Active ingredient indicator
    "strength" : { Ratio } // Quantity of ingredient present
  }],
  "batch" : { // Details about packaged medications
    "lotNumber" : "<string>", // Identifier assigned to batch
    "expirationDate" : "<dateTime>" // When batch will expire
```

```
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Medication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Medication.identifier [ Identifier ], ... ; # 0..* Business identifier for this medication
  fhir:Medication.code [ CodeableConcept ]; # 0..1 Codes that identify this medication
  fhir:Medication.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Medication.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:Medication.form [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:Medication.amount [ Ratio ]; # 0..1 Amount of drug in package
  fhir:Medication.ingredient [ # 0..* Active or inactive ingredient
    # Medication.ingredient.item[x] : 1..1 The actual ingredient or content. One of these 2
      fhir:Medication.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:Medication.ingredient.itemReference [ Reference(Substance|Medication) ]
    fhir:Medication.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:Medication.ingredient.strength [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:Medication.batch [ # 0..1 Details about packaged medications
    fhir:Medication.batch.lotNumber [ string ]; # 0..1 Identifier assigned to batch
    fhir:Medication.batch.expirationDate [ dateTime ]; # 0..1 When batch will expire
  ];
]
```

Changes since Release 3

Medication
Medication.identifier | Added Element
Medication.status | Change value set from http://hl7.org/fhir/ValueSet/medication-status to http://hl7.org/fhir/ValueSet/medication-status|4.0.1 Now marked as Modifier
Medication.amount | Added Element
Medication.ingredient.strength | Added Element
Medication.batch | Added Element
Medication.batch.lotNumber | Added Element
Medication.batch.expirationDate | Added Element
Medication.isBrand | deleted
Medication.isOverTheCounter | deleted
Medication.ingredient.amount | deleted
Medication.package | deleted
Medication.image | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 23 tests that all execute ok. 22 fail round-trip testing and 22 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.5.2.1 Terminology Bindings

Path | Definition | Type | Reference
Medication.code | A coded concept that defines the type of a medication. | Example | SNOMEDCTMedicationCodes
Medication.status | A coded concept defining if the medication is in active use. | Required | Medication Status Codes
Medication.form | A coded concept defining the form of a medication. | Example | SNOMEDCTFormCodes

11.5.3 Best practices for using 'Medication'

Medication does not have a status. If Medication was used to support a formulary use case, then an extension can be used to convey formulary statuses, such as active (e.g. the medication can be ordered) or inactive (e.g. the medication can be documented, but not ordered). Pharmacy is evaluating formulary use cases. Feedback is encouraged to the Pharmacy working group.

11.5.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Returns medications for a specific code | Medication.code | 13 Resources
expiration-date | date | Returns medications in a batch with this expiration date | Medication.batch.expirationDate
form | token | Returns medications for a specific dose form | Medication.form
identifier | token | Returns medications with this external identifier | Medication.identifier

ingredient | reference | Returns medications for this ingredient reference | (Medication.ingredient.item as Reference) (Medication, Substance)
ingredient-code | token | Returns medications for this ingredient code | (Medication.ingredient.item as CodeableConcept)
lot-number | token | Returns medications in a batch with this lot number | Medication.batch.lotNumber
manufacturer | reference | Returns medications made or sold for this manufacturer | Medication.manufacturer (Organization)
status | token | Returns medications for this status | Medication.status

11.5.5 Resource Medication - Examples

URL: https://hl7.org/fhir/R4/medication-examples.html

11.5.5 Resource Medication - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

The following are examples of Medication resources where the medication is a manufactured product:

• IV injection (inpatient order) - Manufactured Product - Vancomycin

• IV Injection - Manufactured Product - Piperacillin

• IV Chemotherapy Example - Manufactured Product - Alemtuzumab

• Oral Chemotherapy Example - Manufactured Product - Myleran

• Eye Drop Example - Manufactured Product - Timoptic

• IV Chemotherapy Example - Manufactured Product - Adcetris

• Subcutaneous injection Example - Manufactured Product - Novolog

• As needed Oral tablet - Manufactured Product - Percocet

• Over the Counter Product - Manufactured Product - Tylenol PM

• Oral Chemotherapy Example - Manufactured Product - Xeloda - includes a reference to a substance in ingredient element

The following are examples of Medication resources where the medication is a generic formulation (i.e., represents an ingredient or set of ingredients plus dosage form)

• Generic Formulation - Oxycodone

• Generic Formulation - Prednisone 5mg tablet

• Generic Formulation - Nystatin 100,000u/ml

• Generic Formulation - Lorazepam 2mg/ml injection

• Generic Formulation - Alprazolam 0.25mg tablet

• Generic Formulation - Mometasone Furorate 0.05mg/Actuat

• Generic Formulation - Fully populated example of a Medication - Chlorthalidone

• Generic Formulation - Oral Chemotherapy Example - Capecitabine – includes a reference to a substance in the Ingredient element

• Generic Formulation - Azithromycin 250mg oral capsule - includes a reference to a substance in the Ingredient element

The following are examples of Medication resources where the medication is a compounded (aka extemporaneous or magistral) product

• Compounded Preparation - KCl in D5W Admixture - includes a reference to a substance in the ingredient element

• Compounded Preparation - TPN Solution - includes a reference to a substance in the ingredient element

• Compounded Preparation - HC 1%, Salicylic Acid 5% in Glaxal Base - includes a reference to a substance in the ingredient element

The following is an example where the medication resource is minimally populated:

• Name Element Only populated - Amoxicillin 250mg/5ml Suspension

Example Name | id | Format
Name element only populated - Amoxicillin 250mg/5ml Suspension | medicationexample1 | XML | JSON | Turtle
Oral Chemotherapy Example - Manufactured Product - Xeloda - includes a reference to a substance in ingredient element | medexample015 | XML | JSON | Turtle
IV injection (inpatient order) - Manufactured Product - Vancomycin | med0301 | XML | JSON | Turtle
IV Injection with a Rate Example - Manufactured Product - Piperacillin | med0302 | XML | JSON | Turtle
IV Chemotherapy Example - Manufactured Product - Alemtuzumab | med0303 | XML | JSON | Turtle
Oral Chemotherapy Example - Manufactured Product - Myleran | med0304 | XML | JSON | Turtle
Eye Drop Example - Manufactured Product - Timoptic | med0305 | XML | JSON | Turtle
IV Chemotherapy Example - Manufactured Product - Adcetris | med0306 | XML | JSON | Turtle
Subcutaneous injection Example - Manufactured Product - Novolog | med0307 | XML | JSON | Turtle
As needed Oral tablet - Manufactured Product - Percocet | med0308 | XML | JSON | Turtle

As needed dosage range - Manufactured Product - Tylenol No 1 Caplet | med0309 | XML | JSON | Turtle
Generic Formulation - Oxycodone | med0310 | XML | JSON | Turtle
Generic Formulation - Prednisone 5mg tablet | med0311 | XML | JSON | Turtle
Generic Formulation - Nystatin 100,000 u/ml suspension | med0312 | XML | JSON | Turtle
Generic Formulation - Lorazepam 2mg/ml injection | med0313 | XML | JSON | Turtle
Generic Formulation - Alprazolam 0.25mg tablet | med0314 | XML | JSON | Turtle
Generic Formulation - Mometasone Furoate 0.05mg/Actuat | med0315 | XML | JSON | Turtle
Generic Formulation - Fully populated example of a Medication - Chlorthalidone | med0316 | XML | JSON | Turtle
Compounded Preparation - KCl in D5W Admixture - includes a reference to a substance in the ingredient element | med0317 | XML | JSON | Turtle
Compounded Preparation - TPN Solution - includes a reference to a substance in the ingredient element | med0318 | XML | JSON | Turtle
Compounded Preparation - HC 1%, Salicylic Acid 5% in Glaxal Base - includes a reference to a substance in the ingredient element | med0319 | XML | JSON | Turtle
Generic Formulation - Azithromycin 250mg oral capsule - includes a reference to a substance in the Ingredient element | med0320 | XML | JSON | Turtle
Generic Formulation - Oral Chemotherapy Example - Capecitabine – includes a reference to a substance in the Ingredient element | med0321 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.5.6 Resource Medication - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medication-definitions.html

11.5.6 Resource Medication - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Medication resource.

Medication
Element Id | Medication
Definition | This resource is primarily used for the identification and definition of a medication for the purposes of prescribing, dispensing, and administering a medication as well as for making statements about medication use.
Cardinality | 0..*
Type | DomainResource
Medication.identifier
Element Id | Medication.identifier
Definition | Business identifier for this medication.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
Comments | The serial number could be included as an identifier.
Medication.code
Element Id | Medication.code
Definition | A code (or set of codes) that specify this medication, or a textual description if no code is available. Usage note: This could be a standard medication code such as a code from RxNorm, SNOMED CT, IDMP etc. It could also be a national or local formulary code, optionally with translations to other code systems.
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept
Summary | true
Comments | Depending on the context of use, the code that was actually selected by the user (prescriber, dispenser, etc.) will have the coding.userSelected set to true. As described in the coding datatype: "A coding may be marked as a "userSelected" if a user selected the particular coded value in a user interface (e.g. the user selects an item in a pick-list). If a user selected coding exists, it is the preferred choice for performing translations etc. Other codes can only be literal translations to alternative code systems, or codes at a lower level of granularity (e.g. a generic code for a vendor-specific primary one).
Medication.status
Element Id | Medication.status
Definition | A code to indicate if the medication is in active use.
Cardinality | 0..1
Terminology Binding | Medication status codes (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes.)
Summary | true
Comments | This status is intended to identify if the medication in a local system is in active use within a drug database or inventory. For example, a pharmacy system may create a new drug file record for a compounded product "ABC Hospital Special Cream" with an active status. At some point in the future, it may be determined that the drug record was created with an error and the status is changed to "entered in error". This status is not intended to specify if a medication is part of a particular formulary. It is possible that the drug record may be referenced by multiple formularies or catalogues and each of those entries would have a separate status.
Medication.manufacturer

Element Id | Medication.manufacturer
Definition | Describes the details of the manufacturer of the medication product. This is not intended to represent the distributor of a medication product.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
Medication.form
Element Id | Medication.form
Definition | Describes the form of the item. Powder; tablets; capsule.
Cardinality | 0..1
Terminology Binding | SNOMED CT Form Codes (Example)
Type | CodeableConcept
Comments | When Medication is referenced from MedicationRequest, this is the ordered form. When Medication is referenced within MedicationDispense, this is the dispensed form. When Medication is referenced within MedicationAdministration, this is administered form.
Medication.amount
Element Id | Medication.amount
Definition | Specific amount of the drug in the packaged product. For example, when specifying a product that has the same strength (For example, Insulin glargine 100 unit per mL solution for injection), this attribute provides additional clarification of the package amount (For example, 3 mL, 10mL, etc.).
Cardinality | 0..1
Type | Ratio
Summary | true
Medication.ingredient
Element Id | Medication.ingredient
Definition | Identifies a particular constituent of interest in the product.
Cardinality | 0..*
Comments | The ingredients need not be a complete list. If an ingredient is not specified, this does not indicate whether an ingredient is present or absent. If an ingredient is specified it does not mean that all ingredients are specified. It is possible to specify both inactive and active ingredients.
Medication.ingredient.item[x]
Element Id | Medication.ingredient.item[x]
Definition | The actual ingredient - either a substance (simple ingredient) or another medication of a medication.
Cardinality | 1..1
Type | CodeableConcept|Reference(Substance | Medication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The ingredient may reference a substance (for example, amoxicillin) or another medication (for example in the case of a compounded product, Glaxal Base).
Medication.ingredient.isActive
Element Id | Medication.ingredient.isActive
Definition | Indication of whether this ingredient affects the therapeutic action of the drug.
Cardinality | 0..1
Type | boolean
Requirements | True indicates that the ingredient affects the therapeutic action of the drug (i.e. active). False indicates that the ingredient does not affect the therapeutic action of the drug (i.e. inactive).
Medication.ingredient.strength
Element Id | Medication.ingredient.strength
Definition | Specifies how many (or how much) of the items there are in this Medication. For example, 250 mg per tablet. This is expressed as a ratio where the numerator is 250mg and the denominator is 1 tablet.
Cardinality | 0..1
Type | Ratio
Medication.batch
Element Id | Medication.batch
Definition | Information that only applies to packages (not products).
Cardinality | 0..1
Medication.batch.lotNumber
Element Id | Medication.batch.lotNumber
Definition | The assigned lot number of a batch of the specified product.
Cardinality | 0..1
Type | string
Medication.batch.expirationDate
Element Id | Medication.batch.expirationDate
Definition | When this specific batch of product will expire.
Cardinality | 0..1
Type | dateTime

11.5.7 Resource Medication - Mappings

URL: https://hl7.org/fhir/R4/medication-mappings.html

11.5.7 Resource Medication - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mapping to NCPDP SCRIPT 10.6

Mappings for the medication resource (see Mappings to Other Standards for further information & status).

11.5.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Medication
identifier | FiveWs.identifier
code | FiveWs.class
manufacturer | FiveWs.actor

11.5.7.2 HL7 v2 Mapping (http://hl7.org/v2)

Medication
identifier
code | RXO-1.1-Requested Give Code.code / RXE-2.1-Give Code.code / RXD-2.1-Dispense/Give Code.code / RXG-4.1-Give Code.code /RXA-5.1-Administered Code.code / RXC-2.1 Component Code
status
manufacturer | RXD-20-Substance Manufacturer Name / RXG-21-Substance Manufacturer Name / RXA-17-Substance Manufacturer Name
form | RXO-5-Requested Dosage Form / RXE-6-Give Dosage Form / RXD-6-Actual Dosage Form / RXG-8-Give Dosage Form / RXA-8-Administered Dosage Form
amount
ingredient
item[x] | RXC-2-Component Code if medication: RXO-1-Requested Give Code / RXE-2-Give Code / RXD-2-Dispense/Give Code / RXG-4-Give Code / RXA-5-Administered Code
isActive
strength | RXC-3-Component Amount & RXC-4-Component Units if medication: RXO-2-Requested Give Amount - Minimum & RXO-4-Requested Give Units / RXO-3-Requested Give Amount - Maximum & RXO-4-Requested Give Units / RXO-11-Requested Dispense Amount & RXO-12-Requested Dispense Units / RXE-3-Give Amount - Minimum & RXE-5-Give Units / RXE-4-Give Amount - Maximum & RXE-5-Give Units / RXE-10-Dispense Amount & RXE-10-Dispense Units
batch
lotNumber | RXA-15 Substance Lot Number / RXG-19 Substance Lot Number
expirationDate | RXA-16 Substance Expiration Date / RXG-20 Substance Expiration Date

11.5.7.3 RIM Mapping (http://hl7.org/v3)

Medication | ManufacturedProduct[classCode=ADMM]
identifier | .id
code | .code
status | .statusCode
manufacturer | .player.scopingRole[typeCode=MANU].scoper
form | .formCode
amount | .quantity
ingredient | .scopesRole[typeCode=INGR]
item[x] | .player
isActive | NA
strength | .quantity
batch | .player[classCode=CONT]
lotNumber | .id
expirationDate | participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].expirationTime

11.5.7.4 Mapping to NCPDP SCRIPT 10.6 (http://ncpdp.org/SCRIPT10_6)

Medication | NewRx/MedicationPrescribed -or- RxFill/MedicationDispensed -or- RxHistoryResponse/MedicationDispensed -or- RxHistoryResponse/MedicationPrescribed
identifier
code | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system = //element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
status
manufacturer | no mapping
form | coding.code = //element(*,DrugCodedType)/FormCode coding.system = //element(*,DrugCodedType)/FormSourceCode
amount
ingredient
item[x] | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system = //element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
isActive
strength | //element(*,DrugCodedType)/Strength
batch | no mapping

lotNumber | no mapping
expirationDate | no mapping

11.5.8 Resource Medication - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medication-profiles.html

11.5.8 Resource Medication - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Medication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.5.8.1 Profiles

No Profiles defined for this resource

11.5.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.5.8.3 Search Extensions

No Search Extensions defined for this resource

11.5.9 Resource Medication - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/medication-version-maps.html

No validation errors - all conversions are clean

11.5.9 Resource Medication - R3/R4 Conversions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Medication.

Functional status for this map: 23 tests that all execute ok. 22 fail round-trip testing and 22 r3 resources are invalid (0 errors). (see documentation)

11.5.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Medication3to4" = "R3 to R4 Conversions for Medication"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Medication" alias MedicationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Medication" alias Medication as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Medication(source src : MedicationR3, target tgt : Medication) extends DomainResource <<type+>> {
  src.code -> tgt.code;
  src.status as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.status',  vt.value = v;
  src.isBrand as v ->  tgt.extension as vt,  vt.url = 'hhttp://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.isBrand',  vt.value = v;
  src.isOverTheCounter as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationOTC',  vt.value = v;
  src.manufacturer -> tgt.manufacturer;
  src.form -> tgt.form;
  src.ingredient as vs0 -> tgt.ingredient as vt0 then ingredient(vs0, vt0);
  src.package as vs0 then package(vs0, tgt);
  src.image as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MedicationImage',  vt.value = v;
}

group ingredient(source src, target tgt) extends BackboneElement {
```

```
  src.item : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.item : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
  src.isActive -> tgt.isActive;
  src.amount -> tgt.strength;
}

group package(source src, target tgt) {
  src.container as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.container',  vt.value = v;
  src.content as vs0 ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.content' then content(vs0, vt);
  src.batch as vs0 -> tgt.batch as vt0 then batch(vs0, vt0);
}

group content(source src, target tgt) extends Element {
  src.item as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.content',  vt.value = v;
  src.amount as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.content.amount',  vt.value = v;
}

group batch(source src, target tgt) extends BackboneElement {
  src.lotNumber -> tgt.lotNumber;
  src.expirationDate -> tgt.expirationDate;
}
```

11.5.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Medication4to3" = "R4 to R3 Conversion for Medication"

uses "http://hl7.org/fhir/StructureDefinition/Medication" alias Medication as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Medication" alias MedicationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Medication(source src : Medication, target tgt : MedicationR3) extends DomainResource <<type+>> {
  src.code -> tgt.code;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.status' then {
    e.value -> tgt.status;
  };
  src.extension as e where url = 'hhttp://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.isBrand' then {
    e.value -> tgt.isBrand;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationOTC' then {
    e.value -> tgt.isOverTheCounter;
  };
  src.manufacturer -> tgt.manufacturer;
  src.form -> tgt.form;
  src.ingredient as vs0 -> tgt.ingredient as vt0 then ingredient(vs0, vt0);
  src where batch.exists() or extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.container').exists() or extension.where(url =
    'http://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.package.content').exists() -> tgt.package as vt0
    then package(src, vt0) "batch";
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-MedicationImage' then {
    e.value -> tgt.image;
  };
}

group ingredient(source src, target tgt) extends BackboneElement {
  src.item : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.item : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
  src.isActive -> tgt.isActive;
  src.strength -> tgt.amount;
}

group package(source src, target tgt) {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.package.container'
    then {
    e.value -> tgt.container;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.package.content' ->
    tgt.content as vt0 then content(e, vt0);
  src.batch as vs0 -> tgt.batch as vt0 then batch(vs0, vt0);
}

group content(source src, target tgt) extends Element {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Medication.package.content' then
     {
    e.value : CodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
    e.value : Reference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Medication.package.content.amount' then {
    e.value -> tgt.amount;
  };
}

group batch(source src, target tgt) extends BackboneElement {
  src.lotNumber -> tgt.lotNumber;
  src.expirationDate -> tgt.expirationDate;
```

}

11.5.9.3 R4 Validation Errors

11.5.10 Medication HL7 Extensions

URL: https://hl7.org/fhir/R4/medication-extensions.html

Defines common extensions used with or related to the Medication resource

11.5.10 Medication HL7 Extensions

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

11.5.10.1 Content

11.5.11 JSON Schema for Medication

URL: https://hl7.org/fhir/R4/medication.schema.json.html

11.5.11 JSON Schema for Medication

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Medication, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Medication",
  "$ref": "#/definitions/Medication",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Medication": {
      "description": "This resource is primarily used for the identification and definition of a medication for the
    purposes of prescribing, dispensing, and administering a medication as well as for making statements about
    medication use.",
      "properties": {
        "resourceType": {
          "description": "This is a Medication resource",
          "const": "Medication"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifier for this medication.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "code": {
      "description": "A code (or set of codes) that specify this medication, or a textual description if no code is
available. Usage note: This could be a standard medication code such as a code from RxNorm, SNOMED CT, IDMP etc. It
could also be a national or local formulary code, optionally with translations to other code systems.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "A code to indicate if the medication is in active use.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "manufacturer": {
      "description": "Describes the details of the manufacturer of the medication product.  This is not intended to
represent the distributor of a medication product.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "form": {
      "description": "Describes the form of the item.  Powder; tablets; capsule.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amount": {
      "description": "Specific amount of the drug in the packaged product.  For example, when specifying a product
that has the same strength (For example, Insulin glargine 100 unit per mL solution for injection), this attribute
provides additional clarification of the package amount (For example, 3 mL, 10mL, etc.).",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "ingredient": {
      "description": "Identifies a particular constituent of interest in the product.",
      "items": {
        "$ref": "#/definitions/Medication_Ingredient"
      },
      "type": "array"
    },
    "batch": {
      "description": "Information that only applies to packages (not products).",
      "$ref": "#/definitions/Medication_Batch"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Medication_Ingredient": {
  "description": "This resource is primarily used for the identification and definition of a medication for the
purposes of prescribing, dispensing, and administering a medication as well as for making statements about
medication use.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "itemCodeableConcept": {
      "description": "The actual ingredient - either a substance (simple ingredient) or another medication of a
medication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "itemReference": {
      "description": "The actual ingredient - either a substance (simple ingredient) or another medication of a
medication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "isActive": {
      "description": "Indication of whether this ingredient affects the therapeutic action of the drug.",
      "$ref": "#/definitions/boolean"
    },
    "_isActive": {
      "description": "Extensions for isActive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "strength": {
      "description": "Specifies how many (or how much) of the items there are in this Medication.  For example, 250
mg per tablet.  This is expressed as a ratio where the numerator is 250mg and the denominator is 1 tablet.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    }
  },
  "additionalProperties": false
},
"Medication_Batch": {
  "description": "This resource is primarily used for the identification and definition of a medication for the
purposes of prescribing, dispensing, and administering a medication as well as for making statements about
medication use.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "lotNumber": {
      "description": "The assigned lot number of a batch of the specified product.",
      "$ref": "#/definitions/string"
    },
    "_lotNumber": {
      "description": "Extensions for lotNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expirationDate": {
```

```
          "description": "When this specific batch of product will expire.",
          "$ref": "#/definitions/dateTime"
        },
        "_expirationDate": {
          "description": "Extensions for expirationDate",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.5.12 Resource Medication - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medication-dependencies.html

11.5.12 Resource Medication - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Medication FMM level 3

Name | Card. | Type | Dependency Analysis
Medication | | DomainResource
identifier | 0..* | Identifier
code | 0..1 | CodeableConcept
status | 0..1 | code
manufacturer | 0..1 | Reference(Organization)
form | 0..1 | CodeableConcept
amount | 0..1 | Ratio
ingredient | 0..* | BackboneElement
item[x] | 1..1
itemCodeableConcept | | CodeableConcept
itemReference | | Reference(Substance | Medication
isActive | 0..1 | boolean
strength | 0..1 | Ratio
batch | 0..1 | BackboneElement
lotNumber | 0..1 | string
expirationDate | 0..1 | dateTime

11.5.13 Value Set Sc-valueset-medication-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medication-status.html

11.5.13 Value Set Sc-valueset-medication-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.5.14 Medication Status CodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medication-status)

Mapping from http://hl7.org/fhir/ValueSet/medication-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "Medication Status Codes"

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
active | equivalent | active
inactive | equivalent | inactive

11.6 Resource MedicationKnowledge - Content

URL: https://hl7.org/fhir/R4/medicationknowledge.html
Pharmacy Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Information about a medication that is used to support knowledge.

11.6.1 Scope and Usage

Note that this content is preliminary has not undergone proper review by the appropriate Workgroups.

This resource supports use cases for creation of and querying for drug information including attributes such as drug classifications, images of medications, drug costs and/or coverages, etc. This resource can be used to return drug information as part of a formulary or a catalogue.

This resource is referenced by itself

11.6.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationKnowledge | TU | | DomainResource | Definition of Medication Knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
code | Σ | 0..1 | CodeableConcept | Code that identifies this medication SNOMED CT Medication Codes (Example)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error Medication knowledge status codes (Required)
manufacturer | Σ | 0..1 | Reference(Organization) | Manufacturer of the item
doseForm | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
amount | Σ | 0..1 | SimpleQuantity | Amount of drug in package
synonym | Σ | 0..* | string | Additional names for a medication
relatedMedicationKnowledge | | 0..* | BackboneElement | Associated or related medication information
type | | 1..1 | CodeableConcept | Category of medicationKnowledge
reference | | 1..* | Reference(MedicationKnowledge) | Associated documentation about the associated medication knowledge
associatedMedication | | 0..* | Reference(Medication) | A medication resource that is associated with this medication
productType | | 0..* | CodeableConcept | Category of the medication or product
monograph | | 0..* | BackboneElement | Associated documentation about the medication
type | | 0..1 | CodeableConcept | The category of medication document
source | | 0..1 | Reference(DocumentReference | Media) | Associated documentation about the medication
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | Medication(s) or substance(s) contained in the medication
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance)
isActive | | 0..1 | boolean | Active ingredient indicator
strength | | 0..1 | Ratio | Quantity of ingredient present
preparationInstruction | | 0..1 | markdown | The instructions for preparing the medication
intendedRoute | | 0..* | CodeableConcept | The intended or approved route of administration SNOMED CT Route Codes (Example)
cost | | 0..* | BackboneElement | The pricing of the medication
type | | 1..1 | CodeableConcept | The category of the cost information
source | | 0..1 | string | The source or owner for the price information
cost | | 1..1 | Money | The price of the medication
monitoringProgram | | 0..* | BackboneElement | Program under which a medication is reviewed
type | | 0..1 | CodeableConcept | Type of program under which the medication is monitored
name | | 0..1 | string | Name of the reviewing program
administrationGuidelines | | 0..* | BackboneElement | Guidelines for administration of the medication
dosage | | 0..* | BackboneElement | Dosage for the medication for the specific guidelines
type | | 1..1 | CodeableConcept | Type of dosage
dosage | | 1..* | Dosage | Dosage for the medication for the specific guidelines
indication[x] | | 0..1 | | Indication for use that apply to the specific administration guidelines
indicationCodeableConcept | | | CodeableConcept
indicationReference | | | Reference(ObservationDefinition)
patientCharacteristics | | 0..* | BackboneElement | Characteristics of the patient that are relevant to the administration guidelines
characteristic[x] | | 1..1 | | Specific characteristic that is relevant to the administration guideline
characteristicCodeableConcept | | | CodeableConcept
characteristicQuantity | | | SimpleQuantity
value | | 0..* | string | The specific characteristic
medicineClassification | | 0..* | BackboneElement | Categorization of the medication within a formulary or classification system
type | | 1..1 | CodeableConcept | The type of category for the medication (for example, therapeutic classification, therapeutic sub-classification)
classification | | 0..* | CodeableConcept | Specific category assigned to the medication
packaging | | 0..1 | BackboneElement | Details about packaged medications
type | | 0..1 | CodeableConcept | A code that defines the specific type of packaging that the medication can be found in Medication knowledge package type codes (Example)
quantity | | 0..1 | SimpleQuantity | The number of product units the package would contain if fully loaded
drugCharacteristic | | 0..* | BackboneElement | Specifies descriptive properties of the medicine

type | | 0..1 | CodeableConcept | Code specifying the type of characteristic of medication Medication knowledge characteristic codes (Example)
value[x] | | 0..1 | | Description of the characteristic
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueQuantity | | | SimpleQuantity
valueBase64Binary | | | base64Binary
contraindication | | 0..* | Reference(DetectedIssue) | Potential clinical issue with or between medication(s)
regulatory | | 0..* | BackboneElement | Regulatory information about a medication
regulatoryAuthority | | 1..1 | Reference(Organization) | Specifies the authority of the regulation
substitution | | 0..* | BackboneElement | Specifies if changes are allowed when dispensing a medication from a regulatory perspective
type | | 1..1 | CodeableConcept | Specifies the type of substitution allowed
allowed | | 1..1 | boolean | Specifies if regulation allows for changes in the medication when dispensing
schedule | | 0..* | BackboneElement | Specifies the schedule of a medication in jurisdiction
schedule | | 1..1 | CodeableConcept | Specifies the specific drug schedule
maxDispense | | 0..1 | BackboneElement | The maximum number of units of the medication that can be dispensed in a period
quantity | | 1..1 | SimpleQuantity | The maximum number of units of the medication that can be dispensed
period | | 0..1 | Duration | The period that applies to the maximum number of units
kinetics | | 0..* | BackboneElement | The time course of drug absorption, distribution, metabolism and excretion of a medication from the body
areaUnderCurve | | 0..* | SimpleQuantity | The drug concentration measured at certain discrete points in time
lethalDose50 | | 0..* | SimpleQuantity | The median lethal dose of a drug
halfLifePeriod | | 0..1 | Duration | Time required for concentration in the body to decrease by half
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationKnowledge xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <code><!-- 0..1 CodeableConcept Code that identifies this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <doseForm><!-- 0..1 CodeableConcept powder | tablets | capsule + --></doseForm>
 <amount><!-- 0..1 Quantity(SimpleQuantity) Amount of drug in package --></amount>
 <synonym value="[string]"/><!-- 0..* Additional names for a medication -->
 <relatedMedicationKnowledge>  <!-- 0..* Associated or related medication information -->
  <type><!-- 1..1 CodeableConcept Category of medicationKnowledge --></type>
  <reference><!-- 1..* Reference(MedicationKnowledge) Associated documentation about the associated medication knowledge
     --></reference>
 </relatedMedicationKnowledge>
 <associatedMedication><!-- 0..* Reference(Medication) A medication resource that is associated with this medication
    --></associatedMedication>
 <productType><!-- 0..* CodeableConcept Category of the medication or product --></productType>
 <monograph>  <!-- 0..* Associated documentation about the medication -->
  <type><!-- 0..1 CodeableConcept The category of medication document --></type>
  <source><!-- 0..1 Reference(DocumentReference|Media) Associated documentation about the medication --></source>
 </monograph>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance) Medication(s) or substance(s) contained in the medication
    --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <strength><!-- 0..1 Ratio Quantity of ingredient present --></strength>
 </ingredient>
 <preparationInstruction value="[markdown]"/><!-- 0..1 The instructions for preparing the medication -->
 <intendedRoute><!-- 0..* CodeableConcept The intended or approved route of administration --></intendedRoute>
 <cost>  <!-- 0..* The pricing of the medication -->
  <type><!-- 1..1 CodeableConcept The category of the cost information --></type>
  <source value="[string]"/><!-- 0..1 The source or owner for the price information -->
  <cost><!-- 1..1 Money The price of the medication --></cost>
 </cost>
 <monitoringProgram>  <!-- 0..* Program under which a medication is reviewed -->
  <type><!-- 0..1 CodeableConcept Type of program under which the medication is monitored --></type>
  <name value="[string]"/><!-- 0..1 Name of the reviewing program -->
 </monitoringProgram>
 <administrationGuidelines>  <!-- 0..* Guidelines for administration of the medication -->
  <dosage>  <!-- 0..* Dosage for the medication for the specific guidelines -->
   <type><!-- 1..1 CodeableConcept Type of dosage --></type>
   <dosage><!-- 1..* Dosage Dosage for the medication for the specific guidelines --></dosage>
  </dosage>
  <indication[x]><!-- 0..1 CodeableConcept|Reference(ObservationDefinition) Indication for use that apply to the
    specific administration guidelines --></indication[x]>
  <patientCharacteristics>  <!-- 0..* Characteristics of the patient that are relevant to the administration guidelines
    -->
   <characteristic[x]><!-- 1..1 CodeableConcept|Quantity(SimpleQuantity) Specific characteristic that is relevant to the
     administration guideline --></characteristic[x]>
   <value value="[string]"/><!-- 0..* The specific characteristic -->
  </patientCharacteristics>
 </administrationGuidelines>
 <medicineClassification>  <!-- 0..* Categorization of the medication within a formulary or classification system -->
  <type><!-- 1..1 CodeableConcept The type of category for the medication (for example, therapeutic classification,
    therapeutic sub-classification) --></type>
```

```
  <classification><!-- 0..* CodeableConcept Specific category assigned to the medication --></classification>
 </medicineClassification>
 <packaging>  <!-- 0..1 Details about packaged medications -->
  <type><!-- 0..1 CodeableConcept A code that defines the specific type of packaging that the medication can be found in
     --></type>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The number of product units the package would contain if fully loaded
    --></quantity>
 </packaging>
 <drugCharacteristic>  <!-- 0..* Specifies descriptive properties of the medicine -->
  <type><!-- 0..1 CodeableConcept Code specifying the type of characteristic of medication --></type>
  <value[x]><!-- 0..1 CodeableConcept|string|Quantity(SimpleQuantity)|
    base64Binary Description of the characteristic --></value[x]>
 </drugCharacteristic>
 <contraindication><!-- 0..* Reference(DetectedIssue) Potential clinical issue with or between medication(s)
    --></contraindication>
 <regulatory>  <!-- 0..* Regulatory information about a medication -->
  <regulatoryAuthority><!-- 1..1 Reference(Organization) Specifies the authority of the regulation
    --></regulatoryAuthority>
  <substitution>  <!-- 0..* Specifies if changes are allowed when dispensing a medication from a regulatory perspective
    -->
   <type><!-- 1..1 CodeableConcept Specifies the type of substitution allowed --></type>
   <allowed value="[boolean]"/><!-- 1..1 Specifies if regulation allows for changes in the medication when dispensing
    -->
  </substitution>
  <schedule>  <!-- 0..* Specifies the schedule of a medication in jurisdiction -->
   <schedule><!-- 1..1 CodeableConcept Specifies the specific drug schedule --></schedule>
  </schedule>
  <maxDispense>  <!-- 0..1 The maximum number of units of the medication that can be dispensed in a period -->
   <quantity><!-- 1..1 Quantity(SimpleQuantity) The maximum number of units of the medication that can be dispensed
    --></quantity>
   <period><!-- 0..1 Duration The period that applies to the maximum number of units --></period>
  </maxDispense>
 </regulatory>
 <kinetics>  <!-- 0..* The time course of drug absorption, distribution, metabolism and excretion of a medication from
    the body -->
  <areaUnderCurve><!-- 0..* Quantity(SimpleQuantity) The drug concentration measured at certain discrete points in time
    --></areaUnderCurve>
  <lethalDose50><!-- 0..* Quantity(SimpleQuantity) The median lethal dose of a drug --></lethalDose50>
  <halfLifePeriod><!-- 0..1 Duration Time required for concentration in the body to decrease by half
    --></halfLifePeriod>
 </kinetics>
</MedicationKnowledge>
```

JSON Template

```
{
  "resourceType" : "MedicationKnowledge",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "code" : { CodeableConcept }, // Code that identifies this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "doseForm" : { CodeableConcept }, // powder | tablets | capsule +
  "amount" : { Quantity(SimpleQuantity) }, // Amount of drug in package
  "synonym" : ["<string>"], // Additional names for a medication
  "relatedMedicationKnowledge" : [{ // Associated or related medication information
    "type" : { CodeableConcept }, // R!  Category of medicationKnowledge
    "reference" : [{ Reference(MedicationKnowledge) }] // R!  Associated documentation about the associated medication
    knowledge
  }],
  "associatedMedication" : [{ Reference(Medication) }], // A medication resource that is associated with this medication
  "productType" : [{ CodeableConcept }], // Category of the medication or product
  "monograph" : [{ // Associated documentation about the medication
    "type" : { CodeableConcept }, // The category of medication document
    "source" : { Reference(DocumentReference|Media) } // Associated documentation about the medication
  }],
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: Medication(s) or substance(s) contained in the medication. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance) },
    "isActive" : <boolean>, // Active ingredient indicator
    "strength" : { Ratio } // Quantity of ingredient present
  }],
  "preparationInstruction" : "<markdown>", // The instructions for preparing the medication
  "intendedRoute" : [{ CodeableConcept }], // The intended or approved route of administration
  "cost" : [{ // The pricing of the medication
    "type" : { CodeableConcept }, // R!  The category of the cost information
    "source" : "<string>", // The source or owner for the price information
    "cost" : { Money } // R!  The price of the medication
  }],
  "monitoringProgram" : [{ // Program under which a medication is reviewed
    "type" : { CodeableConcept }, // Type of program under which the medication is monitored
    "name" : "<string>" // Name of the reviewing program
  }],
  "administrationGuidelines" : [{ // Guidelines for administration of the medication
    "dosage" : [{ // Dosage for the medication for the specific guidelines
      "type" : { CodeableConcept }, // R!  Type of dosage
      "dosage" : [{ Dosage }] // R!  Dosage for the medication for the specific guidelines
    }],
    // indication[x]: Indication for use that apply to the specific administration guidelines. One of these 2:
```

```
    "indicationCodeableConcept" : { CodeableConcept },
    "indicationReference" : { Reference(ObservationDefinition) },
    "patientCharacteristics" : [{ // Characteristics of the patient that are relevant to the administration guidelines
      // characteristic[x]: Specific characteristic that is relevant to the administration guideline. One of these 2:
      "characteristicCodeableConcept" : { CodeableConcept },
      "characteristicQuantity" : { Quantity(SimpleQuantity) },
      "value" : ["<string>"] // The specific characteristic
    }]
  }],
  "medicineClassification" : [{ // Categorization of the medication within a formulary or classification system
    "type" : { CodeableConcept }, // R!  The type of category for the medication (for example, therapeutic
    classification, therapeutic sub-classification)
    "classification" : [{ CodeableConcept }] // Specific category assigned to the medication
  }],
  "packaging" : { // Details about packaged medications
    "type" : { CodeableConcept }, // A code that defines the specific type of packaging that the medication can be found
     in
    "quantity" : { Quantity(SimpleQuantity) } // The number of product units the package would contain if fully loaded
  },
  "drugCharacteristic" : [{ // Specifies descriptive properties of the medicine
    "type" : { CodeableConcept }, // Code specifying the type of characteristic of medication
    // value[x]: Description of the characteristic. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueString" : "<string>"
    "valueQuantity" : { Quantity(SimpleQuantity) }
    "valueBase64Binary" : "<base64Binary>"
  }],
  "contraindication" : [{ Reference(DetectedIssue) }], // Potential clinical issue with or between medication(s)
  "regulatory" : [{ // Regulatory information about a medication
    "regulatoryAuthority" : { Reference(Organization) }, // R!  Specifies the authority of the regulation
    "substitution" : [{ // Specifies if changes are allowed when dispensing a medication from a regulatory perspective
      "type" : { CodeableConcept }, // R!  Specifies the type of substitution allowed
      "allowed" : <boolean> // R!  Specifies if regulation allows for changes in the medication when dispensing
    }],
    "schedule" : [{ // Specifies the schedule of a medication in jurisdiction
      "schedule" : { CodeableConcept } // R!  Specifies the specific drug schedule
    }],
    "maxDispense" : { // The maximum number of units of the medication that can be dispensed in a period
      "quantity" : { Quantity(SimpleQuantity) }, // R!  The maximum number of units of the medication that can be
    dispensed
      "period" : { Duration } // The period that applies to the maximum number of units
    }
  }],
  "kinetics" : [{ // The time course of drug absorption, distribution, metabolism and excretion of a medication from the
     body
    "areaUnderCurve" : [{ Quantity(SimpleQuantity) }], // The drug concentration measured at certain discrete points in
    time
    "lethalDose50" : [{ Quantity(SimpleQuantity) }], // The median lethal dose of a drug
    "halfLifePeriod" : { Duration } // Time required for concentration in the body to decrease by half
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationKnowledge;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationKnowledge.code [ CodeableConcept ]; # 0..1 Code that identifies this medication
  fhir:MedicationKnowledge.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:MedicationKnowledge.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:MedicationKnowledge.doseForm [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:MedicationKnowledge.amount [ Quantity(SimpleQuantity) ]; # 0..1 Amount of drug in package
  fhir:MedicationKnowledge.synonym [ string ], ... ; # 0..* Additional names for a medication
  fhir:MedicationKnowledge.relatedMedicationKnowledge [ # 0..* Associated or related medication information
    fhir:MedicationKnowledge.relatedMedicationKnowledge.type [ CodeableConcept ]; # 1..1 Category of medicationKnowledge
    fhir:MedicationKnowledge.relatedMedicationKnowledge.reference [ Reference(MedicationKnowledge) ], ... ; # 1..*
    Associated documentation about the associated medication knowledge
  ], ...;
  fhir:MedicationKnowledge.associatedMedication [ Reference(Medication) ], ... ; # 0..* A medication resource that is
    associated with this medication
  fhir:MedicationKnowledge.productType [ CodeableConcept ], ... ; # 0..* Category of the medication or product
  fhir:MedicationKnowledge.monograph [ # 0..* Associated documentation about the medication
    fhir:MedicationKnowledge.monograph.type [ CodeableConcept ]; # 0..1 The category of medication document
    fhir:MedicationKnowledge.monograph.source [ Reference(DocumentReference|Media) ]; # 0..1 Associated documentation
    about the medication
  ], ...;
  fhir:MedicationKnowledge.ingredient [ # 0..* Active or inactive ingredient
    # MedicationKnowledge.ingredient.item[x] : 1..1 Medication(s) or substance(s) contained in the medication. One of
    these 2
      fhir:MedicationKnowledge.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:MedicationKnowledge.ingredient.itemReference [ Reference(Substance) ]
    fhir:MedicationKnowledge.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:MedicationKnowledge.ingredient.strength [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:MedicationKnowledge.preparationInstruction [ markdown ]; # 0..1 The instructions for preparing the medication
```

```
fhir:MedicationKnowledge.intendedRoute [ CodeableConcept ], ... ; # 0..* The intended or approved route of
  administration
fhir:MedicationKnowledge.cost [ # 0..* The pricing of the medication
  fhir:MedicationKnowledge.cost.type [ CodeableConcept ]; # 1..1 The category of the cost information
  fhir:MedicationKnowledge.cost.source [ string ]; # 0..1 The source or owner for the price information
  fhir:MedicationKnowledge.cost.cost [ Money ]; # 1..1 The price of the medication
], ...;
fhir:MedicationKnowledge.monitoringProgram [ # 0..* Program under which a medication is reviewed
  fhir:MedicationKnowledge.monitoringProgram.type [ CodeableConcept ]; # 0..1 Type of program under which the
  medication is monitored
  fhir:MedicationKnowledge.monitoringProgram.name [ string ]; # 0..1 Name of the reviewing program
], ...;
fhir:MedicationKnowledge.administrationGuidelines [ # 0..* Guidelines for administration of the medication
  fhir:MedicationKnowledge.administrationGuidelines.dosage [ # 0..* Dosage for the medication for the specific
  guidelines
    fhir:MedicationKnowledge.administrationGuidelines.dosage.type [ CodeableConcept ]; # 1..1 Type of dosage
    fhir:MedicationKnowledge.administrationGuidelines.dosage.dosage [ Dosage ], ... ; # 1..* Dosage for the medication
   for the specific guidelines
  ], ...;
  # MedicationKnowledge.administrationGuidelines.indication[x] : 0..1 Indication for use that apply to the specific
  administration guidelines. One of these 2
    fhir:MedicationKnowledge.administrationGuidelines.indicationCodeableConcept [ CodeableConcept ]
    fhir:MedicationKnowledge.administrationGuidelines.indicationReference [ Reference(ObservationDefinition) ]
  fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics [ # 0..* Characteristics of the patient
  that are relevant to the administration guidelines
    # MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristic[x] : 1..1 Specific
  characteristic that is relevant to the administration guideline. One of these 2
      fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristicCodeableConcept [
  CodeableConcept ]
      fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristicSimpleQuantity [
  Quantity(SimpleQuantity) ]
    fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.value [ string ], ... ; # 0..* The
  specific characteristic
  ], ...;
], ...;
fhir:MedicationKnowledge.medicineClassification [ # 0..* Categorization of the medication within a formulary or
  classification system
  fhir:MedicationKnowledge.medicineClassification.type [ CodeableConcept ]; # 1..1 The type of category for the
  medication (for example, therapeutic classification, therapeutic sub-classification)
  fhir:MedicationKnowledge.medicineClassification.classification [ CodeableConcept ], ... ; # 0..* Specific category
  assigned to the medication
], ...;
fhir:MedicationKnowledge.packaging [ # 0..1 Details about packaged medications
  fhir:MedicationKnowledge.packaging.type [ CodeableConcept ]; # 0..1 A code that defines the specific type of
  packaging that the medication can be found in
  fhir:MedicationKnowledge.packaging.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The number of product units the
  package would contain if fully loaded
];
fhir:MedicationKnowledge.drugCharacteristic [ # 0..* Specifies descriptive properties of the medicine
  fhir:MedicationKnowledge.drugCharacteristic.type [ CodeableConcept ]; # 0..1 Code specifying the type of
  characteristic of medication
  # MedicationKnowledge.drugCharacteristic.value[x] : 0..1 Description of the characteristic. One of these 4
    fhir:MedicationKnowledge.drugCharacteristic.valueCodeableConcept [ CodeableConcept ]
    fhir:MedicationKnowledge.drugCharacteristic.valueString [ string ]
    fhir:MedicationKnowledge.drugCharacteristic.valueSimpleQuantity [ Quantity(SimpleQuantity) ]
    fhir:MedicationKnowledge.drugCharacteristic.valueBase64Binary [ base64Binary ]
], ...;
fhir:MedicationKnowledge.contraindication [ Reference(DetectedIssue) ], ... ; # 0..* Potential clinical issue with or
  between medication(s)
fhir:MedicationKnowledge.regulatory [ # 0..* Regulatory information about a medication
  fhir:MedicationKnowledge.regulatory.regulatoryAuthority [ Reference(Organization) ]; # 1..1 Specifies the authority
  of the regulation
  fhir:MedicationKnowledge.regulatory.substitution [ # 0..* Specifies if changes are allowed when dispensing a
  medication from a regulatory perspective
    fhir:MedicationKnowledge.regulatory.substitution.type [ CodeableConcept ]; # 1..1 Specifies the type of
  substitution allowed
    fhir:MedicationKnowledge.regulatory.substitution.allowed [ boolean ]; # 1..1 Specifies if regulation allows for
  changes in the medication when dispensing
  ], ...;
  fhir:MedicationKnowledge.regulatory.schedule [ # 0..* Specifies the schedule of a medication in jurisdiction
    fhir:MedicationKnowledge.regulatory.schedule.schedule [ CodeableConcept ]; # 1..1 Specifies the specific drug
  schedule
  ], ...;
  fhir:MedicationKnowledge.regulatory.maxDispense [ # 0..1 The maximum number of units of the medication that can be
  dispensed in a period
    fhir:MedicationKnowledge.regulatory.maxDispense.quantity [ Quantity(SimpleQuantity) ]; # 1..1 The maximum number
  of units of the medication that can be dispensed
    fhir:MedicationKnowledge.regulatory.maxDispense.period [ Duration ]; # 0..1 The period that applies to the maximum
   number of units
  ];
], ...;
fhir:MedicationKnowledge.kinetics [ # 0..* The time course of drug absorption, distribution, metabolism and excretion
  of a medication from the body
  fhir:MedicationKnowledge.kinetics.areaUnderCurve [ Quantity(SimpleQuantity) ], ... ; # 0..* The drug concentration
  measured at certain discrete points in time
  fhir:MedicationKnowledge.kinetics.lethalDose50 [ Quantity(SimpleQuantity) ], ... ; # 0..* The median lethal dose of
  a drug
  fhir:MedicationKnowledge.kinetics.halfLifePeriod [ Duration ]; # 0..1 Time required for concentration in the body to
   decrease by half
```

```
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicationKnowledge | TU | | DomainResource | Definition of Medication Knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
code | Σ | 0..1 | CodeableConcept | Code that identifies this medication SNOMED CT Medication Codes (Example)
status | ?!Σ | 0..1 | code | active | inactive | entered-in-error Medication knowledge status codes (Required)
manufacturer | Σ | 0..1 | Reference(Organization) | Manufacturer of the item
doseForm | | 0..1 | CodeableConcept | powder | tablets | capsule + SNOMED CT Form Codes (Example)
amount | Σ | 0..1 | SimpleQuantity | Amount of drug in package
synonym | Σ | 0..* | string | Additional names for a medication
relatedMedicationKnowledge | | 0..* | BackboneElement | Associated or related medication information
type | | 1..1 | CodeableConcept | Category of medicationKnowledge
reference | | 1..* | Reference(MedicationKnowledge) | Associated documentation about the associated medication knowledge
associatedMedication | | 0..* | Reference(Medication) | A medication resource that is associated with this medication
productType | | 0..* | CodeableConcept | Category of the medication or product
monograph | | 0..* | BackboneElement | Associated documentation about the medication
type | | 0..1 | CodeableConcept | The category of medication document
source | | 0..1 | Reference(DocumentReference | Media) | Associated documentation about the medication
ingredient | | 0..* | BackboneElement | Active or inactive ingredient
item[x] | | 1..1 | | Medication(s) or substance(s) contained in the medication
itemCodeableConcept | | | CodeableConcept
itemReference | | | Reference(Substance)
isActive | | 0..1 | boolean | Active ingredient indicator
strength | | 0..1 | Ratio | Quantity of ingredient present
preparationInstruction | | 0..1 | markdown | The instructions for preparing the medication
intendedRoute | | 0..* | CodeableConcept | The intended or approved route of administration SNOMED CT Route Codes (Example)
cost | | 0..* | BackboneElement | The pricing of the medication
type | | 1..1 | CodeableConcept | The category of the cost information
source | | 0..1 | string | The source or owner for the price information
cost | | 1..1 | Money | The price of the medication
monitoringProgram | | 0..* | BackboneElement | Program under which a medication is reviewed
type | | 0..1 | CodeableConcept | Type of program under which the medication is monitored
name | | 0..1 | string | Name of the reviewing program
administrationGuidelines | | 0..* | BackboneElement | Guidelines for administration of the medication
dosage | | 0..* | BackboneElement | Dosage for the medication for the specific guidelines
type | | 1..1 | CodeableConcept | Type of dosage
dosage | | 1..* | Dosage | Dosage for the medication for the specific guidelines
indication[x] | | 0..1 | | Indication for use that apply to the specific administration guidelines
indicationCodeableConcept | | | CodeableConcept
indicationReference | | | Reference(ObservationDefinition)
patientCharacteristics | | 0..* | BackboneElement | Characteristics of the patient that are relevant to the administration guidelines
characteristic[x] | | 1..1 | | Specific characteristic that is relevant to the administration guideline
characteristicCodeableConcept | | | CodeableConcept
characteristicQuantity | | | SimpleQuantity
value | | 0..* | string | The specific characteristic
medicineClassification | | 0..* | BackboneElement | Categorization of the medication within a formulary or classification system
type | | 1..1 | CodeableConcept | The type of category for the medication (for example, therapeutic classification, therapeutic sub-classification)
classification | | 0..* | CodeableConcept | Specific category assigned to the medication
packaging | | 0..1 | BackboneElement | Details about packaged medications
type | | 0..1 | CodeableConcept | A code that defines the specific type of packaging that the medication can be found in Medication knowledge package type codes (Example)
quantity | | 0..1 | SimpleQuantity | The number of product units the package would contain if fully loaded
drugCharacteristic | | 0..* | BackboneElement | Specifies descriptive properties of the medicine
type | | 0..1 | CodeableConcept | Code specifying the type of characteristic of medication Medication knowledge characteristic codes (Example)
value[x] | | 0..1 | | Description of the characteristic
valueCodeableConcept | | | CodeableConcept
valueString | | | string
valueQuantity | | | SimpleQuantity
valueBase64Binary | | | base64Binary
contraindication | | 0..* | Reference(DetectedIssue) | Potential clinical issue with or between medication(s)
regulatory | | 0..* | BackboneElement | Regulatory information about a medication
regulatoryAuthority | | 1..1 | Reference(Organization) | Specifies the authority of the regulation
substitution | | 0..* | BackboneElement | Specifies if changes are allowed when dispensing a medication from a regulatory perspective

type | | 1..1 | CodeableConcept | Specifies the type of substitution allowed
allowed | | 1..1 | boolean | Specifies if regulation allows for changes in the medication when dispensing
schedule | | 0..* | BackboneElement | Specifies the schedule of a medication in jurisdiction
schedule | | 1..1 | CodeableConcept | Specifies the specific drug schedule
maxDispense | | 0..1 | BackboneElement | The maximum number of units of the medication that can be dispensed in a period
quantity | | 1..1 | SimpleQuantity | The maximum number of units of the medication that can be dispensed
period | | 0..1 | Duration | The period that applies to the maximum number of units
kinetics | | 0..* | BackboneElement | The time course of drug absorption, distribution, metabolism and excretion of a medication from the body
areaUnderCurve | | 0..* | SimpleQuantity | The drug concentration measured at certain discrete points in time
lethalDose50 | | 0..* | SimpleQuantity | The median lethal dose of a drug
halfLifePeriod | | 0..1 | Duration | Time required for concentration in the body to decrease by half
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicationKnowledge xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <code><!-- 0..1 CodeableConcept Code that identifies this medication --></code>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <manufacturer><!-- 0..1 Reference(Organization) Manufacturer of the item --></manufacturer>
 <doseForm><!-- 0..1 CodeableConcept powder | tablets | capsule + --></doseForm>
 <amount><!-- 0..1 Quantity(SimpleQuantity) Amount of drug in package --></amount>
 <synonym value="[string]"/><!-- 0..* Additional names for a medication -->
 <relatedMedicationKnowledge>  <!-- 0..* Associated or related medication information -->
  <type><!-- 1..1 CodeableConcept Category of medicationKnowledge --></type>
  <reference><!-- 1..* Reference(MedicationKnowledge) Associated documentation about the associated medication knowledge
     --></reference>
 </relatedMedicationKnowledge>
 <associatedMedication><!-- 0..* Reference(Medication) A medication resource that is associated with this medication
    --></associatedMedication>
 <productType><!-- 0..* CodeableConcept Category of the medication or product --></productType>
 <monograph>  <!-- 0..* Associated documentation about the medication -->
  <type><!-- 0..1 CodeableConcept The category of medication document --></type>
  <source><!-- 0..1 Reference(DocumentReference|Media) Associated documentation about the medication --></source>
 </monograph>
 <ingredient>  <!-- 0..* Active or inactive ingredient -->
  <item[x]><!-- 1..1 CodeableConcept|Reference(Substance) Medication(s) or substance(s) contained in the medication
    --></item[x]>
  <isActive value="[boolean]"/><!-- 0..1 Active ingredient indicator -->
  <strength><!-- 0..1 Ratio Quantity of ingredient present --></strength>
 </ingredient>
 <preparationInstruction value="[markdown]"/><!-- 0..1 The instructions for preparing the medication -->
 <intendedRoute><!-- 0..* CodeableConcept The intended or approved route of administration --></intendedRoute>
 <cost>  <!-- 0..* The pricing of the medication -->
  <type><!-- 1..1 CodeableConcept The category of the cost information --></type>
  <source value="[string]"/><!-- 0..1 The source or owner for the price information -->
  <cost><!-- 1..1 Money The price of the medication --></cost>
 </cost>
 <monitoringProgram>  <!-- 0..* Program under which a medication is reviewed -->
  <type><!-- 0..1 CodeableConcept Type of program under which the medication is monitored --></type>
  <name value="[string]"/><!-- 0..1 Name of the reviewing program -->
 </monitoringProgram>
 <administrationGuidelines>  <!-- 0..* Guidelines for administration of the medication -->
  <dosage>  <!-- 0..* Dosage for the medication for the specific guidelines -->
   <type><!-- 1..1 CodeableConcept Type of dosage --></type>
   <dosage><!-- 1..* Dosage Dosage for the medication for the specific guidelines --></dosage>
  </dosage>
  <indication[x]><!-- 0..1 CodeableConcept|Reference(ObservationDefinition) Indication for use that apply to the
    specific administration guidelines --></indication[x]>
  <patientCharacteristics>  <!-- 0..* Characteristics of the patient that are relevant to the administration guidelines
    -->
   <characteristic[x]><!-- 1..1 CodeableConcept|Quantity(SimpleQuantity) Specific characteristic that is relevant to the
     administration guideline --></characteristic[x]>
   <value value="[string]"/><!-- 0..* The specific characteristic -->
  </patientCharacteristics>
 </administrationGuidelines>
 <medicineClassification>  <!-- 0..* Categorization of the medication within a formulary or classification system -->
  <type><!-- 1..1 CodeableConcept The type of category for the medication (for example, therapeutic classification,
    therapeutic sub-classification) --></type>
  <classification><!-- 0..* CodeableConcept Specific category assigned to the medication --></classification>
 </medicineClassification>
 <packaging>  <!-- 0..1 Details about packaged medications -->
  <type><!-- 0..1 CodeableConcept A code that defines the specific type of packaging that the medication can be found in
     --></type>
  <quantity><!-- 0..1 Quantity(SimpleQuantity) The number of product units the package would contain if fully loaded
    --></quantity>
 </packaging>
 <drugCharacteristic>  <!-- 0..* Specifies descriptive properties of the medicine -->
  <type><!-- 0..1 CodeableConcept Code specifying the type of characteristic of medication --></type>
  <value[x]><!-- 0..1 CodeableConcept|string|Quantity(SimpleQuantity)|
    base64Binary Description of the characteristic --></value[x]>
 </drugCharacteristic>
 <contraindication><!-- 0..* Reference(DetectedIssue) Potential clinical issue with or between medication(s)
```

```
    --></contraindication>
 <regulatory>  <!-- 0..* Regulatory information about a medication -->
  <regulatoryAuthority><!-- 1..1 Reference(Organization) Specifies the authority of the regulation
    --></regulatoryAuthority>
  <substitution>  <!-- 0..* Specifies if changes are allowed when dispensing a medication from a regulatory perspective
    -->
   <type><!-- 1..1 CodeableConcept Specifies the type of substitution allowed --></type>
   <allowed value="[boolean]"/><!-- 1..1 Specifies if regulation allows for changes in the medication when dispensing
    -->
  </substitution>
  <schedule>  <!-- 0..* Specifies the schedule of a medication in jurisdiction -->
   <schedule><!-- 1..1 CodeableConcept Specifies the specific drug schedule --></schedule>
  </schedule>
  <maxDispense>  <!-- 0..1 The maximum number of units of the medication that can be dispensed in a period -->
   <quantity><!-- 1..1 Quantity(SimpleQuantity) The maximum number of units of the medication that can be dispensed
    --></quantity>
   <period><!-- 0..1 Duration The period that applies to the maximum number of units --></period>
  </maxDispense>
 </regulatory>
 <kinetics>  <!-- 0..* The time course of drug absorption, distribution, metabolism and excretion of a medication from
    the body -->
  <areaUnderCurve><!-- 0..* Quantity(SimpleQuantity) The drug concentration measured at certain discrete points in time
    --></areaUnderCurve>
  <lethalDose50><!-- 0..* Quantity(SimpleQuantity) The median lethal dose of a drug --></lethalDose50>
  <halfLifePeriod><!-- 0..1 Duration Time required for concentration in the body to decrease by half
    --></halfLifePeriod>
 </kinetics>
</MedicationKnowledge>
```

JSON Template

```
{
  "resourceType" : "MedicationKnowledge",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "code" : { CodeableConcept }, // Code that identifies this medication
  "status" : "<code>", // active | inactive | entered-in-error
  "manufacturer" : { Reference(Organization) }, // Manufacturer of the item
  "doseForm" : { CodeableConcept }, // powder | tablets | capsule +
  "amount" : { Quantity(SimpleQuantity) }, // Amount of drug in package
  "synonym" : ["<string>"], // Additional names for a medication
  "relatedMedicationKnowledge" : [{ // Associated or related medication information
    "type" : { CodeableConcept }, // R!  Category of medicationKnowledge
    "reference" : [{ Reference(MedicationKnowledge) }] // R!  Associated documentation about the associated medication
    knowledge
  }],
  "associatedMedication" : [{ Reference(Medication) }], // A medication resource that is associated with this medication
  "productType" : [{ CodeableConcept }], // Category of the medication or product
  "monograph" : [{ // Associated documentation about the medication
    "type" : { CodeableConcept }, // The category of medication document
    "source" : { Reference(DocumentReference|Media) } // Associated documentation about the medication
  }],
  "ingredient" : [{ // Active or inactive ingredient
    // item[x]: Medication(s) or substance(s) contained in the medication. One of these 2:
    "itemCodeableConcept" : { CodeableConcept },
    "itemReference" : { Reference(Substance) },
    "isActive" : <boolean>, // Active ingredient indicator
    "strength" : { Ratio } // Quantity of ingredient present
  }],
  "preparationInstruction" : "<markdown>", // The instructions for preparing the medication
  "intendedRoute" : [{ CodeableConcept }], // The intended or approved route of administration
  "cost" : [{ // The pricing of the medication
    "type" : { CodeableConcept }, // R!  The category of the cost information
    "source" : "<string>", // The source or owner for the price information
    "cost" : { Money } // R!  The price of the medication
  }],
  "monitoringProgram" : [{ // Program under which a medication is reviewed
    "type" : { CodeableConcept }, // Type of program under which the medication is monitored
    "name" : "<string>" // Name of the reviewing program
  }],
  "administrationGuidelines" : [{ // Guidelines for administration of the medication
    "dosage" : [{ // Dosage for the medication for the specific guidelines
      "type" : { CodeableConcept }, // R!  Type of dosage
      "dosage" : [{ Dosage }] // R!  Dosage for the medication for the specific guidelines
    }],
    // indication[x]: Indication for use that apply to the specific administration guidelines. One of these 2:
    "indicationCodeableConcept" : { CodeableConcept },
    "indicationReference" : { Reference(ObservationDefinition) },
    "patientCharacteristics" : [{ // Characteristics of the patient that are relevant to the administration guidelines
      // characteristic[x]: Specific characteristic that is relevant to the administration guideline. One of these 2:
      "characteristicCodeableConcept" : { CodeableConcept },
      "characteristicQuantity" : { Quantity(SimpleQuantity) },
      "value" : ["<string>"] // The specific characteristic
    }]
  }],
  "medicineClassification" : [{ // Categorization of the medication within a formulary or classification system
    "type" : { CodeableConcept }, // R!  The type of category for the medication (for example, therapeutic
    classification, therapeutic sub-classification)
    "classification" : [{ CodeableConcept }] // Specific category assigned to the medication
  }],
```

```
  "packaging" : { // Details about packaged medications
    "type" : { CodeableConcept }, // A code that defines the specific type of packaging that the medication can be found
     in
    "quantity" : { Quantity(SimpleQuantity) } // The number of product units the package would contain if fully loaded
  },
  "drugCharacteristic" : [{ // Specifies descriptive properties of the medicine
    "type" : { CodeableConcept }, // Code specifying the type of characteristic of medication
    // value[x]: Description of the characteristic. One of these 4:
    "valueCodeableConcept" : { CodeableConcept }
    "valueString" : "<string>"
    "valueQuantity" : { Quantity(SimpleQuantity) }
    "valueBase64Binary" : "<base64Binary>"
  }],
  "contraindication" : [{ Reference(DetectedIssue) }], // Potential clinical issue with or between medication(s)
  "regulatory" : [{ // Regulatory information about a medication
    "regulatoryAuthority" : { Reference(Organization) }, // R!  Specifies the authority of the regulation
    "substitution" : [{ // Specifies if changes are allowed when dispensing a medication from a regulatory perspective
      "type" : { CodeableConcept }, // R!  Specifies the type of substitution allowed
      "allowed" : <boolean> // R!  Specifies if regulation allows for changes in the medication when dispensing
    }],
    "schedule" : [{ // Specifies the schedule of a medication in jurisdiction
      "schedule" : { CodeableConcept } // R!  Specifies the specific drug schedule
    }],
    "maxDispense" : { // The maximum number of units of the medication that can be dispensed in a period
      "quantity" : { Quantity(SimpleQuantity) }, // R!  The maximum number of units of the medication that can be
    dispensed
      "period" : { Duration } // The period that applies to the maximum number of units
    }
  }],
  "kinetics" : [{ // The time course of drug absorption, distribution, metabolism and excretion of a medication from the
     body
    "areaUnderCurve" : [{ Quantity(SimpleQuantity) }], // The drug concentration measured at certain discrete points in
    time
    "lethalDose50" : [{ Quantity(SimpleQuantity) }], // The median lethal dose of a drug
    "halfLifePeriod" : { Duration } // Time required for concentration in the body to decrease by half
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicationKnowledge;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicationKnowledge.code [ CodeableConcept ]; # 0..1 Code that identifies this medication
  fhir:MedicationKnowledge.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:MedicationKnowledge.manufacturer [ Reference(Organization) ]; # 0..1 Manufacturer of the item
  fhir:MedicationKnowledge.doseForm [ CodeableConcept ]; # 0..1 powder | tablets | capsule +
  fhir:MedicationKnowledge.amount [ Quantity(SimpleQuantity) ]; # 0..1 Amount of drug in package
  fhir:MedicationKnowledge.synonym [ string ], ... ; # 0..* Additional names for a medication
  fhir:MedicationKnowledge.relatedMedicationKnowledge [ # 0..* Associated or related medication information
    fhir:MedicationKnowledge.relatedMedicationKnowledge.type [ CodeableConcept ]; # 1..1 Category of medicationKnowledge
    fhir:MedicationKnowledge.relatedMedicationKnowledge.reference [ Reference(MedicationKnowledge) ], ... ; # 1..*
    Associated documentation about the associated medication knowledge
  ], ...;
  fhir:MedicationKnowledge.associatedMedication [ Reference(Medication) ], ... ; # 0..* A medication resource that is
    associated with this medication
  fhir:MedicationKnowledge.productType [ CodeableConcept ], ... ; # 0..* Category of the medication or product
  fhir:MedicationKnowledge.monograph [ # 0..* Associated documentation about the medication
    fhir:MedicationKnowledge.monograph.type [ CodeableConcept ]; # 0..1 The category of medication document
    fhir:MedicationKnowledge.monograph.source [ Reference(DocumentReference|Media) ]; # 0..1 Associated documentation
    about the medication
  ], ...;
  fhir:MedicationKnowledge.ingredient [ # 0..* Active or inactive ingredient
    # MedicationKnowledge.ingredient.item[x] : 1..1 Medication(s) or substance(s) contained in the medication. One of
    these 2
      fhir:MedicationKnowledge.ingredient.itemCodeableConcept [ CodeableConcept ]
      fhir:MedicationKnowledge.ingredient.itemReference [ Reference(Substance) ]
    fhir:MedicationKnowledge.ingredient.isActive [ boolean ]; # 0..1 Active ingredient indicator
    fhir:MedicationKnowledge.ingredient.strength [ Ratio ]; # 0..1 Quantity of ingredient present
  ], ...;
  fhir:MedicationKnowledge.preparationInstruction [ markdown ]; # 0..1 The instructions for preparing the medication
  fhir:MedicationKnowledge.intendedRoute [ CodeableConcept ], ... ; # 0..* The intended or approved route of
    administration
  fhir:MedicationKnowledge.cost [ # 0..* The pricing of the medication
    fhir:MedicationKnowledge.cost.type [ CodeableConcept ]; # 1..1 The category of the cost information
    fhir:MedicationKnowledge.cost.source [ string ]; # 0..1 The source or owner for the price information
    fhir:MedicationKnowledge.cost.cost [ Money ]; # 1..1 The price of the medication
  ], ...;
  fhir:MedicationKnowledge.monitoringProgram [ # 0..* Program under which a medication is reviewed
    fhir:MedicationKnowledge.monitoringProgram.type [ CodeableConcept ]; # 0..1 Type of program under which the
    medication is monitored
    fhir:MedicationKnowledge.monitoringProgram.name [ string ]; # 0..1 Name of the reviewing program
  ], ...;
  fhir:MedicationKnowledge.administrationGuidelines [ # 0..* Guidelines for administration of the medication
    fhir:MedicationKnowledge.administrationGuidelines.dosage [ # 0..* Dosage for the medication for the specific
```

```
    guidelines
      fhir:MedicationKnowledge.administrationGuidelines.dosage.type [ CodeableConcept ]; # 1..1 Type of dosage
      fhir:MedicationKnowledge.administrationGuidelines.dosage.dosage [ Dosage ], ... ; # 1..* Dosage for the medication
     for the specific guidelines
    ], ...;
    # MedicationKnowledge.administrationGuidelines.indication[x] : 0..1 Indication for use that apply to the specific
    administration guidelines. One of these 2
      fhir:MedicationKnowledge.administrationGuidelines.indicationCodeableConcept [ CodeableConcept ]
      fhir:MedicationKnowledge.administrationGuidelines.indicationReference [ Reference(ObservationDefinition) ]
    fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics [ # 0..* Characteristics of the patient
    that are relevant to the administration guidelines
      # MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristic[x] : 1..1 Specific
    characteristic that is relevant to the administration guideline. One of these 2
        fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristicCodeableConcept [
    CodeableConcept ]
        fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristicSimpleQuantity [
    Quantity(SimpleQuantity) ]
      fhir:MedicationKnowledge.administrationGuidelines.patientCharacteristics.value [ string ], ... ; # 0..* The
    specific characteristic
    ], ...;
  ], ...;
  fhir:MedicationKnowledge.medicineClassification [ # 0..* Categorization of the medication within a formulary or
    classification system
    fhir:MedicationKnowledge.medicineClassification.type [ CodeableConcept ]; # 1..1 The type of category for the
    medication (for example, therapeutic classification, therapeutic sub-classification)
    fhir:MedicationKnowledge.medicineClassification.classification [ CodeableConcept ], ... ; # 0..* Specific category
    assigned to the medication
  ], ...;
  fhir:MedicationKnowledge.packaging [ # 0..1 Details about packaged medications
    fhir:MedicationKnowledge.packaging.type [ CodeableConcept ]; # 0..1 A code that defines the specific type of
    packaging that the medication can be found in
    fhir:MedicationKnowledge.packaging.quantity [ Quantity(SimpleQuantity) ]; # 0..1 The number of product units the
    package would contain if fully loaded
  ];
  fhir:MedicationKnowledge.drugCharacteristic [ # 0..* Specifies descriptive properties of the medicine
    fhir:MedicationKnowledge.drugCharacteristic.type [ CodeableConcept ]; # 0..1 Code specifying the type of
    characteristic of medication
    # MedicationKnowledge.drugCharacteristic.value[x] : 0..1 Description of the characteristic. One of these 4
      fhir:MedicationKnowledge.drugCharacteristic.valueCodeableConcept [ CodeableConcept ]
      fhir:MedicationKnowledge.drugCharacteristic.valueString [ string ]
      fhir:MedicationKnowledge.drugCharacteristic.valueSimpleQuantity [ Quantity(SimpleQuantity) ]
      fhir:MedicationKnowledge.drugCharacteristic.valueBase64Binary [ base64Binary ]
  ], ...;
  fhir:MedicationKnowledge.contraindication [ Reference(DetectedIssue) ], ... ; # 0..* Potential clinical issue with or
    between medication(s)
  fhir:MedicationKnowledge.regulatory [ # 0..* Regulatory information about a medication
    fhir:MedicationKnowledge.regulatory.regulatoryAuthority [ Reference(Organization) ]; # 1..1 Specifies the authority
    of the regulation
    fhir:MedicationKnowledge.regulatory.substitution [ # 0..* Specifies if changes are allowed when dispensing a
    medication from a regulatory perspective
      fhir:MedicationKnowledge.regulatory.substitution.type [ CodeableConcept ]; # 1..1 Specifies the type of
    substitution allowed
      fhir:MedicationKnowledge.regulatory.substitution.allowed [ boolean ]; # 1..1 Specifies if regulation allows for
    changes in the medication when dispensing
    ], ...;
    fhir:MedicationKnowledge.regulatory.schedule [ # 0..* Specifies the schedule of a medication in jurisdiction
      fhir:MedicationKnowledge.regulatory.schedule.schedule [ CodeableConcept ]; # 1..1 Specifies the specific drug
    schedule
    ], ...;
    fhir:MedicationKnowledge.regulatory.maxDispense [ # 0..1 The maximum number of units of the medication that can be
    dispensed in a period
      fhir:MedicationKnowledge.regulatory.maxDispense.quantity [ Quantity(SimpleQuantity) ]; # 1..1 The maximum number
    of units of the medication that can be dispensed
      fhir:MedicationKnowledge.regulatory.maxDispense.period [ Duration ]; # 0..1 The period that applies to the maximum
     number of units
    ];
  ], ...;
  fhir:MedicationKnowledge.kinetics [ # 0..* The time course of drug absorption, distribution, metabolism and excretion
    of a medication from the body
    fhir:MedicationKnowledge.kinetics.areaUnderCurve [ Quantity(SimpleQuantity) ], ... ; # 0..* The drug concentration
    measured at certain discrete points in time
    fhir:MedicationKnowledge.kinetics.lethalDose50 [ Quantity(SimpleQuantity) ], ... ; # 0..* The median lethal dose of
    a drug
    fhir:MedicationKnowledge.kinetics.halfLifePeriod [ Duration ]; # 0..1 Time required for concentration in the body to
     decrease by half
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.6.2.1 Terminology Bindings

Path | Definition | Type | Reference

MedicationKnowledge.code | A coded concept that defines the type of a medication. | Example | SNOMEDCTMedicationCodes
MedicationKnowledge.status | A coded concept defining if the medication is in active use. | Required | MedicationKnowledge Status Codes
MedicationKnowledge.doseForm | A coded concept defining the form of a medication. | Example | SNOMEDCTFormCodes
MedicationKnowledge.intendedRoute | A coded concept defining the intended route of administration. | Example | SNOMEDCTRouteCodes
MedicationKnowledge.packaging.type | A coded concept defining the type of packaging of a medication. | Example | medicationKnowledge Package Type Codes
MedicationKnowledge.drugCharacteristic.type | A coded concept defining the characteristic types of a medication. | Example | medicationKnowledge Characteristic Codes

11.6.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
classification | token | Specific category assigned to the medication | MedicationKnowledge.medicineClassification.classification
classification-type | token | The type of category for the medication (for example, therapeutic classification, therapeutic sub-classification) | MedicationKnowledge.medicineClassification.type
code | token | Code that identifies this medication | MedicationKnowledge.code
doseform | token | powder | tablets | capsule + | MedicationKnowledge.doseForm
ingredient | reference | Medication(s) or substance(s) contained in the medication | (MedicationKnowledge.ingredient.item as Reference) (Substance)
ingredient-code | token | Medication(s) or substance(s) contained in the medication | (MedicationKnowledge.ingredient.item as CodeableConcept)
manufacturer | reference | Manufacturer of the item | MedicationKnowledge.manufacturer (Organization)
monitoring-program-name | token | Name of the reviewing program | MedicationKnowledge.monitoringProgram.name
monitoring-program-type | token | Type of program under which the medication is monitored | MedicationKnowledge.monitoringProgram.type
monograph | reference | Associated documentation about the medication | MedicationKnowledge.monograph.source (Media, DocumentReference)
monograph-type | token | The category of medication document | MedicationKnowledge.monograph.type
source-cost | token | The source or owner for the price information | MedicationKnowledge.cost.source
status | token | active | inactive | entered-in-error | MedicationKnowledge.status

11.6.4 Resource MedicationKnowledge - Examples

URL: https://hl7.org/fhir/R4/medicationknowledge-examples.html

11.6.4 Resource MedicationKnowledge - Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicationknowledge | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.6.5 Resource MedicationKnowledge - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicationknowledge-definitions.html

11.6.5 Resource MedicationKnowledge - Detailed Descriptions

Pharmacy Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicationKnowledge resource.

MedicationKnowledge
Element Id | MedicationKnowledge
Definition | Information about a medication that is used to support knowledge.
Cardinality | 0..*
Type | DomainResource
MedicationKnowledge.code
Element Id | MedicationKnowledge.code
Definition | A code that specifies this medication, or a textual description if no code is available. Usage note: This could be a standard medication code such as a code from RxNorm, SNOMED CT, IDMP etc. It could also be a national or local formulary code, optionally with translations to other code systems.
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept
Summary | true

Comments | Depending on the context of use, the code that was actually selected by the user (prescriber, dispenser, etc.) will have the coding.userSelected set to true. As described in the coding datatype: "A coding may be marked as a "userSelected" if a user selected the particular coded value in a user interface (e.g. the user selects an item in a pick-list). If a user selected coding exists, it is the preferred choice for performing translations etc. Other codes can only be literal translations to alternative code systems, or codes at a lower level of granularity (e.g. a generic code for a vendor-specific primary one).
MedicationKnowledge.status
Element Id | MedicationKnowledge.status
Definition | A code to indicate if the medication is in active use. The status refers to the validity about the information of the medication and not to its medicinal properties.
Cardinality | 0..1
Terminology Binding | Medication knowledge status codes (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes.)
Summary | true
Comments | This status is intended to identify if the medication in a local system is in active use within a drug database or inventory. For example, a pharmacy system may create a new drug file record for a compounded product "ABC Hospital Special Cream" with an active status. At some point in the future, it may be determined that the drug record was created with an error and the status is changed to "entered in error". This status is not intended to specify if a medication is part of a particular formulary. It is possible that the drug record may be referenced by multiple formularies or catalogues and each of those entries would have a separate status.
MedicationKnowledge.manufacturer
Element Id | MedicationKnowledge.manufacturer
Definition | Describes the details of the manufacturer of the medication product. This is not intended to represent the distributor of a medication product.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
MedicationKnowledge.doseForm
Element Id | MedicationKnowledge.doseForm
Definition | Describes the form of the item. Powder; tablets; capsule.
Cardinality | 0..1
Terminology Binding | SNOMED CT Form Codes (Example)
Type | CodeableConcept
Comments | When Medication is referenced from MedicationRequest, this is the ordered form. When Medication is referenced within MedicationDispense, this is the dispensed form. When Medication is referenced within MedicationAdministration, this is administered form.
MedicationKnowledge.amount
Element Id | MedicationKnowledge.amount
Definition | Specific amount of the drug in the packaged product. For example, when specifying a product that has the same strength (For example, Insulin glargine 100 unit per mL solution for injection), this attribute provides additional clarification of the package amount (For example, 3 mL, 10mL, etc.).
Cardinality | 0..1
Type | SimpleQuantity
Summary | true
Comments | This is the quantity of medication in a package. To specify the strength of the medication, the Ingredient.strength attribute is used.
MedicationKnowledge.synonym
Element Id | MedicationKnowledge.synonym
Definition | Additional names for a medication, for example, the name(s) given to a medication in different countries. For example, acetaminophen and paracetamol or salbutamol and albuterol.
Cardinality | 0..*
Type | string
Summary | true
MedicationKnowledge.relatedMedicationKnowledge
Element Id | MedicationKnowledge.relatedMedicationKnowledge
Definition | Associated or related knowledge about a medication.
Cardinality | 0..*
MedicationKnowledge.relatedMedicationKnowledge.type
Element Id | MedicationKnowledge.relatedMedicationKnowledge.type
Definition | The category of the associated medication knowledge reference.
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.relatedMedicationKnowledge.reference
Element Id | MedicationKnowledge.relatedMedicationKnowledge.reference
Definition | Associated documentation about the associated medication knowledge.
Cardinality | 1..*
Type | Reference(MedicationKnowledge)
MedicationKnowledge.associatedMedication
Element Id | MedicationKnowledge.associatedMedication
Definition | Associated or related medications. For example, if the medication is a branded product (e.g. Crestor), this is the Therapeutic Moeity (e.g. Rosuvastatin) or if this is a generic medication (e.g. Rosuvastatin), this would link to a branded product (e.g. Crestor).
Cardinality | 0..*
Type | Reference(Medication)

MedicationKnowledge.productType
Element Id | MedicationKnowledge.productType
Definition | Category of the medication or product (e.g. branded product, therapeutic moeity, generic product, innovator product, etc.).
Cardinality | 0..*
Type | CodeableConcept
MedicationKnowledge.monograph
Element Id | MedicationKnowledge.monograph
Definition | Associated documentation about the medication.
Cardinality | 0..*
MedicationKnowledge.monograph.type
Element Id | MedicationKnowledge.monograph.type
Definition | The category of documentation about the medication. (e.g. professional monograph, patient education monograph).
Cardinality | 0..1
Type | CodeableConcept
MedicationKnowledge.monograph.source
Element Id | MedicationKnowledge.monograph.source
Definition | Associated documentation about the medication.
Cardinality | 0..1
Type | Reference(DocumentReference | Media)
MedicationKnowledge.ingredient
Element Id | MedicationKnowledge.ingredient
Definition | Identifies a particular constituent of interest in the product.
Cardinality | 0..*
MedicationKnowledge.ingredient.item[x]
Element Id | MedicationKnowledge.ingredient.item[x]
Definition | The actual ingredient - either a substance (simple ingredient) or another medication.
Cardinality | 1..1
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
MedicationKnowledge.ingredient.isActive
Element Id | MedicationKnowledge.ingredient.isActive
Definition | Indication of whether this ingredient affects the therapeutic action of the drug.
Cardinality | 0..1
Type | boolean
Requirements | True indicates that the ingredient affects the therapeutic action of the drug (i.e. active). False indicates that the ingredient does not affect the therapeutic action of the drug (i.e. inactive).
MedicationKnowledge.ingredient.strength
Element Id | MedicationKnowledge.ingredient.strength
Definition | Specifies how many (or how much) of the items there are in this Medication. For example, 250 mg per tablet. This is expressed as a ratio where the numerator is 250mg and the denominator is 1 tablet.
Cardinality | 0..1
Type | Ratio
MedicationKnowledge.preparationInstruction
Element Id | MedicationKnowledge.preparationInstruction
Definition | The instructions for preparing the medication.
Cardinality | 0..1
Type | markdown
MedicationKnowledge.intendedRoute
Element Id | MedicationKnowledge.intendedRoute
Definition | The intended or approved route of administration.
Cardinality | 0..*
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
MedicationKnowledge.cost
Element Id | MedicationKnowledge.cost
Definition | The price of the medication.
Cardinality | 0..*
MedicationKnowledge.cost.type
Element Id | MedicationKnowledge.cost.type
Definition | The category of the cost information. For example, manufacturers' cost, patient cost, claim reimbursement cost, actual acquisition cost.
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.cost.source
Element Id | MedicationKnowledge.cost.source
Definition | The source or owner that assigns the price to the medication.
Cardinality | 0..1
Type | string
MedicationKnowledge.cost.cost
Element Id | MedicationKnowledge.cost.cost
Definition | The price of the medication.
Cardinality | 1..1

Type | Money
MedicationKnowledge.monitoringProgram
Element Id | MedicationKnowledge.monitoringProgram
Definition | The program under which the medication is reviewed.
Cardinality | 0..*
MedicationKnowledge.monitoringProgram.type
Element Id | MedicationKnowledge.monitoringProgram.type
Definition | Type of program under which the medication is monitored.
Cardinality | 0..1
Type | CodeableConcept
MedicationKnowledge.monitoringProgram.name
Element Id | MedicationKnowledge.monitoringProgram.name
Definition | Name of the reviewing program.
Cardinality | 0..1
Type | string
MedicationKnowledge.administrationGuidelines
Element Id | MedicationKnowledge.administrationGuidelines
Definition | Guidelines for the administration of the medication.
Cardinality | 0..*
MedicationKnowledge.administrationGuidelines.dosage
Element Id | MedicationKnowledge.administrationGuidelines.dosage
Definition | Dosage for the medication for the specific guidelines.
Cardinality | 0..*
MedicationKnowledge.administrationGuidelines.dosage.type
Element Id | MedicationKnowledge.administrationGuidelines.dosage.type
Definition | The type of dosage (for example, prophylaxis, maintenance, therapeutic, etc.).
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.administrationGuidelines.dosage.dosage
Element Id | MedicationKnowledge.administrationGuidelines.dosage.dosage
Definition | Dosage for the medication for the specific guidelines.
Cardinality | 1..*
Type | Dosage
MedicationKnowledge.administrationGuidelines.indication[x]
Element Id | MedicationKnowledge.administrationGuidelines.indication[x]
Definition | Indication for use that apply to the specific administration guidelines.
Cardinality | 0..1
Type | CodeableConcept|Reference(ObservationDefinition)
[x] Note | See Choice of Data Types for further information about how to use [x]
MedicationKnowledge.administrationGuidelines.patientCharacteristics
Element Id | MedicationKnowledge.administrationGuidelines.patientCharacteristics
Definition | Characteristics of the patient that are relevant to the administration guidelines (for example, height, weight, gender, etc.).
Cardinality | 0..*
MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristic[x]
Element Id | MedicationKnowledge.administrationGuidelines.patientCharacteristics.characteristic[x]
Definition | Specific characteristic that is relevant to the administration guideline (e.g. height, weight, gender).
Cardinality | 1..1
Type | CodeableConcept|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
MedicationKnowledge.administrationGuidelines.patientCharacteristics.value
Element Id | MedicationKnowledge.administrationGuidelines.patientCharacteristics.value
Definition | The specific characteristic (e.g. height, weight, gender, etc.).
Cardinality | 0..*
Type | string
MedicationKnowledge.medicineClassification
Element Id | MedicationKnowledge.medicineClassification
Definition | Categorization of the medication within a formulary or classification system.
Cardinality | 0..*
MedicationKnowledge.medicineClassification.type
Element Id | MedicationKnowledge.medicineClassification.type
Definition | The type of category for the medication (for example, therapeutic classification, therapeutic sub-classification).
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.medicineClassification.classification
Element Id | MedicationKnowledge.medicineClassification.classification
Definition | Specific category assigned to the medication (e.g. anti-infective, anti-hypertensive, antibiotic, etc.).
Cardinality | 0..*
Type | CodeableConcept
MedicationKnowledge.packaging
Element Id | MedicationKnowledge.packaging
Definition | Information that only applies to packages (not products).

Cardinality | 0..1
MedicationKnowledge.packaging.type
Element Id | MedicationKnowledge.packaging.type
Definition | A code that defines the specific type of packaging that the medication can be found in (e.g. blister sleeve, tube, bottle).
Cardinality | 0..1
Terminology Binding | Medication knowledge package type codes (Example)
Type | CodeableConcept
MedicationKnowledge.packaging.quantity
Element Id | MedicationKnowledge.packaging.quantity
Definition | The number of product units the package would contain if fully loaded.
Cardinality | 0..1
Type | SimpleQuantity
MedicationKnowledge.drugCharacteristic
Element Id | MedicationKnowledge.drugCharacteristic
Definition | Specifies descriptive properties of the medicine, such as color, shape, imprints, etc.
Cardinality | 0..*
MedicationKnowledge.drugCharacteristic.type
Element Id | MedicationKnowledge.drugCharacteristic.type
Definition | A code specifying which characteristic of the medicine is being described (for example, colour, shape, imprint).
Cardinality | 0..1
Terminology Binding | Medication knowledge characteristic codes (Example)
Type | CodeableConcept
MedicationKnowledge.drugCharacteristic.value[x]
Element Id | MedicationKnowledge.drugCharacteristic.value[x]
Definition | Description of the characteristic.
Cardinality | 0..1
Type | CodeableConcept|string|SimpleQuantity|base64Binary
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The description should be provided as a CodeableConcept, SimpleQuantity or an image. The description can be a string only when these others are not available.
MedicationKnowledge.contraindication
Element Id | MedicationKnowledge.contraindication
Definition | Potential clinical issue with or between medication(s) (for example, drug-drug interaction, drug-disease contraindication, drug-allergy interaction, etc.).
Cardinality | 0..*
Type | Reference(DetectedIssue)
MedicationKnowledge.regulatory
Element Id | MedicationKnowledge.regulatory
Definition | Regulatory information about a medication.
Cardinality | 0..*
MedicationKnowledge.regulatory.regulatoryAuthority
Element Id | MedicationKnowledge.regulatory.regulatoryAuthority
Definition | The authority that is specifying the regulations.
Cardinality | 1..1
Type | Reference(Organization)
MedicationKnowledge.regulatory.substitution
Element Id | MedicationKnowledge.regulatory.substitution
Definition | Specifies if changes are allowed when dispensing a medication from a regulatory perspective.
Cardinality | 0..*
MedicationKnowledge.regulatory.substitution.type
Element Id | MedicationKnowledge.regulatory.substitution.type
Definition | Specifies the type of substitution allowed.
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.regulatory.substitution.allowed
Element Id | MedicationKnowledge.regulatory.substitution.allowed
Definition | Specifies if regulation allows for changes in the medication when dispensing.
Cardinality | 1..1
Type | boolean
MedicationKnowledge.regulatory.schedule
Element Id | MedicationKnowledge.regulatory.schedule
Definition | Specifies the schedule of a medication in jurisdiction.
Cardinality | 0..*
MedicationKnowledge.regulatory.schedule.schedule
Element Id | MedicationKnowledge.regulatory.schedule.schedule
Definition | Specifies the specific drug schedule.
Cardinality | 1..1
Type | CodeableConcept
MedicationKnowledge.regulatory.maxDispense
Element Id | MedicationKnowledge.regulatory.maxDispense
Definition | The maximum number of units of the medication that can be dispensed in a period.

Cardinality | 0..1
MedicationKnowledge.regulatory.maxDispense.quantity
Element Id | MedicationKnowledge.regulatory.maxDispense.quantity
Definition | The maximum number of units of the medication that can be dispensed.
Cardinality | 1..1
Type | SimpleQuantity
MedicationKnowledge.regulatory.maxDispense.period
Element Id | MedicationKnowledge.regulatory.maxDispense.period
Definition | The period that applies to the maximum number of units.
Cardinality | 0..1
Type | Duration
MedicationKnowledge.kinetics
Element Id | MedicationKnowledge.kinetics
Definition | The time course of drug absorption, distribution, metabolism and excretion of a medication from the body.
Cardinality | 0..*
MedicationKnowledge.kinetics.areaUnderCurve
Element Id | MedicationKnowledge.kinetics.areaUnderCurve
Definition | The drug concentration measured at certain discrete points in time.
Cardinality | 0..*
Type | SimpleQuantity
MedicationKnowledge.kinetics.lethalDose50
Element Id | MedicationKnowledge.kinetics.lethalDose50
Definition | The median lethal dose of a drug.
Cardinality | 0..*
Type | SimpleQuantity
MedicationKnowledge.kinetics.halfLifePeriod
Element Id | MedicationKnowledge.kinetics.halfLifePeriod
Definition | The time required for any specified property (e.g., the concentration of a substance in the body) to decrease by half.
Cardinality | 0..1
Type | Duration

11.6.6 Resource MedicationKnowledge - Mappings

URL: https://hl7.org/fhir/R4/medicationknowledge-mappings.html

11.6.6 Resource MedicationKnowledge - Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mapping to NCPDP SCRIPT 10.6

Mappings for the medicationknowledge resource (see Mappings to Other Standards for further information & status).

11.6.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MedicationKnowledge
code | FiveWs.class
manufacturer | FiveWs.actor

11.6.6.2 HL7 v2 Mapping (http://hl7.org/v2)

MedicationKnowledge
code | RXO-1.1-Requested Give Code.code / RXE-2.1-Give Code.code / RXD-2.1-Dispense/Give Code.code / RXG-4.1-Give Code.code /RXA-5.1-Administered Code.code / RXC-2.1 Component Code
status
manufacturer | RXD-20-Substance Manufacturer Name / RXG-21-Substance Manufacturer Name / RXA-17-Substance Manufacturer Name
doseForm | RXO-5-Requested Dosage Form / RXE-6-Give Dosage Form / RXD-6-Actual Dosage Form / RXG-8-Give Dosage Form / RXA-8-Administered Dosage Form
amount
synonym
relatedMedicationKnowledge
type
reference
associatedMedication
productType
monograph
type

source
ingredient
item[x] | RXC-2-Component Code if medication: RXO-1-Requested Give Code / RXE-2-Give Code / RXD-2-Dispense/Give Code / RXG-4-Give Code / RXA-5-Administered Code
isActive
strength | RXC-3-Component Amount & RXC-4-Component Units if medication: RXO-2-Requested Give Amount - Minimum & RXO-4-Requested Give Units / RXO-3-Requested Give Amount - Maximum & RXO-4-Requested Give Units / RXO-11-Requested Dispense Amount & RXO-12-Requested Dispense Units / RXE-3-Give Amount - Minimum & RXE-5-Give Units / RXE-4-Give Amount - Maximum & RXE-5-Give Units / RXE-10-Dispense Amount & RXE-10-Dispense Units
preparationInstruction
intendedRoute
cost
type
source
cost
monitoringProgram
type
name
administrationGuidelines
dosage
type
dosage
indication[x]
patientCharacteristics
characteristic[x]
value
medicineClassification
type
classification
packaging
type
quantity
drugCharacteristic
type
value[x]
contraindication
regulatory
regulatoryAuthority
substitution
type
allowed
schedule
schedule
maxDispense
quantity
period
kinetics
areaUnderCurve
lethalDose50
halfLifePeriod

11.6.6.3 RIM Mapping (http://hl7.org/v3)

MedicationKnowledge | Todo
code | .code
status | .statusCode
manufacturer | .player.scopingRole[typeCode=MANU].scoper
doseForm | .formCode
amount | .quantity
item[x] | .player
isActive | NA
strength | .quantity

11.6.6.4 Mapping to NCPDP SCRIPT 10.6 (http://ncpdp.org/SCRIPT10_6)

MedicationKnowledge
code | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system = //element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
status
manufacturer | no mapping
doseForm | coding.code = //element(*,DrugCodedType)/FormCode coding.system = //element(*,DrugCodedType)/FormSourceCode
amount
synonym

relatedMedicationKnowledge
type
reference
associatedMedication
productType
monograph
type
source
ingredient
item[x] | coding.code = //element(*,MedicationType)/DrugCoded/ProductCode coding.system = //element(*,MedicationType)/DrugCoded/ProductCodeQualifier coding.display = //element(*,MedicationType)/DrugDescription
isActive
strength | //element(*,DrugCodedType)/Strength
preparationInstruction
intendedRoute
cost
type
source
cost
monitoringProgram
type
name
administrationGuidelines
dosage
type
dosage
indication[x]
patientCharacteristics
characteristic[x]
value
medicineClassification
type
classification
packaging
type
quantity
drugCharacteristic
type
value[x]
contraindication
regulatory
regulatoryAuthority
substitution
type
allowed
schedule
schedule
maxDispense
quantity
period
kinetics
areaUnderCurve
lethalDose50
halfLifePeriod

11.6.7 Resource MedicationKnowledge - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicationknowledge-profiles.html

11.6.7 Resource MedicationKnowledge - Extensions & Profiles

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicationKnowledge resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.6.7.1 Profiles

No Profiles defined for this resource

11.6.7.2 Extensions

No Extensions defined for this resource

Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.6.7.3 Search Extensions

No Search Extensions defined for this resource

11.6.8 JSON Schema for MedicationKnowledge

URL: https://hl7.org/fhir/R4/medicationknowledge.schema.json.html

11.6.8 JSON Schema for MedicationKnowledge

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicationKnowledge, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicationKnowledge",
  "$ref": "#/definitions/MedicationKnowledge",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicationKnowledge": {
      "description": "Information about a medication that is used to support knowledge.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicationKnowledge resource",
          "const": "MedicationKnowledge"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
```

```
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code that specifies this medication, or a textual description if no code is available. Usage
 note: This could be a standard medication code such as a code from RxNorm, SNOMED CT, IDMP etc. It could also be a
national or local formulary code, optionally with translations to other code systems.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "A code to indicate if the medication is in active use.  The status refers to the validity
about the information of the medication and not to its medicinal properties.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "manufacturer": {
      "description": "Describes the details of the manufacturer of the medication product.  This is not intended to
represent the distributor of a medication product.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "doseForm": {
      "description": "Describes the form of the item.  Powder; tablets; capsule.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amount": {
      "description": "Specific amount of the drug in the packaged product.  For example, when specifying a product
that has the same strength (For example, Insulin glargine 100 unit per mL solution for injection), this attribute
provides additional clarification of the package amount (For example, 3 mL, 10mL, etc.).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "synonym": {
      "description": "Additional names for a medication, for example, the name(s) given to a medication in different
 countries.  For example, acetaminophen and paracetamol or salbutamol and albuterol.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_synonym": {
      "description": "Extensions for synonym",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "relatedMedicationKnowledge": {
      "description": "Associated or related knowledge about a medication.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_RelatedMedicationKnowledge"
      },
      "type": "array"
    },
    "associatedMedication": {
      "description": "Associated or related medications.  For example, if the medication is a branded product (e.g.
Crestor), this is the Therapeutic Moeity (e.g. Rosuvastatin) or if this is a generic medication (e.g. Rosuvastatin),
 this would link to a branded product (e.g. Crestor).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "productType": {
      "description": "Category of the medication or product (e.g. branded product, therapeutic moeity, generic
```

```
product, innovator product, etc.).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "monograph": {
      "description": "Associated documentation about the medication.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Monograph"
      },
      "type": "array"
    },
    "ingredient": {
      "description": "Identifies a particular constituent of interest in the product.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Ingredient"
      },
      "type": "array"
    },
    "preparationInstruction": {
      "description": "The instructions for preparing the medication.",
      "$ref": "#/definitions/markdown"
    },
    "_preparationInstruction": {
      "description": "Extensions for preparationInstruction",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "intendedRoute": {
      "description": "The intended or approved route of administration.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "cost": {
      "description": "The price of the medication.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Cost"
      },
      "type": "array"
    },
    "monitoringProgram": {
      "description": "The program under which the medication is reviewed.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_MonitoringProgram"
      },
      "type": "array"
    },
    "administrationGuidelines": {
      "description": "Guidelines for the administration of the medication.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_AdministrationGuidelines"
      },
      "type": "array"
    },
    "medicineClassification": {
      "description": "Categorization of the medication within a formulary or classification system.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_MedicineClassification"
      },
      "type": "array"
    },
    "packaging": {
      "description": "Information that only applies to packages (not products).",
      "$ref": "#/definitions/MedicationKnowledge_Packaging"
    },
    "drugCharacteristic": {
      "description": "Specifies descriptive properties of the medicine, such as color, shape, imprints, etc.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_DrugCharacteristic"
      },
      "type": "array"
    },
    "contraindication": {
      "description": "Potential clinical issue with or between medication(s) (for example, drug-drug interaction,
drug-disease contraindication, drug-allergy interaction, etc.).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "regulatory": {
      "description": "Regulatory information about a medication.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Regulatory"
      },
      "type": "array"
    },
```

```
    "kinetics": {
      "description": "The time course of drug absorption, distribution, metabolism and excretion of a medication
from the body.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Kinetics"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"MedicationKnowledge_RelatedMedicationKnowledge": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The category of the associated medication knowledge reference.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reference": {
      "description": "Associated documentation about the associated medication knowledge.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "reference",
    "type"
  ]
},
"MedicationKnowledge_Monograph": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
```

```
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The category of documentation about the medication. (e.g. professional monograph, patient
education monograph).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "source": {
      "description": "Associated documentation about the medication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_Ingredient": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "itemCodeableConcept": {
      "description": "The actual ingredient - either a substance (simple ingredient) or another medication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "itemReference": {
      "description": "The actual ingredient - either a substance (simple ingredient) or another medication.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "isActive": {
      "description": "Indication of whether this ingredient affects the therapeutic action of the drug.",
      "$ref": "#/definitions/boolean"
    },
    "_isActive": {
      "description": "Extensions for isActive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "strength": {
      "description": "Specifies how many (or how much) of the items there are in this Medication.  For example, 250
mg per tablet.  This is expressed as a ratio where the numerator is 250mg and the denominator is 1 tablet.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_Cost": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The category of the cost information.  For example, manufacturers\u0027 cost, patient cost,
claim reimbursement cost, actual acquisition cost.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "source": {
      "description": "The source or owner that assigns the price to the medication.",
      "$ref": "#/definitions/string"
    },
    "_source": {
      "description": "Extensions for source",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cost": {
      "description": "The price of the medication.",
      "$ref": "Money.schema.json#/definitions/Money"
    }
  },
  "additionalProperties": false,
  "required": [
    "cost",
    "type"
  ]
},
"MedicationKnowledge_MonitoringProgram": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of program under which the medication is monitored.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "name": {
      "description": "Name of the reviewing program.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
```

```
"MedicationKnowledge_AdministrationGuidelines": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "dosage": {
      "description": "Dosage for the medication for the specific guidelines.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Dosage"
      },
      "type": "array"
    },
    "indicationCodeableConcept": {
      "description": "Indication for use that apply to the specific administration guidelines.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "indicationReference": {
      "description": "Indication for use that apply to the specific administration guidelines.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "patientCharacteristics": {
      "description": "Characteristics of the patient that are relevant to the administration guidelines (for
example, height, weight, gender, etc.).",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_PatientCharacteristics"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_Dosage": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
    },
    "type": {
      "description": "The type of dosage (for example, prophylaxis, maintenance, therapeutic, etc.).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "dosage": {
      "description": "Dosage for the medication for the specific guidelines.",
      "items": {
        "$ref": "Dosage.schema.json#/definitions/Dosage"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "dosage",
    "type"
  ]
},
"MedicationKnowledge_PatientCharacteristics": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "characteristicCodeableConcept": {
      "description": "Specific characteristic that is relevant to the administration guideline (e.g. height, weight,
 gender).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "characteristicQuantity": {
      "description": "Specific characteristic that is relevant to the administration guideline (e.g. height, weight,
 gender).",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "value": {
      "description": "The specific characteristic (e.g. height, weight, gender, etc.).",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_value": {
      "description": "Extensions for value",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_MedicineClassification": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of category for the medication (for example, therapeutic classification, therapeutic
sub-classification).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "classification": {
      "description": "Specific category assigned to the medication (e.g. anti-infective, anti-hypertensive,
antibiotic, etc.).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"MedicationKnowledge_Packaging": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "A code that defines the specific type of packaging that the medication can be found in (e.g.
blister sleeve, tube, bottle).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "quantity": {
      "description": "The number of product units the package would contain if fully loaded.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_DrugCharacteristic": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
```

```
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "A code specifying which characteristic of the medicine is being described (for example,
colour, shape, imprint).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueCodeableConcept": {
      "description": "Description of the characteristic.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "valueString": {
      "description": "Description of the characteristic.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueQuantity": {
      "description": "Description of the characteristic.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "valueBase64Binary": {
      "description": "Description of the characteristic.",
      "pattern": "^(\\s*([0-9a-zA-Z\\+/\u003d]){4}\\s*)+$",
      "type": "string"
    },
    "_valueBase64Binary": {
      "description": "Extensions for valueBase64Binary",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"MedicationKnowledge_Regulatory": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "regulatoryAuthority": {
      "description": "The authority that is specifying the regulations.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "substitution": {
      "description": "Specifies if changes are allowed when dispensing a medication from a regulatory perspective.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Substitution"
      },
      "type": "array"
    },
    "schedule": {
      "description": "Specifies the schedule of a medication in jurisdiction.",
      "items": {
        "$ref": "#/definitions/MedicationKnowledge_Schedule"
      },
      "type": "array"
    },
    "maxDispense": {
      "description": "The maximum number of units of the medication that can be dispensed in a period.",
      "$ref": "#/definitions/MedicationKnowledge_MaxDispense"
    }
  },
  "additionalProperties": false,
  "required": [
    "regulatoryAuthority"
  ]
},
"MedicationKnowledge_Substitution": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Specifies the type of substitution allowed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "allowed": {
      "description": "Specifies if regulation allows for changes in the medication when dispensing.",
      "$ref": "#/definitions/boolean"
    },
    "_allowed": {
      "description": "Extensions for allowed",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"MedicationKnowledge_Schedule": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
```

```
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "schedule": {
      "description": "Specifies the specific drug schedule.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false,
  "required": [
    "schedule"
  ]
},
"MedicationKnowledge_MaxDispense": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "quantity": {
      "description": "The maximum number of units of the medication that can be dispensed.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "period": {
      "description": "The period that applies to the maximum number of units.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    }
  },
  "additionalProperties": false,
  "required": [
    "quantity"
  ]
},
"MedicationKnowledge_Kinetics": {
  "description": "Information about a medication that is used to support knowledge.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "areaUnderCurve": {
          "description": "The drug concentration measured at certain discrete points in time.",
          "items": {
            "$ref": "Quantity.schema.json#/definitions/Quantity"
          },
          "type": "array"
        },
        "lethalDose50": {
          "description": "The median lethal dose of a drug.",
          "items": {
            "$ref": "Quantity.schema.json#/definitions/Quantity"
          },
          "type": "array"
        },
        "halfLifePeriod": {
          "description": "The time required for any specified property (e.g., the concentration of a substance in the
    body) to decrease by half.",
          "$ref": "Duration.schema.json#/definitions/Duration"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.6.9 Resource MedicationKnowledge - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicationknowledge-dependencies.html

11.6.9 Resource MedicationKnowledge - Maturity Dependency View

Pharmacy Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicationKnowledge FMM level 0

Name | Card. | Type | Dependency Analysis
MedicationKnowledge | | DomainResource
code | 0..1 | CodeableConcept
status | 0..1 | code
manufacturer | 0..1 | Reference(Organization)
doseForm | 0..1 | CodeableConcept
amount | 0..1 | SimpleQuantity
synonym | 0..* | string
relatedMedicationKnowledge | 0..* | BackboneElement
type | 1..1 | CodeableConcept
reference | 1..* | Reference(MedicationKnowledge)
associatedMedication | 0..* | Reference(Medication)
productType | 0..* | CodeableConcept
monograph | 0..* | BackboneElement
type | 0..1 | CodeableConcept
source | 0..1 | Reference(DocumentReference | Media)
ingredient | 0..* | BackboneElement
item[x] | 1..1
itemCodeableConcept | | CodeableConcept
itemReference | | Reference(Substance
isActive | 0..1 | boolean
strength | 0..1 | Ratio

preparationInstruction | 0..1 | markdown
intendedRoute | 0..* | CodeableConcept
cost | 0..* | BackboneElement
type | 1..1 | CodeableConcept
source | 0..1 | string
cost | 1..1 | Money
monitoringProgram | 0..* | BackboneElement
type | 0..1 | CodeableConcept
name | 0..1 | string
administrationGuidelines | 0..* | BackboneElement
dosage | 0..* | BackboneElement
type | 1..1 | CodeableConcept
dosage | 1..* | Dosage
indication[x] | 0..1
indicationCodeableConcept | | CodeableConcept
indicationReference | | Reference(ObservationDefinition
patientCharacteristics | 0..* | BackboneElement
characteristic[x] | 1..1
characteristicCodeableConcept | | CodeableConcept
characteristicQuantity | | SimpleQuantity
value | 0..* | string
medicineClassification | 0..* | BackboneElement
type | 1..1 | CodeableConcept
classification | 0..* | CodeableConcept
packaging | 0..1 | BackboneElement
type | 0..1 | CodeableConcept
quantity | 0..1 | SimpleQuantity
drugCharacteristic | 0..* | BackboneElement
type | 0..1 | CodeableConcept
value[x] | 0..1
valueCodeableConcept | | CodeableConcept
valueString | | string
valueQuantity | | SimpleQuantity
valueBase64Binary | | base64Binary
contraindication | 0..* | Reference(DetectedIssue)
regulatory | 0..* | BackboneElement
regulatoryAuthority | 1..1 | Reference(Organization)
substitution | 0..* | BackboneElement
type | 1..1 | CodeableConcept
allowed | 1..1 | boolean
schedule | 0..* | BackboneElement
schedule | 1..1 | CodeableConcept
maxDispense | 0..1 | BackboneElement
quantity | 1..1 | SimpleQuantity
period | 0..1 | Duration
kinetics | 0..* | BackboneElement
areaUnderCurve | 0..* | SimpleQuantity
lethalDose50 | 0..* | SimpleQuantity
halfLifePeriod | 0..1 | Duration

11.6.10 Value Set Sc-valueset-medicationknowledge-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-medicationknowledge-status.html

11.6.10 Value Set Sc-valueset-medicationknowledge-status - Status Code Mapping

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.6.11 MedicationKnowledge Status CodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-medicationknowledge-status)

Mapping from http://hl7.org/fhir/ValueSet/medicationknowledge-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "MedicationKnowledge Status Codes"

Source Code | Equivalence | Destination Code

entered-in-error | equivalent | error
active | equivalent | active
inactive | equivalent | inactive

11.7 Resource Immunization - Content

URL: https://hl7.org/fhir/R4/immunization.html
Public Health and Emergency Response Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Describes the event of a patient being administered a vaccine or a record of an immunization as reported by a patient, a clinician or another party.

11.7.1 Scope and Usage

The Immunization resource is intended to cover the recording of current and historical administration of vaccines to patients across all healthcare disciplines in all care settings and all regions. This includes immunization of both humans and animals but does not include the administration of non-vaccine agents, even those that may have or claim to have immunological effects. While the terms "immunization" and "vaccination" are not clinically identical, for the purposes of the FHIR resources, the terms are used synonymously.

Additionally, the Immunization resource is expected to cover key concepts related to the creation, revision and querying of a patient's immunization history. This resource - through consultation with the Public Health work group - is believed to meet key use cases and information requirements as defined in the existing HL7 v2 immunization implementation guide, HL7 v3 POIZ domain and Immunization Domain Analysis Model.

11.7.2 Boundaries and Relationships

Administration of vaccines is intended to be handled using the Immunization resource. MedicationAdministration is intended for tracking the administration of non-vaccine medications. Some systems treat immunizations in the same way as any other medication administration. Such systems SHOULD use an immunization resource to represent these. If systems need to use a MedicationAdministration resource to capture vaccinations for workflow or other reasons, they SHOULD also expose an equivalent Immunization instance.

Immunization.reaction may be an indication of an allergy or intolerance. If this is deemed to be the case, a separate AllergyIntolerance resource instance should be created to indicate it, as most systems will not query against past Immunization.reaction.

This resource is referenced by AdverseEvent, ChargeItem, ImmunizationEvaluation, ImmunizationRecommendation and Observation

11.7.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Immunization | TU | | DomainResource | Immunization event information Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | completed | entered-in-error | not-done Immunization Status Codes (Required)
statusReason | | 0..1 | CodeableConcept | Reason not done Immunization Status Reason Codes (Example)
vaccineCode | Σ | 1..1 | CodeableConcept | Vaccine product administered Vaccine Administered Value Set (Example)
patient | Σ | 1..1 | Reference(Patient) | Who was immunized
encounter | | 0..1 | Reference(Encounter) | Encounter immunization was part of
occurrence[x] | Σ | 1..1 | | Vaccine administration date
occurrenceDateTime | | | dateTime
occurrenceString | | | string
recorded | | 0..1 | dateTime | When the immunization was first captured in the subject's record
primarySource | Σ | 0..1 | boolean | Indicates context the data was recorded in
reportOrigin | | 0..1 | CodeableConcept | Indicates the source of a secondarily reported record Immunization Origin Codes (Example)
location | | 0..1 | Reference(Location) | Where immunization occurred
manufacturer | | 0..1 | Reference(Organization) | Vaccine manufacturer

lotNumber | | 0..1 | string | Vaccine lot number
expirationDate | | 0..1 | date | Vaccine expiration date
site | | 0..1 | CodeableConcept | Body site vaccine was administered Codes for Immunization Site of Administration (Example)
route | | 0..1 | CodeableConcept | How vaccine entered body Immunization Route Codes (Example)
doseQuantity | | 0..1 | SimpleQuantity | Amount of vaccine administered
performer | Σ | 0..* | BackboneElement | Who performed event
function | Σ | 0..1 | CodeableConcept | What type of performance was done Immunization Function Codes (Extensible)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization) | Individual or organization who was performing
note | Σ | 0..* | Annotation | Additional immunization notes
reasonCode | | 0..* | CodeableConcept | Why immunization occurred Immunization Reason Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Why immunization occurred
isSubpotent | ?!Σ | 0..1 | boolean | Dose potency
subpotentReason | | 0..* | CodeableConcept | Reason for being subpotent Immunization Subpotent Reason (Example)
education | I | 0..* | BackboneElement | Educational material presented to patient + Rule: One of documentType or reference SHALL be present
documentType | | 0..1 | string | Educational material document identifier
reference | | 0..1 | uri | Educational material reference pointer
publicationDate | | 0..1 | dateTime | Educational material publication date
presentationDate | | 0..1 | dateTime | Educational material presentation date
programEligibility | | 0..* | CodeableConcept | Patient eligibility for a vaccination program Immunization Program Eligibility (Example)
fundingSource | | 0..1 | CodeableConcept | Funding source for the vaccine Immunization Funding Source (Example)
reaction | | 0..* | BackboneElement | Details of a reaction that follows immunization
date | | 0..1 | dateTime | When reaction started
detail | | 0..1 | Reference(Observation) | Additional information on reaction
reported | | 0..1 | boolean | Indicates self-reported reaction
protocolApplied | | 0..* | BackboneElement | Protocol followed by the provider
series | | 0..1 | string | Name of vaccine series
authority | | 0..1 | Reference(Organization) | Who is responsible for publishing the recommendations
targetDisease | | 0..* | CodeableConcept | Vaccine preventatable disease being targetted Immunization Target Disease Codes (Example)
doseNumber[x] | | 1..1 | | Dose number within series
doseNumberPositiveInt | | | positiveInt
doseNumberString | | | string
seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity
seriesDosesPositiveInt | | | positiveInt
seriesDosesString | | | string
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Immunization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error | not-done -->
 <statusReason><!-- 0..1 CodeableConcept Reason not done --></statusReason>
 <vaccineCode><!-- 1..1 CodeableConcept Vaccine product administered --></vaccineCode>
 <patient><!-- 1..1 Reference(Patient) Who was immunized --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter immunization was part of --></encounter>
 <occurrence[x]><!-- 1..1 dateTime|string Vaccine administration date --></occurrence[x]>
 <recorded value="[dateTime]"/><!-- 0..1 When the immunization was first captured in the subject's record -->
 <primarySource value="[boolean]"/><!-- 0..1 Indicates context the data was recorded in -->
 <reportOrigin><!-- 0..1 CodeableConcept Indicates the source of a secondarily reported record --></reportOrigin>
 <location><!-- 0..1 Reference(Location) Where immunization occurred --></location>
 <manufacturer><!-- 0..1 Reference(Organization) Vaccine manufacturer --></manufacturer>
 <lotNumber value="[string]"/><!-- 0..1 Vaccine lot number -->
 <expirationDate value="[date]"/><!-- 0..1 Vaccine expiration date -->
 <site><!-- 0..1 CodeableConcept Body site vaccine  was administered --></site>
 <route><!-- 0..1 CodeableConcept How vaccine entered body --></route>
 <doseQuantity><!-- 0..1 Quantity(SimpleQuantity) Amount of vaccine administered --></doseQuantity>
 <performer>  <!-- 0..* Who performed event -->
  <function><!-- 0..1 CodeableConcept What type of performance was done --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization) Individual or organization who was performing
     --></actor>
 </performer>
 <note><!-- 0..* Annotation Additional immunization notes --></note>
 <reasonCode><!-- 0..* CodeableConcept Why immunization occurred --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Why immunization occurred
    --></reasonReference>
 <isSubpotent value="[boolean]"/><!-- 0..1 Dose potency -->
 <subpotentReason><!-- 0..* CodeableConcept Reason for being subpotent --></subpotentReason>
 <education>  <!-- 0..* Educational material presented to patient -->
  <documentType value="[string]"/><!-- 0..1 Educational material document identifier -->
  <reference value="[uri]"/><!-- 0..1 Educational material reference pointer -->
  <publicationDate value="[dateTime]"/><!-- 0..1 Educational material publication date -->
  <presentationDate value="[dateTime]"/><!-- 0..1 Educational material presentation date -->
 </education>
 <programEligibility><!-- 0..* CodeableConcept Patient eligibility for a vaccination program --></programEligibility>
 <fundingSource><!-- 0..1 CodeableConcept Funding source for the vaccine --></fundingSource>
```

```
 <reaction>  <!-- 0..* Details of a reaction that follows immunization -->
  <date value="[dateTime]"/><!-- 0..1 When reaction started -->
  <detail><!-- 0..1 Reference(Observation) Additional information on reaction --></detail>
  <reported value="[boolean]"/><!-- 0..1 Indicates self-reported reaction -->
 </reaction>
 <protocolApplied>  <!-- 0..* Protocol followed by the provider -->
  <series value="[string]"/><!-- 0..1 Name of vaccine series -->
  <authority><!-- 0..1 Reference(Organization) Who is responsible for publishing the recommendations --></authority>
  <targetDisease><!-- 0..* CodeableConcept Vaccine preventatable disease being targetted --></targetDisease>
  <doseNumber[x]><!-- 1..1 positiveInt|string Dose number within series --></doseNumber[x]>
  <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
 </protocolApplied>
</Immunization>
```

JSON Template

```
{
  "resourceType" : "Immunization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error | not-done
  "statusReason" : { CodeableConcept }, // Reason not done
  "vaccineCode" : { CodeableConcept }, // R!  Vaccine product administered
  "patient" : { Reference(Patient) }, // R!  Who was immunized
  "encounter" : { Reference(Encounter) }, // Encounter immunization was part of
  // occurrence[x]: Vaccine administration date. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrenceString" : "<string>",
  "recorded" : "<dateTime>", // When the immunization was first captured in the subject's record
  "primarySource" : <boolean>, // Indicates context the data was recorded in
  "reportOrigin" : { CodeableConcept }, // Indicates the source of a secondarily reported record
  "location" : { Reference(Location) }, // Where immunization occurred
  "manufacturer" : { Reference(Organization) }, // Vaccine manufacturer
  "lotNumber" : "<string>", // Vaccine lot number
  "expirationDate" : "<date>", // Vaccine expiration date
  "site" : { CodeableConcept }, // Body site vaccine  was administered
  "route" : { CodeableConcept }, // How vaccine entered body
  "doseQuantity" : { Quantity(SimpleQuantity) }, // Amount of vaccine administered
  "performer" : [{ // Who performed event
    "function" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|PractitionerRole|Organization) } // R!  Individual or organization who was
    performing
  }],
  "note" : [{ Annotation }], // Additional immunization notes
  "reasonCode" : [{ CodeableConcept }], // Why immunization occurred
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Why immunization occurred
  "isSubpotent" : <boolean>, // Dose potency
  "subpotentReason" : [{ CodeableConcept }], // Reason for being subpotent
  "education" : [{ // Educational material presented to patient
    "documentType" : "<string>", // Educational material document identifier
    "reference" : "<uri>", // Educational material reference pointer
    "publicationDate" : "<dateTime>", // Educational material publication date
    "presentationDate" : "<dateTime>" // Educational material presentation date
  }],
  "programEligibility" : [{ CodeableConcept }], // Patient eligibility for a vaccination program
  "fundingSource" : { CodeableConcept }, // Funding source for the vaccine
  "reaction" : [{ // Details of a reaction that follows immunization
    "date" : "<dateTime>", // When reaction started
    "detail" : { Reference(Observation) }, // Additional information on reaction
    "reported" : <boolean> // Indicates self-reported reaction
  }],
  "protocolApplied" : [{ // Protocol followed by the provider
    "series" : "<string>", // Name of vaccine series
    "authority" : { Reference(Organization) }, // Who is responsible for publishing the recommendations
    "targetDisease" : [{ CodeableConcept }], // Vaccine preventatable disease being targetted
    // doseNumber[x]: Dose number within series. One of these 2:
    "doseNumberPositiveInt" : "<positiveInt>",
    "doseNumberString" : "<string>",
    // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
    "seriesDosesPositiveInt" : "<positiveInt>"
    "seriesDosesString" : "<string>"
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Immunization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Immunization.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Immunization.status [ code ]; # 1..1 completed | entered-in-error | not-done
  fhir:Immunization.statusReason [ CodeableConcept ]; # 0..1 Reason not done
  fhir:Immunization.vaccineCode [ CodeableConcept ]; # 1..1 Vaccine product administered
  fhir:Immunization.patient [ Reference(Patient) ]; # 1..1 Who was immunized
  fhir:Immunization.encounter [ Reference(Encounter) ]; # 0..1 Encounter immunization was part of
```

```
  # Immunization.occurrence[x] : 1..1 Vaccine administration date. One of these 2
    fhir:Immunization.occurrenceDateTime [ dateTime ]
    fhir:Immunization.occurrenceString [ string ]
  fhir:Immunization.recorded [ dateTime ]; # 0..1 When the immunization was first captured in the subject's record
  fhir:Immunization.primarySource [ boolean ]; # 0..1 Indicates context the data was recorded in
  fhir:Immunization.reportOrigin [ CodeableConcept ]; # 0..1 Indicates the source of a secondarily reported record
  fhir:Immunization.location [ Reference(Location) ]; # 0..1 Where immunization occurred
  fhir:Immunization.manufacturer [ Reference(Organization) ]; # 0..1 Vaccine manufacturer
  fhir:Immunization.lotNumber [ string ]; # 0..1 Vaccine lot number
  fhir:Immunization.expirationDate [ date ]; # 0..1 Vaccine expiration date
  fhir:Immunization.site [ CodeableConcept ]; # 0..1 Body site vaccine  was administered
  fhir:Immunization.route [ CodeableConcept ]; # 0..1 How vaccine entered body
  fhir:Immunization.doseQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of vaccine administered
  fhir:Immunization.performer [ # 0..* Who performed event
    fhir:Immunization.performer.function [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:Immunization.performer.actor [ Reference(Practitioner|PractitionerRole|Organization) ]; # 1..1 Individual or
    organization who was performing
  ], ...;
  fhir:Immunization.note [ Annotation ], ... ; # 0..* Additional immunization notes
  fhir:Immunization.reasonCode [ CodeableConcept ], ... ; # 0..* Why immunization occurred
  fhir:Immunization.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..* Why immunization
     occurred
  fhir:Immunization.isSubpotent [ boolean ]; # 0..1 Dose potency
  fhir:Immunization.subpotentReason [ CodeableConcept ], ... ; # 0..* Reason for being subpotent
  fhir:Immunization.education [ # 0..* Educational material presented to patient
    fhir:Immunization.education.documentType [ string ]; # 0..1 Educational material document identifier
    fhir:Immunization.education.reference [ uri ]; # 0..1 Educational material reference pointer
    fhir:Immunization.education.publicationDate [ dateTime ]; # 0..1 Educational material publication date
    fhir:Immunization.education.presentationDate [ dateTime ]; # 0..1 Educational material presentation date
  ], ...;
  fhir:Immunization.programEligibility [ CodeableConcept ], ... ; # 0..* Patient eligibility for a vaccination program
  fhir:Immunization.fundingSource [ CodeableConcept ]; # 0..1 Funding source for the vaccine
  fhir:Immunization.reaction [ # 0..* Details of a reaction that follows immunization
    fhir:Immunization.reaction.date [ dateTime ]; # 0..1 When reaction started
    fhir:Immunization.reaction.detail [ Reference(Observation) ]; # 0..1 Additional information on reaction
    fhir:Immunization.reaction.reported [ boolean ]; # 0..1 Indicates self-reported reaction
  ], ...;
  fhir:Immunization.protocolApplied [ # 0..* Protocol followed by the provider
    fhir:Immunization.protocolApplied.series [ string ]; # 0..1 Name of vaccine series
    fhir:Immunization.protocolApplied.authority [ Reference(Organization) ]; # 0..1 Who is responsible for publishing
    the recommendations
    fhir:Immunization.protocolApplied.targetDisease [ CodeableConcept ], ... ; # 0..* Vaccine preventatable disease
    being targetted
    # Immunization.protocolApplied.doseNumber[x] : 1..1 Dose number within series. One of these 2
      fhir:Immunization.protocolApplied.doseNumberPositiveInt [ positiveInt ]
      fhir:Immunization.protocolApplied.doseNumberString [ string ]
    # Immunization.protocolApplied.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of these 2
      fhir:Immunization.protocolApplied.seriesDosesPositiveInt [ positiveInt ]
      fhir:Immunization.protocolApplied.seriesDosesString [ string ]
  ], ...;
]
```

Changes since R3

Immunization

Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/immunization-status to http://hl7.org/fhir/ValueSet/immunization-status|4.0.1

Immunization.statusReason | Renamed from notGiven to statusReason Min Cardinality changed from 1 to 0 Type changed from boolean to CodeableConcept No longer marked as Modifier

Immunization.occurrence[x] | Renamed from date to occurrence[x] Min Cardinality changed from 0 to 1 Add Type string

Immunization.recorded | Added Element

Immunization.primarySource | Min Cardinality changed from 1 to 0

Immunization.performer | Renamed from practitioner to performer

Immunization.performer.function | Moved from Immunization.practitioner.role to function Change value set from http://hl7.org/fhir/ValueSet/immunization-role to http://hl7.org/fhir/ValueSet/immunization-function

Immunization.performer.actor | Moved from Immunization.practitioner to Immunization.performer Type Reference: Added Target Types PractitionerRole, Organization

Immunization.reasonCode | Moved from Immunization.explanation.reason to reasonCode

Immunization.reasonReference | Added Element

Immunization.isSubpotent | Added Element

Immunization.subpotentReason | Added Element

Immunization.education | Added Element

Immunization.education.documentType | Added Element

Immunization.education.reference | Added Element

Immunization.education.publicationDate | Added Element

Immunization.education.presentationDate | Added Element

Immunization.programEligibility | Added Element

Immunization.fundingSource | Added Element

Immunization.protocolApplied | Renamed from vaccinationProtocol to protocolApplied

Immunization.protocolApplied.series | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied

Immunization.protocolApplied.authority | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied

Immunization.protocolApplied.targetDisease | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied Min Cardinality changed from 1 to 0

Immunization.protocolApplied.doseNumber[x] | Moved from Immunization.vaccinationProtocol.doseSequence to doseNumber[x] Min Cardinality changed from 0 to 1 Add Type string
Immunization.protocolApplied.seriesDoses[x] | Moved from Immunization.vaccinationProtocol.seriesDoses to seriesDoses[x] Add Type string
Immunization.explanation | deleted
Immunization.vaccinationProtocol.description | deleted
Immunization.vaccinationProtocol.doseStatus | deleted
Immunization.vaccinationProtocol.doseStatusReason | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Immunization | TU | | DomainResource | Immunization event information Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | completed | entered-in-error | not-done Immunization Status Codes (Required)
statusReason | | 0..1 | CodeableConcept | Reason not done Immunization Status Reason Codes (Example)
vaccineCode | Σ | 1..1 | CodeableConcept | Vaccine product administered Vaccine Administered Value Set (Example)
patient | Σ | 1..1 | Reference(Patient) | Who was immunized
encounter | | 0..1 | Reference(Encounter) | Encounter immunization was part of
occurrence[x] | Σ | 1..1 | | Vaccine administration date
occurrenceDateTime | | | dateTime
occurrenceString | | | string
recorded | | 0..1 | dateTime | When the immunization was first captured in the subject's record
primarySource | Σ | 0..1 | boolean | Indicates context the data was recorded in
reportOrigin | | 0..1 | CodeableConcept | Indicates the source of a secondarily reported record Immunization Origin Codes (Example)
location | | 0..1 | Reference(Location) | Where immunization occurred
manufacturer | | 0..1 | Reference(Organization) | Vaccine manufacturer
lotNumber | | 0..1 | string | Vaccine lot number
expirationDate | | 0..1 | date | Vaccine expiration date
site | | 0..1 | CodeableConcept | Body site vaccine was administered Codes for Immunization Site of Administration (Example)
route | | 0..1 | CodeableConcept | How vaccine entered body Immunization Route Codes (Example)
doseQuantity | | 0..1 | SimpleQuantity | Amount of vaccine administered
performer | Σ | 0..* | BackboneElement | Who performed event
function | Σ | 0..1 | CodeableConcept | What type of performance was done Immunization Function Codes (Extensible)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization) | Individual or organization who was performing
note | Σ | 0..* | Annotation | Additional immunization notes
reasonCode | | 0..* | CodeableConcept | Why immunization occurred Immunization Reason Codes (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport) | Why immunization occurred
isSubpotent | ?!Σ | 0..1 | boolean | Dose potency
subpotentReason | | 0..* | CodeableConcept | Reason for being subpotent Immunization Subpotent Reason (Example)
education | I | 0..* | BackboneElement | Educational material presented to patient + Rule: One of documentType or reference SHALL be present
documentType | | 0..1 | string | Educational material document identifier
reference | | 0..1 | uri | Educational material reference pointer
publicationDate | | 0..1 | dateTime | Educational material publication date
presentationDate | | 0..1 | dateTime | Educational material presentation date
programEligibility | | 0..* | CodeableConcept | Patient eligibility for a vaccination program Immunization Program Eligibility (Example)
fundingSource | | 0..1 | CodeableConcept | Funding source for the vaccine Immunization Funding Source (Example)
reaction | | 0..* | BackboneElement | Details of a reaction that follows immunization
date | | 0..1 | dateTime | When reaction started
detail | | 0..1 | Reference(Observation) | Additional information on reaction
reported | | 0..1 | boolean | Indicates self-reported reaction
protocolApplied | | 0..* | BackboneElement | Protocol followed by the provider
series | | 0..1 | string | Name of vaccine series
authority | | 0..1 | Reference(Organization) | Who is responsible for publishing the recommendations
targetDisease | | 0..* | CodeableConcept | Vaccine preventatable disease being targetted Immunization Target Disease Codes (Example)
doseNumber[x] | | 1..1 | | Dose number within series
doseNumberPositiveInt | | | positiveInt
doseNumberString | | | string
seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity
seriesDosesPositiveInt | | | positiveInt
seriesDosesString | | | string
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Immunization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error | not-done -->
 <statusReason><!-- 0..1 CodeableConcept Reason not done --></statusReason>
 <vaccineCode><!-- 1..1 CodeableConcept Vaccine product administered --></vaccineCode>
 <patient><!-- 1..1 Reference(Patient) Who was immunized --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter immunization was part of --></encounter>
 <occurrence[x]><!-- 1..1 dateTime|string Vaccine administration date --></occurrence[x]>
 <recorded value="[dateTime]"/><!-- 0..1 When the immunization was first captured in the subject's record -->
 <primarySource value="[boolean]"/><!-- 0..1 Indicates context the data was recorded in -->
 <reportOrigin><!-- 0..1 CodeableConcept Indicates the source of a secondarily reported record --></reportOrigin>
 <location><!-- 0..1 Reference(Location) Where immunization occurred --></location>
 <manufacturer><!-- 0..1 Reference(Organization) Vaccine manufacturer --></manufacturer>
 <lotNumber value="[string]"/><!-- 0..1 Vaccine lot number -->
 <expirationDate value="[date]"/><!-- 0..1 Vaccine expiration date -->
 <site><!-- 0..1 CodeableConcept Body site vaccine  was administered --></site>
 <route><!-- 0..1 CodeableConcept How vaccine entered body --></route>
 <doseQuantity><!-- 0..1 Quantity(SimpleQuantity) Amount of vaccine administered --></doseQuantity>
 <performer>  <!-- 0..* Who performed event -->
  <function><!-- 0..1 CodeableConcept What type of performance was done --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization) Individual or organization who was performing
    --></actor>
 </performer>
 <note><!-- 0..* Annotation Additional immunization notes --></note>
 <reasonCode><!-- 0..* CodeableConcept Why immunization occurred --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport) Why immunization occurred
    --></reasonReference>
 <isSubpotent value="[boolean]"/><!-- 0..1 Dose potency -->
 <subpotentReason><!-- 0..* CodeableConcept Reason for being subpotent --></subpotentReason>
 <education>  <!-- 0..* Educational material presented to patient -->
  <documentType value="[string]"/><!-- 0..1 Educational material document identifier -->
  <reference value="[uri]"/><!-- 0..1 Educational material reference pointer -->
  <publicationDate value="[dateTime]"/><!-- 0..1 Educational material publication date -->
  <presentationDate value="[dateTime]"/><!-- 0..1 Educational material presentation date -->
 </education>
 <programEligibility><!-- 0..* CodeableConcept Patient eligibility for a vaccination program --></programEligibility>
 <fundingSource><!-- 0..1 CodeableConcept Funding source for the vaccine --></fundingSource>
 <reaction>  <!-- 0..* Details of a reaction that follows immunization -->
  <date value="[dateTime]"/><!-- 0..1 When reaction started -->
  <detail><!-- 0..1 Reference(Observation) Additional information on reaction --></detail>
  <reported value="[boolean]"/><!-- 0..1 Indicates self-reported reaction -->
 </reaction>
 <protocolApplied>  <!-- 0..* Protocol followed by the provider -->
  <series value="[string]"/><!-- 0..1 Name of vaccine series -->
  <authority><!-- 0..1 Reference(Organization) Who is responsible for publishing the recommendations --></authority>
  <targetDisease><!-- 0..* CodeableConcept Vaccine preventatable disease being targetted --></targetDisease>
  <doseNumber[x]><!-- 1..1 positiveInt|string Dose number within series --></doseNumber[x]>
  <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
 </protocolApplied>
</Immunization>
```

JSON Template

```
{
  "resourceType" : "Immunization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error | not-done
  "statusReason" : { CodeableConcept }, // Reason not done
  "vaccineCode" : { CodeableConcept }, // R!  Vaccine product administered
  "patient" : { Reference(Patient) }, // R!  Who was immunized
  "encounter" : { Reference(Encounter) }, // Encounter immunization was part of
  // occurrence[x]: Vaccine administration date. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrenceString" : "<string>",
  "recorded" : "<dateTime>", // When the immunization was first captured in the subject's record
  "primarySource" : <boolean>, // Indicates context the data was recorded in
  "reportOrigin" : { CodeableConcept }, // Indicates the source of a secondarily reported record
  "location" : { Reference(Location) }, // Where immunization occurred
  "manufacturer" : { Reference(Organization) }, // Vaccine manufacturer
  "lotNumber" : "<string>", // Vaccine lot number
  "expirationDate" : "<date>", // Vaccine expiration date
  "site" : { CodeableConcept }, // Body site vaccine  was administered
  "route" : { CodeableConcept }, // How vaccine entered body
  "doseQuantity" : { Quantity(SimpleQuantity) }, // Amount of vaccine administered
  "performer" : [{ // Who performed event
    "function" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|PractitionerRole|Organization) } // R!  Individual or organization who was
    performing
  }],
  "note" : [{ Annotation }], // Additional immunization notes
  "reasonCode" : [{ CodeableConcept }], // Why immunization occurred
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport) }], // Why immunization occurred
  "isSubpotent" : <boolean>, // Dose potency
  "subpotentReason" : [{ CodeableConcept }], // Reason for being subpotent
```

```
  "education" : [{ // Educational material presented to patient
    "documentType" : "<string>", // Educational material document identifier
    "reference" : "<uri>", // Educational material reference pointer
    "publicationDate" : "<dateTime>", // Educational material publication date
    "presentationDate" : "<dateTime>" // Educational material presentation date
  }],
  "programEligibility" : [{ CodeableConcept }], // Patient eligibility for a vaccination program
  "fundingSource" : { CodeableConcept }, // Funding source for the vaccine
  "reaction" : [{ // Details of a reaction that follows immunization
    "date" : "<dateTime>", // When reaction started
    "detail" : { Reference(Observation) }, // Additional information on reaction
    "reported" : <boolean> // Indicates self-reported reaction
  }],
  "protocolApplied" : [{ // Protocol followed by the provider
    "series" : "<string>", // Name of vaccine series
    "authority" : { Reference(Organization) }, // Who is responsible for publishing the recommendations
    "targetDisease" : [{ CodeableConcept }], // Vaccine preventatable disease being targetted
    // doseNumber[x]: Dose number within series. One of these 2:
    "doseNumberPositiveInt" : "<positiveInt>",
    "doseNumberString" : "<string>",
    // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
    "seriesDosesPositiveInt" : "<positiveInt>"
    "seriesDosesString" : "<string>"
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Immunization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Immunization.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Immunization.status [ code ]; # 1..1 completed | entered-in-error | not-done
  fhir:Immunization.statusReason [ CodeableConcept ]; # 0..1 Reason not done
  fhir:Immunization.vaccineCode [ CodeableConcept ]; # 1..1 Vaccine product administered
  fhir:Immunization.patient [ Reference(Patient) ]; # 1..1 Who was immunized
  fhir:Immunization.encounter [ Reference(Encounter) ]; # 0..1 Encounter immunization was part of
  # Immunization.occurrence[x] : 1..1 Vaccine administration date. One of these 2
    fhir:Immunization.occurrenceDateTime [ dateTime ]
    fhir:Immunization.occurrenceString [ string ]
  fhir:Immunization.recorded [ dateTime ]; # 0..1 When the immunization was first captured in the subject's record
  fhir:Immunization.primarySource [ boolean ]; # 0..1 Indicates context the data was recorded in
  fhir:Immunization.reportOrigin [ CodeableConcept ]; # 0..1 Indicates the source of a secondarily reported record
  fhir:Immunization.location [ Reference(Location) ]; # 0..1 Where immunization occurred
  fhir:Immunization.manufacturer [ Reference(Organization) ]; # 0..1 Vaccine manufacturer
  fhir:Immunization.lotNumber [ string ]; # 0..1 Vaccine lot number
  fhir:Immunization.expirationDate [ date ]; # 0..1 Vaccine expiration date
  fhir:Immunization.site [ CodeableConcept ]; # 0..1 Body site vaccine  was administered
  fhir:Immunization.route [ CodeableConcept ]; # 0..1 How vaccine entered body
  fhir:Immunization.doseQuantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of vaccine administered
  fhir:Immunization.performer [ # 0..* Who performed event
    fhir:Immunization.performer.function [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:Immunization.performer.actor [ Reference(Practitioner|PractitionerRole|Organization) ]; # 1..1 Individual or
    organization who was performing
  ], ...;
  fhir:Immunization.note [ Annotation ], ... ; # 0..* Additional immunization notes
  fhir:Immunization.reasonCode [ CodeableConcept ], ... ; # 0..* Why immunization occurred
  fhir:Immunization.reasonReference [ Reference(Condition|Observation|DiagnosticReport) ], ... ; # 0..* Why immunization
     occurred
  fhir:Immunization.isSubpotent [ boolean ]; # 0..1 Dose potency
  fhir:Immunization.subpotentReason [ CodeableConcept ], ... ; # 0..* Reason for being subpotent
  fhir:Immunization.education [ # 0..* Educational material presented to patient
    fhir:Immunization.education.documentType [ string ]; # 0..1 Educational material document identifier
    fhir:Immunization.education.reference [ uri ]; # 0..1 Educational material reference pointer
    fhir:Immunization.education.publicationDate [ dateTime ]; # 0..1 Educational material publication date
    fhir:Immunization.education.presentationDate [ dateTime ]; # 0..1 Educational material presentation date
  ], ...;
  fhir:Immunization.programEligibility [ CodeableConcept ], ... ; # 0..* Patient eligibility for a vaccination program
  fhir:Immunization.fundingSource [ CodeableConcept ]; # 0..1 Funding source for the vaccine
  fhir:Immunization.reaction [ # 0..* Details of a reaction that follows immunization
    fhir:Immunization.reaction.date [ dateTime ]; # 0..1 When reaction started
    fhir:Immunization.reaction.detail [ Reference(Observation) ]; # 0..1 Additional information on reaction
    fhir:Immunization.reaction.reported [ boolean ]; # 0..1 Indicates self-reported reaction
  ], ...;
  fhir:Immunization.protocolApplied [ # 0..* Protocol followed by the provider
    fhir:Immunization.protocolApplied.series [ string ]; # 0..1 Name of vaccine series
    fhir:Immunization.protocolApplied.authority [ Reference(Organization) ]; # 0..1 Who is responsible for publishing
    the recommendations
    fhir:Immunization.protocolApplied.targetDisease [ CodeableConcept ], ... ; # 0..* Vaccine preventatable disease
    being targetted
    # Immunization.protocolApplied.doseNumber[x] : 1..1 Dose number within series. One of these 2
      fhir:Immunization.protocolApplied.doseNumberPositiveInt [ positiveInt ]
      fhir:Immunization.protocolApplied.doseNumberString [ string ]
    # Immunization.protocolApplied.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of these 2
      fhir:Immunization.protocolApplied.seriesDosesPositiveInt [ positiveInt ]
```

```
      fhir:Immunization.protocolApplied.seriesDosesString [ string ]
  ], ...;
]
```

Changes since Release 3

Immunization
Immunization.status | Change value set from http://hl7.org/fhir/ValueSet/immunization-status to http://hl7.org/fhir/ValueSet/immunization-status|4.0.1
Immunization.statusReason | Renamed from notGiven to statusReason Min Cardinality changed from 1 to 0 Type changed from boolean to CodeableConcept No longer marked as Modifier
Immunization.occurrence[x] | Renamed from date to occurrence[x] Min Cardinality changed from 0 to 1 Add Type string
Immunization.recorded | Added Element
Immunization.primarySource | Min Cardinality changed from 1 to 0
Immunization.performer | Renamed from practitioner to performer
Immunization.performer.function | Moved from Immunization.practitioner.role to function Change value set from http://hl7.org/fhir/ValueSet/immunization-role to http://hl7.org/fhir/ValueSet/immunization-function
Immunization.performer.actor | Moved from Immunization.practitioner to Immunization.performer Type Reference: Added Target Types PractitionerRole, Organization
Immunization.reasonCode | Moved from Immunization.explanation.reason to reasonCode
Immunization.reasonReference | Added Element
Immunization.isSubpotent | Added Element
Immunization.subpotentReason | Added Element
Immunization.education | Added Element
Immunization.education.documentType | Added Element
Immunization.education.reference | Added Element
Immunization.education.publicationDate | Added Element
Immunization.education.presentationDate | Added Element
Immunization.programEligibility | Added Element
Immunization.fundingSource | Added Element
Immunization.protocolApplied | Renamed from vaccinationProtocol to protocolApplied
Immunization.protocolApplied.series | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied
Immunization.protocolApplied.authority | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied
Immunization.protocolApplied.targetDisease | Moved from Immunization.vaccinationProtocol to Immunization.protocolApplied Min Cardinality changed from 1 to 0
Immunization.protocolApplied.doseNumber[x] | Moved from Immunization.vaccinationProtocol.doseSequence to doseNumber[x] Min Cardinality changed from 0 to 1 Add Type string
Immunization.protocolApplied.seriesDoses[x] | Moved from Immunization.vaccinationProtocol.seriesDoses to seriesDoses[x] Add Type string
Immunization.explanation | deleted
Immunization.vaccinationProtocol.description | deleted
Immunization.vaccinationProtocol.doseStatus | deleted
Immunization.vaccinationProtocol.doseStatusReason | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
Immunization.status | A set of codes indicating the current status of an Immunization. | Required | ImmunizationStatusCodes
Immunization.statusReason | The reason why a vaccine was not administered. | Example | ImmunizationStatusReasonCodes
Immunization.vaccineCode | The code for vaccine product administered. | Example | VaccineAdministeredValueSet
Immunization.reportOrigin | The source of the data for a record which is not from a primary source. | Example | ImmunizationOriginCodes
Immunization.site | The site at which the vaccine was administered. | Example | CodesForImmunizationSiteOfAdministration
Immunization.route | The route by which the vaccine was administered. | Example | ImmunizationRouteCodes
Immunization.performer.function | The role a practitioner or organization plays in the immunization event. | Extensible | ImmunizationFunctionCodes
Immunization.reasonCode | The reason why a vaccine was administered. | Example | ImmunizationReasonCodes
Immunization.subpotentReason | The reason why a dose is considered to be subpotent. | Example | ImmunizationSubpotentReason
Immunization.programEligibility | The patient's eligibility for a vaccation program. | Example | ImmunizationProgramEligibility
Immunization.fundingSource | The source of funding used to purchase the vaccine administered. | Example | ImmunizationFundingSource
Immunization.protocolApplied.targetDisease | The vaccine preventable disease the dose is being administered for. | Example | ImmunizationTargetDiseaseCodes

11.7.3.2 Constraints

id | Level | Location | Description | Expression

imm-1 | Rule | Immunization.education | One of documentType or reference SHALL be present | documentType.exists() or reference.exists()

11.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Vaccination (non)-Administration Date | Immunization.occurrence | 17 Resources
identifier | token | Business identifier | Immunization.identifier | 30 Resources
location | reference | The service delivery location or facility in which the vaccine was / was to be administered | Immunization.location (Location)
lot-number | string | Vaccine Lot Number | Immunization.lotNumber
manufacturer | reference | Vaccine Manufacturer | Immunization.manufacturer (Organization)
patient | reference | The patient for the vaccination record | Immunization.patient (Patient) | 33 Resources
performer | reference | The practitioner or organization who played a role in the vaccination | Immunization.performer.actor (Practitioner, Organization, PractitionerRole)
reaction | reference | Additional information on reaction | Immunization.reaction.detail (Observation)
reaction-date | date | When reaction started | Immunization.reaction.date
reason-code | token | Reason why the vaccine was administered | Immunization.reasonCode
reason-reference | reference | Why immunization occurred | Immunization.reasonReference (Condition, Observation, DiagnosticReport)
series | string | The series being followed by the provider | Immunization.protocolApplied.series
status | token | Immunization event status | Immunization.status
status-reason | token | Reason why the vaccine was not administered | Immunization.statusReason
target-disease | token | The target disease the dose is being administered against | Immunization.protocolApplied.targetDisease
vaccine-code | token | Vaccine Product Administered | Immunization.vaccineCode

11.7.5 Resource Immunization - Examples

URL: https://hl7.org/fhir/R4/immunization-examples.html

11.7.5 Resource Immunization - Examples

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Example Name | id | Format
Example of Immunization | example | XML | JSON | Turtle
Not Given Immunization Example | notGiven | XML | JSON | Turtle
Example of a Historical Example | historical | XML | JSON | Turtle
Example of a Subpotent Dose | subpotent | XML | JSON | Turtle
Example where the provider indicates the protocol being followed | protocol | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.7.6 Resource Immunization - Detailed Descriptions

URL: https://hl7.org/fhir/R4/immunization-definitions.html

11.7.6 Resource Immunization - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the Immunization resource.

Immunization
Element Id | Immunization
Definition | Describes the event of a patient being administered a vaccine or a record of an immunization as reported by a patient, a clinician or another party.
Cardinality | 0..*
Type | DomainResource
Immunization.identifier
Element Id | Immunization.identifier
Definition | A unique identifier assigned to this immunization record.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Immunization.status
Element Id | Immunization.status
Definition | Indicates the current status of the immunization event.
Cardinality | 1..1
Terminology Binding | Immunization Status Codes (Required)
Type | code

Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains statuses entered-in-error and not-done which means that the resource should not be treated as valid)
Summary | true
Comments | Will generally be set to show that the immunization has been completed or not done. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Immunization.statusReason
Element Id | Immunization.statusReason
Definition | Indicates the reason the immunization event was not performed.
Cardinality | 0..1
Terminology Binding | Immunization Status Reason Codes (Example)
Type | CodeableConcept
Comments | This is generally only used for the status of "not-done". The reason for performing the immunization event is captured in reasonCode, not here.
Immunization.vaccineCode
Element Id | Immunization.vaccineCode
Definition | Vaccine that was administered or was to be administered.
Cardinality | 1..1
Terminology Binding | Vaccine Administered Value Set (Example)
Type | CodeableConcept
Summary | true
Immunization.patient
Element Id | Immunization.patient
Definition | The patient who either received or did not receive the immunization.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true
Immunization.encounter
Element Id | Immunization.encounter
Definition | The visit or admission or other contact between patient and health care provider the immunization was performed as part of.
Cardinality | 0..1
Type | Reference(Encounter)
Immunization.occurrence[x]
Element Id | Immunization.occurrence[x]
Definition | Date vaccine administered or was to be administered.
Cardinality | 1..1
Type | dateTime|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | When immunizations are given a specific date and time should always be known. When immunizations are patient reported, a specific date might not be known. Although partial dates are allowed, an adult patient might not be able to recall the year a childhood immunization was given. An exact date is always preferable, but the use of the String data type is acceptable when an exact date is not known. A small number of vaccines (e.g. live oral typhoid vaccine) are given as a series of patient self-administered dose over a span of time. In cases like this, often, only the first dose (typically a provider supervised dose) is recorded with the occurrence indicating the date/time of the first dose.
Immunization.recorded
Element Id | Immunization.recorded
Definition | The date the occurrence of the immunization was first captured in the record - potentially significantly after the occurrence of the event.
Cardinality | 0..1
Type | dateTime
Summary | false
Immunization.primarySource
Element Id | Immunization.primarySource
Definition | An indication that the content of the record is based on information from the person who administered the vaccine. This reflects the context under which the data was originally recorded.
Cardinality | 0..1
Type | boolean
Summary | true
Comments | Reflects the “reliability” of the content.
Immunization.reportOrigin
Element Id | Immunization.reportOrigin
Definition | The source of the data when the report of the immunization event is not based on information from the person who administered the vaccine.
Cardinality | 0..1
Terminology Binding | Immunization Origin Codes (Example)
Type | CodeableConcept
Comments | Should not be populated if primarySource = True, not required even if primarySource = False.
Immunization.location
Element Id | Immunization.location
Definition | The service delivery location where the vaccine administration occurred.
Cardinality | 0..1
Type | Reference(Location)

Immunization.manufacturer
Element Id | Immunization.manufacturer
Definition | Name of vaccine manufacturer.
Cardinality | 0..1
Type | Reference(Organization)
Immunization.lotNumber
Element Id | Immunization.lotNumber
Definition | Lot number of the vaccine product.
Cardinality | 0..1
Type | string
Immunization.expirationDate
Element Id | Immunization.expirationDate
Definition | Date vaccine batch expires.
Cardinality | 0..1
Type | date
Immunization.site
Element Id | Immunization.site
Definition | Body site where vaccine was administered.
Cardinality | 0..1
Terminology Binding | Codes for Immunization Site of Administration (Example)
Type | CodeableConcept
Immunization.route
Element Id | Immunization.route
Definition | The path by which the vaccine product is taken into the body.
Cardinality | 0..1
Terminology Binding | Immunization Route Codes (Example)
Type | CodeableConcept
Immunization.doseQuantity
Element Id | Immunization.doseQuantity
Definition | The quantity of vaccine product that was administered.
Cardinality | 0..1
Type | SimpleQuantity
Immunization.performer
Element Id | Immunization.performer
Definition | Indicates who performed the immunization event.
Cardinality | 0..*
Summary | true
Immunization.performer.function
Element Id | Immunization.performer.function
Definition | Describes the type of performance (e.g. ordering provider, administering provider, etc.).
Cardinality | 0..1
Terminology Binding | Immunization Function Codes (Extensible)
Type | CodeableConcept
Summary | true
Immunization.performer.actor
Element Id | Immunization.performer.actor
Definition | The practitioner or organization who performed the action.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Organization)
Summary | true
Comments | When the individual practitioner who performed the action is known, it is best to send.
Immunization.note
Element Id | Immunization.note
Definition | Extra information about the immunization that is not conveyed by the other attributes.
Cardinality | 0..*
Type | Annotation
Summary | true
Immunization.reasonCode
Element Id | Immunization.reasonCode
Definition | Reasons why the vaccine was administered.
Cardinality | 0..*
Terminology Binding | Immunization Reason Codes (Example)
Type | CodeableConcept
Immunization.reasonReference
Element Id | Immunization.reasonReference
Definition | Condition, Observation or DiagnosticReport that supports why the immunization was administered.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport)
Immunization.isSubpotent
Element Id | Immunization.isSubpotent
Definition | Indication if a dose is considered to be subpotent. By default, a dose should be considered to be potent.
Cardinality | 0..1

Type | boolean
Is Modifier | true (Reason: This element is labeled as a modifier because an immunization event with a subpotent vaccine doesn't protect the patient the same way as a potent dose.)
Meaning if Missing | By default, a dose should be considered to be potent.
Summary | true
Comments | Typically, the recognition of the dose being sub-potent is retrospective, after the administration (ex. notification of a manufacturer recall after administration). However, in the case of a partial administration (the patient moves unexpectedly and only some of the dose is actually administered), subpotency may be recognized immediately, but it is still important to record the event.
Immunization.subpotentReason
Element Id | Immunization.subpotentReason
Definition | Reason why a dose is considered to be subpotent.
Cardinality | 0..*
Terminology Binding | Immunization Subpotent Reason (Example)
Type | CodeableConcept
Immunization.education
Element Id | Immunization.education
Definition | Educational material presented to the patient (or guardian) at the time of vaccine administration.
Cardinality | 0..*
Invariants | Defined on this element imm-1 Rule One of documentType or reference SHALL be present documentType.exists() or reference.exists()
Defined on this element
imm-1 | Rule | One of documentType or reference SHALL be present | documentType.exists() or reference.exists()
Immunization.education.documentType
Element Id | Immunization.education.documentType
Definition | Identifier of the material presented to the patient.
Cardinality | 0..1
Type | string
Immunization.education.reference
Element Id | Immunization.education.reference
Definition | Reference pointer to the educational material given to the patient if the information was on line.
Cardinality | 0..1
Type | uri
Immunization.education.publicationDate
Element Id | Immunization.education.publicationDate
Definition | Date the educational material was published.
Cardinality | 0..1
Type | dateTime
Immunization.education.presentationDate
Element Id | Immunization.education.presentationDate
Definition | Date the educational material was given to the patient.
Cardinality | 0..1
Type | dateTime
Immunization.programEligibility
Element Id | Immunization.programEligibility
Definition | Indicates a patient's eligibility for a funding program.
Cardinality | 0..*
Terminology Binding | Immunization Program Eligibility (Example)
Type | CodeableConcept
Immunization.fundingSource
Element Id | Immunization.fundingSource
Definition | Indicates the source of the vaccine actually administered. This may be different than the patient eligibility (e.g. the patient may be eligible for a publically purchased vaccine but due to inventory issues, vaccine purchased with private funds was actually administered).
Cardinality | 0..1
Terminology Binding | Immunization Funding Source (Example)
Type | CodeableConcept
Immunization.reaction
Element Id | Immunization.reaction
Definition | Categorical data indicating that an adverse event is associated in time to an immunization.
Cardinality | 0..*
Comments | A reaction may be an indication of an allergy or intolerance and, if this is determined to be the case, it should be recorded as a new AllergyIntolerance resource instance as most systems will not query against past Immunization.reaction elements.
Immunization.reaction.date
Element Id | Immunization.reaction.date
Definition | Date of reaction to the immunization.
Cardinality | 0..1
Type | dateTime
Immunization.reaction.detail
Element Id | Immunization.reaction.detail
Definition | Details of the reaction.
Cardinality | 0..1

Type | Reference(Observation)
Immunization.reaction.reported
Element Id | Immunization.reaction.reported
Definition | Self-reported indicator.
Cardinality | 0..1
Type | boolean
Immunization.protocolApplied
Element Id | Immunization.protocolApplied
Definition | The protocol (set of recommendations) being followed by the provider who administered the dose.
Cardinality | 0..*
Immunization.protocolApplied.series
Element Id | Immunization.protocolApplied.series
Definition | One possible path to achieve presumed immunity against a disease - within the context of an authority.
Cardinality | 0..1
Type | string
Immunization.protocolApplied.authority
Element Id | Immunization.protocolApplied.authority
Definition | Indicates the authority who published the protocol (e.g. ACIP) that is being followed.
Cardinality | 0..1
Type | Reference(Organization)
Immunization.protocolApplied.targetDisease
Element Id | Immunization.protocolApplied.targetDisease
Definition | The vaccine preventable disease the dose is being administered against.
Cardinality | 0..*
Terminology Binding | Immunization Target Disease Codes (Example)
Type | CodeableConcept
Immunization.protocolApplied.doseNumber[x]
Element Id | Immunization.protocolApplied.doseNumber[x]
Definition | Nominal position in a series.
Cardinality | 1..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The use of an integer is preferred if known. A string should only be used in cases where an integer is not available (such as when documenting a recurring booster dose).
Immunization.protocolApplied.seriesDoses[x]
Element Id | Immunization.protocolApplied.seriesDoses[x]
Definition | The recommended number of doses to achieve immunity.
Cardinality | 0..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The use of an integer is preferred if known. A string should only be used in cases where an integer is not available (such as when documenting a recurring booster dose).

11.7.7 Resource Immunization - Mappings

URL: https://hl7.org/fhir/R4/immunization-mappings.html

11.7.7 Resource Immunization - Mappings

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

CDA (R2)

RIM Mapping

Mappings for the immunization resource (see Mappings to Other Standards for further information & status).

11.7.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Immunization
identifier | FiveWs.identifier
status | FiveWs.status
vaccineCode | FiveWs.what[x]
patient | FiveWs.subject[x]
encounter | FiveWs.context
occurrence[x] | FiveWs.done[x]
recorded | FiveWs.recorded
primarySource | FiveWs.source
reportOrigin | FiveWs.source

location | FiveWs.where[x]
actor | FiveWs.actor

11.7.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Immunization | Event
identifier | Event.identifier
status | Event.status
statusReason | Event.statusReason
vaccineCode | Event.code
patient | Event.subject
encounter | Event.context
occurrence[x] | Event.occurrence[x]
location | Event.location
performer | Event.performer
function | Event.performer.function
actor | Event.performer.actor
note | Event.note
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference

11.7.7.3 HL7 v2 Mapping (http://hl7.org/v2)

Immunization | VXU_V04
identifier
status
statusReason
vaccineCode | RXA-5
patient | PID-3
encounter | PV1-19
occurrence[x] | RXA-3
recorded
primarySource | RXA-9
reportOrigin | RXA-9
location | RXA-27 (or RXA-11, deprecated as of v2.7)
manufacturer | RXA-17
lotNumber | RXA-15
expirationDate | RXA-16
site | RXR-2
route | RXR-1
doseQuantity | RXA-6 / RXA-7
performer | ORC-12 / RXA-10
function
actor
note | OBX-5 : OBX-3 = 48767-8
reasonCode
reasonReference
isSubpotent | RXA-20 = PA (partial administration)
subpotentReason
education
documentType | OBX-5 : OBX-3 = 69764-9
reference
publicationDate | OBX-5 : OBX-3 = 29768-9
presentationDate | OBX-5 : OBX-3 = 29769-7
programEligibility | OBX-5 : OBX-3 = 64994-7
fundingSource
reaction | OBX-3
date | OBX-14 (ideally this would be reported in an IAM segment, but IAM is not part of the HL7 v2 VXU message - most likely would appear in OBX segments if at all)
detail | OBX-5
reported | (HL7 v2 doesn't seem to provide for this)
protocolApplied
series
authority
targetDisease
doseNumber[x]
seriesDoses[x]

11.7.7.4 CDA (R2) (http://hl7.org/v3/cda)

Immunization
identifier | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/id
status
statusReason

vaccineCode | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manufacturedProduct/manufacturedMaterial/realmCode/code
patient
encounter
occurrence[x] | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/effectiveTime/value
recorded
primarySource
reportOrigin
location
manufacturer | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manufacturedProduct/manufacuturerOrganization/name
lotNumber | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/consumable/manufacturedProduct/manufacturedMaterial/lotNumberText
expirationDate
site | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/approachSiteCode/code
route | ClinicalDocument/component/StructuredBody/component/section/entry/substanceAdministration/routeCode/code
doseQuantity
performer
function
actor
note
reasonCode
reasonReference
isSubpotent
subpotentReason
education
documentType
reference
publicationDate
presentationDate
programEligibility
fundingSource
reaction
date
detail
reported
protocolApplied
series
authority
targetDisease
doseNumber[x]
seriesDoses[x]

11.7.7.5 RIM Mapping (http://hl7.org/v3)

Immunization | SubstanceAdministration
identifier | .id
status | statusCode
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN].reasonCOde
vaccineCode | .code
patient | .partipication[ttypeCode=].role
encounter | component->EncounterEvent
occurrence[x] | .effectiveTime
recorded | .participation[typeCode=AUT].time
primarySource | immunization.uncertaintycode (if primary source=false, uncertainty=U)
reportOrigin | .participation[typeCode=INF].role[classCode=PAT] (this syntax for self-reported)
.participation[typeCode=INF].role[classCode=LIC] (this syntax for health care professional)
.participation[typeCode=INF].role[classCode=PRS] (this syntax for family member)
location | .participation[typeCode=LOC].COCT_MT240000UV
manufacturer | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=ORG]
lotNumber | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].id
expirationDate | .participation[typeCode=CSM].role[classCode=INST].scopedRole.scoper[classCode=MMAT].expirationTime
site | observation.targetSiteCode
route | .routeCode
doseQuantity | .doseQuantity
performer | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
function | .participation.functionCode
actor | .player
note | note
reasonCode | [actionNegationInd=false].reasonCode
reasonReference | N/A
isSubpotent | N/A

subpotentReason | N/A
education | N/A
documentType | N/A
reference | N/A
publicationDate | N/A
presentationDate | N/A
programEligibility | N/A
fundingSource | N/A
reaction | Observation[classCode=obs].code
date | .effectiveTime
detail | .value
reported | .participation[typeCode=INF].role[classCode=PAT] (this syntax for self-reported=true)
protocolApplied | N/A
series | N/A
authority | N/A
targetDisease | N/A
doseNumber[x] | N/A
seriesDoses[x] | N/A

11.7.8 Resource Immunization - Extensions & Profiles

URL: https://hl7.org/fhir/R4/immunization-profiles.html

11.7.8 Resource Immunization - Extensions & Profiles

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

This table lists profiles and extensions for the Immunization resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.7.8.1 Profiles

No Profiles defined for this resource

11.7.8.2 Extensions

workflow-researchStudy | Immunization | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.7.8.3 Search Extensions

No Search Extensions defined for this resource

11.7.9 Resource Immunization - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/immunization-version-maps.html

No validation errors - all conversions are clean

11.7.9 Resource Immunization - R3/R4 Conversions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner

R3 : R4 Conversion maps for Immunization.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

11.7.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Immunization3to4" = "R3 to R4 Conversions for Immunization"
```

```
uses "http://hl7.org/fhir/3.0/StructureDefinition/Immunization" alias ImmunizationR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Immunization" alias Immunization as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Immunization(source src : ImmunizationR3, target tgt : Immunization) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.notGiven where value = true -> tgt.status = 'not-done';
  src.notGiven as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.notGiven',  vt.value = v;
  src.vaccineCode -> tgt.vaccineCode;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.date as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.primarySource -> tgt.primarySource;
  src.reportOrigin -> tgt.reportOrigin;
  src.location -> tgt.location;
  src.manufacturer -> tgt.manufacturer;
  src.lotNumber -> tgt.lotNumber;
  src.expirationDate -> tgt.expirationDate;
  src.site -> tgt.site;
  src.route -> tgt.route;
  src.doseQuantity -> tgt.doseQuantity;
  src.practitioner as vs -> tgt.performer as vt then practitioner(vs, vt);
  src.note -> tgt.note;
  src.explanation as s then {
    s.reason -> tgt.reasonCode;
    s.reasonNotGiven -> tgt.reasonCode;
  };
  src.reaction as vs -> tgt.reaction as vt then reaction(vs, vt);
  src.vaccinationProtocol as vs -> tgt.protocolApplied as vt then protocol(vs, vt);
}

group practitioner(source src, target tgt) extends BackboneElement {
  src.role -> tgt.function;
  src.actor -> tgt.actor;
}

group reaction(source src, target tgt) extends BackboneElement {
  src.date -> tgt.date;
  src.detail -> tgt.detail;
  src.reported -> tgt.reported;
}

group protocol(source src, target tgt) extends BackboneElement {
  src.doseSequence as vs -> tgt.doseNumber = create('positiveInt') as vt then positiveInt(vs, vt);
  src.description as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.description',  vt.value = v;
  src.authority -> tgt.authority;
  src.series -> tgt.series;
  src.seriesDoses as vs -> tgt.seriesDoses = create('positiveInt') as vt then positiveInt(vs, vt);
  src.targetDisease -> tgt.targetDisease;
  src.doseStatus as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.doseStatus',  vt.value = v;
  src.doseStatusReason as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.doseStatusReason',  vt.value = v;
}
```

11.7.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Immunization4to3" = "R4 to R3 Conversion for Immunization"

uses "http://hl7.org/fhir/StructureDefinition/Immunization" alias Immunization as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Immunization" alias ImmunizationR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Immunization(source src : Immunization, target tgt : ImmunizationR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status where value != 'not-done' -> tgt.status;
  src.status where value = 'not-done' ->  tgt.status = 'completed',  tgt.notGiven = true;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Immunization.notGiven' then {
    e.value -> tgt.notGiven;
  };
  src.vaccineCode -> tgt.vaccineCode;
  src.patient -> tgt.patient;
  src.encounter -> tgt.encounter;
  src.occurrence : dateTime as vs -> tgt.date as vt then dateTime(vs, vt);
  src.primarySource -> tgt.primarySource;
  src.reportOrigin -> tgt.reportOrigin;
  src.location -> tgt.location;
  src.manufacturer -> tgt.manufacturer;
  src.lotNumber -> tgt.lotNumber;
  src.expirationDate -> tgt.expirationDate;
  src.site -> tgt.site;
  src.route -> tgt.route;
  src.doseQuantity -> tgt.doseQuantity;
  src.performer as vs -> tgt.practitioner as vt then practitioner(vs, vt);
  src.note -> tgt.note;
```

```
  src.reasonCode as vs where src.status != 'not-done' ->  tgt.explanation as t,  t.reason = vs;
  src.reasonCode as vs where src.status = 'not-done' ->  tgt.explanation as t,  t.reasonNotGiven = vs;
  src.reaction as vs -> tgt.reaction as vt then reaction(vs, vt);
  src.protocolApplied as vs -> tgt.vaccinationProtocol as vt then protocol(vs, vt);
}

group practitioner(source src, target tgt) extends BackboneElement {
  src.function -> tgt.role;
  src.actor -> tgt.actor;
}

group reaction(source src, target tgt) extends BackboneElement {
  src.date -> tgt.date;
  src.detail -> tgt.detail;
  src.reported -> tgt.reported;
}

group protocol(source src, target tgt) extends BackboneElement {
  src.doseNumber : positiveInt as vs -> tgt.doseSequence as vt then positiveInt(vs, vt);
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.description' then {
    e.value -> tgt.description;
  };
  src.authority -> tgt.authority;
  src.series -> tgt.series;
  src.seriesDoses : positiveInt as vs -> tgt.seriesDoses as vt then positiveInt(vs, vt);
  src.targetDisease -> tgt.targetDisease;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.doseStatus' then {
    e.value -> tgt.doseStatus;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Immunization.vaccinationProtocol.doseStatusReason' then {
    e.value -> tgt.doseStatusReason;
  };
}
```

11.7.9.3 R4 Validation Errors

11.7.10 Immunization HL7 Extensions

URL: https://hl7.org/fhir/R4/immunization-extensions.html

Defines common extensions used with or related to the Immunization resource

11.7.10 Immunization HL7 Extensions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative

11.7.10.1 Content

11.7.11 JSON Schema for Immunization

URL: https://hl7.org/fhir/R4/immunization.schema.json.html

11.7.11 JSON Schema for Immunization

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is a representation of the json schema for Immunization, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Immunization",
  "$ref": "#/definitions/Immunization",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Immunization": {
      "description": "Describes the event of a patient being administered a vaccine or a record of an immunization as
    reported by a patient, a clinician or another party.",
      "properties": {
        "resourceType": {
          "description": "This is a Immunization resource",
          "const": "Immunization"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
```

```
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique identifier assigned to this immunization record.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the current status of the immunization event.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Indicates the reason the immunization event was not performed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "vaccineCode": {
      "description": "Vaccine that was administered or was to be administered.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "patient": {
      "description": "The patient who either received or did not receive the immunization.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The visit or admission or other contact between patient and health care provider the immunization was performed as part of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "occurrenceDateTime": {
```

```
      "description": "Date vaccine administered or was to be administered.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrenceString": {
      "description": "Date vaccine administered or was to be administered.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_occurrenceString": {
      "description": "Extensions for occurrenceString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recorded": {
      "description": "The date the occurrence of the immunization was first captured in the record - potentially
significantly after the occurrence of the event.",
      "$ref": "#/definitions/dateTime"
    },
    "_recorded": {
      "description": "Extensions for recorded",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "primarySource": {
      "description": "An indication that the content of the record is based on information from the person who
administered the vaccine. This reflects the context under which the data was originally recorded.",
      "$ref": "#/definitions/boolean"
    },
    "_primarySource": {
      "description": "Extensions for primarySource",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reportOrigin": {
      "description": "The source of the data when the report of the immunization event is not based on information
from the person who administered the vaccine.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "location": {
      "description": "The service delivery location where the vaccine administration occurred.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "manufacturer": {
      "description": "Name of vaccine manufacturer.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "lotNumber": {
      "description": "Lot number of the  vaccine product.",
      "$ref": "#/definitions/string"
    },
    "_lotNumber": {
      "description": "Extensions for lotNumber",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expirationDate": {
      "description": "Date vaccine batch expires.",
      "$ref": "#/definitions/date"
    },
    "_expirationDate": {
      "description": "Extensions for expirationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "site": {
      "description": "Body site where vaccine was administered.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "route": {
      "description": "The path by which the vaccine product is taken into the body.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "doseQuantity": {
      "description": "The quantity of vaccine product that was administered.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "performer": {
      "description": "Indicates who performed the immunization event.",
      "items": {
        "$ref": "#/definitions/Immunization_Performer"
      },
      "type": "array"
    },
    "note": {
      "description": "Extra information about the immunization that is not conveyed by the other attributes.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
```

```
      "type": "array"
    },
    "reasonCode": {
      "description": "Reasons why the vaccine was administered.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Condition, Observation or DiagnosticReport that supports why the immunization was
administered.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "isSubpotent": {
      "description": "Indication if a dose is considered to be subpotent. By default, a dose should be considered to
 be potent.",
      "$ref": "#/definitions/boolean"
    },
    "_isSubpotent": {
      "description": "Extensions for isSubpotent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subpotentReason": {
      "description": "Reason why a dose is considered to be subpotent.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "education": {
      "description": "Educational material presented to the patient (or guardian) at the time of vaccine
administration.",
      "items": {
        "$ref": "#/definitions/Immunization_Education"
      },
      "type": "array"
    },
    "programEligibility": {
      "description": "Indicates a patient\u0027s eligibility for a funding program.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "fundingSource": {
      "description": "Indicates the source of the vaccine actually administered. This may be different than the
patient eligibility (e.g. the patient may be eligible for a publically purchased vaccine but due to inventory
issues, vaccine purchased with private funds was actually administered).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reaction": {
      "description": "Categorical data indicating that an adverse event is associated in time to an immunization.",
      "items": {
        "$ref": "#/definitions/Immunization_Reaction"
      },
      "type": "array"
    },
    "protocolApplied": {
      "description": "The protocol (set of recommendations) being followed by the provider who administered the
dose.",
      "items": {
        "$ref": "#/definitions/Immunization_ProtocolApplied"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "vaccineCode",
    "resourceType"
  ]
},
"Immunization_Performer": {
  "description": "Describes the event of a patient being administered a vaccine or a record of an immunization as
reported by a patient, a clinician or another party.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "function": {
      "description": "Describes the type of performance (e.g. ordering provider, administering provider, etc.).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "actor": {
      "description": "The practitioner or organization who performed the action.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "actor"
  ]
},
"Immunization_Education": {
  "description": "Describes the event of a patient being administered a vaccine or a record of an immunization as
reported by a patient, a clinician or another party.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "documentType": {
      "description": "Identifier of the material presented to the patient.",
      "$ref": "#/definitions/string"
    },
    "_documentType": {
      "description": "Extensions for documentType",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reference": {
      "description": "Reference pointer to the educational material given to the patient if the information was on
line.",
      "$ref": "#/definitions/uri"
    },
    "_reference": {
      "description": "Extensions for reference",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publicationDate": {
      "description": "Date the educational material was published.",
      "$ref": "#/definitions/dateTime"
    },
```

```
    "_publicationDate": {
      "description": "Extensions for publicationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "presentationDate": {
      "description": "Date the educational material was given to the patient.",
      "$ref": "#/definitions/dateTime"
    },
    "_presentationDate": {
      "description": "Extensions for presentationDate",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Immunization_Reaction": {
  "description": "Describes the event of a patient being administered a vaccine or a record of an immunization as
reported by a patient, a clinician or another party.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "date": {
      "description": "Date of reaction to the immunization.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "detail": {
      "description": "Details of the reaction.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reported": {
      "description": "Self-reported indicator.",
      "$ref": "#/definitions/boolean"
    },
    "_reported": {
      "description": "Extensions for reported",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"Immunization_ProtocolApplied": {
  "description": "Describes the event of a patient being administered a vaccine or a record of an immunization as
reported by a patient, a clinician or another party.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "series": {
        "description": "One possible path to achieve presumed immunity against a disease - within the context of an
    authority.",
        "$ref": "#/definitions/string"
      },
      "_series": {
        "description": "Extensions for series",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "authority": {
        "description": "Indicates the authority who published the protocol (e.g. ACIP) that is being followed.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "targetDisease": {
        "description": "The vaccine preventable disease the dose is being administered against.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "doseNumberPositiveInt": {
        "description": "Nominal position in a series.",
        "pattern": "^[1-9][0-9]*$",
        "type": "number"
      },
      "_doseNumberPositiveInt": {
        "description": "Extensions for doseNumberPositiveInt",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "doseNumberString": {
        "description": "Nominal position in a series.",
        "pattern": "^[ \\r\\n\\t\\S]+$",
        "type": "string"
      },
      "_doseNumberString": {
        "description": "Extensions for doseNumberString",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "seriesDosesPositiveInt": {
        "description": "The recommended number of doses to achieve immunity.",
        "pattern": "^[1-9][0-9]*$",
        "type": "number"
      },
      "_seriesDosesPositiveInt": {
        "description": "Extensions for seriesDosesPositiveInt",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "seriesDosesString": {
        "description": "The recommended number of doses to achieve immunity.",
        "pattern": "^[ \\r\\n\\t\\S]+$",
        "type": "string"
      },
      "_seriesDosesString": {
        "description": "Extensions for seriesDosesString",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false
  }
 }
}
```


11.7.12 Resource Immunization - Maturity Dependency View

URL: https://hl7.org/fhir/R4/immunization-dependencies.html

11.7.12 Resource Immunization - Maturity Dependency View

Public Health and Emergency Response Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner

Dependency Graph for Immunization FMM level 3

Name | Card. | Type | Dependency Analysis

Immunization | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
vaccineCode | 1..1 | CodeableConcept
patient | 1..1 | Reference(Patient)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
occurrence[x] | 1..1
occurrenceDateTime | | dateTime
occurrenceString | | string
recorded | 0..1 | dateTime
primarySource | 0..1 | boolean
reportOrigin | 0..1 | CodeableConcept
location | 0..1 | Reference(Location)
manufacturer | 0..1 | Reference(Organization)
lotNumber | 0..1 | string
expirationDate | 0..1 | date
site | 0..1 | CodeableConcept
route | 0..1 | CodeableConcept
doseQuantity | 0..1 | SimpleQuantity
performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Organization) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
note | 0..* | Annotation
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport)
isSubpotent | 0..1 | boolean
subpotentReason | 0..* | CodeableConcept
education | 0..* | BackboneElement
documentType | 0..1 | string
reference | 0..1 | uri
publicationDate | 0..1 | dateTime
presentationDate | 0..1 | dateTime
programEligibility | 0..* | CodeableConcept
fundingSource | 0..1 | CodeableConcept
reaction | 0..* | BackboneElement
date | 0..1 | dateTime
detail | 0..1 | Reference(Observation)
reported | 0..1 | boolean
protocolApplied | 0..* | BackboneElement
series | 0..1 | string
authority | 0..1 | Reference(Organization)
targetDisease | 0..* | CodeableConcept
doseNumber[x] | 1..1
doseNumberPositiveInt | | positiveInt
doseNumberString | | string
seriesDoses[x] | 0..1
seriesDosesPositiveInt | | positiveInt
seriesDosesString | | string

11.7.13 Value Set Sc-valueset-immunization-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-immunization-status.html

11.7.13 Value Set Sc-valueset-immunization-status - Status Code Mapping

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.7.14 ImmunizationStatusCodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-immunization-status)

Mapping from http://hl7.org/fhir/ValueSet/immunization-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the current status of the administered dose of vaccine."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
completed | equivalent | complete
not-done | equivalent | abandoned

11.8 Resource ImmunizationEvaluation - Content

URL: https://hl7.org/fhir/R4/immunizationevaluation.html
Public Health and Emergency Response Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Patient

Describes a comparison of an immunization event against published recommendations to determine if the administration is "valid" in relation to those recommendations.

11.8.1 Scope and Usage

The ImmunizationEvaluation resource is intended to cover communicating the results of an evaluation of a vaccine administration event (documented using the Immunization resource) against a set of published recommendations (protocols). Evaluating administered doses in a patient's immunization history is central to ensuring that the patient if fully protected against vaccine-preventable diseases. It is also a necessary prerequisite to generating a customized set of immunization recommendations for the patient.

11.8.2 Boundaries and Relationships

The ImmunizationEvaluation resource communicates the results of an evaluation of an instance of the Immunization resource relative to a set of published recommendations. It is intended to communicate if the vaccine administration event was "valid" or "not valid" relative to the recommendations. Once all vaccine administration events in a patient's record have been evaluated, it is then possible to generate a set of vaccination related recommendations, unique for the patient. These recommendations are communicated using the ImmunizationRecommedation resource.

As the ImmunizationEvaluation resource is focused on the evaluation of a single vaccine administration event, it does not include data regarding the overall patient status relative to the recommendations used in the evaluation. For example, a total of 3 doses of hepatitis B vaccine may be required for patient protection, but an instance of the ImmunizationEvaluation resource will be concerned with only one of the doses (although other instances of ImmunizationEvaluation should exist for the other doses). The ImmunizationRecommendation resource is used to convey the overall patient status (eg. a dose is due or the patient is complete) relative to the hepatitis B series.

11.8.3 Background and Context

Through FHIR Release 3, the Immunization resource contained a "protocol" backbone element which functioned as an evaluation of the vaccine administration event. Beginning in Release 4, this new resource replaces the Immunization.protocol element.

This resource is referenced by ImmunizationRecommendation

11.8.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImmunizationEvaluation | TU | | DomainResource | Immunization evaluation information Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | completed | entered-in-error Immunization Evaluation Status Codes (Required)
patient | Σ | 1..1 | Reference(Patient) | Who this evaluation is for
date | | 0..1 | dateTime | Date evaluation was performed
authority | | 0..1 | Reference(Organization) | Who is responsible for publishing the recommendations
targetDisease | Σ | 1..1 | CodeableConcept | Evaluation target disease Immunization Evaluation Target Disease Codes (Example)
immunizationEvent | Σ | 1..1 | Reference(Immunization) | Immunization being evaluated
doseStatus | Σ | 1..1 | CodeableConcept | Status of the dose relative to published recommendations Immunization Evaluation Dose Status codes (Example)
doseStatusReason | | 0..* | CodeableConcept | Reason for the dose status Immunization Evaluation Dose Status Reason codes (Example)

description | | 0..1 | string | Evaluation notes
series | | 0..1 | string | Name of vaccine series
doseNumber[x] | | 0..1 | | Dose number within series
doseNumberPositiveInt | | | positiveInt
doseNumberString | | | string
seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity
seriesDosesPositiveInt | | | positiveInt
seriesDosesString | | | string
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationEvaluation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error -->
 <patient><!-- 1..1 Reference(Patient) Who this evaluation is for --></patient>
 <date value="[dateTime]"/><!-- 0..1 Date evaluation was performed -->
 <authority><!-- 0..1 Reference(Organization) Who is responsible for publishing the recommendations --></authority>
 <targetDisease><!-- 1..1 CodeableConcept Evaluation target disease --></targetDisease>
 <immunizationEvent><!-- 1..1 Reference(Immunization) Immunization being evaluated --></immunizationEvent>
 <doseStatus><!-- 1..1 CodeableConcept Status of the dose relative to published recommendations --></doseStatus>
 <doseStatusReason><!-- 0..* CodeableConcept Reason for the dose status --></doseStatusReason>
 <description value="[string]"/><!-- 0..1 Evaluation notes -->
 <series value="[string]"/><!-- 0..1 Name of vaccine series -->
 <doseNumber[x]><!-- 0..1 positiveInt|string Dose number within series --></doseNumber[x]>
 <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
</ImmunizationEvaluation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationEvaluation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error
  "patient" : { Reference(Patient) }, // R!  Who this evaluation is for
  "date" : "<dateTime>", // Date evaluation was performed
  "authority" : { Reference(Organization) }, // Who is responsible for publishing the recommendations
  "targetDisease" : { CodeableConcept }, // R!  Evaluation target disease
  "immunizationEvent" : { Reference(Immunization) }, // R!  Immunization being evaluated
  "doseStatus" : { CodeableConcept }, // R!  Status of the dose relative to published recommendations
  "doseStatusReason" : [{ CodeableConcept }], // Reason for the dose status
  "description" : "<string>", // Evaluation notes
  "series" : "<string>", // Name of vaccine series
  // doseNumber[x]: Dose number within series. One of these 2:
  "doseNumberPositiveInt" : "<positiveInt>",
  "doseNumberString" : "<string>",
  // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
  "seriesDosesPositiveInt" : "<positiveInt>"
  "seriesDosesString" : "<string>"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationEvaluation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationEvaluation.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ImmunizationEvaluation.status [ code ]; # 1..1 completed | entered-in-error
  fhir:ImmunizationEvaluation.patient [ Reference(Patient) ]; # 1..1 Who this evaluation is for
  fhir:ImmunizationEvaluation.date [ dateTime ]; # 0..1 Date evaluation was performed
  fhir:ImmunizationEvaluation.authority [ Reference(Organization) ]; # 0..1 Who is responsible for publishing the
    recommendations
  fhir:ImmunizationEvaluation.targetDisease [ CodeableConcept ]; # 1..1 Evaluation target disease
  fhir:ImmunizationEvaluation.immunizationEvent [ Reference(Immunization) ]; # 1..1 Immunization being evaluated
  fhir:ImmunizationEvaluation.doseStatus [ CodeableConcept ]; # 1..1 Status of the dose relative to published
    recommendations
  fhir:ImmunizationEvaluation.doseStatusReason [ CodeableConcept ], ... ; # 0..* Reason for the dose status
  fhir:ImmunizationEvaluation.description [ string ]; # 0..1 Evaluation notes
  fhir:ImmunizationEvaluation.series [ string ]; # 0..1 Name of vaccine series
  # ImmunizationEvaluation.doseNumber[x] : 0..1 Dose number within series. One of these 2
    fhir:ImmunizationEvaluation.doseNumberPositiveInt [ positiveInt ]
    fhir:ImmunizationEvaluation.doseNumberString [ string ]
  # ImmunizationEvaluation.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of these 2
    fhir:ImmunizationEvaluation.seriesDosesPositiveInt [ positiveInt ]
    fhir:ImmunizationEvaluation.seriesDosesString [ string ]
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
ImmunizationEvaluation | TU | | DomainResource | Immunization evaluation information Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | completed | entered-in-error Immunization Evaluation Status Codes (Required)
patient | Σ | 1..1 | Reference(Patient) | Who this evaluation is for
date | | 0..1 | dateTime | Date evaluation was performed
authority | | 0..1 | Reference(Organization) | Who is responsible for publishing the recommendations
targetDisease | Σ | 1..1 | CodeableConcept | Evaluation target disease Immunization Evaluation Target Disease Codes (Example)
immunizationEvent | Σ | 1..1 | Reference(Immunization) | Immunization being evaluated
doseStatus | Σ | 1..1 | CodeableConcept | Status of the dose relative to published recommendations Immunization Evaluation Dose Status codes (Example)
doseStatusReason | | 0..* | CodeableConcept | Reason for the dose status Immunization Evaluation Dose Status Reason codes (Example)
description | | 0..1 | string | Evaluation notes
series | | 0..1 | string | Name of vaccine series
doseNumber[x] | | 0..1 | | Dose number within series
doseNumberPositiveInt | | | positiveInt
doseNumberString | | | string
seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity
seriesDosesPositiveInt | | | positiveInt
seriesDosesString | | | string
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationEvaluation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 completed | entered-in-error -->
 <patient><!-- 1..1 Reference(Patient) Who this evaluation is for --></patient>
 <date value="[dateTime]"/><!-- 0..1 Date evaluation was performed -->
 <authority><!-- 0..1 Reference(Organization) Who is responsible for publishing the recommendations --></authority>
 <targetDisease><!-- 1..1 CodeableConcept Evaluation target disease --></targetDisease>
 <immunizationEvent><!-- 1..1 Reference(Immunization) Immunization being evaluated --></immunizationEvent>
 <doseStatus><!-- 1..1 CodeableConcept Status of the dose relative to published recommendations --></doseStatus>
 <doseStatusReason><!-- 0..* CodeableConcept Reason for the dose status --></doseStatusReason>
 <description value="[string]"/><!-- 0..1 Evaluation notes -->
 <series value="[string]"/><!-- 0..1 Name of vaccine series -->
 <doseNumber[x]><!-- 0..1 positiveInt|string Dose number within series --></doseNumber[x]>
 <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
</ImmunizationEvaluation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationEvaluation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  completed | entered-in-error
  "patient" : { Reference(Patient) }, // R!  Who this evaluation is for
  "date" : "<dateTime>", // Date evaluation was performed
  "authority" : { Reference(Organization) }, // Who is responsible for publishing the recommendations
  "targetDisease" : { CodeableConcept }, // R!  Evaluation target disease
  "immunizationEvent" : { Reference(Immunization) }, // R!  Immunization being evaluated
  "doseStatus" : { CodeableConcept }, // R!  Status of the dose relative to published recommendations
  "doseStatusReason" : [{ CodeableConcept }], // Reason for the dose status
  "description" : "<string>", // Evaluation notes
  "series" : "<string>", // Name of vaccine series
  // doseNumber[x]: Dose number within series. One of these 2:
  "doseNumberPositiveInt" : "<positiveInt>",
  "doseNumberString" : "<string>",
  // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
  "seriesDosesPositiveInt" : "<positiveInt>"
  "seriesDosesString" : "<string>"
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationEvaluation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationEvaluation.identifier [ Identifier ], ... ; # 0..* Business identifier
```

```
  fhir:ImmunizationEvaluation.status [ code ]; # 1..1 completed | entered-in-error
  fhir:ImmunizationEvaluation.patient [ Reference(Patient) ]; # 1..1 Who this evaluation is for
  fhir:ImmunizationEvaluation.date [ dateTime ]; # 0..1 Date evaluation was performed
  fhir:ImmunizationEvaluation.authority [ Reference(Organization) ]; # 0..1 Who is responsible for publishing the
    recommendations
  fhir:ImmunizationEvaluation.targetDisease [ CodeableConcept ]; # 1..1 Evaluation target disease
  fhir:ImmunizationEvaluation.immunizationEvent [ Reference(Immunization) ]; # 1..1 Immunization being evaluated
  fhir:ImmunizationEvaluation.doseStatus [ CodeableConcept ]; # 1..1 Status of the dose relative to published
    recommendations
  fhir:ImmunizationEvaluation.doseStatusReason [ CodeableConcept ], ... ; # 0..* Reason for the dose status
  fhir:ImmunizationEvaluation.description [ string ]; # 0..1 Evaluation notes
  fhir:ImmunizationEvaluation.series [ string ]; # 0..1 Name of vaccine series
  # ImmunizationEvaluation.doseNumber[x] : 0..1 Dose number within series. One of these 2
    fhir:ImmunizationEvaluation.doseNumberPositiveInt [ positiveInt ]
    fhir:ImmunizationEvaluation.doseNumberString [ string ]
  # ImmunizationEvaluation.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of these 2
    fhir:ImmunizationEvaluation.seriesDosesPositiveInt [ positiveInt ]
    fhir:ImmunizationEvaluation.seriesDosesString [ string ]
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.8.4.1 Terminology Bindings

Path | Definition | Type | Reference
ImmunizationEvaluation.status | The status of the evaluation being done. | Required | ImmunizationEvaluationStatusCodes
ImmunizationEvaluation.targetDisease | The vaccine preventable disease the dose is being evaluated against. | Example | ImmunizationEvaluationTargetDiseaseCodes
ImmunizationEvaluation.doseStatus | The status of the administered dose relative to the published recommendations for the target disease. | Example | ImmunizationEvaluationDoseStatusCodes
ImmunizationEvaluation.doseStatusReason | The reason the dose status was assigned. | Example | ImmunizationEvaluationDoseStatusReasonCodes

11.8.5 Cardinality of Related Resources

A single vaccine administration event (as communicated via the Immunization resource) may be evaluated against multiple published recommendations, against the same set of published recommendations at different points in time or against published recommendations for different vaccine-preventable diseases. That is a single Immunization record may be linked to multiple ImmunizationEvaluation records.

For example a dose of Twinrix contains antigen against both Hepatitis A and Hepatitis B. The evaluation of such a dose should happen against recommendations for both types of Hepatitis. Furthermore, the dose may be evaluated against both national and local recommendation or may be evaluated against either set of recommendations at multiple points in time (yearly as the patient comes in for standard "well person" appointment with their care provider).

Typically a single "recommendation" is prepared for the patient based on the evaluation of all of the vaccine administration events in the patient's history. This "recommendation" may include individual recommendations for multiple vaccines/vaccine groups. For example, a recommendation may be generated that includes recommendations for both influenza and diphtheria/tetanus vaccinations. Such recommendations use the ImmunizationRecommedation resource.

11.8.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Date the evaluation was generated | ImmunizationEvaluation.date
dose-status | token | The status of the dose relative to published recommendations | ImmunizationEvaluation.doseStatus
identifier | token | ID of the evaluation | ImmunizationEvaluation.identifier
immunization-event | reference | The vaccine administration event being evaluated | ImmunizationEvaluation.immunizationEvent (Immunization)
patient | reference | The patient being evaluated | ImmunizationEvaluation.patient (Patient)
status | token | Immunization evaluation status | ImmunizationEvaluation.status
target-disease | token | The vaccine preventable disease being evaluated against | ImmunizationEvaluation.targetDisease

11.8.7 Resource ImmunizationEvaluation - Examples

URL: https://hl7.org/fhir/R4/immunizationevaluation-examples.html

11.8.7 Resource ImmunizationEvaluation - Examples

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
Example of a valid dose | example | XML | JSON | Turtle
Example of an invalid dose | notValid | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.8.8 Resource ImmunizationEvaluation - Detailed Descriptions

URL: https://hl7.org/fhir/R4/immunizationevaluation-definitions.html

11.8.8 Resource ImmunizationEvaluation - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the ImmunizationEvaluation resource.

ImmunizationEvaluation
Element Id | ImmunizationEvaluation
Definition | Describes a comparison of an immunization event against published recommendations to determine if the administration is "valid" in relation to those recommendations.
Cardinality | 0..*
Type | DomainResource
ImmunizationEvaluation.identifier
Element Id | ImmunizationEvaluation.identifier
Definition | A unique identifier assigned to this immunization evaluation record.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
ImmunizationEvaluation.status
Element Id | ImmunizationEvaluation.status
Definition | Indicates the current status of the evaluation of the vaccination administration event.
Cardinality | 1..1
Terminology Binding | Immunization Evaluation Status Codes (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid).)
Summary | true
ImmunizationEvaluation.patient
Element Id | ImmunizationEvaluation.patient
Definition | The individual for whom the evaluation is being done.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true
ImmunizationEvaluation.date
Element Id | ImmunizationEvaluation.date
Definition | The date the evaluation of the vaccine administration event was performed.
Cardinality | 0..1
Type | dateTime
ImmunizationEvaluation.authority
Element Id | ImmunizationEvaluation.authority
Definition | Indicates the authority who published the protocol (e.g. ACIP).
Cardinality | 0..1
Type | Reference(Organization)
ImmunizationEvaluation.targetDisease
Element Id | ImmunizationEvaluation.targetDisease
Definition | The vaccine preventable disease the dose is being evaluated against.
Cardinality | 1..1
Terminology Binding | Immunization Evaluation Target Disease Codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationEvaluation.immunizationEvent
Element Id | ImmunizationEvaluation.immunizationEvent
Definition | The vaccine administration event being evaluated.
Cardinality | 1..1
Type | Reference(Immunization)
Summary | true
ImmunizationEvaluation.doseStatus
Element Id | ImmunizationEvaluation.doseStatus
Definition | Indicates if the dose is valid or not valid with respect to the published recommendations.

Cardinality | 1..1
Terminology Binding | Immunization Evaluation Dose Status codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationEvaluation.doseStatusReason
Element Id | ImmunizationEvaluation.doseStatusReason
Definition | Provides an explanation as to why the vaccine administration event is valid or not relative to the published recommendations.
Cardinality | 0..*
Terminology Binding | Immunization Evaluation Dose Status Reason codes (Example)
Type | CodeableConcept
ImmunizationEvaluation.description
Element Id | ImmunizationEvaluation.description
Definition | Additional information about the evaluation.
Cardinality | 0..1
Type | string
ImmunizationEvaluation.series
Element Id | ImmunizationEvaluation.series
Definition | One possible path to achieve presumed immunity against a disease - within the context of an authority.
Cardinality | 0..1
Type | string
ImmunizationEvaluation.doseNumber[x]
Element Id | ImmunizationEvaluation.doseNumber[x]
Definition | Nominal position in a series.
Cardinality | 0..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The use of an integer is preferred if known. A string should only be used in cases where an integer is not available (such as when documenting a recurring booster dose).
ImmunizationEvaluation.seriesDoses[x]
Element Id | ImmunizationEvaluation.seriesDoses[x]
Definition | The recommended number of doses to achieve immunity.
Cardinality | 0..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The use of an integer is preferred if known. A string should only be used in cases where an integer is not available (such as when documenting a recurring booster dose).

11.8.9 Resource ImmunizationEvaluation - Mappings

URL: https://hl7.org/fhir/R4/immunizationevaluation-mappings.html

11.8.9 Resource ImmunizationEvaluation - Mappings

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

Mappings for the immunizationevaluation resource (see Mappings to Other Standards for further information & status).

11.8.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ImmunizationEvaluation
identifier | FiveWs.identifier
status | FiveWs.status
patient | FiveWs.subject[x]
date | FiveWs.init

11.8.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ImmunizationEvaluation | Event
identifier | Event.identifier
status | Event.status
patient | Event.subject
date | Event.occurrenceDateTime
description | Event.note

11.8.9.3 HL7 v2 Mapping (http://hl7.org/v2)

ImmunizationEvaluation | RSP_K11

identifier
status
patient | PID-3
date | RXA-3
authority | OBX-5 : OBX-3 = (59779-9)
targetDisease | OBX-5 : OBX-3 = (30956-7 | 38890-0)
immunizationEvent
doseStatus | OBX-5 : OBX-3 = (59781-5)
doseStatusReason | OBX-5 : OBX-3 = (30982-3)
description | OBX-5 : OBX-3 = (48767-8)
series
doseNumber[x] | OBX-5 : OBX-3 = (30973-2)
seriesDoses[x] | OBX-5 : OBX-3 = (59782-3)

11.8.10 Resource ImmunizationEvaluation - Extensions & Profiles

URL: https://hl7.org/fhir/R4/immunizationevaluation-profiles.html

11.8.10 Resource ImmunizationEvaluation - Extensions & Profiles

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the ImmunizationEvaluation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.8.10.1 Profiles

No Profiles defined for this resource

11.8.10.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.8.10.3 Search Extensions

No Search Extensions defined for this resource

11.8.11 JSON Schema for ImmunizationEvaluation

URL: https://hl7.org/fhir/R4/immunizationevaluation.schema.json.html

11.8.11 JSON Schema for ImmunizationEvaluation

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for ImmunizationEvaluation, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ImmunizationEvaluation",
  "$ref": "#/definitions/ImmunizationEvaluation",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ImmunizationEvaluation": {
      "description": "Describes a comparison of an immunization event against published recommendations to determine if
    the administration is \"valid\" in relation to those  recommendations.",
      "properties": {
        "resourceType": {
          "description": "This is a ImmunizationEvaluation resource",
          "const": "ImmunizationEvaluation"
        },
```

```
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique identifier assigned to this immunization evaluation record.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the current status of the evaluation of the vaccination administration event.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "The individual for whom the evaluation is being done.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date the evaluation of the vaccine administration event was performed.",
      "$ref": "#/definitions/dateTime"
```

```
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "authority": {
      "description": "Indicates the authority who published the protocol (e.g. ACIP).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "targetDisease": {
      "description": "The vaccine preventable disease the dose is being evaluated against.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "immunizationEvent": {
      "description": "The vaccine administration event being evaluated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "doseStatus": {
      "description": "Indicates if the dose is valid or not valid with respect to the published recommendations.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "doseStatusReason": {
      "description": "Provides an explanation as to why the vaccine administration event is valid or not relative to
 the published recommendations.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "description": {
      "description": "Additional information about the evaluation.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "series": {
      "description": "One possible path to achieve presumed immunity against a disease - within the context of an
authority.",
      "$ref": "#/definitions/string"
    },
    "_series": {
      "description": "Extensions for series",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doseNumberPositiveInt": {
      "description": "Nominal position in a series.",
      "pattern": "^[1-9][0-9]*$",
      "type": "number"
    },
    "_doseNumberPositiveInt": {
      "description": "Extensions for doseNumberPositiveInt",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doseNumberString": {
      "description": "Nominal position in a series.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_doseNumberString": {
      "description": "Extensions for doseNumberString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "seriesDosesPositiveInt": {
      "description": "The recommended number of doses to achieve immunity.",
      "pattern": "^[1-9][0-9]*$",
      "type": "number"
    },
    "_seriesDosesPositiveInt": {
      "description": "Extensions for seriesDosesPositiveInt",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "seriesDosesString": {
      "description": "The recommended number of doses to achieve immunity.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_seriesDosesString": {
      "description": "Extensions for seriesDosesString",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "doseStatus",
    "patient",
    "targetDisease",
    "immunizationEvent",
```

```
        "resourceType"
      ]
    }
  }
}
```

11.8.12 Resource ImmunizationEvaluation - Maturity Dependency View

URL: https://hl7.org/fhir/R4/immunizationevaluation-dependencies.html

11.8.12 Resource ImmunizationEvaluation - Maturity Dependency View

Public Health and Emergency Response Work Group | Maturity Level: 0 | Trial Use | Compartments: Patient

Dependency Graph for ImmunizationEvaluation FMM level 0

Name | Card. | Type | Dependency Analysis
ImmunizationEvaluation | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
patient | 1..1 | Reference(Patient)
date | 0..1 | dateTime
authority | 0..1 | Reference(Organization)
targetDisease | 1..1 | CodeableConcept
immunizationEvent | 1..1 | Reference(Immunization)
doseStatus | 1..1 | CodeableConcept
doseStatusReason | 0..* | CodeableConcept
description | 0..1 | string
series | 0..1 | string
doseNumber[x] | 0..1
doseNumberPositiveInt | | positiveInt
doseNumberString | | string
seriesDoses[x] | 0..1
seriesDosesPositiveInt | | positiveInt
seriesDosesString | | string

11.8.13 Value Set Sc-valueset-immunization-evaluation-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-immunization-evaluation-status.html

11.8.13 Value Set Sc-valueset-immunization-evaluation-status - Status Code Mapping

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

11.8.14 ImmunizationEvaluationStatusCodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-immunization-evaluation-status)

Mapping from http://hl7.org/fhir/ValueSet/immunization-evaluation-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the current status of the evaluation for vaccine administration event."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
completed | equivalent | complete

11.9 Resource ImmunizationRecommendation - Content

URL: https://hl7.org/fhir/R4/immunizationrecommendation.html
Public Health and Emergency Response Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

A patient's point-in-time set of recommendations (i.e. forecasting) according to a published schedule with optional supporting justification.

11.9.1 Scope and Usage

The ImmunizationRecommendation resource is intended to cover communication of a specified patient's immunization recommendations and status across all healthcare disciplines in all care settings and all regions. The set of customized (for the patient) recommendations is based on the comparison of the patient's immunization history with a set of published recommendations (protocols).

Additionally, the ImmunizationRecommendation resource is expected to cover key concepts related to the querying of a patient's immunization recommendations and status. This resource - through consultation with the Public Health work group - is believed to meet key use cases and information requirements as defined in the existing HL7 v3 POIZ domain and Immunization Domain Analysis Model.

11.9.2 Boundaries and Relationships

This resource references the following resources:

• Patient

• Organization

• Immunization

• ImmunizationEvaluation

This resource is referenced by Appointment, DiagnosticReport, Encounter, MedicationRequest and Observation

11.9.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImmunizationRecommendation | TU | | DomainResource | Guidance or advice relating to an immunization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
patient | Σ | 1..1 | Reference(Patient) | Who this profile is for
date | Σ | 1..1 | dateTime | Date recommendation(s) created
authority | | 0..1 | Reference(Organization) | Who is responsible for protocol
recommendation | ΣI | 1..* | BackboneElement | Vaccine administration recommendations + Rule: One of vaccineCode or targetDisease SHALL be present
vaccineCode | Σ | 0..* | CodeableConcept | Vaccine or vaccine group recommendation applies to Vaccine Administered Value Set (Example)
targetDisease | Σ | 0..1 | CodeableConcept | Disease to be immunized against Immunization Recommendation Target Disease Codes (Example)
contraindicatedVaccineCode | Σ | 0..* | CodeableConcept | Vaccine which is contraindicated to fulfill the recommendation Vaccine Administered Value Set (Example)
forecastStatus | ?!Σ | 1..1 | CodeableConcept | Vaccine recommendation status Immunization Recommendation Status Codes (Example)
forecastReason | Σ | 0..* | CodeableConcept | Vaccine administration status reason Immunization Recommendation Reason Codes (Example)
dateCriterion | | 0..* | BackboneElement | Dates governing proposed immunization
code | | 1..1 | CodeableConcept | Type of date Immunization Recommendation Date Criterion Codes (Example)
value | | 1..1 | dateTime | Recommended date
description | | 0..1 | string | Protocol details
series | | 0..1 | string | Name of vaccination series
doseNumber[x] | Σ | 0..1 | | Recommended dose number within series
doseNumberPositiveInt | | | positiveInt
doseNumberString | | | string
seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity
seriesDosesPositiveInt | | | positiveInt
seriesDosesString | | | string
supportingImmunization | | 0..* | Reference(Immunization | ImmunizationEvaluation) | Past immunizations supporting recommendation
supportingPatientInformation | | 0..* | Reference(Any) | Patient observations supporting recommendation
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationRecommendation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <patient><!-- 1..1 Reference(Patient) Who this profile is for --></patient>
 <date value="[dateTime]"/><!-- 1..1 Date recommendation(s) created -->
 <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
 <recommendation>  <!-- 1..* Vaccine administration recommendations -->
  <vaccineCode><!-- 0..* CodeableConcept Vaccine  or vaccine group recommendation applies to --></vaccineCode>
  <targetDisease><!-- 0..1 CodeableConcept Disease to be immunized against --></targetDisease>
  <contraindicatedVaccineCode><!-- 0..* CodeableConcept Vaccine which is contraindicated to fulfill the recommendation
    --></contraindicatedVaccineCode>
  <forecastStatus><!-- 1..1 CodeableConcept Vaccine recommendation status --></forecastStatus>
  <forecastReason><!-- 0..* CodeableConcept Vaccine administration status reason --></forecastReason>
  <dateCriterion>  <!-- 0..* Dates governing proposed immunization -->
   <code><!-- 1..1 CodeableConcept Type of date --></code>
   <value value="[dateTime]"/><!-- 1..1 Recommended date -->
  </dateCriterion>
  <description value="[string]"/><!-- 0..1 Protocol details -->
  <series value="[string]"/><!-- 0..1 Name of vaccination series -->
  <doseNumber[x]><!-- 0..1 positiveInt|string Recommended dose number within series --></doseNumber[x]>
  <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
  <supportingImmunization><!-- 0..* Reference(Immunization|ImmunizationEvaluation) Past immunizations supporting
    recommendation --></supportingImmunization>
  <supportingPatientInformation><!-- 0..* Reference(Any) Patient observations supporting recommendation
    --></supportingPatientInformation>
 </recommendation>
</ImmunizationRecommendation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationRecommendation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "patient" : { Reference(Patient) }, // R!  Who this profile is for
  "date" : "<dateTime>", // R!  Date recommendation(s) created
  "authority" : { Reference(Organization) }, // Who is responsible for protocol
  "recommendation" : [{ // R!  Vaccine administration recommendations
    "vaccineCode" : [{ CodeableConcept }], // Vaccine  or vaccine group recommendation applies to
    "targetDisease" : { CodeableConcept }, // Disease to be immunized against
    "contraindicatedVaccineCode" : [{ CodeableConcept }], // Vaccine which is contraindicated to fulfill the
    recommendation
    "forecastStatus" : { CodeableConcept }, // R!  Vaccine recommendation status
    "forecastReason" : [{ CodeableConcept }], // Vaccine administration status reason
    "dateCriterion" : [{ // Dates governing proposed immunization
      "code" : { CodeableConcept }, // R!  Type of date
      "value" : "<dateTime>" // R!  Recommended date
    }],
    "description" : "<string>", // Protocol details
    "series" : "<string>", // Name of vaccination series
    // doseNumber[x]: Recommended dose number within series. One of these 2:
    "doseNumberPositiveInt" : "<positiveInt>",
    "doseNumberString" : "<string>",
    // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
    "seriesDosesPositiveInt" : "<positiveInt>",
    "seriesDosesString" : "<string>",
    "supportingImmunization" : [{ Reference(Immunization|ImmunizationEvaluation) }], // Past immunizations supporting
    recommendation
    "supportingPatientInformation" : [{ Reference(Any) }] // Patient observations supporting recommendation
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationRecommendation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationRecommendation.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ImmunizationRecommendation.patient [ Reference(Patient) ]; # 1..1 Who this profile is for
  fhir:ImmunizationRecommendation.date [ dateTime ]; # 1..1 Date recommendation(s) created
  fhir:ImmunizationRecommendation.authority [ Reference(Organization) ]; # 0..1 Who is responsible for protocol
  fhir:ImmunizationRecommendation.recommendation [ # 1..* Vaccine administration recommendations
    fhir:ImmunizationRecommendation.recommendation.vaccineCode [ CodeableConcept ], ... ; # 0..* Vaccine  or vaccine
    group recommendation applies to
    fhir:ImmunizationRecommendation.recommendation.targetDisease [ CodeableConcept ]; # 0..1 Disease to be immunized
    against
    fhir:ImmunizationRecommendation.recommendation.contraindicatedVaccineCode [ CodeableConcept ], ... ; # 0..* Vaccine
    which is contraindicated to fulfill the recommendation
    fhir:ImmunizationRecommendation.recommendation.forecastStatus [ CodeableConcept ]; # 1..1 Vaccine recommendation
    status
    fhir:ImmunizationRecommendation.recommendation.forecastReason [ CodeableConcept ], ... ; # 0..* Vaccine
    administration status reason
    fhir:ImmunizationRecommendation.recommendation.dateCriterion [ # 0..* Dates governing proposed immunization
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.code [ CodeableConcept ]; # 1..1 Type of date
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.value [ dateTime ]; # 1..1 Recommended date
    ], ...;
```

```
    fhir:ImmunizationRecommendation.recommendation.description [ string ]; # 0..1 Protocol details
    fhir:ImmunizationRecommendation.recommendation.series [ string ]; # 0..1 Name of vaccination series
    # ImmunizationRecommendation.recommendation.doseNumber[x] : 0..1 Recommended dose number within series. One of these
     2
      fhir:ImmunizationRecommendation.recommendation.doseNumberPositiveInt [ positiveInt ]
      fhir:ImmunizationRecommendation.recommendation.doseNumberString [ string ]
    # ImmunizationRecommendation.recommendation.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of
    these 2
      fhir:ImmunizationRecommendation.recommendation.seriesDosesPositiveInt [ positiveInt ]
      fhir:ImmunizationRecommendation.recommendation.seriesDosesString [ string ]
    fhir:ImmunizationRecommendation.recommendation.supportingImmunization [
    Reference(Immunization|ImmunizationEvaluation) ], ... ; # 0..* Past immunizations supporting recommendation
    fhir:ImmunizationRecommendation.recommendation.supportingPatientInformation [ Reference(Any) ], ... ; # 0..* Patient
     observations supporting recommendation
  ], ...;
]
```

Changes since R3

ImmunizationRecommendation

ImmunizationRecommendation.date | Moved from ImmunizationRecommendation.recommendation to ImmunizationRecommendation

ImmunizationRecommendation.authority | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation

ImmunizationRecommendation.recommendation.vaccineCode | Max Cardinality changed from 1 to *

ImmunizationRecommendation.recommendation.contraindicatedVaccineCode | Added Element

ImmunizationRecommendation.recommendation.forecastStatus | Now marked as Modifier

ImmunizationRecommendation.recommendation.forecastReason | Added Element

ImmunizationRecommendation.recommendation.description | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation

ImmunizationRecommendation.recommendation.series | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation

ImmunizationRecommendation.recommendation.doseNumber[x] | Renamed from doseNumber to doseNumber[x] Add Type string

ImmunizationRecommendation.recommendation.seriesDoses[x] | Moved from ImmunizationRecommendation.recommendation.protocol.doseSequence to seriesDoses[x] Add Type string

ImmunizationRecommendation.recommendation.supportingImmunization | Type Reference: Added Target Type ImmunizationEvaluation

ImmunizationRecommendation.recommendation.supportingPatientInformation | Type Reference: Added Target Type Resource Type Reference: Removed Target Types Observation, AllergyIntolerance

ImmunizationRecommendation.recommendation.protocol | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

ImmunizationRecommendation | TU | | DomainResource | Guidance or advice relating to an immunization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business identifier

patient | Σ | 1..1 | Reference(Patient) | Who this profile is for

date | Σ | 1..1 | dateTime | Date recommendation(s) created

authority | | 0..1 | Reference(Organization) | Who is responsible for protocol

recommendation | ΣI | 1..* | BackboneElement | Vaccine administration recommendations + Rule: One of vaccineCode or targetDisease SHALL be present

vaccineCode | Σ | 0..* | CodeableConcept | Vaccine or vaccine group recommendation applies to Vaccine Administered Value Set (Example)

targetDisease | Σ | 0..1 | CodeableConcept | Disease to be immunized against Immunization Recommendation Target Disease Codes (Example)

contraindicatedVaccineCode | Σ | 0..* | CodeableConcept | Vaccine which is contraindicated to fulfill the recommendation Vaccine Administered Value Set (Example)

forecastStatus | ?!Σ | 1..1 | CodeableConcept | Vaccine recommendation status Immunization Recommendation Status Codes (Example)

forecastReason | Σ | 0..* | CodeableConcept | Vaccine administration status reason Immunization Recommendation Reason Codes (Example)

dateCriterion | | 0..* | BackboneElement | Dates governing proposed immunization

code | | 1..1 | CodeableConcept | Type of date Immunization Recommendation Date Criterion Codes (Example)

value | | 1..1 | dateTime | Recommended date

description | | 0..1 | string | Protocol details

series | | 0..1 | string | Name of vaccination series

doseNumber[x] | Σ | 0..1 | | Recommended dose number within series

doseNumberPositiveInt | | | positiveInt

doseNumberString | | | string

seriesDoses[x] | | 0..1 | | Recommended number of doses for immunity

seriesDosesPositiveInt | | | positiveInt

seriesDosesString | | | string

supportingImmunization | | 0..* | Reference(Immunization | ImmunizationEvaluation) | Past immunizations supporting recommendation
supportingPatientInformation | | 0..* | Reference(Any) | Patient observations supporting recommendation
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImmunizationRecommendation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <patient><!-- 1..1 Reference(Patient) Who this profile is for --></patient>
 <date value="[dateTime]"/><!-- 1..1 Date recommendation(s) created -->
 <authority><!-- 0..1 Reference(Organization) Who is responsible for protocol --></authority>
 <recommendation>  <!-- 1..* Vaccine administration recommendations -->
  <vaccineCode><!-- 0..* CodeableConcept Vaccine  or vaccine group recommendation applies to --></vaccineCode>
  <targetDisease><!-- 0..1 CodeableConcept Disease to be immunized against --></targetDisease>
  <contraindicatedVaccineCode><!-- 0..* CodeableConcept Vaccine which is contraindicated to fulfill the recommendation
    --></contraindicatedVaccineCode>
  <forecastStatus><!-- 1..1 CodeableConcept Vaccine recommendation status --></forecastStatus>
  <forecastReason><!-- 0..* CodeableConcept Vaccine administration status reason --></forecastReason>
  <dateCriterion>  <!-- 0..* Dates governing proposed immunization -->
   <code><!-- 1..1 CodeableConcept Type of date --></code>
   <value value="[dateTime]"/><!-- 1..1 Recommended date -->
  </dateCriterion>
  <description value="[string]"/><!-- 0..1 Protocol details -->
  <series value="[string]"/><!-- 0..1 Name of vaccination series -->
  <doseNumber[x]><!-- 0..1 positiveInt|string Recommended dose number within series --></doseNumber[x]>
  <seriesDoses[x]><!-- 0..1 positiveInt|string Recommended number of doses for immunity --></seriesDoses[x]>
  <supportingImmunization><!-- 0..* Reference(Immunization|ImmunizationEvaluation) Past immunizations supporting
    recommendation --></supportingImmunization>
  <supportingPatientInformation><!-- 0..* Reference(Any) Patient observations supporting recommendation
    --></supportingPatientInformation>
 </recommendation>
</ImmunizationRecommendation>
```

JSON Template

```
{
  "resourceType" : "ImmunizationRecommendation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "patient" : { Reference(Patient) }, // R!  Who this profile is for
  "date" : "<dateTime>", // R!  Date recommendation(s) created
  "authority" : { Reference(Organization) }, // Who is responsible for protocol
  "recommendation" : [{ // R!  Vaccine administration recommendations
    "vaccineCode" : [{ CodeableConcept }], // Vaccine  or vaccine group recommendation applies to
    "targetDisease" : { CodeableConcept }, // Disease to be immunized against
    "contraindicatedVaccineCode" : [{ CodeableConcept }], // Vaccine which is contraindicated to fulfill the
    recommendation
    "forecastStatus" : { CodeableConcept }, // R!  Vaccine recommendation status
    "forecastReason" : [{ CodeableConcept }], // Vaccine administration status reason
    "dateCriterion" : [{ // Dates governing proposed immunization
      "code" : { CodeableConcept }, // R!  Type of date
      "value" : "<dateTime>" // R!  Recommended date
    }],
    "description" : "<string>", // Protocol details
    "series" : "<string>", // Name of vaccination series
    // doseNumber[x]: Recommended dose number within series. One of these 2:
    "doseNumberPositiveInt" : "<positiveInt>",
    "doseNumberString" : "<string>",
    // seriesDoses[x]: Recommended number of doses for immunity. One of these 2:
    "seriesDosesPositiveInt" : "<positiveInt>",
    "seriesDosesString" : "<string>",
    "supportingImmunization" : [{ Reference(Immunization|ImmunizationEvaluation) }], // Past immunizations supporting
    recommendation
    "supportingPatientInformation" : [{ Reference(Any) }] // Patient observations supporting recommendation
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImmunizationRecommendation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImmunizationRecommendation.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ImmunizationRecommendation.patient [ Reference(Patient) ]; # 1..1 Who this profile is for
  fhir:ImmunizationRecommendation.date [ dateTime ]; # 1..1 Date recommendation(s) created
  fhir:ImmunizationRecommendation.authority [ Reference(Organization) ]; # 0..1 Who is responsible for protocol
  fhir:ImmunizationRecommendation.recommendation [ # 1..* Vaccine administration recommendations
    fhir:ImmunizationRecommendation.recommendation.vaccineCode [ CodeableConcept ], ... ; # 0..* Vaccine  or vaccine
    group recommendation applies to
    fhir:ImmunizationRecommendation.recommendation.targetDisease [ CodeableConcept ]; # 0..1 Disease to be immunized
```

```
    against
    fhir:ImmunizationRecommendation.recommendation.contraindicatedVaccineCode [ CodeableConcept ], ... ; # 0..* Vaccine
    which is contraindicated to fulfill the recommendation
    fhir:ImmunizationRecommendation.recommendation.forecastStatus [ CodeableConcept ]; # 1..1 Vaccine recommendation
    status
    fhir:ImmunizationRecommendation.recommendation.forecastReason [ CodeableConcept ], ... ; # 0..* Vaccine
    administration status reason
    fhir:ImmunizationRecommendation.recommendation.dateCriterion [ # 0..* Dates governing proposed immunization
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.code [ CodeableConcept ]; # 1..1 Type of date
      fhir:ImmunizationRecommendation.recommendation.dateCriterion.value [ dateTime ]; # 1..1 Recommended date
    ], ...;
    fhir:ImmunizationRecommendation.recommendation.description [ string ]; # 0..1 Protocol details
    fhir:ImmunizationRecommendation.recommendation.series [ string ]; # 0..1 Name of vaccination series
    # ImmunizationRecommendation.recommendation.doseNumber[x] : 0..1 Recommended dose number within series. One of these
     2
      fhir:ImmunizationRecommendation.recommendation.doseNumberPositiveInt [ positiveInt ]
      fhir:ImmunizationRecommendation.recommendation.doseNumberString [ string ]
    # ImmunizationRecommendation.recommendation.seriesDoses[x] : 0..1 Recommended number of doses for immunity. One of
    these 2
      fhir:ImmunizationRecommendation.recommendation.seriesDosesPositiveInt [ positiveInt ]
      fhir:ImmunizationRecommendation.recommendation.seriesDosesString [ string ]
    fhir:ImmunizationRecommendation.recommendation.supportingImmunization [
    Reference(Immunization|ImmunizationEvaluation) ], ... ; # 0..* Past immunizations supporting recommendation
    fhir:ImmunizationRecommendation.recommendation.supportingPatientInformation [ Reference(Any) ], ... ; # 0..* Patient
     observations supporting recommendation
  ], ...;
]
```

Changes since Release 3

ImmunizationRecommendation
ImmunizationRecommendation.date | Moved from ImmunizationRecommendation.recommendation to ImmunizationRecommendation
ImmunizationRecommendation.authority | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation
ImmunizationRecommendation.recommendation.vaccineCode | Max Cardinality changed from 1 to *
ImmunizationRecommendation.recommendation.contraindicatedVaccineCode | Added Element
ImmunizationRecommendation.recommendation.forecastStatus | Now marked as Modifier
ImmunizationRecommendation.recommendation.forecastReason | Added Element
ImmunizationRecommendation.recommendation.description | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation
ImmunizationRecommendation.recommendation.series | Moved from ImmunizationRecommendation.recommendation.protocol to ImmunizationRecommendation.recommendation
ImmunizationRecommendation.recommendation.doseNumber[x] | Renamed from doseNumber to doseNumber[x] Add Type string
ImmunizationRecommendation.recommendation.seriesDoses[x] | Moved from ImmunizationRecommendation.recommendation.protocol.doseSequence to seriesDoses[x] Add Type string
ImmunizationRecommendation.recommendation.supportingImmunization | Type Reference: Added Target Type ImmunizationEvaluation
ImmunizationRecommendation.recommendation.supportingPatientInformation | Type Reference: Added Target Type Resource Type Reference: Removed Target Types Observation, AllergyIntolerance
ImmunizationRecommendation.recommendation.protocol | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.9.3.1 Terminology Bindings

Path | Definition | Type | Reference
ImmunizationRecommendation.recommendation.vaccineCode
ImmunizationRecommendation.recommendation.contraindicatedVaccineCode | The type of vaccine administered. | Example | VaccineAdministeredValueSet
ImmunizationRecommendation.recommendation.targetDisease | The disease that the recommended vaccination targets. | Example | ImmunizationRecommendationTargetDiseaseCodes
ImmunizationRecommendation.recommendation.forecastStatus | The patient's status with respect to a vaccination protocol. | Example | ImmunizationRecommendationStatusCodes
ImmunizationRecommendation.recommendation.forecastReason | The reason for the patient's status with respect to a vaccination protocol. | Example | ImmunizationRecommendationReasonCodes
ImmunizationRecommendation.recommendation.dateCriterion.code | Classifies date criterion with respect to conveying information about a patient's vaccination status (e.g. due date, latest to give date, etc.). | Example | ImmunizationRecommendationDateCriterionCodes

11.9.3.2 Constraints

id | Level | Location | Description | Expression
imr-1 | Rule | ImmunizationRecommendation.recommendation | One of vaccineCode or targetDisease SHALL be present | vaccineCode.exists() or targetDisease.exists()

11.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Date recommendation(s) created | ImmunizationRecommendation.date
identifier | token | Business identifier | ImmunizationRecommendation.identifier
information | reference | Patient observations supporting recommendation | ImmunizationRecommendation.recommendation.supportingPatientInformation (Any)
patient | reference | Who this profile is for | ImmunizationRecommendation.patient (Patient)
status | token | Vaccine recommendation status | ImmunizationRecommendation.recommendation.forecastStatus
support | reference | Past immunizations supporting recommendation | ImmunizationRecommendation.recommendation.supportingImmunization (Immunization, ImmunizationEvaluation)
target-disease | token | Disease to be immunized against | ImmunizationRecommendation.recommendation.targetDisease
vaccine-type | token | Vaccine or vaccine group recommendation applies to | ImmunizationRecommendation.recommendation.vaccineCode

11.9.5 Resource ImmunizationRecommendation - Examples

URL: https://hl7.org/fhir/R4/immunizationrecommendation-examples.html

11.9.5 Resource ImmunizationRecommendation - Examples

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
Example of immunizationrecommendation using a vaccine code | example | XML | JSON | Turtle
Example of immunizationrecommendation using a target disease | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.9.6 Resource ImmunizationRecommendation - Detailed Descriptions

URL: https://hl7.org/fhir/R4/immunizationrecommendation-definitions.html

11.9.6 Resource ImmunizationRecommendation - Detailed Descriptions

Public Health and Emergency Response Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the ImmunizationRecommendation resource.

ImmunizationRecommendation
Element Id | ImmunizationRecommendation
Definition | A patient's point-in-time set of recommendations (i.e. forecasting) according to a published schedule with optional supporting justification.
Cardinality | 0..*
Type | DomainResource
ImmunizationRecommendation.identifier
Element Id | ImmunizationRecommendation.identifier
Definition | A unique identifier assigned to this particular recommendation record.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
ImmunizationRecommendation.patient
Element Id | ImmunizationRecommendation.patient
Definition | The patient the recommendation(s) are for.
Cardinality | 1..1
Type | Reference(Patient)
Summary | true
ImmunizationRecommendation.date
Element Id | ImmunizationRecommendation.date
Definition | The date the immunization recommendation(s) were created.
Cardinality | 1..1
Type | dateTime
Summary | true
ImmunizationRecommendation.authority
Element Id | ImmunizationRecommendation.authority
Definition | Indicates the authority who published the protocol (e.g. ACIP).
Cardinality | 0..1
Type | Reference(Organization)

ImmunizationRecommendation.recommendation
Element Id | ImmunizationRecommendation.recommendation
Definition | Vaccine administration recommendations.
Cardinality | 1..*
Summary | true
Invariants | Defined on this element imr-1 Rule One of vaccineCode or targetDisease SHALL be present vaccineCode.exists() or targetDisease.exists()
Defined on this element
imr-1 | Rule | One of vaccineCode or targetDisease SHALL be present | vaccineCode.exists() or targetDisease.exists()
ImmunizationRecommendation.recommendation.vaccineCode
Element Id | ImmunizationRecommendation.recommendation.vaccineCode
Definition | Vaccine(s) or vaccine group that pertain to the recommendation.
Cardinality | 0..*
Terminology Binding | Vaccine Administered Value Set (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.targetDisease
Element Id | ImmunizationRecommendation.recommendation.targetDisease
Definition | The targeted disease for the recommendation.
Cardinality | 0..1
Terminology Binding | Immunization Recommendation Target Disease Codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.contraindicatedVaccineCode
Element Id | ImmunizationRecommendation.recommendation.contraindicatedVaccineCode
Definition | Vaccine(s) which should not be used to fulfill the recommendation.
Cardinality | 0..*
Terminology Binding | Vaccine Administered Value Set (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.forecastStatus
Element Id | ImmunizationRecommendation.recommendation.forecastStatus
Definition | Indicates the patient status with respect to the path to immunity for the target disease.
Cardinality | 1..1
Terminology Binding | Immunization Recommendation Status Codes (Example)
Type | CodeableConcept
Is Modifier | true (Reason: This element is labeled as a modifier because the forecastStatus indicates whether a patient needs additional doses (due, overdue, etc) or not (contraindicated, complete, etc).)
Summary | true
ImmunizationRecommendation.recommendation.forecastReason
Element Id | ImmunizationRecommendation.recommendation.forecastReason
Definition | The reason for the assigned forecast status.
Cardinality | 0..*
Terminology Binding | Immunization Recommendation Reason Codes (Example)
Type | CodeableConcept
Summary | true
ImmunizationRecommendation.recommendation.dateCriterion
Element Id | ImmunizationRecommendation.recommendation.dateCriterion
Definition | Vaccine date recommendations. For example, earliest date to administer, latest date to administer, etc.
Cardinality | 0..*
ImmunizationRecommendation.recommendation.dateCriterion.code
Element Id | ImmunizationRecommendation.recommendation.dateCriterion.code
Definition | Date classification of recommendation. For example, earliest date to give, latest date to give, etc.
Cardinality | 1..1
Terminology Binding | Immunization Recommendation Date Criterion Codes (Example)
Type | CodeableConcept
ImmunizationRecommendation.recommendation.dateCriterion.value
Element Id | ImmunizationRecommendation.recommendation.dateCriterion.value
Definition | The date whose meaning is specified by dateCriterion.code.
Cardinality | 1..1
Type | dateTime
ImmunizationRecommendation.recommendation.description
Element Id | ImmunizationRecommendation.recommendation.description
Definition | Contains the description about the protocol under which the vaccine was administered.
Cardinality | 0..1
Type | string
ImmunizationRecommendation.recommendation.series
Element Id | ImmunizationRecommendation.recommendation.series
Definition | One possible path to achieve presumed immunity against a disease - within the context of an authority.
Cardinality | 0..1
Type | string
ImmunizationRecommendation.recommendation.doseNumber[x]

Element Id | ImmunizationRecommendation.recommendation.doseNumber[x]
Definition | Nominal position of the recommended dose in a series (e.g. dose 2 is the next recommended dose).
Cardinality | 0..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | The use of an integer is prefered if known. A string should only be used in cases where an interger is not available (such as when documenting a recurring booster dose).
ImmunizationRecommendation.recommendation.seriesDoses[x]
Element Id | ImmunizationRecommendation.recommendation.seriesDoses[x]
Definition | The recommended number of doses to achieve immunity.
Cardinality | 0..1
Type | positiveInt|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The use of an integer is prefered if known. A string should only be used in cases where an interger is not available (such as when documenting a recurring booster dose).
ImmunizationRecommendation.recommendation.supportingImmunization
Element Id | ImmunizationRecommendation.recommendation.supportingImmunization
Definition | Immunization event history and/or evaluation that supports the status and recommendation.
Cardinality | 0..*
Type | Reference(Immunization | ImmunizationEvaluation)
ImmunizationRecommendation.recommendation.supportingPatientInformation
Element Id | ImmunizationRecommendation.recommendation.supportingPatientInformation
Definition | Patient Information that supports the status and recommendation. This includes patient observations, adverse reactions and allergy/intolerance information.
Cardinality | 0..*
Type | Reference(Any)

11.9.7 Resource ImmunizationRecommendation - Mappings

URL: https://hl7.org/fhir/R4/immunizationrecommendation-mappings.html

11.9.7 Resource ImmunizationRecommendation - Mappings

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the immunizationrecommendation resource (see Mappings to Other Standards for further information & status).

11.9.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ImmunizationRecommendation
identifier | FiveWs.identifier
patient | FiveWs.subject[x]

11.9.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ImmunizationRecommendation | Request
identifier | Request.identifier
patient | Request.subject
date | Request.authoredOn
vaccineCode | Request.code
dateCriterion | Request.occurrence[x]
supportingPatientInformation | Request.supportingInfo

11.9.7.3 HL7 v2 Mapping (http://hl7.org/v2)

ImmunizationRecommendation | VXU_V04
identifier
patient | PID-3
date | OBX-15
authority | OBX-5 : OBX-3 = 59779-9
recommendation
vaccineCode | OBX-5 : OBX-3 = (30956-7 | 38890-0)
targetDisease | OBX-5 : OBX-3 = (30956-7 | 38890-0)
contraindicatedVaccineCode
forecastStatus | OBX-5 : OBX-3 = 59783-1
forecastReason | OBX-5 : OBX-3 = 30982-3

dateCriterion | OBX-3: 30980-7 = Next dose recommended, 59777-3 = Latest next dose should be given, 59778-1 = Dose is overdue
code | OBX-3
value | OBX-5
description | OBX-5 : OBX-3 = 30956-7
series | RXA-5-administered code
doseNumber[x] | OBX-5 : OBX-3 = 30973-2
seriesDoses[x] | OBX-5 : OBX-3 = 59782-3
supportingImmunization | ? No clear way to associate in HL7 v2 or IG
supportingPatientInformation | OBX If OBX is associated with (RXA : RXA-5-administered code = '998^No vaccine administered^CVX'), could contain the reason the vaccine was not administered

11.9.7.4 RIM Mapping (http://hl7.org/v3)

ImmunizationRecommendation | SubstanceAdministration[moodCode=RMD]
identifier | .id
patient | .partipication[typeCode=SUBJ].role
date | .partipication[typeCode=AUTH].time
authority | .participation[typeCode=AUT]
recommendation | .outboundRelationship[typeCode='COMP'].target[classCode=SBADM, moodCode=RMD]
vaccineCode | .partipication[typeCode=CSM].role[classCode=ADMM].code
targetDisease | .partipication[typeCode=CSM].role[classCode=ADMM].code
forecastStatus | .inboundRelationship[typeCode=SUBJ] .source[classCode=OBS, moodCode=EVN, code="Immunization Forecast Status"].value
forecastReason | N/A
dateCriterion | .outboundRelationship[typeCode='LIMIT'].target[classCode=OBS, moodCode=EVN, isCriterionInd=true]
code | .code
value | .value
description | .text
series | .title
doseNumber[x] | .inboundRelationship[typeCode=COMP, .source[classCode=SBADM, moodCode=INT]].sequenceNumber
seriesDoses[x] | .inboundRelationship[typeCode=COMP, .source[classCode=SBADM, moodCode=DEF]].sequenceNumber
supportingImmunization | .outboundRelationship[typeCode=RSON].target[classCode=SBADM, moodCode=EVN, code="immunization"]
supportingPatientInformation | .outboundRelationship[typeCode=RSON].target[classCode=OBS, moodCode=EVN]

11.9.8 Resource ImmunizationRecommendation - Extensions & Profiles

URL: https://hl7.org/fhir/R4/immunizationrecommendation-profiles.html

11.9.8 Resource ImmunizationRecommendation - Extensions & Profiles

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the ImmunizationRecommendation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.9.8.1 Profiles

No Profiles defined for this resource

11.9.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.9.8.3 Search Extensions

No Search Extensions defined for this resource

11.9.9 Resource ImmunizationRecommendation - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/immunizationrecommendation-version-maps.html

No validation errors - all conversions are clean

11.9.9 Resource ImmunizationRecommendation - R3/R4 Conversions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for ImmunizationRecommendation.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

11.9.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ImmunizationRecommendation3to4" = "R3 to R4 Conversions for
    ImmunizationRecommendation"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendationR3 as
    source
uses "http://hl7.org/fhir/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendation as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ImmunizationRecommendation(source src : ImmunizationRecommendationR3, target tgt : ImmunizationRecommendation)
    extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.patient -> tgt.patient;
  src.recommendation as s -> tgt.recommendation as t then recommendation(s, t, tgt);
}

group recommendation(source src, target tgt, target res) extends BackboneElement {
  src.date -> res.date;
  src.vaccineCode -> tgt.vaccineCode;
  src.targetDisease -> tgt.targetDisease;
  src.doseNumber as vs -> tgt.doseNumber = create('positiveInt') as vt then positiveInt(vs, vt);
  src.forecastStatus -> tgt.forecastStatus;
  src.dateCriterion as s -> tgt.dateCriterion as t then dateCriterion(s, t);
  src.protocol as s then {
    s.description -> tgt.description;
    s.doseSequence as vs -> tgt.seriesDoses = create('positiveInt') as vt then positiveInt(vs, vt);
    s.authority -> res.authority;
    s.series -> tgt.series;
  };
  src.series -> tgt.series;
  src.supportingImmunization -> tgt.supportingImmunization;
  src.supportingPatientInformation -> tgt.supportingPatientInformation;
}

group dateCriterion(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value -> tgt.value;
}
```

11.9.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ImmunizationRecommendation4to3" = "R4 to R3 Conversion for
    ImmunizationRecommendation"

uses "http://hl7.org/fhir/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendation as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ImmunizationRecommendation" alias ImmunizationRecommendationR3 as
    target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ImmunizationRecommendation(source src : ImmunizationRecommendationR3, target tgt : ImmunizationRecommendation)
    extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.patient -> tgt.patient;
  src.recommendation as s -> tgt.recommendation as t then recommendation(s, t, src);
}

group recommendation(source src, target tgt, source res) extends BackboneElement {
  res.date -> tgt.date;
  src.vaccineCode -> tgt.vaccineCode;
  src.targetDisease -> tgt.targetDisease;
  src.doseNumber : positiveInt as vs -> tgt.doseNumber as vt then positiveInt(vs, vt);
  src.forecastStatus -> tgt.forecastStatus;
  src.dateCriterion as s -> tgt.dateCriterion as t then dateCriterion(s, t);
  src.description as v ->  tgt.protocol as t,  t.description = v;
  src.seriesDoses : positiveInt as v ->  tgt.protocol as t,  t.doseSequence = v;
  res.authority as v ->  tgt.protocol as t,  t.authority = v;
  src.series as v ->  tgt.protocol as t,  t.series = v;
  src.supportingImmunization -> tgt.supportingImmunization;
  src.supportingPatientInformation -> tgt.supportingPatientInformation;
```

```
}

group dateCriterion(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value -> tgt.value;
}
```

11.9.9.3 R4 Validation Errors

11.9.10 ImmunizationRecommendation HL7 Extensions

URL: https://hl7.org/fhir/R4/immunizationrecommendation-extensions.html

Defines common extensions used with or related to the ImmunizationRecommendation resource

11.9.10 ImmunizationRecommendation HL7 Extensions

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative

11.9.10.1 Content

11.9.11 JSON Schema for ImmunizationRecommendation

URL: https://hl7.org/fhir/R4/immunizationrecommendation.schema.json.html

11.9.11 JSON Schema for ImmunizationRecommendation

Public Health and Emergency Response Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for ImmunizationRecommendation, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ImmunizationRecommendation",
  "$ref": "#/definitions/ImmunizationRecommendation",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ImmunizationRecommendation": {
      "description": "A patient\u0027s point-in-time set of recommendations (i.e. forecasting) according to a published
    schedule with optional supporting justification.",
      "properties": {
        "resourceType": {
          "description": "This is a ImmunizationRecommendation resource",
          "const": "ImmunizationRecommendation"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
```

```
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A unique identifier assigned to this particular recommendation record.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "patient": {
      "description": "The patient the recommendation(s) are for.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date the immunization recommendation(s) were created.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "authority": {
      "description": "Indicates the authority who published the protocol (e.g. ACIP).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "recommendation": {
      "description": "Vaccine administration recommendations.",
      "items": {
        "$ref": "#/definitions/ImmunizationRecommendation_Recommendation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "recommendation",
    "resourceType"
  ]
},
"ImmunizationRecommendation_Recommendation": {
  "description": "A patient\u0027s point-in-time set of recommendations (i.e. forecasting) according to a published
schedule with optional supporting justification.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
```

```
 of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "vaccineCode": {
      "description": "Vaccine(s) or vaccine group that pertain to the recommendation.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "targetDisease": {
      "description": "The targeted disease for the recommendation.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "contraindicatedVaccineCode": {
      "description": "Vaccine(s) which should not be used to fulfill the recommendation.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "forecastStatus": {
      "description": "Indicates the patient status with respect to the path to immunity for the target disease.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "forecastReason": {
      "description": "The reason for the assigned forecast status.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "dateCriterion": {
      "description": "Vaccine date recommendations.  For example, earliest date to administer, latest date to administer, etc.",
      "items": {
        "$ref": "#/definitions/ImmunizationRecommendation_DateCriterion"
      },
      "type": "array"
    },
    "description": {
      "description": "Contains the description about the protocol under which the vaccine was administered.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "series": {
      "description": "One possible path to achieve presumed immunity against a disease - within the context of an authority.",
      "$ref": "#/definitions/string"
    },
    "_series": {
      "description": "Extensions for series",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doseNumberPositiveInt": {
      "description": "Nominal position of the recommended dose in a series (e.g. dose 2 is the next recommended dose).",
      "pattern": "^[1-9][0-9]*$",
      "type": "number"
    },
    "_doseNumberPositiveInt": {
      "description": "Extensions for doseNumberPositiveInt",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "doseNumberString": {
      "description": "Nominal position of the recommended dose in a series (e.g. dose 2 is the next recommended dose).",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_doseNumberString": {
      "description": "Extensions for doseNumberString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "seriesDosesPositiveInt": {
      "description": "The recommended number of doses to achieve immunity.",
      "pattern": "^[1-9][0-9]*$",
      "type": "number"
    },
    "_seriesDosesPositiveInt": {
      "description": "Extensions for seriesDosesPositiveInt",
```

```
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "seriesDosesString": {
        "description": "The recommended number of doses to achieve immunity.",
        "pattern": "^[ \\r\\n\\t\\S]+$",
        "type": "string"
      },
      "_seriesDosesString": {
        "description": "Extensions for seriesDosesString",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "supportingImmunization": {
        "description": "Immunization event history and/or evaluation that supports the status and recommendation.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "supportingPatientInformation": {
        "description": "Patient Information that supports the status and recommendation.  This includes patient
    observations, adverse reactions and allergy/intolerance information.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "forecastStatus"
    ]
  },
  "ImmunizationRecommendation_DateCriterion": {
    "description": "A patient\u0027s point-in-time set of recommendations (i.e. forecasting) according to a published
    schedule with optional supporting justification.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "code": {
        "description": "Date classification of recommendation.  For example, earliest date to give, latest date to
    give, etc.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "value": {
        "description": "The date whose meaning is specified by dateCriterion.code.",
        "$ref": "#/definitions/dateTime"
      },
      "_value": {
        "description": "Extensions for value",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false,
    "required": [
      "code"
    ]
  }
 }
}
```

11.9.12 Resource ImmunizationRecommendation - Maturity Dependency View

URL: https://hl7.org/fhir/R4/immunizationrecommendation-dependencies.html

11.9.12 Resource ImmunizationRecommendation - Maturity Dependency View

Public Health and Emergency Response Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Dependency Graph for ImmunizationRecommendation FMM level 1

Name | Card. | Type | Dependency Analysis
ImmunizationRecommendation | | DomainResource
identifier | 0..* | Identifier
patient | 1..1 | Reference(Patient)
date | 1..1 | dateTime
authority | 0..1 | Reference(Organization)
recommendation | 1..* | BackboneElement
vaccineCode | 0..* | CodeableConcept
targetDisease | 0..1 | CodeableConcept
contraindicatedVaccineCode | 0..* | CodeableConcept
forecastStatus | 1..1 | CodeableConcept
forecastReason | 0..* | CodeableConcept
dateCriterion | 0..* | BackboneElement
code | 1..1 | CodeableConcept
value | 1..1 | dateTime
description | 0..1 | string
series | 0..1 | string
doseNumber[x] | 0..1
doseNumberPositiveInt | | positiveInt
doseNumberString | | string
seriesDoses[x] | 0..1
seriesDosesPositiveInt | | positiveInt
seriesDosesString | | string
supportingImmunization | 0..* | Reference(Immunization | ImmunizationEvaluation) | Type Warning: (ImmunizationEvaluation = FMM0-Trial Use vs. Element = FMM1-Trial Use)
supportingPatientInformation | 0..* | Reference(Any)

11.10 Dosage

URL: https://hl7.org/fhir/R4/dosage.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

The Dosage structure defines general dosage instruction information typically represented in medication requests, medication dispenses and medication statements.

Note: the Dosage structure allows modifier extensions.

11.10.0.1 Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Dosage | ΣTU | | BackBoneElement | How the medication is/was taken or should be taken Elements defined in Ancestors: id, extension, modifierExtension
sequence | Σ | 0..1 | integer | The order of the dosage instructions
text | Σ | 0..1 | string | Free text dosage instructions e.g. SIG
additionalInstruction | Σ | 0..* | CodeableConcept | Supplemental instruction or warnings to the patient - e.g. "with meals", "may cause drowsiness" SNOMED CT Additional Dosage Instructions (Example)
patientInstruction | Σ | 0..1 | string | Patient or consumer oriented instructions
timing | Σ | 0..1 | Timing | When medication should be administered
asNeeded[x] | Σ | 0..1 | | Take "as needed" (for x) SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept

site | Σ | 0..1 | CodeableConcept | Body site to administer to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | Σ | 0..1 | CodeableConcept | How drug should enter body SNOMED CT Route Codes (Example)
method | Σ | 0..1 | CodeableConcept | Technique for administering medication SNOMED CT Administration Method Codes (Example)
doseAndRate | Σ | 0..* | Element | Amount of medication administered
type | Σ | 0..1 | CodeableConcept | The kind of dose or rate specified DoseAndRateType (Example)
dose[x] | Σ | 0..1 | | Amount of medication per dose
doseRange | | | Range
doseQuantity | | | SimpleQuantity
rate[x] | Σ | 0..1 | | Amount of medication per unit of time
rateRatio | | | Ratio
rateRange | | | Range
rateQuantity | | | SimpleQuantity
maxDosePerPeriod | Σ | 0..1 | Ratio | Upper limit on medication per unit of time
maxDosePerAdministration | Σ | 0..1 | SimpleQuantity | Upper limit on medication per administration
maxDosePerLifetime | Σ | 0..1 | SimpleQuantity | Upper limit on medication per lifetime of the patient
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Dosage xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <sequence value="[integer]"/><!-- 0..1 The order of the dosage instructions -->
 <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
 <additionalInstruction><!-- 0..* CodeableConcept Supplemental instruction or warnings to the patient - e.g. "with
    meals", "may cause drowsiness" --></additionalInstruction>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer oriented instructions -->
 <timing><!-- 0..1 Timing When medication should be administered --></timing>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Take "as needed" (for x) --></asNeeded[x]>
 <site><!-- 0..1 CodeableConcept Body site to administer to --></site>
 <route><!-- 0..1 CodeableConcept How drug should enter body --></route>
 <method><!-- 0..1 CodeableConcept Technique for administering medication --></method>
 <doseAndRate>  <!-- 0..* Amount of medication administered -->
  <type><!-- 0..1 CodeableConcept The kind of dose or rate specified --></type>
  <dose[x]><!-- 0..1 Range|Quantity(SimpleQuantity) Amount of medication per dose --></dose[x]>
  <rate[x]><!-- 0..1 Ratio|Range|Quantity(SimpleQuantity) Amount of medication per unit of time --></rate[x]>
 </doseAndRate>
 <maxDosePerPeriod><!-- 0..1 Ratio Upper limit on medication per unit of time --></maxDosePerPeriod>
 <maxDosePerAdministration><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per administration
    --></maxDosePerAdministration>
 <maxDosePerLifetime><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per lifetime of the patient
    --></maxDosePerLifetime>
</Dosage>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "sequence" : <integer>, // The order of the dosage instructions
  "text" : "<string>", // Free text dosage instructions e.g. SIG
  "additionalInstruction" : [{ CodeableConcept }], // Supplemental instruction or warnings to the patient - e.g. "with
    meals", "may cause drowsiness"
  "patientInstruction" : "<string>", // Patient or consumer oriented instructions
  "timing" : { Timing }, // When medication should be administered
  // asNeeded[x]: Take "as needed" (for x). One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "site" : { CodeableConcept }, // Body site to administer to
  "route" : { CodeableConcept }, // How drug should enter body
  "method" : { CodeableConcept }, // Technique for administering medication
  "doseAndRate" : [{ // Amount of medication administered
    "type" : { CodeableConcept }, // The kind of dose or rate specified
    // dose[x]: Amount of medication per dose. One of these 2:
    "doseRange" : { Range },
    "doseQuantity" : { Quantity(SimpleQuantity) },
    // rate[x]: Amount of medication per unit of time. One of these 3:
    "rateRatio" : { Ratio }
    "rateRange" : { Range }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  }],
  "maxDosePerPeriod" : { Ratio }, // Upper limit on medication per unit of time
  "maxDosePerAdministration" : { Quantity(SimpleQuantity) }, // Upper limit on medication per administration
  "maxDosePerLifetime" : { Quantity(SimpleQuantity) } // Upper limit on medication per lifetime of the patient
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:Dosage.sequence [ integer ]; # 0..1 The order of the dosage instructions
  fhir:Dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
  fhir:Dosage.additionalInstruction [ CodeableConcept ], ... ; # 0..* Supplemental instruction or warnings to the
    patient - e.g. "with meals", "may cause drowsiness"
```

```
  fhir:Dosage.patientInstruction [ string ]; # 0..1 Patient or consumer oriented instructions
  fhir:Dosage.timing [ Timing ]; # 0..1 When medication should be administered
  # Dosage.asNeeded[x] : 0..1 Take "as needed" (for x). One of these 2
    fhir:Dosage.asNeededBoolean [ boolean ]
    fhir:Dosage.asNeededCodeableConcept [ CodeableConcept ]
  fhir:Dosage.site [ CodeableConcept ]; # 0..1 Body site to administer to
  fhir:Dosage.route [ CodeableConcept ]; # 0..1 How drug should enter body
  fhir:Dosage.method [ CodeableConcept ]; # 0..1 Technique for administering medication
  fhir:Dosage.doseAndRate [ # 0..* Amount of medication administered
    fhir:Dosage.doseAndRate.type [ CodeableConcept ]; # 0..1 The kind of dose or rate specified
    # Dosage.doseAndRate.dose[x] : 0..1 Amount of medication per dose. One of these 2
      fhir:Dosage.doseAndRate.doseRange [ Range ]
      fhir:Dosage.doseAndRate.doseSimpleQuantity [ Quantity(SimpleQuantity) ]
    # Dosage.doseAndRate.rate[x] : 0..1 Amount of medication per unit of time. One of these 3
      fhir:Dosage.doseAndRate.rateRatio [ Ratio ]
      fhir:Dosage.doseAndRate.rateRange [ Range ]
      fhir:Dosage.doseAndRate.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ], ...;
  fhir:Dosage.maxDosePerPeriod [ Ratio ]; # 0..1 Upper limit on medication per unit of time
  fhir:Dosage.maxDosePerAdministration [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per administration
  fhir:Dosage.maxDosePerLifetime [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per lifetime of the
    patient
]
```

Changes since Release 3

Dosage
Dosage.doseAndRate | Added Element
Dosage.doseAndRate.type | Added Element
Dosage.doseAndRate.dose[x] | Added Element
Dosage.doseAndRate.rate[x] | Added Element
Dosage.dose[x] | deleted
Dosage.rate[x] | deleted

See the Full Difference for further information

Structure

Name | Flags | Card. | Type | Description & Constraints
Dosage | ΣTU | | BackBoneElement | How the medication is/was taken or should be taken Elements defined in Ancestors: id, extension, modifierExtension
sequence | Σ | 0..1 | integer | The order of the dosage instructions
text | Σ | 0..1 | string | Free text dosage instructions e.g. SIG
additionalInstruction | Σ | 0..* | CodeableConcept | Supplemental instruction or warnings to the patient - e.g. "with meals", "may cause drowsiness" SNOMED CT Additional Dosage Instructions (Example)
patientInstruction | Σ | 0..1 | string | Patient or consumer oriented instructions
timing | Σ | 0..1 | Timing | When medication should be administered
asNeeded[x] | Σ | 0..1 | | Take "as needed" (for x) SNOMED CT Medication As Needed Reason Codes (Example)
asNeededBoolean | | | boolean
asNeededCodeableConcept | | | CodeableConcept
site | Σ | 0..1 | CodeableConcept | Body site to administer to SNOMED CT Anatomical Structure for Administration Site Codes (Example)
route | Σ | 0..1 | CodeableConcept | How drug should enter body SNOMED CT Route Codes (Example)
method | Σ | 0..1 | CodeableConcept | Technique for administering medication SNOMED CT Administration Method Codes (Example)
doseAndRate | Σ | 0..* | Element | Amount of medication administered
type | Σ | 0..1 | CodeableConcept | The kind of dose or rate specified DoseAndRateType (Example)
dose[x] | Σ | 0..1 | | Amount of medication per dose
doseRange | | | Range
doseQuantity | | | SimpleQuantity
rate[x] | Σ | 0..1 | | Amount of medication per unit of time
rateRatio | | | Ratio
rateRange | | | Range
rateQuantity | | | SimpleQuantity
maxDosePerPeriod | Σ | 0..1 | Ratio | Upper limit on medication per unit of time
maxDosePerAdministration | Σ | 0..1 | SimpleQuantity | Upper limit on medication per administration
maxDosePerLifetime | Σ | 0..1 | SimpleQuantity | Upper limit on medication per lifetime of the patient
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Dosage xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <sequence value="[integer]"/><!-- 0..1 The order of the dosage instructions -->
 <text value="[string]"/><!-- 0..1 Free text dosage instructions e.g. SIG -->
 <additionalInstruction><!-- 0..* CodeableConcept Supplemental instruction or warnings to the patient - e.g. "with
    meals", "may cause drowsiness" --></additionalInstruction>
 <patientInstruction value="[string]"/><!-- 0..1 Patient or consumer oriented instructions -->
 <timing><!-- 0..1 Timing When medication should be administered --></timing>
 <asNeeded[x]><!-- 0..1 boolean|CodeableConcept Take "as needed" (for x) --></asNeeded[x]>
 <site><!-- 0..1 CodeableConcept Body site to administer to --></site>
 <route><!-- 0..1 CodeableConcept How drug should enter body --></route>
```

```
 <method><!-- 0..1 CodeableConcept Technique for administering medication --></method>
 <doseAndRate>  <!-- 0..* Amount of medication administered -->
  <type><!-- 0..1 CodeableConcept The kind of dose or rate specified --></type>
  <dose[x]><!-- 0..1 Range|Quantity(SimpleQuantity) Amount of medication per dose --></dose[x]>
  <rate[x]><!-- 0..1 Ratio|Range|Quantity(SimpleQuantity) Amount of medication per unit of time --></rate[x]>
 </doseAndRate>
 <maxDosePerPeriod><!-- 0..1 Ratio Upper limit on medication per unit of time --></maxDosePerPeriod>
 <maxDosePerAdministration><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per administration
    --></maxDosePerAdministration>
 <maxDosePerLifetime><!-- 0..1 Quantity(SimpleQuantity) Upper limit on medication per lifetime of the patient
    --></maxDosePerLifetime>
</Dosage>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "sequence" : <integer>, // The order of the dosage instructions
  "text" : "<string>", // Free text dosage instructions e.g. SIG
  "additionalInstruction" : [{ CodeableConcept }], // Supplemental instruction or warnings to the patient - e.g. "with
    meals", "may cause drowsiness"
  "patientInstruction" : "<string>", // Patient or consumer oriented instructions
  "timing" : { Timing }, // When medication should be administered
  // asNeeded[x]: Take "as needed" (for x). One of these 2:
  "asNeededBoolean" : <boolean>,
  "asNeededCodeableConcept" : { CodeableConcept },
  "site" : { CodeableConcept }, // Body site to administer to
  "route" : { CodeableConcept }, // How drug should enter body
  "method" : { CodeableConcept }, // Technique for administering medication
  "doseAndRate" : [{ // Amount of medication administered
    "type" : { CodeableConcept }, // The kind of dose or rate specified
    // dose[x]: Amount of medication per dose. One of these 2:
    "doseRange" : { Range },
    "doseQuantity" : { Quantity(SimpleQuantity) },
    // rate[x]: Amount of medication per unit of time. One of these 3:
    "rateRatio" : { Ratio }
    "rateRange" : { Range }
    "rateQuantity" : { Quantity(SimpleQuantity) }
  }],
  "maxDosePerPeriod" : { Ratio }, // Upper limit on medication per unit of time
  "maxDosePerAdministration" : { Quantity(SimpleQuantity) }, // Upper limit on medication per administration
  "maxDosePerLifetime" : { Quantity(SimpleQuantity) } // Upper limit on medication per lifetime of the patient
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:Dosage.sequence [ integer ]; # 0..1 The order of the dosage instructions
  fhir:Dosage.text [ string ]; # 0..1 Free text dosage instructions e.g. SIG
  fhir:Dosage.additionalInstruction [ CodeableConcept ], ... ; # 0..* Supplemental instruction or warnings to the
    patient - e.g. "with meals", "may cause drowsiness"
  fhir:Dosage.patientInstruction [ string ]; # 0..1 Patient or consumer oriented instructions
  fhir:Dosage.timing [ Timing ]; # 0..1 When medication should be administered
  # Dosage.asNeeded[x] : 0..1 Take "as needed" (for x). One of these 2
    fhir:Dosage.asNeededBoolean [ boolean ]
    fhir:Dosage.asNeededCodeableConcept [ CodeableConcept ]
  fhir:Dosage.site [ CodeableConcept ]; # 0..1 Body site to administer to
  fhir:Dosage.route [ CodeableConcept ]; # 0..1 How drug should enter body
  fhir:Dosage.method [ CodeableConcept ]; # 0..1 Technique for administering medication
  fhir:Dosage.doseAndRate [ # 0..* Amount of medication administered
    fhir:Dosage.doseAndRate.type [ CodeableConcept ]; # 0..1 The kind of dose or rate specified
    # Dosage.doseAndRate.dose[x] : 0..1 Amount of medication per dose. One of these 2
      fhir:Dosage.doseAndRate.doseRange [ Range ]
      fhir:Dosage.doseAndRate.doseSimpleQuantity [ Quantity(SimpleQuantity) ]
    # Dosage.doseAndRate.rate[x] : 0..1 Amount of medication per unit of time. One of these 3
      fhir:Dosage.doseAndRate.rateRatio [ Ratio ]
      fhir:Dosage.doseAndRate.rateRange [ Range ]
      fhir:Dosage.doseAndRate.rateSimpleQuantity [ Quantity(SimpleQuantity) ]
  ], ...;
  fhir:Dosage.maxDosePerPeriod [ Ratio ]; # 0..1 Upper limit on medication per unit of time
  fhir:Dosage.maxDosePerAdministration [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per administration
  fhir:Dosage.maxDosePerLifetime [ Quantity(SimpleQuantity) ]; # 0..1 Upper limit on medication per lifetime of the
    patient
]
```

Changes since Release 3

Dosage
Dosage.doseAndRate | Added Element
Dosage.doseAndRate.type | Added Element
Dosage.doseAndRate.dose[x] | Added Element
Dosage.doseAndRate.rate[x] | Added Element
Dosage.dose[x] | deleted
Dosage.rate[x] | deleted

See the Full Difference for further information

Constraints

Dosage is used in the following places: ActivityDefinition, MedicationDispense, MedicationKnowledge, MedicationRequest and MedicationStatement

11.10.0.2 Interpretation of Dosage in different contexts

11.10.1 Examples

URL: https://hl7.org/fhir/R4/dosage-examples.html

11.10.1 Dosage Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicationRequest examples for examples using Dosage.

11.10.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/dosage-definitions.html

11.10.2 Dosage Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.10.2.1 Dosage

Dosage
Standards Status | This element has a standards status of "Trial Use"
Element Id | Dosage
Definition | Indicates how the medication is/was taken or should be taken by the patient.
Cardinality | 0..*
Type | BackboneElement
Summary | true
Dosage.sequence
Element Id | Dosage.sequence
Definition | Indicates the order in which the dosage instructions should be applied or interpreted.
Cardinality | 0..1
Type | integer
Requirements | If the sequence number of multiple Dosages is the same, then it is implied that the instructions are to be treated as concurrent. If the sequence number is different, then the Dosages are intended to be sequential.
Summary | true
Dosage.text
Element Id | Dosage.text
Definition | Free text dosage instructions e.g. SIG.
Cardinality | 0..1
Type | string
Requirements | Free text dosage instructions can be used for cases where the instructions are too complex to code. The content of this attribute does not include the name or description of the medication. When coded instructions are present, the free text instructions may still be present for display to humans taking or administering the medication. It is expected that the text instructions will always be populated. If the dosage.timing attribute is also populated, then the dosage.text should reflect the same information as the timing. Additional information about administration or preparation of the medication should be included as text.
Summary | true
Dosage.additionalInstruction
Element Id | Dosage.additionalInstruction
Definition | Supplemental instructions to the patient on how to take the medication (e.g. "with meals" or"take half to one hour before food") or warnings for the patient about the medication (e.g. "may cause drowsiness" or "avoid exposure of skin to direct sunlight or sunlamps").
Cardinality | 0..*
Terminology Binding | SNOMED CT Additional Dosage Instructions (Example)
Type | CodeableConcept
Requirements | Additional instruction is intended to be coded, but where no code exists, the element could include text. For example, "Swallow with plenty of water" which might or might not be coded.
Summary | true
Comments | Information about administration or preparation of the medication (e.g. "infuse as rapidly as possibly via intraperitoneal port" or "immediately following drug x") should be populated in dosage.text.
Dosage.patientInstruction
Element Id | Dosage.patientInstruction
Definition | Instructions in terms that are understood by the patient or consumer.
Cardinality | 0..1
Type | string
Summary | true
Dosage.timing
Element Id | Dosage.timing
Definition | When medication should be administered.
Cardinality | 0..1

Type | Timing
Requirements | The timing schedule for giving the medication to the patient. This data type allows many different expressions. For example: "Every 8 hours"; "Three times a day"; "1/2 an hour before breakfast for 10 days from 23-Dec 2011:"; "15 Oct 2013, 17 Oct 2013 and 1 Nov 2013". Sometimes, a rate can imply duration when expressed as total volume / duration (e.g. 500mL/2 hours implies a duration of 2 hours). However, when rate doesn't imply duration (e.g. 250mL/hour), then the timing.repeat.duration is needed to convey the infuse over time period.
Summary | true
Comments | This attribute might not always be populated while the Dosage.text is expected to be populated. If both are populated, then the Dosage.text should reflect the content of the Dosage.timing.
Dosage.asNeeded[x]
Element Id | Dosage.asNeeded[x]
Definition | Indicates whether the Medication is only taken when needed within a specific dosing schedule (Boolean option), or it indicates the precondition for taking the Medication (CodeableConcept).
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication As Needed Reason Codes (Example)
Type | boolean|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Can express "as needed" without a reason by setting the Boolean = True. In this case the CodeableConcept is not populated. Or you can express "as needed" with a reason by including the CodeableConcept. In this case the Boolean is assumed to be True. If you set the Boolean to False, then the dose is given according to the schedule and is not "prn" or "as needed".
Dosage.site
Element Id | Dosage.site
Definition | Body site to administer to.
Cardinality | 0..1
Terminology Binding | SNOMED CT Anatomical Structure for Administration Site Codes (Example)
Type | CodeableConcept
Requirements | A coded specification of the anatomic site where the medication first enters the body.
Summary | true
Comments | If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension bodySite. May be a summary code, or a reference to a very precise definition of the location, or both.
Dosage.route
Element Id | Dosage.route
Definition | How drug should enter body.
Cardinality | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
Requirements | A code specifying the route or physiological path of administration of a therapeutic agent into or onto a patient's body.
Summary | true
Dosage.method
Element Id | Dosage.method
Definition | Technique for administering medication.
Cardinality | 0..1
Terminology Binding | SNOMED CT Administration Method Codes (Example)
Type | CodeableConcept
Requirements | A coded value indicating the method by which the medication is introduced into or onto the body. Most commonly used for injections. For examples, Slow Push; Deep IV.
Summary | true
Comments | Terminologies used often pre-coordinate this term with the route and or form of administration.
Dosage.doseAndRate
Element Id | Dosage.doseAndRate
Definition | The amount of medication administered.
Cardinality | 0..*
Summary | true
Dosage.doseAndRate.type
Element Id | Dosage.doseAndRate.type
Definition | The kind of dose or rate specified, for example, ordered or calculated.
Cardinality | 0..1
Terminology Binding | DoseAndRateType (Example)
Type | CodeableConcept
Requirements | If the type is not populated, assume to be "ordered".
Summary | true
Dosage.doseAndRate.dose[x]
Element Id | Dosage.doseAndRate.dose[x]
Definition | Amount of medication per dose.
Cardinality | 0..1
Type | Range|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The amount of therapeutic or other substance given at one administration event.
Summary | true
Comments | Note that this specifies the quantity of the specified medication, not the quantity for each active ingredient(s). Each ingredient amount can be communicated in the Medication resource. For example, if one wants to communicate that a tablet was

375 mg, where the dose was one tablet, you can use the Medication resource to document that the tablet was comprised of 375 mg of drug XYZ. Alternatively if the dose was 375 mg, then you may only need to use the Medication resource to indicate this was a tablet. If the example were an IV such as dopamine and you wanted to communicate that 400mg of dopamine was mixed in 500 ml of some IV solution, then this would all be communicated in the Medication resource. If the administration is not intended to be instantaneous (rate is present or timing has a duration), this can be specified to convey the total amount to be administered over the period of time as indicated by the schedule e.g. 500 ml in dose, with timing used to convey that this should be done over 4 hours.
Dosage.doseAndRate.rate[x]
Element Id | Dosage.doseAndRate.rate[x]
Definition | Amount of medication per unit of time.
Cardinality | 0..1
Type | Ratio|Range|SimpleQuantity
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Identifies the speed with which the medication was or will be introduced into the patient. Typically the rate for an infusion e.g. 100 ml per 1 hour or 100 ml/hr. May also be expressed as a rate per unit of time e.g. 500 ml per 2 hours. Other examples: 200 mcg/min or 200 mcg/1 minute; 1 liter/8 hours. Sometimes, a rate can imply duration when expressed as total volume / duration (e.g. 500mL/2 hours implies a duration of 2 hours). However, when rate doesn't imply duration (e.g. 250mL/hour), then the timing.repeat.duration is needed to convey the infuse over time period.
Summary | true
Comments | It is possible to supply both a rate and a doseQuantity to provide full details about how the medication is to be administered and supplied. If the rate is intended to change over time, depending on local rules/regulations, each change should be captured as a new version of the MedicationRequest with an updated rate, or captured with a new MedicationRequest with the new rate. It is possible to specify a rate over time (for example, 100 ml/hour) using either the rateRatio and rateQuantity. The rateQuantity approach requires systems to have the capability to parse UCUM grammer where ml/hour is included rather than a specific ratio where the time is specified as the denominator. Where a rate such as 500ml over 2 hours is specified, the use of rateRatio may be more semantically correct than specifying using a rateQuantity of 250 mg/hour.
Dosage.maxDosePerPeriod
Element Id | Dosage.maxDosePerPeriod
Definition | Upper limit on medication per unit of time.
Cardinality | 0..1
Type | Ratio
Requirements | The maximum total quantity of a therapeutic substance that may be administered to a subject over the period of time. For example, 1000mg in 24 hours.
Summary | true
Comments | This is intended for use as an adjunct to the dosage when there is an upper cap. For example "2 tablets every 4 hours to a maximum of 8/day".
Dosage.maxDosePerAdministration
Element Id | Dosage.maxDosePerAdministration
Definition | Upper limit on medication per administration.
Cardinality | 0..1
Type | SimpleQuantity
Requirements | The maximum total quantity of a therapeutic substance that may be administered to a subject per administration.
Summary | true
Comments | This is intended for use as an adjunct to the dosage when there is an upper cap. For example, a body surface area related dose with a maximum amount, such as 1.5 mg/m2 (maximum 2 mg) IV over 5 – 10 minutes would have doseQuantity of 1.5 mg/m2 and maxDosePerAdministration of 2 mg.
Dosage.maxDosePerLifetime
Element Id | Dosage.maxDosePerLifetime
Definition | Upper limit on medication per lifetime of the patient.
Cardinality | 0..1
Type | SimpleQuantity
Requirements | The maximum total quantity of a therapeutic substance that may be administered per lifetime of the subject.
Summary | true

11.10.3 Mappings

URL: https://hl7.org/fhir/R4/dosage-mappings.html

11.10.3 Dosage Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for Dosage (see Mappings to Other Standards for further information & status).

11.10.3.1 HL7 v2 Mapping (http://hl7.org/v2)

Dosage
sequence | TQ1-1
text | RXO-6; RXE-21
additionalInstruction | RXO-7
patientInstruction | RXO-7
timing
asNeeded[x] | TQ1-9
site | RXR-2
route | RXR-1
method | RXR-4

doseAndRate | TQ1-2
type | RXO-21; RXE-23
dose[x] | RXO-2, RXE-3
rate[x] | RXE22, RXE23, RXE-24
maxDosePerPeriod | RXO-23, RXE-19
maxDosePerAdministration
maxDosePerLifetime

11.10.3.2 RIM Mapping (http://hl7.org/v3)

Dosage | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=INT]
sequence | .text
text | .text
additionalInstruction | .text
patientInstruction | .text
timing | .effectiveTime
asNeeded[x] | .outboundRelationship[typeCode=PRCN].target[classCode=OBS, moodCode=EVN, code="as needed"].value=boolean or codable concept
site | .approachSiteCode
route | .routeCode
method | .doseQuantity
dose[x] | .doseQuantity
rate[x] | .rateQuantity
maxDosePerPeriod | .maxDoseQuantity
maxDosePerAdministration | not supported
maxDosePerLifetime | not supported

11.11 Resource MedicinalProduct - Content

URL: https://hl7.org/fhir/R4/medicinalproduct.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed definition of a medicinal product, typically for uses other than direct patient care (e.g. regulatory use).

11.11.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

Any pharmaceutical product or combination of pharmaceutical products that may be administered to human beings (or animals) for treating or preventing disease, with the aim/purpose of making a medical diagnosis or to restore, correct or modify physiological functions.

A Medicinal Product may contain in the packaging one or more manufactured items and one or more pharmaceutical products. In certain regions a Medicinal Product may also be defined as any substance or combination of substances which may be used to make a medical diagnosis.

This applies to proprietary medicinal products for human use intended to be placed on the market and to industrially manufactured medicinal products, the marketing of which has been authorised by a Medicines Regulatory Agency. However, the provisions do not apply to: i) medicinal products prepared according to prescription (e.g. prepared in a pharmacy from a prescription intended for a specific patient), ii) medicinal products prepared in accordance with an official formula (e.g. prepared in a pharmacy in accordance with the instructions in a pharmacopoeia and intended to be given direct to the patient by the pharmacy), iii) medicinal products intended for research and development trials (see 11.2 Investigational Medicinal Product) and to iv) intermediate products intended for subsequent processing by an authorised manufacturer. [from ENV 13607 and ENV 12610]

This resource is referenced by MedicinalProductAuthorization, MedicinalProductContraindication, MedicinalProductIndication, MedicinalProductInteraction, MedicinalProductPackaged and MedicinalProductUndesirableEffect

11.11.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProduct | ΣTU | | DomainResource | Detailed definition of a medicinal product, typically for uses other than direct patient care (e.g. regulatory use) Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for this product. Could be an MPID
type | Σ | 0..1 | CodeableConcept | Regulatory type, e.g. Investigational or Authorized
domain | Σ | 0..1 | Coding | If this medicine applies to human or veterinary uses
combinedPharmaceuticalDoseForm | Σ | 0..1 | CodeableConcept | The dose form for a single part product, or combined form of a multiple part product
legalStatusOfSupply | Σ | 0..1 | CodeableConcept | The legal status of supply of the medicinal product as classified by the regulator
additionalMonitoringIndicator | Σ | 0..1 | CodeableConcept | Whether the Medicinal Product is subject to additional monitoring for regulatory reasons
specialMeasures | Σ | 0..* | string | Whether the Medicinal Product is subject to special measures for regulatory reasons
paediatricUseIndicator | Σ | 0..1 | CodeableConcept | If authorised for use in children
productClassification | Σ | 0..* | CodeableConcept | Allows the product to be classified by various systems
marketingStatus | Σ | 0..* | MarketingStatus | Marketing status of the medicinal product, in contrast to marketing authorizaton
pharmaceuticalProduct | Σ | 0..* | Reference(MedicinalProductPharmaceutical) | Pharmaceutical aspects of product
packagedMedicinalProduct | Σ | 0..* | Reference(MedicinalProductPackaged) | Package representation for the product
attachedDocument | Σ | 0..* | Reference(DocumentReference) | Supporting documentation, typically for regulatory submission
masterFile | Σ | 0..* | Reference(DocumentReference) | A master file for to the medicinal product (e.g. Pharmacovigilance System Master File)
contact | Σ | 0..* | Reference(Organization | PractitionerRole) | A product specific contact, person (in a role), or an organization
clinicalTrial | Σ | 0..* | Reference(ResearchStudy) | Clinical trials or studies that this product is involved in
name | Σ | 1..* | BackboneElement | The product's name, including full name and possibly coded parts
productName | Σ | 1..1 | string | The full product name
namePart | Σ | 0..* | BackboneElement | Coding words or phrases of the name
part | Σ | 1..1 | string | A fragment of a product name
type | Σ | 1..1 | Coding | Idenifying type for this part of the name (e.g. strength part)
countryLanguage | Σ | 0..* | BackboneElement | Country where the name applies
country | Σ | 1..1 | CodeableConcept | Country code for where this name applies
jurisdiction | Σ | 0..1 | CodeableConcept | Jurisdiction code for where this name applies
language | Σ | 1..1 | CodeableConcept | Language code for this name
crossReference | Σ | 0..* | Identifier | Reference to another product, e.g. for linking authorised to investigational product
manufacturingBusinessOperation | Σ | 0..* | BackboneElement | An operation applied to the product, for manufacturing or adminsitrative purpose
operationType | Σ | 0..1 | CodeableConcept | The type of manufacturing operation
authorisationReferenceNumber | Σ | 0..1 | Identifier | Regulatory authorization reference number
effectiveDate | Σ | 0..1 | dateTime | Regulatory authorization date
confidentialityIndicator | Σ | 0..1 | CodeableConcept | To indicate if this proces is commercially confidential
manufacturer | Σ | 0..* | Reference(Organization) | The manufacturer or establishment associated with the process
regulator | Σ | 0..1 | Reference(Organization) | A regulator which oversees the operation
specialDesignation | Σ | 0..* | BackboneElement | Indicates if the medicinal product has an orphan designation for the treatment of a rare disease
identifier | Σ | 0..* | Identifier | Identifier for the designation, or procedure number
type | Σ | 0..1 | CodeableConcept | The type of special designation, e.g. orphan drug, minor use
intendedUse | Σ | 0..1 | CodeableConcept | The intended use of the product, e.g. prevention, treatment
indication[x] | Σ | 0..1 | | Condition for which the medicinal use applies
indicationCodeableConcept | | | CodeableConcept
indicationReference | | | Reference(MedicinalProductIndication)
status | Σ | 0..1 | CodeableConcept | For example granted, pending, expired or withdrawn
date | Σ | 0..1 | dateTime | Date when the designation was granted
species | Σ | 0..1 | CodeableConcept | Animal species for which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProduct xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for this product. Could be an MPID --></identifier>
 <type><!-- 0..1 CodeableConcept Regulatory type, e.g. Investigational or Authorized --></type>
 <domain><!-- 0..1 Coding If this medicine applies to human or veterinary uses --></domain>
 <combinedPharmaceuticalDoseForm><!-- 0..1 CodeableConcept The dose form for a single part product, or combined form of
    a multiple part product --></combinedPharmaceuticalDoseForm>
 <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply of the medicinal product as classified by the
     regulator --></legalStatusOfSupply>
 <additionalMonitoringIndicator><!-- 0..1 CodeableConcept Whether the Medicinal Product is subject to additional
    monitoring for regulatory reasons --></additionalMonitoringIndicator>
 <specialMeasures value="[string]"/><!-- 0..* Whether the Medicinal Product is subject to special measures for
    regulatory reasons -->
 <paediatricUseIndicator><!-- 0..1 CodeableConcept If authorised for use in children --></paediatricUseIndicator>
 <productClassification><!-- 0..* CodeableConcept Allows the product to be classified by various systems
    --></productClassification>
 <marketingStatus><!-- 0..* MarketingStatus Marketing status of the medicinal product, in contrast to marketing
    authorizaton --></marketingStatus>
```

```
 <pharmaceuticalProduct><!-- 0..* Reference(MedicinalProductPharmaceutical) Pharmaceutical aspects of product
    --></pharmaceuticalProduct>
 <packagedMedicinalProduct><!-- 0..* Reference(MedicinalProductPackaged) Package representation for the product
    --></packagedMedicinalProduct>
 <attachedDocument><!-- 0..* Reference(DocumentReference) Supporting documentation, typically for regulatory submission
    --></attachedDocument>
 <masterFile><!-- 0..* Reference(DocumentReference) A master file for to the medicinal product (e.g. Pharmacovigilance
    System Master File) --></masterFile>
 <contact><!-- 0..* Reference(Organization|PractitionerRole) A product specific contact, person (in a role), or an
    organization --></contact>
 <clinicalTrial><!-- 0..* Reference(ResearchStudy) Clinical trials or studies that this product is involved in
    --></clinicalTrial>
 <name>  <!-- 1..* The product's name, including full name and possibly coded parts -->
  <productName value="[string]"/><!-- 1..1 The full product name -->
  <namePart>  <!-- 0..* Coding words or phrases of the name -->
   <part value="[string]"/><!-- 1..1 A fragment of a product name -->
   <type><!-- 1..1 Coding Idenifying type for this part of the name (e.g. strength part) --></type>
  </namePart>
  <countryLanguage>  <!-- 0..* Country where the name applies -->
   <country><!-- 1..1 CodeableConcept Country code for where this name applies --></country>
   <jurisdiction><!-- 0..1 CodeableConcept Jurisdiction code for where this name applies --></jurisdiction>
   <language><!-- 1..1 CodeableConcept Language code for this name --></language>
  </countryLanguage>
 </name>
 <crossReference><!-- 0..* Identifier Reference to another product, e.g. for linking authorised to investigational
    product --></crossReference>
 <manufacturingBusinessOperation>  <!-- 0..* An operation applied to the product, for manufacturing or adminsitrative
    purpose -->
  <operationType><!-- 0..1 CodeableConcept The type of manufacturing operation --></operationType>
  <authorisationReferenceNumber><!-- 0..1 Identifier Regulatory authorization reference number
    --></authorisationReferenceNumber>
  <effectiveDate value="[dateTime]"/><!-- 0..1 Regulatory authorization date -->
  <confidentialityIndicator><!-- 0..1 CodeableConcept To indicate if this proces is commercially confidential
    --></confidentialityIndicator>
  <manufacturer><!-- 0..* Reference(Organization) The manufacturer or establishment associated with the process
    --></manufacturer>
  <regulator><!-- 0..1 Reference(Organization) A regulator which oversees the operation --></regulator>
 </manufacturingBusinessOperation>
 <specialDesignation>  <!-- 0..* Indicates if the medicinal product has an orphan designation for the treatment of a
    rare disease -->
  <identifier><!-- 0..* Identifier Identifier for the designation, or procedure number --></identifier>
  <type><!-- 0..1 CodeableConcept The type of special designation, e.g. orphan drug, minor use --></type>
  <intendedUse><!-- 0..1 CodeableConcept The intended use of the product, e.g. prevention, treatment --></intendedUse>
  <indication[x]><!-- 0..1 CodeableConcept|Reference(MedicinalProductIndication) Condition for which the medicinal use
    applies --></indication[x]>
  <status><!-- 0..1 CodeableConcept For example granted, pending, expired or withdrawn --></status>
  <date value="[dateTime]"/><!-- 0..1 Date when the designation was granted -->
  <species><!-- 0..1 CodeableConcept Animal species for which this applies --></species>
 </specialDesignation>
</MedicinalProduct>
```

JSON Template

```
{
  "resourceType" : "MedicinalProduct",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for this product. Could be an MPID
  "type" : { CodeableConcept }, // Regulatory type, e.g. Investigational or Authorized
  "domain" : { Coding }, // If this medicine applies to human or veterinary uses
  "combinedPharmaceuticalDoseForm" : { CodeableConcept }, // The dose form for a single part product, or combined form
    of a multiple part product
  "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply of the medicinal product as classified by
    the regulator
  "additionalMonitoringIndicator" : { CodeableConcept }, // Whether the Medicinal Product is subject to additional
    monitoring for regulatory reasons
  "specialMeasures" : ["<string>"], // Whether the Medicinal Product is subject to special measures for regulatory
    reasons
  "paediatricUseIndicator" : { CodeableConcept }, // If authorised for use in children
  "productClassification" : [{ CodeableConcept }], // Allows the product to be classified by various systems
  "marketingStatus" : [{ MarketingStatus }], // Marketing status of the medicinal product, in contrast to marketing
    authorizaton
  "pharmaceuticalProduct" : [{ Reference(MedicinalProductPharmaceutical) }], // Pharmaceutical aspects of product
  "packagedMedicinalProduct" : [{ Reference(MedicinalProductPackaged) }], // Package representation for the product
  "attachedDocument" : [{ Reference(DocumentReference) }], // Supporting documentation, typically for regulatory
    submission
  "masterFile" : [{ Reference(DocumentReference) }], // A master file for to the medicinal product (e.g.
    Pharmacovigilance System Master File)
  "contact" : [{ Reference(Organization|PractitionerRole) }], // A product specific contact, person (in a role), or an
    organization
  "clinicalTrial" : [{ Reference(ResearchStudy) }], // Clinical trials or studies that this product is involved in
  "name" : [{ // R!  The product's name, including full name and possibly coded parts
    "productName" : "<string>", // R!  The full product name
    "namePart" : [{ // Coding words or phrases of the name
      "part" : "<string>", // R!  A fragment of a product name
      "type" : { Coding } // R!  Idenifying type for this part of the name (e.g. strength part)
    }],
    "countryLanguage" : [{ // Country where the name applies
      "country" : { CodeableConcept }, // R!  Country code for where this name applies
      "jurisdiction" : { CodeableConcept }, // Jurisdiction code for where this name applies
```

```
      "language" : { CodeableConcept } // R!  Language code for this name
    }]
  }],
  "crossReference" : [{ Identifier }], // Reference to another product, e.g. for linking authorised to investigational
    product
  "manufacturingBusinessOperation" : [{ // An operation applied to the product, for manufacturing or adminsitrative
    purpose
    "operationType" : { CodeableConcept }, // The type of manufacturing operation
    "authorisationReferenceNumber" : { Identifier }, // Regulatory authorization reference number
    "effectiveDate" : "<dateTime>", // Regulatory authorization date
    "confidentialityIndicator" : { CodeableConcept }, // To indicate if this proces is commercially confidential
    "manufacturer" : [{ Reference(Organization) }], // The manufacturer or establishment associated with the process
    "regulator" : { Reference(Organization) } // A regulator which oversees the operation
  }],
  "specialDesignation" : [{ // Indicates if the medicinal product has an orphan designation for the treatment of a rare
    disease
    "identifier" : [{ Identifier }], // Identifier for the designation, or procedure number
    "type" : { CodeableConcept }, // The type of special designation, e.g. orphan drug, minor use
    "intendedUse" : { CodeableConcept }, // The intended use of the product, e.g. prevention, treatment
    // indication[x]: Condition for which the medicinal use applies. One of these 2:
    "indicationCodeableConcept" : { CodeableConcept },
    "indicationReference" : { Reference(MedicinalProductIndication) },
    "status" : { CodeableConcept }, // For example granted, pending, expired or withdrawn
    "date" : "<dateTime>", // Date when the designation was granted
    "species" : { CodeableConcept } // Animal species for which this applies
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProduct;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProduct.identifier [ Identifier ], ... ; # 0..* Business identifier for this product. Could be an MPID
  fhir:MedicinalProduct.type [ CodeableConcept ]; # 0..1 Regulatory type, e.g. Investigational or Authorized
  fhir:MedicinalProduct.domain [ Coding ]; # 0..1 If this medicine applies to human or veterinary uses
  fhir:MedicinalProduct.combinedPharmaceuticalDoseForm [ CodeableConcept ]; # 0..1 The dose form for a single part
    product, or combined form of a multiple part product
  fhir:MedicinalProduct.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The legal status of supply of the medicinal
    product as classified by the regulator
  fhir:MedicinalProduct.additionalMonitoringIndicator [ CodeableConcept ]; # 0..1 Whether the Medicinal Product is
    subject to additional monitoring for regulatory reasons
  fhir:MedicinalProduct.specialMeasures [ string ], ... ; # 0..* Whether the Medicinal Product is subject to special
    measures for regulatory reasons
  fhir:MedicinalProduct.paediatricUseIndicator [ CodeableConcept ]; # 0..1 If authorised for use in children
  fhir:MedicinalProduct.productClassification [ CodeableConcept ], ... ; # 0..* Allows the product to be classified by
    various systems
  fhir:MedicinalProduct.marketingStatus [ MarketingStatus ], ... ; # 0..* Marketing status of the medicinal product, in
    contrast to marketing authorizaton
  fhir:MedicinalProduct.pharmaceuticalProduct [ Reference(MedicinalProductPharmaceutical) ], ... ; # 0..* Pharmaceutical
     aspects of product
  fhir:MedicinalProduct.packagedMedicinalProduct [ Reference(MedicinalProductPackaged) ], ... ; # 0..* Package
    representation for the product
  fhir:MedicinalProduct.attachedDocument [ Reference(DocumentReference) ], ... ; # 0..* Supporting documentation,
    typically for regulatory submission
  fhir:MedicinalProduct.masterFile [ Reference(DocumentReference) ], ... ; # 0..* A master file for to the medicinal
    product (e.g. Pharmacovigilance System Master File)
  fhir:MedicinalProduct.contact [ Reference(Organization|PractitionerRole) ], ... ; # 0..* A product specific contact,
    person (in a role), or an organization
  fhir:MedicinalProduct.clinicalTrial [ Reference(ResearchStudy) ], ... ; # 0..* Clinical trials or studies that this
    product is involved in
  fhir:MedicinalProduct.name [ # 1..* The product's name, including full name and possibly coded parts
    fhir:MedicinalProduct.name.productName [ string ]; # 1..1 The full product name
    fhir:MedicinalProduct.name.namePart [ # 0..* Coding words or phrases of the name
      fhir:MedicinalProduct.name.namePart.part [ string ]; # 1..1 A fragment of a product name
      fhir:MedicinalProduct.name.namePart.type [ Coding ]; # 1..1 Idenifying type for this part of the name (e.g.
    strength part)
    ], ...;
    fhir:MedicinalProduct.name.countryLanguage [ # 0..* Country where the name applies
      fhir:MedicinalProduct.name.countryLanguage.country [ CodeableConcept ]; # 1..1 Country code for where this name
    applies
      fhir:MedicinalProduct.name.countryLanguage.jurisdiction [ CodeableConcept ]; # 0..1 Jurisdiction code for where
    this name applies
      fhir:MedicinalProduct.name.countryLanguage.language [ CodeableConcept ]; # 1..1 Language code for this name
    ], ...;
  ], ...;
  fhir:MedicinalProduct.crossReference [ Identifier ], ... ; # 0..* Reference to another product, e.g. for linking
    authorised to investigational product
  fhir:MedicinalProduct.manufacturingBusinessOperation [ # 0..* An operation applied to the product, for manufacturing
    or adminsitrative purpose
    fhir:MedicinalProduct.manufacturingBusinessOperation.operationType [ CodeableConcept ]; # 0..1 The type of
    manufacturing operation
    fhir:MedicinalProduct.manufacturingBusinessOperation.authorisationReferenceNumber [ Identifier ]; # 0..1 Regulatory
    authorization reference number
    fhir:MedicinalProduct.manufacturingBusinessOperation.effectiveDate [ dateTime ]; # 0..1 Regulatory authorization
```

```
    date
    fhir:MedicinalProduct.manufacturingBusinessOperation.confidentialityIndicator [ CodeableConcept ]; # 0..1 To
    indicate if this proces is commercially confidential
    fhir:MedicinalProduct.manufacturingBusinessOperation.manufacturer [ Reference(Organization) ], ... ; # 0..* The
    manufacturer or establishment associated with the process
    fhir:MedicinalProduct.manufacturingBusinessOperation.regulator [ Reference(Organization) ]; # 0..1 A regulator which
     oversees the operation
  ], ...;
  fhir:MedicinalProduct.specialDesignation [ # 0..* Indicates if the medicinal product has an orphan designation for the
     treatment of a rare disease
    fhir:MedicinalProduct.specialDesignation.identifier [ Identifier ], ... ; # 0..* Identifier for the designation, or
    procedure number
    fhir:MedicinalProduct.specialDesignation.type [ CodeableConcept ]; # 0..1 The type of special designation, e.g.
    orphan drug, minor use
    fhir:MedicinalProduct.specialDesignation.intendedUse [ CodeableConcept ]; # 0..1 The intended use of the product,
    e.g. prevention, treatment
    # MedicinalProduct.specialDesignation.indication[x] : 0..1 Condition for which the medicinal use applies. One of
    these 2
      fhir:MedicinalProduct.specialDesignation.indicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProduct.specialDesignation.indicationReference [ Reference(MedicinalProductIndication) ]
    fhir:MedicinalProduct.specialDesignation.status [ CodeableConcept ]; # 0..1 For example granted, pending, expired or
     withdrawn
    fhir:MedicinalProduct.specialDesignation.date [ dateTime ]; # 0..1 Date when the designation was granted
    fhir:MedicinalProduct.specialDesignation.species [ CodeableConcept ]; # 0..1 Animal species for which this applies
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProduct | ΣTU | | DomainResource | Detailed definition of a medicinal product, typically for uses other than direct patient care (e.g. regulatory use) Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business identifier for this product. Could be an MPID

type | Σ | 0..1 | CodeableConcept | Regulatory type, e.g. Investigational or Authorized

domain | Σ | 0..1 | Coding | If this medicine applies to human or veterinary uses

combinedPharmaceuticalDoseForm | Σ | 0..1 | CodeableConcept | The dose form for a single part product, or combined form of a multiple part product

legalStatusOfSupply | Σ | 0..1 | CodeableConcept | The legal status of supply of the medicinal product as classified by the regulator

additionalMonitoringIndicator | Σ | 0..1 | CodeableConcept | Whether the Medicinal Product is subject to additional monitoring for regulatory reasons

specialMeasures | Σ | 0..* | string | Whether the Medicinal Product is subject to special measures for regulatory reasons

paediatricUseIndicator | Σ | 0..1 | CodeableConcept | If authorised for use in children

productClassification | Σ | 0..* | CodeableConcept | Allows the product to be classified by various systems

marketingStatus | Σ | 0..* | MarketingStatus | Marketing status of the medicinal product, in contrast to marketing authorizaton

pharmaceuticalProduct | Σ | 0..* | Reference(MedicinalProductPharmaceutical) | Pharmaceutical aspects of product

packagedMedicinalProduct | Σ | 0..* | Reference(MedicinalProductPackaged) | Package representation for the product

attachedDocument | Σ | 0..* | Reference(DocumentReference) | Supporting documentation, typically for regulatory submission

masterFile | Σ | 0..* | Reference(DocumentReference) | A master file for to the medicinal product (e.g. Pharmacovigilance System Master File)

contact | Σ | 0..* | Reference(Organization | PractitionerRole) | A product specific contact, person (in a role), or an organization

clinicalTrial | Σ | 0..* | Reference(ResearchStudy) | Clinical trials or studies that this product is involved in

name | Σ | 1..* | BackboneElement | The product's name, including full name and possibly coded parts

productName | Σ | 1..1 | string | The full product name

namePart | Σ | 0..* | BackboneElement | Coding words or phrases of the name

part | Σ | 1..1 | string | A fragment of a product name

type | Σ | 1..1 | Coding | Idenifying type for this part of the name (e.g. strength part)

countryLanguage | Σ | 0..* | BackboneElement | Country where the name applies

country | Σ | 1..1 | CodeableConcept | Country code for where this name applies

jurisdiction | Σ | 0..1 | CodeableConcept | Jurisdiction code for where this name applies

language | Σ | 1..1 | CodeableConcept | Language code for this name

crossReference | Σ | 0..* | Identifier | Reference to another product, e.g. for linking authorised to investigational product

manufacturingBusinessOperation | Σ | 0..* | BackboneElement | An operation applied to the product, for manufacturing or adminsitrative purpose

operationType | Σ | 0..1 | CodeableConcept | The type of manufacturing operation

authorisationReferenceNumber | Σ | 0..1 | Identifier | Regulatory authorization reference number

effectiveDate | Σ | 0..1 | dateTime | Regulatory authorization date

confidentialityIndicator | Σ | 0..1 | CodeableConcept | To indicate if this proces is commercially confidential

manufacturer | Σ | 0..* | Reference(Organization) | The manufacturer or establishment associated with the process

regulator | Σ | 0..1 | Reference(Organization) | A regulator which oversees the operation

specialDesignation | Σ | 0..* | BackboneElement | Indicates if the medicinal product has an orphan designation for the treatment of a rare disease

identifier | Σ | 0..* | Identifier | Identifier for the designation, or procedure number

type | Σ | 0..1 | CodeableConcept | The type of special designation, e.g. orphan drug, minor use

intendedUse | Σ | 0..1 | CodeableConcept | The intended use of the product, e.g. prevention, treatment

indication[x] | Σ | 0..1 | | Condition for which the medicinal use applies
indicationCodeableConcept | | | CodeableConcept
indicationReference | | | Reference(MedicinalProductIndication)
status | Σ | 0..1 | CodeableConcept | For example granted, pending, expired or withdrawn
date | Σ | 0..1 | dateTime | Date when the designation was granted
species | Σ | 0..1 | CodeableConcept | Animal species for which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProduct xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for this product. Could be an MPID --></identifier>
 <type><!-- 0..1 CodeableConcept Regulatory type, e.g. Investigational or Authorized --></type>
 <domain><!-- 0..1 Coding If this medicine applies to human or veterinary uses --></domain>
 <combinedPharmaceuticalDoseForm><!-- 0..1 CodeableConcept The dose form for a single part product, or combined form of
    a multiple part product --></combinedPharmaceuticalDoseForm>
 <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply of the medicinal product as classified by the
     regulator --></legalStatusOfSupply>
 <additionalMonitoringIndicator><!-- 0..1 CodeableConcept Whether the Medicinal Product is subject to additional
    monitoring for regulatory reasons --></additionalMonitoringIndicator>
 <specialMeasures value="[string]"/><!-- 0..* Whether the Medicinal Product is subject to special measures for
    regulatory reasons -->
 <paediatricUseIndicator><!-- 0..1 CodeableConcept If authorised for use in children --></paediatricUseIndicator>
 <productClassification><!-- 0..* CodeableConcept Allows the product to be classified by various systems
    --></productClassification>
 <marketingStatus><!-- 0..* MarketingStatus Marketing status of the medicinal product, in contrast to marketing
    authorizaton --></marketingStatus>
 <pharmaceuticalProduct><!-- 0..* Reference(MedicinalProductPharmaceutical) Pharmaceutical aspects of product
    --></pharmaceuticalProduct>
 <packagedMedicinalProduct><!-- 0..* Reference(MedicinalProductPackaged) Package representation for the product
    --></packagedMedicinalProduct>
 <attachedDocument><!-- 0..* Reference(DocumentReference) Supporting documentation, typically for regulatory submission
    --></attachedDocument>
 <masterFile><!-- 0..* Reference(DocumentReference) A master file for to the medicinal product (e.g. Pharmacovigilance
    System Master File) --></masterFile>
 <contact><!-- 0..* Reference(Organization|PractitionerRole) A product specific contact, person (in a role), or an
    organization --></contact>
 <clinicalTrial><!-- 0..* Reference(ResearchStudy) Clinical trials or studies that this product is involved in
    --></clinicalTrial>
 <name>  <!-- 1..* The product's name, including full name and possibly coded parts -->
  <productName value="[string]"/><!-- 1..1 The full product name -->
  <namePart>  <!-- 0..* Coding words or phrases of the name -->
   <part value="[string]"/><!-- 1..1 A fragment of a product name -->
   <type><!-- 1..1 Coding Idenifying type for this part of the name (e.g. strength part) --></type>
  </namePart>
  <countryLanguage>  <!-- 0..* Country where the name applies -->
   <country><!-- 1..1 CodeableConcept Country code for where this name applies --></country>
   <jurisdiction><!-- 0..1 CodeableConcept Jurisdiction code for where this name applies --></jurisdiction>
   <language><!-- 1..1 CodeableConcept Language code for this name --></language>
  </countryLanguage>
 </name>
 <crossReference><!-- 0..* Identifier Reference to another product, e.g. for linking authorised to investigational
    product --></crossReference>
 <manufacturingBusinessOperation>  <!-- 0..* An operation applied to the product, for manufacturing or adminsitrative
    purpose -->
  <operationType><!-- 0..1 CodeableConcept The type of manufacturing operation --></operationType>
  <authorisationReferenceNumber><!-- 0..1 Identifier Regulatory authorization reference number
    --></authorisationReferenceNumber>
  <effectiveDate value="[dateTime]"/><!-- 0..1 Regulatory authorization date -->
  <confidentialityIndicator><!-- 0..1 CodeableConcept To indicate if this proces is commercially confidential
    --></confidentialityIndicator>
  <manufacturer><!-- 0..* Reference(Organization) The manufacturer or establishment associated with the process
    --></manufacturer>
  <regulator><!-- 0..1 Reference(Organization) A regulator which oversees the operation --></regulator>
 </manufacturingBusinessOperation>
 <specialDesignation>  <!-- 0..* Indicates if the medicinal product has an orphan designation for the treatment of a
    rare disease -->
  <identifier><!-- 0..* Identifier Identifier for the designation, or procedure number --></identifier>
  <type><!-- 0..1 CodeableConcept The type of special designation, e.g. orphan drug, minor use --></type>
  <intendedUse><!-- 0..1 CodeableConcept The intended use of the product, e.g. prevention, treatment --></intendedUse>
  <indication[x]><!-- 0..1 CodeableConcept|Reference(MedicinalProductIndication) Condition for which the medicinal use
    applies --></indication[x]>
  <status><!-- 0..1 CodeableConcept For example granted, pending, expired or withdrawn --></status>
  <date value="[dateTime]"/><!-- 0..1 Date when the designation was granted -->
  <species><!-- 0..1 CodeableConcept Animal species for which this applies --></species>
 </specialDesignation>
</MedicinalProduct>
```

JSON Template

```
{
  "resourceType" : "MedicinalProduct",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for this product. Could be an MPID
  "type" : { CodeableConcept }, // Regulatory type, e.g. Investigational or Authorized
```

```
  "domain" : { Coding }, // If this medicine applies to human or veterinary uses
  "combinedPharmaceuticalDoseForm" : { CodeableConcept }, // The dose form for a single part product, or combined form
    of a multiple part product
  "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply of the medicinal product as classified by
    the regulator
  "additionalMonitoringIndicator" : { CodeableConcept }, // Whether the Medicinal Product is subject to additional
    monitoring for regulatory reasons
  "specialMeasures" : ["<string>"], // Whether the Medicinal Product is subject to special measures for regulatory
    reasons
  "paediatricUseIndicator" : { CodeableConcept }, // If authorised for use in children
  "productClassification" : [{ CodeableConcept }], // Allows the product to be classified by various systems
  "marketingStatus" : [{ MarketingStatus }], // Marketing status of the medicinal product, in contrast to marketing
    authorizaton
  "pharmaceuticalProduct" : [{ Reference(MedicinalProductPharmaceutical) }], // Pharmaceutical aspects of product
  "packagedMedicinalProduct" : [{ Reference(MedicinalProductPackaged) }], // Package representation for the product
  "attachedDocument" : [{ Reference(DocumentReference) }], // Supporting documentation, typically for regulatory
    submission
  "masterFile" : [{ Reference(DocumentReference) }], // A master file for to the medicinal product (e.g.
    Pharmacovigilance System Master File)
  "contact" : [{ Reference(Organization|PractitionerRole) }], // A product specific contact, person (in a role), or an
    organization
  "clinicalTrial" : [{ Reference(ResearchStudy) }], // Clinical trials or studies that this product is involved in
  "name" : [{ // R!  The product's name, including full name and possibly coded parts
    "productName" : "<string>", // R!  The full product name
    "namePart" : [{ // Coding words or phrases of the name
      "part" : "<string>", // R!  A fragment of a product name
      "type" : { Coding } // R!  Idenifying type for this part of the name (e.g. strength part)
    }],
    "countryLanguage" : [{ // Country where the name applies
      "country" : { CodeableConcept }, // R!  Country code for where this name applies
      "jurisdiction" : { CodeableConcept }, // Jurisdiction code for where this name applies
      "language" : { CodeableConcept } // R!  Language code for this name
    }]
  }],
  "crossReference" : [{ Identifier }], // Reference to another product, e.g. for linking authorised to investigational
    product
  "manufacturingBusinessOperation" : [{ // An operation applied to the product, for manufacturing or adminsitrative
    purpose
    "operationType" : { CodeableConcept }, // The type of manufacturing operation
    "authorisationReferenceNumber" : { Identifier }, // Regulatory authorization reference number
    "effectiveDate" : "<dateTime>", // Regulatory authorization date
    "confidentialityIndicator" : { CodeableConcept }, // To indicate if this proces is commercially confidential
    "manufacturer" : [{ Reference(Organization) }], // The manufacturer or establishment associated with the process
    "regulator" : { Reference(Organization) } // A regulator which oversees the operation
  }],
  "specialDesignation" : [{ // Indicates if the medicinal product has an orphan designation for the treatment of a rare
    disease
    "identifier" : [{ Identifier }], // Identifier for the designation, or procedure number
    "type" : { CodeableConcept }, // The type of special designation, e.g. orphan drug, minor use
    "intendedUse" : { CodeableConcept }, // The intended use of the product, e.g. prevention, treatment
    // indication[x]: Condition for which the medicinal use applies. One of these 2:
    "indicationCodeableConcept" : { CodeableConcept },
    "indicationReference" : { Reference(MedicinalProductIndication) },
    "status" : { CodeableConcept }, // For example granted, pending, expired or withdrawn
    "date" : "<dateTime>", // Date when the designation was granted
    "species" : { CodeableConcept } // Animal species for which this applies
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProduct;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProduct.identifier [ Identifier ], ... ; # 0..* Business identifier for this product. Could be an MPID
  fhir:MedicinalProduct.type [ CodeableConcept ]; # 0..1 Regulatory type, e.g. Investigational or Authorized
  fhir:MedicinalProduct.domain [ Coding ]; # 0..1 If this medicine applies to human or veterinary uses
  fhir:MedicinalProduct.combinedPharmaceuticalDoseForm [ CodeableConcept ]; # 0..1 The dose form for a single part
    product, or combined form of a multiple part product
  fhir:MedicinalProduct.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The legal status of supply of the medicinal
    product as classified by the regulator
  fhir:MedicinalProduct.additionalMonitoringIndicator [ CodeableConcept ]; # 0..1 Whether the Medicinal Product is
    subject to additional monitoring for regulatory reasons
  fhir:MedicinalProduct.specialMeasures [ string ], ... ; # 0..* Whether the Medicinal Product is subject to special
    measures for regulatory reasons
  fhir:MedicinalProduct.paediatricUseIndicator [ CodeableConcept ]; # 0..1 If authorised for use in children
  fhir:MedicinalProduct.productClassification [ CodeableConcept ], ... ; # 0..* Allows the product to be classified by
    various systems
  fhir:MedicinalProduct.marketingStatus [ MarketingStatus ], ... ; # 0..* Marketing status of the medicinal product, in
    contrast to marketing authorizaton
  fhir:MedicinalProduct.pharmaceuticalProduct [ Reference(MedicinalProductPharmaceutical) ], ... ; # 0..* Pharmaceutical
     aspects of product
  fhir:MedicinalProduct.packagedMedicinalProduct [ Reference(MedicinalProductPackaged) ], ... ; # 0..* Package
    representation for the product
  fhir:MedicinalProduct.attachedDocument [ Reference(DocumentReference) ], ... ; # 0..* Supporting documentation,
```

```
    typically for regulatory submission
  fhir:MedicinalProduct.masterFile [ Reference(DocumentReference) ], ... ; # 0..* A master file for to the medicinal
    product (e.g. Pharmacovigilance System Master File)
  fhir:MedicinalProduct.contact [ Reference(Organization|PractitionerRole) ], ... ; # 0..* A product specific contact,
    person (in a role), or an organization
  fhir:MedicinalProduct.clinicalTrial [ Reference(ResearchStudy) ], ... ; # 0..* Clinical trials or studies that this
    product is involved in
  fhir:MedicinalProduct.name [ # 1..* The product's name, including full name and possibly coded parts
    fhir:MedicinalProduct.name.productName [ string ]; # 1..1 The full product name
    fhir:MedicinalProduct.name.namePart [ # 0..* Coding words or phrases of the name
      fhir:MedicinalProduct.name.namePart.part [ string ]; # 1..1 A fragment of a product name
      fhir:MedicinalProduct.name.namePart.type [ Coding ]; # 1..1 Idenifying type for this part of the name (e.g.
    strength part)
    ], ...;
    fhir:MedicinalProduct.name.countryLanguage [ # 0..* Country where the name applies
      fhir:MedicinalProduct.name.countryLanguage.country [ CodeableConcept ]; # 1..1 Country code for where this name
    applies
      fhir:MedicinalProduct.name.countryLanguage.jurisdiction [ CodeableConcept ]; # 0..1 Jurisdiction code for where
    this name applies
      fhir:MedicinalProduct.name.countryLanguage.language [ CodeableConcept ]; # 1..1 Language code for this name
    ], ...;
  ], ...;
  fhir:MedicinalProduct.crossReference [ Identifier ], ... ; # 0..* Reference to another product, e.g. for linking
    authorised to investigational product
  fhir:MedicinalProduct.manufacturingBusinessOperation [ # 0..* An operation applied to the product, for manufacturing
    or adminsitrative purpose
    fhir:MedicinalProduct.manufacturingBusinessOperation.operationType [ CodeableConcept ]; # 0..1 The type of
    manufacturing operation
    fhir:MedicinalProduct.manufacturingBusinessOperation.authorisationReferenceNumber [ Identifier ]; # 0..1 Regulatory
    authorization reference number
    fhir:MedicinalProduct.manufacturingBusinessOperation.effectiveDate [ dateTime ]; # 0..1 Regulatory authorization
    date
    fhir:MedicinalProduct.manufacturingBusinessOperation.confidentialityIndicator [ CodeableConcept ]; # 0..1 To
    indicate if this proces is commercially confidential
    fhir:MedicinalProduct.manufacturingBusinessOperation.manufacturer [ Reference(Organization) ], ... ; # 0..* The
    manufacturer or establishment associated with the process
    fhir:MedicinalProduct.manufacturingBusinessOperation.regulator [ Reference(Organization) ]; # 0..1 A regulator which
     oversees the operation
  ], ...;
  fhir:MedicinalProduct.specialDesignation [ # 0..* Indicates if the medicinal product has an orphan designation for the
     treatment of a rare disease
    fhir:MedicinalProduct.specialDesignation.identifier [ Identifier ], ... ; # 0..* Identifier for the designation, or
    procedure number
    fhir:MedicinalProduct.specialDesignation.type [ CodeableConcept ]; # 0..1 The type of special designation, e.g.
    orphan drug, minor use
    fhir:MedicinalProduct.specialDesignation.intendedUse [ CodeableConcept ]; # 0..1 The intended use of the product,
    e.g. prevention, treatment
    # MedicinalProduct.specialDesignation.indication[x] : 0..1 Condition for which the medicinal use applies. One of
    these 2
      fhir:MedicinalProduct.specialDesignation.indicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProduct.specialDesignation.indicationReference [ Reference(MedicinalProductIndication) ]
    fhir:MedicinalProduct.specialDesignation.status [ CodeableConcept ]; # 0..1 For example granted, pending, expired or
     withdrawn
    fhir:MedicinalProduct.specialDesignation.date [ dateTime ]; # 0..1 Date when the designation was granted
    fhir:MedicinalProduct.specialDesignation.species [ CodeableConcept ]; # 0..1 Animal species for which this applies
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.11.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | Business identifier for this product. Could be an MPID | MedicinalProduct.identifier
name | string | The full product name | MedicinalProduct.name.productName
name-language | token | Language code for this name | MedicinalProduct.name.countryLanguage.language

11.11.4 Resource MedicinalProduct - Examples

URL: https://hl7.org/fhir/R4/medicinalproduct-examples.html

11.11.4 Resource MedicinalProduct - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of medicinalproduct | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.11.5 Resource MedicinalProduct - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproduct-definitions.html

11.11.5 Resource MedicinalProduct - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProduct resource.

MedicinalProduct
Element Id | MedicinalProduct
Definition | Detailed definition of a medicinal product, typically for uses other than direct patient care (e.g. regulatory use).
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProduct.identifier
Element Id | MedicinalProduct.identifier
Definition | Business identifier for this product. Could be an MPID.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Alternate Names | MPID
Summary | true
MedicinalProduct.type
Element Id | MedicinalProduct.type
Definition | Regulatory type, e.g. Investigational or Authorized.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.domain
Element Id | MedicinalProduct.domain
Definition | If this medicine applies to human or veterinary uses.
Cardinality | 0..1
Type | Coding
Summary | true
MedicinalProduct.combinedPharmaceuticalDoseForm
Element Id | MedicinalProduct.combinedPharmaceuticalDoseForm
Definition | The dose form for a single part product, or combined form of a multiple part product.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.legalStatusOfSupply
Element Id | MedicinalProduct.legalStatusOfSupply
Definition | The legal status of supply of the medicinal product as classified by the regulator.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.additionalMonitoringIndicator
Element Id | MedicinalProduct.additionalMonitoringIndicator
Definition | Whether the Medicinal Product is subject to additional monitoring for regulatory reasons.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.specialMeasures
Element Id | MedicinalProduct.specialMeasures
Definition | Whether the Medicinal Product is subject to special measures for regulatory reasons.
Cardinality | 0..*
Type | string
Summary | true
MedicinalProduct.paediatricUseIndicator
Element Id | MedicinalProduct.paediatricUseIndicator
Definition | If authorised for use in children.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.productClassification
Element Id | MedicinalProduct.productClassification
Definition | Allows the product to be classified by various systems.

Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProduct.marketingStatus
Element Id | MedicinalProduct.marketingStatus
Definition | Marketing status of the medicinal product, in contrast to marketing authorizaton.
Cardinality | 0..*
Type | MarketingStatus
Summary | true
MedicinalProduct.pharmaceuticalProduct
Element Id | MedicinalProduct.pharmaceuticalProduct
Definition | Pharmaceutical aspects of product.
Cardinality | 0..*
Type | Reference(MedicinalProductPharmaceutical)
Summary | true
MedicinalProduct.packagedMedicinalProduct
Element Id | MedicinalProduct.packagedMedicinalProduct
Definition | Package representation for the product.
Cardinality | 0..*
Type | Reference(MedicinalProductPackaged)
Summary | true
MedicinalProduct.attachedDocument
Element Id | MedicinalProduct.attachedDocument
Definition | Supporting documentation, typically for regulatory submission.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
MedicinalProduct.masterFile
Element Id | MedicinalProduct.masterFile
Definition | A master file for to the medicinal product (e.g. Pharmacovigilance System Master File).
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
MedicinalProduct.contact
Element Id | MedicinalProduct.contact
Definition | A product specific contact, person (in a role), or an organization.
Cardinality | 0..*
Type | Reference(Organization | PractitionerRole)
Summary | true
MedicinalProduct.clinicalTrial
Element Id | MedicinalProduct.clinicalTrial
Definition | Clinical trials or studies that this product is involved in.
Cardinality | 0..*
Type | Reference(ResearchStudy)
Summary | true
MedicinalProduct.name
Element Id | MedicinalProduct.name
Definition | The product's name, including full name and possibly coded parts.
Cardinality | 1..*
Summary | true
MedicinalProduct.name.productName
Element Id | MedicinalProduct.name.productName
Definition | The full product name.
Cardinality | 1..1
Type | string
Summary | true
MedicinalProduct.name.namePart
Element Id | MedicinalProduct.name.namePart
Definition | Coding words or phrases of the name.
Cardinality | 0..*
Summary | true
MedicinalProduct.name.namePart.part
Element Id | MedicinalProduct.name.namePart.part
Definition | A fragment of a product name.
Cardinality | 1..1
Type | string
Summary | true
MedicinalProduct.name.namePart.type
Element Id | MedicinalProduct.name.namePart.type
Definition | Idenifying type for this part of the name (e.g. strength part).
Cardinality | 1..1
Type | Coding

Summary | true
MedicinalProduct.name.countryLanguage
Element Id | MedicinalProduct.name.countryLanguage
Definition | Country where the name applies.
Cardinality | 0..*
Summary | true
MedicinalProduct.name.countryLanguage.country
Element Id | MedicinalProduct.name.countryLanguage.country
Definition | Country code for where this name applies.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProduct.name.countryLanguage.jurisdiction
Element Id | MedicinalProduct.name.countryLanguage.jurisdiction
Definition | Jurisdiction code for where this name applies.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.name.countryLanguage.language
Element Id | MedicinalProduct.name.countryLanguage.language
Definition | Language code for this name.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProduct.crossReference
Element Id | MedicinalProduct.crossReference
Definition | Reference to another product, e.g. for linking authorised to investigational product.
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProduct.manufacturingBusinessOperation
Element Id | MedicinalProduct.manufacturingBusinessOperation
Definition | An operation applied to the product, for manufacturing or adminsitrative purpose.
Cardinality | 0..*
Summary | true
MedicinalProduct.manufacturingBusinessOperation.operationType
Element Id | MedicinalProduct.manufacturingBusinessOperation.operationType
Definition | The type of manufacturing operation.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.manufacturingBusinessOperation.authorisationReferenceNumber
Element Id | MedicinalProduct.manufacturingBusinessOperation.authorisationReferenceNumber
Definition | Regulatory authorization reference number.
Cardinality | 0..1
Type | Identifier
Summary | true
MedicinalProduct.manufacturingBusinessOperation.effectiveDate
Element Id | MedicinalProduct.manufacturingBusinessOperation.effectiveDate
Definition | Regulatory authorization date.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProduct.manufacturingBusinessOperation.confidentialityIndicator
Element Id | MedicinalProduct.manufacturingBusinessOperation.confidentialityIndicator
Definition | To indicate if this proces is commercially confidential.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.manufacturingBusinessOperation.manufacturer
Element Id | MedicinalProduct.manufacturingBusinessOperation.manufacturer
Definition | The manufacturer or establishment associated with the process.
Cardinality | 0..*
Type | Reference(Organization)
Summary | true
MedicinalProduct.manufacturingBusinessOperation.regulator
Element Id | MedicinalProduct.manufacturingBusinessOperation.regulator
Definition | A regulator which oversees the operation.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
MedicinalProduct.specialDesignation

Element Id | MedicinalProduct.specialDesignation
Definition | Indicates if the medicinal product has an orphan designation for the treatment of a rare disease.
Cardinality | 0..*
Summary | true
MedicinalProduct.specialDesignation.identifier
Element Id | MedicinalProduct.specialDesignation.identifier
Definition | Identifier for the designation, or procedure number.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProduct.specialDesignation.type
Element Id | MedicinalProduct.specialDesignation.type
Definition | The type of special designation, e.g. orphan drug, minor use.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.specialDesignation.intendedUse
Element Id | MedicinalProduct.specialDesignation.intendedUse
Definition | The intended use of the product, e.g. prevention, treatment.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.specialDesignation.indication[x]
Element Id | MedicinalProduct.specialDesignation.indication[x]
Definition | Condition for which the medicinal use applies.
Cardinality | 0..1
Type | CodeableConcept|Reference(MedicinalProductIndication)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicinalProduct.specialDesignation.status
Element Id | MedicinalProduct.specialDesignation.status
Definition | For example granted, pending, expired or withdrawn.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProduct.specialDesignation.date
Element Id | MedicinalProduct.specialDesignation.date
Definition | Date when the designation was granted.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProduct.specialDesignation.species
Element Id | MedicinalProduct.specialDesignation.species
Definition | Animal species for which this applies.
Cardinality | 0..1
Type | CodeableConcept
Summary | true

11.11.6 Resource MedicinalProduct - Mappings

URL: https://hl7.org/fhir/R4/medicinalproduct-mappings.html

11.11.6 Resource MedicinalProduct - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproduct resource (see Mappings to Other Standards for further information & status).

11.11.7 Resource MedicinalProduct - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproduct-profiles.html

11.11.7 Resource MedicinalProduct - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProduct resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.11.7.1 Profiles

No Profiles defined for this resource

11.11.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.11.7.3 Search Extensions

No Search Extensions defined for this resource

11.11.8 Resource MedicinalProduct - Operations

URL: https://hl7.org/fhir/R4/medicinalproduct-operations.html

11.11.8 Resource MedicinalProduct - Operations

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 1 operation operations associated with it:

$everything | Fetch Product Record

For more information about operations, including how they are invoked, see Operations.

11.11.8.1 Operation $everything on MedicinalProduct

URL: https://hl7.org/fhir/R4/medicinalproduct-operation-everything.html

11.11.8.1 Operation $everything on MedicinalProduct

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: N/A

This operation is used to return all the information related to one or more products described in the resource or context on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the product resource(s) itself is returned, along with any other resources that the server has that are related to the products(s), and that are available for the given user. This is typically the marketing authorisations, ingredients, packages, therapeutic indications and so on. The server also returns whatever resources are needed to support the records - e.g. linked organizations, document references etc.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/MedicinalProduct-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/MedicinalProduct/$everything

URL: [base]/MedicinalProduct/[id]/$everything

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
_since | 0..1 | instant | | | Resources updated after this period will be included in the response. The intent of this parameter is to allow a client to request only records that have changed since the last request, based on either the return header time, or or (for asynchronous use), the transaction time
_count | 0..1 | integer | | | See discussion below on the utility of paging through the results of the $everything operation
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply performing a search and using _include and _revinclude are:

• unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging)

• the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones).

This frees the client from needing to determine what it could or should ask for, particularly with regard to included resources. It also makes for a much shorter and easier to construct query string. Servers should consider returning appropriate Provenance and AuditTrail on the returned resources, even though these are not directly part of the product data.

When this operation is used to access multiple product records at once, the return bundle could be rather a lot of data; servers may choose to require that such requests are made asynchronously, and associated with bulk data formats. Alternatively, clients may choose to page through the result set (or servers may require this). Paging through the results is done the same as for Searching, using the _count parameter, and Bundle links. Implementers should note that paging will be slower than simply returning all the results at once (more network traffic, multiple latency delays) but may be required in order not to exhaust available memory reading or writing the whole response in a single package. Unlike searching, there is no inherent user-display order for the $everything operation. Servers might consider sorting the returned resources in descending order of last record update, but are not required to do so.

The _since parameter is provided to support periodic queries to get additional information that has changed about the product since the last query. This means that the _since parameter is based on record time. The value of the _since parameter should be set to the time from the server. If using direct response, this is the timestamp in the response header. If using the async interface, this is the transaction timestamp in the json response. Servers should ensure that the timestamps a managed such that the client does not miss any changes. Clients should be able to handle getting the same response more than once in the case that the transaction falls on a time boundary. Clients should ensure that the other query parameters are constant to ensure a coherent set of records when doing periodic queries.

For more information about operations, including how they are invoked, see Operations.

11.11.9 JSON Schema for MedicinalProduct

URL: https://hl7.org/fhir/R4/medicinalproduct.schema.json.html

11.11.9 JSON Schema for MedicinalProduct

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProduct, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProduct",
  "$ref": "#/definitions/MedicinalProduct",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProduct": {
      "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care
    (e.g. regulatory use).",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProduct resource",
          "const": "MedicinalProduct"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
```

```
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifier for this product. Could be an MPID.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "Regulatory type, e.g. Investigational or Authorized.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "domain": {
      "description": "If this medicine applies to human or veterinary uses.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "combinedPharmaceuticalDoseForm": {
      "description": "The dose form for a single part product, or combined form of a multiple part product.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "legalStatusOfSupply": {
      "description": "The legal status of supply of the medicinal product as classified by the regulator.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "additionalMonitoringIndicator": {
      "description": "Whether the Medicinal Product is subject to additional monitoring for regulatory reasons.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "specialMeasures": {
      "description": "Whether the Medicinal Product is subject to special measures for regulatory reasons.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_specialMeasures": {
      "description": "Extensions for specialMeasures",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "paediatricUseIndicator": {
      "description": "If authorised for use in children.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "productClassification": {
      "description": "Allows the product to be classified by various systems.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
```

```
    },
    "marketingStatus": {
      "description": "Marketing status of the medicinal product, in contrast to marketing authorizaton.",
      "items": {
        "$ref": "MarketingStatus.schema.json#/definitions/MarketingStatus"
      },
      "type": "array"
    },
    "pharmaceuticalProduct": {
      "description": "Pharmaceutical aspects of product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "packagedMedicinalProduct": {
      "description": "Package representation for the product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "attachedDocument": {
      "description": "Supporting documentation, typically for regulatory submission.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "masterFile": {
      "description": "A master file for to the medicinal product (e.g. Pharmacovigilance System Master File).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "contact": {
      "description": "A product specific contact, person (in a role), or an organization.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "clinicalTrial": {
      "description": "Clinical trials or studies that this product is involved in.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "name": {
      "description": "The product\u0027s name, including full name and possibly coded parts.",
      "items": {
        "$ref": "#/definitions/MedicinalProduct_Name"
      },
      "type": "array"
    },
    "crossReference": {
      "description": "Reference to another product, e.g. for linking authorised to investigational product.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "manufacturingBusinessOperation": {
      "description": "An operation applied to the product, for manufacturing or adminsitrative purpose.",
      "items": {
        "$ref": "#/definitions/MedicinalProduct_ManufacturingBusinessOperation"
      },
      "type": "array"
    },
    "specialDesignation": {
      "description": "Indicates if the medicinal product has an orphan designation for the treatment of a rare
disease.",
      "items": {
        "$ref": "#/definitions/MedicinalProduct_SpecialDesignation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "name",
    "resourceType"
  ]
},
"MedicinalProduct_Name": {
  "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care
(e.g. regulatory use).",
```

```
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "productName": {
      "description": "The full product name.",
      "$ref": "#/definitions/string"
    },
    "_productName": {
      "description": "Extensions for productName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "namePart": {
      "description": "Coding words or phrases of the name.",
      "items": {
        "$ref": "#/definitions/MedicinalProduct_NamePart"
      },
      "type": "array"
    },
    "countryLanguage": {
      "description": "Country where the name applies.",
      "items": {
        "$ref": "#/definitions/MedicinalProduct_CountryLanguage"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MedicinalProduct_NamePart": {
  "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care (e.g. regulatory use).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "part": {
```

```
      "description": "A fragment of a product name.",
      "$ref": "#/definitions/string"
    },
    "_part": {
      "description": "Extensions for part",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Idenifying type for this part of the name (e.g. strength part).",
      "$ref": "Coding.schema.json#/definitions/Coding"
    }
  },
  "additionalProperties": false,
  "required": [
    "type"
  ]
},
"MedicinalProduct_CountryLanguage": {
  "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care
(e.g. regulatory use).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "country": {
      "description": "Country code for where this name applies.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "jurisdiction": {
      "description": "Jurisdiction code for where this name applies.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "language": {
      "description": "Language code for this name.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false,
  "required": [
    "country",
    "language"
  ]
},
"MedicinalProduct_ManufacturingBusinessOperation": {
  "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care
(e.g. regulatory use).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "operationType": {
      "description": "The type of manufacturing operation.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "authorisationReferenceNumber": {
      "description": "Regulatory authorization reference number.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "effectiveDate": {
      "description": "Regulatory authorization date.",
      "$ref": "#/definitions/dateTime"
    },
    "_effectiveDate": {
      "description": "Extensions for effectiveDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "confidentialityIndicator": {
      "description": "To indicate if this proces is commercially confidential.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "manufacturer": {
      "description": "The manufacturer or establishment associated with the process.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "regulator": {
      "description": "A regulator which oversees the operation.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"MedicinalProduct_SpecialDesignation": {
  "description": "Detailed definition of a medicinal product, typically for uses other than direct patient care
(e.g. regulatory use).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for the designation, or procedure number.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of special designation, e.g. orphan drug, minor use.",
```

```
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "intendedUse": {
          "description": "The intended use of the product, e.g. prevention, treatment.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "indicationCodeableConcept": {
          "description": "Condition for which the medicinal use applies.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "indicationReference": {
          "description": "Condition for which the medicinal use applies.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "status": {
          "description": "For example granted, pending, expired or withdrawn.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "date": {
          "description": "Date when the designation was granted.",
          "$ref": "#/definitions/dateTime"
        },
        "_date": {
          "description": "Extensions for date",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "species": {
          "description": "Animal species for which this applies.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.11.10 Resource MedicinalProduct - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproduct-dependencies.html

11.11.10 Resource MedicinalProduct - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProduct FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProduct | | DomainResource
identifier | 0..* | Identifier
type | 0..1 | CodeableConcept
domain | 0..1 | Coding
combinedPharmaceuticalDoseForm | 0..1 | CodeableConcept
legalStatusOfSupply | 0..1 | CodeableConcept
additionalMonitoringIndicator | 0..1 | CodeableConcept
specialMeasures | 0..* | string
paediatricUseIndicator | 0..1 | CodeableConcept
productClassification | 0..* | CodeableConcept
marketingStatus | 0..* | MarketingStatus | Type Error: (MarketingStatus = FMM5-Draft vs. Element = FMM0-Trial Use)
pharmaceuticalProduct | 0..* | Reference(MedicinalProductPharmaceutical)
packagedMedicinalProduct | 0..* | Reference(MedicinalProductPackaged)
attachedDocument | 0..* | Reference(DocumentReference)
masterFile | 0..* | Reference(DocumentReference)
contact | 0..* | Reference(Organization | PractitionerRole)
clinicalTrial | 0..* | Reference(ResearchStudy)
name | 1..* | BackboneElement
productName | 1..1 | string
namePart | 0..* | BackboneElement
part | 1..1 | string
type | 1..1 | Coding
countryLanguage | 0..* | BackboneElement
country | 1..1 | CodeableConcept
jurisdiction | 0..1 | CodeableConcept
language | 1..1 | CodeableConcept
crossReference | 0..* | Identifier
manufacturingBusinessOperation | 0..* | BackboneElement
operationType | 0..1 | CodeableConcept
authorisationReferenceNumber | 0..1 | Identifier
effectiveDate | 0..1 | dateTime

confidentialityIndicator | 0..1 | CodeableConcept
manufacturer | 0..* | Reference(Organization)
regulator | 0..1 | Reference(Organization)
specialDesignation | 0..* | BackboneElement
identifier | 0..* | Identifier
type | 0..1 | CodeableConcept
intendedUse | 0..1 | CodeableConcept
indication[x] | 0..1
indicationCodeableConcept | | CodeableConcept
indicationReference | | Reference(MedicinalProductIndication
status | 0..1 | CodeableConcept
date | 0..1 | dateTime
species | 0..1 | CodeableConcept

11.12 Resource MedicinalProductAuthorization - Content

URL: https://hl7.org/fhir/R4/medicinalproductauthorization.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The regulatory authorization of a medicinal product.

11.12.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

MedicinalProductAuthorization is a resource covering the Marketing Authorization of a Medicinal Product, from a regulatory point of view.

This resource is referenced by MedicinalProductPackaged

11.12.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductAuthorization | ΣTU | | DomainResource | The regulatory authorization of a medicinal product Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier for the marketing authorization, as assigned by a regulator
subject | Σ | 0..1 | Reference(MedicinalProduct | MedicinalProductPackaged) | The medicinal product that is being authorized
country | Σ | 0..* | CodeableConcept | The country in which the marketing authorization has been granted
jurisdiction | Σ | 0..* | CodeableConcept | Jurisdiction within a country
status | Σ | 0..1 | CodeableConcept | The status of the marketing authorization
statusDate | Σ | 0..1 | dateTime | The date at which the given status has become applicable
restoreDate | Σ | 0..1 | dateTime | The date when a suspended the marketing or the marketing authorization of the product is anticipated to be restored
validityPeriod | Σ | 0..1 | Period | The beginning of the time period in which the marketing authorization is in the specific status shall be specified A complete date consisting of day, month and year shall be specified using the ISO 8601 date format
dataExclusivityPeriod | Σ | 0..1 | Period | A period of time after authorization before generic product applicatiosn can be submitted
dateOfFirstAuthorization | Σ | 0..1 | dateTime | The date when the first authorization was granted by a Medicines Regulatory Agency
internationalBirthDate | Σ | 0..1 | dateTime | Date of first marketing authorization for a company's new medicinal product in any country in the World
legalBasis | Σ | 0..1 | CodeableConcept | The legal framework against which this authorization is granted
jurisdictionalAuthorization | Σ | 0..* | BackboneElement | Authorization in areas within a country
identifier | Σ | 0..* | Identifier | The assigned number for the marketing authorization
country | Σ | 0..1 | CodeableConcept | Country of authorization
jurisdiction | Σ | 0..* | CodeableConcept | Jurisdiction within a country
legalStatusOfSupply | Σ | 0..1 | CodeableConcept | The legal status of supply in a jurisdiction or region
validityPeriod | Σ | 0..1 | Period | The start and expected end date of the authorization
holder | Σ | 0..1 | Reference(Organization) | Marketing Authorization Holder
regulator | Σ | 0..1 | Reference(Organization) | Medicines Regulatory Agency
procedure | Σ | 0..1 | BackboneElement | The regulatory procedure for granting or amending a marketing authorization
identifier | Σ | 0..1 | Identifier | Identifier for this procedure
type | Σ | 1..1 | CodeableConcept | Type of procedure

date[x] | Σ | 0..1 | | Date of procedure
datePeriod | | | Period
dateDateTime | | | dateTime
application | Σ | 0..* | see procedure | Applcations submitted to obtain a marketing authorization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductAuthorization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for the marketing authorization, as assigned by a regulator
    --></identifier>
 <subject><!-- 0..1 Reference(MedicinalProduct|MedicinalProductPackaged) The medicinal product that is being authorized
    --></subject>
 <country><!-- 0..* CodeableConcept The country in which the marketing authorization has been granted --></country>
 <jurisdiction><!-- 0..* CodeableConcept Jurisdiction within a country --></jurisdiction>
 <status><!-- 0..1 CodeableConcept The status of the marketing authorization --></status>
 <statusDate value="[dateTime]"/><!-- 0..1 The date at which the given status has become applicable -->
 <restoreDate value="[dateTime]"/><!-- 0..1 The date when a suspended the marketing or the marketing authorization of
    the product is anticipated to be restored -->
 <validityPeriod><!-- 0..1 Period The beginning of the time period in which the marketing authorization is in the
    specific status shall be specified A complete date consisting of day, month and year shall be specified using the
    ISO 8601 date format --></validityPeriod>
 <dataExclusivityPeriod><!-- 0..1 Period A period of time after authorization before generic product applicatiosn can be
     submitted --></dataExclusivityPeriod>
 <dateOfFirstAuthorization value="[dateTime]"/><!-- 0..1 The date when the first authorization was granted by a
    Medicines Regulatory Agency -->
 <internationalBirthDate value="[dateTime]"/><!-- 0..1 Date of first marketing authorization for a company's new
    medicinal product in any country in the World -->
 <legalBasis><!-- 0..1 CodeableConcept The legal framework against which this authorization is granted --></legalBasis>
 <jurisdictionalAuthorization>  <!-- 0..* Authorization in areas within a country -->
  <identifier><!-- 0..* Identifier The assigned number for the marketing authorization --></identifier>
  <country><!-- 0..1 CodeableConcept Country of authorization --></country>
  <jurisdiction><!-- 0..* CodeableConcept Jurisdiction within a country --></jurisdiction>
  <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply in a jurisdiction or region
    --></legalStatusOfSupply>
  <validityPeriod><!-- 0..1 Period The start and expected end date of the authorization --></validityPeriod>
 </jurisdictionalAuthorization>
 <holder><!-- 0..1 Reference(Organization) Marketing Authorization Holder --></holder>
 <regulator><!-- 0..1 Reference(Organization) Medicines Regulatory Agency --></regulator>
 <procedure>  <!-- 0..1 The regulatory procedure for granting or amending a marketing authorization -->
  <identifier><!-- 0..1 Identifier Identifier for this procedure --></identifier>
  <type><!-- 1..1 CodeableConcept Type of procedure --></type>
  <date[x]><!-- 0..1 Period|dateTime Date of procedure --></date[x]>
  <application><!-- 0..* Content as for MedicinalProductAuthorization.procedure Applcations submitted to obtain a
    marketing authorization --></application>
 </procedure>
</MedicinalProductAuthorization>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductAuthorization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for the marketing authorization, as assigned by a regulator
  "subject" : { Reference(MedicinalProduct|MedicinalProductPackaged) }, // The medicinal product that is being
    authorized
  "country" : [{ CodeableConcept }], // The country in which the marketing authorization has been granted
  "jurisdiction" : [{ CodeableConcept }], // Jurisdiction within a country
  "status" : { CodeableConcept }, // The status of the marketing authorization
  "statusDate" : "<dateTime>", // The date at which the given status has become applicable
  "restoreDate" : "<dateTime>", // The date when a suspended the marketing or the marketing authorization of the product
     is anticipated to be restored
  "validityPeriod" : { Period }, // The beginning of the time period in which the marketing authorization is in the
    specific status shall be specified A complete date consisting of day, month and year shall be specified using the
    ISO 8601 date format
  "dataExclusivityPeriod" : { Period }, // A period of time after authorization before generic product applicatiosn can
    be submitted
  "dateOfFirstAuthorization" : "<dateTime>", // The date when the first authorization was granted by a Medicines
    Regulatory Agency
  "internationalBirthDate" : "<dateTime>", // Date of first marketing authorization for a company's new medicinal
    product in any country in the World
  "legalBasis" : { CodeableConcept }, // The legal framework against which this authorization is granted
  "jurisdictionalAuthorization" : [{ // Authorization in areas within a country
    "identifier" : [{ Identifier }], // The assigned number for the marketing authorization
    "country" : { CodeableConcept }, // Country of authorization
    "jurisdiction" : [{ CodeableConcept }], // Jurisdiction within a country
    "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply in a jurisdiction or region
    "validityPeriod" : { Period } // The start and expected end date of the authorization
  }],
  "holder" : { Reference(Organization) }, // Marketing Authorization Holder
  "regulator" : { Reference(Organization) }, // Medicines Regulatory Agency
  "procedure" : { // The regulatory procedure for granting or amending a marketing authorization
    "identifier" : { Identifier }, // Identifier for this procedure
    "type" : { CodeableConcept }, // R!  Type of procedure
    // date[x]: Date of procedure. One of these 2:
```

```
    "datePeriod" : { Period },
    "dateDateTime" : "<dateTime>",
    "application" : [{ Content as for MedicinalProductAuthorization.procedure }] // Applcations submitted to obtain a
    marketing authorization
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductAuthorization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductAuthorization.identifier [ Identifier ], ... ; # 0..* Business identifier for the marketing
    authorization, as assigned by a regulator
  fhir:MedicinalProductAuthorization.subject [ Reference(MedicinalProduct|MedicinalProductPackaged) ]; # 0..1 The
    medicinal product that is being authorized
  fhir:MedicinalProductAuthorization.country [ CodeableConcept ], ... ; # 0..* The country in which the marketing
    authorization has been granted
  fhir:MedicinalProductAuthorization.jurisdiction [ CodeableConcept ], ... ; # 0..* Jurisdiction within a country
  fhir:MedicinalProductAuthorization.status [ CodeableConcept ]; # 0..1 The status of the marketing authorization
  fhir:MedicinalProductAuthorization.statusDate [ dateTime ]; # 0..1 The date at which the given status has become
    applicable
  fhir:MedicinalProductAuthorization.restoreDate [ dateTime ]; # 0..1 The date when a suspended the marketing or the
    marketing authorization of the product is anticipated to be restored
  fhir:MedicinalProductAuthorization.validityPeriod [ Period ]; # 0..1 The beginning of the time period in which the
    marketing authorization is in the specific status shall be specified A complete date consisting of day, month and
    year shall be specified using the ISO 8601 date format
  fhir:MedicinalProductAuthorization.dataExclusivityPeriod [ Period ]; # 0..1 A period of time after authorization
    before generic product applicatiosn can be submitted
  fhir:MedicinalProductAuthorization.dateOfFirstAuthorization [ dateTime ]; # 0..1 The date when the first authorization
     was granted by a Medicines Regulatory Agency
  fhir:MedicinalProductAuthorization.internationalBirthDate [ dateTime ]; # 0..1 Date of first marketing authorization
    for a company's new medicinal product in any country in the World
  fhir:MedicinalProductAuthorization.legalBasis [ CodeableConcept ]; # 0..1 The legal framework against which this
    authorization is granted
  fhir:MedicinalProductAuthorization.jurisdictionalAuthorization [ # 0..* Authorization in areas within a country
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.identifier [ Identifier ], ... ; # 0..* The assigned
    number for the marketing authorization
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.country [ CodeableConcept ]; # 0..1 Country of
    authorization
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.jurisdiction [ CodeableConcept ], ... ; # 0..*
    Jurisdiction within a country
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The
    legal status of supply in a jurisdiction or region
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.validityPeriod [ Period ]; # 0..1 The start and
    expected end date of the authorization
  ], ...;
  fhir:MedicinalProductAuthorization.holder [ Reference(Organization) ]; # 0..1 Marketing Authorization Holder
  fhir:MedicinalProductAuthorization.regulator [ Reference(Organization) ]; # 0..1 Medicines Regulatory Agency
  fhir:MedicinalProductAuthorization.procedure [ # 0..1 The regulatory procedure for granting or amending a marketing
    authorization
    fhir:MedicinalProductAuthorization.procedure.identifier [ Identifier ]; # 0..1 Identifier for this procedure
    fhir:MedicinalProductAuthorization.procedure.type [ CodeableConcept ]; # 1..1 Type of procedure
    # MedicinalProductAuthorization.procedure.date[x] : 0..1 Date of procedure. One of these 2
      fhir:MedicinalProductAuthorization.procedure.datePeriod [ Period ]
      fhir:MedicinalProductAuthorization.procedure.dateDateTime [ dateTime ]
    fhir:MedicinalProductAuthorization.procedure.application [ See MedicinalProductAuthorization.procedure ], ... ; #
    0..* Applcations submitted to obtain a marketing authorization
  ];
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProductAuthorization | ΣTU | | DomainResource | The regulatory authorization of a medicinal product Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business identifier for the marketing authorization, as assigned by a regulator

subject | Σ | 0..1 | Reference(MedicinalProduct | MedicinalProductPackaged) | The medicinal product that is being authorized

country | Σ | 0..* | CodeableConcept | The country in which the marketing authorization has been granted

jurisdiction | Σ | 0..* | CodeableConcept | Jurisdiction within a country

status | Σ | 0..1 | CodeableConcept | The status of the marketing authorization

statusDate | Σ | 0..1 | dateTime | The date at which the given status has become applicable

restoreDate | Σ | 0..1 | dateTime | The date when a suspended the marketing or the marketing authorization of the product is anticipated to be restored

validityPeriod | Σ | 0..1 | Period | The beginning of the time period in which the marketing authorization is in the specific status shall be specified A complete date consisting of day, month and year shall be specified using the ISO 8601 date format

dataExclusivityPeriod | Σ | 0..1 | Period | A period of time after authorization before generic product applicatiosn can be submitted

dateOfFirstAuthorization | Σ | 0..1 | dateTime | The date when the first authorization was granted by a Medicines Regulatory Agency
internationalBirthDate | Σ | 0..1 | dateTime | Date of first marketing authorization for a company's new medicinal product in any country in the World
legalBasis | Σ | 0..1 | CodeableConcept | The legal framework against which this authorization is granted
jurisdictionalAuthorization | Σ | 0..* | BackboneElement | Authorization in areas within a country
identifier | Σ | 0..* | Identifier | The assigned number for the marketing authorization
country | Σ | 0..1 | CodeableConcept | Country of authorization
jurisdiction | Σ | 0..* | CodeableConcept | Jurisdiction within a country
legalStatusOfSupply | Σ | 0..1 | CodeableConcept | The legal status of supply in a jurisdiction or region
validityPeriod | Σ | 0..1 | Period | The start and expected end date of the authorization
holder | Σ | 0..1 | Reference(Organization) | Marketing Authorization Holder
regulator | Σ | 0..1 | Reference(Organization) | Medicines Regulatory Agency
procedure | Σ | 0..1 | BackboneElement | The regulatory procedure for granting or amending a marketing authorization
identifier | Σ | 0..1 | Identifier | Identifier for this procedure
type | Σ | 1..1 | CodeableConcept | Type of procedure
date[x] | Σ | 0..1 | | Date of procedure
datePeriod | | | Period
dateDateTime | | | dateTime
application | Σ | 0..* | see procedure | Applcations submitted to obtain a marketing authorization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductAuthorization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier for the marketing authorization, as assigned by a regulator
    --></identifier>
 <subject><!-- 0..1 Reference(MedicinalProduct|MedicinalProductPackaged) The medicinal product that is being authorized
    --></subject>
 <country><!-- 0..* CodeableConcept The country in which the marketing authorization has been granted --></country>
 <jurisdiction><!-- 0..* CodeableConcept Jurisdiction within a country --></jurisdiction>
 <status><!-- 0..1 CodeableConcept The status of the marketing authorization --></status>
 <statusDate value="[dateTime]"/><!-- 0..1 The date at which the given status has become applicable -->
 <restoreDate value="[dateTime]"/><!-- 0..1 The date when a suspended the marketing or the marketing authorization of
    the product is anticipated to be restored -->
 <validityPeriod><!-- 0..1 Period The beginning of the time period in which the marketing authorization is in the
    specific status shall be specified A complete date consisting of day, month and year shall be specified using the
    ISO 8601 date format --></validityPeriod>
 <dataExclusivityPeriod><!-- 0..1 Period A period of time after authorization before generic product applicatiosn can be
     submitted --></dataExclusivityPeriod>
 <dateOfFirstAuthorization value="[dateTime]"/><!-- 0..1 The date when the first authorization was granted by a
    Medicines Regulatory Agency -->
 <internationalBirthDate value="[dateTime]"/><!-- 0..1 Date of first marketing authorization for a company's new
    medicinal product in any country in the World -->
 <legalBasis><!-- 0..1 CodeableConcept The legal framework against which this authorization is granted --></legalBasis>
 <jurisdictionalAuthorization>  <!-- 0..* Authorization in areas within a country -->
  <identifier><!-- 0..* Identifier The assigned number for the marketing authorization --></identifier>
  <country><!-- 0..1 CodeableConcept Country of authorization --></country>
  <jurisdiction><!-- 0..* CodeableConcept Jurisdiction within a country --></jurisdiction>
  <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply in a jurisdiction or region
    --></legalStatusOfSupply>
  <validityPeriod><!-- 0..1 Period The start and expected end date of the authorization --></validityPeriod>
 </jurisdictionalAuthorization>
 <holder><!-- 0..1 Reference(Organization) Marketing Authorization Holder --></holder>
 <regulator><!-- 0..1 Reference(Organization) Medicines Regulatory Agency --></regulator>
 <procedure>  <!-- 0..1 The regulatory procedure for granting or amending a marketing authorization -->
  <identifier><!-- 0..1 Identifier Identifier for this procedure --></identifier>
  <type><!-- 1..1 CodeableConcept Type of procedure --></type>
  <date[x]><!-- 0..1 Period|dateTime Date of procedure --></date[x]>
  <application><!-- 0..* Content as for MedicinalProductAuthorization.procedure Applcations submitted to obtain a
    marketing authorization --></application>
 </procedure>
</MedicinalProductAuthorization>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductAuthorization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier for the marketing authorization, as assigned by a regulator
  "subject" : { Reference(MedicinalProduct|MedicinalProductPackaged) }, // The medicinal product that is being
    authorized
  "country" : [{ CodeableConcept }], // The country in which the marketing authorization has been granted
  "jurisdiction" : [{ CodeableConcept }], // Jurisdiction within a country
  "status" : { CodeableConcept }, // The status of the marketing authorization
  "statusDate" : "<dateTime>", // The date at which the given status has become applicable
  "restoreDate" : "<dateTime>", // The date when a suspended the marketing or the marketing authorization of the product
     is anticipated to be restored
  "validityPeriod" : { Period }, // The beginning of the time period in which the marketing authorization is in the
    specific status shall be specified A complete date consisting of day, month and year shall be specified using the
    ISO 8601 date format
```

```
  "dataExclusivityPeriod" : { Period }, // A period of time after authorization before generic product applicatiosn can
    be submitted
  "dateOfFirstAuthorization" : "<dateTime>", // The date when the first authorization was granted by a Medicines
    Regulatory Agency
  "internationalBirthDate" : "<dateTime>", // Date of first marketing authorization for a company's new medicinal
    product in any country in the World
  "legalBasis" : { CodeableConcept }, // The legal framework against which this authorization is granted
  "jurisdictionalAuthorization" : [{ // Authorization in areas within a country
    "identifier" : [{ Identifier }], // The assigned number for the marketing authorization
    "country" : { CodeableConcept }, // Country of authorization
    "jurisdiction" : [{ CodeableConcept }], // Jurisdiction within a country
    "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply in a jurisdiction or region
    "validityPeriod" : { Period } // The start and expected end date of the authorization
  }],
  "holder" : { Reference(Organization) }, // Marketing Authorization Holder
  "regulator" : { Reference(Organization) }, // Medicines Regulatory Agency
  "procedure" : { // The regulatory procedure for granting or amending a marketing authorization
    "identifier" : { Identifier }, // Identifier for this procedure
    "type" : { CodeableConcept }, // R!  Type of procedure
    // date[x]: Date of procedure. One of these 2:
    "datePeriod" : { Period },
    "dateDateTime" : "<dateTime>",
    "application" : [{ Content as for MedicinalProductAuthorization.procedure }] // Applcations submitted to obtain a
    marketing authorization
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductAuthorization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductAuthorization.identifier [ Identifier ], ... ; # 0..* Business identifier for the marketing
    authorization, as assigned by a regulator
  fhir:MedicinalProductAuthorization.subject [ Reference(MedicinalProduct|MedicinalProductPackaged) ]; # 0..1 The
    medicinal product that is being authorized
  fhir:MedicinalProductAuthorization.country [ CodeableConcept ], ... ; # 0..* The country in which the marketing
    authorization has been granted
  fhir:MedicinalProductAuthorization.jurisdiction [ CodeableConcept ], ... ; # 0..* Jurisdiction within a country
  fhir:MedicinalProductAuthorization.status [ CodeableConcept ]; # 0..1 The status of the marketing authorization
  fhir:MedicinalProductAuthorization.statusDate [ dateTime ]; # 0..1 The date at which the given status has become
    applicable
  fhir:MedicinalProductAuthorization.restoreDate [ dateTime ]; # 0..1 The date when a suspended the marketing or the
    marketing authorization of the product is anticipated to be restored
  fhir:MedicinalProductAuthorization.validityPeriod [ Period ]; # 0..1 The beginning of the time period in which the
    marketing authorization is in the specific status shall be specified A complete date consisting of day, month and
    year shall be specified using the ISO 8601 date format
  fhir:MedicinalProductAuthorization.dataExclusivityPeriod [ Period ]; # 0..1 A period of time after authorization
    before generic product applicatiosn can be submitted
  fhir:MedicinalProductAuthorization.dateOfFirstAuthorization [ dateTime ]; # 0..1 The date when the first authorization
     was granted by a Medicines Regulatory Agency
  fhir:MedicinalProductAuthorization.internationalBirthDate [ dateTime ]; # 0..1 Date of first marketing authorization
    for a company's new medicinal product in any country in the World
  fhir:MedicinalProductAuthorization.legalBasis [ CodeableConcept ]; # 0..1 The legal framework against which this
    authorization is granted
  fhir:MedicinalProductAuthorization.jurisdictionalAuthorization [ # 0..* Authorization in areas within a country
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.identifier [ Identifier ], ... ; # 0..* The assigned
    number for the marketing authorization
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.country [ CodeableConcept ]; # 0..1 Country of
    authorization
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.jurisdiction [ CodeableConcept ], ... ; # 0..*
    Jurisdiction within a country
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The
    legal status of supply in a jurisdiction or region
    fhir:MedicinalProductAuthorization.jurisdictionalAuthorization.validityPeriod [ Period ]; # 0..1 The start and
    expected end date of the authorization
  ], ...;
  fhir:MedicinalProductAuthorization.holder [ Reference(Organization) ]; # 0..1 Marketing Authorization Holder
  fhir:MedicinalProductAuthorization.regulator [ Reference(Organization) ]; # 0..1 Medicines Regulatory Agency
  fhir:MedicinalProductAuthorization.procedure [ # 0..1 The regulatory procedure for granting or amending a marketing
    authorization
    fhir:MedicinalProductAuthorization.procedure.identifier [ Identifier ]; # 0..1 Identifier for this procedure
    fhir:MedicinalProductAuthorization.procedure.type [ CodeableConcept ]; # 1..1 Type of procedure
    # MedicinalProductAuthorization.procedure.date[x] : 0..1 Date of procedure. One of these 2
      fhir:MedicinalProductAuthorization.procedure.datePeriod [ Period ]
      fhir:MedicinalProductAuthorization.procedure.dateDateTime [ dateTime ]
    fhir:MedicinalProductAuthorization.procedure.application [ See MedicinalProductAuthorization.procedure ], ... ; #
    0..* Applcations submitted to obtain a marketing authorization
  ];
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.12.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
country | token | The country in which the marketing authorization has been granted | MedicinalProductAuthorization.country
holder | reference | Marketing Authorization Holder | MedicinalProductAuthorization.holder (Organization)
identifier | token | Business identifier for the marketing authorization, as assigned by a regulator | MedicinalProductAuthorization.identifier
status | token | The status of the marketing authorization | MedicinalProductAuthorization.status
subject | reference | The medicinal product that is being authorized | MedicinalProductAuthorization.subject (MedicinalProductPackaged, MedicinalProduct)

11.12.4 Resource MedicinalProductAuthorization - Examples

URL: https://hl7.org/fhir/R4/medicinalproductauthorization-examples.html

11.12.4 Resource MedicinalProductAuthorization - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductauthorization | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.12.5 Resource MedicinalProductAuthorization - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductauthorization-definitions.html

11.12.5 Resource MedicinalProductAuthorization - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductAuthorization resource.

MedicinalProductAuthorization
Element Id | MedicinalProductAuthorization
Definition | The regulatory authorization of a medicinal product.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductAuthorization.identifier
Element Id | MedicinalProductAuthorization.identifier
Definition | Business identifier for the marketing authorization, as assigned by a regulator.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProductAuthorization.subject
Element Id | MedicinalProductAuthorization.subject
Definition | The medicinal product that is being authorized.
Cardinality | 0..1
Type | Reference(MedicinalProduct | MedicinalProductPackaged)
Summary | true
MedicinalProductAuthorization.country
Element Id | MedicinalProductAuthorization.country
Definition | The country in which the marketing authorization has been granted.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.jurisdiction
Element Id | MedicinalProductAuthorization.jurisdiction
Definition | Jurisdiction within a country.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.status

Element Id | MedicinalProductAuthorization.status
Definition | The status of the marketing authorization.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.statusDate
Element Id | MedicinalProductAuthorization.statusDate
Definition | The date at which the given status has become applicable.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProductAuthorization.restoreDate
Element Id | MedicinalProductAuthorization.restoreDate
Definition | The date when a suspended the marketing or the marketing authorization of the product is anticipated to be restored.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProductAuthorization.validityPeriod
Element Id | MedicinalProductAuthorization.validityPeriod
Definition | The beginning of the time period in which the marketing authorization is in the specific status shall be specified A complete date consisting of day, month and year shall be specified using the ISO 8601 date format.
Cardinality | 0..1
Type | Period
Summary | true
MedicinalProductAuthorization.dataExclusivityPeriod
Element Id | MedicinalProductAuthorization.dataExclusivityPeriod
Definition | A period of time after authorization before generic product applicatiosn can be submitted.
Cardinality | 0..1
Type | Period
Summary | true
MedicinalProductAuthorization.dateOfFirstAuthorization
Element Id | MedicinalProductAuthorization.dateOfFirstAuthorization
Definition | The date when the first authorization was granted by a Medicines Regulatory Agency.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProductAuthorization.internationalBirthDate
Element Id | MedicinalProductAuthorization.internationalBirthDate
Definition | Date of first marketing authorization for a company's new medicinal product in any country in the World.
Cardinality | 0..1
Type | dateTime
Summary | true
MedicinalProductAuthorization.legalBasis
Element Id | MedicinalProductAuthorization.legalBasis
Definition | The legal framework against which this authorization is granted.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.jurisdictionalAuthorization
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization
Definition | Authorization in areas within a country.
Cardinality | 0..*
Summary | true
MedicinalProductAuthorization.jurisdictionalAuthorization.identifier
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization.identifier
Definition | The assigned number for the marketing authorization.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProductAuthorization.jurisdictionalAuthorization.country
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization.country
Definition | Country of authorization.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.jurisdictionalAuthorization.jurisdiction
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization.jurisdiction
Definition | Jurisdiction within a country.
Cardinality | 0..*
Type | CodeableConcept
Summary | true

MedicinalProductAuthorization.jurisdictionalAuthorization.legalStatusOfSupply
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization.legalStatusOfSupply
Definition | The legal status of supply in a jurisdiction or region.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.jurisdictionalAuthorization.validityPeriod
Element Id | MedicinalProductAuthorization.jurisdictionalAuthorization.validityPeriod
Definition | The start and expected end date of the authorization.
Cardinality | 0..1
Type | Period
Summary | true
MedicinalProductAuthorization.holder
Element Id | MedicinalProductAuthorization.holder
Definition | Marketing Authorization Holder.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
MedicinalProductAuthorization.regulator
Element Id | MedicinalProductAuthorization.regulator
Definition | Medicines Regulatory Agency.
Cardinality | 0..1
Type | Reference(Organization)
Summary | true
MedicinalProductAuthorization.procedure
Element Id | MedicinalProductAuthorization.procedure
Definition | The regulatory procedure for granting or amending a marketing authorization.
Cardinality | 0..1
Summary | true
MedicinalProductAuthorization.procedure.identifier
Element Id | MedicinalProductAuthorization.procedure.identifier
Definition | Identifier for this procedure.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
MedicinalProductAuthorization.procedure.type
Element Id | MedicinalProductAuthorization.procedure.type
Definition | Type of procedure.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductAuthorization.procedure.date[x]
Element Id | MedicinalProductAuthorization.procedure.date[x]
Definition | Date of procedure.
Cardinality | 0..1
Type | Period|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicinalProductAuthorization.procedure.application
Element Id | MedicinalProductAuthorization.procedure.application
Definition | Applcations submitted to obtain a marketing authorization.
Cardinality | 0..*
Type | See MedicinalProductAuthorization.procedure
Summary | true

11.12.6 Resource MedicinalProductAuthorization - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductauthorization-mappings.html

11.12.6 Resource MedicinalProductAuthorization - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductauthorization resource (see Mappings to Other Standards for further information & status).

11.12.7 Resource MedicinalProductAuthorization - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductauthorization-profiles.html

11.12.7 Resource MedicinalProductAuthorization - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductAuthorization resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.12.7.1 Profiles

No Profiles defined for this resource

11.12.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.12.7.3 Search Extensions

No Search Extensions defined for this resource

11.12.8 JSON Schema for MedicinalProductAuthorization

URL: https://hl7.org/fhir/R4/medicinalproductauthorization.schema.json.html

11.12.8 JSON Schema for MedicinalProductAuthorization

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductAuthorization, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductAuthorization",
  "$ref": "#/definitions/MedicinalProductAuthorization",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductAuthorization": {
      "description": "The regulatory authorization of a medicinal product.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductAuthorization resource",
          "const": "MedicinalProductAuthorization"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
```

```
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifier for the marketing authorization, as assigned by a regulator.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "subject": {
      "description": "The medicinal product that is being authorized.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "country": {
      "description": "The country in which the marketing authorization has been granted.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "Jurisdiction within a country.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "status": {
      "description": "The status of the marketing authorization.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "statusDate": {
      "description": "The date at which the given status has become applicable.",
      "$ref": "#/definitions/dateTime"
    },
    "_statusDate": {
      "description": "Extensions for statusDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "restoreDate": {
      "description": "The date when a suspended the marketing or the marketing authorization of the product is
anticipated to be restored.",
      "$ref": "#/definitions/dateTime"
    },
    "_restoreDate": {
      "description": "Extensions for restoreDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "validityPeriod": {
      "description": "The beginning of the time period in which the marketing authorization is in the specific
status shall be specified A complete date consisting of day, month and year shall be specified using the ISO 8601
date format.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "dataExclusivityPeriod": {
      "description": "A period of time after authorization before generic product applicatiosn can be submitted.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "dateOfFirstAuthorization": {
      "description": "The date when the first authorization was granted by a Medicines Regulatory Agency.",
      "$ref": "#/definitions/dateTime"
    },
    "_dateOfFirstAuthorization": {
      "description": "Extensions for dateOfFirstAuthorization",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "internationalBirthDate": {
      "description": "Date of first marketing authorization for a company\u0027s new medicinal product in any
country in the World.",
      "$ref": "#/definitions/dateTime"
    },
    "_internationalBirthDate": {
      "description": "Extensions for internationalBirthDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "legalBasis": {
      "description": "The legal framework against which this authorization is granted.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "jurisdictionalAuthorization": {
      "description": "Authorization in areas within a country.",
      "items": {
        "$ref": "#/definitions/MedicinalProductAuthorization_JurisdictionalAuthorization"
      },
      "type": "array"
    },
    "holder": {
      "description": "Marketing Authorization Holder.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "regulator": {
      "description": "Medicines Regulatory Agency.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "procedure": {
      "description": "The regulatory procedure for granting or amending a marketing authorization.",
      "$ref": "#/definitions/MedicinalProductAuthorization_Procedure"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"MedicinalProductAuthorization_JurisdictionalAuthorization": {
  "description": "The regulatory authorization of a medicinal product.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
```

```
      "description": "The assigned number for the marketing authorization.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "country": {
      "description": "Country of authorization.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "jurisdiction": {
      "description": "Jurisdiction within a country.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "legalStatusOfSupply": {
      "description": "The legal status of supply in a jurisdiction or region.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "validityPeriod": {
      "description": "The start and expected end date of the authorization.",
      "$ref": "Period.schema.json#/definitions/Period"
    }
  },
  "additionalProperties": false
},
"MedicinalProductAuthorization_Procedure": {
  "description": "The regulatory authorization of a medicinal product.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier for this procedure.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "type": {
      "description": "Type of procedure.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "datePeriod": {
      "description": "Date of procedure.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "dateDateTime": {
      "description": "Date of procedure.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_dateDateTime": {
      "description": "Extensions for dateDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "application": {
      "description": "Applcations submitted to obtain a marketing authorization.",
      "items": {
        "$ref": "#/definitions/MedicinalProductAuthorization_Procedure"
      },
      "type": "array"
    }
```

```
      },
      "additionalProperties": false,
      "required": [
        "type"
      ]
    }
  }
}
```

11.12.9 Resource MedicinalProductAuthorization - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductauthorization-dependencies.html

11.12.9 Resource MedicinalProductAuthorization - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductAuthorization FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductAuthorization | | DomainResource
identifier | 0..* | Identifier
subject | 0..1 | Reference(MedicinalProduct | MedicinalProductPackaged)
country | 0..* | CodeableConcept
jurisdiction | 0..* | CodeableConcept
status | 0..1 | CodeableConcept
statusDate | 0..1 | dateTime
restoreDate | 0..1 | dateTime
validityPeriod | 0..1 | Period
dataExclusivityPeriod | 0..1 | Period
dateOfFirstAuthorization | 0..1 | dateTime
internationalBirthDate | 0..1 | dateTime
legalBasis | 0..1 | CodeableConcept
jurisdictionalAuthorization | 0..* | BackboneElement
identifier | 0..* | Identifier
country | 0..1 | CodeableConcept
jurisdiction | 0..* | CodeableConcept
legalStatusOfSupply | 0..1 | CodeableConcept
validityPeriod | 0..1 | Period
holder | 0..1 | Reference(Organization)
regulator | 0..1 | Reference(Organization)
procedure | 0..1 | BackboneElement
identifier | 0..1 | Identifier
type | 1..1 | CodeableConcept
date[x] | 0..1
datePeriod | | Period
dateDateTime | | dateTime
application | 0..* | see procedure

11.13 Resource MedicinalProductContraindication - Content

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The clinical particulars - indications, contraindications etc. of a medicinal product, including for regulatory purposes.

11.13.1 Scope and Usage

A reason why a medicinal product is not to be used in a certain context

11.13.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProductContraindication | ΣTU | | DomainResource | MedicinalProductContraindication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

subject | Σ | 0..* | Reference(MedicinalProduct | Medication) | The medication for which this is an indication

disease | Σ | 0..1 | CodeableConcept | The disease, symptom or procedure for the contraindication

diseaseStatus | Σ | 0..1 | CodeableConcept | The status of the disease or symptom for the contraindication

comorbidity | Σ | 0..* | CodeableConcept | A comorbidity (concurrent condition) or coinfection

therapeuticIndication | Σ | 0..* | Reference(MedicinalProductIndication) | Information about the use of the medicinal product in relation to other therapies as part of the indication

otherTherapy | Σ | 0..* | BackboneElement | Information about the use of the medicinal product in relation to other therapies described as part of the indication

therapyRelationshipType | Σ | 1..1 | CodeableConcept | The type of relationship between the medicinal product indication or contraindication and another therapy

medication[x] | Σ | 1..1 | | Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication

medicationCodeableConcept | | | CodeableConcept

medicationReference | | | Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)

population | Σ | 0..* | Population | The population group to which this applies

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductContraindication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <disease><!-- 0..1 CodeableConcept The disease, symptom or procedure for the contraindication --></disease>
 <diseaseStatus><!-- 0..1 CodeableConcept The status of the disease or symptom for the contraindication
    --></diseaseStatus>
 <comorbidity><!-- 0..* CodeableConcept A comorbidity (concurrent condition) or coinfection --></comorbidity>
 <therapeuticIndication><!-- 0..* Reference(MedicinalProductIndication) Information about the use of the medicinal
    product in relation to other therapies as part of the indication --></therapeuticIndication>
 <otherTherapy>  <!-- 0..* Information about the use of the medicinal product in relation to other therapies described
    as part of the indication -->
  <therapyRelationshipType><!-- 1..1 CodeableConcept The type of relationship between the medicinal product indication
    or contraindication and another therapy --></therapyRelationshipType>
  <medication[x]><!-- 1..1 CodeableConcept|Reference(MedicinalProduct|Medication|
    Substance|SubstanceSpecification) Reference to a specific medication (active substance, medicinal product or class
    of products) as part of an indication or contraindication --></medication[x]>
 </otherTherapy>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductContraindication>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductContraindication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "disease" : { CodeableConcept }, // The disease, symptom or procedure for the contraindication
  "diseaseStatus" : { CodeableConcept }, // The status of the disease or symptom for the contraindication
  "comorbidity" : [{ CodeableConcept }], // A comorbidity (concurrent condition) or coinfection
  "therapeuticIndication" : [{ Reference(MedicinalProductIndication) }], // Information about the use of the medicinal
    product in relation to other therapies as part of the indication
  "otherTherapy" : [{ // Information about the use of the medicinal product in relation to other therapies described as
    part of the indication
    "therapyRelationshipType" : { CodeableConcept }, // R!  The type of relationship between the medicinal product
    indication or contraindication and another therapy
    // medication[x]:
      Reference to a specific medication (active substance, medicinal product or class of products) as part of an
    indication or contraindication. One of these 2:
    "medicationCodeableConcept" : { CodeableConcept }
    "medicationReference" : { Reference(MedicinalProduct|Medication|Substance|
    SubstanceSpecification) }
  }],
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductContraindication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductContraindication.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication
    for which this is an indication
  fhir:MedicinalProductContraindication.disease [ CodeableConcept ]; # 0..1 The disease, symptom or procedure for the
    contraindication
```

```
  fhir:MedicinalProductContraindication.diseaseStatus [ CodeableConcept ]; # 0..1 The status of the disease or symptom
    for the contraindication
  fhir:MedicinalProductContraindication.comorbidity [ CodeableConcept ], ... ; # 0..* A comorbidity (concurrent
    condition) or coinfection
  fhir:MedicinalProductContraindication.therapeuticIndication [ Reference(MedicinalProductIndication) ], ... ; # 0..*
    Information about the use of the medicinal product in relation to other therapies as part of the indication
  fhir:MedicinalProductContraindication.otherTherapy [ # 0..* Information about the use of the medicinal product in
    relation to other therapies described as part of the indication
    fhir:MedicinalProductContraindication.otherTherapy.therapyRelationshipType [ CodeableConcept ]; # 1..1 The type of
    relationship between the medicinal product indication or contraindication and another therapy
    # MedicinalProductContraindication.otherTherapy.medication[x] : 1..1 Reference to a specific medication (active
    substance, medicinal product or class of products) as part of an indication or contraindication. One of these 2
      fhir:MedicinalProductContraindication.otherTherapy.medicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProductContraindication.otherTherapy.medicationReference [
    Reference(MedicinalProduct|Medication|Substance|SubstanceSpecification) ]
  ], ...;
  fhir:MedicinalProductContraindication.population [ Population ], ... ; # 0..* The population group to which this
    applies
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
MedicinalProductContraindication	ΣTU		DomainResource	MedicinalProductContraindication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject	Σ	0..*	Reference(MedicinalProduct | Medication)	The medication for which this is an indication
disease	Σ	0..1	CodeableConcept	The disease, symptom or procedure for the contraindication
diseaseStatus	Σ	0..1	CodeableConcept	The status of the disease or symptom for the contraindication
comorbidity	Σ	0..*	CodeableConcept	A comorbidity (concurrent condition) or coinfection
therapeuticIndication	Σ	0..*	Reference(MedicinalProductIndication)	Information about the use of the medicinal product in relation to other therapies as part of the indication
otherTherapy	Σ	0..*	BackboneElement	Information about the use of the medicinal product in relation to other therapies described as part of the indication
therapyRelationshipType	Σ	1..1	CodeableConcept	The type of relationship between the medicinal product indication or contraindication and another therapy
medication[x]	Σ	1..1		Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication
medicationCodeableConcept			CodeableConcept	
medicationReference			Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)	
population	Σ	0..*	Population	The population group to which this applies

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductContraindication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <disease><!-- 0..1 CodeableConcept The disease, symptom or procedure for the contraindication --></disease>
 <diseaseStatus><!-- 0..1 CodeableConcept The status of the disease or symptom for the contraindication
    --></diseaseStatus>
 <comorbidity><!-- 0..* CodeableConcept A comorbidity (concurrent condition) or coinfection --></comorbidity>
 <therapeuticIndication><!-- 0..* Reference(MedicinalProductIndication) Information about the use of the medicinal
    product in relation to other therapies as part of the indication --></therapeuticIndication>
 <otherTherapy>  <!-- 0..* Information about the use of the medicinal product in relation to other therapies described
    as part of the indication -->
  <therapyRelationshipType><!-- 1..1 CodeableConcept The type of relationship between the medicinal product indication
    or contraindication and another therapy --></therapyRelationshipType>
  <medication[x]><!-- 1..1 CodeableConcept|Reference(MedicinalProduct|Medication|
    Substance|SubstanceSpecification) Reference to a specific medication (active substance, medicinal product or class
    of products) as part of an indication or contraindication --></medication[x]>
 </otherTherapy>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductContraindication>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductContraindication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "disease" : { CodeableConcept }, // The disease, symptom or procedure for the contraindication
  "diseaseStatus" : { CodeableConcept }, // The status of the disease or symptom for the contraindication
  "comorbidity" : [{ CodeableConcept }], // A comorbidity (concurrent condition) or coinfection
  "therapeuticIndication" : [{ Reference(MedicinalProductIndication) }], // Information about the use of the medicinal
    product in relation to other therapies as part of the indication
  "otherTherapy" : [{ // Information about the use of the medicinal product in relation to other therapies described as
    part of the indication
    "therapyRelationshipType" : { CodeableConcept }, // R!  The type of relationship between the medicinal product
    indication or contraindication and another therapy
```

```
    // medication[x]:
      Reference to a specific medication (active substance, medicinal product or class of products) as part of an
    indication or contraindication. One of these 2:
    "medicationCodeableConcept" : { CodeableConcept }
    "medicationReference" : { Reference(MedicinalProduct|Medication|Substance|
    SubstanceSpecification) }
  }],
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductContraindication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductContraindication.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication
    for which this is an indication
  fhir:MedicinalProductContraindication.disease [ CodeableConcept ]; # 0..1 The disease, symptom or procedure for the
    contraindication
  fhir:MedicinalProductContraindication.diseaseStatus [ CodeableConcept ]; # 0..1 The status of the disease or symptom
    for the contraindication
  fhir:MedicinalProductContraindication.comorbidity [ CodeableConcept ], ... ; # 0..* A comorbidity (concurrent
    condition) or coinfection
  fhir:MedicinalProductContraindication.therapeuticIndication [ Reference(MedicinalProductIndication) ], ... ; # 0..*
    Information about the use of the medicinal product in relation to other therapies as part of the indication
  fhir:MedicinalProductContraindication.otherTherapy [ # 0..* Information about the use of the medicinal product in
    relation to other therapies described as part of the indication
    fhir:MedicinalProductContraindication.otherTherapy.therapyRelationshipType [ CodeableConcept ]; # 1..1 The type of
    relationship between the medicinal product indication or contraindication and another therapy
    # MedicinalProductContraindication.otherTherapy.medication[x] : 1..1 Reference to a specific medication (active
    substance, medicinal product or class of products) as part of an indication or contraindication. One of these 2
      fhir:MedicinalProductContraindication.otherTherapy.medicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProductContraindication.otherTherapy.medicationReference [
    Reference(MedicinalProduct|Medication|Substance|SubstanceSpecification) ]
  ], ...;
  fhir:MedicinalProductContraindication.population [ Population ], ... ; # 0..* The population group to which this
    applies
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.13.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
subject | reference | The medication for which this is an contraindication | MedicinalProductContraindication.subject (Medication, MedicinalProduct)

11.13.4 Resource MedicinalProductContraindication - Examples

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication-examples.html

11.13.4 Resource MedicinalProductContraindication - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductcontraindication | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.13.5 Resource MedicinalProductContraindication - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication-definitions.html

11.13.5 Resource MedicinalProductContraindication - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductContraindication resource.

MedicinalProductContraindication
Element Id | MedicinalProductContraindication
Definition | The clinical particulars - indications, contraindications etc. of a medicinal product, including for regulatory purposes.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductContraindication.subject
Element Id | MedicinalProductContraindication.subject
Definition | The medication for which this is an indication.
Cardinality | 0..*
Type | Reference(MedicinalProduct | Medication)
Summary | true
MedicinalProductContraindication.disease
Element Id | MedicinalProductContraindication.disease
Definition | The disease, symptom or procedure for the contraindication.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductContraindication.diseaseStatus
Element Id | MedicinalProductContraindication.diseaseStatus
Definition | The status of the disease or symptom for the contraindication.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductContraindication.comorbidity
Element Id | MedicinalProductContraindication.comorbidity
Definition | A comorbidity (concurrent condition) or coinfection.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductContraindication.therapeuticIndication
Element Id | MedicinalProductContraindication.therapeuticIndication
Definition | Information about the use of the medicinal product in relation to other therapies as part of the indication.
Cardinality | 0..*
Type | Reference(MedicinalProductIndication)
Summary | true
MedicinalProductContraindication.otherTherapy
Element Id | MedicinalProductContraindication.otherTherapy
Definition | Information about the use of the medicinal product in relation to other therapies described as part of the indication.
Cardinality | 0..*
Summary | true
MedicinalProductContraindication.otherTherapy.therapyRelationshipType
Element Id | MedicinalProductContraindication.otherTherapy.therapyRelationshipType
Definition | The type of relationship between the medicinal product indication or contraindication and another therapy.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductContraindication.otherTherapy.medication[x]
Element Id | MedicinalProductContraindication.otherTherapy.medication[x]
Definition | Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication.
Cardinality | 1..1
Type | CodeableConcept|Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicinalProductContraindication.population
Element Id | MedicinalProductContraindication.population
Definition | The population group to which this applies.
Cardinality | 0..*
Type | Population
Summary | true

11.13.6 Resource MedicinalProductContraindication - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication-mappings.html

11.13.6 Resource MedicinalProductContraindication - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductcontraindication resource (see Mappings to Other Standards for further information & status).

11.13.7 Resource MedicinalProductContraindication - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication-profiles.html

11.13.7 Resource MedicinalProductContraindication - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductContraindication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.13.7.1 Profiles

No Profiles defined for this resource

11.13.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.13.7.3 Search Extensions

No Search Extensions defined for this resource

11.13.8 JSON Schema for MedicinalProductContraindication

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication.schema.json.html

11.13.8 JSON Schema for MedicinalProductContraindication

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductContraindication, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductContraindication",
  "$ref": "#/definitions/MedicinalProductContraindication",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductContraindication": {
      "description": "The clinical particulars - indications, contraindications etc. of a medicinal product, including
    for regulatory purposes.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductContraindication resource",
          "const": "MedicinalProductContraindication"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
```

```
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "subject": {
      "description": "The medication for which this is an indication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "disease": {
      "description": "The disease, symptom or procedure for the contraindication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "diseaseStatus": {
      "description": "The status of the disease or symptom for the contraindication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "comorbidity": {
      "description": "A comorbidity (concurrent condition) or coinfection.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "therapeuticIndication": {
      "description": "Information about the use of the medicinal product in relation to other therapies as part of
the indication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "otherTherapy": {
      "description": "Information about the use of the medicinal product in relation to other therapies described as
```

```
     part of the indication.",
          "items": {
            "$ref": "#/definitions/MedicinalProductContraindication_OtherTherapy"
          },
          "type": "array"
        },
        "population": {
          "description": "The population group to which this applies.",
          "items": {
            "$ref": "Population.schema.json#/definitions/Population"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    },
    "MedicinalProductContraindication_OtherTherapy": {
      "description": "The clinical particulars - indications, contraindications etc. of a medicinal product, including
    for regulatory purposes.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "therapyRelationshipType": {
          "description": "The type of relationship between the medicinal product indication or contraindication and
    another therapy.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "medicationCodeableConcept": {
          "description": "Reference to a specific medication (active substance, medicinal product or class of products)
    as part of an indication or contraindication.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "medicationReference": {
          "description": "Reference to a specific medication (active substance, medicinal product or class of products)
    as part of an indication or contraindication.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "therapyRelationshipType"
      ]
    }
  }
}
```


11.13.9 Resource MedicinalProductContraindication - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductcontraindication-dependencies.html

11.13.9 Resource MedicinalProductContraindication - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductContraindication FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductContraindication | | DomainResource
subject | 0..* | Reference(MedicinalProduct | Medication)
disease | 0..1 | CodeableConcept
diseaseStatus | 0..1 | CodeableConcept
comorbidity | 0..* | CodeableConcept
therapeuticIndication | 0..* | Reference(MedicinalProductIndication)
otherTherapy | 0..* | BackboneElement
therapyRelationshipType | 1..1 | CodeableConcept
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification
population | 0..* | Population | Type Error: (Population = FMM5-Draft vs. Element = FMM0-Trial Use)

11.14 Resource MedicinalProductIndication - Content

URL: https://hl7.org/fhir/R4/medicinalproductindication.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Indication for the Medicinal Product.

11.14.1 Scope and Usage

A medical situation in which a medicinal product is appropriate for use

This resource is referenced by MedicinalProduct and MedicinalProductContraindication

11.14.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductIndication | ΣTU | | DomainResource | MedicinalProductIndication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject | Σ | 0..* | Reference(MedicinalProduct | Medication) | The medication for which this is an indication
diseaseSymptomProcedure | Σ | 0..1 | CodeableConcept | The disease, symptom or procedure that is the indication for treatment
diseaseStatus | Σ | 0..1 | CodeableConcept | The status of the disease or symptom for which the indication applies
comorbidity | Σ | 0..* | CodeableConcept | Comorbidity (concurrent condition) or co-infection as part of the indication
intendedEffect | Σ | 0..1 | CodeableConcept | The intended effect, aim or strategy to be achieved by the indication
duration | Σ | 0..1 | Quantity | Timing or duration information as part of the indication
otherTherapy | Σ | 0..* | BackboneElement | Information about the use of the medicinal product in relation to other therapies described as part of the indication
therapyRelationshipType | Σ | 1..1 | CodeableConcept | The type of relationship between the medicinal product indication or contraindication and another therapy
medication[x] | Σ | 1..1 | | Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)
undesirableEffect | Σ | 0..* | Reference(MedicinalProductUndesirableEffect) | Describe the undesirable effects of the medicinal product
population | Σ | 0..* | Population | The population group to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductIndication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <diseaseSymptomProcedure><!-- 0..1 CodeableConcept The disease, symptom or procedure that is the indication for
    treatment --></diseaseSymptomProcedure>
 <diseaseStatus><!-- 0..1 CodeableConcept The status of the disease or symptom for which the indication applies
    --></diseaseStatus>
 <comorbidity><!-- 0..* CodeableConcept Comorbidity (concurrent condition) or co-infection as part of the indication
    --></comorbidity>
```

```
 <intendedEffect><!-- 0..1 CodeableConcept The intended effect, aim or strategy to be achieved by the indication
    --></intendedEffect>
 <duration><!-- 0..1 Quantity Timing or duration information as part of the indication --></duration>
 <otherTherapy>  <!-- 0..* Information about the use of the medicinal product in relation to other therapies described
    as part of the indication -->
  <therapyRelationshipType><!-- 1..1 CodeableConcept The type of relationship between the medicinal product indication
    or contraindication and another therapy --></therapyRelationshipType>
  <medication[x]><!-- 1..1 CodeableConcept|Reference(MedicinalProduct|Medication|
    Substance|SubstanceSpecification) Reference to a specific medication (active substance, medicinal product or class
    of products) as part of an indication or contraindication --></medication[x]>
 </otherTherapy>
 <undesirableEffect><!-- 0..* Reference(MedicinalProductUndesirableEffect) Describe the undesirable effects of the
    medicinal product --></undesirableEffect>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductIndication>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductIndication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "diseaseSymptomProcedure" : { CodeableConcept }, // The disease, symptom or procedure that is the indication for
    treatment
  "diseaseStatus" : { CodeableConcept }, // The status of the disease or symptom for which the indication applies
  "comorbidity" : [{ CodeableConcept }], // Comorbidity (concurrent condition) or co-infection as part of the indication
  "intendedEffect" : { CodeableConcept }, // The intended effect, aim or strategy to be achieved by the indication
  "duration" : { Quantity }, // Timing or duration information as part of the indication
  "otherTherapy" : [{ // Information about the use of the medicinal product in relation to other therapies described as
    part of the indication
    "therapyRelationshipType" : { CodeableConcept }, // R!  The type of relationship between the medicinal product
    indication or contraindication and another therapy
    // medication[x]:
      Reference to a specific medication (active substance, medicinal product or class of products) as part of an
    indication or contraindication. One of these 2:
    "medicationCodeableConcept" : { CodeableConcept }
    "medicationReference" : { Reference(MedicinalProduct|Medication|Substance|
    SubstanceSpecification) }
  }],
  "undesirableEffect" : [{ Reference(MedicinalProductUndesirableEffect) }], // Describe the undesirable effects of the
    medicinal product
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductIndication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductIndication.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication for
    which this is an indication
  fhir:MedicinalProductIndication.diseaseSymptomProcedure [ CodeableConcept ]; # 0..1 The disease, symptom or procedure
    that is the indication for treatment
  fhir:MedicinalProductIndication.diseaseStatus [ CodeableConcept ]; # 0..1 The status of the disease or symptom for
    which the indication applies
  fhir:MedicinalProductIndication.comorbidity [ CodeableConcept ], ... ; # 0..* Comorbidity (concurrent condition) or
    co-infection as part of the indication
  fhir:MedicinalProductIndication.intendedEffect [ CodeableConcept ]; # 0..1 The intended effect, aim or strategy to be
    achieved by the indication
  fhir:MedicinalProductIndication.duration [ Quantity ]; # 0..1 Timing or duration information as part of the indication
  fhir:MedicinalProductIndication.otherTherapy [ # 0..* Information about the use of the medicinal product in relation
    to other therapies described as part of the indication
    fhir:MedicinalProductIndication.otherTherapy.therapyRelationshipType [ CodeableConcept ]; # 1..1 The type of
    relationship between the medicinal product indication or contraindication and another therapy
    # MedicinalProductIndication.otherTherapy.medication[x] : 1..1 Reference to a specific medication (active substance,
     medicinal product or class of products) as part of an indication or contraindication. One of these 2
      fhir:MedicinalProductIndication.otherTherapy.medicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProductIndication.otherTherapy.medicationReference [
    Reference(MedicinalProduct|Medication|Substance|SubstanceSpecification) ]
  ], ...;
  fhir:MedicinalProductIndication.undesirableEffect [ Reference(MedicinalProductUndesirableEffect) ], ... ; # 0..*
    Describe the undesirable effects of the medicinal product
  fhir:MedicinalProductIndication.population [ Population ], ... ; # 0..* The population group to which this applies
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProductIndication | ΣTU | | | DomainResource | MedicinalProductIndication Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

subject | Σ | 0..* | Reference(MedicinalProduct | Medication) | The medication for which this is an indication
diseaseSymptomProcedure | Σ | 0..1 | CodeableConcept | The disease, symptom or procedure that is the indication for treatment
diseaseStatus | Σ | 0..1 | CodeableConcept | The status of the disease or symptom for which the indication applies
comorbidity | Σ | 0..* | CodeableConcept | Comorbidity (concurrent condition) or co-infection as part of the indication
intendedEffect | Σ | 0..1 | CodeableConcept | The intended effect, aim or strategy to be achieved by the indication
duration | Σ | 0..1 | Quantity | Timing or duration information as part of the indication
otherTherapy | Σ | 0..* | BackboneElement | Information about the use of the medicinal product in relation to other therapies described as part of the indication
therapyRelationshipType | Σ | 1..1 | CodeableConcept | The type of relationship between the medicinal product indication or contraindication and another therapy
medication[x] | Σ | 1..1 | | Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication
medicationCodeableConcept | | | CodeableConcept
medicationReference | | | Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)
undesirableEffect | Σ | 0..* | Reference(MedicinalProductUndesirableEffect) | Describe the undesirable effects of the medicinal product
population | Σ | 0..* | Population | The population group to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductIndication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <diseaseSymptomProcedure><!-- 0..1 CodeableConcept The disease, symptom or procedure that is the indication for
    treatment --></diseaseSymptomProcedure>
 <diseaseStatus><!-- 0..1 CodeableConcept The status of the disease or symptom for which the indication applies
    --></diseaseStatus>
 <comorbidity><!-- 0..* CodeableConcept Comorbidity (concurrent condition) or co-infection as part of the indication
    --></comorbidity>
 <intendedEffect><!-- 0..1 CodeableConcept The intended effect, aim or strategy to be achieved by the indication
    --></intendedEffect>
 <duration><!-- 0..1 Quantity Timing or duration information as part of the indication --></duration>
 <otherTherapy>  <!-- 0..* Information about the use of the medicinal product in relation to other therapies described
    as part of the indication -->
  <therapyRelationshipType><!-- 1..1 CodeableConcept The type of relationship between the medicinal product indication
    or contraindication and another therapy --></therapyRelationshipType>
  <medication[x]><!-- 1..1 CodeableConcept|Reference(MedicinalProduct|Medication|
    Substance|SubstanceSpecification) Reference to a specific medication (active substance, medicinal product or class
    of products) as part of an indication or contraindication --></medication[x]>
 </otherTherapy>
 <undesirableEffect><!-- 0..* Reference(MedicinalProductUndesirableEffect) Describe the undesirable effects of the
    medicinal product --></undesirableEffect>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductIndication>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductIndication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "diseaseSymptomProcedure" : { CodeableConcept }, // The disease, symptom or procedure that is the indication for
    treatment
  "diseaseStatus" : { CodeableConcept }, // The status of the disease or symptom for which the indication applies
  "comorbidity" : [{ CodeableConcept }], // Comorbidity (concurrent condition) or co-infection as part of the indication
  "intendedEffect" : { CodeableConcept }, // The intended effect, aim or strategy to be achieved by the indication
  "duration" : { Quantity }, // Timing or duration information as part of the indication
  "otherTherapy" : [{ // Information about the use of the medicinal product in relation to other therapies described as
    part of the indication
    "therapyRelationshipType" : { CodeableConcept }, // R!  The type of relationship between the medicinal product
    indication or contraindication and another therapy
    // medication[x]:
      Reference to a specific medication (active substance, medicinal product or class of products) as part of an
    indication or contraindication. One of these 2:
    "medicationCodeableConcept" : { CodeableConcept }
    "medicationReference" : { Reference(MedicinalProduct|Medication|Substance|
    SubstanceSpecification) }
  }],
  "undesirableEffect" : [{ Reference(MedicinalProductUndesirableEffect) }], // Describe the undesirable effects of the
    medicinal product
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductIndication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
```

```
  fhir:MedicinalProductIndication.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication for
    which this is an indication
  fhir:MedicinalProductIndication.diseaseSymptomProcedure [ CodeableConcept ]; # 0..1 The disease, symptom or procedure
    that is the indication for treatment
  fhir:MedicinalProductIndication.diseaseStatus [ CodeableConcept ]; # 0..1 The status of the disease or symptom for
    which the indication applies
  fhir:MedicinalProductIndication.comorbidity [ CodeableConcept ], ... ; # 0..* Comorbidity (concurrent condition) or
    co-infection as part of the indication
  fhir:MedicinalProductIndication.intendedEffect [ CodeableConcept ]; # 0..1 The intended effect, aim or strategy to be
    achieved by the indication
  fhir:MedicinalProductIndication.duration [ Quantity ]; # 0..1 Timing or duration information as part of the indication
  fhir:MedicinalProductIndication.otherTherapy [ # 0..* Information about the use of the medicinal product in relation
    to other therapies described as part of the indication
    fhir:MedicinalProductIndication.otherTherapy.therapyRelationshipType [ CodeableConcept ]; # 1..1 The type of
    relationship between the medicinal product indication or contraindication and another therapy
    # MedicinalProductIndication.otherTherapy.medication[x] : 1..1 Reference to a specific medication (active substance,
     medicinal product or class of products) as part of an indication or contraindication. One of these 2
      fhir:MedicinalProductIndication.otherTherapy.medicationCodeableConcept [ CodeableConcept ]
      fhir:MedicinalProductIndication.otherTherapy.medicationReference [
    Reference(MedicinalProduct|Medication|Substance|SubstanceSpecification) ]
  ], ...;
  fhir:MedicinalProductIndication.undesirableEffect [ Reference(MedicinalProductUndesirableEffect) ], ... ; # 0..*
    Describe the undesirable effects of the medicinal product
  fhir:MedicinalProductIndication.population [ Population ], ... ; # 0..* The population group to which this applies
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.14.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
subject | reference | The medication for which this is an indication | MedicinalProductIndication.subject (Medication, MedicinalProduct)

11.14.4 Resource MedicinalProductIndication - Examples

URL: https://hl7.org/fhir/R4/medicinalproductindication-examples.html

11.14.4 Resource MedicinalProductIndication - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductindication | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.14.5 Resource MedicinalProductIndication - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductindication-definitions.html

11.14.5 Resource MedicinalProductIndication - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductIndication resource.

MedicinalProductIndication
Element Id | MedicinalProductIndication
Definition | Indication for the Medicinal Product.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductIndication.subject
Element Id | MedicinalProductIndication.subject
Definition | The medication for which this is an indication.
Cardinality | 0..*
Type | Reference(MedicinalProduct | Medication)
Summary | true

MedicinalProductIndication.diseaseSymptomProcedure
Element Id | MedicinalProductIndication.diseaseSymptomProcedure
Definition | The disease, symptom or procedure that is the indication for treatment.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductIndication.diseaseStatus
Element Id | MedicinalProductIndication.diseaseStatus
Definition | The status of the disease or symptom for which the indication applies.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductIndication.comorbidity
Element Id | MedicinalProductIndication.comorbidity
Definition | Comorbidity (concurrent condition) or co-infection as part of the indication.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductIndication.intendedEffect
Element Id | MedicinalProductIndication.intendedEffect
Definition | The intended effect, aim or strategy to be achieved by the indication.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductIndication.duration
Element Id | MedicinalProductIndication.duration
Definition | Timing or duration information as part of the indication.
Cardinality | 0..1
Type | Quantity
Summary | true
MedicinalProductIndication.otherTherapy
Element Id | MedicinalProductIndication.otherTherapy
Definition | Information about the use of the medicinal product in relation to other therapies described as part of the indication.
Cardinality | 0..*
Summary | true
MedicinalProductIndication.otherTherapy.therapyRelationshipType
Element Id | MedicinalProductIndication.otherTherapy.therapyRelationshipType
Definition | The type of relationship between the medicinal product indication or contraindication and another therapy.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductIndication.otherTherapy.medication[x]
Element Id | MedicinalProductIndication.otherTherapy.medication[x]
Definition | Reference to a specific medication (active substance, medicinal product or class of products) as part of an indication or contraindication.
Cardinality | 1..1
Type | CodeableConcept|Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicinalProductIndication.undesirableEffect
Element Id | MedicinalProductIndication.undesirableEffect
Definition | Describe the undesirable effects of the medicinal product.
Cardinality | 0..*
Type | Reference(MedicinalProductUndesirableEffect)
Summary | true
MedicinalProductIndication.population
Element Id | MedicinalProductIndication.population
Definition | The population group to which this applies.
Cardinality | 0..*
Type | Population
Summary | true

11.14.6 Resource MedicinalProductIndication - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductindication-mappings.html

11.14.6 Resource MedicinalProductIndication - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductindication resource (see Mappings to Other Standards for further information & status).

11.14.7 Resource MedicinalProductIndication - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductindication-profiles.html

11.14.7 Resource MedicinalProductIndication - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductIndication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.14.7.1 Profiles

No Profiles defined for this resource

11.14.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.14.7.3 Search Extensions

No Search Extensions defined for this resource

11.14.8 JSON Schema for MedicinalProductIndication

URL: https://hl7.org/fhir/R4/medicinalproductindication.schema.json.html

11.14.8 JSON Schema for MedicinalProductIndication

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductIndication, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductIndication",
  "$ref": "#/definitions/MedicinalProductIndication",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductIndication": {
      "description": "Indication for the Medicinal Product.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductIndication resource",
          "const": "MedicinalProductIndication"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "subject": {
      "description": "The medication for which this is an indication.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "diseaseSymptomProcedure": {
      "description": "The disease, symptom or procedure that is the indication for treatment.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "diseaseStatus": {
      "description": "The status of the disease or symptom for which the indication applies.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "comorbidity": {
      "description": "Comorbidity (concurrent condition) or co-infection as part of the indication.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "intendedEffect": {
      "description": "The intended effect, aim or strategy to be achieved by the indication.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "duration": {
      "description": "Timing or duration information as part of the indication.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "otherTherapy": {
      "description": "Information about the use of the medicinal product in relation to other therapies described as
 part of the indication.",
      "items": {
        "$ref": "#/definitions/MedicinalProductIndication_OtherTherapy"
      },
      "type": "array"
    },
    "undesirableEffect": {
      "description": "Describe the undesirable effects of the medicinal product.",
```

```
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "population": {
        "description": "The population group to which this applies.",
        "items": {
          "$ref": "Population.schema.json#/definitions/Population"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  },
  "MedicinalProductIndication_OtherTherapy": {
    "description": "Indication for the Medicinal Product.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
  value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
  definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the element and that modifies the understanding of the element in which it is contained and/or the understanding of
  the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
  use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
   of the definition of the extension. Applications processing a resource are required to check for modifier
  extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
  (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "therapyRelationshipType": {
        "description": "The type of relationship between the medicinal product indication or contraindication and
  another therapy.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "medicationCodeableConcept": {
        "description": "Reference to a specific medication (active substance, medicinal product or class of products)
  as part of an indication or contraindication.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "medicationReference": {
        "description": "Reference to a specific medication (active substance, medicinal product or class of products)
  as part of an indication or contraindication.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      }
    },
    "additionalProperties": false,
    "required": [
      "therapyRelationshipType"
    ]
  }
 }
}
```


11.14.9 Resource MedicinalProductIndication - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductindication-dependencies.html

11.14.9 Resource MedicinalProductIndication - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductIndication FMM level 0

Name | Card. | Type | Dependency Analysis

MedicinalProductIndication | | DomainResource
subject | 0..* | Reference(MedicinalProduct | Medication)
diseaseSymptomProcedure | 0..1 | CodeableConcept
diseaseStatus | 0..1 | CodeableConcept
comorbidity | 0..* | CodeableConcept
intendedEffect | 0..1 | CodeableConcept
duration | 0..1 | Quantity
otherTherapy | 0..* | BackboneElement
therapyRelationshipType | 1..1 | CodeableConcept
medication[x] | 1..1
medicationCodeableConcept | | CodeableConcept
medicationReference | | Reference(MedicinalProduct | Medication | Substance | SubstanceSpecification
undesirableEffect | 0..* | Reference(MedicinalProductUndesirableEffect)
population | 0..* | Population | Type Error: (Population = FMM5-Draft vs. Element = FMM0-Trial Use)

11.15 Resource MedicinalProductInteraction - Content

URL: https://hl7.org/fhir/R4/medicinalproductinteraction.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The interactions of the medicinal product with other medicinal products, or other forms of interactions.

11.15.1 Scope and Usage

For detailing potential interactions associated with usage of a medicinal product

11.15.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductInteraction | ΣTU | | DomainResource | MedicinalProductInteraction Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject | Σ | 0..* | Reference(MedicinalProduct | Medication | Substance) | The medication for which this is a described interaction
description | Σ | 0..1 | string | The interaction described
interactant | Σ | 0..* | BackboneElement | The specific medication, food or laboratory test that interacts
item[x] | Σ | 1..1 | | The specific medication, food or laboratory test that interacts
itemReference | | | Reference(MedicinalProduct | Medication | Substance | ObservationDefinition)
itemCodeableConcept | | | CodeableConcept
type | Σ | 0..1 | CodeableConcept | The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab test interaction
effect | Σ | 0..1 | CodeableConcept | The effect of the interaction, for example "reduced gastric absorption of primary medication"
incidence | Σ | 0..1 | CodeableConcept | The incidence of the interaction, e.g. theoretical, observed
management | Σ | 0..1 | CodeableConcept | Actions for managing the interaction
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductInteraction xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication|Substance) The medication for which this is a described
    interaction --></subject>
 <description value="[string]"/><!-- 0..1 The interaction described -->
 <interactant>  <!-- 0..* The specific medication, food or laboratory test that interacts -->
  <item[x]><!-- 1..1 Reference(MedicinalProduct|Medication|Substance|
    ObservationDefinition)|CodeableConcept The specific medication, food or laboratory test that interacts --></item[x]>
 </interactant>
 <type><!-- 0..1 CodeableConcept The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab
     test interaction --></type>
 <effect><!-- 0..1 CodeableConcept The effect of the interaction, for example "reduced gastric absorption of primary
    medication" --></effect>
 <incidence><!-- 0..1 CodeableConcept The incidence of the interaction, e.g. theoretical, observed --></incidence>
 <management><!-- 0..1 CodeableConcept Actions for managing the interaction --></management>
</MedicinalProductInteraction>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductInteraction",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication|Substance) }], // The medication for which this is a described
    interaction
  "description" : "<string>", // The interaction described
  "interactant" : [{ // The specific medication, food or laboratory test that interacts
    // item[x]: The specific medication, food or laboratory test that interacts. One of these 2:
    "itemReference" : { Reference(MedicinalProduct|Medication|Substance|
    ObservationDefinition) }
    "itemCodeableConcept" : { CodeableConcept }
  }],
  "type" : { CodeableConcept }, // The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-
    lab test interaction
  "effect" : { CodeableConcept }, // The effect of the interaction, for example "reduced gastric absorption of primary
    medication"
  "incidence" : { CodeableConcept }, // The incidence of the interaction, e.g. theoretical, observed
  "management" : { CodeableConcept } // Actions for managing the interaction
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductInteraction;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductInteraction.subject [ Reference(MedicinalProduct|Medication|Substance) ], ... ; # 0..* The
    medication for which this is a described interaction
  fhir:MedicinalProductInteraction.description [ string ]; # 0..1 The interaction described
  fhir:MedicinalProductInteraction.interactant [ # 0..* The specific medication, food or laboratory test that interacts
    # MedicinalProductInteraction.interactant.item[x] : 1..1 The specific medication, food or laboratory test that
    interacts. One of these 2
      fhir:MedicinalProductInteraction.interactant.itemReference [
    Reference(MedicinalProduct|Medication|Substance|ObservationDefinition) ]
      fhir:MedicinalProductInteraction.interactant.itemCodeableConcept [ CodeableConcept ]
  ], ...;
  fhir:MedicinalProductInteraction.type [ CodeableConcept ]; # 0..1 The type of the interaction e.g. drug-drug
    interaction, drug-food interaction, drug-lab test interaction
  fhir:MedicinalProductInteraction.effect [ CodeableConcept ]; # 0..1 The effect of the interaction, for example
    "reduced gastric absorption of primary medication"
  fhir:MedicinalProductInteraction.incidence [ CodeableConcept ]; # 0..1 The incidence of the interaction, e.g.
    theoretical, observed
  fhir:MedicinalProductInteraction.management [ CodeableConcept ]; # 0..1 Actions for managing the interaction
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
MedicinalProductInteraction	ΣTU		DomainResource	MedicinalProductInteraction Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject	Σ	0..*	Reference(MedicinalProduct | Medication | Substance)	The medication for which this is a described interaction
description	Σ	0..1	string	The interaction described
interactant	Σ	0..*	BackboneElement	The specific medication, food or laboratory test that interacts
item[x]	Σ	1..1		The specific medication, food or laboratory test that interacts
itemReference			Reference(MedicinalProduct | Medication | Substance | ObservationDefinition)	
itemCodeableConcept			CodeableConcept	
type	Σ	0..1	CodeableConcept	The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab test interaction
effect	Σ	0..1	CodeableConcept	The effect of the interaction, for example "reduced gastric absorption of primary medication"
incidence	Σ	0..1	CodeableConcept	The incidence of the interaction, e.g. theoretical, observed
management	Σ	0..1	CodeableConcept	Actions for managing the interaction

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductInteraction xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication|Substance) The medication for which this is a described
    interaction --></subject>
 <description value="[string]"/><!-- 0..1 The interaction described -->
 <interactant>  <!-- 0..* The specific medication, food or laboratory test that interacts -->
  <item[x]><!-- 1..1 Reference(MedicinalProduct|Medication|Substance|
    ObservationDefinition)|CodeableConcept The specific medication, food or laboratory test that interacts --></item[x]>
 </interactant>
 <type><!-- 0..1 CodeableConcept The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab
     test interaction --></type>
```

```
 <effect><!-- 0..1 CodeableConcept The effect of the interaction, for example "reduced gastric absorption of primary
    medication" --></effect>
 <incidence><!-- 0..1 CodeableConcept The incidence of the interaction, e.g. theoretical, observed --></incidence>
 <management><!-- 0..1 CodeableConcept Actions for managing the interaction --></management>
</MedicinalProductInteraction>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductInteraction",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication|Substance) }], // The medication for which this is a described
    interaction
  "description" : "<string>", // The interaction described
  "interactant" : [{ // The specific medication, food or laboratory test that interacts
    // item[x]: The specific medication, food or laboratory test that interacts. One of these 2:
    "itemReference" : { Reference(MedicinalProduct|Medication|Substance|
    ObservationDefinition) }
    "itemCodeableConcept" : { CodeableConcept }
  }],
  "type" : { CodeableConcept }, // The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-
    lab test interaction
  "effect" : { CodeableConcept }, // The effect of the interaction, for example "reduced gastric absorption of primary
    medication"
  "incidence" : { CodeableConcept }, // The incidence of the interaction, e.g. theoretical, observed
  "management" : { CodeableConcept } // Actions for managing the interaction
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductInteraction;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductInteraction.subject [ Reference(MedicinalProduct|Medication|Substance) ], ... ; # 0..* The
    medication for which this is a described interaction
  fhir:MedicinalProductInteraction.description [ string ]; # 0..1 The interaction described
  fhir:MedicinalProductInteraction.interactant [ # 0..* The specific medication, food or laboratory test that interacts
    # MedicinalProductInteraction.interactant.item[x] : 1..1 The specific medication, food or laboratory test that
    interacts. One of these 2
      fhir:MedicinalProductInteraction.interactant.itemReference [
    Reference(MedicinalProduct|Medication|Substance|ObservationDefinition) ]
      fhir:MedicinalProductInteraction.interactant.itemCodeableConcept [ CodeableConcept ]
  ], ...;
  fhir:MedicinalProductInteraction.type [ CodeableConcept ]; # 0..1 The type of the interaction e.g. drug-drug
    interaction, drug-food interaction, drug-lab test interaction
  fhir:MedicinalProductInteraction.effect [ CodeableConcept ]; # 0..1 The effect of the interaction, for example
    "reduced gastric absorption of primary medication"
  fhir:MedicinalProductInteraction.incidence [ CodeableConcept ]; # 0..1 The incidence of the interaction, e.g.
    theoretical, observed
  fhir:MedicinalProductInteraction.management [ CodeableConcept ]; # 0..1 Actions for managing the interaction
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.15.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

subject | reference | The medication for which this is an interaction | MedicinalProductInteraction.subject (Medication, Substance, MedicinalProduct)

11.15.4 Resource MedicinalProductInteraction - Examples

URL: https://hl7.org/fhir/R4/medicinalproductinteraction-examples.html

11.15.4 Resource MedicinalProductInteraction - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of medicinalproductinteraction | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.15.5 Resource MedicinalProductInteraction - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductinteraction-definitions.html

11.15.5 Resource MedicinalProductInteraction - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductInteraction resource.

MedicinalProductInteraction
Element Id | MedicinalProductInteraction
Definition | The interactions of the medicinal product with other medicinal products, or other forms of interactions.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductInteraction.subject
Element Id | MedicinalProductInteraction.subject
Definition | The medication for which this is a described interaction.
Cardinality | 0..*
Type | Reference(MedicinalProduct | Medication | Substance)
Summary | true
MedicinalProductInteraction.description
Element Id | MedicinalProductInteraction.description
Definition | The interaction described.
Cardinality | 0..1
Type | string
Summary | true
MedicinalProductInteraction.interactant
Element Id | MedicinalProductInteraction.interactant
Definition | The specific medication, food or laboratory test that interacts.
Cardinality | 0..*
Summary | true
MedicinalProductInteraction.interactant.item[x]
Element Id | MedicinalProductInteraction.interactant.item[x]
Definition | The specific medication, food or laboratory test that interacts.
Cardinality | 1..1
Type | Reference(MedicinalProduct | Medication | Substance | ObservationDefinition)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
MedicinalProductInteraction.type
Element Id | MedicinalProductInteraction.type
Definition | The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab test interaction.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductInteraction.effect
Element Id | MedicinalProductInteraction.effect
Definition | The effect of the interaction, for example "reduced gastric absorption of primary medication".
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductInteraction.incidence
Element Id | MedicinalProductInteraction.incidence
Definition | The incidence of the interaction, e.g. theoretical, observed.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductInteraction.management
Element Id | MedicinalProductInteraction.management
Definition | Actions for managing the interaction.
Cardinality | 0..1
Type | CodeableConcept
Summary | true

11.15.6 Resource MedicinalProductInteraction - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductinteraction-mappings.html

11.15.6 Resource MedicinalProductInteraction - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductinteraction resource (see Mappings to Other Standards for further information & status).

11.15.7 Resource MedicinalProductInteraction - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductinteraction-profiles.html

11.15.7 Resource MedicinalProductInteraction - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductInteraction resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.15.7.1 Profiles

No Profiles defined for this resource

11.15.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.15.7.3 Search Extensions

No Search Extensions defined for this resource

11.15.8 JSON Schema for MedicinalProductInteraction

URL: https://hl7.org/fhir/R4/medicinalproductinteraction.schema.json.html

11.15.8 JSON Schema for MedicinalProductInteraction

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductInteraction, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductInteraction",
  "$ref": "#/definitions/MedicinalProductInteraction",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductInteraction": {
      "description": "The interactions of the medicinal product with other medicinal products, or other forms of
    interactions.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductInteraction resource",
          "const": "MedicinalProductInteraction"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
```

```
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "subject": {
      "description": "The medication for which this is a described interaction.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "description": {
      "description": "The interaction described.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "interactant": {
      "description": "The specific medication, food or laboratory test that interacts.",
      "items": {
        "$ref": "#/definitions/MedicinalProductInteraction_Interactant"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of the interaction e.g. drug-drug interaction, drug-food interaction, drug-lab test
interaction.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "effect": {
      "description": "The effect of the interaction, for example \"reduced gastric absorption of primary
medication\".",
```

```
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "incidence": {
          "description": "The incidence of the interaction, e.g. theoretical, observed.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "management": {
          "description": "Actions for managing the interaction.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType"
      ]
    },
    "MedicinalProductInteraction_Interactant": {
      "description": "The interactions of the medicinal product with other medicinal products, or other forms of
    interactions.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "itemReference": {
          "description": "The specific medication, food or laboratory test that interacts.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "itemCodeableConcept": {
          "description": "The specific medication, food or laboratory test that interacts.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.15.9 Resource MedicinalProductInteraction - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductinteraction-dependencies.html

11.15.9 Resource MedicinalProductInteraction - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductInteraction FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductInteraction | | DomainResource
subject | 0..* | Reference(MedicinalProduct | Medication | Substance)
description | 0..1 | string
interactant | 0..* | BackboneElement
item[x] | 1..1
itemReference | | Reference(MedicinalProduct | Medication | Substance | ObservationDefinition
itemCodeableConcept | | CodeableConcept
type | 0..1 | CodeableConcept
effect | 0..1 | CodeableConcept

incidence | 0..1 | CodeableConcept
management | 0..1 | CodeableConcept

11.16 Resource MedicinalProductUndesirableEffect - Content

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Describe the undesirable effects of the medicinal product.

11.16.1 Scope and Usage

A potential negative outcome from the use of a medicinal product. This is part of the definional information about a product and is not realted to any particular actual use of a medication. See also AdverseEvent, which is for actual events (or near misses), and AllergyIntolerance.

This resource is referenced by MedicinalProductIndication

11.16.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductUndesirableEffect | ΣTU | | DomainResource | MedicinalProductUndesirableEffect Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject | Σ | 0..* | Reference(MedicinalProduct | Medication) | The medication for which this is an indication
symptomConditionEffect | Σ | 0..1 | CodeableConcept | The symptom, condition or undesirable effect
classification | Σ | 0..1 | CodeableConcept | Classification of the effect
frequencyOfOccurrence | Σ | 0..1 | CodeableConcept | The frequency of occurrence of the effect
population | Σ | 0..* | Population | The population group to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductUndesirableEffect xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <symptomConditionEffect><!-- 0..1 CodeableConcept The symptom, condition or undesirable effect
    --></symptomConditionEffect>
 <classification><!-- 0..1 CodeableConcept Classification of the effect --></classification>
 <frequencyOfOccurrence><!-- 0..1 CodeableConcept The frequency of occurrence of the effect --></frequencyOfOccurrence>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductUndesirableEffect>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductUndesirableEffect",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "symptomConditionEffect" : { CodeableConcept }, // The symptom, condition or undesirable effect
  "classification" : { CodeableConcept }, // Classification of the effect
  "frequencyOfOccurrence" : { CodeableConcept }, // The frequency of occurrence of the effect
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductUndesirableEffect;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductUndesirableEffect.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication
     for which this is an indication
  fhir:MedicinalProductUndesirableEffect.symptomConditionEffect [ CodeableConcept ]; # 0..1 The symptom, condition or
    undesirable effect
```

```
  fhir:MedicinalProductUndesirableEffect.classification [ CodeableConcept ]; # 0..1 Classification of the effect
  fhir:MedicinalProductUndesirableEffect.frequencyOfOccurrence [ CodeableConcept ]; # 0..1 The frequency of occurrence
    of the effect
  fhir:MedicinalProductUndesirableEffect.population [ Population ], ... ; # 0..* The population group to which this
    applies
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductUndesirableEffect | ΣTU | | DomainResource | MedicinalProductUndesirableEffect Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
subject | Σ | 0..* | Reference(MedicinalProduct | Medication) | The medication for which this is an indication
symptomConditionEffect | Σ | 0..1 | CodeableConcept | The symptom, condition or undesirable effect
classification | Σ | 0..1 | CodeableConcept | Classification of the effect
frequencyOfOccurrence | Σ | 0..1 | CodeableConcept | The frequency of occurrence of the effect
population | Σ | 0..* | Population | The population group to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductUndesirableEffect xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <subject><!-- 0..* Reference(MedicinalProduct|Medication) The medication for which this is an indication --></subject>
 <symptomConditionEffect><!-- 0..1 CodeableConcept The symptom, condition or undesirable effect
    --></symptomConditionEffect>
 <classification><!-- 0..1 CodeableConcept Classification of the effect --></classification>
 <frequencyOfOccurrence><!-- 0..1 CodeableConcept The frequency of occurrence of the effect --></frequencyOfOccurrence>
 <population><!-- 0..* Population The population group to which this applies --></population>
</MedicinalProductUndesirableEffect>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductUndesirableEffect",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "subject" : [{ Reference(MedicinalProduct|Medication) }], // The medication for which this is an indication
  "symptomConditionEffect" : { CodeableConcept }, // The symptom, condition or undesirable effect
  "classification" : { CodeableConcept }, // Classification of the effect
  "frequencyOfOccurrence" : { CodeableConcept }, // The frequency of occurrence of the effect
  "population" : [{ Population }] // The population group to which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductUndesirableEffect;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductUndesirableEffect.subject [ Reference(MedicinalProduct|Medication) ], ... ; # 0..* The medication
     for which this is an indication
  fhir:MedicinalProductUndesirableEffect.symptomConditionEffect [ CodeableConcept ]; # 0..1 The symptom, condition or
    undesirable effect
  fhir:MedicinalProductUndesirableEffect.classification [ CodeableConcept ]; # 0..1 Classification of the effect
  fhir:MedicinalProductUndesirableEffect.frequencyOfOccurrence [ CodeableConcept ]; # 0..1 The frequency of occurrence
    of the effect
  fhir:MedicinalProductUndesirableEffect.population [ Population ], ... ; # 0..* The population group to which this
    applies
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.16.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
subject | reference | The medication for which this is an undesirable effect | MedicinalProductUndesirableEffect.subject (Medication, MedicinalProduct)

11.16.4 Resource MedicinalProductUndesirableEffect - Examples

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect-examples.html

11.16.4 Resource MedicinalProductUndesirableEffect - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductundesirableeffect | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.16.5 Resource MedicinalProductUndesirableEffect - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect-definitions.html

11.16.5 Resource MedicinalProductUndesirableEffect - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductUndesirableEffect resource.

MedicinalProductUndesirableEffect
Element Id | MedicinalProductUndesirableEffect
Definition | Describe the undesirable effects of the medicinal product.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductUndesirableEffect.subject
Element Id | MedicinalProductUndesirableEffect.subject
Definition | The medication for which this is an indication.
Cardinality | 0..*
Type | Reference(MedicinalProduct | Medication)
Summary | true
MedicinalProductUndesirableEffect.symptomConditionEffect
Element Id | MedicinalProductUndesirableEffect.symptomConditionEffect
Definition | The symptom, condition or undesirable effect.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductUndesirableEffect.classification
Element Id | MedicinalProductUndesirableEffect.classification
Definition | Classification of the effect.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductUndesirableEffect.frequencyOfOccurrence
Element Id | MedicinalProductUndesirableEffect.frequencyOfOccurrence
Definition | The frequency of occurrence of the effect.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductUndesirableEffect.population
Element Id | MedicinalProductUndesirableEffect.population
Definition | The population group to which this applies.
Cardinality | 0..*
Type | Population
Summary | true

11.16.6 Resource MedicinalProductUndesirableEffect - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect-mappings.html

11.16.6 Resource MedicinalProductUndesirableEffect - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductundesirableeffect resource (see Mappings to Other Standards for further information & status).

11.16.7 Resource MedicinalProductUndesirableEffect - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect-profiles.html

11.16.7 Resource MedicinalProductUndesirableEffect - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductUndesirableEffect resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.16.7.1 Profiles

No Profiles defined for this resource

11.16.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.16.7.3 Search Extensions

No Search Extensions defined for this resource

11.16.8 JSON Schema for MedicinalProductUndesirableEffect

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect.schema.json.html

11.16.8 JSON Schema for MedicinalProductUndesirableEffect

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductUndesirableEffect, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductUndesirableEffect",
  "$ref": "#/definitions/MedicinalProductUndesirableEffect",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductUndesirableEffect": {
      "description": "Describe the undesirable effects of the medicinal product.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductUndesirableEffect resource",
          "const": "MedicinalProductUndesirableEffect"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
```

```
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "subject": {
        "description": "The medication for which this is an indication.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "symptomConditionEffect": {
        "description": "The symptom, condition or undesirable effect.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "classification": {
        "description": "Classification of the effect.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "frequencyOfOccurrence": {
        "description": "The frequency of occurrence of the effect.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "population": {
        "description": "The population group to which this applies.",
        "items": {
          "$ref": "Population.schema.json#/definitions/Population"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  }
 }
}
```

11.16.9 Resource MedicinalProductUndesirableEffect - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductundesirableeffect-dependencies.html

11.16.9 Resource MedicinalProductUndesirableEffect - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductUndesirableEffect FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductUndesirableEffect | | DomainResource
subject | 0..* | Reference(MedicinalProduct | Medication)
symptomConditionEffect | 0..1 | CodeableConcept
classification | 0..1 | CodeableConcept
frequencyOfOccurrence | 0..1 | CodeableConcept
population | 0..* | Population | Type Error: (Population = FMM5-Draft vs. Element = FMM0-Trial Use)

11.17 Resource MedicinalProductIngredient - Content

URL: https://hl7.org/fhir/R4/medicinalproductingredient.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

An ingredient of a manufactured item or pharmaceutical product.

11.17.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

Material used in the preparation of a medicinal/pharmaceutical product

The ingredient is part of a Medicinal Product, either alone or in combination with other ingredients. The ingredient is also a component of a pharmaceutical product. Ingredient is equal to a substance with the indication of the specific role it is playing in the product.

This resource is referenced by MedicinalProductManufactured and MedicinalProductPharmaceutical

11.17.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductIngredient | ΣTU | | DomainResource | An ingredient of a manufactured item or pharmaceutical product Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Identifier for the ingredient
role | Σ | 1..1 | CodeableConcept | Ingredient role e.g. Active ingredient, excipient
allergenicIndicator | Σ | 0..1 | boolean | If the ingredient is a known or suspected allergen
manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of this Ingredient
specifiedSubstance | Σ | 0..* | BackboneElement | A specified substance that comprises this ingredient
code | Σ | 1..1 | CodeableConcept | The specified substance
group | Σ | 1..1 | CodeableConcept | The group of specified substance, e.g. group 1 to 4
confidentiality | Σ | 0..1 | CodeableConcept | Confidentiality level of the specified substance as the ingredient
strength | Σ | 0..* | BackboneElement | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product
presentation | Σ | 1..1 | Ratio | The quantity of substance in the unit of presentation, or in the volume (or mass) of the single pharmaceutical product or manufactured item
presentationLowLimit | ?!Σ | 0..1 | Ratio | A lower limit for the quantity of substance in the unit of presentation. For use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the upper limit
concentration | Σ | 0..1 | Ratio | The strength per unitary volume (or mass)
concentrationLowLimit | ?!Σ | 0..1 | Ratio | A lower limit for the strength per unitary volume (or mass), for when there is a range. The concentration attribute then becomes the upper limit
measurementPoint | Σ | 0..1 | string | For when strength is measured at a particular point or distance

country | Σ | 0..* | CodeableConcept | The country or countries for which the strength range applies
referenceStrength | Σ | 0..* | BackboneElement | Strength expressed in terms of a reference substance
substance | Σ | 0..1 | CodeableConcept | Relevant reference substance
strength | Σ | 1..1 | Ratio | Strength expressed in terms of a reference substance
strengthLowLimit | Σ | 0..1 | Ratio | Strength expressed in terms of a reference substance
measurementPoint | Σ | 0..1 | string | For when strength is measured at a particular point or distance
country | Σ | 0..* | CodeableConcept | The country or countries for which the strength range applies
substance | Σ | 0..1 | BackboneElement | The ingredient substance
code | Σ | 1..1 | CodeableConcept | The ingredient substance
strength | Σ | 0..* | see strength | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductIngredient xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier for the ingredient --></identifier>
 <role><!-- 1..1 CodeableConcept Ingredient role e.g. Active ingredient, excipient --></role>
 <allergenicIndicator value="[boolean]"/><!-- 0..1 If the ingredient is a known or suspected allergen -->
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Ingredient --></manufacturer>
 <specifiedSubstance>  <!-- 0..* A specified substance that comprises this ingredient -->
  <code><!-- 1..1 CodeableConcept The specified substance --></code>
  <group><!-- 1..1 CodeableConcept The group of specified substance, e.g. group 1 to 4 --></group>
  <confidentiality><!-- 0..1 CodeableConcept Confidentiality level of the specified substance as the ingredient
    --></confidentiality>
  <strength>  <!-- 0..* Quantity of the substance or specified substance present in the manufactured item or
    pharmaceutical product -->
   <presentation><!-- 1..1 Ratio The quantity of substance in the unit of presentation, or in the volume (or mass) of
    the single pharmaceutical product or manufactured item --></presentation>
   <presentationLowLimit><!-- 0..1 Ratio A lower limit for the quantity of substance in the unit of presentation. For
    use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the upper
    limit --></presentationLowLimit>
   <concentration><!-- 0..1 Ratio The strength per unitary volume (or mass) --></concentration>
   <concentrationLowLimit><!-- 0..1 Ratio A lower limit for the strength per unitary volume (or mass), for when there is
     a range. The concentration attribute then becomes the upper limit --></concentrationLowLimit>
   <measurementPoint value="[string]"/><!-- 0..1 For when strength is measured at a particular point or distance -->
   <country><!-- 0..* CodeableConcept The country or countries for which the strength range applies --></country>
   <referenceStrength>  <!-- 0..* Strength expressed in terms of a reference substance -->
    <substance><!-- 0..1 CodeableConcept Relevant reference substance --></substance>
    <strength><!-- 1..1 Ratio Strength expressed in terms of a reference substance --></strength>
    <strengthLowLimit><!-- 0..1 Ratio Strength expressed in terms of a reference substance --></strengthLowLimit>
    <measurementPoint value="[string]"/><!-- 0..1 For when strength is measured at a particular point or distance -->
    <country><!-- 0..* CodeableConcept The country or countries for which the strength range applies --></country>
   </referenceStrength>
  </strength>
 </specifiedSubstance>
 <substance>  <!-- 0..1 The ingredient substance -->
  <code><!-- 1..1 CodeableConcept The ingredient substance --></code>
  <strength><!-- 0..* Content as for MedicinalProductIngredient.specifiedSubstance.strength Quantity of the substance or
     specified substance present in the manufactured item or pharmaceutical product --></strength>
 </substance>
</MedicinalProductIngredient>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductIngredient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier for the ingredient
  "role" : { CodeableConcept }, // R!  Ingredient role e.g. Active ingredient, excipient
  "allergenicIndicator" : <boolean>, // If the ingredient is a known or suspected allergen
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of this Ingredient
  "specifiedSubstance" : [{ // A specified substance that comprises this ingredient
    "code" : { CodeableConcept }, // R!  The specified substance
    "group" : { CodeableConcept }, // R!  The group of specified substance, e.g. group 1 to 4
    "confidentiality" : { CodeableConcept }, // Confidentiality level of the specified substance as the ingredient
    "strength" : [{ // Quantity of the substance or specified substance present in the manufactured item or
    pharmaceutical product
      "presentation" : { Ratio }, // R!  The quantity of substance in the unit of presentation, or in the volume (or
    mass) of the single pharmaceutical product or manufactured item
      "presentationLowLimit" : { Ratio }, // A lower limit for the quantity of substance in the unit of presentation.
    For use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the
    upper limit
      "concentration" : { Ratio }, // The strength per unitary volume (or mass)
      "concentrationLowLimit" : { Ratio }, // A lower limit for the strength per unitary volume (or mass), for when
    there is a range. The concentration attribute then becomes the upper limit
      "measurementPoint" : "<string>", // For when strength is measured at a particular point or distance
      "country" : [{ CodeableConcept }], // The country or countries for which the strength range applies
      "referenceStrength" : [{ // Strength expressed in terms of a reference substance
        "substance" : { CodeableConcept }, // Relevant reference substance
        "strength" : { Ratio }, // R!  Strength expressed in terms of a reference substance
        "strengthLowLimit" : { Ratio }, // Strength expressed in terms of a reference substance
        "measurementPoint" : "<string>", // For when strength is measured at a particular point or distance
```

```
        "country" : [{ CodeableConcept }] // The country or countries for which the strength range applies
      }]
    }]
  }],
  "substance" : { // The ingredient substance
    "code" : { CodeableConcept }, // R!  The ingredient substance
    "strength" : [{ Content as for MedicinalProductIngredient.specifiedSubstance.strength }] // Quantity of the
    substance or specified substance present in the manufactured item or pharmaceutical product
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductIngredient;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductIngredient.identifier [ Identifier ]; # 0..1 Identifier for the ingredient
  fhir:MedicinalProductIngredient.role [ CodeableConcept ]; # 1..1 Ingredient role e.g. Active ingredient, excipient
  fhir:MedicinalProductIngredient.allergenicIndicator [ boolean ]; # 0..1 If the ingredient is a known or suspected
    allergen
  fhir:MedicinalProductIngredient.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of this Ingredient
  fhir:MedicinalProductIngredient.specifiedSubstance [ # 0..* A specified substance that comprises this ingredient
    fhir:MedicinalProductIngredient.specifiedSubstance.code [ CodeableConcept ]; # 1..1 The specified substance
    fhir:MedicinalProductIngredient.specifiedSubstance.group [ CodeableConcept ]; # 1..1 The group of specified
    substance, e.g. group 1 to 4
    fhir:MedicinalProductIngredient.specifiedSubstance.confidentiality [ CodeableConcept ]; # 0..1 Confidentiality level
     of the specified substance as the ingredient
    fhir:MedicinalProductIngredient.specifiedSubstance.strength [ # 0..* Quantity of the substance or specified
    substance present in the manufactured item or pharmaceutical product
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.presentation [ Ratio ]; # 1..1 The quantity of
    substance in the unit of presentation, or in the volume (or mass) of the single pharmaceutical product or
    manufactured item
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.presentationLowLimit [ Ratio ]; # 0..1 A lower limit
    for the quantity of substance in the unit of presentation. For use when there is a range of strengths, this is the
    lower limit, with the presentation attribute becoming the upper limit
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.concentration [ Ratio ]; # 0..1 The strength per
    unitary volume (or mass)
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.concentrationLowLimit [ Ratio ]; # 0..1 A lower limit
    for the strength per unitary volume (or mass), for when there is a range. The concentration attribute then becomes
    the upper limit
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.measurementPoint [ string ]; # 0..1 For when strength
    is measured at a particular point or distance
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.country [ CodeableConcept ], ... ; # 0..* The country
    or countries for which the strength range applies
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength [ # 0..* Strength expressed in terms
     of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.substance [ CodeableConcept ]; #
    0..1 Relevant reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strength [ Ratio ]; # 1..1
    Strength expressed in terms of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strengthLowLimit [ Ratio ]; # 0..1
     Strength expressed in terms of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.measurementPoint [ string ]; #
    0..1 For when strength is measured at a particular point or distance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.country [ CodeableConcept ], ... ;
     # 0..* The country or countries for which the strength range applies
      ], ...;
    ], ...;
  ], ...;
  fhir:MedicinalProductIngredient.substance [ # 0..1 The ingredient substance
    fhir:MedicinalProductIngredient.substance.code [ CodeableConcept ]; # 1..1 The ingredient substance
    fhir:MedicinalProductIngredient.substance.strength [ See MedicinalProductIngredient.specifiedSubstance.strength ],
    ... ; # 0..* Quantity of the substance or specified substance present in the manufactured item or pharmaceutical
    product
  ];
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductIngredient | ΣTU | | DomainResource | An ingredient of a manufactured item or pharmaceutical product Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Identifier for the ingredient
role | Σ | 1..1 | CodeableConcept | Ingredient role e.g. Active ingredient, excipient
allergenicIndicator | Σ | 0..1 | boolean | If the ingredient is a known or suspected allergen
manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of this Ingredient
specifiedSubstance | Σ | 0..* | BackboneElement | A specified substance that comprises this ingredient
code | Σ | 1..1 | CodeableConcept | The specified substance
group | Σ | 1..1 | CodeableConcept | The group of specified substance, e.g. group 1 to 4

confidentiality | Σ | 0..1 | CodeableConcept | Confidentiality level of the specified substance as the ingredient
strength | Σ | 0..* | BackboneElement | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product
presentation | Σ | 1..1 | Ratio | The quantity of substance in the unit of presentation, or in the volume (or mass) of the single pharmaceutical product or manufactured item
presentationLowLimit | ?!Σ | 0..1 | Ratio | A lower limit for the quantity of substance in the unit of presentation. For use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the upper limit
concentration | Σ | 0..1 | Ratio | The strength per unitary volume (or mass)
concentrationLowLimit | ?!Σ | 0..1 | Ratio | A lower limit for the strength per unitary volume (or mass), for when there is a range. The concentration attribute then becomes the upper limit
measurementPoint | Σ | 0..1 | string | For when strength is measured at a particular point or distance
country | Σ | 0..* | CodeableConcept | The country or countries for which the strength range applies
referenceStrength | Σ | 0..* | BackboneElement | Strength expressed in terms of a reference substance
substance | Σ | 0..1 | CodeableConcept | Relevant reference substance
strength | Σ | 1..1 | Ratio | Strength expressed in terms of a reference substance
strengthLowLimit | Σ | 0..1 | Ratio | Strength expressed in terms of a reference substance
measurementPoint | Σ | 0..1 | string | For when strength is measured at a particular point or distance
country | Σ | 0..* | CodeableConcept | The country or countries for which the strength range applies
substance | Σ | 0..1 | BackboneElement | The ingredient substance
code | Σ | 1..1 | CodeableConcept | The ingredient substance
strength | Σ | 0..* | see strength | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductIngredient xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier for the ingredient --></identifier>
 <role><!-- 1..1 CodeableConcept Ingredient role e.g. Active ingredient, excipient --></role>
 <allergenicIndicator value="[boolean]"/><!-- 0..1 If the ingredient is a known or suspected allergen -->
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Ingredient --></manufacturer>
 <specifiedSubstance>  <!-- 0..* A specified substance that comprises this ingredient -->
  <code><!-- 1..1 CodeableConcept The specified substance --></code>
  <group><!-- 1..1 CodeableConcept The group of specified substance, e.g. group 1 to 4 --></group>
  <confidentiality><!-- 0..1 CodeableConcept Confidentiality level of the specified substance as the ingredient
    --></confidentiality>
  <strength>  <!-- 0..* Quantity of the substance or specified substance present in the manufactured item or
    pharmaceutical product -->
   <presentation><!-- 1..1 Ratio The quantity of substance in the unit of presentation, or in the volume (or mass) of
    the single pharmaceutical product or manufactured item --></presentation>
   <presentationLowLimit><!-- 0..1 Ratio A lower limit for the quantity of substance in the unit of presentation. For
    use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the upper
    limit --></presentationLowLimit>
   <concentration><!-- 0..1 Ratio The strength per unitary volume (or mass) --></concentration>
   <concentrationLowLimit><!-- 0..1 Ratio A lower limit for the strength per unitary volume (or mass), for when there is
     a range. The concentration attribute then becomes the upper limit --></concentrationLowLimit>
   <measurementPoint value="[string]"/><!-- 0..1 For when strength is measured at a particular point or distance -->
   <country><!-- 0..* CodeableConcept The country or countries for which the strength range applies --></country>
   <referenceStrength>  <!-- 0..* Strength expressed in terms of a reference substance -->
    <substance><!-- 0..1 CodeableConcept Relevant reference substance --></substance>
    <strength><!-- 1..1 Ratio Strength expressed in terms of a reference substance --></strength>
    <strengthLowLimit><!-- 0..1 Ratio Strength expressed in terms of a reference substance --></strengthLowLimit>
    <measurementPoint value="[string]"/><!-- 0..1 For when strength is measured at a particular point or distance -->
    <country><!-- 0..* CodeableConcept The country or countries for which the strength range applies --></country>
   </referenceStrength>
  </strength>
 </specifiedSubstance>
 <substance>  <!-- 0..1 The ingredient substance -->
  <code><!-- 1..1 CodeableConcept The ingredient substance --></code>
  <strength><!-- 0..* Content as for MedicinalProductIngredient.specifiedSubstance.strength Quantity of the substance or
     specified substance present in the manufactured item or pharmaceutical product --></strength>
 </substance>
</MedicinalProductIngredient>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductIngredient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier for the ingredient
  "role" : { CodeableConcept }, // R!  Ingredient role e.g. Active ingredient, excipient
  "allergenicIndicator" : <boolean>, // If the ingredient is a known or suspected allergen
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of this Ingredient
  "specifiedSubstance" : [{ // A specified substance that comprises this ingredient
    "code" : { CodeableConcept }, // R!  The specified substance
    "group" : { CodeableConcept }, // R!  The group of specified substance, e.g. group 1 to 4
    "confidentiality" : { CodeableConcept }, // Confidentiality level of the specified substance as the ingredient
    "strength" : [{ // Quantity of the substance or specified substance present in the manufactured item or
    pharmaceutical product
      "presentation" : { Ratio }, // R!  The quantity of substance in the unit of presentation, or in the volume (or
    mass) of the single pharmaceutical product or manufactured item
```

```
      "presentationLowLimit" : { Ratio }, // A lower limit for the quantity of substance in the unit of presentation.
    For use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the
    upper limit
      "concentration" : { Ratio }, // The strength per unitary volume (or mass)
      "concentrationLowLimit" : { Ratio }, // A lower limit for the strength per unitary volume (or mass), for when
    there is a range. The concentration attribute then becomes the upper limit
      "measurementPoint" : "<string>", // For when strength is measured at a particular point or distance
      "country" : [{ CodeableConcept }], // The country or countries for which the strength range applies
      "referenceStrength" : [{ // Strength expressed in terms of a reference substance
        "substance" : { CodeableConcept }, // Relevant reference substance
        "strength" : { Ratio }, // R!  Strength expressed in terms of a reference substance
        "strengthLowLimit" : { Ratio }, // Strength expressed in terms of a reference substance
        "measurementPoint" : "<string>", // For when strength is measured at a particular point or distance
        "country" : [{ CodeableConcept }] // The country or countries for which the strength range applies
      }]
    }]
  }],
  "substance" : { // The ingredient substance
    "code" : { CodeableConcept }, // R!  The ingredient substance
    "strength" : [{ Content as for MedicinalProductIngredient.specifiedSubstance.strength }] // Quantity of the
    substance or specified substance present in the manufactured item or pharmaceutical product
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductIngredient;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductIngredient.identifier [ Identifier ]; # 0..1 Identifier for the ingredient
  fhir:MedicinalProductIngredient.role [ CodeableConcept ]; # 1..1 Ingredient role e.g. Active ingredient, excipient
  fhir:MedicinalProductIngredient.allergenicIndicator [ boolean ]; # 0..1 If the ingredient is a known or suspected
    allergen
  fhir:MedicinalProductIngredient.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of this Ingredient
  fhir:MedicinalProductIngredient.specifiedSubstance [ # 0..* A specified substance that comprises this ingredient
    fhir:MedicinalProductIngredient.specifiedSubstance.code [ CodeableConcept ]; # 1..1 The specified substance
    fhir:MedicinalProductIngredient.specifiedSubstance.group [ CodeableConcept ]; # 1..1 The group of specified
    substance, e.g. group 1 to 4
    fhir:MedicinalProductIngredient.specifiedSubstance.confidentiality [ CodeableConcept ]; # 0..1 Confidentiality level
     of the specified substance as the ingredient
    fhir:MedicinalProductIngredient.specifiedSubstance.strength [ # 0..* Quantity of the substance or specified
    substance present in the manufactured item or pharmaceutical product
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.presentation [ Ratio ]; # 1..1 The quantity of
    substance in the unit of presentation, or in the volume (or mass) of the single pharmaceutical product or
    manufactured item
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.presentationLowLimit [ Ratio ]; # 0..1 A lower limit
    for the quantity of substance in the unit of presentation. For use when there is a range of strengths, this is the
    lower limit, with the presentation attribute becoming the upper limit
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.concentration [ Ratio ]; # 0..1 The strength per
    unitary volume (or mass)
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.concentrationLowLimit [ Ratio ]; # 0..1 A lower limit
    for the strength per unitary volume (or mass), for when there is a range. The concentration attribute then becomes
    the upper limit
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.measurementPoint [ string ]; # 0..1 For when strength
    is measured at a particular point or distance
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.country [ CodeableConcept ], ... ; # 0..* The country
    or countries for which the strength range applies
      fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength [ # 0..* Strength expressed in terms
     of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.substance [ CodeableConcept ]; #
    0..1 Relevant reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strength [ Ratio ]; # 1..1
    Strength expressed in terms of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strengthLowLimit [ Ratio ]; # 0..1
     Strength expressed in terms of a reference substance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.measurementPoint [ string ]; #
    0..1 For when strength is measured at a particular point or distance
        fhir:MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.country [ CodeableConcept ], ... ;
     # 0..* The country or countries for which the strength range applies
      ], ...;
    ], ...;
  ], ...;
  fhir:MedicinalProductIngredient.substance [ # 0..1 The ingredient substance
    fhir:MedicinalProductIngredient.substance.code [ CodeableConcept ]; # 1..1 The ingredient substance
    fhir:MedicinalProductIngredient.substance.strength [ See MedicinalProductIngredient.specifiedSubstance.strength ],
    ... ; # 0..* Quantity of the substance or specified substance present in the manufactured item or pharmaceutical
    product
  ];
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.17.3 Resource MedicinalProductIngredient - Examples

URL: https://hl7.org/fhir/R4/medicinalproductingredient-examples.html

11.17.3 Resource MedicinalProductIngredient - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductingredient | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.17.4 Resource MedicinalProductIngredient - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductingredient-definitions.html

11.17.4 Resource MedicinalProductIngredient - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductIngredient resource.

MedicinalProductIngredient
Element Id | MedicinalProductIngredient
Definition | An ingredient of a manufactured item or pharmaceutical product.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductIngredient.identifier
Element Id | MedicinalProductIngredient.identifier
Definition | The identifier(s) of this Ingredient that are assigned by business processes and/or used to refer to it when a direct URL reference to the resource itself is not appropriate.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
MedicinalProductIngredient.role
Element Id | MedicinalProductIngredient.role
Definition | Ingredient role e.g. Active ingredient, excipient.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.allergenicIndicator
Element Id | MedicinalProductIngredient.allergenicIndicator
Definition | If the ingredient is a known or suspected allergen.
Cardinality | 0..1
Type | boolean
Summary | true
MedicinalProductIngredient.manufacturer
Element Id | MedicinalProductIngredient.manufacturer
Definition | Manufacturer of this Ingredient.
Cardinality | 0..*
Type | Reference(Organization)
Summary | true
MedicinalProductIngredient.specifiedSubstance
Element Id | MedicinalProductIngredient.specifiedSubstance
Definition | A specified substance that comprises this ingredient.
Cardinality | 0..*
Summary | true
MedicinalProductIngredient.specifiedSubstance.code
Element Id | MedicinalProductIngredient.specifiedSubstance.code
Definition | The specified substance.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.specifiedSubstance.group
Element Id | MedicinalProductIngredient.specifiedSubstance.group

Definition | The group of specified substance, e.g. group 1 to 4.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.specifiedSubstance.confidentiality
Element Id | MedicinalProductIngredient.specifiedSubstance.confidentiality
Definition | Confidentiality level of the specified substance as the ingredient.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength
Element Id | MedicinalProductIngredient.specifiedSubstance.strength
Definition | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product.
Cardinality | 0..*
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.presentation
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.presentation
Definition | The quantity of substance in the unit of presentation, or in the volume (or mass) of the single pharmaceutical product or manufactured item.
Cardinality | 1..1
Type | Ratio
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.presentationLowLimit
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.presentationLowLimit
Definition | A lower limit for the quantity of substance in the unit of presentation. For use when there is a range of strengths, this is the lower limit, with the presentation attribute becoming the upper limit.
Cardinality | 0..1
Type | Ratio
Is Modifier | true (Reason: If this is present it implies that the whole strength is expressed as a range, due to uncertainty or variation)
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.concentration
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.concentration
Definition | The strength per unitary volume (or mass).
Cardinality | 0..1
Type | Ratio
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.concentrationLowLimit
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.concentrationLowLimit
Definition | A lower limit for the strength per unitary volume (or mass), for when there is a range. The concentration attribute then becomes the upper limit.
Cardinality | 0..1
Type | Ratio
Is Modifier | true (Reason: If this is present it implies that the whole strength is expressed as a range, due to uncertainty or variation)
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.measurementPoint
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.measurementPoint
Definition | For when strength is measured at a particular point or distance.
Cardinality | 0..1
Type | string
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.country
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.country
Definition | The country or countries for which the strength range applies.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength
Definition | Strength expressed in terms of a reference substance.
Cardinality | 0..*
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.substance
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.substance
Definition | Relevant reference substance.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strength
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strength
Definition | Strength expressed in terms of a reference substance.
Cardinality | 1..1

Type | Ratio
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strengthLowLimit
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.strengthLowLimit
Definition | Strength expressed in terms of a reference substance.
Cardinality | 0..1
Type | Ratio
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.measurementPoint
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.measurementPoint
Definition | For when strength is measured at a particular point or distance.
Cardinality | 0..1
Type | string
Summary | true
MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.country
Element Id | MedicinalProductIngredient.specifiedSubstance.strength.referenceStrength.country
Definition | The country or countries for which the strength range applies.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.substance
Element Id | MedicinalProductIngredient.substance
Definition | The ingredient substance.
Cardinality | 0..1
Summary | true
MedicinalProductIngredient.substance.code
Element Id | MedicinalProductIngredient.substance.code
Definition | The ingredient substance.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductIngredient.substance.strength
Element Id | MedicinalProductIngredient.substance.strength
Definition | Quantity of the substance or specified substance present in the manufactured item or pharmaceutical product.
Cardinality | 0..*
Type | See MedicinalProductIngredient.specifiedSubstance.strength
Summary | true

11.17.5 Resource MedicinalProductIngredient - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductingredient-mappings.html

11.17.5 Resource MedicinalProductIngredient - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductingredient resource (see Mappings to Other Standards for further information & status).

11.17.6 Resource MedicinalProductIngredient - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductingredient-profiles.html

11.17.6 Resource MedicinalProductIngredient - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductIngredient resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.17.6.1 Profiles

No Profiles defined for this resource

11.17.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.17.6.3 Search Extensions

No Search Extensions defined for this resource

11.17.7 JSON Schema for MedicinalProductIngredient

URL: https://hl7.org/fhir/R4/medicinalproductingredient.schema.json.html

11.17.7 JSON Schema for MedicinalProductIngredient

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductIngredient, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductIngredient",
  "$ref": "#/definitions/MedicinalProductIngredient",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductIngredient": {
      "description": "An ingredient of a manufactured item or pharmaceutical product.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductIngredient resource",
          "const": "MedicinalProductIngredient"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "The identifier(s) of this Ingredient that are assigned by business processes and/or used to
refer to it when a direct URL reference to the resource itself is not appropriate.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "role": {
      "description": "Ingredient role e.g. Active ingredient, excipient.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "allergenicIndicator": {
      "description": "If the ingredient is a known or suspected allergen.",
      "$ref": "#/definitions/boolean"
    },
    "_allergenicIndicator": {
      "description": "Extensions for allergenicIndicator",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "manufacturer": {
      "description": "Manufacturer of this Ingredient.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "specifiedSubstance": {
      "description": "A specified substance that comprises this ingredient.",
      "items": {
        "$ref": "#/definitions/MedicinalProductIngredient_SpecifiedSubstance"
      },
      "type": "array"
    },
    "substance": {
      "description": "The ingredient substance.",
      "$ref": "#/definitions/MedicinalProductIngredient_Substance"
    }
  },
  "additionalProperties": false,
  "required": [
    "role",
    "resourceType"
  ]
},
"MedicinalProductIngredient_SpecifiedSubstance": {
  "description": "An ingredient of a manufactured item or pharmaceutical product.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
```

```
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The specified substance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "group": {
      "description": "The group of specified substance, e.g. group 1 to 4.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "confidentiality": {
      "description": "Confidentiality level of the specified substance as the ingredient.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "strength": {
      "description": "Quantity of the substance or specified substance present in the manufactured item or
pharmaceutical product.",
      "items": {
        "$ref": "#/definitions/MedicinalProductIngredient_Strength"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code",
    "group"
  ]
},
"MedicinalProductIngredient_Strength": {
  "description": "An ingredient of a manufactured item or pharmaceutical product.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "presentation": {
      "description": "The quantity of substance in the unit of presentation, or in the volume (or mass) of the
single pharmaceutical product or manufactured item.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "presentationLowLimit": {
      "description": "A lower limit for the quantity of substance in the unit of presentation. For use when there is
 a range of strengths, this is the lower limit, with the presentation attribute becoming the upper limit.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "concentration": {
      "description": "The strength per unitary volume (or mass).",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "concentrationLowLimit": {
      "description": "A lower limit for the strength per unitary volume (or mass), for when there is a range. The
concentration attribute then becomes the upper limit.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "measurementPoint": {
      "description": "For when strength is measured at a particular point or distance.",
      "$ref": "#/definitions/string"
    },
```

```
    "_measurementPoint": {
      "description": "Extensions for measurementPoint",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "country": {
      "description": "The country or countries for which the strength range applies.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "referenceStrength": {
      "description": "Strength expressed in terms of a reference substance.",
      "items": {
        "$ref": "#/definitions/MedicinalProductIngredient_ReferenceStrength"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "presentation"
  ]
},
"MedicinalProductIngredient_ReferenceStrength": {
  "description": "An ingredient of a manufactured item or pharmaceutical product.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "substance": {
      "description": "Relevant reference substance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "strength": {
      "description": "Strength expressed in terms of a reference substance.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "strengthLowLimit": {
      "description": "Strength expressed in terms of a reference substance.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "measurementPoint": {
      "description": "For when strength is measured at a particular point or distance.",
      "$ref": "#/definitions/string"
    },
    "_measurementPoint": {
      "description": "Extensions for measurementPoint",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "country": {
      "description": "The country or countries for which the strength range applies.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "strength"
  ]
},
```

```
    "MedicinalProductIngredient_Substance": {
      "description": "An ingredient of a manufactured item or pharmaceutical product.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "code": {
          "description": "The ingredient substance.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "strength": {
          "description": "Quantity of the substance or specified substance present in the manufactured item or
    pharmaceutical product.",
          "items": {
            "$ref": "#/definitions/MedicinalProductIngredient_Strength"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "code"
      ]
    }
  }
}
```


11.17.8 Resource MedicinalProductIngredient - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductingredient-dependencies.html

11.17.8 Resource MedicinalProductIngredient - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductIngredient FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductIngredient | | DomainResource
identifier | 0..1 | Identifier
role | 1..1 | CodeableConcept
allergenicIndicator | 0..1 | boolean
manufacturer | 0..* | Reference(Organization)
specifiedSubstance | 0..* | BackboneElement
code | 1..1 | CodeableConcept
group | 1..1 | CodeableConcept
confidentiality | 0..1 | CodeableConcept
strength | 0..* | BackboneElement
presentation | 1..1 | Ratio
presentationLowLimit | 0..1 | Ratio
concentration | 0..1 | Ratio
concentrationLowLimit | 0..1 | Ratio
measurementPoint | 0..1 | string
country | 0..* | CodeableConcept
referenceStrength | 0..* | BackboneElement

substance | 0..1 | CodeableConcept
strength | 1..1 | Ratio
strengthLowLimit | 0..1 | Ratio
measurementPoint | 0..1 | string
country | 0..* | CodeableConcept
substance | 0..1 | BackboneElement
code | 1..1 | CodeableConcept
strength | 0..* | see strength

11.18 Resource MedicinalProductManufactured - Content

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The manufactured item as contained in the packaged medicinal product.

11.18.1 Scope and Usage

Any pharmaceutical product or combination of pharmaceutical products that may be administered to human beings (or animals) for treating or preventing disease, with the aim/purpose of making a medical diagnosis or to restore, correct or modify physiological functions.

A Medicinal Product may contain in the packaging one or more manufactured items and one or more pharmaceutical products. In certain regions a Medicinal Product may also be defined as any substance or combination of substances which may be used to make a medical diagnosis.

This applies to proprietary medicinal products for human use intended to be placed on the market and to industrially manufactured medicinal products, the marketing of which has been authorised by a Medicines Regulatory Agency. However, the provisions do not apply to: i) medicinal products prepared according to prescription (e.g. prepared in a pharmacy from a prescription intended for a specific patient), ii) medicinal products prepared in accordance with an official formula (e.g. prepared in a pharmacy in accordance with the instructions in a pharmacopoeia and intended to be given direct to the patient by the pharmacy), iii) medicinal products intended for research and development trials (see 11.2 Investigational Medicinal Product) and to iv) intermediate products intended for subsequent processing by an authorised manufacturer. [from ENV 13607 and ENV 12610]

This resource is referenced by MedicinalProductPackaged

11.18.2 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductManufactured | ΣTU | | DomainResource | The manufactured item as contained in the packaged medicinal product
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
manufacturedDoseForm | Σ | 1..1 | CodeableConcept | Dose form as manufactured and before any transformation into the pharmaceutical product
unitOfPresentation | Σ | 0..1 | CodeableConcept | The “real world” units in which the quantity of the manufactured item is described
quantity | Σ | 1..1 | Quantity | The quantity or "count number" of the manufactured item
manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of the item (Note that this should be named "manufacturer" but it currently causes technical issues)
ingredient | Σ | 0..* | Reference(MedicinalProductIngredient) | Ingredient
physicalCharacteristics | Σ | 0..1 | ProdCharacteristic | Dimensions, color etc.
otherCharacteristics | Σ | 0..* | CodeableConcept | Other codeable characteristics
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductManufactured xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <manufacturedDoseForm><!-- 1..1 CodeableConcept Dose form as manufactured and before any transformation into the
    pharmaceutical product --></manufacturedDoseForm>
 <unitOfPresentation><!-- 0..1 CodeableConcept The â€œreal worldâ€ units in which the quantity of the manufactured item
     is described --></unitOfPresentation>
```

```
 <quantity><!-- 1..1 Quantity The quantity or "count number" of the manufactured item --></quantity>
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of the item (Note that this should be named "manufacturer"
     but it currently causes technical issues) --></manufacturer>
 <ingredient><!-- 0..* Reference(MedicinalProductIngredient) Ingredient --></ingredient>
 <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
 <otherCharacteristics><!-- 0..* CodeableConcept Other codeable characteristics --></otherCharacteristics>
</MedicinalProductManufactured>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductManufactured",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "manufacturedDoseForm" : { CodeableConcept }, // R!  Dose form as manufactured and before any transformation into the
    pharmaceutical product
  "unitOfPresentation" : { CodeableConcept }, // The â€œreal worldâ€ units in which the quantity of the manufactured
    item is described
  "quantity" : { Quantity }, // R!  The quantity or "count number" of the manufactured item
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of the item (Note that this should be named
    "manufacturer" but it currently causes technical issues)
  "ingredient" : [{ Reference(MedicinalProductIngredient) }], // Ingredient
  "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
  "otherCharacteristics" : [{ CodeableConcept }] // Other codeable characteristics
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductManufactured;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductManufactured.manufacturedDoseForm [ CodeableConcept ]; # 1..1 Dose form as manufactured and
    before any transformation into the pharmaceutical product
  fhir:MedicinalProductManufactured.unitOfPresentation [ CodeableConcept ]; # 0..1 The â€œreal worldâ€ units in which
    the quantity of the manufactured item is described
  fhir:MedicinalProductManufactured.quantity [ Quantity ]; # 1..1 The quantity or "count number" of the manufactured
    item
  fhir:MedicinalProductManufactured.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of the item
    (Note that this should be named "manufacturer" but it currently causes technical issues)
  fhir:MedicinalProductManufactured.ingredient [ Reference(MedicinalProductIngredient) ], ... ; # 0..* Ingredient
  fhir:MedicinalProductManufactured.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color etc.
  fhir:MedicinalProductManufactured.otherCharacteristics [ CodeableConcept ], ... ; # 0..* Other codeable
    characteristics
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProductManufactured | ΣTU | | DomainResource | The manufactured item as contained in the packaged medicinal product
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

manufacturedDoseForm | Σ | 1..1 | CodeableConcept | Dose form as manufactured and before any transformation into the pharmaceutical product

unitOfPresentation | Σ | 0..1 | CodeableConcept | The “real world” units in which the quantity of the manufactured item is described

quantity | Σ | 1..1 | Quantity | The quantity or "count number" of the manufactured item

manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of the item (Note that this should be named "manufacturer" but it currently causes technical issues)

ingredient | Σ | 0..* | Reference(MedicinalProductIngredient) | Ingredient

physicalCharacteristics | Σ | 0..1 | ProdCharacteristic | Dimensions, color etc.

otherCharacteristics | Σ | 0..* | CodeableConcept | Other codeable characteristics

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductManufactured xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <manufacturedDoseForm><!-- 1..1 CodeableConcept Dose form as manufactured and before any transformation into the
    pharmaceutical product --></manufacturedDoseForm>
 <unitOfPresentation><!-- 0..1 CodeableConcept The â€œreal worldâ€ units in which the quantity of the manufactured item
     is described --></unitOfPresentation>
 <quantity><!-- 1..1 Quantity The quantity or "count number" of the manufactured item --></quantity>
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of the item (Note that this should be named "manufacturer"
     but it currently causes technical issues) --></manufacturer>
 <ingredient><!-- 0..* Reference(MedicinalProductIngredient) Ingredient --></ingredient>
 <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
 <otherCharacteristics><!-- 0..* CodeableConcept Other codeable characteristics --></otherCharacteristics>
</MedicinalProductManufactured>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductManufactured",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "manufacturedDoseForm" : { CodeableConcept }, // R!  Dose form as manufactured and before any transformation into the
    pharmaceutical product
  "unitOfPresentation" : { CodeableConcept }, // The â€œreal worldâ€ units in which the quantity of the manufactured
    item is described
  "quantity" : { Quantity }, // R!  The quantity or "count number" of the manufactured item
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of the item (Note that this should be named
    "manufacturer" but it currently causes technical issues)
  "ingredient" : [{ Reference(MedicinalProductIngredient) }], // Ingredient
  "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
  "otherCharacteristics" : [{ CodeableConcept }] // Other codeable characteristics
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductManufactured;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductManufactured.manufacturedDoseForm [ CodeableConcept ]; # 1..1 Dose form as manufactured and
    before any transformation into the pharmaceutical product
  fhir:MedicinalProductManufactured.unitOfPresentation [ CodeableConcept ]; # 0..1 The â€œreal worldâ€ units in which
    the quantity of the manufactured item is described
  fhir:MedicinalProductManufactured.quantity [ Quantity ]; # 1..1 The quantity or "count number" of the manufactured
    item
  fhir:MedicinalProductManufactured.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of the item
    (Note that this should be named "manufacturer" but it currently causes technical issues)
  fhir:MedicinalProductManufactured.ingredient [ Reference(MedicinalProductIngredient) ], ... ; # 0..* Ingredient
  fhir:MedicinalProductManufactured.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color etc.
  fhir:MedicinalProductManufactured.otherCharacteristics [ CodeableConcept ], ... ; # 0..* Other codeable
    characteristics
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.18.3 Resource MedicinalProductManufactured - Examples

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured-examples.html

11.18.3 Resource MedicinalProductManufactured - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductmanufactured | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.18.4 Resource MedicinalProductManufactured - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured-definitions.html

11.18.4 Resource MedicinalProductManufactured - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductManufactured resource.

MedicinalProductManufactured
Element Id | MedicinalProductManufactured
Definition | The manufactured item as contained in the packaged medicinal product.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductManufactured.manufacturedDoseForm
Element Id | MedicinalProductManufactured.manufacturedDoseForm

Definition | Dose form as manufactured and before any transformation into the pharmaceutical product.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductManufactured.unitOfPresentation
Element Id | MedicinalProductManufactured.unitOfPresentation
Definition | The “real world” units in which the quantity of the manufactured item is described.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductManufactured.quantity
Element Id | MedicinalProductManufactured.quantity
Definition | The quantity or "count number" of the manufactured item.
Cardinality | 1..1
Type | Quantity
Summary | true
MedicinalProductManufactured.manufacturer
Element Id | MedicinalProductManufactured.manufacturer
Definition | Manufacturer of the item (Note that this should be named "manufacturer" but it currently causes technical issues).
Cardinality | 0..*
Type | Reference(Organization)
Summary | true
MedicinalProductManufactured.ingredient
Element Id | MedicinalProductManufactured.ingredient
Definition | Ingredient.
Cardinality | 0..*
Type | Reference(MedicinalProductIngredient)
Summary | true
MedicinalProductManufactured.physicalCharacteristics
Element Id | MedicinalProductManufactured.physicalCharacteristics
Definition | Dimensions, color etc.
Cardinality | 0..1
Type | ProdCharacteristic
Summary | true
MedicinalProductManufactured.otherCharacteristics
Element Id | MedicinalProductManufactured.otherCharacteristics
Definition | Other codeable characteristics.
Cardinality | 0..*
Type | CodeableConcept
Summary | true

11.18.5 Resource MedicinalProductManufactured - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured-mappings.html

11.18.5 Resource MedicinalProductManufactured - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductmanufactured resource (see Mappings to Other Standards for further information & status).

11.18.6 Resource MedicinalProductManufactured - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured-profiles.html

11.18.6 Resource MedicinalProductManufactured - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductManufactured resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.18.6.1 Profiles

No Profiles defined for this resource

11.18.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements

resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.18.6.3 Search Extensions

No Search Extensions defined for this resource

11.18.7 JSON Schema for MedicinalProductManufactured

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured.schema.json.html

11.18.7 JSON Schema for MedicinalProductManufactured

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductManufactured, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductManufactured",
  "$ref": "#/definitions/MedicinalProductManufactured",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductManufactured": {
      "description": "The manufactured item as contained in the packaged medicinal product.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductManufactured resource",
          "const": "MedicinalProductManufactured"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
```

```
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "manufacturedDoseForm": {
          "description": "Dose form as manufactured and before any transformation into the pharmaceutical product.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "unitOfPresentation": {
          "description": "The “real world” units in which the quantity of the manufactured item is described.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "quantity": {
          "description": "The quantity or \"count number\" of the manufactured item.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "manufacturer": {
          "description": "Manufacturer of the item (Note that this should be named \"manufacturer\" but it currently
    causes technical issues).",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "ingredient": {
          "description": "Ingredient.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "physicalCharacteristics": {
          "description": "Dimensions, color etc.",
          "$ref": "ProdCharacteristic.schema.json#/definitions/ProdCharacteristic"
        },
        "otherCharacteristics": {
          "description": "Other codeable characteristics.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "quantity",
        "manufacturedDoseForm",
        "resourceType"
      ]
    }
  }
}
```


11.18.8 Resource MedicinalProductManufactured - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductmanufactured-dependencies.html

11.18.8 Resource MedicinalProductManufactured - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductManufactured FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductManufactured | | DomainResource
manufacturedDoseForm | 1..1 | CodeableConcept
unitOfPresentation | 0..1 | CodeableConcept
quantity | 1..1 | Quantity
manufacturer | 0..* | Reference(Organization)
ingredient | 0..* | Reference(MedicinalProductIngredient)
physicalCharacteristics | 0..1 | ProdCharacteristic | Type Error: (ProdCharacteristic = FMM5-Draft vs. Element = FMM0-Trial Use)
otherCharacteristics | 0..* | CodeableConcept

11.19 Resource MedicinalProductPackaged - Content

URL: https://hl7.org/fhir/R4/medicinalproductpackaged.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A medicinal product in a container or package.

11.19.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

Medicinal Product in a container being part of a package, representing the entirety that has been packaged for sale or supply

This resource is referenced by MedicinalProduct and MedicinalProductAuthorization

11.19.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductPackaged | ΣTU | | DomainResource | A medicinal product in a container or package Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier
subject | Σ | 0..* | Reference(MedicinalProduct) | The product with this is a pack for
description | Σ | 0..1 | string | Textual description
legalStatusOfSupply | Σ | 0..1 | CodeableConcept | The legal status of supply of the medicinal product as classified by the regulator
marketingStatus | Σ | 0..* | MarketingStatus | Marketing information
marketingAuthorization | Σ | 0..1 | Reference(MedicinalProductAuthorization) | Manufacturer of this Package Item
manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of this Package Item
batchIdentifier | Σ | 0..* | BackboneElement | Batch numbering
outerPackaging | Σ | 1..1 | Identifier | A number appearing on the outer packaging of a specific batch
immediatePackaging | Σ | 0..1 | Identifier | A number appearing on the immediate packaging (and not the outer packaging)
packageItem | Σ | 1..* | BackboneElement | A packaging item, as a contained for medicine, possibly with other packaging items within
identifier | Σ | 0..* | Identifier | Including possibly Data Carrier Identifier
type | Σ | 1..1 | CodeableConcept | The physical type of the container of the medicine
quantity | Σ | 1..1 | Quantity | The quantity of this package in the medicinal product, at the current level of packaging. The outermost is always 1
material | Σ | 0..* | CodeableConcept | Material type of the package item
alternateMaterial | Σ | 0..* | CodeableConcept | A possible alternate material for the packaging
device | Σ | 0..* | Reference(DeviceDefinition) | A device accompanying a medicinal product
manufacturedItem | Σ | 0..* | Reference(MedicinalProductManufactured) | The manufactured item as contained in the packaged medicinal product
packageItem | Σ | 0..* | see packageItem | Allows containers within containers
physicalCharacteristics | Σ | 0..1 | ProdCharacteristic | Dimensions, color etc.
otherCharacteristics | Σ | 0..* | CodeableConcept | Other codeable characteristics
shelfLifeStorage | Σ | 0..* | ProductShelfLife | Shelf Life and storage information
manufacturer | Σ | 0..* | Reference(Organization) | Manufacturer of this Package Item
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductPackaged xmlns="http://hl7.org/fhir">
```

```
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <subject><!-- 0..* Reference(MedicinalProduct) The product with this is a pack for --></subject>
 <description value="[string]"/><!-- 0..1 Textual description -->
 <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply of the medicinal product as classified by the
     regulator --></legalStatusOfSupply>
 <marketingStatus><!-- 0..* MarketingStatus Marketing information --></marketingStatus>
 <marketingAuthorization><!-- 0..1 Reference(MedicinalProductAuthorization) Manufacturer of this Package Item
    --></marketingAuthorization>
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Package Item --></manufacturer>
 <batchIdentifier>  <!-- 0..* Batch numbering -->
  <outerPackaging><!-- 1..1 Identifier A number appearing on the outer packaging of a specific batch
    --></outerPackaging>
  <immediatePackaging><!-- 0..1 Identifier A number appearing on the immediate packaging (and not the outer packaging)
    --></immediatePackaging>
 </batchIdentifier>
 <packageItem>  <!-- 1..* A packaging item, as a contained for medicine, possibly with other packaging items within -->
  <identifier><!-- 0..* Identifier Including possibly Data Carrier Identifier --></identifier>
  <type><!-- 1..1 CodeableConcept The physical type of the container of the medicine --></type>
  <quantity><!-- 1..1 Quantity The quantity of this package in the medicinal product, at the current level of packaging.
     The outermost is always 1 --></quantity>
  <material><!-- 0..* CodeableConcept Material type of the package item --></material>
  <alternateMaterial><!-- 0..* CodeableConcept A possible alternate material for the packaging --></alternateMaterial>
  <device><!-- 0..* Reference(DeviceDefinition) A device accompanying a medicinal product --></device>
  <manufacturedItem><!-- 0..* Reference(MedicinalProductManufactured) The manufactured item as contained in the packaged
     medicinal product --></manufacturedItem>
  <packageItem><!-- 0..* Content as for MedicinalProductPackaged.packageItem Allows containers within containers
    --></packageItem>
  <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
  <otherCharacteristics><!-- 0..* CodeableConcept Other codeable characteristics --></otherCharacteristics>
  <shelfLifeStorage><!-- 0..* ProductShelfLife Shelf Life and storage information --></shelfLifeStorage>
  <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Package Item --></manufacturer>
 </packageItem>
</MedicinalProductPackaged>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductPackaged",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "subject" : [{ Reference(MedicinalProduct) }], // The product with this is a pack for
  "description" : "<string>", // Textual description
  "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply of the medicinal product as classified by
    the regulator
  "marketingStatus" : [{ MarketingStatus }], // Marketing information
  "marketingAuthorization" : { Reference(MedicinalProductAuthorization) }, // Manufacturer of this Package Item
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of this Package Item
  "batchIdentifier" : [{ // Batch numbering
    "outerPackaging" : { Identifier }, // R!  A number appearing on the outer packaging of a specific batch
    "immediatePackaging" : { Identifier } // A number appearing on the immediate packaging (and not the outer packaging)
  }],
  "packageItem" : [{ // R!  A packaging item, as a contained for medicine, possibly with other packaging items within
    "identifier" : [{ Identifier }], // Including possibly Data Carrier Identifier
    "type" : { CodeableConcept }, // R!  The physical type of the container of the medicine
    "quantity" : { Quantity }, // R!  The quantity of this package in the medicinal product, at the current level of
    packaging. The outermost is always 1
    "material" : [{ CodeableConcept }], // Material type of the package item
    "alternateMaterial" : [{ CodeableConcept }], // A possible alternate material for the packaging
    "device" : [{ Reference(DeviceDefinition) }], // A device accompanying a medicinal product
    "manufacturedItem" : [{ Reference(MedicinalProductManufactured) }], // The manufactured item as contained in the
    packaged medicinal product
    "packageItem" : [{ Content as for MedicinalProductPackaged.packageItem }], // Allows containers within containers
    "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
    "otherCharacteristics" : [{ CodeableConcept }], // Other codeable characteristics
    "shelfLifeStorage" : [{ ProductShelfLife }], // Shelf Life and storage information
    "manufacturer" : [{ Reference(Organization) }] // Manufacturer of this Package Item
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductPackaged;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductPackaged.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:MedicinalProductPackaged.subject [ Reference(MedicinalProduct) ], ... ; # 0..* The product with this is a pack
    for
  fhir:MedicinalProductPackaged.description [ string ]; # 0..1 Textual description
  fhir:MedicinalProductPackaged.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The legal status of supply of the
    medicinal product as classified by the regulator
  fhir:MedicinalProductPackaged.marketingStatus [ MarketingStatus ], ... ; # 0..* Marketing information
  fhir:MedicinalProductPackaged.marketingAuthorization [ Reference(MedicinalProductAuthorization) ]; # 0..1 Manufacturer
     of this Package Item
```

```
  fhir:MedicinalProductPackaged.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of this Package Item
  fhir:MedicinalProductPackaged.batchIdentifier [ # 0..* Batch numbering
    fhir:MedicinalProductPackaged.batchIdentifier.outerPackaging [ Identifier ]; # 1..1 A number appearing on the outer
    packaging of a specific batch
    fhir:MedicinalProductPackaged.batchIdentifier.immediatePackaging [ Identifier ]; # 0..1 A number appearing on the
    immediate packaging (and not the outer packaging)
  ], ...;
  fhir:MedicinalProductPackaged.packageItem [ # 1..* A packaging item, as a contained for medicine, possibly with other
    packaging items within
    fhir:MedicinalProductPackaged.packageItem.identifier [ Identifier ], ... ; # 0..* Including possibly Data Carrier
    Identifier
    fhir:MedicinalProductPackaged.packageItem.type [ CodeableConcept ]; # 1..1 The physical type of the container of the
     medicine
    fhir:MedicinalProductPackaged.packageItem.quantity [ Quantity ]; # 1..1 The quantity of this package in the
    medicinal product, at the current level of packaging. The outermost is always 1
    fhir:MedicinalProductPackaged.packageItem.material [ CodeableConcept ], ... ; # 0..* Material type of the package
    item
    fhir:MedicinalProductPackaged.packageItem.alternateMaterial [ CodeableConcept ], ... ; # 0..* A possible alternate
    material for the packaging
    fhir:MedicinalProductPackaged.packageItem.device [ Reference(DeviceDefinition) ], ... ; # 0..* A device accompanying
     a medicinal product
    fhir:MedicinalProductPackaged.packageItem.manufacturedItem [ Reference(MedicinalProductManufactured) ], ... ; # 0..*
     The manufactured item as contained in the packaged medicinal product
    fhir:MedicinalProductPackaged.packageItem.packageItem [ See MedicinalProductPackaged.packageItem ], ... ; # 0..*
    Allows containers within containers
    fhir:MedicinalProductPackaged.packageItem.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color
    etc.
    fhir:MedicinalProductPackaged.packageItem.otherCharacteristics [ CodeableConcept ], ... ; # 0..* Other codeable
    characteristics
    fhir:MedicinalProductPackaged.packageItem.shelfLifeStorage [ ProductShelfLife ], ... ; # 0..* Shelf Life and storage
     information
    fhir:MedicinalProductPackaged.packageItem.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of
    this Package Item
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
MedicinalProductPackaged	ΣTU		DomainResource	A medicinal product in a container or package Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Unique identifier
subject	Σ	0..*	Reference(MedicinalProduct)	The product with this is a pack for
description	Σ	0..1	string	Textual description
legalStatusOfSupply	Σ	0..1	CodeableConcept	The legal status of supply of the medicinal product as classified by the regulator
marketingStatus	Σ	0..*	MarketingStatus	Marketing information
marketingAuthorization	Σ	0..1	Reference(MedicinalProductAuthorization)	Manufacturer of this Package Item
manufacturer	Σ	0..*	Reference(Organization)	Manufacturer of this Package Item
batchIdentifier	Σ	0..*	BackboneElement	Batch numbering
outerPackaging	Σ	1..1	Identifier	A number appearing on the outer packaging of a specific batch
immediatePackaging	Σ	0..1	Identifier	A number appearing on the immediate packaging (and not the outer packaging)
packageItem	Σ	1..*	BackboneElement	A packaging item, as a contained for medicine, possibly with other packaging items within
identifier	Σ	0..*	Identifier	Including possibly Data Carrier Identifier
type	Σ	1..1	CodeableConcept	The physical type of the container of the medicine
quantity	Σ	1..1	Quantity	The quantity of this package in the medicinal product, at the current level of packaging. The outermost is always 1
material	Σ	0..*	CodeableConcept	Material type of the package item
alternateMaterial	Σ	0..*	CodeableConcept	A possible alternate material for the packaging
device	Σ	0..*	Reference(DeviceDefinition)	A device accompanying a medicinal product
manufacturedItem	Σ	0..*	Reference(MedicinalProductManufactured)	The manufactured item as contained in the packaged medicinal product
packageItem	Σ	0..*	see packageItem	Allows containers within containers
physicalCharacteristics	Σ	0..1	ProdCharacteristic	Dimensions, color etc.
otherCharacteristics	Σ	0..*	CodeableConcept	Other codeable characteristics
shelfLifeStorage	Σ	0..*	ProductShelfLife	Shelf Life and storage information
manufacturer	Σ	0..*	Reference(Organization)	Manufacturer of this Package Item
Documentation for this format				

UML Diagram (Legend)

XML Template

```
<MedicinalProductPackaged xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <subject><!-- 0..* Reference(MedicinalProduct) The product with this is a pack for --></subject>
 <description value="[string]"/><!-- 0..1 Textual description -->
```

```
 <legalStatusOfSupply><!-- 0..1 CodeableConcept The legal status of supply of the medicinal product as classified by the
     regulator --></legalStatusOfSupply>
 <marketingStatus><!-- 0..* MarketingStatus Marketing information --></marketingStatus>
 <marketingAuthorization><!-- 0..1 Reference(MedicinalProductAuthorization) Manufacturer of this Package Item
    --></marketingAuthorization>
 <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Package Item --></manufacturer>
 <batchIdentifier>  <!-- 0..* Batch numbering -->
  <outerPackaging><!-- 1..1 Identifier A number appearing on the outer packaging of a specific batch
    --></outerPackaging>
  <immediatePackaging><!-- 0..1 Identifier A number appearing on the immediate packaging (and not the outer packaging)
    --></immediatePackaging>
 </batchIdentifier>
 <packageItem>  <!-- 1..* A packaging item, as a contained for medicine, possibly with other packaging items within -->
  <identifier><!-- 0..* Identifier Including possibly Data Carrier Identifier --></identifier>
  <type><!-- 1..1 CodeableConcept The physical type of the container of the medicine --></type>
  <quantity><!-- 1..1 Quantity The quantity of this package in the medicinal product, at the current level of packaging.
     The outermost is always 1 --></quantity>
  <material><!-- 0..* CodeableConcept Material type of the package item --></material>
  <alternateMaterial><!-- 0..* CodeableConcept A possible alternate material for the packaging --></alternateMaterial>
  <device><!-- 0..* Reference(DeviceDefinition) A device accompanying a medicinal product --></device>
  <manufacturedItem><!-- 0..* Reference(MedicinalProductManufactured) The manufactured item as contained in the packaged
     medicinal product --></manufacturedItem>
  <packageItem><!-- 0..* Content as for MedicinalProductPackaged.packageItem Allows containers within containers
    --></packageItem>
  <physicalCharacteristics><!-- 0..1 ProdCharacteristic Dimensions, color etc. --></physicalCharacteristics>
  <otherCharacteristics><!-- 0..* CodeableConcept Other codeable characteristics --></otherCharacteristics>
  <shelfLifeStorage><!-- 0..* ProductShelfLife Shelf Life and storage information --></shelfLifeStorage>
  <manufacturer><!-- 0..* Reference(Organization) Manufacturer of this Package Item --></manufacturer>
 </packageItem>
</MedicinalProductPackaged>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductPackaged",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "subject" : [{ Reference(MedicinalProduct) }], // The product with this is a pack for
  "description" : "<string>", // Textual description
  "legalStatusOfSupply" : { CodeableConcept }, // The legal status of supply of the medicinal product as classified by
    the regulator
  "marketingStatus" : [{ MarketingStatus }], // Marketing information
  "marketingAuthorization" : { Reference(MedicinalProductAuthorization) }, // Manufacturer of this Package Item
  "manufacturer" : [{ Reference(Organization) }], // Manufacturer of this Package Item
  "batchIdentifier" : [{ // Batch numbering
    "outerPackaging" : { Identifier }, // R!  A number appearing on the outer packaging of a specific batch
    "immediatePackaging" : { Identifier } // A number appearing on the immediate packaging (and not the outer packaging)
  }],
  "packageItem" : [{ // R!  A packaging item, as a contained for medicine, possibly with other packaging items within
    "identifier" : [{ Identifier }], // Including possibly Data Carrier Identifier
    "type" : { CodeableConcept }, // R!  The physical type of the container of the medicine
    "quantity" : { Quantity }, // R!  The quantity of this package in the medicinal product, at the current level of
    packaging. The outermost is always 1
    "material" : [{ CodeableConcept }], // Material type of the package item
    "alternateMaterial" : [{ CodeableConcept }], // A possible alternate material for the packaging
    "device" : [{ Reference(DeviceDefinition) }], // A device accompanying a medicinal product
    "manufacturedItem" : [{ Reference(MedicinalProductManufactured) }], // The manufactured item as contained in the
    packaged medicinal product
    "packageItem" : [{ Content as for MedicinalProductPackaged.packageItem }], // Allows containers within containers
    "physicalCharacteristics" : { ProdCharacteristic }, // Dimensions, color etc.
    "otherCharacteristics" : [{ CodeableConcept }], // Other codeable characteristics
    "shelfLifeStorage" : [{ ProductShelfLife }], // Shelf Life and storage information
    "manufacturer" : [{ Reference(Organization) }] // Manufacturer of this Package Item
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductPackaged;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductPackaged.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:MedicinalProductPackaged.subject [ Reference(MedicinalProduct) ], ... ; # 0..* The product with this is a pack
    for
  fhir:MedicinalProductPackaged.description [ string ]; # 0..1 Textual description
  fhir:MedicinalProductPackaged.legalStatusOfSupply [ CodeableConcept ]; # 0..1 The legal status of supply of the
    medicinal product as classified by the regulator
  fhir:MedicinalProductPackaged.marketingStatus [ MarketingStatus ], ... ; # 0..* Marketing information
  fhir:MedicinalProductPackaged.marketingAuthorization [ Reference(MedicinalProductAuthorization) ]; # 0..1 Manufacturer
     of this Package Item
  fhir:MedicinalProductPackaged.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of this Package Item
  fhir:MedicinalProductPackaged.batchIdentifier [ # 0..* Batch numbering
    fhir:MedicinalProductPackaged.batchIdentifier.outerPackaging [ Identifier ]; # 1..1 A number appearing on the outer
    packaging of a specific batch
    fhir:MedicinalProductPackaged.batchIdentifier.immediatePackaging [ Identifier ]; # 0..1 A number appearing on the
```

```
    immediate packaging (and not the outer packaging)
  ], ...;
  fhir:MedicinalProductPackaged.packageItem [ # 1..* A packaging item, as a contained for medicine, possibly with other
    packaging items within
    fhir:MedicinalProductPackaged.packageItem.identifier [ Identifier ], ... ; # 0..* Including possibly Data Carrier
    Identifier
    fhir:MedicinalProductPackaged.packageItem.type [ CodeableConcept ]; # 1..1 The physical type of the container of the
     medicine
    fhir:MedicinalProductPackaged.packageItem.quantity [ Quantity ]; # 1..1 The quantity of this package in the
    medicinal product, at the current level of packaging. The outermost is always 1
    fhir:MedicinalProductPackaged.packageItem.material [ CodeableConcept ], ... ; # 0..* Material type of the package
    item
    fhir:MedicinalProductPackaged.packageItem.alternateMaterial [ CodeableConcept ], ... ; # 0..* A possible alternate
    material for the packaging
    fhir:MedicinalProductPackaged.packageItem.device [ Reference(DeviceDefinition) ], ... ; # 0..* A device accompanying
     a medicinal product
    fhir:MedicinalProductPackaged.packageItem.manufacturedItem [ Reference(MedicinalProductManufactured) ], ... ; # 0..*
     The manufactured item as contained in the packaged medicinal product
    fhir:MedicinalProductPackaged.packageItem.packageItem [ See MedicinalProductPackaged.packageItem ], ... ; # 0..*
    Allows containers within containers
    fhir:MedicinalProductPackaged.packageItem.physicalCharacteristics [ ProdCharacteristic ]; # 0..1 Dimensions, color
    etc.
    fhir:MedicinalProductPackaged.packageItem.otherCharacteristics [ CodeableConcept ], ... ; # 0..* Other codeable
    characteristics
    fhir:MedicinalProductPackaged.packageItem.shelfLifeStorage [ ProductShelfLife ], ... ; # 0..* Shelf Life and storage
     information
    fhir:MedicinalProductPackaged.packageItem.manufacturer [ Reference(Organization) ], ... ; # 0..* Manufacturer of
    this Package Item
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.19.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | Unique identifier | MedicinalProductPackaged.identifier
subject | reference | The product with this is a pack for | MedicinalProductPackaged.subject (MedicinalProduct)

11.19.4 Resource MedicinalProductPackaged - Examples

URL: https://hl7.org/fhir/R4/medicinalproductpackaged-examples.html

11.19.4 Resource MedicinalProductPackaged - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductpackaged | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.19.5 Resource MedicinalProductPackaged - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductpackaged-definitions.html

11.19.5 Resource MedicinalProductPackaged - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductPackaged resource.

MedicinalProductPackaged
Element Id | MedicinalProductPackaged
Definition | A medicinal product in a container or package.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductPackaged.identifier
Element Id | MedicinalProductPackaged.identifier

Definition | Unique identifier.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProductPackaged.subject
Element Id | MedicinalProductPackaged.subject
Definition | The product with this is a pack for.
Cardinality | 0..*
Type | Reference(MedicinalProduct)
Summary | true
MedicinalProductPackaged.description
Element Id | MedicinalProductPackaged.description
Definition | Textual description.
Cardinality | 0..1
Type | string
Summary | true
MedicinalProductPackaged.legalStatusOfSupply
Element Id | MedicinalProductPackaged.legalStatusOfSupply
Definition | The legal status of supply of the medicinal product as classified by the regulator.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductPackaged.marketingStatus
Element Id | MedicinalProductPackaged.marketingStatus
Definition | Marketing information.
Cardinality | 0..*
Type | MarketingStatus
Summary | true
MedicinalProductPackaged.marketingAuthorization
Element Id | MedicinalProductPackaged.marketingAuthorization
Definition | Manufacturer of this Package Item.
Cardinality | 0..1
Type | Reference(MedicinalProductAuthorization)
Summary | true
MedicinalProductPackaged.manufacturer
Element Id | MedicinalProductPackaged.manufacturer
Definition | Manufacturer of this Package Item.
Cardinality | 0..*
Type | Reference(Organization)
Summary | true
MedicinalProductPackaged.batchIdentifier
Element Id | MedicinalProductPackaged.batchIdentifier
Definition | Batch numbering.
Cardinality | 0..*
Summary | true
MedicinalProductPackaged.batchIdentifier.outerPackaging
Element Id | MedicinalProductPackaged.batchIdentifier.outerPackaging
Definition | A number appearing on the outer packaging of a specific batch.
Cardinality | 1..1
Type | Identifier
Summary | true
MedicinalProductPackaged.batchIdentifier.immediatePackaging
Element Id | MedicinalProductPackaged.batchIdentifier.immediatePackaging
Definition | A number appearing on the immediate packaging (and not the outer packaging).
Cardinality | 0..1
Type | Identifier
Summary | true
MedicinalProductPackaged.packageItem
Element Id | MedicinalProductPackaged.packageItem
Definition | A packaging item, as a contained for medicine, possibly with other packaging items within.
Cardinality | 1..*
Summary | true
MedicinalProductPackaged.packageItem.identifier
Element Id | MedicinalProductPackaged.packageItem.identifier
Definition | Including possibly Data Carrier Identifier.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
MedicinalProductPackaged.packageItem.type
Element Id | MedicinalProductPackaged.packageItem.type

Definition | The physical type of the container of the medicine.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPackaged.packageItem.quantity
Element Id | MedicinalProductPackaged.packageItem.quantity
Definition | The quantity of this package in the medicinal product, at the current level of packaging. The outermost is always 1.
Cardinality | 1..1
Type | Quantity
Summary | true
MedicinalProductPackaged.packageItem.material
Element Id | MedicinalProductPackaged.packageItem.material
Definition | Material type of the package item.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductPackaged.packageItem.alternateMaterial
Element Id | MedicinalProductPackaged.packageItem.alternateMaterial
Definition | A possible alternate material for the packaging.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductPackaged.packageItem.device
Element Id | MedicinalProductPackaged.packageItem.device
Definition | A device accompanying a medicinal product.
Cardinality | 0..*
Type | Reference(DeviceDefinition)
Summary | true
MedicinalProductPackaged.packageItem.manufacturedItem
Element Id | MedicinalProductPackaged.packageItem.manufacturedItem
Definition | The manufactured item as contained in the packaged medicinal product.
Cardinality | 0..*
Type | Reference(MedicinalProductManufactured)
Summary | true
MedicinalProductPackaged.packageItem.packageItem
Element Id | MedicinalProductPackaged.packageItem.packageItem
Definition | Allows containers within containers.
Cardinality | 0..*
Type | See MedicinalProductPackaged.packageItem
Summary | true
MedicinalProductPackaged.packageItem.physicalCharacteristics
Element Id | MedicinalProductPackaged.packageItem.physicalCharacteristics
Definition | Dimensions, color etc.
Cardinality | 0..1
Type | ProdCharacteristic
Summary | true
MedicinalProductPackaged.packageItem.otherCharacteristics
Element Id | MedicinalProductPackaged.packageItem.otherCharacteristics
Definition | Other codeable characteristics.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
MedicinalProductPackaged.packageItem.shelfLifeStorage
Element Id | MedicinalProductPackaged.packageItem.shelfLifeStorage
Definition | Shelf Life and storage information.
Cardinality | 0..*
Type | ProductShelfLife
Summary | true
MedicinalProductPackaged.packageItem.manufacturer
Element Id | MedicinalProductPackaged.packageItem.manufacturer
Definition | Manufacturer of this Package Item.
Cardinality | 0..*
Type | Reference(Organization)
Summary | true

11.19.6 Resource MedicinalProductPackaged - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductpackaged-mappings.html

11.19.6 Resource MedicinalProductPackaged - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductpackaged resource (see Mappings to Other Standards for further information & status).

11.19.7 Resource MedicinalProductPackaged - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductpackaged-profiles.html

11.19.7 Resource MedicinalProductPackaged - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductPackaged resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.19.7.1 Profiles

No Profiles defined for this resource

11.19.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.19.7.3 Search Extensions

No Search Extensions defined for this resource

11.19.8 JSON Schema for MedicinalProductPackaged

URL: https://hl7.org/fhir/R4/medicinalproductpackaged.schema.json.html

11.19.8 JSON Schema for MedicinalProductPackaged

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductPackaged, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductPackaged",
  "$ref": "#/definitions/MedicinalProductPackaged",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductPackaged": {
      "description": "A medicinal product in a container or package.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductPackaged resource",
          "const": "MedicinalProductPackaged"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
```

```
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Unique identifier.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "subject": {
      "description": "The product with this is a pack for.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "description": {
      "description": "Textual description.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "legalStatusOfSupply": {
      "description": "The legal status of supply of the medicinal product as classified by the regulator.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "marketingStatus": {
      "description": "Marketing information.",
      "items": {
        "$ref": "MarketingStatus.schema.json#/definitions/MarketingStatus"
      },
      "type": "array"
    },
```

```
    "marketingAuthorization": {
      "description": "Manufacturer of this Package Item.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "manufacturer": {
      "description": "Manufacturer of this Package Item.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "batchIdentifier": {
      "description": "Batch numbering.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPackaged_BatchIdentifier"
      },
      "type": "array"
    },
    "packageItem": {
      "description": "A packaging item, as a contained for medicine, possibly with other packaging items within.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPackaged_PackageItem"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "packageItem",
    "resourceType"
  ]
},
"MedicinalProductPackaged_BatchIdentifier": {
  "description": "A medicinal product in a container or package.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "outerPackaging": {
      "description": "A number appearing on the outer packaging of a specific batch.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "immediatePackaging": {
      "description": "A number appearing on the immediate packaging (and not the outer packaging).",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    }
  },
  "additionalProperties": false,
  "required": [
    "outerPackaging"
  ]
},
"MedicinalProductPackaged_PackageItem": {
  "description": "A medicinal product in a container or package.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Including possibly Data Carrier Identifier.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "type": {
      "description": "The physical type of the container of the medicine.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "quantity": {
      "description": "The quantity of this package in the medicinal product, at the current level of packaging. The
outermost is always 1.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "material": {
      "description": "Material type of the package item.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "alternateMaterial": {
      "description": "A possible alternate material for the packaging.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "device": {
      "description": "A device accompanying a medicinal product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "manufacturedItem": {
      "description": "The manufactured item as contained in the packaged medicinal product.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "packageItem": {
      "description": "Allows containers within containers.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPackaged_PackageItem"
      },
      "type": "array"
    },
    "physicalCharacteristics": {
      "description": "Dimensions, color etc.",
      "$ref": "ProdCharacteristic.schema.json#/definitions/ProdCharacteristic"
    },
    "otherCharacteristics": {
      "description": "Other codeable characteristics.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "shelfLifeStorage": {
      "description": "Shelf Life and storage information.",
      "items": {
        "$ref": "ProductShelfLife.schema.json#/definitions/ProductShelfLife"
      },
      "type": "array"
```

```
        },
        "manufacturer": {
          "description": "Manufacturer of this Package Item.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "quantity",
        "type"
      ]
    }
  }
}
```

11.19.9 Resource MedicinalProductPackaged - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductpackaged-dependencies.html

11.19.9 Resource MedicinalProductPackaged - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductPackaged FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductPackaged | | DomainResource
identifier | 0..* | Identifier
subject | 0..* | Reference(MedicinalProduct)
description | 0..1 | string
legalStatusOfSupply | 0..1 | CodeableConcept
marketingStatus | 0..* | MarketingStatus | Type Error: (MarketingStatus = FMM5-Draft vs. Element = FMM0-Trial Use)
marketingAuthorization | 0..1 | Reference(MedicinalProductAuthorization)
manufacturer | 0..* | Reference(Organization)
batchIdentifier | 0..* | BackboneElement
outerPackaging | 1..1 | Identifier
immediatePackaging | 0..1 | Identifier
packageItem | 1..* | BackboneElement
identifier | 0..* | Identifier
type | 1..1 | CodeableConcept
quantity | 1..1 | Quantity
material | 0..* | CodeableConcept
alternateMaterial | 0..* | CodeableConcept
device | 0..* | Reference(DeviceDefinition)
manufacturedItem | 0..* | Reference(MedicinalProductManufactured)
packageItem | 0..* | see packageItem
physicalCharacteristics | 0..1 | ProdCharacteristic | Type Error: (ProdCharacteristic = FMM5-Draft vs. Element = FMM0-Trial Use)
otherCharacteristics | 0..* | CodeableConcept
shelfLifeStorage | 0..* | ProductShelfLife | Type Error: (ProductShelfLife = FMM5-Draft vs. Element = FMM0-Trial Use)
manufacturer | 0..* | Reference(Organization)

11.20 Resource MedicinalProductPharmaceutical - Content

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A pharmaceutical product described in terms of its composition and dose form.

11.20.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

Qualitative and quantitative composition of a Medicinal Product in the dose form approved for administration in line with the regulated product information

Note that in many instances, the pharmaceutical product is equal to the manufactured item. However, there are instances where the manufactured item must undergo a transformation before being administered to the patient (as the pharmaceutical product) and the two are not equal.

This resource is referenced by MedicinalProduct

11.20.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MedicinalProductPharmaceutical | ΣTU | | DomainResource | A pharmaceutical product described in terms of its composition and dose form Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | An identifier for the pharmaceutical medicinal product
administrableDoseForm | Σ | 1..1 | CodeableConcept | The administrable dose form, after necessary reconstitution
unitOfPresentation | Σ | 0..1 | CodeableConcept | Todo
ingredient | Σ | 0..* | Reference(MedicinalProductIngredient) | Ingredient
device | Σ | 0..* | Reference(DeviceDefinition) | Accompanying device
characteristics | Σ | 0..* | BackboneElement | Characteristics e.g. a products onset of action
code | Σ | 1..1 | CodeableConcept | A coded characteristic
status | Σ | 0..1 | CodeableConcept | The status of characteristic e.g. assigned or pending
routeOfAdministration | Σ | 1..* | BackboneElement | The path by which the pharmaceutical product is taken into or makes contact with the body
code | Σ | 1..1 | CodeableConcept | Coded expression for the route
firstDose | Σ | 0..1 | Quantity | The first dose (dose quantity) administered in humans can be specified, for a product under investigation, using a numerical value and its unit of measurement
maxSingleDose | Σ | 0..1 | Quantity | The maximum single dose that can be administered as per the protocol of a clinical trial can be specified using a numerical value and its unit of measurement
maxDosePerDay | Σ | 0..1 | Quantity | The maximum dose per day (maximum dose quantity to be administered in any one 24-h period) that can be administered as per the protocol referenced in the clinical trial authorisation
maxDosePerTreatmentPeriod | Σ | 0..1 | Ratio | The maximum dose per treatment period that can be administered as per the protocol referenced in the clinical trial authorisation
maxTreatmentPeriod | Σ | 0..1 | Duration | The maximum treatment period during which an Investigational Medicinal Product can be administered as per the protocol referenced in the clinical trial authorisation
targetSpecies | Σ | 0..* | BackboneElement | A species for which this route applies
code | Σ | 1..1 | CodeableConcept | Coded expression for the species
withdrawalPeriod | Σ | 0..* | BackboneElement | A species specific time during which consumption of animal product is not appropriate
tissue | Σ | 1..1 | CodeableConcept | Coded expression for the type of tissue for which the withdrawal period applues, e.g. meat, milk
value | Σ | 1..1 | Quantity | A value for the time
supportingInformation | Σ | 0..1 | string | Extra information about the withdrawal period
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductPharmaceutical xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for the pharmaceutical medicinal product --></identifier>
 <administrableDoseForm><!-- 1..1 CodeableConcept The administrable dose form, after necessary reconstitution
    --></administrableDoseForm>
 <unitOfPresentation><!-- 0..1 CodeableConcept Todo --></unitOfPresentation>
 <ingredient><!-- 0..* Reference(MedicinalProductIngredient) Ingredient --></ingredient>
 <device><!-- 0..* Reference(DeviceDefinition) Accompanying device --></device>
 <characteristics>  <!-- 0..* Characteristics e.g. a products onset of action -->
  <code><!-- 1..1 CodeableConcept A coded characteristic --></code>
  <status><!-- 0..1 CodeableConcept The status of characteristic e.g. assigned or pending --></status>
 </characteristics>
 <routeOfAdministration>  <!-- 1..* The path by which the pharmaceutical product is taken into or makes contact with the
     body -->
  <code><!-- 1..1 CodeableConcept Coded expression for the route --></code>
  <firstDose><!-- 0..1 Quantity The first dose (dose quantity) administered in humans can be specified, for a product
    under investigation, using a numerical value and its unit of measurement --></firstDose>
  <maxSingleDose><!-- 0..1 Quantity The maximum single dose that can be administered as per the protocol of a clinical
    trial can be specified using a numerical value and its unit of measurement --></maxSingleDose>
  <maxDosePerDay><!-- 0..1 Quantity The maximum dose per day (maximum dose quantity to be administered in any one 24-h
    period) that can be administered as per the protocol referenced in the clinical trial authorisation
    --></maxDosePerDay>
  <maxDosePerTreatmentPeriod><!-- 0..1 Ratio The maximum dose per treatment period that can be administered as per the
    protocol referenced in the clinical trial authorisation --></maxDosePerTreatmentPeriod>
  <maxTreatmentPeriod><!-- 0..1 Duration The maximum treatment period during which an Investigational Medicinal Product
    can be administered as per the protocol referenced in the clinical trial authorisation --></maxTreatmentPeriod>
  <targetSpecies>  <!-- 0..* A species for which this route applies -->
   <code><!-- 1..1 CodeableConcept Coded expression for the species --></code>
   <withdrawalPeriod>  <!-- 0..* A species specific time during which consumption of animal product is not appropriate
    -->
    <tissue><!-- 1..1 CodeableConcept Coded expression for the type of tissue for which the withdrawal period applues,
```

```
    e.g. meat, milk --></tissue>
    <value><!-- 1..1 Quantity A value for the time --></value>
    <supportingInformation value="[string]"/><!-- 0..1 Extra information about the withdrawal period -->
   </withdrawalPeriod>
  </targetSpecies>
 </routeOfAdministration>
</MedicinalProductPharmaceutical>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductPharmaceutical",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for the pharmaceutical medicinal product
  "administrableDoseForm" : { CodeableConcept }, // R!  The administrable dose form, after necessary reconstitution
  "unitOfPresentation" : { CodeableConcept }, // Todo
  "ingredient" : [{ Reference(MedicinalProductIngredient) }], // Ingredient
  "device" : [{ Reference(DeviceDefinition) }], // Accompanying device
  "characteristics" : [{ // Characteristics e.g. a products onset of action
    "code" : { CodeableConcept }, // R!  A coded characteristic
    "status" : { CodeableConcept } // The status of characteristic e.g. assigned or pending
  }],
  "routeOfAdministration" : [{ // R!  The path by which the pharmaceutical product is taken into or makes contact with
    the body
    "code" : { CodeableConcept }, // R!  Coded expression for the route
    "firstDose" : { Quantity }, // The first dose (dose quantity) administered in humans can be specified, for a product
     under investigation, using a numerical value and its unit of measurement
    "maxSingleDose" : { Quantity }, // The maximum single dose that can be administered as per the protocol of a
    clinical trial can be specified using a numerical value and its unit of measurement
    "maxDosePerDay" : { Quantity }, // The maximum dose per day (maximum dose quantity to be administered in any one
    24-h period) that can be administered as per the protocol referenced in the clinical trial authorisation
    "maxDosePerTreatmentPeriod" : { Ratio }, // The maximum dose per treatment period that can be administered as per
    the protocol referenced in the clinical trial authorisation
    "maxTreatmentPeriod" : { Duration }, // The maximum treatment period during which an Investigational Medicinal
    Product can be administered as per the protocol referenced in the clinical trial authorisation
    "targetSpecies" : [{ // A species for which this route applies
      "code" : { CodeableConcept }, // R!  Coded expression for the species
      "withdrawalPeriod" : [{ // A species specific time during which consumption of animal product is not appropriate
        "tissue" : { CodeableConcept }, // R!  Coded expression for the type of tissue for which the withdrawal period
    applues, e.g. meat, milk
        "value" : { Quantity }, // R!  A value for the time
        "supportingInformation" : "<string>" // Extra information about the withdrawal period
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductPharmaceutical;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductPharmaceutical.identifier [ Identifier ], ... ; # 0..* An identifier for the pharmaceutical
    medicinal product
  fhir:MedicinalProductPharmaceutical.administrableDoseForm [ CodeableConcept ]; # 1..1 The administrable dose form,
    after necessary reconstitution
  fhir:MedicinalProductPharmaceutical.unitOfPresentation [ CodeableConcept ]; # 0..1 Todo
  fhir:MedicinalProductPharmaceutical.ingredient [ Reference(MedicinalProductIngredient) ], ... ; # 0..* Ingredient
  fhir:MedicinalProductPharmaceutical.device [ Reference(DeviceDefinition) ], ... ; # 0..* Accompanying device
  fhir:MedicinalProductPharmaceutical.characteristics [ # 0..* Characteristics e.g. a products onset of action
    fhir:MedicinalProductPharmaceutical.characteristics.code [ CodeableConcept ]; # 1..1 A coded characteristic
    fhir:MedicinalProductPharmaceutical.characteristics.status [ CodeableConcept ]; # 0..1 The status of characteristic
    e.g. assigned or pending
  ], ...;
  fhir:MedicinalProductPharmaceutical.routeOfAdministration [ # 1..* The path by which the pharmaceutical product is
    taken into or makes contact with the body
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.code [ CodeableConcept ]; # 1..1 Coded expression for the
    route
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.firstDose [ Quantity ]; # 0..1 The first dose (dose
    quantity) administered in humans can be specified, for a product under investigation, using a numerical value and
    its unit of measurement
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxSingleDose [ Quantity ]; # 0..1 The maximum single dose
     that can be administered as per the protocol of a clinical trial can be specified using a numerical value and its
    unit of measurement
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerDay [ Quantity ]; # 0..1 The maximum dose per
    day (maximum dose quantity to be administered in any one 24-h period) that can be administered as per the protocol
    referenced in the clinical trial authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerTreatmentPeriod [ Ratio ]; # 0..1 The maximum
    dose per treatment period that can be administered as per the protocol referenced in the clinical trial
    authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxTreatmentPeriod [ Duration ]; # 0..1 The maximum
    treatment period during which an Investigational Medicinal Product can be administered as per the protocol
    referenced in the clinical trial authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies [ # 0..* A species for which this route
    applies
```

```
      fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code [ CodeableConcept ]; # 1..1 Coded
    expression for the species
      fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod [ # 0..* A species
    specific time during which consumption of animal product is not appropriate
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.tissue [
    CodeableConcept ]; # 1..1 Coded expression for the type of tissue for which the withdrawal period applues, e.g.
    meat, milk
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.value [ Quantity ]; #
    1..1 A value for the time
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.supportingInformation [
     string ]; # 0..1 Extra information about the withdrawal period
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

MedicinalProductPharmaceutical | ΣTU | | DomainResource | A pharmaceutical product described in terms of its composition and dose form Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | An identifier for the pharmaceutical medicinal product

administrableDoseForm | Σ | 1..1 | CodeableConcept | The administrable dose form, after necessary reconstitution

unitOfPresentation | Σ | 0..1 | CodeableConcept | Todo

ingredient | Σ | 0..* | Reference(MedicinalProductIngredient) | Ingredient

device | Σ | 0..* | Reference(DeviceDefinition) | Accompanying device

characteristics | Σ | 0..* | BackboneElement | Characteristics e.g. a products onset of action

code | Σ | 1..1 | CodeableConcept | A coded characteristic

status | Σ | 0..1 | CodeableConcept | The status of characteristic e.g. assigned or pending

routeOfAdministration | Σ | 1..* | BackboneElement | The path by which the pharmaceutical product is taken into or makes contact with the body

code | Σ | 1..1 | CodeableConcept | Coded expression for the route

firstDose | Σ | 0..1 | Quantity | The first dose (dose quantity) administered in humans can be specified, for a product under investigation, using a numerical value and its unit of measurement

maxSingleDose | Σ | 0..1 | Quantity | The maximum single dose that can be administered as per the protocol of a clinical trial can be specified using a numerical value and its unit of measurement

maxDosePerDay | Σ | 0..1 | Quantity | The maximum dose per day (maximum dose quantity to be administered in any one 24-h period) that can be administered as per the protocol referenced in the clinical trial authorisation

maxDosePerTreatmentPeriod | Σ | 0..1 | Ratio | The maximum dose per treatment period that can be administered as per the protocol referenced in the clinical trial authorisation

maxTreatmentPeriod | Σ | 0..1 | Duration | The maximum treatment period during which an Investigational Medicinal Product can be administered as per the protocol referenced in the clinical trial authorisation

targetSpecies | Σ | 0..* | BackboneElement | A species for which this route applies

code | Σ | 1..1 | CodeableConcept | Coded expression for the species

withdrawalPeriod | Σ | 0..* | BackboneElement | A species specific time during which consumption of animal product is not appropriate

tissue | Σ | 1..1 | CodeableConcept | Coded expression for the type of tissue for which the withdrawal period applues, e.g. meat, milk

value | Σ | 1..1 | Quantity | A value for the time

supportingInformation | Σ | 0..1 | string | Extra information about the withdrawal period

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MedicinalProductPharmaceutical xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for the pharmaceutical medicinal product --></identifier>
 <administrableDoseForm><!-- 1..1 CodeableConcept The administrable dose form, after necessary reconstitution
    --></administrableDoseForm>
 <unitOfPresentation><!-- 0..1 CodeableConcept Todo --></unitOfPresentation>
 <ingredient><!-- 0..* Reference(MedicinalProductIngredient) Ingredient --></ingredient>
 <device><!-- 0..* Reference(DeviceDefinition) Accompanying device --></device>
 <characteristics>  <!-- 0..* Characteristics e.g. a products onset of action -->
  <code><!-- 1..1 CodeableConcept A coded characteristic --></code>
  <status><!-- 0..1 CodeableConcept The status of characteristic e.g. assigned or pending --></status>
 </characteristics>
 <routeOfAdministration>  <!-- 1..* The path by which the pharmaceutical product is taken into or makes contact with the
     body -->
  <code><!-- 1..1 CodeableConcept Coded expression for the route --></code>
  <firstDose><!-- 0..1 Quantity The first dose (dose quantity) administered in humans can be specified, for a product
    under investigation, using a numerical value and its unit of measurement --></firstDose>
  <maxSingleDose><!-- 0..1 Quantity The maximum single dose that can be administered as per the protocol of a clinical
    trial can be specified using a numerical value and its unit of measurement --></maxSingleDose>
  <maxDosePerDay><!-- 0..1 Quantity The maximum dose per day (maximum dose quantity to be administered in any one 24-h
    period) that can be administered as per the protocol referenced in the clinical trial authorisation
    --></maxDosePerDay>
```

```
  <maxDosePerTreatmentPeriod><!-- 0..1 Ratio The maximum dose per treatment period that can be administered as per the
    protocol referenced in the clinical trial authorisation --></maxDosePerTreatmentPeriod>
  <maxTreatmentPeriod><!-- 0..1 Duration The maximum treatment period during which an Investigational Medicinal Product
    can be administered as per the protocol referenced in the clinical trial authorisation --></maxTreatmentPeriod>
  <targetSpecies>  <!-- 0..* A species for which this route applies -->
   <code><!-- 1..1 CodeableConcept Coded expression for the species --></code>
   <withdrawalPeriod>  <!-- 0..* A species specific time during which consumption of animal product is not appropriate
    -->
    <tissue><!-- 1..1 CodeableConcept Coded expression for the type of tissue for which the withdrawal period applues,
    e.g. meat, milk --></tissue>
    <value><!-- 1..1 Quantity A value for the time --></value>
    <supportingInformation value="[string]"/><!-- 0..1 Extra information about the withdrawal period -->
   </withdrawalPeriod>
  </targetSpecies>
 </routeOfAdministration>
</MedicinalProductPharmaceutical>
```

JSON Template

```
{
  "resourceType" : "MedicinalProductPharmaceutical",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for the pharmaceutical medicinal product
  "administrableDoseForm" : { CodeableConcept }, // R!  The administrable dose form, after necessary reconstitution
  "unitOfPresentation" : { CodeableConcept }, // Todo
  "ingredient" : [{ Reference(MedicinalProductIngredient) }], // Ingredient
  "device" : [{ Reference(DeviceDefinition) }], // Accompanying device
  "characteristics" : [{ // Characteristics e.g. a products onset of action
    "code" : { CodeableConcept }, // R!  A coded characteristic
    "status" : { CodeableConcept } // The status of characteristic e.g. assigned or pending
  }],
  "routeOfAdministration" : [{ // R!  The path by which the pharmaceutical product is taken into or makes contact with
    the body
    "code" : { CodeableConcept }, // R!  Coded expression for the route
    "firstDose" : { Quantity }, // The first dose (dose quantity) administered in humans can be specified, for a product
     under investigation, using a numerical value and its unit of measurement
    "maxSingleDose" : { Quantity }, // The maximum single dose that can be administered as per the protocol of a
    clinical trial can be specified using a numerical value and its unit of measurement
    "maxDosePerDay" : { Quantity }, // The maximum dose per day (maximum dose quantity to be administered in any one
    24-h period) that can be administered as per the protocol referenced in the clinical trial authorisation
    "maxDosePerTreatmentPeriod" : { Ratio }, // The maximum dose per treatment period that can be administered as per
    the protocol referenced in the clinical trial authorisation
    "maxTreatmentPeriod" : { Duration }, // The maximum treatment period during which an Investigational Medicinal
    Product can be administered as per the protocol referenced in the clinical trial authorisation
    "targetSpecies" : [{ // A species for which this route applies
      "code" : { CodeableConcept }, // R!  Coded expression for the species
      "withdrawalPeriod" : [{ // A species specific time during which consumption of animal product is not appropriate
        "tissue" : { CodeableConcept }, // R!  Coded expression for the type of tissue for which the withdrawal period
    applues, e.g. meat, milk
        "value" : { Quantity }, // R!  A value for the time
        "supportingInformation" : "<string>" // Extra information about the withdrawal period
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MedicinalProductPharmaceutical;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MedicinalProductPharmaceutical.identifier [ Identifier ], ... ; # 0..* An identifier for the pharmaceutical
    medicinal product
  fhir:MedicinalProductPharmaceutical.administrableDoseForm [ CodeableConcept ]; # 1..1 The administrable dose form,
    after necessary reconstitution
  fhir:MedicinalProductPharmaceutical.unitOfPresentation [ CodeableConcept ]; # 0..1 Todo
  fhir:MedicinalProductPharmaceutical.ingredient [ Reference(MedicinalProductIngredient) ], ... ; # 0..* Ingredient
  fhir:MedicinalProductPharmaceutical.device [ Reference(DeviceDefinition) ], ... ; # 0..* Accompanying device
  fhir:MedicinalProductPharmaceutical.characteristics [ # 0..* Characteristics e.g. a products onset of action
    fhir:MedicinalProductPharmaceutical.characteristics.code [ CodeableConcept ]; # 1..1 A coded characteristic
    fhir:MedicinalProductPharmaceutical.characteristics.status [ CodeableConcept ]; # 0..1 The status of characteristic
    e.g. assigned or pending
  ], ...;
  fhir:MedicinalProductPharmaceutical.routeOfAdministration [ # 1..* The path by which the pharmaceutical product is
    taken into or makes contact with the body
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.code [ CodeableConcept ]; # 1..1 Coded expression for the
    route
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.firstDose [ Quantity ]; # 0..1 The first dose (dose
    quantity) administered in humans can be specified, for a product under investigation, using a numerical value and
    its unit of measurement
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxSingleDose [ Quantity ]; # 0..1 The maximum single dose
     that can be administered as per the protocol of a clinical trial can be specified using a numerical value and its
    unit of measurement
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerDay [ Quantity ]; # 0..1 The maximum dose per
    day (maximum dose quantity to be administered in any one 24-h period) that can be administered as per the protocol
```

```
    referenced in the clinical trial authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerTreatmentPeriod [ Ratio ]; # 0..1 The maximum
    dose per treatment period that can be administered as per the protocol referenced in the clinical trial
    authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.maxTreatmentPeriod [ Duration ]; # 0..1 The maximum
    treatment period during which an Investigational Medicinal Product can be administered as per the protocol
    referenced in the clinical trial authorisation
    fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies [ # 0..* A species for which this route
    applies
      fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code [ CodeableConcept ]; # 1..1 Coded
    expression for the species
      fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod [ # 0..* A species
    specific time during which consumption of animal product is not appropriate
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.tissue [
    CodeableConcept ]; # 1..1 Coded expression for the type of tissue for which the withdrawal period applues, e.g.
    meat, milk
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.value [ Quantity ]; #
    1..1 A value for the time
        fhir:MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.supportingInformation [
     string ]; # 0..1 Extra information about the withdrawal period
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.20.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | An identifier for the pharmaceutical medicinal product | MedicinalProductPharmaceutical.identifier
route | token | Coded expression for the route | MedicinalProductPharmaceutical.routeOfAdministration.code
target-species | token | Coded expression for the species | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code

11.20.4 Resource MedicinalProductPharmaceutical - Examples

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical-examples.html

11.20.4 Resource MedicinalProductPharmaceutical - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of medicinalproductpharmaceutical | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.20.5 Resource MedicinalProductPharmaceutical - Detailed Descriptions

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical-definitions.html

11.20.5 Resource MedicinalProductPharmaceutical - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the MedicinalProductPharmaceutical resource.

MedicinalProductPharmaceutical
Element Id | MedicinalProductPharmaceutical
Definition | A pharmaceutical product described in terms of its composition and dose form.
Cardinality | 0..*
Type | DomainResource
Summary | true
MedicinalProductPharmaceutical.identifier
Element Id | MedicinalProductPharmaceutical.identifier
Definition | An identifier for the pharmaceutical medicinal product.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*

Type | Identifier
Summary | true
MedicinalProductPharmaceutical.administrableDoseForm
Element Id | MedicinalProductPharmaceutical.administrableDoseForm
Definition | The administrable dose form, after necessary reconstitution.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.unitOfPresentation
Element Id | MedicinalProductPharmaceutical.unitOfPresentation
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.ingredient
Element Id | MedicinalProductPharmaceutical.ingredient
Definition | Ingredient.
Cardinality | 0..*
Type | Reference(MedicinalProductIngredient)
Summary | true
MedicinalProductPharmaceutical.device
Element Id | MedicinalProductPharmaceutical.device
Definition | Accompanying device.
Cardinality | 0..*
Type | Reference(DeviceDefinition)
Summary | true
MedicinalProductPharmaceutical.characteristics
Element Id | MedicinalProductPharmaceutical.characteristics
Definition | Characteristics e.g. a products onset of action.
Cardinality | 0..*
Summary | true
MedicinalProductPharmaceutical.characteristics.code
Element Id | MedicinalProductPharmaceutical.characteristics.code
Definition | A coded characteristic.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.characteristics.status
Element Id | MedicinalProductPharmaceutical.characteristics.status
Definition | The status of characteristic e.g. assigned or pending.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration
Element Id | MedicinalProductPharmaceutical.routeOfAdministration
Definition | The path by which the pharmaceutical product is taken into or makes contact with the body.
Cardinality | 1..*
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.code
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.code
Definition | Coded expression for the route.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.firstDose
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.firstDose
Definition | The first dose (dose quantity) administered in humans can be specified, for a product under investigation, using a numerical value and its unit of measurement.
Cardinality | 0..1
Type | Quantity
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.maxSingleDose
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.maxSingleDose
Definition | The maximum single dose that can be administered as per the protocol of a clinical trial can be specified using a numerical value and its unit of measurement.
Cardinality | 0..1
Type | Quantity
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerDay
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerDay
Definition | The maximum dose per day (maximum dose quantity to be administered in any one 24-h period) that can be administered as per the protocol referenced in the clinical trial authorisation.

Cardinality | 0..1
Type | Quantity
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerTreatmentPeriod
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.maxDosePerTreatmentPeriod
Definition | The maximum dose per treatment period that can be administered as per the protocol referenced in the clinical trial authorisation.
Cardinality | 0..1
Type | Ratio
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.maxTreatmentPeriod
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.maxTreatmentPeriod
Definition | The maximum treatment period during which an Investigational Medicinal Product can be administered as per the protocol referenced in the clinical trial authorisation.
Cardinality | 0..1
Type | Duration
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies
Definition | A species for which this route applies.
Cardinality | 0..*
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code
Definition | Coded expression for the species.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod
Definition | A species specific time during which consumption of animal product is not appropriate.
Cardinality | 0..*
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.tissue
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.tissue
Definition | Coded expression for the type of tissue for which the withdrawal period applues, e.g. meat, milk.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.value
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.value
Definition | A value for the time.
Cardinality | 1..1
Type | Quantity
Summary | true
MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.supportingInformation
Element Id | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.withdrawalPeriod.supportingInformation
Definition | Extra information about the withdrawal period.
Cardinality | 0..1
Type | string
Summary | true

11.20.6 Resource MedicinalProductPharmaceutical - Mappings

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical-mappings.html

11.20.6 Resource MedicinalProductPharmaceutical - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the medicinalproductpharmaceutical resource (see Mappings to Other Standards for further information & status).

11.20.7 Resource MedicinalProductPharmaceutical - Extensions & Profiles

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical-profiles.html

11.20.7 Resource MedicinalProductPharmaceutical - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the MedicinalProductPharmaceutical resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.20.7.1 Profiles

No Profiles defined for this resource

11.20.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.20.7.3 Search Extensions

No Search Extensions defined for this resource

11.20.8 JSON Schema for MedicinalProductPharmaceutical

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical.schema.json.html

11.20.8 JSON Schema for MedicinalProductPharmaceutical

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for MedicinalProductPharmaceutical, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MedicinalProductPharmaceutical",
  "$ref": "#/definitions/MedicinalProductPharmaceutical",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MedicinalProductPharmaceutical": {
      "description": "A pharmaceutical product described in terms of its composition and dose form.",
      "properties": {
        "resourceType": {
          "description": "This is a MedicinalProductPharmaceutical resource",
          "const": "MedicinalProductPharmaceutical"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
```

```
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "An identifier for the pharmaceutical medicinal product.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "administrableDoseForm": {
      "description": "The administrable dose form, after necessary reconstitution.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "unitOfPresentation": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "ingredient": {
      "description": "Ingredient.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "device": {
      "description": "Accompanying device.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "characteristics": {
      "description": "Characteristics e.g. a products onset of action.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPharmaceutical_Characteristics"
      },
      "type": "array"
    },
    "routeOfAdministration": {
      "description": "The path by which the pharmaceutical product is taken into or makes contact with the body.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPharmaceutical_RouteOfAdministration"

      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "administrableDoseForm",
    "routeOfAdministration",
    "resourceType"
  ]
},
```

```
"MedicinalProductPharmaceutical_Characteristics": {
  "description": "A pharmaceutical product described in terms of its composition and dose form.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A coded characteristic.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The status of characteristic e.g. assigned or pending.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false,
  "required": [
    "code"
  ]
},
"MedicinalProductPharmaceutical_RouteOfAdministration": {
  "description": "A pharmaceutical product described in terms of its composition and dose form.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Coded expression for the route.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "firstDose": {
      "description": "The first dose (dose quantity) administered in humans can be specified, for a product under
investigation, using a numerical value and its unit of measurement.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "maxSingleDose": {
      "description": "The maximum single dose that can be administered as per the protocol of a clinical trial can
```

```
be specified using a numerical value and its unit of measurement.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "maxDosePerDay": {
      "description": "The maximum dose per day (maximum dose quantity to be administered in any one 24-h period)
that can be administered as per the protocol referenced in the clinical trial authorisation.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "maxDosePerTreatmentPeriod": {
      "description": "The maximum dose per treatment period that can be administered as per the protocol referenced
in the clinical trial authorisation.",
      "$ref": "Ratio.schema.json#/definitions/Ratio"
    },
    "maxTreatmentPeriod": {
      "description": "The maximum treatment period during which an Investigational Medicinal Product can be
administered as per the protocol referenced in the clinical trial authorisation.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "targetSpecies": {
      "description": "A species for which this route applies.",

      "items": {
        "$ref": "#/definitions/MedicinalProductPharmaceutical_TargetSpecies"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code"
  ]
},
"MedicinalProductPharmaceutical_TargetSpecies": {
  "description": "A pharmaceutical product described in terms of its composition and dose form.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Coded expression for the species.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "withdrawalPeriod": {
      "description": "A species specific time during which consumption of animal product is not appropriate.",
      "items": {
        "$ref": "#/definitions/MedicinalProductPharmaceutical_WithdrawalPeriod"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code"
  ]
},
"MedicinalProductPharmaceutical_WithdrawalPeriod": {
  "description": "A pharmaceutical product described in terms of its composition and dose form.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "tissue": {
      "description": "Coded expression for the type of tissue for which the withdrawal period applues, e.g. meat,
    milk.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "value": {
      "description": "A value for the time.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "supportingInformation": {
      "description": "Extra information about the withdrawal period.",
      "$ref": "#/definitions/string"
    },
    "_supportingInformation": {
      "description": "Extensions for supportingInformation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "tissue",
    "value"
  ]
 }
 }
}
```


11.20.9 Resource MedicinalProductPharmaceutical - Maturity Dependency View

URL: https://hl7.org/fhir/R4/medicinalproductpharmaceutical-dependencies.html

11.20.9 Resource MedicinalProductPharmaceutical - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for MedicinalProductPharmaceutical FMM level 0

Name | Card. | Type | Dependency Analysis
MedicinalProductPharmaceutical | | DomainResource
identifier | 0..* | Identifier
administrableDoseForm | 1..1 | CodeableConcept
unitOfPresentation | 0..1 | CodeableConcept
ingredient | 0..* | Reference(MedicinalProductIngredient)
device | 0..* | Reference(DeviceDefinition)
characteristics | 0..* | BackboneElement
code | 1..1 | CodeableConcept
status | 0..1 | CodeableConcept
routeOfAdministration | 1..* | BackboneElement
code | 1..1 | CodeableConcept
firstDose | 0..1 | Quantity
maxSingleDose | 0..1 | Quantity
maxDosePerDay | 0..1 | Quantity
maxDosePerTreatmentPeriod | 0..1 | Ratio
maxTreatmentPeriod | 0..1 | Duration
targetSpecies | 0..* | BackboneElement
code | 1..1 | CodeableConcept

withdrawalPeriod | 0..* | BackboneElement
tissue | 1..1 | CodeableConcept
value | 1..1 | Quantity
supportingInformation | 0..1 | string

11.21 Resource SubstanceSpecification - Content

URL: https://hl7.org/fhir/R4/substancespecification.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The detailed description of a substance, typically at a level beyond what is used for prescribing.

11.21.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

This resource is referenced by MedicinalProductContraindication, MedicinalProductIndication and itself

11.21.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceSpecification | ΣTU | | DomainResource | The detailed description of a substance, typically at a level beyond what is used for prescribing Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Identifier by which this substance is known
type | Σ | 0..1 | CodeableConcept | High level categorization, e.g. polymer or nucleic acid
status | Σ | 0..1 | CodeableConcept | Status of substance within the catalogue e.g. approved
domain | Σ | 0..1 | CodeableConcept | If the substance applies to only human or veterinary use
description | Σ | 0..1 | string | Textual description of the substance
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
comment | Σ | 0..1 | string | Textual comment about this record of a substance
moiety | Σ | 0..* | BackboneElement | Moiety, for structural modifications
role | Σ | 0..1 | CodeableConcept | Role that the moiety is playing
identifier | Σ | 0..1 | Identifier | Identifier by which this moiety substance is known
name | Σ | 0..1 | string | Textual name for this moiety substance
stereochemistry | Σ | 0..1 | CodeableConcept | Stereochemistry type
opticalActivity | Σ | 0..1 | CodeableConcept | Optical activity type
molecularFormula | Σ | 0..1 | string | Molecular formula
amount[x] | Σ | 0..1 | | Quantitative value for this moiety
amountQuantity | | | Quantity
amountString | | | string
property | Σ | 0..* | BackboneElement | General specifications for this substance, including how it is related to other substances
category | Σ | 0..1 | CodeableConcept | A category for this property, e.g. Physical, Chemical, Enzymatic
code | Σ | 0..1 | CodeableConcept | Property type e.g. viscosity, pH, isoelectric point
parameters | Σ | 0..1 | string | Parameters that were used in the measurement of a property (e.g. for viscosity: measured at 20C with a pH of 7.1)
definingSubstance[x] | Σ | 0..1 | | A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol)
definingSubstanceReference | | | Reference(SubstanceSpecification | Substance)
definingSubstanceCodeableConcept | | | CodeableConcept
amount[x] | Σ | 0..1 | | Quantitative value for this property
amountQuantity | | | Quantity
amountString | | | string
referenceInformation | Σ | 0..1 | Reference(SubstanceReferenceInformation) | General information detailing this substance
structure | Σ | 0..1 | BackboneElement | Structural information
stereochemistry | Σ | 0..1 | CodeableConcept | Stereochemistry type
opticalActivity | Σ | 0..1 | CodeableConcept | Optical activity type
molecularFormula | Σ | 0..1 | string | Molecular formula
molecularFormulaByMoiety | Σ | 0..1 | string | Specified per moiety according to the Hill system, i.e. first C, then H, then alphabetical, each moiety separated by a dot
isotope | Σ | 0..* | BackboneElement | Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio
identifier | Σ | 0..1 | Identifier | Substance identifier for each non-natural or radioisotope
name | Σ | 0..1 | CodeableConcept | Substance name for each non-natural or radioisotope

substitution | Σ | 0..1 | CodeableConcept | The type of isotopic substitution present in a single substance
halfLife | Σ | 0..1 | Quantity | Half life - for a non-natural nuclide
molecularWeight | Σ | 0..1 | BackboneElement | The molecular weight or weight range (for proteins, polymers or nucleic acids)
method | Σ | 0..1 | CodeableConcept | The method by which the molecular weight was determined
type | Σ | 0..1 | CodeableConcept | Type of molecular weight such as exact, average (also known as. number average), weight average
amount | Σ | 0..1 | Quantity | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field
molecularWeight | Σ | 0..1 | see molecularWeight | The molecular weight or weight range (for proteins, polymers or nucleic acids)
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
representation | Σ | 0..* | BackboneElement | Molecular structural representation
type | Σ | 0..1 | CodeableConcept | The type of structure (e.g. Full, Partial, Representative)
representation | Σ | 0..1 | string | The structural representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX
attachment | Σ | 0..1 | Attachment | An attached file with the structural representation
code | Σ | 0..* | BackboneElement | Codes associated with the substance
code | Σ | 0..1 | CodeableConcept | The specific code
status | Σ | 0..1 | CodeableConcept | Status of the code assignment
statusDate | Σ | 0..1 | dateTime | The date at which the code status is changed as part of the terminology maintenance
comment | Σ | 0..1 | string | Any comment can be provided in this field, if necessary
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
name | Σ | 0..* | BackboneElement | Names applicable to this substance
name | Σ | 1..1 | string | The actual name
type | Σ | 0..1 | CodeableConcept | Name type
status | Σ | 0..1 | CodeableConcept | The status of the name
preferred | Σ | 0..1 | boolean | If this is the preferred name for this substance
language | Σ | 0..* | CodeableConcept | Language of the name
domain | Σ | 0..* | CodeableConcept | The use context of this name for example if there is a different name a drug active ingredient as opposed to a food colour additive
jurisdiction | Σ | 0..* | CodeableConcept | The jurisdiction where this name applies
synonym | Σ | 0..* | see name | A synonym of this name
translation | Σ | 0..* | see name | A translation for this name
official | Σ | 0..* | BackboneElement | Details of the official nature of this name
authority | Σ | 0..1 | CodeableConcept | Which authority uses this official name
status | Σ | 0..1 | CodeableConcept | The status of the official name
date | Σ | 0..1 | dateTime | Date of official name change
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
molecularWeight | Σ | 0..* | see molecularWeight | The molecular weight or weight range (for proteins, polymers or nucleic acids)
relationship | Σ | 0..* | BackboneElement | A link between this substance and another, with details of the relationship
substance[x] | Σ | 0..1 | | A pointer to another substance, as a resource or just a representational code
substanceReference | | | Reference(SubstanceSpecification)
substanceCodeableConcept | | | CodeableConcept
relationship | Σ | 0..1 | CodeableConcept | For example "salt to parent", "active moiety", "starting material"
isDefining | Σ | 0..1 | boolean | For example where an enzyme strongly bonds with a particular substance, this is a defining relationship for that enzyme, out of several possible substance relationships
amount[x] | Σ | 0..1 | | A numeric factor for the relationship, for instance to express that the salt of a substance has some percentage of the active substance in relation to some other
amountQuantity | | | Quantity
amountRange | | | Range
amountRatio | | | Ratio
amountString | | | string
amountRatioLowLimit | Σ | 0..1 | Ratio | For use when the numeric
amountType | Σ | 0..1 | CodeableConcept | An operator for the amount, for example "average", "approximately", "less than"
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
nucleicAcid | Σ | 0..1 | Reference(SubstanceNucleicAcid) | Data items specific to nucleic acids
polymer | Σ | 0..1 | Reference(SubstancePolymer) | Data items specific to polymers
protein | Σ | 0..1 | Reference(SubstanceProtein) | Data items specific to proteins
sourceMaterial | Σ | 0..1 | Reference(SubstanceSourceMaterial) | Material or taxonomic/anatomical source for the substance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceSpecification xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier by which this substance is known --></identifier>
 <type><!-- 0..1 CodeableConcept High level categorization, e.g. polymer or nucleic acid --></type>
 <status><!-- 0..1 CodeableConcept Status of substance within the catalogue e.g. approved --></status>
 <domain><!-- 0..1 CodeableConcept If the substance applies to only human or veterinary use --></domain>
 <description value="[string]"/><!-- 0..1 Textual description of the substance -->
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 <comment value="[string]"/><!-- 0..1 Textual comment about this record of a substance -->
 <moiety>  <!-- 0..* Moiety, for structural modifications -->
  <role><!-- 0..1 CodeableConcept Role that the moiety is playing --></role>
  <identifier><!-- 0..1 Identifier Identifier by which this moiety substance is known --></identifier>
  <name value="[string]"/><!-- 0..1 Textual name for this moiety substance -->
```

```
 <stereochemistry><!-- 0..1 CodeableConcept Stereochemistry type --></stereochemistry>
 <opticalActivity><!-- 0..1 CodeableConcept Optical activity type --></opticalActivity>
 <molecularFormula value="[string]"/><!-- 0..1 Molecular formula -->
 <amount[x]><!-- 0..1 Quantity|string Quantitative value for this moiety --></amount[x]>
</moiety>
<property>  <!-- 0..* General specifications for this substance, including how it is related to other substances -->
 <category><!-- 0..1 CodeableConcept A category for this property, e.g. Physical, Chemical, Enzymatic --></category>
 <code><!-- 0..1 CodeableConcept Property type e.g. viscosity, pH, isoelectric point --></code>
 <parameters value="[string]"/><!-- 0..1 Parameters that were used in the measurement of a property (e.g. for
   viscosity: measured at 20C with a pH of 7.1) -->
 <definingSubstance[x]><!-- 0..1 Reference(SubstanceSpecification|Substance)|
   CodeableConcept A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol)
   --></definingSubstance[x]>
 <amount[x]><!-- 0..1 Quantity|string Quantitative value for this property --></amount[x]>
</property>
<referenceInformation><!-- 0..1 Reference(SubstanceReferenceInformation) General information detailing this substance
   --></referenceInformation>
<structure>  <!-- 0..1 Structural information -->
 <stereochemistry><!-- 0..1 CodeableConcept Stereochemistry type --></stereochemistry>
 <opticalActivity><!-- 0..1 CodeableConcept Optical activity type --></opticalActivity>
 <molecularFormula value="[string]"/><!-- 0..1 Molecular formula -->
 <molecularFormulaByMoiety value="[string]"/><!-- 0..1 Specified per moiety according to the Hill system, i.e. first C,
    then H, then alphabetical, each moiety separated by a dot -->
 <isotope>  <!-- 0..* Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio -->
  <identifier><!-- 0..1 Identifier Substance identifier for each non-natural or radioisotope --></identifier>
  <name><!-- 0..1 CodeableConcept Substance name for each non-natural or radioisotope --></name>
  <substitution><!-- 0..1 CodeableConcept The type of isotopic substitution present in a single substance
   --></substitution>
  <halfLife><!-- 0..1 Quantity Half life - for a non-natural nuclide --></halfLife>
  <molecularWeight>  <!-- 0..1 The molecular weight or weight range (for proteins, polymers or nucleic acids) -->
   <method><!-- 0..1 CodeableConcept The method by which the molecular weight was determined --></method>
   <type><!-- 0..1 CodeableConcept Type of molecular weight such as exact, average (also known as. number average),
   weight average --></type>
   <amount><!-- 0..1 Quantity Used to capture quantitative values for a variety of elements. If only limits are given,
   the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be
   captured in this field --></amount>
  </molecularWeight>
 </isotope>
 <molecularWeight><!-- 0..1 Content as for SubstanceSpecification.structure.isotope.molecularWeight The molecular
   weight or weight range (for proteins, polymers or nucleic acids) --></molecularWeight>
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 <representation>  <!-- 0..* Molecular structural representation -->
  <type><!-- 0..1 CodeableConcept The type of structure (e.g. Full, Partial, Representative) --></type>
  <representation value="[string]"/><!-- 0..1 The structural representation as text string in a format e.g. InChI,
   SMILES, MOLFILE, CDX -->
  <attachment><!-- 0..1 Attachment An attached file with the structural representation --></attachment>
 </representation>
</structure>
<code>  <!-- 0..* Codes associated with the substance -->
 <code><!-- 0..1 CodeableConcept The specific code --></code>
 <status><!-- 0..1 CodeableConcept Status of the code assignment --></status>
 <statusDate value="[dateTime]"/><!-- 0..1 The date at which the code status is changed as part of the terminology
   maintenance -->
 <comment value="[string]"/><!-- 0..1 Any comment can be provided in this field, if necessary -->
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
</code>
<name>  <!-- 0..* Names applicable to this substance -->
 <name value="[string]"/><!-- 1..1 The actual name -->
 <type><!-- 0..1 CodeableConcept Name type --></type>
 <status><!-- 0..1 CodeableConcept The status of the name --></status>
 <preferred value="[boolean]"/><!-- 0..1 If this is the preferred name for this substance -->
 <language><!-- 0..* CodeableConcept Language of the name --></language>
 <domain><!-- 0..* CodeableConcept The use context of this name for example if there is a different name a drug active
   ingredient as opposed to a food colour additive --></domain>
 <jurisdiction><!-- 0..* CodeableConcept The jurisdiction where this name applies --></jurisdiction>
 <synonym><!-- 0..* Content as for SubstanceSpecification.name A synonym of this name --></synonym>
 <translation><!-- 0..* Content as for SubstanceSpecification.name A translation for this name --></translation>
 <official>  <!-- 0..* Details of the official nature of this name -->
  <authority><!-- 0..1 CodeableConcept Which authority uses this official name --></authority>
  <status><!-- 0..1 CodeableConcept The status of the official name --></status>
  <date value="[dateTime]"/><!-- 0..1 Date of official name change -->
 </official>
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
</name>
<molecularWeight><!-- 0..* Content as for SubstanceSpecification.structure.isotope.molecularWeight The molecular weight
    or weight range (for proteins, polymers or nucleic acids) --></molecularWeight>
<relationship>  <!-- 0..* A link between this substance and another, with details of the relationship -->
 <substance[x]><!-- 0..1 Reference(SubstanceSpecification)|CodeableConcept A pointer to another substance, as a
   resource or just a representational code --></substance[x]>
 <relationship><!-- 0..1 CodeableConcept For example "salt to parent", "active moiety", "starting material"
   --></relationship>
 <isDefining value="[boolean]"/><!-- 0..1 For example where an enzyme strongly bonds with a particular substance, this
   is a defining relationship for that enzyme, out of several possible substance relationships -->
 <amount[x]><!-- 0..1 Quantity|Range|Ratio|string A numeric factor for the relationship, for instance to express that
   the salt of a substance has some percentage of the active substance in relation to some other --></amount[x]>
 <amountRatioLowLimit><!-- 0..1 Ratio For use when the numeric --></amountRatioLowLimit>
 <amountType><!-- 0..1 CodeableConcept An operator for the amount, for example "average", "approximately", "less than"
   --></amountType>
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
```

```
 </relationship>
 <nucleicAcid><!-- 0..1 Reference(SubstanceNucleicAcid) Data items specific to nucleic acids --></nucleicAcid>
 <polymer><!-- 0..1 Reference(SubstancePolymer) Data items specific to polymers --></polymer>
 <protein><!-- 0..1 Reference(SubstanceProtein) Data items specific to proteins --></protein>
 <sourceMaterial><!-- 0..1 Reference(SubstanceSourceMaterial) Material or taxonomic/anatomical source for the substance
    --></sourceMaterial>
</SubstanceSpecification>
```

JSON Template

```
{
  "resourceType" : "SubstanceSpecification",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier by which this substance is known
  "type" : { CodeableConcept }, // High level categorization, e.g. polymer or nucleic acid
  "status" : { CodeableConcept }, // Status of substance within the catalogue e.g. approved
  "domain" : { CodeableConcept }, // If the substance applies to only human or veterinary use
  "description" : "<string>", // Textual description of the substance
  "source" : [{ Reference(DocumentReference) }], // Supporting literature
  "comment" : "<string>", // Textual comment about this record of a substance
  "moiety" : [{ // Moiety, for structural modifications
    "role" : { CodeableConcept }, // Role that the moiety is playing
    "identifier" : { Identifier }, // Identifier by which this moiety substance is known
    "name" : "<string>", // Textual name for this moiety substance
    "stereochemistry" : { CodeableConcept }, // Stereochemistry type
    "opticalActivity" : { CodeableConcept }, // Optical activity type
    "molecularFormula" : "<string>", // Molecular formula
    // amount[x]: Quantitative value for this moiety. One of these 2:
    "amountQuantity" : { Quantity }
    "amountString" : "<string>"
  }],
  "property" : [{ // General specifications for this substance, including how it is related to other substances
    "category" : { CodeableConcept }, // A category for this property, e.g. Physical, Chemical, Enzymatic
    "code" : { CodeableConcept }, // Property type e.g. viscosity, pH, isoelectric point
    "parameters" : "<string>", // Parameters that were used in the measurement of a property (e.g. for viscosity:
    measured at 20C with a pH of 7.1)
    // definingSubstance[x]:
      A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol). One of these 2:
    "definingSubstanceReference" : { Reference(SubstanceSpecification|Substance) },
    "definingSubstanceCodeableConcept" : { CodeableConcept },
    // amount[x]: Quantitative value for this property. One of these 2:
    "amountQuantity" : { Quantity }
    "amountString" : "<string>"
  }],
  "referenceInformation" : { Reference(SubstanceReferenceInformation) }, // General information detailing this substance
  "structure" : { // Structural information
    "stereochemistry" : { CodeableConcept }, // Stereochemistry type
    "opticalActivity" : { CodeableConcept }, // Optical activity type
    "molecularFormula" : "<string>", // Molecular formula
    "molecularFormulaByMoiety" : "<string>", // Specified per moiety according to the Hill system, i.e. first C, then H,
     then alphabetical, each moiety separated by a dot
    "isotope" : [{ // Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio
      "identifier" : { Identifier }, // Substance identifier for each non-natural or radioisotope
      "name" : { CodeableConcept }, // Substance name for each non-natural or radioisotope
      "substitution" : { CodeableConcept }, // The type of isotopic substitution present in a single substance
      "halfLife" : { Quantity }, // Half life - for a non-natural nuclide
      "molecularWeight" : { // The molecular weight or weight range (for proteins, polymers or nucleic acids)
        "method" : { CodeableConcept }, // The method by which the molecular weight was determined
        "type" : { CodeableConcept }, // Type of molecular weight such as exact, average (also known as. number
    average), weight average
        "amount" : { Quantity } // Used to capture quantitative values for a variety of elements. If only limits are
    given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it
    would be captured in this field
      }
    }],
    "molecularWeight" : { Content as for SubstanceSpecification.structure.isotope.molecularWeight }, // The molecular
    weight or weight range (for proteins, polymers or nucleic acids)
    "source" : [{ Reference(DocumentReference) }], // Supporting literature
    "representation" : [{ // Molecular structural representation
      "type" : { CodeableConcept }, // The type of structure (e.g. Full, Partial, Representative)
      "representation" : "<string>", // The structural representation as text string in a format e.g. InChI, SMILES,
    MOLFILE, CDX
      "attachment" : { Attachment } // An attached file with the structural representation
    }]
  },
  "code" : [{ // Codes associated with the substance
    "code" : { CodeableConcept }, // The specific code
    "status" : { CodeableConcept }, // Status of the code assignment
    "statusDate" : "<dateTime>", // The date at which the code status is changed as part of the terminology maintenance
    "comment" : "<string>", // Any comment can be provided in this field, if necessary
    "source" : [{ Reference(DocumentReference) }] // Supporting literature
  }],
  "name" : [{ // Names applicable to this substance
    "name" : "<string>", // R!  The actual name
    "type" : { CodeableConcept }, // Name type
    "status" : { CodeableConcept }, // The status of the name
    "preferred" : <boolean>, // If this is the preferred name for this substance
    "language" : [{ CodeableConcept }], // Language of the name
    "domain" : [{ CodeableConcept }], // The use context of this name for example if there is a different name a drug
```

```
    active ingredient as opposed to a food colour additive
    "jurisdiction" : [{ CodeableConcept }], // The jurisdiction where this name applies
    "synonym" : [{ Content as for SubstanceSpecification.name }], // A synonym of this name
    "translation" : [{ Content as for SubstanceSpecification.name }], // A translation for this name
    "official" : [{ // Details of the official nature of this name
      "authority" : { CodeableConcept }, // Which authority uses this official name
      "status" : { CodeableConcept }, // The status of the official name
      "date" : "<dateTime>" // Date of official name change
    }],
    "source" : [{ Reference(DocumentReference) }] // Supporting literature
  }],
  "molecularWeight" : [{ Content as for SubstanceSpecification.structure.isotope.molecularWeight }], // The molecular
    weight or weight range (for proteins, polymers or nucleic acids)
  "relationship" : [{ // A link between this substance and another, with details of the relationship
    // substance[x]: A pointer to another substance, as a resource or just a representational code. One of these 2:
    "substanceReference" : { Reference(SubstanceSpecification) },
    "substanceCodeableConcept" : { CodeableConcept },
    "relationship" : { CodeableConcept }, // For example "salt to parent", "active moiety", "starting material"
    "isDefining" : <boolean>, // For example where an enzyme strongly bonds with a particular substance, this is a
    defining relationship for that enzyme, out of several possible substance relationships
    // amount[x]:
      A numeric factor for the relationship, for instance to express that the salt of a substance has some percentage of
     the active substance in relation to some other. One of these 4:
    "amountQuantity" : { Quantity },
    "amountRange" : { Range },
    "amountRatio" : { Ratio },
    "amountString" : "<string>",
    "amountRatioLowLimit" : { Ratio }, // For use when the numeric
    "amountType" : { CodeableConcept }, // An operator for the amount, for example "average", "approximately", "less
    than"
    "source" : [{ Reference(DocumentReference) }] // Supporting literature
  }],
  "nucleicAcid" : { Reference(SubstanceNucleicAcid) }, // Data items specific to nucleic acids
  "polymer" : { Reference(SubstancePolymer) }, // Data items specific to polymers
  "protein" : { Reference(SubstanceProtein) }, // Data items specific to proteins
  "sourceMaterial" : { Reference(SubstanceSourceMaterial) } // Material or taxonomic/anatomical source for the substance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceSpecification;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceSpecification.identifier [ Identifier ]; # 0..1 Identifier by which this substance is known
  fhir:SubstanceSpecification.type [ CodeableConcept ]; # 0..1 High level categorization, e.g. polymer or nucleic acid
  fhir:SubstanceSpecification.status [ CodeableConcept ]; # 0..1 Status of substance within the catalogue e.g. approved
  fhir:SubstanceSpecification.domain [ CodeableConcept ]; # 0..1 If the substance applies to only human or veterinary
    use
  fhir:SubstanceSpecification.description [ string ]; # 0..1 Textual description of the substance
  fhir:SubstanceSpecification.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  fhir:SubstanceSpecification.comment [ string ]; # 0..1 Textual comment about this record of a substance
  fhir:SubstanceSpecification.moiety [ # 0..* Moiety, for structural modifications
    fhir:SubstanceSpecification.moiety.role [ CodeableConcept ]; # 0..1 Role that the moiety is playing
    fhir:SubstanceSpecification.moiety.identifier [ Identifier ]; # 0..1 Identifier by which this moiety substance is
    known
    fhir:SubstanceSpecification.moiety.name [ string ]; # 0..1 Textual name for this moiety substance
    fhir:SubstanceSpecification.moiety.stereochemistry [ CodeableConcept ]; # 0..1 Stereochemistry type
    fhir:SubstanceSpecification.moiety.opticalActivity [ CodeableConcept ]; # 0..1 Optical activity type
    fhir:SubstanceSpecification.moiety.molecularFormula [ string ]; # 0..1 Molecular formula
    # SubstanceSpecification.moiety.amount[x] : 0..1 Quantitative value for this moiety. One of these 2
      fhir:SubstanceSpecification.moiety.amountQuantity [ Quantity ]
      fhir:SubstanceSpecification.moiety.amountString [ string ]
  ], ...;
  fhir:SubstanceSpecification.property [ # 0..* General specifications for this substance, including how it is related
    to other substances
    fhir:SubstanceSpecification.property.category [ CodeableConcept ]; # 0..1 A category for this property, e.g.
    Physical, Chemical, Enzymatic
    fhir:SubstanceSpecification.property.code [ CodeableConcept ]; # 0..1 Property type e.g. viscosity, pH, isoelectric
    point
    fhir:SubstanceSpecification.property.parameters [ string ]; # 0..1 Parameters that were used in the measurement of a
     property (e.g. for viscosity: measured at 20C with a pH of 7.1)
    # SubstanceSpecification.property.definingSubstance[x] : 0..1 A substance upon which a defining property depends
    (e.g. for solubility: in water, in alcohol). One of these 2
      fhir:SubstanceSpecification.property.definingSubstanceReference [ Reference(SubstanceSpecification|Substance) ]
      fhir:SubstanceSpecification.property.definingSubstanceCodeableConcept [ CodeableConcept ]
    # SubstanceSpecification.property.amount[x] : 0..1 Quantitative value for this property. One of these 2
      fhir:SubstanceSpecification.property.amountQuantity [ Quantity ]
      fhir:SubstanceSpecification.property.amountString [ string ]
  ], ...;
  fhir:SubstanceSpecification.referenceInformation [ Reference(SubstanceReferenceInformation) ]; # 0..1 General
    information detailing this substance
  fhir:SubstanceSpecification.structure [ # 0..1 Structural information
    fhir:SubstanceSpecification.structure.stereochemistry [ CodeableConcept ]; # 0..1 Stereochemistry type
    fhir:SubstanceSpecification.structure.opticalActivity [ CodeableConcept ]; # 0..1 Optical activity type
    fhir:SubstanceSpecification.structure.molecularFormula [ string ]; # 0..1 Molecular formula
```

```
  fhir:SubstanceSpecification.structure.molecularFormulaByMoiety [ string ]; # 0..1 Specified per moiety according to
  the Hill system, i.e. first C, then H, then alphabetical, each moiety separated by a dot
  fhir:SubstanceSpecification.structure.isotope [ # 0..* Applicable for single substances that contain a radionuclide
  or a non-natural isotopic ratio
    fhir:SubstanceSpecification.structure.isotope.identifier [ Identifier ]; # 0..1 Substance identifier for each non-
  natural or radioisotope
    fhir:SubstanceSpecification.structure.isotope.name [ CodeableConcept ]; # 0..1 Substance name for each non-natural
   or radioisotope
    fhir:SubstanceSpecification.structure.isotope.substitution [ CodeableConcept ]; # 0..1 The type of isotopic
  substitution present in a single substance
    fhir:SubstanceSpecification.structure.isotope.halfLife [ Quantity ]; # 0..1 Half life - for a non-natural nuclide
    fhir:SubstanceSpecification.structure.isotope.molecularWeight [ # 0..1 The molecular weight or weight range (for
  proteins, polymers or nucleic acids)
      fhir:SubstanceSpecification.structure.isotope.molecularWeight.method [ CodeableConcept ]; # 0..1 The method by
  which the molecular weight was determined
      fhir:SubstanceSpecification.structure.isotope.molecularWeight.type [ CodeableConcept ]; # 0..1 Type of molecular
   weight such as exact, average (also known as. number average), weight average
      fhir:SubstanceSpecification.structure.isotope.molecularWeight.amount [ Quantity ]; # 0..1 Used to capture
  quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average.
  If only a single definite value for a given element is given, it would be captured in this field
    ];
  ], ...;
  fhir:SubstanceSpecification.structure.molecularWeight [ See SubstanceSpecification.structure.isotope.molecularWeight
   ]; # 0..1 The molecular weight or weight range (for proteins, polymers or nucleic acids)
  fhir:SubstanceSpecification.structure.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  fhir:SubstanceSpecification.structure.representation [ # 0..* Molecular structural representation
    fhir:SubstanceSpecification.structure.representation.type [ CodeableConcept ]; # 0..1 The type of structure (e.g.
  Full, Partial, Representative)
    fhir:SubstanceSpecification.structure.representation.representation [ string ]; # 0..1 The structural
  representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX
    fhir:SubstanceSpecification.structure.representation.attachment [ Attachment ]; # 0..1 An attached file with the
  structural representation
  ], ...;
];
fhir:SubstanceSpecification.code [ # 0..* Codes associated with the substance
  fhir:SubstanceSpecification.code.code [ CodeableConcept ]; # 0..1 The specific code
  fhir:SubstanceSpecification.code.status [ CodeableConcept ]; # 0..1 Status of the code assignment
  fhir:SubstanceSpecification.code.statusDate [ dateTime ]; # 0..1 The date at which the code status is changed as
  part of the terminology maintenance
  fhir:SubstanceSpecification.code.comment [ string ]; # 0..1 Any comment can be provided in this field, if necessary
  fhir:SubstanceSpecification.code.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
], ...;
fhir:SubstanceSpecification.name [ # 0..* Names applicable to this substance
  fhir:SubstanceSpecification.name.name [ string ]; # 1..1 The actual name
  fhir:SubstanceSpecification.name.type [ CodeableConcept ]; # 0..1 Name type
  fhir:SubstanceSpecification.name.status [ CodeableConcept ]; # 0..1 The status of the name
  fhir:SubstanceSpecification.name.preferred [ boolean ]; # 0..1 If this is the preferred name for this substance
  fhir:SubstanceSpecification.name.language [ CodeableConcept ], ... ; # 0..* Language of the name
  fhir:SubstanceSpecification.name.domain [ CodeableConcept ], ... ; # 0..* The use context of this name for example
  if there is a different name a drug active ingredient as opposed to a food colour additive
  fhir:SubstanceSpecification.name.jurisdiction [ CodeableConcept ], ... ; # 0..* The jurisdiction where this name
  applies
  fhir:SubstanceSpecification.name.synonym [ See SubstanceSpecification.name ], ... ; # 0..* A synonym of this name
  fhir:SubstanceSpecification.name.translation [ See SubstanceSpecification.name ], ... ; # 0..* A translation for
  this name
  fhir:SubstanceSpecification.name.official [ # 0..* Details of the official nature of this name
    fhir:SubstanceSpecification.name.official.authority [ CodeableConcept ]; # 0..1 Which authority uses this official
   name
    fhir:SubstanceSpecification.name.official.status [ CodeableConcept ]; # 0..1 The status of the official name
    fhir:SubstanceSpecification.name.official.date [ dateTime ]; # 0..1 Date of official name change
  ], ...;
  fhir:SubstanceSpecification.name.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
], ...;
fhir:SubstanceSpecification.molecularWeight [ See SubstanceSpecification.structure.isotope.molecularWeight ], ... ; #
  0..* The molecular weight or weight range (for proteins, polymers or nucleic acids)
fhir:SubstanceSpecification.relationship [ # 0..* A link between this substance and another, with details of the
  relationship
  # SubstanceSpecification.relationship.substance[x] : 0..1 A pointer to another substance, as a resource or just a
  representational code. One of these 2
    fhir:SubstanceSpecification.relationship.substanceReference [ Reference(SubstanceSpecification) ]
    fhir:SubstanceSpecification.relationship.substanceCodeableConcept [ CodeableConcept ]
  fhir:SubstanceSpecification.relationship.relationship [ CodeableConcept ]; # 0..1 For example "salt to parent",
  "active moiety", "starting material"
  fhir:SubstanceSpecification.relationship.isDefining [ boolean ]; # 0..1 For example where an enzyme strongly bonds
  with a particular substance, this is a defining relationship for that enzyme, out of several possible substance
  relationships
  # SubstanceSpecification.relationship.amount[x] : 0..1 A numeric factor for the relationship, for instance to
  express that the salt of a substance has some percentage of the active substance in relation to some other. One of
  these 4
    fhir:SubstanceSpecification.relationship.amountQuantity [ Quantity ]
    fhir:SubstanceSpecification.relationship.amountRange [ Range ]
    fhir:SubstanceSpecification.relationship.amountRatio [ Ratio ]
    fhir:SubstanceSpecification.relationship.amountString [ string ]
  fhir:SubstanceSpecification.relationship.amountRatioLowLimit [ Ratio ]; # 0..1 For use when the numeric
  fhir:SubstanceSpecification.relationship.amountType [ CodeableConcept ]; # 0..1 An operator for the amount, for
  example "average", "approximately", "less than"
  fhir:SubstanceSpecification.relationship.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
], ...;
fhir:SubstanceSpecification.nucleicAcid [ Reference(SubstanceNucleicAcid) ]; # 0..1 Data items specific to nucleic
```

```
    acids
  fhir:SubstanceSpecification.polymer [ Reference(SubstancePolymer) ]; # 0..1 Data items specific to polymers
  fhir:SubstanceSpecification.protein [ Reference(SubstanceProtein) ]; # 0..1 Data items specific to proteins
  fhir:SubstanceSpecification.sourceMaterial [ Reference(SubstanceSourceMaterial) ]; # 0..1 Material or
    taxonomic/anatomical source for the substance
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceSpecification | ΣTU | | DomainResource | The detailed description of a substance, typically at a level beyond what is used for prescribing Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Identifier by which this substance is known
type | Σ | 0..1 | CodeableConcept | High level categorization, e.g. polymer or nucleic acid
status | Σ | 0..1 | CodeableConcept | Status of substance within the catalogue e.g. approved
domain | Σ | 0..1 | CodeableConcept | If the substance applies to only human or veterinary use
description | Σ | 0..1 | string | Textual description of the substance
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
comment | Σ | 0..1 | string | Textual comment about this record of a substance
moiety | Σ | 0..* | BackboneElement | Moiety, for structural modifications
role | Σ | 0..1 | CodeableConcept | Role that the moiety is playing
identifier | Σ | 0..1 | Identifier | Identifier by which this moiety substance is known
name | Σ | 0..1 | string | Textual name for this moiety substance
stereochemistry | Σ | 0..1 | CodeableConcept | Stereochemistry type
opticalActivity | Σ | 0..1 | CodeableConcept | Optical activity type
molecularFormula | Σ | 0..1 | string | Molecular formula
amount[x] | Σ | 0..1 | | Quantitative value for this moiety
amountQuantity | | | Quantity
amountString | | | string
property | Σ | 0..* | BackboneElement | General specifications for this substance, including how it is related to other substances
category | Σ | 0..1 | CodeableConcept | A category for this property, e.g. Physical, Chemical, Enzymatic
code | Σ | 0..1 | CodeableConcept | Property type e.g. viscosity, pH, isoelectric point
parameters | Σ | 0..1 | string | Parameters that were used in the measurement of a property (e.g. for viscosity: measured at 20C with a pH of 7.1)
definingSubstance[x] | Σ | 0..1 | | A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol)
definingSubstanceReference | | | Reference(SubstanceSpecification | Substance)
definingSubstanceCodeableConcept | | | CodeableConcept
amount[x] | Σ | 0..1 | | Quantitative value for this property
amountQuantity | | | Quantity
amountString | | | string
referenceInformation | Σ | 0..1 | Reference(SubstanceReferenceInformation) | General information detailing this substance
structure | Σ | 0..1 | BackboneElement | Structural information
stereochemistry | Σ | 0..1 | CodeableConcept | Stereochemistry type
opticalActivity | Σ | 0..1 | CodeableConcept | Optical activity type
molecularFormula | Σ | 0..1 | string | Molecular formula
molecularFormulaByMoiety | Σ | 0..1 | string | Specified per moiety according to the Hill system, i.e. first C, then H, then alphabetical, each moiety separated by a dot
isotope | Σ | 0..* | BackboneElement | Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio
identifier | Σ | 0..1 | Identifier | Substance identifier for each non-natural or radioisotope
name | Σ | 0..1 | CodeableConcept | Substance name for each non-natural or radioisotope
substitution | Σ | 0..1 | CodeableConcept | The type of isotopic substitution present in a single substance
halfLife | Σ | 0..1 | Quantity | Half life - for a non-natural nuclide
molecularWeight | Σ | 0..1 | BackboneElement | The molecular weight or weight range (for proteins, polymers or nucleic acids)
method | Σ | 0..1 | CodeableConcept | The method by which the molecular weight was determined
type | Σ | 0..1 | CodeableConcept | Type of molecular weight such as exact, average (also known as. number average), weight average
amount | Σ | 0..1 | Quantity | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field
molecularWeight | Σ | 0..1 | see molecularWeight | The molecular weight or weight range (for proteins, polymers or nucleic acids)
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
representation | Σ | 0..* | BackboneElement | Molecular structural representation
type | Σ | 0..1 | CodeableConcept | The type of structure (e.g. Full, Partial, Representative)
representation | Σ | 0..1 | string | The structural representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX
attachment | Σ | 0..1 | Attachment | An attached file with the structural representation
code | Σ | 0..* | BackboneElement | Codes associated with the substance
code | Σ | 0..1 | CodeableConcept | The specific code
status | Σ | 0..1 | CodeableConcept | Status of the code assignment
statusDate | Σ | 0..1 | dateTime | The date at which the code status is changed as part of the terminology maintenance
comment | Σ | 0..1 | string | Any comment can be provided in this field, if necessary
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
name | Σ | 0..* | BackboneElement | Names applicable to this substance

name | Σ | 1..1 | string | The actual name
type | Σ | 0..1 | CodeableConcept | Name type
status | Σ | 0..1 | CodeableConcept | The status of the name
preferred | Σ | 0..1 | boolean | If this is the preferred name for this substance
language | Σ | 0..* | CodeableConcept | Language of the name
domain | Σ | 0..* | CodeableConcept | The use context of this name for example if there is a different name a drug active ingredient as opposed to a food colour additive
jurisdiction | Σ | 0..* | CodeableConcept | The jurisdiction where this name applies
synonym | Σ | 0..* | see name | A synonym of this name
translation | Σ | 0..* | see name | A translation for this name
official | Σ | 0..* | BackboneElement | Details of the official nature of this name
authority | Σ | 0..1 | CodeableConcept | Which authority uses this official name
status | Σ | 0..1 | CodeableConcept | The status of the official name
date | Σ | 0..1 | dateTime | Date of official name change
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
molecularWeight | Σ | 0..* | see molecularWeight | The molecular weight or weight range (for proteins, polymers or nucleic acids)
relationship | Σ | 0..* | BackboneElement | A link between this substance and another, with details of the relationship
substance[x] | Σ | 0..1 | | A pointer to another substance, as a resource or just a representational code
substanceReference | | | Reference(SubstanceSpecification)
substanceCodeableConcept | | | CodeableConcept
relationship | Σ | 0..1 | CodeableConcept | For example "salt to parent", "active moiety", "starting material"
isDefining | Σ | 0..1 | boolean | For example where an enzyme strongly bonds with a particular substance, this is a defining relationship for that enzyme, out of several possible substance relationships
amount[x] | Σ | 0..1 | | A numeric factor for the relationship, for instance to express that the salt of a substance has some percentage of the active substance in relation to some other
amountQuantity | | | Quantity
amountRange | | | Range
amountRatio | | | Ratio
amountString | | | string
amountRatioLowLimit | Σ | 0..1 | Ratio | For use when the numeric
amountType | Σ | 0..1 | CodeableConcept | An operator for the amount, for example "average", "approximately", "less than"
source | Σ | 0..* | Reference(DocumentReference) | Supporting literature
nucleicAcid | Σ | 0..1 | Reference(SubstanceNucleicAcid) | Data items specific to nucleic acids
polymer | Σ | 0..1 | Reference(SubstancePolymer) | Data items specific to polymers
protein | Σ | 0..1 | Reference(SubstanceProtein) | Data items specific to proteins
sourceMaterial | Σ | 0..1 | Reference(SubstanceSourceMaterial) | Material or taxonomic/anatomical source for the substance
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceSpecification xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier by which this substance is known --></identifier>
 <type><!-- 0..1 CodeableConcept High level categorization, e.g. polymer or nucleic acid --></type>
 <status><!-- 0..1 CodeableConcept Status of substance within the catalogue e.g. approved --></status>
 <domain><!-- 0..1 CodeableConcept If the substance applies to only human or veterinary use --></domain>
 <description value="[string]"/><!-- 0..1 Textual description of the substance -->
 <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 <comment value="[string]"/><!-- 0..1 Textual comment about this record of a substance -->
 <moiety>  <!-- 0..* Moiety, for structural modifications -->
  <role><!-- 0..1 CodeableConcept Role that the moiety is playing --></role>
  <identifier><!-- 0..1 Identifier Identifier by which this moiety substance is known --></identifier>
  <name value="[string]"/><!-- 0..1 Textual name for this moiety substance -->
  <stereochemistry><!-- 0..1 CodeableConcept Stereochemistry type --></stereochemistry>
  <opticalActivity><!-- 0..1 CodeableConcept Optical activity type --></opticalActivity>
  <molecularFormula value="[string]"/><!-- 0..1 Molecular formula -->
  <amount[x]><!-- 0..1 Quantity|string Quantitative value for this moiety --></amount[x]>
 </moiety>
 <property>  <!-- 0..* General specifications for this substance, including how it is related to other substances -->
  <category><!-- 0..1 CodeableConcept A category for this property, e.g. Physical, Chemical, Enzymatic --></category>
  <code><!-- 0..1 CodeableConcept Property type e.g. viscosity, pH, isoelectric point --></code>
  <parameters value="[string]"/><!-- 0..1 Parameters that were used in the measurement of a property (e.g. for
    viscosity: measured at 20C with a pH of 7.1) -->
  <definingSubstance[x]><!-- 0..1 Reference(SubstanceSpecification|Substance)|
    CodeableConcept A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol)
    --></definingSubstance[x]>
  <amount[x]><!-- 0..1 Quantity|string Quantitative value for this property --></amount[x]>
 </property>
 <referenceInformation><!-- 0..1 Reference(SubstanceReferenceInformation) General information detailing this substance
    --></referenceInformation>
 <structure>  <!-- 0..1 Structural information -->
  <stereochemistry><!-- 0..1 CodeableConcept Stereochemistry type --></stereochemistry>
  <opticalActivity><!-- 0..1 CodeableConcept Optical activity type --></opticalActivity>
  <molecularFormula value="[string]"/><!-- 0..1 Molecular formula -->
  <molecularFormulaByMoiety value="[string]"/><!-- 0..1 Specified per moiety according to the Hill system, i.e. first C,
     then H, then alphabetical, each moiety separated by a dot -->
  <isotope>  <!-- 0..* Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio -->
   <identifier><!-- 0..1 Identifier Substance identifier for each non-natural or radioisotope --></identifier>
   <name><!-- 0..1 CodeableConcept Substance name for each non-natural or radioisotope --></name>
```

```
   <substitution><!-- 0..1 CodeableConcept The type of isotopic substitution present in a single substance
    --></substitution>
   <halfLife><!-- 0..1 Quantity Half life - for a non-natural nuclide --></halfLife>
   <molecularWeight>  <!-- 0..1 The molecular weight or weight range (for proteins, polymers or nucleic acids) -->
    <method><!-- 0..1 CodeableConcept The method by which the molecular weight was determined --></method>
    <type><!-- 0..1 CodeableConcept Type of molecular weight such as exact, average (also known as. number average),
    weight average --></type>
    <amount><!-- 0..1 Quantity Used to capture quantitative values for a variety of elements. If only limits are given,
    the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be
    captured in this field --></amount>
   </molecularWeight>
  </isotope>
  <molecularWeight><!-- 0..1 Content as for SubstanceSpecification.structure.isotope.molecularWeight The molecular
    weight or weight range (for proteins, polymers or nucleic acids) --></molecularWeight>
  <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
  <representation>  <!-- 0..* Molecular structural representation -->
   <type><!-- 0..1 CodeableConcept The type of structure (e.g. Full, Partial, Representative) --></type>
   <representation value="[string]"/><!-- 0..1 The structural representation as text string in a format e.g. InChI,
    SMILES, MOLFILE, CDX -->
   <attachment><!-- 0..1 Attachment An attached file with the structural representation --></attachment>
  </representation>
 </structure>
 <code>  <!-- 0..* Codes associated with the substance -->
  <code><!-- 0..1 CodeableConcept The specific code --></code>
  <status><!-- 0..1 CodeableConcept Status of the code assignment --></status>
  <statusDate value="[dateTime]"/><!-- 0..1 The date at which the code status is changed as part of the terminology
    maintenance -->
  <comment value="[string]"/><!-- 0..1 Any comment can be provided in this field, if necessary -->
  <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 </code>
 <name>  <!-- 0..* Names applicable to this substance -->
  <name value="[string]"/><!-- 1..1 The actual name -->
  <type><!-- 0..1 CodeableConcept Name type --></type>
  <status><!-- 0..1 CodeableConcept The status of the name --></status>
  <preferred value="[boolean]"/><!-- 0..1 If this is the preferred name for this substance -->
  <language><!-- 0..* CodeableConcept Language of the name --></language>
  <domain><!-- 0..* CodeableConcept The use context of this name for example if there is a different name a drug active
    ingredient as opposed to a food colour additive --></domain>
  <jurisdiction><!-- 0..* CodeableConcept The jurisdiction where this name applies --></jurisdiction>
  <synonym><!-- 0..* Content as for SubstanceSpecification.name A synonym of this name --></synonym>
  <translation><!-- 0..* Content as for SubstanceSpecification.name A translation for this name --></translation>
  <official>  <!-- 0..* Details of the official nature of this name -->
   <authority><!-- 0..1 CodeableConcept Which authority uses this official name --></authority>
   <status><!-- 0..1 CodeableConcept The status of the official name --></status>
   <date value="[dateTime]"/><!-- 0..1 Date of official name change -->
  </official>
  <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 </name>
 <molecularWeight><!-- 0..* Content as for SubstanceSpecification.structure.isotope.molecularWeight The molecular weight
     or weight range (for proteins, polymers or nucleic acids) --></molecularWeight>
 <relationship>  <!-- 0..* A link between this substance and another, with details of the relationship -->
  <substance[x]><!-- 0..1 Reference(SubstanceSpecification)|CodeableConcept A pointer to another substance, as a
    resource or just a representational code --></substance[x]>
  <relationship><!-- 0..1 CodeableConcept For example "salt to parent", "active moiety", "starting material"
    --></relationship>
  <isDefining value="[boolean]"/><!-- 0..1 For example where an enzyme strongly bonds with a particular substance, this
    is a defining relationship for that enzyme, out of several possible substance relationships -->
  <amount[x]><!-- 0..1 Quantity|Range|Ratio|string A numeric factor for the relationship, for instance to express that
    the salt of a substance has some percentage of the active substance in relation to some other --></amount[x]>
  <amountRatioLowLimit><!-- 0..1 Ratio For use when the numeric --></amountRatioLowLimit>
  <amountType><!-- 0..1 CodeableConcept An operator for the amount, for example "average", "approximately", "less than"
    --></amountType>
  <source><!-- 0..* Reference(DocumentReference) Supporting literature --></source>
 </relationship>
 <nucleicAcid><!-- 0..1 Reference(SubstanceNucleicAcid) Data items specific to nucleic acids --></nucleicAcid>
 <polymer><!-- 0..1 Reference(SubstancePolymer) Data items specific to polymers --></polymer>
 <protein><!-- 0..1 Reference(SubstanceProtein) Data items specific to proteins --></protein>
 <sourceMaterial><!-- 0..1 Reference(SubstanceSourceMaterial) Material or taxonomic/anatomical source for the substance
    --></sourceMaterial>
</SubstanceSpecification>
```

JSON Template

```
{
  "resourceType" : "SubstanceSpecification",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier by which this substance is known
  "type" : { CodeableConcept }, // High level categorization, e.g. polymer or nucleic acid
  "status" : { CodeableConcept }, // Status of substance within the catalogue e.g. approved
  "domain" : { CodeableConcept }, // If the substance applies to only human or veterinary use
  "description" : "<string>", // Textual description of the substance
  "source" : [{ Reference(DocumentReference) }], // Supporting literature
  "comment" : "<string>", // Textual comment about this record of a substance
  "moiety" : [{ // Moiety, for structural modifications
    "role" : { CodeableConcept }, // Role that the moiety is playing
    "identifier" : { Identifier }, // Identifier by which this moiety substance is known
    "name" : "<string>", // Textual name for this moiety substance
    "stereochemistry" : { CodeableConcept }, // Stereochemistry type
    "opticalActivity" : { CodeableConcept }, // Optical activity type
```

```
  "molecularFormula" : "<string>", // Molecular formula
  // amount[x]: Quantitative value for this moiety. One of these 2:
  "amountQuantity" : { Quantity }
  "amountString" : "<string>"
}],
"property" : [{ // General specifications for this substance, including how it is related to other substances
  "category" : { CodeableConcept }, // A category for this property, e.g. Physical, Chemical, Enzymatic
  "code" : { CodeableConcept }, // Property type e.g. viscosity, pH, isoelectric point
  "parameters" : "<string>", // Parameters that were used in the measurement of a property (e.g. for viscosity:
  measured at 20C with a pH of 7.1)
  // definingSubstance[x]:
    A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol). One of these 2:
  "definingSubstanceReference" : { Reference(SubstanceSpecification|Substance) },
  "definingSubstanceCodeableConcept" : { CodeableConcept },
  // amount[x]: Quantitative value for this property. One of these 2:
  "amountQuantity" : { Quantity }
  "amountString" : "<string>"
}],
"referenceInformation" : { Reference(SubstanceReferenceInformation) }, // General information detailing this substance
"structure" : { // Structural information
  "stereochemistry" : { CodeableConcept }, // Stereochemistry type
  "opticalActivity" : { CodeableConcept }, // Optical activity type
  "molecularFormula" : "<string>", // Molecular formula
  "molecularFormulaByMoiety" : "<string>", // Specified per moiety according to the Hill system, i.e. first C, then H,
   then alphabetical, each moiety separated by a dot
  "isotope" : [{ // Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio
    "identifier" : { Identifier }, // Substance identifier for each non-natural or radioisotope
    "name" : { CodeableConcept }, // Substance name for each non-natural or radioisotope
    "substitution" : { CodeableConcept }, // The type of isotopic substitution present in a single substance
    "halfLife" : { Quantity }, // Half life - for a non-natural nuclide
    "molecularWeight" : { // The molecular weight or weight range (for proteins, polymers or nucleic acids)
      "method" : { CodeableConcept }, // The method by which the molecular weight was determined
      "type" : { CodeableConcept }, // Type of molecular weight such as exact, average (also known as. number
  average), weight average
      "amount" : { Quantity } // Used to capture quantitative values for a variety of elements. If only limits are
  given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it
  would be captured in this field
    }
  }],
  "molecularWeight" : { Content as for SubstanceSpecification.structure.isotope.molecularWeight }, // The molecular
  weight or weight range (for proteins, polymers or nucleic acids)
  "source" : [{ Reference(DocumentReference) }], // Supporting literature
  "representation" : [{ // Molecular structural representation
    "type" : { CodeableConcept }, // The type of structure (e.g. Full, Partial, Representative)
    "representation" : "<string>", // The structural representation as text string in a format e.g. InChI, SMILES,
  MOLFILE, CDX
    "attachment" : { Attachment } // An attached file with the structural representation
  }]
},
"code" : [{ // Codes associated with the substance
  "code" : { CodeableConcept }, // The specific code
  "status" : { CodeableConcept }, // Status of the code assignment
  "statusDate" : "<dateTime>", // The date at which the code status is changed as part of the terminology maintenance
  "comment" : "<string>", // Any comment can be provided in this field, if necessary
  "source" : [{ Reference(DocumentReference) }] // Supporting literature
}],
"name" : [{ // Names applicable to this substance
  "name" : "<string>", // R!  The actual name
  "type" : { CodeableConcept }, // Name type
  "status" : { CodeableConcept }, // The status of the name
  "preferred" : <boolean>, // If this is the preferred name for this substance
  "language" : [{ CodeableConcept }], // Language of the name
  "domain" : [{ CodeableConcept }], // The use context of this name for example if there is a different name a drug
  active ingredient as opposed to a food colour additive
  "jurisdiction" : [{ CodeableConcept }], // The jurisdiction where this name applies
  "synonym" : [{ Content as for SubstanceSpecification.name }], // A synonym of this name
  "translation" : [{ Content as for SubstanceSpecification.name }], // A translation for this name
  "official" : [{ // Details of the official nature of this name
    "authority" : { CodeableConcept }, // Which authority uses this official name
    "status" : { CodeableConcept }, // The status of the official name
    "date" : "<dateTime>" // Date of official name change
  }],
  "source" : [{ Reference(DocumentReference) }] // Supporting literature
}],
"molecularWeight" : [{ Content as for SubstanceSpecification.structure.isotope.molecularWeight }], // The molecular
  weight or weight range (for proteins, polymers or nucleic acids)
"relationship" : [{ // A link between this substance and another, with details of the relationship
  // substance[x]: A pointer to another substance, as a resource or just a representational code. One of these 2:
  "substanceReference" : { Reference(SubstanceSpecification) },
  "substanceCodeableConcept" : { CodeableConcept },
  "relationship" : { CodeableConcept }, // For example "salt to parent", "active moiety", "starting material"
  "isDefining" : <boolean>, // For example where an enzyme strongly bonds with a particular substance, this is a
  defining relationship for that enzyme, out of several possible substance relationships
  // amount[x]:
    A numeric factor for the relationship, for instance to express that the salt of a substance has some percentage of
   the active substance in relation to some other. One of these 4:
  "amountQuantity" : { Quantity },
  "amountRange" : { Range },
  "amountRatio" : { Ratio },
```

```
    "amountString" : "<string>",
    "amountRatioLowLimit" : { Ratio }, // For use when the numeric
    "amountType" : { CodeableConcept }, // An operator for the amount, for example "average", "approximately", "less
    than"
    "source" : [{ Reference(DocumentReference) }] // Supporting literature
  }],
  "nucleicAcid" : { Reference(SubstanceNucleicAcid) }, // Data items specific to nucleic acids
  "polymer" : { Reference(SubstancePolymer) }, // Data items specific to polymers
  "protein" : { Reference(SubstanceProtein) }, // Data items specific to proteins
  "sourceMaterial" : { Reference(SubstanceSourceMaterial) } // Material or taxonomic/anatomical source for the substance
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceSpecification;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceSpecification.identifier [ Identifier ]; # 0..1 Identifier by which this substance is known
  fhir:SubstanceSpecification.type [ CodeableConcept ]; # 0..1 High level categorization, e.g. polymer or nucleic acid
  fhir:SubstanceSpecification.status [ CodeableConcept ]; # 0..1 Status of substance within the catalogue e.g. approved
  fhir:SubstanceSpecification.domain [ CodeableConcept ]; # 0..1 If the substance applies to only human or veterinary
    use
  fhir:SubstanceSpecification.description [ string ]; # 0..1 Textual description of the substance
  fhir:SubstanceSpecification.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  fhir:SubstanceSpecification.comment [ string ]; # 0..1 Textual comment about this record of a substance
  fhir:SubstanceSpecification.moiety [ # 0..* Moiety, for structural modifications
    fhir:SubstanceSpecification.moiety.role [ CodeableConcept ]; # 0..1 Role that the moiety is playing
    fhir:SubstanceSpecification.moiety.identifier [ Identifier ]; # 0..1 Identifier by which this moiety substance is
    known
    fhir:SubstanceSpecification.moiety.name [ string ]; # 0..1 Textual name for this moiety substance
    fhir:SubstanceSpecification.moiety.stereochemistry [ CodeableConcept ]; # 0..1 Stereochemistry type
    fhir:SubstanceSpecification.moiety.opticalActivity [ CodeableConcept ]; # 0..1 Optical activity type
    fhir:SubstanceSpecification.moiety.molecularFormula [ string ]; # 0..1 Molecular formula
    # SubstanceSpecification.moiety.amount[x] : 0..1 Quantitative value for this moiety. One of these 2
      fhir:SubstanceSpecification.moiety.amountQuantity [ Quantity ]
      fhir:SubstanceSpecification.moiety.amountString [ string ]
  ], ...;
  fhir:SubstanceSpecification.property [ # 0..* General specifications for this substance, including how it is related
    to other substances
    fhir:SubstanceSpecification.property.category [ CodeableConcept ]; # 0..1 A category for this property, e.g.
    Physical, Chemical, Enzymatic
    fhir:SubstanceSpecification.property.code [ CodeableConcept ]; # 0..1 Property type e.g. viscosity, pH, isoelectric
    point
    fhir:SubstanceSpecification.property.parameters [ string ]; # 0..1 Parameters that were used in the measurement of a
     property (e.g. for viscosity: measured at 20C with a pH of 7.1)
    # SubstanceSpecification.property.definingSubstance[x] : 0..1 A substance upon which a defining property depends
    (e.g. for solubility: in water, in alcohol). One of these 2
      fhir:SubstanceSpecification.property.definingSubstanceReference [ Reference(SubstanceSpecification|Substance) ]
      fhir:SubstanceSpecification.property.definingSubstanceCodeableConcept [ CodeableConcept ]
    # SubstanceSpecification.property.amount[x] : 0..1 Quantitative value for this property. One of these 2
      fhir:SubstanceSpecification.property.amountQuantity [ Quantity ]
      fhir:SubstanceSpecification.property.amountString [ string ]
  ], ...;
  fhir:SubstanceSpecification.referenceInformation [ Reference(SubstanceReferenceInformation) ]; # 0..1 General
    information detailing this substance
  fhir:SubstanceSpecification.structure [ # 0..1 Structural information
    fhir:SubstanceSpecification.structure.stereochemistry [ CodeableConcept ]; # 0..1 Stereochemistry type
    fhir:SubstanceSpecification.structure.opticalActivity [ CodeableConcept ]; # 0..1 Optical activity type
    fhir:SubstanceSpecification.structure.molecularFormula [ string ]; # 0..1 Molecular formula
    fhir:SubstanceSpecification.structure.molecularFormulaByMoiety [ string ]; # 0..1 Specified per moiety according to
    the Hill system, i.e. first C, then H, then alphabetical, each moiety separated by a dot
    fhir:SubstanceSpecification.structure.isotope [ # 0..* Applicable for single substances that contain a radionuclide
    or a non-natural isotopic ratio
      fhir:SubstanceSpecification.structure.isotope.identifier [ Identifier ]; # 0..1 Substance identifier for each non-
    natural or radioisotope
      fhir:SubstanceSpecification.structure.isotope.name [ CodeableConcept ]; # 0..1 Substance name for each non-natural
     or radioisotope
      fhir:SubstanceSpecification.structure.isotope.substitution [ CodeableConcept ]; # 0..1 The type of isotopic
    substitution present in a single substance
      fhir:SubstanceSpecification.structure.isotope.halfLife [ Quantity ]; # 0..1 Half life - for a non-natural nuclide
      fhir:SubstanceSpecification.structure.isotope.molecularWeight [ # 0..1 The molecular weight or weight range (for
    proteins, polymers or nucleic acids)
        fhir:SubstanceSpecification.structure.isotope.molecularWeight.method [ CodeableConcept ]; # 0..1 The method by
    which the molecular weight was determined
        fhir:SubstanceSpecification.structure.isotope.molecularWeight.type [ CodeableConcept ]; # 0..1 Type of molecular
     weight such as exact, average (also known as. number average), weight average
        fhir:SubstanceSpecification.structure.isotope.molecularWeight.amount [ Quantity ]; # 0..1 Used to capture
    quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average.
    If only a single definite value for a given element is given, it would be captured in this field
      ];
    ], ...;
    fhir:SubstanceSpecification.structure.molecularWeight [ See SubstanceSpecification.structure.isotope.molecularWeight
     ]; # 0..1 The molecular weight or weight range (for proteins, polymers or nucleic acids)
    fhir:SubstanceSpecification.structure.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
    fhir:SubstanceSpecification.structure.representation [ # 0..* Molecular structural representation
```

```
    fhir:SubstanceSpecification.structure.representation.type [ CodeableConcept ]; # 0..1 The type of structure (e.g.
    Full, Partial, Representative)
    fhir:SubstanceSpecification.structure.representation.representation [ string ]; # 0..1 The structural
    representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX
    fhir:SubstanceSpecification.structure.representation.attachment [ Attachment ]; # 0..1 An attached file with the
    structural representation
    ], ...;
  ];
  fhir:SubstanceSpecification.code [ # 0..* Codes associated with the substance
    fhir:SubstanceSpecification.code.code [ CodeableConcept ]; # 0..1 The specific code
    fhir:SubstanceSpecification.code.status [ CodeableConcept ]; # 0..1 Status of the code assignment
    fhir:SubstanceSpecification.code.statusDate [ dateTime ]; # 0..1 The date at which the code status is changed as
    part of the terminology maintenance
    fhir:SubstanceSpecification.code.comment [ string ]; # 0..1 Any comment can be provided in this field, if necessary
    fhir:SubstanceSpecification.code.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  ], ...;
  fhir:SubstanceSpecification.name [ # 0..* Names applicable to this substance
    fhir:SubstanceSpecification.name.name [ string ]; # 1..1 The actual name
    fhir:SubstanceSpecification.name.type [ CodeableConcept ]; # 0..1 Name type
    fhir:SubstanceSpecification.name.status [ CodeableConcept ]; # 0..1 The status of the name
    fhir:SubstanceSpecification.name.preferred [ boolean ]; # 0..1 If this is the preferred name for this substance
    fhir:SubstanceSpecification.name.language [ CodeableConcept ], ... ; # 0..* Language of the name
    fhir:SubstanceSpecification.name.domain [ CodeableConcept ], ... ; # 0..* The use context of this name for example
    if there is a different name a drug active ingredient as opposed to a food colour additive
    fhir:SubstanceSpecification.name.jurisdiction [ CodeableConcept ], ... ; # 0..* The jurisdiction where this name
    applies
    fhir:SubstanceSpecification.name.synonym [ See SubstanceSpecification.name ], ... ; # 0..* A synonym of this name
    fhir:SubstanceSpecification.name.translation [ See SubstanceSpecification.name ], ... ; # 0..* A translation for
    this name
    fhir:SubstanceSpecification.name.official [ # 0..* Details of the official nature of this name
      fhir:SubstanceSpecification.name.official.authority [ CodeableConcept ]; # 0..1 Which authority uses this official
     name
      fhir:SubstanceSpecification.name.official.status [ CodeableConcept ]; # 0..1 The status of the official name
      fhir:SubstanceSpecification.name.official.date [ dateTime ]; # 0..1 Date of official name change
    ], ...;
    fhir:SubstanceSpecification.name.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  ], ...;
  fhir:SubstanceSpecification.molecularWeight [ See SubstanceSpecification.structure.isotope.molecularWeight ], ... ; #
    0..* The molecular weight or weight range (for proteins, polymers or nucleic acids)
  fhir:SubstanceSpecification.relationship [ # 0..* A link between this substance and another, with details of the
    relationship
    # SubstanceSpecification.relationship.substance[x] : 0..1 A pointer to another substance, as a resource or just a
    representational code. One of these 2
      fhir:SubstanceSpecification.relationship.substanceReference [ Reference(SubstanceSpecification) ]
      fhir:SubstanceSpecification.relationship.substanceCodeableConcept [ CodeableConcept ]
    fhir:SubstanceSpecification.relationship.relationship [ CodeableConcept ]; # 0..1 For example "salt to parent",
    "active moiety", "starting material"
    fhir:SubstanceSpecification.relationship.isDefining [ boolean ]; # 0..1 For example where an enzyme strongly bonds
    with a particular substance, this is a defining relationship for that enzyme, out of several possible substance
    relationships
    # SubstanceSpecification.relationship.amount[x] : 0..1 A numeric factor for the relationship, for instance to
    express that the salt of a substance has some percentage of the active substance in relation to some other. One of
    these 4
      fhir:SubstanceSpecification.relationship.amountQuantity [ Quantity ]
      fhir:SubstanceSpecification.relationship.amountRange [ Range ]
      fhir:SubstanceSpecification.relationship.amountRatio [ Ratio ]
      fhir:SubstanceSpecification.relationship.amountString [ string ]
    fhir:SubstanceSpecification.relationship.amountRatioLowLimit [ Ratio ]; # 0..1 For use when the numeric
    fhir:SubstanceSpecification.relationship.amountType [ CodeableConcept ]; # 0..1 An operator for the amount, for
    example "average", "approximately", "less than"
    fhir:SubstanceSpecification.relationship.source [ Reference(DocumentReference) ], ... ; # 0..* Supporting literature
  ], ...;
  fhir:SubstanceSpecification.nucleicAcid [ Reference(SubstanceNucleicAcid) ]; # 0..1 Data items specific to nucleic
    acids
  fhir:SubstanceSpecification.polymer [ Reference(SubstancePolymer) ]; # 0..1 Data items specific to polymers
  fhir:SubstanceSpecification.protein [ Reference(SubstanceProtein) ]; # 0..1 Data items specific to proteins
  fhir:SubstanceSpecification.sourceMaterial [ Reference(SubstanceSourceMaterial) ]; # 0..1 Material or
    taxonomic/anatomical source for the substance
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.21.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
code	token	The specific code	SubstanceSpecification.code.code	

11.21.4 Resource SubstanceSpecification - Examples

URL: https://hl7.org/fhir/R4/substancespecification-examples.html

11.21.4 Resource SubstanceSpecification - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of substancespecification | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.21.5 Resource SubstanceSpecification - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substancespecification-definitions.html

11.21.5 Resource SubstanceSpecification - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstanceSpecification resource.

SubstanceSpecification
Element Id | SubstanceSpecification
Definition | The detailed description of a substance, typically at a level beyond what is used for prescribing.
Cardinality | 0..*
Type | DomainResource
Summary | true
SubstanceSpecification.identifier
Element Id | SubstanceSpecification.identifier
Definition | Identifier by which this substance is known.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceSpecification.type
Element Id | SubstanceSpecification.type
Definition | High level categorization, e.g. polymer or nucleic acid.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.status
Element Id | SubstanceSpecification.status
Definition | Status of substance within the catalogue e.g. approved.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.domain
Element Id | SubstanceSpecification.domain
Definition | If the substance applies to only human or veterinary use.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.description
Element Id | SubstanceSpecification.description
Definition | Textual description of the substance.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.source
Element Id | SubstanceSpecification.source
Definition | Supporting literature.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceSpecification.comment
Element Id | SubstanceSpecification.comment
Definition | Textual comment about this record of a substance.
Cardinality | 0..1
Type | string

Summary | true
SubstanceSpecification.moiety
Element Id | SubstanceSpecification.moiety
Definition | Moiety, for structural modifications.
Cardinality | 0..*
Summary | true
SubstanceSpecification.moiety.role
Element Id | SubstanceSpecification.moiety.role
Definition | Role that the moiety is playing.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.moiety.identifier
Element Id | SubstanceSpecification.moiety.identifier
Definition | Identifier by which this moiety substance is known.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceSpecification.moiety.name
Element Id | SubstanceSpecification.moiety.name
Definition | Textual name for this moiety substance.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.moiety.stereochemistry
Element Id | SubstanceSpecification.moiety.stereochemistry
Definition | Stereochemistry type.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.moiety.opticalActivity
Element Id | SubstanceSpecification.moiety.opticalActivity
Definition | Optical activity type.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.moiety.molecularFormula
Element Id | SubstanceSpecification.moiety.molecularFormula
Definition | Molecular formula.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.moiety.amount[x]
Element Id | SubstanceSpecification.moiety.amount[x]
Definition | Quantitative value for this moiety.
Cardinality | 0..1
Type | Quantity|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceSpecification.property
Element Id | SubstanceSpecification.property
Definition | General specifications for this substance, including how it is related to other substances.
Cardinality | 0..*
Summary | true
SubstanceSpecification.property.category
Element Id | SubstanceSpecification.property.category
Definition | A category for this property, e.g. Physical, Chemical, Enzymatic.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.property.code
Element Id | SubstanceSpecification.property.code
Definition | Property type e.g. viscosity, pH, isoelectric point.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.property.parameters
Element Id | SubstanceSpecification.property.parameters
Definition | Parameters that were used in the measurement of a property (e.g. for viscosity: measured at 20C with a pH of 7.1).
Cardinality | 0..1
Type | string

Summary | true
SubstanceSpecification.property.definingSubstance[x]
Element Id | SubstanceSpecification.property.definingSubstance[x]
Definition | A substance upon which a defining property depends (e.g. for solubility: in water, in alcohol).
Cardinality | 0..1
Type | Reference(SubstanceSpecification | Substance)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceSpecification.property.amount[x]
Element Id | SubstanceSpecification.property.amount[x]
Definition | Quantitative value for this property.
Cardinality | 0..1
Type | Quantity|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceSpecification.referenceInformation
Element Id | SubstanceSpecification.referenceInformation
Definition | General information detailing this substance.
Cardinality | 0..1
Type | Reference(SubstanceReferenceInformation)
Summary | true
SubstanceSpecification.structure
Element Id | SubstanceSpecification.structure
Definition | Structural information.
Cardinality | 0..1
Summary | true
SubstanceSpecification.structure.stereochemistry
Element Id | SubstanceSpecification.structure.stereochemistry
Definition | Stereochemistry type.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.opticalActivity
Element Id | SubstanceSpecification.structure.opticalActivity
Definition | Optical activity type.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.molecularFormula
Element Id | SubstanceSpecification.structure.molecularFormula
Definition | Molecular formula.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.structure.molecularFormulaByMoiety
Element Id | SubstanceSpecification.structure.molecularFormulaByMoiety
Definition | Specified per moiety according to the Hill system, i.e. first C, then H, then alphabetical, each moiety separated by a dot.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.structure.isotope
Element Id | SubstanceSpecification.structure.isotope
Definition | Applicable for single substances that contain a radionuclide or a non-natural isotopic ratio.
Cardinality | 0..*
Summary | true
SubstanceSpecification.structure.isotope.identifier
Element Id | SubstanceSpecification.structure.isotope.identifier
Definition | Substance identifier for each non-natural or radioisotope.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceSpecification.structure.isotope.name
Element Id | SubstanceSpecification.structure.isotope.name
Definition | Substance name for each non-natural or radioisotope.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.isotope.substitution
Element Id | SubstanceSpecification.structure.isotope.substitution
Definition | The type of isotopic substitution present in a single substance.

Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.isotope.halfLife
Element Id | SubstanceSpecification.structure.isotope.halfLife
Definition | Half life - for a non-natural nuclide.
Cardinality | 0..1
Type | Quantity
Summary | true
SubstanceSpecification.structure.isotope.molecularWeight
Element Id | SubstanceSpecification.structure.isotope.molecularWeight
Definition | The molecular weight or weight range (for proteins, polymers or nucleic acids).
Cardinality | 0..1
Summary | true
SubstanceSpecification.structure.isotope.molecularWeight.method
Element Id | SubstanceSpecification.structure.isotope.molecularWeight.method
Definition | The method by which the molecular weight was determined.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.isotope.molecularWeight.type
Element Id | SubstanceSpecification.structure.isotope.molecularWeight.type
Definition | Type of molecular weight such as exact, average (also known as. number average), weight average.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.isotope.molecularWeight.amount
Element Id | SubstanceSpecification.structure.isotope.molecularWeight.amount
Definition | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field.
Cardinality | 0..1
Type | Quantity
Summary | true
SubstanceSpecification.structure.molecularWeight
Element Id | SubstanceSpecification.structure.molecularWeight
Definition | The molecular weight or weight range (for proteins, polymers or nucleic acids).
Cardinality | 0..1
Type | See SubstanceSpecification.structure.isotope.molecularWeight
Summary | true
SubstanceSpecification.structure.source
Element Id | SubstanceSpecification.structure.source
Definition | Supporting literature.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceSpecification.structure.representation
Element Id | SubstanceSpecification.structure.representation
Definition | Molecular structural representation.
Cardinality | 0..*
Summary | true
SubstanceSpecification.structure.representation.type
Element Id | SubstanceSpecification.structure.representation.type
Definition | The type of structure (e.g. Full, Partial, Representative).
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.structure.representation.representation
Element Id | SubstanceSpecification.structure.representation.representation
Definition | The structural representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.structure.representation.attachment
Element Id | SubstanceSpecification.structure.representation.attachment
Definition | An attached file with the structural representation.
Cardinality | 0..1
Type | Attachment
Summary | true
SubstanceSpecification.code
Element Id | SubstanceSpecification.code
Definition | Codes associated with the substance.
Cardinality | 0..*

Summary | true
SubstanceSpecification.code.code
Element Id | SubstanceSpecification.code.code
Definition | The specific code.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.code.status
Element Id | SubstanceSpecification.code.status
Definition | Status of the code assignment.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.code.statusDate
Element Id | SubstanceSpecification.code.statusDate
Definition | The date at which the code status is changed as part of the terminology maintenance.
Cardinality | 0..1
Type | dateTime
Summary | true
SubstanceSpecification.code.comment
Element Id | SubstanceSpecification.code.comment
Definition | Any comment can be provided in this field, if necessary.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSpecification.code.source
Element Id | SubstanceSpecification.code.source
Definition | Supporting literature.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceSpecification.name
Element Id | SubstanceSpecification.name
Definition | Names applicable to this substance.
Cardinality | 0..*
Summary | true
SubstanceSpecification.name.name
Element Id | SubstanceSpecification.name.name
Definition | The actual name.
Cardinality | 1..1
Type | string
Summary | true
SubstanceSpecification.name.type
Element Id | SubstanceSpecification.name.type
Definition | Name type.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.status
Element Id | SubstanceSpecification.name.status
Definition | The status of the name.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.preferred
Element Id | SubstanceSpecification.name.preferred
Definition | If this is the preferred name for this substance.
Cardinality | 0..1
Type | boolean
Summary | true
SubstanceSpecification.name.language
Element Id | SubstanceSpecification.name.language
Definition | Language of the name.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.domain
Element Id | SubstanceSpecification.name.domain
Definition | The use context of this name for example if there is a different name a drug active ingredient as opposed to a food colour additive.
Cardinality | 0..*
Type | CodeableConcept

Summary | true
SubstanceSpecification.name.jurisdiction
Element Id | SubstanceSpecification.name.jurisdiction
Definition | The jurisdiction where this name applies.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.synonym
Element Id | SubstanceSpecification.name.synonym
Definition | A synonym of this name.
Cardinality | 0..*
Type | See SubstanceSpecification.name
Summary | true
SubstanceSpecification.name.translation
Element Id | SubstanceSpecification.name.translation
Definition | A translation for this name.
Cardinality | 0..*
Type | See SubstanceSpecification.name
Summary | true
SubstanceSpecification.name.official
Element Id | SubstanceSpecification.name.official
Definition | Details of the official nature of this name.
Cardinality | 0..*
Summary | true
SubstanceSpecification.name.official.authority
Element Id | SubstanceSpecification.name.official.authority
Definition | Which authority uses this official name.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.official.status
Element Id | SubstanceSpecification.name.official.status
Definition | The status of the official name.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.name.official.date
Element Id | SubstanceSpecification.name.official.date
Definition | Date of official name change.
Cardinality | 0..1
Type | dateTime
Summary | true
SubstanceSpecification.name.source
Element Id | SubstanceSpecification.name.source
Definition | Supporting literature.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceSpecification.molecularWeight
Element Id | SubstanceSpecification.molecularWeight
Definition | The molecular weight or weight range (for proteins, polymers or nucleic acids).
Cardinality | 0..*
Type | See SubstanceSpecification.structure.isotope.molecularWeight
Summary | true
SubstanceSpecification.relationship
Element Id | SubstanceSpecification.relationship
Definition | A link between this substance and another, with details of the relationship.
Cardinality | 0..*
Summary | true
SubstanceSpecification.relationship.substance[x]
Element Id | SubstanceSpecification.relationship.substance[x]
Definition | A pointer to another substance, as a resource or just a representational code.
Cardinality | 0..1
Type | Reference(SubstanceSpecification)|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceSpecification.relationship.relationship
Element Id | SubstanceSpecification.relationship.relationship
Definition | For example "salt to parent", "active moiety", "starting material".
Cardinality | 0..1
Type | CodeableConcept
Summary | true

SubstanceSpecification.relationship.isDefining
Element Id | SubstanceSpecification.relationship.isDefining
Definition | For example where an enzyme strongly bonds with a particular substance, this is a defining relationship for that enzyme, out of several possible substance relationships.
Cardinality | 0..1
Type | boolean
Summary | true
SubstanceSpecification.relationship.amount[x]
Element Id | SubstanceSpecification.relationship.amount[x]
Definition | A numeric factor for the relationship, for instance to express that the salt of a substance has some percentage of the active substance in relation to some other.
Cardinality | 0..1
Type | Quantity|Range|Ratio|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceSpecification.relationship.amountRatioLowLimit
Element Id | SubstanceSpecification.relationship.amountRatioLowLimit
Definition | For use when the numeric.
Cardinality | 0..1
Type | Ratio
Summary | true
SubstanceSpecification.relationship.amountType
Element Id | SubstanceSpecification.relationship.amountType
Definition | An operator for the amount, for example "average", "approximately", "less than".
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSpecification.relationship.source
Element Id | SubstanceSpecification.relationship.source
Definition | Supporting literature.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceSpecification.nucleicAcid
Element Id | SubstanceSpecification.nucleicAcid
Definition | Data items specific to nucleic acids.
Cardinality | 0..1
Type | Reference(SubstanceNucleicAcid)
Summary | true
SubstanceSpecification.polymer
Element Id | SubstanceSpecification.polymer
Definition | Data items specific to polymers.
Cardinality | 0..1
Type | Reference(SubstancePolymer)
Summary | true
SubstanceSpecification.protein
Element Id | SubstanceSpecification.protein
Definition | Data items specific to proteins.
Cardinality | 0..1
Type | Reference(SubstanceProtein)
Summary | true
SubstanceSpecification.sourceMaterial
Element Id | SubstanceSpecification.sourceMaterial
Definition | Material or taxonomic/anatomical source for the substance.
Cardinality | 0..1
Type | Reference(SubstanceSourceMaterial)
Summary | true

11.21.6 Resource SubstanceSpecification - Mappings

URL: https://hl7.org/fhir/R4/substancespecification-mappings.html

11.21.6 Resource SubstanceSpecification - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substancespecification resource (see Mappings to Other Standards for further information & status).

11.21.7 Resource SubstanceSpecification - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substancespecification-profiles.html

11.21.7 Resource SubstanceSpecification - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstanceSpecification resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.21.7.1 Profiles

No Profiles defined for this resource

11.21.7.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.21.7.3 Search Extensions

No Search Extensions defined for this resource

11.21.8 JSON Schema for SubstanceSpecification

URL: https://hl7.org/fhir/R4/substancespecification.schema.json.html

11.21.8 JSON Schema for SubstanceSpecification

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstanceSpecification, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstanceSpecification",
  "$ref": "#/definitions/SubstanceSpecification",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstanceSpecification": {
      "description": "The detailed description of a substance, typically at a level beyond what is used for
    prescribing.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstanceSpecification resource",
          "const": "SubstanceSpecification"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
```

```
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Identifier by which this substance is known.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "type": {
      "description": "High level categorization, e.g. polymer or nucleic acid.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "Status of substance within the catalogue e.g. approved.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "domain": {
      "description": "If the substance applies to only human or veterinary use.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "Textual description of the substance.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "source": {
      "description": "Supporting literature.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "comment": {
      "description": "Textual comment about this record of a substance.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "moiety": {
      "description": "Moiety, for structural modifications.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Moiety"
```

```
      },
      "type": "array"
    },
    "property": {
      "description": "General specifications for this substance, including how it is related to other substances.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Property"
      },
      "type": "array"
    },
    "referenceInformation": {
      "description": "General information detailing this substance.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "structure": {
      "description": "Structural information.",
      "$ref": "#/definitions/SubstanceSpecification_Structure"
    },
    "code": {
      "description": "Codes associated with the substance.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Code"
      },
      "type": "array"
    },
    "name": {
      "description": "Names applicable to this substance.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Name"
      },
      "type": "array"
    },
    "molecularWeight": {
      "description": "The molecular weight or weight range (for proteins, polymers or nucleic acids).",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_MolecularWeight"
      },
      "type": "array"
    },
    "relationship": {
      "description": "A link between this substance and another, with details of the relationship.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Relationship"
      },
      "type": "array"
    },
    "nucleicAcid": {
      "description": "Data items specific to nucleic acids.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "polymer": {
      "description": "Data items specific to polymers.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "protein": {
      "description": "Data items specific to proteins.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "sourceMaterial": {
      "description": "Material or taxonomic/anatomical source for the substance.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SubstanceSpecification_Moiety": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "role": {
      "description": "Role that the moiety is playing.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "identifier": {
      "description": "Identifier by which this moiety substance is known.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "name": {
      "description": "Textual name for this moiety substance.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "stereochemistry": {
      "description": "Stereochemistry type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "opticalActivity": {
      "description": "Optical activity type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "molecularFormula": {
      "description": "Molecular formula.",
      "$ref": "#/definitions/string"
    },
    "_molecularFormula": {
      "description": "Extensions for molecularFormula",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "amountQuantity": {
      "description": "Quantitative value for this moiety.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "amountString": {
      "description": "Quantitative value for this moiety.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_amountString": {
      "description": "Extensions for amountString",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Property": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "category": {
      "description": "A category for this property, e.g. Physical, Chemical, Enzymatic.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "code": {
      "description": "Property type e.g. viscosity, pH, isoelectric point.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "parameters": {
      "description": "Parameters that were used in the measurement of a property (e.g. for viscosity: measured at
20C with a pH of 7.1).",
      "$ref": "#/definitions/string"
    },
    "_parameters": {
      "description": "Extensions for parameters",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definingSubstanceReference": {
      "description": "A substance upon which a defining property depends (e.g. for solubility: in water, in
alcohol).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "definingSubstanceCodeableConcept": {
      "description": "A substance upon which a defining property depends (e.g. for solubility: in water, in
alcohol).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amountQuantity": {
      "description": "Quantitative value for this property.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "amountString": {
      "description": "Quantitative value for this property.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_amountString": {
      "description": "Extensions for amountString",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Structure": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "stereochemistry": {
      "description": "Stereochemistry type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "opticalActivity": {
      "description": "Optical activity type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "molecularFormula": {
```

```
      "description": "Molecular formula.",
      "$ref": "#/definitions/string"
    },
    "_molecularFormula": {
      "description": "Extensions for molecularFormula",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "molecularFormulaByMoiety": {
      "description": "Specified per moiety according to the Hill system, i.e. first C, then H, then alphabetical,
each moiety separated by a dot.",
      "$ref": "#/definitions/string"
    },
    "_molecularFormulaByMoiety": {
      "description": "Extensions for molecularFormulaByMoiety",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "isotope": {
      "description": "Applicable for single substances that contain a radionuclide or a non-natural isotopic
ratio.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Isotope"
      },
      "type": "array"
    },
    "molecularWeight": {
      "description": "The molecular weight or weight range (for proteins, polymers or nucleic acids).",
      "$ref": "#/definitions/SubstanceSpecification_MolecularWeight"
    },
    "source": {
      "description": "Supporting literature.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "representation": {
      "description": "Molecular structural representation.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Representation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Isotope": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Substance identifier for each non-natural or radioisotope.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "name": {
      "description": "Substance name for each non-natural or radioisotope.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "substitution": {
      "description": "The type of isotopic substitution present in a single substance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
    },
    "halfLife": {
      "description": "Half life - for a non-natural nuclide.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "molecularWeight": {
      "description": "The molecular weight or weight range (for proteins, polymers or nucleic acids).",
      "$ref": "#/definitions/SubstanceSpecification_MolecularWeight"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_MolecularWeight": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "method": {
      "description": "The method by which the molecular weight was determined.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "type": {
      "description": "Type of molecular weight such as exact, average (also known as. number average), weight
average.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amount": {
      "description": "Used to capture quantitative values for a variety of elements. If only limits are given, the
arithmetic mean would be the average. If only a single definite value for a given element is given, it would be
captured in this field.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Representation": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
```

```
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of structure (e.g. Full, Partial, Representative).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "representation": {
      "description": "The structural representation as text string in a format e.g. InChI, SMILES, MOLFILE, CDX.",
      "$ref": "#/definitions/string"
    },
    "_representation": {
      "description": "Extensions for representation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "attachment": {
      "description": "An attached file with the structural representation.",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Code": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The specific code.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "Status of the code assignment.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "statusDate": {
      "description": "The date at which the code status is changed as part of the terminology maintenance.",
      "$ref": "#/definitions/dateTime"
    },
    "_statusDate": {
      "description": "Extensions for statusDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Any comment can be provided in this field, if necessary.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "source": {
      "description": "Supporting literature.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
```

```
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Name": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The actual name.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Name type.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The status of the name.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "preferred": {
      "description": "If this is the preferred name for this substance.",
      "$ref": "#/definitions/boolean"
    },
    "_preferred": {
      "description": "Extensions for preferred",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "Language of the name.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "domain": {
      "description": "The use context of this name for example if there is a different name a drug active ingredient
 as opposed to a food colour additive.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "The jurisdiction where this name applies.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "synonym": {
      "description": "A synonym of this name.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Name"
      },
      "type": "array"
```

```
    },
    "translation": {
      "description": "A translation for this name.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Name"
      },
      "type": "array"
    },
    "official": {
      "description": "Details of the official nature of this name.",
      "items": {
        "$ref": "#/definitions/SubstanceSpecification_Official"
      },
      "type": "array"
    },
    "source": {
      "description": "Supporting literature.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Official": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "authority": {
      "description": "Which authority uses this official name.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The status of the official name.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "date": {
      "description": "Date of official name change.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SubstanceSpecification_Relationship": {
  "description": "The detailed description of a substance, typically at a level beyond what is used for
prescribing.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "substanceReference": {
          "description": "A pointer to another substance, as a resource or just a representational code.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "substanceCodeableConcept": {
          "description": "A pointer to another substance, as a resource or just a representational code.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "relationship": {
          "description": "For example \"salt to parent\", \"active moiety\", \"starting material\".",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "isDefining": {
          "description": "For example where an enzyme strongly bonds with a particular substance, this is a defining
    relationship for that enzyme, out of several possible substance relationships.",
          "$ref": "#/definitions/boolean"
        },
        "_isDefining": {
          "description": "Extensions for isDefining",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "amountQuantity": {
          "description": "A numeric factor for the relationship, for instance to express that the salt of a substance
    has some percentage of the active substance in relation to some other.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "amountRange": {
          "description": "A numeric factor for the relationship, for instance to express that the salt of a substance
    has some percentage of the active substance in relation to some other.",
          "$ref": "Range.schema.json#/definitions/Range"
        },
        "amountRatio": {
          "description": "A numeric factor for the relationship, for instance to express that the salt of a substance
    has some percentage of the active substance in relation to some other.",
          "$ref": "Ratio.schema.json#/definitions/Ratio"
        },
        "amountString": {
          "description": "A numeric factor for the relationship, for instance to express that the salt of a substance
    has some percentage of the active substance in relation to some other.",
          "pattern": "^[ \\r\\n\\t\\S]+$",
          "type": "string"
        },
        "_amountString": {
          "description": "Extensions for amountString",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "amountRatioLowLimit": {
          "description": "For use when the numeric.",
          "$ref": "Ratio.schema.json#/definitions/Ratio"
        },
        "amountType": {
          "description": "An operator for the amount, for example \"average\", \"approximately\", \"less than\".",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "source": {
          "description": "Supporting literature.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```

11.21.9 Resource SubstanceSpecification - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substancespecification-dependencies.html

11.21.9 Resource SubstanceSpecification - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstanceSpecification FMM level 0

Name | Card. | Type | Dependency Analysis
SubstanceSpecification | | DomainResource
identifier | 0..1 | Identifier
type | 0..1 | CodeableConcept
status | 0..1 | CodeableConcept
domain | 0..1 | CodeableConcept
description | 0..1 | string
source | 0..* | Reference(DocumentReference)
comment | 0..1 | string
moiety | 0..* | BackboneElement
role | 0..1 | CodeableConcept
identifier | 0..1 | Identifier
name | 0..1 | string
stereochemistry | 0..1 | CodeableConcept
opticalActivity | 0..1 | CodeableConcept
molecularFormula | 0..1 | string
amount[x] | 0..1
amountQuantity | | Quantity
amountString | | string
property | 0..* | BackboneElement
category | 0..1 | CodeableConcept
code | 0..1 | CodeableConcept
parameters | 0..1 | string
definingSubstance[x] | 0..1
definingSubstanceReference | | Reference(SubstanceSpecification | Substance
definingSubstanceCodeableConcept | | CodeableConcept
amount[x] | 0..1
amountQuantity | | Quantity
amountString | | string
referenceInformation | 0..1 | Reference(SubstanceReferenceInformation)
structure | 0..1 | BackboneElement
stereochemistry | 0..1 | CodeableConcept
opticalActivity | 0..1 | CodeableConcept
molecularFormula | 0..1 | string
molecularFormulaByMoiety | 0..1 | string
isotope | 0..* | BackboneElement
identifier | 0..1 | Identifier
name | 0..1 | CodeableConcept
substitution | 0..1 | CodeableConcept
halfLife | 0..1 | Quantity
molecularWeight | 0..1 | BackboneElement
method | 0..1 | CodeableConcept
type | 0..1 | CodeableConcept
amount | 0..1 | Quantity
molecularWeight | 0..1 | see molecularWeight
source | 0..* | Reference(DocumentReference)
representation | 0..* | BackboneElement
type | 0..1 | CodeableConcept
representation | 0..1 | string
attachment | 0..1 | Attachment
code | 0..* | BackboneElement
code | 0..1 | CodeableConcept
status | 0..1 | CodeableConcept
statusDate | 0..1 | dateTime
comment | 0..1 | string
source | 0..* | Reference(DocumentReference)
name | 0..* | BackboneElement
name | 1..1 | string
type | 0..1 | CodeableConcept
status | 0..1 | CodeableConcept
preferred | 0..1 | boolean

language | 0..* | CodeableConcept
domain | 0..* | CodeableConcept
jurisdiction | 0..* | CodeableConcept
synonym | 0..* | see name
translation | 0..* | see name
official | 0..* | BackboneElement
authority | 0..1 | CodeableConcept
status | 0..1 | CodeableConcept
date | 0..1 | dateTime
source | 0..* | Reference(DocumentReference)
molecularWeight | 0..* | see molecularWeight
relationship | 0..* | BackboneElement
substance[x] | 0..1
substanceReference | | Reference(SubstanceSpecification
substanceCodeableConcept | | CodeableConcept
relationship | 0..1 | CodeableConcept
isDefining | 0..1 | boolean
amount[x] | 0..1
amountQuantity | | Quantity
amountRange | | Range
amountRatio | | Ratio
amountString | | string
amountRatioLowLimit | 0..1 | Ratio
amountType | 0..1 | CodeableConcept
source | 0..* | Reference(DocumentReference)
nucleicAcid | 0..1 | Reference(SubstanceNucleicAcid)
polymer | 0..1 | Reference(SubstancePolymer)
protein | 0..1 | Reference(SubstanceProtein)
sourceMaterial | 0..1 | Reference(SubstanceSourceMaterial)

11.22 Resource SubstancePolymer - Content

URL: https://hl7.org/fhir/R4/substancepolymer.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Todo.

11.22.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

This resource is referenced by SubstanceSpecification

11.22.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstancePolymer | ΣTU | | DomainResource | Todo Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
class | Σ | 0..1 | CodeableConcept | Todo
geometry | Σ | 0..1 | CodeableConcept | Todo
copolymerConnectivity | Σ | 0..* | CodeableConcept | Todo
modification | Σ | 0..* | string | Todo
monomerSet | Σ | 0..* | BackboneElement | Todo
ratioType | Σ | 0..1 | CodeableConcept | Todo
startingMaterial | Σ | 0..* | BackboneElement | Todo
material | Σ | 0..1 | CodeableConcept | Todo
type | Σ | 0..1 | CodeableConcept | Todo
isDefining | Σ | 0..1 | boolean | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
repeat | Σ | 0..* | BackboneElement | Todo
numberOfUnits | Σ | 0..1 | integer | Todo

averageMolecularFormula | Σ | 0..1 | string | Todo
repeatUnitAmountType | Σ | 0..1 | CodeableConcept | Todo
repeatUnit | Σ | 0..* | BackboneElement | Todo
orientationOfPolymerisation | Σ | 0..1 | CodeableConcept | Todo
repeatUnit | Σ | 0..1 | string | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
degreeOfPolymerisation | Σ | 0..* | BackboneElement | Todo
degree | Σ | 0..1 | CodeableConcept | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
structuralRepresentation | Σ | 0..* | BackboneElement | Todo
type | Σ | 0..1 | CodeableConcept | Todo
representation | Σ | 0..1 | string | Todo
attachment | Σ | 0..1 | Attachment | Todo
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstancePolymer xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <class><!-- 0..1 CodeableConcept Todo --></class>
 <geometry><!-- 0..1 CodeableConcept Todo --></geometry>
 <copolymerConnectivity><!-- 0..* CodeableConcept Todo --></copolymerConnectivity>
 <modification value="[string]"/><!-- 0..* Todo -->
 <monomerSet>  <!-- 0..* Todo -->
  <ratioType><!-- 0..1 CodeableConcept Todo --></ratioType>
  <startingMaterial>  <!-- 0..* Todo -->
   <material><!-- 0..1 CodeableConcept Todo --></material>
   <type><!-- 0..1 CodeableConcept Todo --></type>
   <isDefining value="[boolean]"/><!-- 0..1 Todo -->
   <amount><!-- 0..1 SubstanceAmount Todo --></amount>
  </startingMaterial>
 </monomerSet>
 <repeat>  <!-- 0..* Todo -->
  <numberOfUnits value="[integer]"/><!-- 0..1 Todo -->
  <averageMolecularFormula value="[string]"/><!-- 0..1 Todo -->
  <repeatUnitAmountType><!-- 0..1 CodeableConcept Todo --></repeatUnitAmountType>
  <repeatUnit>  <!-- 0..* Todo -->
   <orientationOfPolymerisation><!-- 0..1 CodeableConcept Todo --></orientationOfPolymerisation>
   <repeatUnit value="[string]"/><!-- 0..1 Todo -->
   <amount><!-- 0..1 SubstanceAmount Todo --></amount>
   <degreeOfPolymerisation>  <!-- 0..* Todo -->
    <degree><!-- 0..1 CodeableConcept Todo --></degree>
    <amount><!-- 0..1 SubstanceAmount Todo --></amount>
   </degreeOfPolymerisation>
   <structuralRepresentation>  <!-- 0..* Todo -->
    <type><!-- 0..1 CodeableConcept Todo --></type>
    <representation value="[string]"/><!-- 0..1 Todo -->
    <attachment><!-- 0..1 Attachment Todo --></attachment>
   </structuralRepresentation>
  </repeatUnit>
 </repeat>
</SubstancePolymer>
```

JSON Template

```
{
  "resourceType" : "SubstancePolymer",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "class" : { CodeableConcept }, // Todo
  "geometry" : { CodeableConcept }, // Todo
  "copolymerConnectivity" : [{ CodeableConcept }], // Todo
  "modification" : ["<string>"], // Todo
  "monomerSet" : [{ // Todo
    "ratioType" : { CodeableConcept }, // Todo
    "startingMaterial" : [{ // Todo
      "material" : { CodeableConcept }, // Todo
      "type" : { CodeableConcept }, // Todo
      "isDefining" : <boolean>, // Todo
      "amount" : { SubstanceAmount } // Todo
    }]
  }],
  "repeat" : [{ // Todo
    "numberOfUnits" : <integer>, // Todo
    "averageMolecularFormula" : "<string>", // Todo
    "repeatUnitAmountType" : { CodeableConcept }, // Todo
    "repeatUnit" : [{ // Todo
      "orientationOfPolymerisation" : { CodeableConcept }, // Todo
      "repeatUnit" : "<string>", // Todo
      "amount" : { SubstanceAmount }, // Todo
      "degreeOfPolymerisation" : [{ // Todo
        "degree" : { CodeableConcept }, // Todo
        "amount" : { SubstanceAmount } // Todo
      }],
      "structuralRepresentation" : [{ // Todo
```

```
        "type" : { CodeableConcept }, // Todo
        "representation" : "<string>", // Todo
        "attachment" : { Attachment } // Todo
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstancePolymer;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstancePolymer.class [ CodeableConcept ]; # 0..1 Todo
  fhir:SubstancePolymer.geometry [ CodeableConcept ]; # 0..1 Todo
  fhir:SubstancePolymer.copolymerConnectivity [ CodeableConcept ], ... ; # 0..* Todo
  fhir:SubstancePolymer.modification [ string ], ... ; # 0..* Todo
  fhir:SubstancePolymer.monomerSet [ # 0..* Todo
    fhir:SubstancePolymer.monomerSet.ratioType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstancePolymer.monomerSet.startingMaterial [ # 0..* Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.material [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.type [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.isDefining [ boolean ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.amount [ SubstanceAmount ]; # 0..1 Todo
    ], ...;
  ], ...;
  fhir:SubstancePolymer.repeat [ # 0..* Todo
    fhir:SubstancePolymer.repeat.numberOfUnits [ integer ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.averageMolecularFormula [ string ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.repeatUnitAmountType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.repeatUnit [ # 0..* Todo
      fhir:SubstancePolymer.repeat.repeatUnit.orientationOfPolymerisation [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.repeatUnit [ string ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.amount [ SubstanceAmount ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation [ # 0..* Todo
        fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.degree [ CodeableConcept ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.amount [ SubstanceAmount ]; # 0..1 Todo
      ], ...;
      fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation [ # 0..* Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.type [ CodeableConcept ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.representation [ string ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.attachment [ Attachment ]; # 0..1 Todo
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstancePolymer | ΣTU | | DomainResource | Todo Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
class | Σ | 0..1 | CodeableConcept | Todo
geometry | Σ | 0..1 | CodeableConcept | Todo
copolymerConnectivity | Σ | 0..* | CodeableConcept | Todo
modification | Σ | 0..* | string | Todo
monomerSet | Σ | 0..* | BackboneElement | Todo
ratioType | Σ | 0..1 | CodeableConcept | Todo
startingMaterial | Σ | 0..* | BackboneElement | Todo
material | Σ | 0..1 | CodeableConcept | Todo
type | Σ | 0..1 | CodeableConcept | Todo
isDefining | Σ | 0..1 | boolean | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
repeat | Σ | 0..* | BackboneElement | Todo
numberOfUnits | Σ | 0..1 | integer | Todo
averageMolecularFormula | Σ | 0..1 | string | Todo
repeatUnitAmountType | Σ | 0..1 | CodeableConcept | Todo
repeatUnit | Σ | 0..* | BackboneElement | Todo
orientationOfPolymerisation | Σ | 0..1 | CodeableConcept | Todo
repeatUnit | Σ | 0..1 | string | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
degreeOfPolymerisation | Σ | 0..* | BackboneElement | Todo
degree | Σ | 0..1 | CodeableConcept | Todo
amount | Σ | 0..1 | SubstanceAmount | Todo
structuralRepresentation | Σ | 0..* | BackboneElement | Todo

type | Σ | 0..1 | CodeableConcept | Todo
representation | Σ | 0..1 | string | Todo
attachment | Σ | 0..1 | Attachment | Todo
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstancePolymer xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <class><!-- 0..1 CodeableConcept Todo --></class>
 <geometry><!-- 0..1 CodeableConcept Todo --></geometry>
 <copolymerConnectivity><!-- 0..* CodeableConcept Todo --></copolymerConnectivity>
 <modification value="[string]"/><!-- 0..* Todo -->
 <monomerSet>  <!-- 0..* Todo -->
  <ratioType><!-- 0..1 CodeableConcept Todo --></ratioType>
  <startingMaterial>  <!-- 0..* Todo -->
   <material><!-- 0..1 CodeableConcept Todo --></material>
   <type><!-- 0..1 CodeableConcept Todo --></type>
   <isDefining value="[boolean]"/><!-- 0..1 Todo -->
   <amount><!-- 0..1 SubstanceAmount Todo --></amount>
  </startingMaterial>
 </monomerSet>
 <repeat>  <!-- 0..* Todo -->
  <numberOfUnits value="[integer]"/><!-- 0..1 Todo -->
  <averageMolecularFormula value="[string]"/><!-- 0..1 Todo -->
  <repeatUnitAmountType><!-- 0..1 CodeableConcept Todo --></repeatUnitAmountType>
  <repeatUnit>  <!-- 0..* Todo -->
   <orientationOfPolymerisation><!-- 0..1 CodeableConcept Todo --></orientationOfPolymerisation>
   <repeatUnit value="[string]"/><!-- 0..1 Todo -->
   <amount><!-- 0..1 SubstanceAmount Todo --></amount>
   <degreeOfPolymerisation>  <!-- 0..* Todo -->
    <degree><!-- 0..1 CodeableConcept Todo --></degree>
    <amount><!-- 0..1 SubstanceAmount Todo --></amount>
   </degreeOfPolymerisation>
   <structuralRepresentation>  <!-- 0..* Todo -->
    <type><!-- 0..1 CodeableConcept Todo --></type>
    <representation value="[string]"/><!-- 0..1 Todo -->
    <attachment><!-- 0..1 Attachment Todo --></attachment>
   </structuralRepresentation>
  </repeatUnit>
 </repeat>
</SubstancePolymer>
```

JSON Template

```
{
  "resourceType" : "SubstancePolymer",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "class" : { CodeableConcept }, // Todo
  "geometry" : { CodeableConcept }, // Todo
  "copolymerConnectivity" : [{ CodeableConcept }], // Todo
  "modification" : ["<string>"], // Todo
  "monomerSet" : [{ // Todo
    "ratioType" : { CodeableConcept }, // Todo
    "startingMaterial" : [{ // Todo
      "material" : { CodeableConcept }, // Todo
      "type" : { CodeableConcept }, // Todo
      "isDefining" : <boolean>, // Todo
      "amount" : { SubstanceAmount } // Todo
    }]
  }],
  "repeat" : [{ // Todo
    "numberOfUnits" : <integer>, // Todo
    "averageMolecularFormula" : "<string>", // Todo
    "repeatUnitAmountType" : { CodeableConcept }, // Todo
    "repeatUnit" : [{ // Todo
      "orientationOfPolymerisation" : { CodeableConcept }, // Todo
      "repeatUnit" : "<string>", // Todo
      "amount" : { SubstanceAmount }, // Todo
      "degreeOfPolymerisation" : [{ // Todo
        "degree" : { CodeableConcept }, // Todo
        "amount" : { SubstanceAmount } // Todo
      }],
      "structuralRepresentation" : [{ // Todo
        "type" : { CodeableConcept }, // Todo
        "representation" : "<string>", // Todo
        "attachment" : { Attachment } // Todo
      }]
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:SubstancePolymer;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstancePolymer.class [ CodeableConcept ]; # 0..1 Todo
  fhir:SubstancePolymer.geometry [ CodeableConcept ]; # 0..1 Todo
  fhir:SubstancePolymer.copolymerConnectivity [ CodeableConcept ], ... ; # 0..* Todo
  fhir:SubstancePolymer.modification [ string ], ... ; # 0..* Todo
  fhir:SubstancePolymer.monomerSet [ # 0..* Todo
    fhir:SubstancePolymer.monomerSet.ratioType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstancePolymer.monomerSet.startingMaterial [ # 0..* Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.material [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.type [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.isDefining [ boolean ]; # 0..1 Todo
      fhir:SubstancePolymer.monomerSet.startingMaterial.amount [ SubstanceAmount ]; # 0..1 Todo
    ], ...;
  ], ...;
  fhir:SubstancePolymer.repeat [ # 0..* Todo
    fhir:SubstancePolymer.repeat.numberOfUnits [ integer ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.averageMolecularFormula [ string ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.repeatUnitAmountType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstancePolymer.repeat.repeatUnit [ # 0..* Todo
      fhir:SubstancePolymer.repeat.repeatUnit.orientationOfPolymerisation [ CodeableConcept ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.repeatUnit [ string ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.amount [ SubstanceAmount ]; # 0..1 Todo
      fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation [ # 0..* Todo
        fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.degree [ CodeableConcept ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.amount [ SubstanceAmount ]; # 0..1 Todo
      ], ...;
      fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation [ # 0..* Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.type [ CodeableConcept ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.representation [ string ]; # 0..1 Todo
        fhir:SubstancePolymer.repeat.repeatUnit.structuralRepresentation.attachment [ Attachment ]; # 0..1 Todo
      ], ...;
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.22.3 Resource SubstancePolymer - Examples

URL: https://hl7.org/fhir/R4/substancepolymer-examples.html

11.22.3 Resource SubstancePolymer - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of substancepolymer | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.22.4 Resource SubstancePolymer - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substancepolymer-definitions.html

11.22.4 Resource SubstancePolymer - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstancePolymer resource.

SubstancePolymer
Element Id | SubstancePolymer
Definition | Todo.
Cardinality | 0..*
Type | DomainResource
Summary | true
SubstancePolymer.class
Element Id | SubstancePolymer.class
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.geometry
Element Id | SubstancePolymer.geometry

Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.copolymerConnectivity
Element Id | SubstancePolymer.copolymerConnectivity
Definition | Todo.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
SubstancePolymer.modification
Element Id | SubstancePolymer.modification
Definition | Todo.
Cardinality | 0..*
Type | string
Summary | true
SubstancePolymer.monomerSet
Element Id | SubstancePolymer.monomerSet
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.monomerSet.ratioType
Element Id | SubstancePolymer.monomerSet.ratioType
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.monomerSet.startingMaterial
Element Id | SubstancePolymer.monomerSet.startingMaterial
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.monomerSet.startingMaterial.material
Element Id | SubstancePolymer.monomerSet.startingMaterial.material
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.monomerSet.startingMaterial.type
Element Id | SubstancePolymer.monomerSet.startingMaterial.type
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.monomerSet.startingMaterial.isDefining
Element Id | SubstancePolymer.monomerSet.startingMaterial.isDefining
Definition | Todo.
Cardinality | 0..1
Type | boolean
Summary | true
SubstancePolymer.monomerSet.startingMaterial.amount
Element Id | SubstancePolymer.monomerSet.startingMaterial.amount
Definition | Todo.
Cardinality | 0..1
Type | SubstanceAmount
Summary | true
SubstancePolymer.repeat
Element Id | SubstancePolymer.repeat
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.repeat.numberOfUnits
Element Id | SubstancePolymer.repeat.numberOfUnits
Definition | Todo.
Cardinality | 0..1
Type | integer
Summary | true
SubstancePolymer.repeat.averageMolecularFormula
Element Id | SubstancePolymer.repeat.averageMolecularFormula
Definition | Todo.
Cardinality | 0..1
Type | string

Summary | true
SubstancePolymer.repeat.repeatUnitAmountType
Element Id | SubstancePolymer.repeat.repeatUnitAmountType
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.repeat.repeatUnit
Element Id | SubstancePolymer.repeat.repeatUnit
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.repeat.repeatUnit.orientationOfPolymerisation
Element Id | SubstancePolymer.repeat.repeatUnit.orientationOfPolymerisation
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.repeat.repeatUnit.repeatUnit
Element Id | SubstancePolymer.repeat.repeatUnit.repeatUnit
Definition | Todo.
Cardinality | 0..1
Type | string
Summary | true
SubstancePolymer.repeat.repeatUnit.amount
Element Id | SubstancePolymer.repeat.repeatUnit.amount
Definition | Todo.
Cardinality | 0..1
Type | SubstanceAmount
Summary | true
SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation
Element Id | SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.degree
Element Id | SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.degree
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.amount
Element Id | SubstancePolymer.repeat.repeatUnit.degreeOfPolymerisation.amount
Definition | Todo.
Cardinality | 0..1
Type | SubstanceAmount
Summary | true
SubstancePolymer.repeat.repeatUnit.structuralRepresentation
Element Id | SubstancePolymer.repeat.repeatUnit.structuralRepresentation
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstancePolymer.repeat.repeatUnit.structuralRepresentation.type
Element Id | SubstancePolymer.repeat.repeatUnit.structuralRepresentation.type
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstancePolymer.repeat.repeatUnit.structuralRepresentation.representation
Element Id | SubstancePolymer.repeat.repeatUnit.structuralRepresentation.representation
Definition | Todo.
Cardinality | 0..1
Type | string
Summary | true
SubstancePolymer.repeat.repeatUnit.structuralRepresentation.attachment
Element Id | SubstancePolymer.repeat.repeatUnit.structuralRepresentation.attachment
Definition | Todo.
Cardinality | 0..1
Type | Attachment
Summary | true

11.22.5 Resource SubstancePolymer - Mappings

URL: https://hl7.org/fhir/R4/substancepolymer-mappings.html

11.22.5 Resource SubstancePolymer - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substancepolymer resource (see Mappings to Other Standards for further information & status).

11.22.6 Resource SubstancePolymer - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substancepolymer-profiles.html

11.22.6 Resource SubstancePolymer - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstancePolymer resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.22.6.1 Profiles

No Profiles defined for this resource

11.22.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.22.6.3 Search Extensions

No Search Extensions defined for this resource

11.22.7 JSON Schema for SubstancePolymer

URL: https://hl7.org/fhir/R4/substancepolymer.schema.json.html

11.22.7 JSON Schema for SubstancePolymer

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstancePolymer, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstancePolymer",
  "$ref": "#/definitions/SubstancePolymer",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstancePolymer": {
      "description": "Todo.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstancePolymer resource",
          "const": "SubstancePolymer"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
```

```
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "class": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "geometry": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "copolymerConnectivity": {
      "description": "Todo.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "modification": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_modification": {
      "description": "Extensions for modification",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "monomerSet": {
```

```
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_MonomerSet"
      },
      "type": "array"
    },
    "repeat": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_Repeat"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SubstancePolymer_MonomerSet": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "ratioType": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "startingMaterial": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_StartingMaterial"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstancePolymer_StartingMaterial": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
```

```
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "material": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "type": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "isDefining": {
      "description": "Todo.",
      "$ref": "#/definitions/boolean"
    },
    "_isDefining": {
      "description": "Extensions for isDefining",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "amount": {
      "description": "Todo.",
      "$ref": "SubstanceAmount.schema.json#/definitions/SubstanceAmount"
    }
  },
  "additionalProperties": false
},
"SubstancePolymer_Repeat": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "numberOfUnits": {
      "description": "Todo.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfUnits": {
      "description": "Extensions for numberOfUnits",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "averageMolecularFormula": {
      "description": "Todo.",
      "$ref": "#/definitions/string"
    },
    "_averageMolecularFormula": {
      "description": "Extensions for averageMolecularFormula",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "repeatUnitAmountType": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "repeatUnit": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_RepeatUnit"
      },
      "type": "array"
```

```
    }
  },
  "additionalProperties": false
},
"SubstancePolymer_RepeatUnit": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "orientationOfPolymerisation": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "repeatUnit": {
      "description": "Todo.",
      "$ref": "#/definitions/string"
    },
    "_repeatUnit": {
      "description": "Extensions for repeatUnit",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "amount": {
      "description": "Todo.",
      "$ref": "SubstanceAmount.schema.json#/definitions/SubstanceAmount"
    },
    "degreeOfPolymerisation": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_DegreeOfPolymerisation"
      },
      "type": "array"
    },
    "structuralRepresentation": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstancePolymer_StructuralRepresentation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstancePolymer_DegreeOfPolymerisation": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "degree": {
          "description": "Todo.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "amount": {
          "description": "Todo.",
          "$ref": "SubstanceAmount.schema.json#/definitions/SubstanceAmount"
        }
      },
      "additionalProperties": false
    },
    "SubstancePolymer_StructuralRepresentation": {
      "description": "Todo.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "type": {
          "description": "Todo.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "representation": {
          "description": "Todo.",
          "$ref": "#/definitions/string"
        },
        "_representation": {
          "description": "Extensions for representation",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "attachment": {
          "description": "Todo.",
          "$ref": "Attachment.schema.json#/definitions/Attachment"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.22.8 Resource SubstancePolymer - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substancepolymer-dependencies.html

11.22.8 Resource SubstancePolymer - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstancePolymer FMM level 0

Name | Card. | Type | Dependency Analysis
SubstancePolymer | | DomainResource
class | 0..1 | CodeableConcept
geometry | 0..1 | CodeableConcept
copolymerConnectivity | 0..* | CodeableConcept
modification | 0..* | string
monomerSet | 0..* | BackboneElement
ratioType | 0..1 | CodeableConcept
startingMaterial | 0..* | BackboneElement
material | 0..1 | CodeableConcept
type | 0..1 | CodeableConcept
isDefining | 0..1 | boolean
amount | 0..1 | SubstanceAmount | Type Error: (SubstanceAmount = FMM5-Draft vs. Element = FMM0-Trial Use)
repeat | 0..* | BackboneElement
numberOfUnits | 0..1 | integer
averageMolecularFormula | 0..1 | string
repeatUnitAmountType | 0..1 | CodeableConcept
repeatUnit | 0..* | BackboneElement
orientationOfPolymerisation | 0..1 | CodeableConcept
repeatUnit | 0..1 | string
amount | 0..1 | SubstanceAmount | Type Error: (SubstanceAmount = FMM5-Draft vs. Element = FMM0-Trial Use)
degreeOfPolymerisation | 0..* | BackboneElement
degree | 0..1 | CodeableConcept
amount | 0..1 | SubstanceAmount | Type Error: (SubstanceAmount = FMM5-Draft vs. Element = FMM0-Trial Use)
structuralRepresentation | 0..* | BackboneElement
type | 0..1 | CodeableConcept
representation | 0..1 | string
attachment | 0..1 | Attachment

11.23 Resource SubstanceProtein - Content

URL: https://hl7.org/fhir/R4/substanceprotein.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators.

11.23.1 Scope and Usage

This resource is referenced by SubstanceSpecification

11.23.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceProtein | ΣTU | | DomainResource | A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
sequenceType | Σ | 0..1 | CodeableConcept | The SubstanceProtein descriptive elements will only be used when a complete or partial amino acid sequence is available or derivable from a nucleic acid sequence
numberOfSubunits | Σ | 0..1 | integer | Number of linear sequences of amino acids linked through peptide bonds. The number of subunits constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be variable
disulfideLinkage | Σ | 0..* | string | The disulphide bond between two cysteine residues either on the same subunit or on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be listed in increasing order

of subunit number and position within subunit followed by the abbreviation of the amino acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective positions

subunit | Σ | 0..* | BackboneElement | This subclause refers to the description of each subunit constituting the SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times

subunit | Σ | 0..1 | integer | Index of primary sequences of amino acids linked through peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts

sequence | Σ | 0..1 | string | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence

length | Σ | 0..1 | integer | Length of linear sequences of amino acids contained in the subunit

sequenceAttachment | Σ | 0..1 | Attachment | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence

nTerminalModificationId | Σ | 0..1 | Identifier | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID

nTerminalModification | Σ | 0..1 | string | The name of the fragment modified at the N-terminal of the SubstanceProtein shall be specified

cTerminalModificationId | Σ | 0..1 | Identifier | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID

cTerminalModification | Σ | 0..1 | string | The modification at the C-terminal shall be specified

Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceProtein xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sequenceType><!-- 0..1 CodeableConcept The SubstanceProtein descriptive elements will only be used when a complete or
    partial amino acid sequence is available or derivable from a nucleic acid sequence --></sequenceType>
 <numberOfSubunits value="[integer]"/><!-- 0..1 Number of linear sequences of amino acids linked through peptide bonds.
    The number of subunits constituting the SubstanceProtein shall be described. It is possible that the number of
    subunits can be variable -->
 <disulfideLinkage value="[string]"/><!-- 0..* The disulphide bond between two cysteine residues either on the same
    subunit or on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein
     shall be listed in increasing order of subunit number and position within subunit followed by the abbreviation of
    the amino acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the
    respective positions -->
 <subunit>  <!-- 0..* This subclause refers to the description of each subunit constituting the SubstanceProtein. A
    subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided
    when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains
    within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be
    ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times -->
  <subunit value="[integer]"/><!-- 0..1 Index of primary sequences of amino acids linked through peptide bonds in order
    of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
     sequences will be repeated and have sequential subscripts -->
  <sequence value="[string]"/><!-- 0..1 The sequence information shall be provided enumerating the amino acids from N-
    to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence -->
  <length value="[integer]"/><!-- 0..1 Length of linear sequences of amino acids contained in the subunit -->
  <sequenceAttachment><!-- 0..1 Attachment The sequence information shall be provided enumerating the amino acids from
    N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence
    --></sequenceAttachment>
  <nTerminalModificationId><!-- 0..1 Identifier Unique identifier for molecular fragment modification based on the ISO
    11238 Substance ID --></nTerminalModificationId>
  <nTerminalModification value="[string]"/><!-- 0..1 The name of the fragment modified at the N-terminal of the
    SubstanceProtein shall be specified -->
  <cTerminalModificationId><!-- 0..1 Identifier Unique identifier for molecular fragment modification based on the ISO
    11238 Substance ID --></cTerminalModificationId>
  <cTerminalModification value="[string]"/><!-- 0..1 The modification at the C-terminal shall be specified -->
 </subunit>
</SubstanceProtein>
```

JSON Template

```
{
  "resourceType" : "SubstanceProtein",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sequenceType" : { CodeableConcept }, // The SubstanceProtein descriptive elements will only be used when a complete
    or partial amino acid sequence is available or derivable from a nucleic acid sequence
  "numberOfSubunits" : <integer>, // Number of linear sequences of amino acids linked through peptide bonds. The number
```

```
    of subunits constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be
    variable
  "disulfideLinkage" : ["<string>"], // The disulphide bond between two cysteine residues either on the same subunit or
    on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be
    listed in increasing order of subunit number and position within subunit followed by the abbreviation of the amino
    acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective
    positions
  "subunit" : [{ // This subclause refers to the description of each subunit constituting the SubstanceProtein. A
    subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided
    when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains
    within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be
    ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times
    "subunit" : <integer>, // Index of primary sequences of amino acids linked through peptide bonds in order of
    decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
    sequences will be repeated and have sequential subscripts
    "sequence" : "<string>", // The sequence information shall be provided enumerating the amino acids from N- to
    C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence
    "length" : <integer>, // Length of linear sequences of amino acids contained in the subunit
    "sequenceAttachment" : { Attachment }, // The sequence information shall be provided enumerating the amino acids
    from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids
    and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated
    sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X
    should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence
    "nTerminalModificationId" : { Identifier }, // Unique identifier for molecular fragment modification based on the
    ISO 11238 Substance ID
    "nTerminalModification" : "<string>", // The name of the fragment modified at the N-terminal of the SubstanceProtein
     shall be specified
    "cTerminalModificationId" : { Identifier }, // Unique identifier for molecular fragment modification based on the
    ISO 11238 Substance ID
    "cTerminalModification" : "<string>" // The modification at the C-terminal shall be specified
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceProtein;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceProtein.sequenceType [ CodeableConcept ]; # 0..1 The SubstanceProtein descriptive elements will only be
    used when a complete or partial amino acid sequence is available or derivable from a nucleic acid sequence
  fhir:SubstanceProtein.numberOfSubunits [ integer ]; # 0..1 Number of linear sequences of amino acids linked through
    peptide bonds. The number of subunits constituting the SubstanceProtein shall be described. It is possible that the
    number of subunits can be variable
  fhir:SubstanceProtein.disulfideLinkage [ string ], ... ; # 0..* The disulphide bond between two cysteine residues
    either on the same subunit or on two different subunits shall be described. The position of the disulfide bonds in
    the SubstanceProtein shall be listed in increasing order of subunit number and position within subunit followed by
    the abbreviation of the amino acids involved. The disulfide linkage positions shall actually contain the amino acid
    Cysteine at the respective positions
  fhir:SubstanceProtein.subunit [ # 0..* This subclause refers to the description of each subunit constituting the
    SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit
    information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are
    not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences
     of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be
    repeated multiple times
    fhir:SubstanceProtein.subunit.subunit [ integer ]; # 0..1 Index of primary sequences of amino acids linked through
    peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight.
    Subunits that have identical sequences will be repeated and have sequential subscripts
    fhir:SubstanceProtein.subunit.sequence [ string ]; # 0..1 The sequence information shall be provided enumerating the
     amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for
    L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the
    translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code
     an X should be used within the sequence. The modified amino acids will be distinguished by their position in the
    sequence
    fhir:SubstanceProtein.subunit.length [ integer ]; # 0..1 Length of linear sequences of amino acids contained in the
    subunit
    fhir:SubstanceProtein.subunit.sequenceAttachment [ Attachment ]; # 0..1 The sequence information shall be provided
    enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall
     be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described
    using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single
    letter code an X should be used within the sequence. The modified amino acids will be distinguished by their
    position in the sequence
    fhir:SubstanceProtein.subunit.nTerminalModificationId [ Identifier ]; # 0..1 Unique identifier for molecular
    fragment modification based on the ISO 11238 Substance ID
    fhir:SubstanceProtein.subunit.nTerminalModification [ string ]; # 0..1 The name of the fragment modified at the
    N-terminal of the SubstanceProtein shall be specified
    fhir:SubstanceProtein.subunit.cTerminalModificationId [ Identifier ]; # 0..1 Unique identifier for molecular
    fragment modification based on the ISO 11238 Substance ID
    fhir:SubstanceProtein.subunit.cTerminalModification [ string ]; # 0..1 The modification at the C-terminal shall be
    specified
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

SubstanceProtein | ΣTU | | DomainResource | A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

sequenceType | Σ | 0..1 | CodeableConcept | The SubstanceProtein descriptive elements will only be used when a complete or partial amino acid sequence is available or derivable from a nucleic acid sequence

numberOfSubunits | Σ | 0..1 | integer | Number of linear sequences of amino acids linked through peptide bonds. The number of subunits constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be variable

disulfideLinkage | Σ | 0..* | string | The disulphide bond between two cysteine residues either on the same subunit or on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be listed in increasing order of subunit number and position within subunit followed by the abbreviation of the amino acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective positions

subunit | Σ | 0..* | BackboneElement | This subclause refers to the description of each subunit constituting the SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times

subunit | Σ | 0..1 | integer | Index of primary sequences of amino acids linked through peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts

sequence | Σ | 0..1 | string | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence

length | Σ | 0..1 | integer | Length of linear sequences of amino acids contained in the subunit

sequenceAttachment | Σ | 0..1 | Attachment | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence

nTerminalModificationId | Σ | 0..1 | Identifier | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID

nTerminalModification | Σ | 0..1 | string | The name of the fragment modified at the N-terminal of the SubstanceProtein shall be specified

cTerminalModificationId | Σ | 0..1 | Identifier | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID

cTerminalModification | Σ | 0..1 | string | The modification at the C-terminal shall be specified

Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceProtein xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sequenceType><!-- 0..1 CodeableConcept The SubstanceProtein descriptive elements will only be used when a complete or
    partial amino acid sequence is available or derivable from a nucleic acid sequence --></sequenceType>
 <numberOfSubunits value="[integer]"/><!-- 0..1 Number of linear sequences of amino acids linked through peptide bonds.
    The number of subunits constituting the SubstanceProtein shall be described. It is possible that the number of
    subunits can be variable -->
 <disulfideLinkage value="[string]"/><!-- 0..* The disulphide bond between two cysteine residues either on the same
    subunit or on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein
     shall be listed in increasing order of subunit number and position within subunit followed by the abbreviation of
    the amino acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the
    respective positions -->
 <subunit>  <!-- 0..* This subclause refers to the description of each subunit constituting the SubstanceProtein. A
    subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided
    when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains
    within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be
    ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times -->
  <subunit value="[integer]"/><!-- 0..1 Index of primary sequences of amino acids linked through peptide bonds in order
    of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
     sequences will be repeated and have sequential subscripts -->
  <sequence value="[string]"/><!-- 0..1 The sequence information shall be provided enumerating the amino acids from N-
    to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence -->
  <length value="[integer]"/><!-- 0..1 Length of linear sequences of amino acids contained in the subunit -->
  <sequenceAttachment><!-- 0..1 Attachment The sequence information shall be provided enumerating the amino acids from
    N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
```

```
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence
    --></sequenceAttachment>
  <nTerminalModificationId><!-- 0..1 Identifier Unique identifier for molecular fragment modification based on the ISO
    11238 Substance ID --></nTerminalModificationId>
  <nTerminalModification value="[string]"/><!-- 0..1 The name of the fragment modified at the N-terminal of the
    SubstanceProtein shall be specified -->
  <cTerminalModificationId><!-- 0..1 Identifier Unique identifier for molecular fragment modification based on the ISO
    11238 Substance ID --></cTerminalModificationId>
  <cTerminalModification value="[string]"/><!-- 0..1 The modification at the C-terminal shall be specified -->
 </subunit>
</SubstanceProtein>
```

JSON Template

```
{
  "resourceType" : "SubstanceProtein",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sequenceType" : { CodeableConcept }, // The SubstanceProtein descriptive elements will only be used when a complete
    or partial amino acid sequence is available or derivable from a nucleic acid sequence
  "numberOfSubunits" : <integer>, // Number of linear sequences of amino acids linked through peptide bonds. The number
    of subunits constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be
    variable
  "disulfideLinkage" : ["<string>"], // The disulphide bond between two cysteine residues either on the same subunit or
    on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be
    listed in increasing order of subunit number and position within subunit followed by the abbreviation of the amino
    acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective
    positions
  "subunit" : [{ // This subclause refers to the description of each subunit constituting the SubstanceProtein. A
    subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided
    when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains
    within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be
    ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times
    "subunit" : <integer>, // Index of primary sequences of amino acids linked through peptide bonds in order of
    decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
    sequences will be repeated and have sequential subscripts
    "sequence" : "<string>", // The sequence information shall be provided enumerating the amino acids from N- to
    C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and
    lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence;
    for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used
    within the sequence. The modified amino acids will be distinguished by their position in the sequence
    "length" : <integer>, // Length of linear sequences of amino acids contained in the subunit
    "sequenceAttachment" : { Attachment }, // The sequence information shall be provided enumerating the amino acids
    from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids
    and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated
    sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X
    should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence
    "nTerminalModificationId" : { Identifier }, // Unique identifier for molecular fragment modification based on the
    ISO 11238 Substance ID
    "nTerminalModification" : "<string>", // The name of the fragment modified at the N-terminal of the SubstanceProtein
     shall be specified
    "cTerminalModificationId" : { Identifier }, // Unique identifier for molecular fragment modification based on the
    ISO 11238 Substance ID
    "cTerminalModification" : "<string>" // The modification at the C-terminal shall be specified
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceProtein;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceProtein.sequenceType [ CodeableConcept ]; # 0..1 The SubstanceProtein descriptive elements will only be
    used when a complete or partial amino acid sequence is available or derivable from a nucleic acid sequence
  fhir:SubstanceProtein.numberOfSubunits [ integer ]; # 0..1 Number of linear sequences of amino acids linked through
    peptide bonds. The number of subunits constituting the SubstanceProtein shall be described. It is possible that the
    number of subunits can be variable
  fhir:SubstanceProtein.disulfideLinkage [ string ], ... ; # 0..* The disulphide bond between two cysteine residues
    either on the same subunit or on two different subunits shall be described. The position of the disulfide bonds in
    the SubstanceProtein shall be listed in increasing order of subunit number and position within subunit followed by
    the abbreviation of the amino acids involved. The disulfide linkage positions shall actually contain the amino acid
    Cysteine at the respective positions
  fhir:SubstanceProtein.subunit [ # 0..* This subclause refers to the description of each subunit constituting the
    SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit
    information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are
    not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences
     of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be
    repeated multiple times
    fhir:SubstanceProtein.subunit.subunit [ integer ]; # 0..1 Index of primary sequences of amino acids linked through
    peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight.
    Subunits that have identical sequences will be repeated and have sequential subscripts
    fhir:SubstanceProtein.subunit.sequence [ string ]; # 0..1 The sequence information shall be provided enumerating the
     amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for
    L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the
    translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code
```

```
     an X should be used within the sequence. The modified amino acids will be distinguished by their position in the
    sequence
    fhir:SubstanceProtein.subunit.length [ integer ]; # 0..1 Length of linear sequences of amino acids contained in the
    subunit
    fhir:SubstanceProtein.subunit.sequenceAttachment [ Attachment ]; # 0..1 The sequence information shall be provided
    enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall
     be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described
    using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single
    letter code an X should be used within the sequence. The modified amino acids will be distinguished by their
    position in the sequence
    fhir:SubstanceProtein.subunit.nTerminalModificationId [ Identifier ]; # 0..1 Unique identifier for molecular
    fragment modification based on the ISO 11238 Substance ID
    fhir:SubstanceProtein.subunit.nTerminalModification [ string ]; # 0..1 The name of the fragment modified at the
    N-terminal of the SubstanceProtein shall be specified
    fhir:SubstanceProtein.subunit.cTerminalModificationId [ Identifier ]; # 0..1 Unique identifier for molecular
    fragment modification based on the ISO 11238 Substance ID
    fhir:SubstanceProtein.subunit.cTerminalModification [ string ]; # 0..1 The modification at the C-terminal shall be
    specified
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.23.3 Resource SubstanceProtein - Examples

URL: https://hl7.org/fhir/R4/substanceprotein-examples.html

11.23.3 Resource SubstanceProtein - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of substanceprotein | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.23.4 Resource SubstanceProtein - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substanceprotein-definitions.html

11.23.4 Resource SubstanceProtein - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstanceProtein resource.

SubstanceProtein

Element Id | SubstanceProtein

Definition | A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators.

Cardinality | 0..*

Type | DomainResource

Summary | true

SubstanceProtein.sequenceType

Element Id | SubstanceProtein.sequenceType

Definition | The SubstanceProtein descriptive elements will only be used when a complete or partial amino acid sequence is available or derivable from a nucleic acid sequence.

Cardinality | 0..1

Type | CodeableConcept

Summary | true

SubstanceProtein.numberOfSubunits

Element Id | SubstanceProtein.numberOfSubunits

Definition | Number of linear sequences of amino acids linked through peptide bonds. The number of subunits constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be variable.

Cardinality | 0..1

Type | integer

Summary | true

SubstanceProtein.disulfideLinkage

Element Id | SubstanceProtein.disulfideLinkage

Definition | The disulphide bond between two cysteine residues either on the same subunit or on two different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be listed in increasing order of subunit number and position within subunit followed by the abbreviation of the amino acids involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective positions.
Cardinality | 0..*
Type | string
Summary | true
SubstanceProtein.subunit
Element Id | SubstanceProtein.subunit
Definition | This subclause refers to the description of each subunit constituting the SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times.
Cardinality | 0..*
Summary | true
SubstanceProtein.subunit.subunit
Element Id | SubstanceProtein.subunit.subunit
Definition | Index of primary sequences of amino acids linked through peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts.
Cardinality | 0..1
Type | integer
Summary | true
SubstanceProtein.subunit.sequence
Element Id | SubstanceProtein.subunit.sequence
Definition | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence.
Cardinality | 0..1
Type | string
Summary | true
SubstanceProtein.subunit.length
Element Id | SubstanceProtein.subunit.length
Definition | Length of linear sequences of amino acids contained in the subunit.
Cardinality | 0..1
Type | integer
Summary | true
SubstanceProtein.subunit.sequenceAttachment
Element Id | SubstanceProtein.subunit.sequenceAttachment
Definition | The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence.
Cardinality | 0..1
Type | Attachment
Summary | true
SubstanceProtein.subunit.nTerminalModificationId
Element Id | SubstanceProtein.subunit.nTerminalModificationId
Definition | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID.
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceProtein.subunit.nTerminalModification
Element Id | SubstanceProtein.subunit.nTerminalModification
Definition | The name of the fragment modified at the N-terminal of the SubstanceProtein shall be specified.
Cardinality | 0..1
Type | string
Summary | true
SubstanceProtein.subunit.cTerminalModificationId
Element Id | SubstanceProtein.subunit.cTerminalModificationId
Definition | Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID.
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceProtein.subunit.cTerminalModification
Element Id | SubstanceProtein.subunit.cTerminalModification
Definition | The modification at the C-terminal shall be specified.
Cardinality | 0..1
Type | string
Summary | true

11.23.5 Resource SubstanceProtein - Mappings

URL: https://hl7.org/fhir/R4/substanceprotein-mappings.html

11.23.5 Resource SubstanceProtein - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substanceprotein resource (see Mappings to Other Standards for further information & status).

11.23.6 Resource SubstanceProtein - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substanceprotein-profiles.html

11.23.6 Resource SubstanceProtein - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstanceProtein resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.23.6.1 Profiles

No Profiles defined for this resource

11.23.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.23.6.3 Search Extensions

No Search Extensions defined for this resource

11.23.7 JSON Schema for SubstanceProtein

URL: https://hl7.org/fhir/R4/substanceprotein.schema.json.html

11.23.7 JSON Schema for SubstanceProtein

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstanceProtein, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstanceProtein",
  "$ref": "#/definitions/SubstanceProtein",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstanceProtein": {
      "description": "A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination
    of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes
     all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or
    prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth
    factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstanceProtein resource",
          "const": "SubstanceProtein"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
```

```
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "sequenceType": {
      "description": "The SubstanceProtein descriptive elements will only be used when a complete or partial amino
acid sequence is available or derivable from a nucleic acid sequence.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "numberOfSubunits": {
      "description": "Number of linear sequences of amino acids linked through peptide bonds. The number of subunits
 constituting the SubstanceProtein shall be described. It is possible that the number of subunits can be variable.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfSubunits": {
      "description": "Extensions for numberOfSubunits",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "disulfideLinkage": {
      "description": "The disulphide bond between two cysteine residues either on the same subunit or on two
different subunits shall be described. The position of the disulfide bonds in the SubstanceProtein shall be listed
in increasing order of subunit number and position within subunit followed by the abbreviation of the amino acids
involved. The disulfide linkage positions shall actually contain the amino acid Cysteine at the respective
positions.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
```

```
    },
    "_disulfideLinkage": {
      "description": "Extensions for disulfideLinkage",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "subunit": {
      "description": "This subclause refers to the description of each subunit constituting the SubstanceProtein. A subunit is a linear sequence of amino acids linked through peptide bonds. The Subunit information shall be provided when the finished SubstanceProtein is a complex of multiple sequences; subunits are not used to delineate domains within a single sequence. Subunits are listed in order of decreasing length; sequences of the same length will be ordered by decreasing molecular weight; subunits that have identical sequences will be repeated multiple times.",
      "items": {
        "$ref": "#/definitions/SubstanceProtein_Subunit"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SubstanceProtein_Subunit": {
  "description": "A SubstanceProtein is defined as a single unit of a linear amino acid sequence, or a combination of subunits that are either covalently linked or have a defined invariant stoichiometric relationship. This includes all synthetic, recombinant and purified SubstanceProteins of defined sequence, whether the use is therapeutic or prophylactic. This set of elements will be used to describe albumins, coagulation factors, cytokines, growth factors, peptide/SubstanceProtein hormones, enzymes, toxins, toxoids, recombinant vaccines, and immunomodulators.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "subunit": {
      "description": "Index of primary sequences of amino acids linked through peptide bonds in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts.",
      "$ref": "#/definitions/integer"
    },
    "_subunit": {
      "description": "Extensions for subunit",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sequence": {
      "description": "The sequence information shall be provided enumerating the amino acids from N- to C-terminal end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic peptide containing amino acids that are not represented with a single letter code an X should be used within the sequence. The modified amino acids will be distinguished by their position in the sequence.",
      "$ref": "#/definitions/string"
    },
    "_sequence": {
      "description": "Extensions for sequence",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "length": {
      "description": "Length of linear sequences of amino acids contained in the subunit.",
      "$ref": "#/definitions/integer"
    },
    "_length": {
      "description": "Extensions for length",
```

```
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "sequenceAttachment": {
        "description": "The sequence information shall be provided enumerating the amino acids from N- to C-terminal
    end using standard single-letter amino acid codes. Uppercase shall be used for L-amino acids and lowercase for
    D-amino acids. Transcribed SubstanceProteins will always be described using the translated sequence; for synthetic
    peptide containing amino acids that are not represented with a single letter code an X should be used within the
    sequence. The modified amino acids will be distinguished by their position in the sequence.",
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "nTerminalModificationId": {
        "description": "Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID.",
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "nTerminalModification": {
        "description": "The name of the fragment modified at the N-terminal of the SubstanceProtein shall be
    specified.",
        "$ref": "#/definitions/string"
      },
      "_nTerminalModification": {
        "description": "Extensions for nTerminalModification",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "cTerminalModificationId": {
        "description": "Unique identifier for molecular fragment modification based on the ISO 11238 Substance ID.",
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "cTerminalModification": {
        "description": "The modification at the C-terminal shall be specified.",
        "$ref": "#/definitions/string"
      },
      "_cTerminalModification": {
        "description": "Extensions for cTerminalModification",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false
  }
 }
}
```


11.23.8 Resource SubstanceProtein - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substanceprotein-dependencies.html

11.23.8 Resource SubstanceProtein - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstanceProtein FMM level 0

Name | Card. | Type | Dependency Analysis
SubstanceProtein | | DomainResource
sequenceType | 0..1 | CodeableConcept
numberOfSubunits | 0..1 | integer
disulfideLinkage | 0..* | string
subunit | 0..* | BackboneElement
subunit | 0..1 | integer
sequence | 0..1 | string
length | 0..1 | integer
sequenceAttachment | 0..1 | Attachment
nTerminalModificationId | 0..1 | Identifier
nTerminalModification | 0..1 | string
cTerminalModificationId | 0..1 | Identifier
cTerminalModification | 0..1 | string

11.24 Resource SubstanceNucleicAcid - Content

URL: https://hl7.org/fhir/R4/substancenucleicacid.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in the 5’-3’ direction.

11.24.1 Scope and Usage

This resource is referenced by SubstanceSpecification

11.24.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

SubstanceNucleicAcid | ΣTU | | DomainResource | Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in the 5’-3’ direction Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

sequenceType | Σ | 0..1 | CodeableConcept | The type of the sequence shall be specified based on a controlled vocabulary

numberOfSubunits | Σ | 0..1 | integer | The number of linear sequences of nucleotides linked through phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit

areaOfHybridisation | Σ | 0..1 | string | The area of hybridisation shall be described if applicable for double stranded RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall be specified in increasing order. The underscore “” shall be used as separator as follows: “Subunitnumber Residue”

oligoNucleotideType | Σ | 0..1 | CodeableConcept | (TBC)

subunit | Σ | 0..* | BackboneElement | Subunits are listed in order of decreasing length; sequences of the same length will be ordered by molecular weight; subunits that have identical sequences will be repeated multiple times

subunit | Σ | 0..1 | integer | Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts

sequence | Σ | 0..1 | string | Actual nucleotide sequence notation from 5' to 3' end using standard single letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be captured

length | Σ | 0..1 | integer | The length of the sequence shall be captured

sequenceAttachment | Σ | 0..1 | Attachment | (TBC)

fivePrime | Σ | 0..1 | CodeableConcept | The nucleotide present at the 5’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the first position in the sequence. A separate representation would be redundant

threePrime | Σ | 0..1 | CodeableConcept | The nucleotide present at the 3’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the last position in the sequence. A separate representation would be redundant

linkage | Σ | 0..* | BackboneElement | The linkages between sugar residues will also be captured

connectivity | Σ | 0..1 | string | The entity that links the sugar residues together should also be captured for nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3’-5’. If the linkage is either 3’-3’ or 5’-5’ this should be specified

identifier | Σ | 0..1 | Identifier | Each linkage will be registered as a fragment and have an ID

name | Σ | 0..1 | string | Each linkage will be registered as a fragment and have at least one name. A single name shall be assigned to each linkage

residueSite | Σ | 0..1 | string | Residues shall be captured as described in 5.3.6.8.3

sugar | Σ | 0..* | BackboneElement | 5.3.6.8.1 Sugar ID (Mandatory)

identifier | Σ | 0..1 | Identifier | The Substance ID of the sugar or sugar-like component that make up the nucleotide

name | Σ | 0..1 | string | The name of the sugar or sugar-like component that make up the nucleotide

residueSite | Σ | 0..1 | string | The residues that contain a given sugar will be captured. The order of given residues will be captured in the 5’-3’direction consistent with the base sequences listed above

Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceNucleicAcid xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sequenceType><!-- 0..1 CodeableConcept The type of the sequence shall be specified based on a controlled vocabulary
    --></sequenceType>
 <numberOfSubunits value="[integer]"/><!-- 0..1 The number of linear sequences of nucleotides linked through
    phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly associated
    typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the assumption is that
    there is 1 subunit -->
 <areaOfHybridisation value="[string]"/><!-- 0..1 The area of hybridisation shall be described if applicable for double
    stranded RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall
    be specified in increasing order. The underscore â€œâ€ shall be used as separator as follows: â€œSubunitnumber
    Residueâ€ -->
 <oligoNucleotideType><!-- 0..1 CodeableConcept (TBC) --></oligoNucleotideType>
 <subunit>  <!-- 0..* Subunits are listed in order of decreasing length; sequences of the same length will be ordered by
     molecular weight; subunits that have identical sequences will be repeated multiple times -->
  <subunit value="[integer]"/><!-- 0..1 Index of linear sequences of nucleic acids in order of decreasing length.
    Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be
    repeated and have sequential subscripts -->
  <sequence value="[string]"/><!-- 0..1 Actual nucleotide sequence notation from 5' to 3' end using standard single
    letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be
```

```
    captured -->
  <length value="[integer]"/><!-- 0..1 The length of the sequence shall be captured -->
  <sequenceAttachment><!-- 0..1 Attachment (TBC) --></sequenceAttachment>
  <fivePrime><!-- 0..1 CodeableConcept The nucleotide present at the 5â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the first position in the sequence. A separate representation would be redundant --></fivePrime>
  <threePrime><!-- 0..1 CodeableConcept The nucleotide present at the 3â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the last position in the sequence. A separate representation would be redundant --></threePrime>
  <linkage>  <!-- 0..* The linkages between sugar residues will also be captured -->
   <connectivity value="[string]"/><!-- 0..1 The entity that links the sugar residues together should also be captured
    for nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic
    oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3â€™-5â€™. If the
    linkage is either 3â€™-3â€™ or 5â€™-5â€™ this should be specified -->
   <identifier><!-- 0..1 Identifier Each linkage will be registered as a fragment and have an ID --></identifier>
   <name value="[string]"/><!-- 0..1 Each linkage will be registered as a fragment and have at least one name. A single
    name shall be assigned to each linkage -->
   <residueSite value="[string]"/><!-- 0..1 Residues shall be captured as described in 5.3.6.8.3 -->
  </linkage>
  <sugar>  <!-- 0..* 5.3.6.8.1 Sugar ID (Mandatory) -->
   <identifier><!-- 0..1 Identifier The Substance ID of the sugar or sugar-like component that make up the nucleotide
    --></identifier>
   <name value="[string]"/><!-- 0..1 The name of the sugar or sugar-like component that make up the nucleotide -->
   <residueSite value="[string]"/><!-- 0..1 The residues that contain a given sugar will be captured. The order of given
     residues will be captured in the 5â€˜-3â€˜direction consistent with the base sequences listed above -->
  </sugar>
 </subunit>
</SubstanceNucleicAcid>
```

JSON Template

```
{
  "resourceType" : "SubstanceNucleicAcid",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sequenceType" : { CodeableConcept }, // The type of the sequence shall be specified based on a controlled vocabulary
  "numberOfSubunits" : <integer>, // The number of linear sequences of nucleotides linked through phosphodiester bonds
    shall be described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-
    Crick base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit
  "areaOfHybridisation" : "<string>", // The area of hybridisation shall be described if applicable for double stranded
    RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall be
    specified in increasing order. The underscore â€œâ€ shall be used as separator as follows: â€œSubunitnumber
    Residueâ€
  "oligoNucleotideType" : { CodeableConcept }, // (TBC)
  "subunit" : [{ // Subunits are listed in order of decreasing length; sequences of the same length will be ordered by
    molecular weight; subunits that have identical sequences will be repeated multiple times
    "subunit" : <integer>, // Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the
     same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have
    sequential subscripts
    "sequence" : "<string>", // Actual nucleotide sequence notation from 5' to 3' end using standard single letter
    codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be
    captured
    "length" : <integer>, // The length of the sequence shall be captured
    "sequenceAttachment" : { Attachment }, // (TBC)
    "fivePrime" : { CodeableConcept }, // The nucleotide present at the 5â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the first position in the sequence. A separate representation would be redundant
    "threePrime" : { CodeableConcept }, // The nucleotide present at the 3â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the last position in the sequence. A separate representation would be redundant
    "linkage" : [{ // The linkages between sugar residues will also be captured
      "connectivity" : "<string>", // The entity that links the sugar residues together should also be captured for
    nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides
    phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3â€™-5â€™. If the linkage is either
    3â€™-3â€™ or 5â€™-5â€™ this should be specified
      "identifier" : { Identifier }, // Each linkage will be registered as a fragment and have an ID
      "name" : "<string>", // Each linkage will be registered as a fragment and have at least one name. A single name
    shall be assigned to each linkage
      "residueSite" : "<string>" // Residues shall be captured as described in 5.3.6.8.3
    }],
    "sugar" : [{ // 5.3.6.8.1 Sugar ID (Mandatory)
      "identifier" : { Identifier }, // The Substance ID of the sugar or sugar-like component that make up the
    nucleotide
      "name" : "<string>", // The name of the sugar or sugar-like component that make up the nucleotide
      "residueSite" : "<string>" // The residues that contain a given sugar will be captured. The order of given
    residues will be captured in the 5â€˜-3â€˜direction consistent with the base sequences listed above
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceNucleicAcid;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceNucleicAcid.sequenceType [ CodeableConcept ]; # 0..1 The type of the sequence shall be specified based
```

```
    on a controlled vocabulary
  fhir:SubstanceNucleicAcid.numberOfSubunits [ integer ]; # 0..1 The number of linear sequences of nucleotides linked
    through phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly
    associated typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the
    assumption is that there is 1 subunit
  fhir:SubstanceNucleicAcid.areaOfHybridisation [ string ]; # 0..1 The area of hybridisation shall be described if
    applicable for double stranded RNA or DNA. The number associated with the subunit followed by the number associated
    to the residue shall be specified in increasing order. The underscore â€œâ€ shall be used as separator as follows:
    â€œSubunitnumber Residueâ€
  fhir:SubstanceNucleicAcid.oligoNucleotideType [ CodeableConcept ]; # 0..1 (TBC)
  fhir:SubstanceNucleicAcid.subunit [ # 0..* Subunits are listed in order of decreasing length; sequences of the same
    length will be ordered by molecular weight; subunits that have identical sequences will be repeated multiple times
    fhir:SubstanceNucleicAcid.subunit.subunit [ integer ]; # 0..1 Index of linear sequences of nucleic acids in order of
     decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
    sequences will be repeated and have sequential subscripts
    fhir:SubstanceNucleicAcid.subunit.sequence [ string ]; # 0..1 Actual nucleotide sequence notation from 5' to 3' end
    using standard single letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate
    linkage should also be captured
    fhir:SubstanceNucleicAcid.subunit.length [ integer ]; # 0..1 The length of the sequence shall be captured
    fhir:SubstanceNucleicAcid.subunit.sequenceAttachment [ Attachment ]; # 0..1 (TBC)
    fhir:SubstanceNucleicAcid.subunit.fivePrime [ CodeableConcept ]; # 0..1 The nucleotide present at the 5â€™ terminal
    shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end,
    the 5â€™ prime nucleotide is the letter at the first position in the sequence. A separate representation would be
    redundant
    fhir:SubstanceNucleicAcid.subunit.threePrime [ CodeableConcept ]; # 0..1 The nucleotide present at the 3â€™ terminal
     shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end,
    the 5â€™ prime nucleotide is the letter at the last position in the sequence. A separate representation would be
    redundant
    fhir:SubstanceNucleicAcid.subunit.linkage [ # 0..* The linkages between sugar residues will also be captured
      fhir:SubstanceNucleicAcid.subunit.linkage.connectivity [ string ]; # 0..1 The entity that links the sugar residues
     together should also be captured for nearly all naturally occurring nucleic acid the linkage is a phosphate group.
    For many synthetic oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be
    3â€™-5â€™. If the linkage is either 3â€™-3â€™ or 5â€™-5â€™ this should be specified
      fhir:SubstanceNucleicAcid.subunit.linkage.identifier [ Identifier ]; # 0..1 Each linkage will be registered as a
    fragment and have an ID
      fhir:SubstanceNucleicAcid.subunit.linkage.name [ string ]; # 0..1 Each linkage will be registered as a fragment
    and have at least one name. A single name shall be assigned to each linkage
      fhir:SubstanceNucleicAcid.subunit.linkage.residueSite [ string ]; # 0..1 Residues shall be captured as described
    in 5.3.6.8.3
    ], ...;
    fhir:SubstanceNucleicAcid.subunit.sugar [ # 0..* 5.3.6.8.1 Sugar ID (Mandatory)
      fhir:SubstanceNucleicAcid.subunit.sugar.identifier [ Identifier ]; # 0..1 The Substance ID of the sugar or sugar-
    like component that make up the nucleotide
      fhir:SubstanceNucleicAcid.subunit.sugar.name [ string ]; # 0..1 The name of the sugar or sugar-like component that
     make up the nucleotide
      fhir:SubstanceNucleicAcid.subunit.sugar.residueSite [ string ]; # 0..1 The residues that contain a given sugar
    will be captured. The order of given residues will be captured in the 5â€˜-3â€˜direction consistent with the base
    sequences listed above
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
SubstanceNucleicAcid	ΣTU		DomainResource	Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in the 5’-3’ direction Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
sequenceType	Σ	0..1	CodeableConcept	The type of the sequence shall be specified based on a controlled vocabulary
numberOfSubunits	Σ	0..1	integer	The number of linear sequences of nucleotides linked through phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit
areaOfHybridisation	Σ	0..1	string	The area of hybridisation shall be described if applicable for double stranded RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall be specified in increasing order. The underscore “” shall be used as separator as follows: “Subunitnumber Residue”
oligoNucleotideType	Σ	0..1	CodeableConcept	(TBC)
subunit	Σ	0..*	BackboneElement	Subunits are listed in order of decreasing length; sequences of the same length will be ordered by molecular weight; subunits that have identical sequences will be repeated multiple times
subunit	Σ	0..1	integer	Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts
sequence	Σ	0..1	string	Actual nucleotide sequence notation from 5' to 3' end using standard single letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be captured
length	Σ	0..1	integer	The length of the sequence shall be captured
sequenceAttachment	Σ	0..1	Attachment	(TBC)
fivePrime	Σ	0..1	CodeableConcept	The nucleotide present at the 5’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the first position in the sequence. A separate representation would be redundant
threePrime	Σ	0..1	CodeableConcept	The nucleotide present at the 3’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the last position in

the sequence. A separate representation would be redundant
linkage | Σ | 0..* | BackboneElement | The linkages between sugar residues will also be captured
connectivity | Σ | 0..1 | string | The entity that links the sugar residues together should also be captured for nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3’-5’. If the linkage is either 3’-3’ or 5’-5’ this should be specified
identifier | Σ | 0..1 | Identifier | Each linkage will be registered as a fragment and have an ID
name | Σ | 0..1 | string | Each linkage will be registered as a fragment and have at least one name. A single name shall be assigned to each linkage
residueSite | Σ | 0..1 | string | Residues shall be captured as described in 5.3.6.8.3
sugar | Σ | 0..* | BackboneElement | 5.3.6.8.1 Sugar ID (Mandatory)
identifier | Σ | 0..1 | Identifier | The Substance ID of the sugar or sugar-like component that make up the nucleotide
name | Σ | 0..1 | string | The name of the sugar or sugar-like component that make up the nucleotide
residueSite | Σ | 0..1 | string | The residues that contain a given sugar will be captured. The order of given residues will be captured in the 5‘-3‘direction consistent with the base sequences listed above
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceNucleicAcid xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sequenceType><!-- 0..1 CodeableConcept The type of the sequence shall be specified based on a controlled vocabulary
    --></sequenceType>
 <numberOfSubunits value="[integer]"/><!-- 0..1 The number of linear sequences of nucleotides linked through
    phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly associated
    typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the assumption is that
    there is 1 subunit -->
 <areaOfHybridisation value="[string]"/><!-- 0..1 The area of hybridisation shall be described if applicable for double
    stranded RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall
    be specified in increasing order. The underscore â€œâ€ shall be used as separator as follows: â€œSubunitnumber
    Residueâ€ -->
 <oligoNucleotideType><!-- 0..1 CodeableConcept (TBC) --></oligoNucleotideType>
 <subunit>  <!-- 0..* Subunits are listed in order of decreasing length; sequences of the same length will be ordered by
     molecular weight; subunits that have identical sequences will be repeated multiple times -->
  <subunit value="[integer]"/><!-- 0..1 Index of linear sequences of nucleic acids in order of decreasing length.
    Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be
    repeated and have sequential subscripts -->
  <sequence value="[string]"/><!-- 0..1 Actual nucleotide sequence notation from 5' to 3' end using standard single
    letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be
    captured -->
  <length value="[integer]"/><!-- 0..1 The length of the sequence shall be captured -->
  <sequenceAttachment><!-- 0..1 Attachment (TBC) --></sequenceAttachment>
  <fivePrime><!-- 0..1 CodeableConcept The nucleotide present at the 5â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the first position in the sequence. A separate representation would be redundant --></fivePrime>
  <threePrime><!-- 0..1 CodeableConcept The nucleotide present at the 3â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
     letter at the last position in the sequence. A separate representation would be redundant --></threePrime>
  <linkage>  <!-- 0..* The linkages between sugar residues will also be captured -->
   <connectivity value="[string]"/><!-- 0..1 The entity that links the sugar residues together should also be captured
    for nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic
    oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3â€™-5â€™. If the
    linkage is either 3â€™-3â€™ or 5â€™-5â€™ this should be specified -->
   <identifier><!-- 0..1 Identifier Each linkage will be registered as a fragment and have an ID --></identifier>
   <name value="[string]"/><!-- 0..1 Each linkage will be registered as a fragment and have at least one name. A single
    name shall be assigned to each linkage -->
   <residueSite value="[string]"/><!-- 0..1 Residues shall be captured as described in 5.3.6.8.3 -->
  </linkage>
  <sugar>  <!-- 0..* 5.3.6.8.1 Sugar ID (Mandatory) -->
   <identifier><!-- 0..1 Identifier The Substance ID of the sugar or sugar-like component that make up the nucleotide
    --></identifier>
   <name value="[string]"/><!-- 0..1 The name of the sugar or sugar-like component that make up the nucleotide -->
   <residueSite value="[string]"/><!-- 0..1 The residues that contain a given sugar will be captured. The order of given
     residues will be captured in the 5â€˜-3â€˜direction consistent with the base sequences listed above -->
  </sugar>
 </subunit>
</SubstanceNucleicAcid>
```

JSON Template

```
{
  "resourceType" : "SubstanceNucleicAcid",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sequenceType" : { CodeableConcept }, // The type of the sequence shall be specified based on a controlled vocabulary
  "numberOfSubunits" : <integer>, // The number of linear sequences of nucleotides linked through phosphodiester bonds
    shall be described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-
    Crick base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit
  "areaOfHybridisation" : "<string>", // The area of hybridisation shall be described if applicable for double stranded
    RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall be
    specified in increasing order. The underscore â€œâ€ shall be used as separator as follows: â€œSubunitnumber
    Residueâ€
  "oligoNucleotideType" : { CodeableConcept }, // (TBC)
  "subunit" : [{ // Subunits are listed in order of decreasing length; sequences of the same length will be ordered by
    molecular weight; subunits that have identical sequences will be repeated multiple times
    "subunit" : <integer>, // Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the
```

```
    same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have
    sequential subscripts
    "sequence" : "<string>", // Actual nucleotide sequence notation from 5' to 3' end using standard single letter
    codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be
    captured
    "length" : <integer>, // The length of the sequence shall be captured
    "sequenceAttachment" : { Attachment }, // (TBC)
    "fivePrime" : { CodeableConcept }, // The nucleotide present at the 5â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
    letter at the first position in the sequence. A separate representation would be redundant
    "threePrime" : { CodeableConcept }, // The nucleotide present at the 3â€™ terminal shall be specified based on a
    controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5â€™ prime nucleotide is the
    letter at the last position in the sequence. A separate representation would be redundant
    "linkage" : [{ // The linkages between sugar residues will also be captured
      "connectivity" : "<string>", // The entity that links the sugar residues together should also be captured for
    nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides
    phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3â€™-5â€™. If the linkage is either
    3â€™-3â€™ or 5â€™-5â€™ this should be specified
      "identifier" : { Identifier }, // Each linkage will be registered as a fragment and have an ID
      "name" : "<string>", // Each linkage will be registered as a fragment and have at least one name. A single name
    shall be assigned to each linkage
      "residueSite" : "<string>" // Residues shall be captured as described in 5.3.6.8.3
    }],
    "sugar" : [{ // 5.3.6.8.1 Sugar ID (Mandatory)
      "identifier" : { Identifier }, // The Substance ID of the sugar or sugar-like component that make up the
    nucleotide
      "name" : "<string>", // The name of the sugar or sugar-like component that make up the nucleotide
      "residueSite" : "<string>" // The residues that contain a given sugar will be captured. The order of given
    residues will be captured in the 5â€˜-3â€˜direction consistent with the base sequences listed above
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceNucleicAcid;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceNucleicAcid.sequenceType [ CodeableConcept ]; # 0..1 The type of the sequence shall be specified based
    on a controlled vocabulary
  fhir:SubstanceNucleicAcid.numberOfSubunits [ integer ]; # 0..1 The number of linear sequences of nucleotides linked
    through phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly
    associated typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the
    assumption is that there is 1 subunit
  fhir:SubstanceNucleicAcid.areaOfHybridisation [ string ]; # 0..1 The area of hybridisation shall be described if
    applicable for double stranded RNA or DNA. The number associated with the subunit followed by the number associated
    to the residue shall be specified in increasing order. The underscore â€œâ€ shall be used as separator as follows:
    â€œSubunitnumber Residueâ€
  fhir:SubstanceNucleicAcid.oligoNucleotideType [ CodeableConcept ]; # 0..1 (TBC)
  fhir:SubstanceNucleicAcid.subunit [ # 0..* Subunits are listed in order of decreasing length; sequences of the same
    length will be ordered by molecular weight; subunits that have identical sequences will be repeated multiple times
    fhir:SubstanceNucleicAcid.subunit.subunit [ integer ]; # 0..1 Index of linear sequences of nucleic acids in order of
     decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical
    sequences will be repeated and have sequential subscripts
    fhir:SubstanceNucleicAcid.subunit.sequence [ string ]; # 0..1 Actual nucleotide sequence notation from 5' to 3' end
    using standard single letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate
    linkage should also be captured
    fhir:SubstanceNucleicAcid.subunit.length [ integer ]; # 0..1 The length of the sequence shall be captured
    fhir:SubstanceNucleicAcid.subunit.sequenceAttachment [ Attachment ]; # 0..1 (TBC)
    fhir:SubstanceNucleicAcid.subunit.fivePrime [ CodeableConcept ]; # 0..1 The nucleotide present at the 5â€™ terminal
    shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end,
    the 5â€™ prime nucleotide is the letter at the first position in the sequence. A separate representation would be
    redundant
    fhir:SubstanceNucleicAcid.subunit.threePrime [ CodeableConcept ]; # 0..1 The nucleotide present at the 3â€™ terminal
     shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end,
    the 5â€™ prime nucleotide is the letter at the last position in the sequence. A separate representation would be
    redundant
    fhir:SubstanceNucleicAcid.subunit.linkage [ # 0..* The linkages between sugar residues will also be captured
      fhir:SubstanceNucleicAcid.subunit.linkage.connectivity [ string ]; # 0..1 The entity that links the sugar residues
     together should also be captured for nearly all naturally occurring nucleic acid the linkage is a phosphate group.
    For many synthetic oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be
    3â€™-5â€™. If the linkage is either 3â€™-3â€™ or 5â€™-5â€™ this should be specified
      fhir:SubstanceNucleicAcid.subunit.linkage.identifier [ Identifier ]; # 0..1 Each linkage will be registered as a
    fragment and have an ID
      fhir:SubstanceNucleicAcid.subunit.linkage.name [ string ]; # 0..1 Each linkage will be registered as a fragment
    and have at least one name. A single name shall be assigned to each linkage
      fhir:SubstanceNucleicAcid.subunit.linkage.residueSite [ string ]; # 0..1 Residues shall be captured as described
    in 5.3.6.8.3
    ], ...;
    fhir:SubstanceNucleicAcid.subunit.sugar [ # 0..* 5.3.6.8.1 Sugar ID (Mandatory)
      fhir:SubstanceNucleicAcid.subunit.sugar.identifier [ Identifier ]; # 0..1 The Substance ID of the sugar or sugar-
    like component that make up the nucleotide
      fhir:SubstanceNucleicAcid.subunit.sugar.name [ string ]; # 0..1 The name of the sugar or sugar-like component that
     make up the nucleotide
      fhir:SubstanceNucleicAcid.subunit.sugar.residueSite [ string ]; # 0..1 The residues that contain a given sugar
```

```
    will be captured. The order of given residues will be captured in the 5â€˜-3â€˜direction consistent with the base
    sequences listed above
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.24.3 Resource SubstanceNucleicAcid - Examples

URL: https://hl7.org/fhir/R4/substancenucleicacid-examples.html

11.24.3 Resource SubstanceNucleicAcid - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of substancenucleicacid | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.24.4 Resource SubstanceNucleicAcid - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substancenucleicacid-definitions.html

11.24.4 Resource SubstanceNucleicAcid - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstanceNucleicAcid resource.

SubstanceNucleicAcid
Element Id | SubstanceNucleicAcid
Definition | Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in the 5’-3’ direction.
Cardinality | 0..*
Type | DomainResource
Summary | true
SubstanceNucleicAcid.sequenceType
Element Id | SubstanceNucleicAcid.sequenceType
Definition | The type of the sequence shall be specified based on a controlled vocabulary.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceNucleicAcid.numberOfSubunits
Element Id | SubstanceNucleicAcid.numberOfSubunits
Definition | The number of linear sequences of nucleotides linked through phosphodiester bonds shall be described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-Crick base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit.
Cardinality | 0..1
Type | integer
Summary | true
SubstanceNucleicAcid.areaOfHybridisation
Element Id | SubstanceNucleicAcid.areaOfHybridisation
Definition | The area of hybridisation shall be described if applicable for double stranded RNA or DNA. The number associated with the subunit followed by the number associated to the residue shall be specified in increasing order. The underscore “” shall be used as separator as follows: “Subunitnumber Residue”.
Cardinality | 0..1
Type | string
Summary | true
SubstanceNucleicAcid.oligoNucleotideType
Element Id | SubstanceNucleicAcid.oligoNucleotideType
Definition | (TBC).
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceNucleicAcid.subunit
Element Id | SubstanceNucleicAcid.subunit

Definition | Subunits are listed in order of decreasing length; sequences of the same length will be ordered by molecular weight; subunits that have identical sequences will be repeated multiple times.
Cardinality | 0..*
Summary | true
SubstanceNucleicAcid.subunit.subunit
Element Id | SubstanceNucleicAcid.subunit.subunit
Definition | Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have sequential subscripts.
Cardinality | 0..1
Type | integer
Summary | true
SubstanceNucleicAcid.subunit.sequence
Element Id | SubstanceNucleicAcid.subunit.sequence
Definition | Actual nucleotide sequence notation from 5' to 3' end using standard single letter codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be captured.
Cardinality | 0..1
Type | string
Summary | true
SubstanceNucleicAcid.subunit.length
Element Id | SubstanceNucleicAcid.subunit.length
Definition | The length of the sequence shall be captured.
Cardinality | 0..1
Type | integer
Summary | true
SubstanceNucleicAcid.subunit.sequenceAttachment
Element Id | SubstanceNucleicAcid.subunit.sequenceAttachment
Definition | (TBC).
Cardinality | 0..1
Type | Attachment
Summary | true
SubstanceNucleicAcid.subunit.fivePrime
Element Id | SubstanceNucleicAcid.subunit.fivePrime
Definition | The nucleotide present at the 5’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the first position in the sequence. A separate representation would be redundant.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceNucleicAcid.subunit.threePrime
Element Id | SubstanceNucleicAcid.subunit.threePrime
Definition | The nucleotide present at the 3’ terminal shall be specified based on a controlled vocabulary. Since the sequence is represented from the 5' to the 3' end, the 5’ prime nucleotide is the letter at the last position in the sequence. A separate representation would be redundant.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceNucleicAcid.subunit.linkage
Element Id | SubstanceNucleicAcid.subunit.linkage
Definition | The linkages between sugar residues will also be captured.
Cardinality | 0..*
Summary | true
SubstanceNucleicAcid.subunit.linkage.connectivity
Element Id | SubstanceNucleicAcid.subunit.linkage.connectivity
Definition | The entity that links the sugar residues together should also be captured for nearly all naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3’-5’. If the linkage is either 3’-3’ or 5’-5’ this should be specified.
Cardinality | 0..1
Type | string
Summary | true
SubstanceNucleicAcid.subunit.linkage.identifier
Element Id | SubstanceNucleicAcid.subunit.linkage.identifier
Definition | Each linkage will be registered as a fragment and have an ID.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceNucleicAcid.subunit.linkage.name
Element Id | SubstanceNucleicAcid.subunit.linkage.name
Definition | Each linkage will be registered as a fragment and have at least one name. A single name shall be assigned to each linkage.
Cardinality | 0..1
Type | string
Summary | true

SubstanceNucleicAcid.subunit.linkage.residueSite
Element Id | SubstanceNucleicAcid.subunit.linkage.residueSite
Definition | Residues shall be captured as described in 5.3.6.8.3.
Cardinality | 0..1
Type | string
Summary | true
SubstanceNucleicAcid.subunit.sugar
Element Id | SubstanceNucleicAcid.subunit.sugar
Definition | 5.3.6.8.1 Sugar ID (Mandatory).
Cardinality | 0..*
Summary | true
SubstanceNucleicAcid.subunit.sugar.identifier
Element Id | SubstanceNucleicAcid.subunit.sugar.identifier
Definition | The Substance ID of the sugar or sugar-like component that make up the nucleotide.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceNucleicAcid.subunit.sugar.name
Element Id | SubstanceNucleicAcid.subunit.sugar.name
Definition | The name of the sugar or sugar-like component that make up the nucleotide.
Cardinality | 0..1
Type | string
Summary | true
SubstanceNucleicAcid.subunit.sugar.residueSite
Element Id | SubstanceNucleicAcid.subunit.sugar.residueSite
Definition | The residues that contain a given sugar will be captured. The order of given residues will be captured in the 5‘-3‘direction consistent with the base sequences listed above.
Cardinality | 0..1
Type | string
Summary | true

11.24.5 Resource SubstanceNucleicAcid - Mappings

URL: https://hl7.org/fhir/R4/substancenucleicacid-mappings.html

11.24.5 Resource SubstanceNucleicAcid - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substancenucleicacid resource (see Mappings to Other Standards for further information & status).

11.24.6 Resource SubstanceNucleicAcid - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substancenucleicacid-profiles.html

11.24.6 Resource SubstanceNucleicAcid - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstanceNucleicAcid resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.24.6.1 Profiles

No Profiles defined for this resource

11.24.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers

request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.24.6.3 Search Extensions

No Search Extensions defined for this resource

11.24.7 JSON Schema for SubstanceNucleicAcid

URL: https://hl7.org/fhir/R4/substancenucleicacid.schema.json.html

11.24.7 JSON Schema for SubstanceNucleicAcid

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstanceNucleicAcid, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstanceNucleicAcid",
  "$ref": "#/definitions/SubstanceNucleicAcid",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstanceNucleicAcid": {
      "description": "Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual
    substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in
    the 5’-3’ direction.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstanceNucleicAcid resource",
          "const": "SubstanceNucleicAcid"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "sequenceType": {
      "description": "The type of the sequence shall be specified based on a controlled vocabulary.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "numberOfSubunits": {
      "description": "The number of linear sequences of nucleotides linked through phosphodiester bonds shall be
described. Subunits would be strands of nucleic acids that are tightly associated typically through Watson-Crick
base pairing. NOTE: If not specified in the reference source, the assumption is that there is 1 subunit.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfSubunits": {
      "description": "Extensions for numberOfSubunits",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "areaOfHybridisation": {
      "description": "The area of hybridisation shall be described if applicable for double stranded RNA or DNA. The
 number associated with the subunit followed by the number associated to the residue shall be specified in
increasing order. The underscore “” shall be used as separator as follows: “Subunitnumber Residue”.",
      "$ref": "#/definitions/string"
    },
    "_areaOfHybridisation": {
      "description": "Extensions for areaOfHybridisation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "oligoNucleotideType": {
      "description": "(TBC).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subunit": {
      "description": "Subunits are listed in order of decreasing length; sequences of the same length will be
ordered by molecular weight; subunits that have identical sequences will be repeated multiple times.",
      "items": {
        "$ref": "#/definitions/SubstanceNucleicAcid_Subunit"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SubstanceNucleicAcid_Subunit": {
  "description": "Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual
substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in
the 5’-3’ direction.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
    },
    "subunit": {
      "description": "Index of linear sequences of nucleic acids in order of decreasing length. Sequences of the
same length will be ordered by molecular weight. Subunits that have identical sequences will be repeated and have
sequential subscripts.",
      "$ref": "#/definitions/integer"
    },
    "_subunit": {
      "description": "Extensions for subunit",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sequence": {
      "description": "Actual nucleotide sequence notation from 5\u0027 to 3\u0027 end using standard single letter
codes. In addition to the base sequence, sugar and type of phosphate or non-phosphate linkage should also be
captured.",
      "$ref": "#/definitions/string"
    },
    "_sequence": {
      "description": "Extensions for sequence",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "length": {
      "description": "The length of the sequence shall be captured.",
      "$ref": "#/definitions/integer"
    },
    "_length": {
      "description": "Extensions for length",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sequenceAttachment": {
      "description": "(TBC).",
      "$ref": "Attachment.schema.json#/definitions/Attachment"
    },
    "fivePrime": {
      "description": "The nucleotide present at the 5’ terminal shall be specified based on a controlled vocabulary.
 Since the sequence is represented from the 5\u0027 to the 3\u0027 end, the 5’ prime nucleotide is the letter at the
 first position in the sequence. A separate representation would be redundant.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "threePrime": {
      "description": "The nucleotide present at the 3’ terminal shall be specified based on a controlled vocabulary.
 Since the sequence is represented from the 5\u0027 to the 3\u0027 end, the 5’ prime nucleotide is the letter at the
 last position in the sequence. A separate representation would be redundant.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "linkage": {
      "description": "The linkages between sugar residues will also be captured.",
      "items": {
        "$ref": "#/definitions/SubstanceNucleicAcid_Linkage"
      },
      "type": "array"
    },
    "sugar": {
      "description": "5.3.6.8.1 Sugar ID (Mandatory).",
      "items": {
        "$ref": "#/definitions/SubstanceNucleicAcid_Sugar"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceNucleicAcid_Linkage": {
  "description": "Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual
substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in
the 5’-3’ direction.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
```

```
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "connectivity": {
      "description": "The entity that links the sugar residues together should also be captured for nearly all
naturally occurring nucleic acid the linkage is a phosphate group. For many synthetic oligonucleotides
phosphorothioate linkages are often seen. Linkage connectivity is assumed to be 3’-5’. If the linkage is either
3’-3’ or 5’-5’ this should be specified.",
      "$ref": "#/definitions/string"
    },
    "_connectivity": {
      "description": "Extensions for connectivity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "Each linkage will be registered as a fragment and have an ID.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "name": {
      "description": "Each linkage will be registered as a fragment and have at least one name. A single name shall
be assigned to each linkage.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "residueSite": {
      "description": "Residues shall be captured as described in 5.3.6.8.3.",
      "$ref": "#/definitions/string"
    },
    "_residueSite": {
      "description": "Extensions for residueSite",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SubstanceNucleicAcid_Sugar": {
  "description": "Nucleic acids are defined by three distinct elements: the base, sugar and linkage. Individual
substance/moiety IDs will be created for each of these elements. The nucleotide sequence will be always entered in
the 5’-3’ direction.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "The Substance ID of the sugar or sugar-like component that make up the nucleotide.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "name": {
      "description": "The name of the sugar or sugar-like component that make up the nucleotide.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "residueSite": {
```

```
          "description": "The residues that contain a given sugar will be captured. The order of given residues will be
    captured in the 5‘-3‘direction consistent with the base sequences listed above.",
          "$ref": "#/definitions/string"
        },
        "_residueSite": {
          "description": "Extensions for residueSite",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


11.24.8 Resource SubstanceNucleicAcid - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substancenucleicacid-dependencies.html

11.24.8 Resource SubstanceNucleicAcid - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstanceNucleicAcid FMM level 0

Name | Card. | Type | Dependency Analysis
SubstanceNucleicAcid | | DomainResource
sequenceType | 0..1 | CodeableConcept
numberOfSubunits | 0..1 | integer
areaOfHybridisation | 0..1 | string
oligoNucleotideType | 0..1 | CodeableConcept
subunit | 0..* | BackboneElement
subunit | 0..1 | integer
sequence | 0..1 | string
length | 0..1 | integer
sequenceAttachment | 0..1 | Attachment
fivePrime | 0..1 | CodeableConcept
threePrime | 0..1 | CodeableConcept
linkage | 0..* | BackboneElement
connectivity | 0..1 | string
identifier | 0..1 | Identifier
name | 0..1 | string
residueSite | 0..1 | string
sugar | 0..* | BackboneElement
identifier | 0..1 | Identifier
name | 0..1 | string
residueSite | 0..1 | string

11.25 Resource SubstanceReferenceInformation - Content

URL: https://hl7.org/fhir/R4/substancereferenceinformation.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Todo.

11.25.1 Scope and Usage

This resource is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft resource not suitable for production implementation

This resource is referenced by SubstanceSpecification

11.25.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints

SubstanceReferenceInformation | ΣTU | | DomainResource | Todo Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
comment | Σ | 0..1 | string | Todo
gene | Σ | 0..* | BackboneElement | Todo
geneSequenceOrigin | Σ | 0..1 | CodeableConcept | Todo
gene | Σ | 0..1 | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
geneElement | Σ | 0..* | BackboneElement | Todo
type | Σ | 0..1 | CodeableConcept | Todo
element | Σ | 0..1 | Identifier | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
classification | Σ | 0..* | BackboneElement | Todo
domain | Σ | 0..1 | CodeableConcept | Todo
classification | Σ | 0..1 | CodeableConcept | Todo
subtype | Σ | 0..* | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
target | Σ | 0..* | BackboneElement | Todo
target | Σ | 0..1 | Identifier | Todo
type | Σ | 0..1 | CodeableConcept | Todo
interaction | Σ | 0..1 | CodeableConcept | Todo
organism | Σ | 0..1 | CodeableConcept | Todo
organismType | Σ | 0..1 | CodeableConcept | Todo
amount[x] | Σ | 0..1 | | Todo
amountQuantity | | | Quantity
amountRange | | | Range
amountString | | | string
amountType | Σ | 0..1 | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceReferenceInformation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <comment value="[string]"/><!-- 0..1 Todo -->
 <gene>  <!-- 0..* Todo -->
  <geneSequenceOrigin><!-- 0..1 CodeableConcept Todo --></geneSequenceOrigin>
  <gene><!-- 0..1 CodeableConcept Todo --></gene>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </gene>
 <geneElement>  <!-- 0..* Todo -->
  <type><!-- 0..1 CodeableConcept Todo --></type>
  <element><!-- 0..1 Identifier Todo --></element>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </geneElement>
 <classification>  <!-- 0..* Todo -->
  <domain><!-- 0..1 CodeableConcept Todo --></domain>
  <classification><!-- 0..1 CodeableConcept Todo --></classification>
  <subtype><!-- 0..* CodeableConcept Todo --></subtype>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </classification>
 <target>  <!-- 0..* Todo -->
  <target><!-- 0..1 Identifier Todo --></target>
  <type><!-- 0..1 CodeableConcept Todo --></type>
  <interaction><!-- 0..1 CodeableConcept Todo --></interaction>
  <organism><!-- 0..1 CodeableConcept Todo --></organism>
  <organismType><!-- 0..1 CodeableConcept Todo --></organismType>
  <amount[x]><!-- 0..1 Quantity|Range|string Todo --></amount[x]>
  <amountType><!-- 0..1 CodeableConcept Todo --></amountType>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </target>
</SubstanceReferenceInformation>
```

JSON Template

```
{
  "resourceType" : "SubstanceReferenceInformation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "comment" : "<string>", // Todo
  "gene" : [{ // Todo
    "geneSequenceOrigin" : { CodeableConcept }, // Todo
    "gene" : { CodeableConcept }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "geneElement" : [{ // Todo
    "type" : { CodeableConcept }, // Todo
    "element" : { Identifier }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "classification" : [{ // Todo
```

```
    "domain" : { CodeableConcept }, // Todo
    "classification" : { CodeableConcept }, // Todo
    "subtype" : [{ CodeableConcept }], // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "target" : [{ // Todo
    "target" : { Identifier }, // Todo
    "type" : { CodeableConcept }, // Todo
    "interaction" : { CodeableConcept }, // Todo
    "organism" : { CodeableConcept }, // Todo
    "organismType" : { CodeableConcept }, // Todo
    // amount[x]: Todo. One of these 3:
    "amountQuantity" : { Quantity },
    "amountRange" : { Range },
    "amountString" : "<string>",
    "amountType" : { CodeableConcept }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceReferenceInformation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceReferenceInformation.comment [ string ]; # 0..1 Todo
  fhir:SubstanceReferenceInformation.gene [ # 0..* Todo
    fhir:SubstanceReferenceInformation.gene.geneSequenceOrigin [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.gene.gene [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.gene.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.geneElement [ # 0..* Todo
    fhir:SubstanceReferenceInformation.geneElement.type [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.geneElement.element [ Identifier ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.geneElement.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.classification [ # 0..* Todo
    fhir:SubstanceReferenceInformation.classification.domain [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.classification.classification [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.classification.subtype [ CodeableConcept ], ... ; # 0..* Todo
    fhir:SubstanceReferenceInformation.classification.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.target [ # 0..* Todo
    fhir:SubstanceReferenceInformation.target.target [ Identifier ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.type [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.interaction [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.organism [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.organismType [ CodeableConcept ]; # 0..1 Todo
    # SubstanceReferenceInformation.target.amount[x] : 0..1 Todo. One of these 3
      fhir:SubstanceReferenceInformation.target.amountQuantity [ Quantity ]
      fhir:SubstanceReferenceInformation.target.amountRange [ Range ]
      fhir:SubstanceReferenceInformation.target.amountString [ string ]
    fhir:SubstanceReferenceInformation.target.amountType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceReferenceInformation | ΣTU | | DomainResource | Todo Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
comment | Σ | 0..1 | string | Todo
gene | Σ | 0..* | BackboneElement | Todo
geneSequenceOrigin | Σ | 0..1 | CodeableConcept | Todo
gene | Σ | 0..1 | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
geneElement | Σ | 0..* | BackboneElement | Todo
type | Σ | 0..1 | CodeableConcept | Todo
element | Σ | 0..1 | Identifier | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
classification | Σ | 0..* | BackboneElement | Todo
domain | Σ | 0..1 | CodeableConcept | Todo
classification | Σ | 0..1 | CodeableConcept | Todo
subtype | Σ | 0..* | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
target | Σ | 0..* | BackboneElement | Todo

target | Σ | 0..1 | Identifier | Todo
type | Σ | 0..1 | CodeableConcept | Todo
interaction | Σ | 0..1 | CodeableConcept | Todo
organism | Σ | 0..1 | CodeableConcept | Todo
organismType | Σ | 0..1 | CodeableConcept | Todo
amount[x] | Σ | 0..1 | | Todo
amountQuantity | | | Quantity
amountRange | | | Range
amountString | | | string
amountType | Σ | 0..1 | CodeableConcept | Todo
source | Σ | 0..* | Reference(DocumentReference) | Todo
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceReferenceInformation xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <comment value="[string]"/><!-- 0..1 Todo -->
 <gene>  <!-- 0..* Todo -->
  <geneSequenceOrigin><!-- 0..1 CodeableConcept Todo --></geneSequenceOrigin>
  <gene><!-- 0..1 CodeableConcept Todo --></gene>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </gene>
 <geneElement>  <!-- 0..* Todo -->
  <type><!-- 0..1 CodeableConcept Todo --></type>
  <element><!-- 0..1 Identifier Todo --></element>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </geneElement>
 <classification>  <!-- 0..* Todo -->
  <domain><!-- 0..1 CodeableConcept Todo --></domain>
  <classification><!-- 0..1 CodeableConcept Todo --></classification>
  <subtype><!-- 0..* CodeableConcept Todo --></subtype>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </classification>
 <target>  <!-- 0..* Todo -->
  <target><!-- 0..1 Identifier Todo --></target>
  <type><!-- 0..1 CodeableConcept Todo --></type>
  <interaction><!-- 0..1 CodeableConcept Todo --></interaction>
  <organism><!-- 0..1 CodeableConcept Todo --></organism>
  <organismType><!-- 0..1 CodeableConcept Todo --></organismType>
  <amount[x]><!-- 0..1 Quantity|Range|string Todo --></amount[x]>
  <amountType><!-- 0..1 CodeableConcept Todo --></amountType>
  <source><!-- 0..* Reference(DocumentReference) Todo --></source>
 </target>
</SubstanceReferenceInformation>
```

JSON Template

```
{
  "resourceType" : "SubstanceReferenceInformation",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "comment" : "<string>", // Todo
  "gene" : [{ // Todo
    "geneSequenceOrigin" : { CodeableConcept }, // Todo
    "gene" : { CodeableConcept }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "geneElement" : [{ // Todo
    "type" : { CodeableConcept }, // Todo
    "element" : { Identifier }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "classification" : [{ // Todo
    "domain" : { CodeableConcept }, // Todo
    "classification" : { CodeableConcept }, // Todo
    "subtype" : [{ CodeableConcept }], // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }],
  "target" : [{ // Todo
    "target" : { Identifier }, // Todo
    "type" : { CodeableConcept }, // Todo
    "interaction" : { CodeableConcept }, // Todo
    "organism" : { CodeableConcept }, // Todo
    "organismType" : { CodeableConcept }, // Todo
    // amount[x]: Todo. One of these 3:
    "amountQuantity" : { Quantity },
    "amountRange" : { Range },
    "amountString" : "<string>",
    "amountType" : { CodeableConcept }, // Todo
    "source" : [{ Reference(DocumentReference) }] // Todo
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceReferenceInformation;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceReferenceInformation.comment [ string ]; # 0..1 Todo
  fhir:SubstanceReferenceInformation.gene [ # 0..* Todo
    fhir:SubstanceReferenceInformation.gene.geneSequenceOrigin [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.gene.gene [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.gene.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.geneElement [ # 0..* Todo
    fhir:SubstanceReferenceInformation.geneElement.type [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.geneElement.element [ Identifier ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.geneElement.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.classification [ # 0..* Todo
    fhir:SubstanceReferenceInformation.classification.domain [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.classification.classification [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.classification.subtype [ CodeableConcept ], ... ; # 0..* Todo
    fhir:SubstanceReferenceInformation.classification.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
  fhir:SubstanceReferenceInformation.target [ # 0..* Todo
    fhir:SubstanceReferenceInformation.target.target [ Identifier ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.type [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.interaction [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.organism [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.organismType [ CodeableConcept ]; # 0..1 Todo
    # SubstanceReferenceInformation.target.amount[x] : 0..1 Todo. One of these 3
      fhir:SubstanceReferenceInformation.target.amountQuantity [ Quantity ]
      fhir:SubstanceReferenceInformation.target.amountRange [ Range ]
      fhir:SubstanceReferenceInformation.target.amountString [ string ]
    fhir:SubstanceReferenceInformation.target.amountType [ CodeableConcept ]; # 0..1 Todo
    fhir:SubstanceReferenceInformation.target.source [ Reference(DocumentReference) ], ... ; # 0..* Todo
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.25.3 Resource SubstanceReferenceInformation - Examples

URL: https://hl7.org/fhir/R4/substancereferenceinformation-examples.html

11.25.3 Resource SubstanceReferenceInformation - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of substancereferenceinformation | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.25.4 Resource SubstanceReferenceInformation - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substancereferenceinformation-definitions.html

11.25.4 Resource SubstanceReferenceInformation - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstanceReferenceInformation resource.

SubstanceReferenceInformation
Element Id | SubstanceReferenceInformation
Definition | Todo.
Cardinality | 0..*
Type | DomainResource
Summary | true
SubstanceReferenceInformation.comment
Element Id | SubstanceReferenceInformation.comment
Definition | Todo.

Cardinality | 0..1
Type | string
Summary | true
SubstanceReferenceInformation.gene
Element Id | SubstanceReferenceInformation.gene
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstanceReferenceInformation.gene.geneSequenceOrigin
Element Id | SubstanceReferenceInformation.gene.geneSequenceOrigin
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.gene.gene
Element Id | SubstanceReferenceInformation.gene.gene
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.gene.source
Element Id | SubstanceReferenceInformation.gene.source
Definition | Todo.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceReferenceInformation.geneElement
Element Id | SubstanceReferenceInformation.geneElement
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstanceReferenceInformation.geneElement.type
Element Id | SubstanceReferenceInformation.geneElement.type
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.geneElement.element
Element Id | SubstanceReferenceInformation.geneElement.element
Definition | Todo.
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceReferenceInformation.geneElement.source
Element Id | SubstanceReferenceInformation.geneElement.source
Definition | Todo.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceReferenceInformation.classification
Element Id | SubstanceReferenceInformation.classification
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstanceReferenceInformation.classification.domain
Element Id | SubstanceReferenceInformation.classification.domain
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.classification.classification
Element Id | SubstanceReferenceInformation.classification.classification
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.classification.subtype
Element Id | SubstanceReferenceInformation.classification.subtype
Definition | Todo.
Cardinality | 0..*
Type | CodeableConcept
Summary | true

SubstanceReferenceInformation.classification.source
Element Id | SubstanceReferenceInformation.classification.source
Definition | Todo.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
SubstanceReferenceInformation.target
Element Id | SubstanceReferenceInformation.target
Definition | Todo.
Cardinality | 0..*
Summary | true
SubstanceReferenceInformation.target.target
Element Id | SubstanceReferenceInformation.target.target
Definition | Todo.
Cardinality | 0..1
Type | Identifier
Summary | true
SubstanceReferenceInformation.target.type
Element Id | SubstanceReferenceInformation.target.type
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.target.interaction
Element Id | SubstanceReferenceInformation.target.interaction
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.target.organism
Element Id | SubstanceReferenceInformation.target.organism
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.target.organismType
Element Id | SubstanceReferenceInformation.target.organismType
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.target.amount[x]
Element Id | SubstanceReferenceInformation.target.amount[x]
Definition | Todo.
Cardinality | 0..1
Type | Quantity|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceReferenceInformation.target.amountType
Element Id | SubstanceReferenceInformation.target.amountType
Definition | Todo.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceReferenceInformation.target.source
Element Id | SubstanceReferenceInformation.target.source
Definition | Todo.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true

11.25.5 Resource SubstanceReferenceInformation - Mappings

URL: https://hl7.org/fhir/R4/substancereferenceinformation-mappings.html

11.25.5 Resource SubstanceReferenceInformation - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substancereferenceinformation resource (see Mappings to Other Standards for further information & status).

11.25.6 Resource SubstanceReferenceInformation - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substancereferenceinformation-profiles.html

11.25.6 Resource SubstanceReferenceInformation - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstanceReferenceInformation resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.25.6.1 Profiles

No Profiles defined for this resource

11.25.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.25.6.3 Search Extensions

No Search Extensions defined for this resource

11.25.7 JSON Schema for SubstanceReferenceInformation

URL: https://hl7.org/fhir/R4/substancereferenceinformation.schema.json.html

11.25.7 JSON Schema for SubstanceReferenceInformation

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstanceReferenceInformation, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstanceReferenceInformation",
  "$ref": "#/definitions/SubstanceReferenceInformation",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstanceReferenceInformation": {
      "description": "Todo.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstanceReferenceInformation resource",
          "const": "SubstanceReferenceInformation"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "comment": {
      "description": "Todo.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "gene": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstanceReferenceInformation_Gene"
      },
      "type": "array"
    },
    "geneElement": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstanceReferenceInformation_GeneElement"
      },
      "type": "array"
    },
    "classification": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstanceReferenceInformation_Classification"
      },
      "type": "array"
    },
    "target": {
      "description": "Todo.",
      "items": {
        "$ref": "#/definitions/SubstanceReferenceInformation_Target"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
```

```
  ]
},
"SubstanceReferenceInformation_Gene": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "geneSequenceOrigin": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "gene": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "source": {
      "description": "Todo.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceReferenceInformation_GeneElement": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "element": {
```

```
      "description": "Todo.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "source": {
      "description": "Todo.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceReferenceInformation_Classification": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "domain": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "classification": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subtype": {
      "description": "Todo.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "source": {
      "description": "Todo.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"SubstanceReferenceInformation_Target": {
  "description": "Todo.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "target": {
      "description": "Todo.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "type": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "interaction": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "organism": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "organismType": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "amountQuantity": {
      "description": "Todo.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "amountRange": {
      "description": "Todo.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "amountString": {
      "description": "Todo.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_amountString": {
      "description": "Extensions for amountString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "amountType": {
      "description": "Todo.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "source": {
      "description": "Todo.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
 }
 }
}
```


11.25.8 Resource SubstanceReferenceInformation - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substancereferenceinformation-dependencies.html

11.25.8 Resource SubstanceReferenceInformation - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstanceReferenceInformation FMM level 0

Name | Card. | Type | Dependency Analysis
SubstanceReferenceInformation | | DomainResource
comment | 0..1 | string
gene | 0..* | BackboneElement
geneSequenceOrigin | 0..1 | CodeableConcept

gene | 0..1 | CodeableConcept
source | 0..* | Reference(DocumentReference)
geneElement | 0..* | BackboneElement
type | 0..1 | CodeableConcept
element | 0..1 | Identifier
source | 0..* | Reference(DocumentReference)
classification | 0..* | BackboneElement
domain | 0..1 | CodeableConcept
classification | 0..1 | CodeableConcept
subtype | 0..* | CodeableConcept
source | 0..* | Reference(DocumentReference)
target | 0..* | BackboneElement
target | 0..1 | Identifier
type | 0..1 | CodeableConcept
interaction | 0..1 | CodeableConcept
organism | 0..1 | CodeableConcept
organismType | 0..1 | CodeableConcept
amount[x] | 0..1
amountQuantity | | Quantity
amountRange | | Range
amountString | | string
amountType | 0..1 | CodeableConcept
source | 0..* | Reference(DocumentReference)

11.26 Resource SubstanceSourceMaterial - Content

URL: https://hl7.org/fhir/R4/substancesourcematerial.html
Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Source material shall capture information on the taxonomic and anatomical origins as well as the fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be described using a controlled vocabulary as required. This information is captured for naturally derived polymers (. starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For Herbal preparations, the fraction information will be captured at the Substance information level and additional information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for further explanation the Substance Class: Structurally Diverse and the herbal annex.

11.26.1 Scope and Usage

This resource is referenced by SubstanceSpecification

11.26.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceSourceMaterial | ΣTU | | DomainResource | Source material shall capture information on the taxonomic and anatomical origins as well as the fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be described using a controlled vocabulary as required. This information is captured for naturally derived polymers (. starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For Herbal preparations, the fraction information will be captured at the Substance information level and additional information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for further explanation the Substance Class: Structurally Diverse and the herbal annex Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
sourceMaterialClass | Σ | 0..1 | CodeableConcept | General high level classification of the source material specific to the origin of the material
sourceMaterialType | Σ | 0..1 | CodeableConcept | The type of the source material shall be specified based on a controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
sourceMaterialState | Σ | 0..1 | CodeableConcept | The state of the source material when extracted
organismId | Σ | 0..1 | Identifier | The unique identifier associated with the source material parent organism shall be specified
organismName | Σ | 0..1 | string | The organism accepted Scientific name shall be provided based on the organism taxonomy

parentSubstanceId | Σ | 0..* | Identifier | The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)

parentSubstanceName | Σ | 0..* | string | The parent substance of the Herbal Drug, or Herbal preparation

countryOfOrigin | Σ | 0..* | CodeableConcept | The country where the plant material is harvested or the countries where the plasma is sourced from as laid down in accordance with the Plasma Master File. For “Plasma-derived substances” the attribute country of origin provides information about the countries used for the manufacturing of the Cryopoor plama or Crioprecipitate

geographicalLocation | Σ | 0..* | string | The place/region where the plant is harvested or the places/regions where the animal source material has its habitat

developmentStage | Σ | 0..1 | CodeableConcept | Stage of life for animals, plants, insects and microorganisms. This information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine serum)

fractionDescription | Σ | 0..* | BackboneElement | Many complex materials are fractions of parts of plants, animals, or minerals. Fraction elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from Plants, fraction information will be captured at the Substance information level (. Oils, Juices and Exudates). Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified Substance Group 1 information level. For plasma-derived products fraction information will be captured at the Substance and the Specified Substance Group 1 levels

fraction | Σ | 0..1 | string | This element is capturing information about the fraction of a plant part, or human plasma for fractionation

materialType | Σ | 0..1 | CodeableConcept | The specific type of the material constituting the component. For Herbal preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1

organism | Σ | 0..1 | BackboneElement | This subclause describes the organism which the substance is derived from. For vaccines, the parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described for the Substance Name: ., Leaf

family | Σ | 0..1 | CodeableConcept | The family of an organism shall be specified

genus | Σ | 0..1 | CodeableConcept | The genus of an organism shall be specified; refers to the Latin epithet of the genus element of the plant/animal scientific name; it is present in names for genera, species and infraspecies

species | Σ | 0..1 | CodeableConcept | The species of an organism shall be specified; refers to the Latin epithet of the species of the plant/animal; it is present in names for species and infraspecies

intraspecificType | Σ | 0..1 | CodeableConcept | The Intraspecific type of an organism shall be specified

intraspecificDescription | Σ | 0..1 | string | The intraspecific description of an organism shall be specified based on a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the antigen in line with the WHO convention

author | Σ | 0..* | BackboneElement | 4.9.13.6.1 Author type (Conditional)

authorType | Σ | 0..1 | CodeableConcept | The type of author of an organism species shall be specified. The parenthetical author of an organism species refers to the first author who published the plant/animal name (of any rank). The primary author of an organism species refers to the first author(s), who validly published the plant/animal name

authorDescription | Σ | 0..1 | string | The author of an organism species shall be specified. The author year of an organism shall also be specified when applicable; refers to the year in which the first author(s) published the infraspecific plant/animal name (of any rank)

hybrid | Σ | 0..1 | BackboneElement | 4.9.13.8.1 Hybrid species maternal organism ID (Optional)

maternalOrganismId | Σ | 0..1 | string | The identifier of the maternal species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal

maternalOrganismName | Σ | 0..1 | string | The name of the maternal species constituting the hybrid organism shall be specified. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal

paternalOrganismId | Σ | 0..1 | string | The identifier of the paternal species constituting the hybrid organism shall be specified based on a controlled vocabulary

paternalOrganismName | Σ | 0..1 | string | The name of the paternal species constituting the hybrid organism shall be specified

hybridType | Σ | 0..1 | CodeableConcept | The hybrid type of an organism shall be specified

organismGeneral | Σ | 0..1 | BackboneElement | 4.9.13.7.1 Kingdom (Conditional)

kingdom | Σ | 0..1 | CodeableConcept | The kingdom of an organism shall be specified

phylum | Σ | 0..1 | CodeableConcept | The phylum of an organism shall be specified

class | Σ | 0..1 | CodeableConcept | The class of an organism shall be specified

order | Σ | 0..1 | CodeableConcept | The order of an organism shall be specified,

partDescription | Σ | 0..* | BackboneElement | To do

part | Σ | 0..1 | CodeableConcept | Entity of anatomical origin of source material within an organism

partLocation | Σ | 0..1 | CodeableConcept | The detailed anatomic location when the part can be extracted from different anatomical locations of the organism. Multiple alternative locations may apply

Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceSourceMaterial xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sourceMaterialClass><!-- 0..1 CodeableConcept General high level classification of the source material specific to the
     origin of the material --></sourceMaterialClass>
 <sourceMaterialType><!-- 0..1 CodeableConcept The type of the source material shall be specified based on a controlled
    vocabulary. For vaccines, this subclause refers to the class of infectious agent --></sourceMaterialType>
 <sourceMaterialState><!-- 0..1 CodeableConcept The state of the source material when extracted
    --></sourceMaterialState>
 <organismId><!-- 0..1 Identifier The unique identifier associated with the source material parent organism shall be
    specified --></organismId>
 <organismName value="[string]"/><!-- 0..1 The organism accepted Scientific name shall be provided based on the organism
     taxonomy -->
 <parentSubstanceId><!-- 0..* Identifier The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the
    substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant) --></parentSubstanceId>
 <parentSubstanceName value="[string]"/><!-- 0..* The parent substance of the Herbal Drug, or Herbal preparation -->
```

```
 <countryOfOrigin><!-- 0..* CodeableConcept The country where the plant material is harvested or the countries where the
     plasma is sourced from as laid down in accordance with the Plasma Master File. For â€œPlasma-derived substancesâ€
    the attribute country of origin provides information about the countries used for the manufacturing of the Cryopoor
    plama or Crioprecipitate --></countryOfOrigin>
 <geographicalLocation value="[string]"/><!-- 0..* The place/region where the plant is harvested or the places/regions
    where the animal source material has its habitat -->
 <developmentStage><!-- 0..1 CodeableConcept Stage of life for animals, plants, insects and microorganisms. This
    information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine
     serum) --></developmentStage>
 <fractionDescription>  <!-- 0..* Many complex materials are fractions of parts of plants, animals, or minerals.
    Fraction elements are often necessary to define both Substances and Specified Group 1 Substances. For substances
    derived from Plants, fraction information will be captured at the Substance information level ( . Oils, Juices and
    Exudates). Additional information for Extracts, such as extraction solvent composition, will be captured at the
    Specified Substance Group 1 information level. For plasma-derived products fraction information will be captured at
    the Substance and the Specified Substance Group 1 levels -->
  <fraction value="[string]"/><!-- 0..1 This element is capturing information about the fraction of a plant part, or
    human plasma for fractionation -->
  <materialType><!-- 0..1 CodeableConcept The specific type of the material constituting the component. For Herbal
    preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1
    --></materialType>
 </fractionDescription>
 <organism>  <!-- 0..1 This subclause describes the organism which the substance is derived from. For vaccines, the
    parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described
     for the Substance Name: ., Leaf -->
  <family><!-- 0..1 CodeableConcept The family of an organism shall be specified --></family>
  <genus><!-- 0..1 CodeableConcept The genus of an organism shall be specified; refers to the Latin epithet of the genus
     element of the plant/animal scientific name; it is present in names for genera, species and infraspecies
    --></genus>
  <species><!-- 0..1 CodeableConcept The species of an organism shall be specified; refers to the Latin epithet of the
    species of the plant/animal; it is present in names for species and infraspecies --></species>
  <intraspecificType><!-- 0..1 CodeableConcept The Intraspecific type of an organism shall be specified
    --></intraspecificType>
  <intraspecificDescription value="[string]"/><!-- 0..1 The intraspecific description of an organism shall be specified
    based on a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of
    the antigen in line with the WHO convention -->
  <author>  <!-- 0..* 4.9.13.6.1 Author type (Conditional) -->
   <authorType><!-- 0..1 CodeableConcept The type of author of an organism species shall be specified. The parenthetical
     author of an organism species refers to the first author who published the plant/animal name (of any rank). The
    primary author of an organism species refers to the first author(s), who validly published the plant/animal name
    --></authorType>
   <authorDescription value="[string]"/><!-- 0..1 The author of an organism species shall be specified. The author year
    of an organism shall also be specified when applicable; refers to the year in which the first author(s) published
    the infraspecific plant/animal name (of any rank) -->
  </author>
  <hybrid>  <!-- 0..1 4.9.13.8.1 Hybrid species maternal organism ID (Optional) -->
   <maternalOrganismId value="[string]"/><!-- 0..1 The identifier of the maternal species constituting the hybrid
    organism shall be specified based on a controlled vocabulary. For plants, the parents arenâ€™t always known, and it
    is unlikely that it will be known which is maternal and which is paternal -->
   <maternalOrganismName value="[string]"/><!-- 0..1 The name of the maternal species constituting the hybrid organism
    shall be specified. For plants, the parents arenâ€™t always known, and it is unlikely that it will be known which is
     maternal and which is paternal -->
   <paternalOrganismId value="[string]"/><!-- 0..1 The identifier of the paternal species constituting the hybrid
    organism shall be specified based on a controlled vocabulary -->
   <paternalOrganismName value="[string]"/><!-- 0..1 The name of the paternal species constituting the hybrid organism
    shall be specified -->
   <hybridType><!-- 0..1 CodeableConcept The hybrid type of an organism shall be specified --></hybridType>
  </hybrid>
  <organismGeneral>  <!-- 0..1 4.9.13.7.1 Kingdom (Conditional) -->
   <kingdom><!-- 0..1 CodeableConcept The kingdom of an organism shall be specified --></kingdom>
   <phylum><!-- 0..1 CodeableConcept The phylum of an organism shall be specified --></phylum>
   <class><!-- 0..1 CodeableConcept The class of an organism shall be specified --></class>
   <order><!-- 0..1 CodeableConcept The order of an organism shall be specified, --></order>
  </organismGeneral>
 </organism>
 <partDescription>  <!-- 0..* To do -->
  <part><!-- 0..1 CodeableConcept Entity of anatomical origin of source material within an organism --></part>
  <partLocation><!-- 0..1 CodeableConcept The detailed anatomic location when the part can be extracted from different
    anatomical locations of the organism. Multiple alternative locations may apply --></partLocation>
 </partDescription>
</SubstanceSourceMaterial>
```

JSON Template

```
{
  "resourceType" : "SubstanceSourceMaterial",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sourceMaterialClass" : { CodeableConcept }, // General high level classification of the source material specific to
    the origin of the material
  "sourceMaterialType" : { CodeableConcept }, // The type of the source material shall be specified based on a
    controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
  "sourceMaterialState" : { CodeableConcept }, // The state of the source material when extracted
  "organismId" : { Identifier }, // The unique identifier associated with the source material parent organism shall be
    specified
  "organismName" : "<string>", // The organism accepted Scientific name shall be provided based on the organism taxonomy
  "parentSubstanceId" : [{ Identifier }], // The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of
    the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)
  "parentSubstanceName" : ["<string>"], // The parent substance of the Herbal Drug, or Herbal preparation
  "countryOfOrigin" : [{ CodeableConcept }], // The country where the plant material is harvested or the countries where
     the plasma is sourced from as laid down in accordance with the Plasma Master File. For â€œPlasma-derived
```

```
    substancesâ€€ the attribute country of origin provides information about the countries used for the manufacturing of
     the Cryopoor plama or Crioprecipitate
  "geographicalLocation" : ["<string>"], // The place/region where the plant is harvested or the places/regions where
    the animal source material has its habitat
  "developmentStage" : { CodeableConcept }, // Stage of life for animals, plants, insects and microorganisms. This
    information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine
     serum)
  "fractionDescription" : [{ // Many complex materials are fractions of parts of plants, animals, or minerals. Fraction
    elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from
     Plants, fraction information will be captured at the Substance information level ( . Oils, Juices and Exudates).
    Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified
    Substance Group 1 information level. For plasma-derived products fraction information will be captured at the
    Substance and the Specified Substance Group 1 levels
    "fraction" : "<string>", // This element is capturing information about the fraction of a plant part, or human
    plasma for fractionation
    "materialType" : { CodeableConcept } // The specific type of the material constituting the component. For Herbal
    preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1
  }],
  "organism" : { // This subclause describes the organism which the substance is derived from. For vaccines, the parent
    organism shall be specified based on these subclause elements. As an example, full taxonomy will be described for
    the Substance Name: ., Leaf
    "family" : { CodeableConcept }, // The family of an organism shall be specified
    "genus" : { CodeableConcept }, // The genus of an organism shall be specified; refers to the Latin epithet of the
    genus element of the plant/animal scientific name; it is present in names for genera, species and infraspecies
    "species" : { CodeableConcept }, // The species of an organism shall be specified; refers to the Latin epithet of
    the species of the plant/animal; it is present in names for species and infraspecies
    "intraspecificType" : { CodeableConcept }, // The Intraspecific type of an organism shall be specified
    "intraspecificDescription" : "<string>", // The intraspecific description of an organism shall be specified based on
     a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the
    antigen in line with the WHO convention
    "author" : [{ // 4.9.13.6.1 Author type (Conditional)
      "authorType" : { CodeableConcept }, // The type of author of an organism species shall be specified. The
    parenthetical author of an organism species refers to the first author who published the plant/animal name (of any
    rank). The primary author of an organism species refers to the first author(s), who validly published the
    plant/animal name
      "authorDescription" : "<string>" // The author of an organism species shall be specified. The author year of an
    organism shall also be specified when applicable; refers to the year in which the first author(s) published the
    infraspecific plant/animal name (of any rank)
    }],
    "hybrid" : { // 4.9.13.8.1 Hybrid species maternal organism ID (Optional)
      "maternalOrganismId" : "<string>", // The identifier of the maternal species constituting the hybrid organism
    shall be specified based on a controlled vocabulary. For plants, the parents arenâ€™t always known, and it is
    unlikely that it will be known which is maternal and which is paternal
      "maternalOrganismName" : "<string>", // The name of the maternal species constituting the hybrid organism shall be
     specified. For plants, the parents arenâ€™t always known, and it is unlikely that it will be known which is
    maternal and which is paternal
      "paternalOrganismId" : "<string>", // The identifier of the paternal species constituting the hybrid organism
    shall be specified based on a controlled vocabulary
      "paternalOrganismName" : "<string>", // The name of the paternal species constituting the hybrid organism shall be
     specified
      "hybridType" : { CodeableConcept } // The hybrid type of an organism shall be specified
    },
    "organismGeneral" : { // 4.9.13.7.1 Kingdom (Conditional)
      "kingdom" : { CodeableConcept }, // The kingdom of an organism shall be specified
      "phylum" : { CodeableConcept }, // The phylum of an organism shall be specified
      "class" : { CodeableConcept }, // The class of an organism shall be specified
      "order" : { CodeableConcept } // The order of an organism shall be specified,
    }
  },
  "partDescription" : [{ // To do
    "part" : { CodeableConcept }, // Entity of anatomical origin of source material within an organism
    "partLocation" : { CodeableConcept } // The detailed anatomic location when the part can be extracted from different
     anatomical locations of the organism. Multiple alternative locations may apply
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SubstanceSourceMaterial;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SubstanceSourceMaterial.sourceMaterialClass [ CodeableConcept ]; # 0..1 General high level classification of the
    source material specific to the origin of the material
  fhir:SubstanceSourceMaterial.sourceMaterialType [ CodeableConcept ]; # 0..1 The type of the source material shall be
    specified based on a controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
  fhir:SubstanceSourceMaterial.sourceMaterialState [ CodeableConcept ]; # 0..1 The state of the source material when
    extracted
  fhir:SubstanceSourceMaterial.organismId [ Identifier ]; # 0..1 The unique identifier associated with the source
    material parent organism shall be specified
  fhir:SubstanceSourceMaterial.organismName [ string ]; # 0..1 The organism accepted Scientific name shall be provided
    based on the organism taxonomy
  fhir:SubstanceSourceMaterial.parentSubstanceId [ Identifier ], ... ; # 0..* The parent of the herbal drug Ginkgo
    biloba, Leaf is the substance ID of the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)
  fhir:SubstanceSourceMaterial.parentSubstanceName [ string ], ... ; # 0..* The parent substance of the Herbal Drug, or
    Herbal preparation
```

```
  fhir:SubstanceSourceMaterial.countryOfOrigin [ CodeableConcept ], ... ; # 0..* The country where the plant material is
     harvested or the countries where the plasma is sourced from as laid down in accordance with the Plasma Master File.
     For â€œPlasma-derived substancesâ€ the attribute country of origin provides information about the countries used
    for the manufacturing of the Cryopoor plama or Crioprecipitate
  fhir:SubstanceSourceMaterial.geographicalLocation [ string ], ... ; # 0..* The place/region where the plant is
    harvested or the places/regions where the animal source material has its habitat
  fhir:SubstanceSourceMaterial.developmentStage [ CodeableConcept ]; # 0..1 Stage of life for animals, plants, insects
    and microorganisms. This information shall be provided only when the substance is significantly different in these
    stages (e.g. foetal bovine serum)
  fhir:SubstanceSourceMaterial.fractionDescription [ # 0..* Many complex materials are fractions of parts of plants,
    animals, or minerals. Fraction elements are often necessary to define both Substances and Specified Group 1
    Substances. For substances derived from Plants, fraction information will be captured at the Substance information
    level ( . Oils, Juices and Exudates). Additional information for Extracts, such as extraction solvent composition,
    will be captured at the Specified Substance Group 1 information level. For plasma-derived products fraction
    information will be captured at the Substance and the Specified Substance Group 1 levels
    fhir:SubstanceSourceMaterial.fractionDescription.fraction [ string ]; # 0..1 This element is capturing information
    about the fraction of a plant part, or human plasma for fractionation
    fhir:SubstanceSourceMaterial.fractionDescription.materialType [ CodeableConcept ]; # 0..1 The specific type of the
    material constituting the component. For Herbal preparations the particulars of the extracts (liquid/dry) is
    described in Specified Substance Group 1
  ], ...;
  fhir:SubstanceSourceMaterial.organism [ # 0..1 This subclause describes the organism which the substance is derived
    from. For vaccines, the parent organism shall be specified based on these subclause elements. As an example, full
    taxonomy will be described for the Substance Name: ., Leaf
    fhir:SubstanceSourceMaterial.organism.family [ CodeableConcept ]; # 0..1 The family of an organism shall be
    specified
    fhir:SubstanceSourceMaterial.organism.genus [ CodeableConcept ]; # 0..1 The genus of an organism shall be specified;
     refers to the Latin epithet of the genus element of the plant/animal scientific name; it is present in names for
    genera, species and infraspecies
    fhir:SubstanceSourceMaterial.organism.species [ CodeableConcept ]; # 0..1 The species of an organism shall be
    specified; refers to the Latin epithet of the species of the plant/animal; it is present in names for species and
    infraspecies
    fhir:SubstanceSourceMaterial.organism.intraspecificType [ CodeableConcept ]; # 0..1 The Intraspecific type of an
    organism shall be specified
    fhir:SubstanceSourceMaterial.organism.intraspecificDescription [ string ]; # 0..1 The intraspecific description of
    an organism shall be specified based on a controlled vocabulary. For Influenza Vaccine, the intraspecific
    description shall contain the syntax of the antigen in line with the WHO convention
    fhir:SubstanceSourceMaterial.organism.author [ # 0..* 4.9.13.6.1 Author type (Conditional)
      fhir:SubstanceSourceMaterial.organism.author.authorType [ CodeableConcept ]; # 0..1 The type of author of an
    organism species shall be specified. The parenthetical author of an organism species refers to the first author who
    published the plant/animal name (of any rank). The primary author of an organism species refers to the first
    author(s), who validly published the plant/animal name
      fhir:SubstanceSourceMaterial.organism.author.authorDescription [ string ]; # 0..1 The author of an organism
    species shall be specified. The author year of an organism shall also be specified when applicable; refers to the
    year in which the first author(s) published the infraspecific plant/animal name (of any rank)
    ], ...;
    fhir:SubstanceSourceMaterial.organism.hybrid [ # 0..1 4.9.13.8.1 Hybrid species maternal organism ID (Optional)
      fhir:SubstanceSourceMaterial.organism.hybrid.maternalOrganismId [ string ]; # 0..1 The identifier of the maternal
    species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the
    parents arenâ€™t always known, and it is unlikely that it will be known which is maternal and which is paternal
      fhir:SubstanceSourceMaterial.organism.hybrid.maternalOrganismName [ string ]; # 0..1 The name of the maternal
    species constituting the hybrid organism shall be specified. For plants, the parents arenâ€™t always known, and it
    is unlikely that it will be known which is maternal and which is paternal
      fhir:SubstanceSourceMaterial.organism.hybrid.paternalOrganismId [ string ]; # 0..1 The identifier of the paternal
    species constituting the hybrid organism shall be specified based on a controlled vocabulary
      fhir:SubstanceSourceMaterial.organism.hybrid.paternalOrganismName [ string ]; # 0..1 The name of the paternal
    species constituting the hybrid organism shall be specified
      fhir:SubstanceSourceMaterial.organism.hybrid.hybridType [ CodeableConcept ]; # 0..1 The hybrid type of an organism
     shall be specified
    ];
    fhir:SubstanceSourceMaterial.organism.organismGeneral [ # 0..1 4.9.13.7.1 Kingdom (Conditional)
      fhir:SubstanceSourceMaterial.organism.organismGeneral.kingdom [ CodeableConcept ]; # 0..1 The kingdom of an
    organism shall be specified
      fhir:SubstanceSourceMaterial.organism.organismGeneral.phylum [ CodeableConcept ]; # 0..1 The phylum of an organism
     shall be specified
      fhir:SubstanceSourceMaterial.organism.organismGeneral.class [ CodeableConcept ]; # 0..1 The class of an organism
    shall be specified
      fhir:SubstanceSourceMaterial.organism.organismGeneral.order [ CodeableConcept ]; # 0..1 The order of an organism
    shall be specified,
    ];
  ];
  fhir:SubstanceSourceMaterial.partDescription [ # 0..* To do
    fhir:SubstanceSourceMaterial.partDescription.part [ CodeableConcept ]; # 0..1 Entity of anatomical origin of source
    material within an organism
    fhir:SubstanceSourceMaterial.partDescription.partLocation [ CodeableConcept ]; # 0..1 The detailed anatomic location
     when the part can be extracted from different anatomical locations of the organism. Multiple alternative locations
    may apply
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

SubstanceSourceMaterial | ΣTU | | DomainResource | Source material shall capture information on the taxonomic and anatomical origins as well as the fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be described using a controlled vocabulary as required. This information is captured for naturally derived polymers (. starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For Herbal preparations, the fraction information will be captured at the Substance information level and additional information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for further explanation the Substance Class: Structurally Diverse and the herbal annex Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
sourceMaterialClass | Σ | 0..1 | CodeableConcept | General high level classification of the source material specific to the origin of the material
sourceMaterialType | Σ | 0..1 | CodeableConcept | The type of the source material shall be specified based on a controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
sourceMaterialState | Σ | 0..1 | CodeableConcept | The state of the source material when extracted
organismId | Σ | 0..1 | Identifier | The unique identifier associated with the source material parent organism shall be specified
organismName | Σ | 0..1 | string | The organism accepted Scientific name shall be provided based on the organism taxonomy
parentSubstanceId | Σ | 0..* | Identifier | The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)
parentSubstanceName | Σ | 0..* | string | The parent substance of the Herbal Drug, or Herbal preparation
countryOfOrigin | Σ | 0..* | CodeableConcept | The country where the plant material is harvested or the countries where the plasma is sourced from as laid down in accordance with the Plasma Master File. For “Plasma-derived substances” the attribute country of origin provides information about the countries used for the manufacturing of the Cryopoor plama or Crioprecipitate
geographicalLocation | Σ | 0..* | string | The place/region where the plant is harvested or the places/regions where the animal source material has its habitat
developmentStage | Σ | 0..1 | CodeableConcept | Stage of life for animals, plants, insects and microorganisms. This information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine serum)
fractionDescription | Σ | 0..* | BackboneElement | Many complex materials are fractions of parts of plants, animals, or minerals. Fraction elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from Plants, fraction information will be captured at the Substance information level (. Oils, Juices and Exudates). Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified Substance Group 1 information level. For plasma-derived products fraction information will be captured at the Substance and the Specified Substance Group 1 levels
fraction | Σ | 0..1 | string | This element is capturing information about the fraction of a plant part, or human plasma for fractionation
materialType | Σ | 0..1 | CodeableConcept | The specific type of the material constituting the component. For Herbal preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1
organism | Σ | 0..1 | BackboneElement | This subclause describes the organism which the substance is derived from. For vaccines, the parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described for the Substance Name: ., Leaf
family | Σ | 0..1 | CodeableConcept | The family of an organism shall be specified
genus | Σ | 0..1 | CodeableConcept | The genus of an organism shall be specified; refers to the Latin epithet of the genus element of the plant/animal scientific name; it is present in names for genera, species and infraspecies
species | Σ | 0..1 | CodeableConcept | The species of an organism shall be specified; refers to the Latin epithet of the species of the plant/animal; it is present in names for species and infraspecies
intraspecificType | Σ | 0..1 | CodeableConcept | The Intraspecific type of an organism shall be specified
intraspecificDescription | Σ | 0..1 | string | The intraspecific description of an organism shall be specified based on a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the antigen in line with the WHO convention
author | Σ | 0..* | BackboneElement | 4.9.13.6.1 Author type (Conditional)
authorType | Σ | 0..1 | CodeableConcept | The type of author of an organism species shall be specified. The parenthetical author of an organism species refers to the first author who published the plant/animal name (of any rank). The primary author of an organism species refers to the first author(s), who validly published the plant/animal name
authorDescription | Σ | 0..1 | string | The author of an organism species shall be specified. The author year of an organism shall also be specified when applicable; refers to the year in which the first author(s) published the infraspecific plant/animal name (of any rank)
hybrid | Σ | 0..1 | BackboneElement | 4.9.13.8.1 Hybrid species maternal organism ID (Optional)
maternalOrganismId | Σ | 0..1 | string | The identifier of the maternal species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal
maternalOrganismName | Σ | 0..1 | string | The name of the maternal species constituting the hybrid organism shall be specified. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal
paternalOrganismId | Σ | 0..1 | string | The identifier of the paternal species constituting the hybrid organism shall be specified based on a controlled vocabulary
paternalOrganismName | Σ | 0..1 | string | The name of the paternal species constituting the hybrid organism shall be specified
hybridType | Σ | 0..1 | CodeableConcept | The hybrid type of an organism shall be specified
organismGeneral | Σ | 0..1 | BackboneElement | 4.9.13.7.1 Kingdom (Conditional)
kingdom | Σ | 0..1 | CodeableConcept | The kingdom of an organism shall be specified
phylum | Σ | 0..1 | CodeableConcept | The phylum of an organism shall be specified
class | Σ | 0..1 | CodeableConcept | The class of an organism shall be specified
order | Σ | 0..1 | CodeableConcept | The order of an organism shall be specified,
partDescription | Σ | 0..* | BackboneElement | To do
part | Σ | 0..1 | CodeableConcept | Entity of anatomical origin of source material within an organism
partLocation | Σ | 0..1 | CodeableConcept | The detailed anatomic location when the part can be extracted from different anatomical locations of the organism. Multiple alternative locations may apply
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceSourceMaterial xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <sourceMaterialClass><!-- 0..1 CodeableConcept General high level classification of the source material specific to the
     origin of the material --></sourceMaterialClass>
 <sourceMaterialType><!-- 0..1 CodeableConcept The type of the source material shall be specified based on a controlled
    vocabulary. For vaccines, this subclause refers to the class of infectious agent --></sourceMaterialType>
 <sourceMaterialState><!-- 0..1 CodeableConcept The state of the source material when extracted
    --></sourceMaterialState>
 <organismId><!-- 0..1 Identifier The unique identifier associated with the source material parent organism shall be
    specified --></organismId>
 <organismName value="[string]"/><!-- 0..1 The organism accepted Scientific name shall be provided based on the organism
     taxonomy -->
 <parentSubstanceId><!-- 0..* Identifier The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the
    substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant) --></parentSubstanceId>
 <parentSubstanceName value="[string]"/><!-- 0..* The parent substance of the Herbal Drug, or Herbal preparation -->
 <countryOfOrigin><!-- 0..* CodeableConcept The country where the plant material is harvested or the countries where the
     plasma is sourced from as laid down in accordance with the Plasma Master File. For â€œPlasma-derived substancesâ€
    the attribute country of origin provides information about the countries used for the manufacturing of the Cryopoor
    plama or Crioprecipitate --></countryOfOrigin>
 <geographicalLocation value="[string]"/><!-- 0..* The place/region where the plant is harvested or the places/regions
    where the animal source material has its habitat -->
 <developmentStage><!-- 0..1 CodeableConcept Stage of life for animals, plants, insects and microorganisms. This
    information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine
     serum) --></developmentStage>
 <fractionDescription>  <!-- 0..* Many complex materials are fractions of parts of plants, animals, or minerals.
    Fraction elements are often necessary to define both Substances and Specified Group 1 Substances. For substances
    derived from Plants, fraction information will be captured at the Substance information level ( . Oils, Juices and
    Exudates). Additional information for Extracts, such as extraction solvent composition, will be captured at the
    Specified Substance Group 1 information level. For plasma-derived products fraction information will be captured at
    the Substance and the Specified Substance Group 1 levels -->
  <fraction value="[string]"/><!-- 0..1 This element is capturing information about the fraction of a plant part, or
    human plasma for fractionation -->
  <materialType><!-- 0..1 CodeableConcept The specific type of the material constituting the component. For Herbal
    preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1
    --></materialType>
 </fractionDescription>
 <organism>  <!-- 0..1 This subclause describes the organism which the substance is derived from. For vaccines, the
    parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described
     for the Substance Name: ., Leaf -->
  <family><!-- 0..1 CodeableConcept The family of an organism shall be specified --></family>
  <genus><!-- 0..1 CodeableConcept The genus of an organism shall be specified; refers to the Latin epithet of the genus
     element of the plant/animal scientific name; it is present in names for genera, species and infraspecies
    --></genus>
  <species><!-- 0..1 CodeableConcept The species of an organism shall be specified; refers to the Latin epithet of the
    species of the plant/animal; it is present in names for species and infraspecies --></species>
  <intraspecificType><!-- 0..1 CodeableConcept The Intraspecific type of an organism shall be specified
    --></intraspecificType>
  <intraspecificDescription value="[string]"/><!-- 0..1 The intraspecific description of an organism shall be specified
    based on a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of
    the antigen in line with the WHO convention -->
  <author>  <!-- 0..* 4.9.13.6.1 Author type (Conditional) -->
   <authorType><!-- 0..1 CodeableConcept The type of author of an organism species shall be specified. The parenthetical
     author of an organism species refers to the first author who published the plant/animal name (of any rank). The
    primary author of an organism species refers to the first author(s), who validly published the plant/animal name
    --></authorType>
   <authorDescription value="[string]"/><!-- 0..1 The author of an organism species shall be specified. The author year
    of an organism shall also be specified when applicable; refers to the year in which the first author(s) published
    the infraspecific plant/animal name (of any rank) -->
  </author>
  <hybrid>  <!-- 0..1 4.9.13.8.1 Hybrid species maternal organism ID (Optional) -->
   <maternalOrganismId value="[string]"/><!-- 0..1 The identifier of the maternal species constituting the hybrid
    organism shall be specified based on a controlled vocabulary. For plants, the parents arenâ€™t always known, and it
    is unlikely that it will be known which is maternal and which is paternal -->
   <maternalOrganismName value="[string]"/><!-- 0..1 The name of the maternal species constituting the hybrid organism
    shall be specified. For plants, the parents arenâ€™t always known, and it is unlikely that it will be known which is
     maternal and which is paternal -->
   <paternalOrganismId value="[string]"/><!-- 0..1 The identifier of the paternal species constituting the hybrid
    organism shall be specified based on a controlled vocabulary -->
   <paternalOrganismName value="[string]"/><!-- 0..1 The name of the paternal species constituting the hybrid organism
    shall be specified -->
   <hybridType><!-- 0..1 CodeableConcept The hybrid type of an organism shall be specified --></hybridType>
  </hybrid>
  <organismGeneral>  <!-- 0..1 4.9.13.7.1 Kingdom (Conditional) -->
   <kingdom><!-- 0..1 CodeableConcept The kingdom of an organism shall be specified --></kingdom>
   <phylum><!-- 0..1 CodeableConcept The phylum of an organism shall be specified --></phylum>
   <class><!-- 0..1 CodeableConcept The class of an organism shall be specified --></class>
   <order><!-- 0..1 CodeableConcept The order of an organism shall be specified, --></order>
  </organismGeneral>
 </organism>
 <partDescription>  <!-- 0..* To do -->
  <part><!-- 0..1 CodeableConcept Entity of anatomical origin of source material within an organism --></part>
  <partLocation><!-- 0..1 CodeableConcept The detailed anatomic location when the part can be extracted from different
    anatomical locations of the organism. Multiple alternative locations may apply --></partLocation>
 </partDescription>
</SubstanceSourceMaterial>
```

JSON Template

```
{
  "resourceType" : "SubstanceSourceMaterial",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "sourceMaterialClass" : { CodeableConcept }, // General high level classification of the source material specific to
    the origin of the material
  "sourceMaterialType" : { CodeableConcept }, // The type of the source material shall be specified based on a
    controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
  "sourceMaterialState" : { CodeableConcept }, // The state of the source material when extracted
  "organismId" : { Identifier }, // The unique identifier associated with the source material parent organism shall be
    specified
  "organismName" : "<string>", // The organism accepted Scientific name shall be provided based on the organism taxonomy
  "parentSubstanceId" : [{ Identifier }], // The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of
    the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)
  "parentSubstanceName" : ["<string>"], // The parent substance of the Herbal Drug, or Herbal preparation
  "countryOfOrigin" : [{ CodeableConcept }], // The country where the plant material is harvested or the countries where
     the plasma is sourced from as laid down in accordance with the Plasma Master File. For â€œPlasma-derived
    substancesâ€ the attribute country of origin provides information about the countries used for the manufacturing of
     the Cryopoor plama or Crioprecipitate
  "geographicalLocation" : ["<string>"], // The place/region where the plant is harvested or the places/regions where
    the animal source material has its habitat
  "developmentStage" : { CodeableConcept }, // Stage of life for animals, plants, insects and microorganisms. This
    information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine
     serum)
  "fractionDescription" : [{ // Many complex materials are fractions of parts of plants, animals, or minerals. Fraction
    elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from
     Plants, fraction information will be captured at the Substance information level ( . Oils, Juices and Exudates).
    Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified
    Substance Group 1 information level. For plasma-derived products fraction information will be captured at the
    Substance and the Specified Substance Group 1 levels
    "fraction" : "<string>", // This element is capturing information about the fraction of a plant part, or human
    plasma for fractionation
    "materialType" : { CodeableConcept } // The specific type of the material constituting the component. For Herbal
    preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1
  }],
  "organism" : { // This subclause describes the organism which the substance is derived from. For vaccines, the parent
    organism shall be specified based on these subclause elements. As an example, full taxonomy will be described for
    the Substance Name: ., Leaf
    "family" : { CodeableConcept }, // The family of an organism shall be specified
    "genus" : { CodeableConcept }, // The genus of an organism shall be specified; refers to the Latin epithet of the
    genus element of the plant/animal scientific name; it is present in names for genera, species and infraspecies
    "species" : { CodeableConcept }, // The species of an organism shall be specified; refers to the Latin epithet of
    the species of the plant/animal; it is present in names for species and infraspecies
    "intraspecificType" : { CodeableConcept }, // The Intraspecific type of an organism shall be specified
    "intraspecificDescription" : "<string>", // The intraspecific description of an organism shall be specified based on
     a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the
    antigen in line with the WHO convention
    "author" : [{ // 4.9.13.6.1 Author type (Conditional)
      "authorType" : { CodeableConcept }, // The type of author of an organism species shall be specified. The
    parenthetical author of an organism species refers to the first author who published the plant/animal name (of any
    rank). The primary author of an organism species refers to the first author(s), who validly published the
    plant/animal name
      "authorDescription" : "<string>" // The author of an organism species shall be specified. The author year of an
    organism shall also be specified when applicable; refers to the year in which the first author(s) published the
    infraspecific plant/animal name (of any rank)
    }],
    "hybrid" : { // 4.9.13.8.1 Hybrid species maternal organism ID (Optional)
      "maternalOrganismId" : "<string>", // The identifier of the maternal species constituting the hybrid organism
    shall be specified based on a controlled vocabulary. For plants, the parents arenâ€™t always known, and it is
    unlikely that it will be known which is maternal and which is paternal
      "maternalOrganismName" : "<string>", // The name of the maternal species constituting the hybrid organism shall be
     specified. For plants, the parents arenâ€™t always known, and it is unlikely that it will be known which is
    maternal and which is paternal
      "paternalOrganismId" : "<string>", // The identifier of the paternal species constituting the hybrid organism
    shall be specified based on a controlled vocabulary
      "paternalOrganismName" : "<string>", // The name of the paternal species constituting the hybrid organism shall be
     specified
      "hybridType" : { CodeableConcept } // The hybrid type of an organism shall be specified
    },
    "organismGeneral" : { // 4.9.13.7.1 Kingdom (Conditional)
      "kingdom" : { CodeableConcept }, // The kingdom of an organism shall be specified
      "phylum" : { CodeableConcept }, // The phylum of an organism shall be specified
      "class" : { CodeableConcept }, // The class of an organism shall be specified
      "order" : { CodeableConcept } // The order of an organism shall be specified,
    }
  },
  "partDescription" : [{ // To do
    "part" : { CodeableConcept }, // Entity of anatomical origin of source material within an organism
    "partLocation" : { CodeableConcept } // The detailed anatomic location when the part can be extracted from different
     anatomical locations of the organism. Multiple alternative locations may apply
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:SubstanceSourceMaterial;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
# from Resource: .id, .meta, .implicitRules, and .language
# from DomainResource: .text, .contained, .extension, and .modifierExtension
fhir:SubstanceSourceMaterial.sourceMaterialClass [ CodeableConcept ]; # 0..1 General high level classification of the
  source material specific to the origin of the material
fhir:SubstanceSourceMaterial.sourceMaterialType [ CodeableConcept ]; # 0..1 The type of the source material shall be
  specified based on a controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent
fhir:SubstanceSourceMaterial.sourceMaterialState [ CodeableConcept ]; # 0..1 The state of the source material when
  extracted
fhir:SubstanceSourceMaterial.organismId [ Identifier ]; # 0..1 The unique identifier associated with the source
  material parent organism shall be specified
fhir:SubstanceSourceMaterial.organismName [ string ]; # 0..1 The organism accepted Scientific name shall be provided
  based on the organism taxonomy
fhir:SubstanceSourceMaterial.parentSubstanceId [ Identifier ], ... ; # 0..* The parent of the herbal drug Ginkgo
  biloba, Leaf is the substance ID of the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant)
fhir:SubstanceSourceMaterial.parentSubstanceName [ string ], ... ; # 0..* The parent substance of the Herbal Drug, or
  Herbal preparation
fhir:SubstanceSourceMaterial.countryOfOrigin [ CodeableConcept ], ... ; # 0..* The country where the plant material is
   harvested or the countries where the plasma is sourced from as laid down in accordance with the Plasma Master File.
   For â€œPlasma-derived substancesâ€ the attribute country of origin provides information about the countries used
  for the manufacturing of the Cryopoor plama or Crioprecipitate
fhir:SubstanceSourceMaterial.geographicalLocation [ string ], ... ; # 0..* The place/region where the plant is
  harvested or the places/regions where the animal source material has its habitat
fhir:SubstanceSourceMaterial.developmentStage [ CodeableConcept ]; # 0..1 Stage of life for animals, plants, insects
  and microorganisms. This information shall be provided only when the substance is significantly different in these
  stages (e.g. foetal bovine serum)
fhir:SubstanceSourceMaterial.fractionDescription [ # 0..* Many complex materials are fractions of parts of plants,
  animals, or minerals. Fraction elements are often necessary to define both Substances and Specified Group 1
  Substances. For substances derived from Plants, fraction information will be captured at the Substance information
  level ( . Oils, Juices and Exudates). Additional information for Extracts, such as extraction solvent composition,
  will be captured at the Specified Substance Group 1 information level. For plasma-derived products fraction
  information will be captured at the Substance and the Specified Substance Group 1 levels
  fhir:SubstanceSourceMaterial.fractionDescription.fraction [ string ]; # 0..1 This element is capturing information
  about the fraction of a plant part, or human plasma for fractionation
  fhir:SubstanceSourceMaterial.fractionDescription.materialType [ CodeableConcept ]; # 0..1 The specific type of the
  material constituting the component. For Herbal preparations the particulars of the extracts (liquid/dry) is
  described in Specified Substance Group 1
], ...;
fhir:SubstanceSourceMaterial.organism [ # 0..1 This subclause describes the organism which the substance is derived
  from. For vaccines, the parent organism shall be specified based on these subclause elements. As an example, full
  taxonomy will be described for the Substance Name: ., Leaf
  fhir:SubstanceSourceMaterial.organism.family [ CodeableConcept ]; # 0..1 The family of an organism shall be
  specified
  fhir:SubstanceSourceMaterial.organism.genus [ CodeableConcept ]; # 0..1 The genus of an organism shall be specified;
   refers to the Latin epithet of the genus element of the plant/animal scientific name; it is present in names for
  genera, species and infraspecies
  fhir:SubstanceSourceMaterial.organism.species [ CodeableConcept ]; # 0..1 The species of an organism shall be
  specified; refers to the Latin epithet of the species of the plant/animal; it is present in names for species and
  infraspecies
  fhir:SubstanceSourceMaterial.organism.intraspecificType [ CodeableConcept ]; # 0..1 The Intraspecific type of an
  organism shall be specified
  fhir:SubstanceSourceMaterial.organism.intraspecificDescription [ string ]; # 0..1 The intraspecific description of
  an organism shall be specified based on a controlled vocabulary. For Influenza Vaccine, the intraspecific
  description shall contain the syntax of the antigen in line with the WHO convention
  fhir:SubstanceSourceMaterial.organism.author [ # 0..* 4.9.13.6.1 Author type (Conditional)
    fhir:SubstanceSourceMaterial.organism.author.authorType [ CodeableConcept ]; # 0..1 The type of author of an
  organism species shall be specified. The parenthetical author of an organism species refers to the first author who
  published the plant/animal name (of any rank). The primary author of an organism species refers to the first
  author(s), who validly published the plant/animal name
    fhir:SubstanceSourceMaterial.organism.author.authorDescription [ string ]; # 0..1 The author of an organism
  species shall be specified. The author year of an organism shall also be specified when applicable; refers to the
  year in which the first author(s) published the infraspecific plant/animal name (of any rank)
  ], ...;
  fhir:SubstanceSourceMaterial.organism.hybrid [ # 0..1 4.9.13.8.1 Hybrid species maternal organism ID (Optional)
    fhir:SubstanceSourceMaterial.organism.hybrid.maternalOrganismId [ string ]; # 0..1 The identifier of the maternal
  species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the
  parents arenâ€™t always known, and it is unlikely that it will be known which is maternal and which is paternal
    fhir:SubstanceSourceMaterial.organism.hybrid.maternalOrganismName [ string ]; # 0..1 The name of the maternal
  species constituting the hybrid organism shall be specified. For plants, the parents arenâ€™t always known, and it
  is unlikely that it will be known which is maternal and which is paternal
    fhir:SubstanceSourceMaterial.organism.hybrid.paternalOrganismId [ string ]; # 0..1 The identifier of the paternal
  species constituting the hybrid organism shall be specified based on a controlled vocabulary
    fhir:SubstanceSourceMaterial.organism.hybrid.paternalOrganismName [ string ]; # 0..1 The name of the paternal
  species constituting the hybrid organism shall be specified
    fhir:SubstanceSourceMaterial.organism.hybrid.hybridType [ CodeableConcept ]; # 0..1 The hybrid type of an organism
   shall be specified
  ];
  fhir:SubstanceSourceMaterial.organism.organismGeneral [ # 0..1 4.9.13.7.1 Kingdom (Conditional)
    fhir:SubstanceSourceMaterial.organism.organismGeneral.kingdom [ CodeableConcept ]; # 0..1 The kingdom of an
  organism shall be specified
    fhir:SubstanceSourceMaterial.organism.organismGeneral.phylum [ CodeableConcept ]; # 0..1 The phylum of an organism
   shall be specified
    fhir:SubstanceSourceMaterial.organism.organismGeneral.class [ CodeableConcept ]; # 0..1 The class of an organism
  shall be specified
    fhir:SubstanceSourceMaterial.organism.organismGeneral.order [ CodeableConcept ]; # 0..1 The order of an organism
  shall be specified,
  ];
];
fhir:SubstanceSourceMaterial.partDescription [ # 0..* To do
  fhir:SubstanceSourceMaterial.partDescription.part [ CodeableConcept ]; # 0..1 Entity of anatomical origin of source
```

```
    material within an organism
    fhir:SubstanceSourceMaterial.partDescription.partLocation [ CodeableConcept ]; # 0..1 The detailed anatomic location
     when the part can be extracted from different anatomical locations of the organism. Multiple alternative locations
    may apply
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

11.26.3 Resource SubstanceSourceMaterial - Examples

URL: https://hl7.org/fhir/R4/substancesourcematerial-examples.html

11.26.3 Resource SubstanceSourceMaterial - Examples

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of substancesourcematerial | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

11.26.4 Resource SubstanceSourceMaterial - Detailed Descriptions

URL: https://hl7.org/fhir/R4/substancesourcematerial-definitions.html

11.26.4 Resource SubstanceSourceMaterial - Detailed Descriptions

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SubstanceSourceMaterial resource.

SubstanceSourceMaterial
Element Id | SubstanceSourceMaterial
Definition | Source material shall capture information on the taxonomic and anatomical origins as well as the fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be described using a controlled vocabulary as required. This information is captured for naturally derived polymers (. starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For Herbal preparations, the fraction information will be captured at the Substance information level and additional information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for further explanation the Substance Class: Structurally Diverse and the herbal annex.
Cardinality | 0..*
Type | DomainResource
Summary | true
SubstanceSourceMaterial.sourceMaterialClass
Element Id | SubstanceSourceMaterial.sourceMaterialClass
Definition | General high level classification of the source material specific to the origin of the material.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.sourceMaterialType
Element Id | SubstanceSourceMaterial.sourceMaterialType
Definition | The type of the source material shall be specified based on a controlled vocabulary. For vaccines, this subclause refers to the class of infectious agent.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.sourceMaterialState
Element Id | SubstanceSourceMaterial.sourceMaterialState
Definition | The state of the source material when extracted.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organismId
Element Id | SubstanceSourceMaterial.organismId
Definition | The unique identifier associated with the source material parent organism shall be specified.
Cardinality | 0..1

Type | Identifier
Summary | true
SubstanceSourceMaterial.organismName
Element Id | SubstanceSourceMaterial.organismName
Definition | The organism accepted Scientific name shall be provided based on the organism taxonomy.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.parentSubstanceId
Element Id | SubstanceSourceMaterial.parentSubstanceId
Definition | The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the substance (fresh) of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant).
Cardinality | 0..*
Type | Identifier
Summary | true
SubstanceSourceMaterial.parentSubstanceName
Element Id | SubstanceSourceMaterial.parentSubstanceName
Definition | The parent substance of the Herbal Drug, or Herbal preparation.
Cardinality | 0..*
Type | string
Summary | true
SubstanceSourceMaterial.countryOfOrigin
Element Id | SubstanceSourceMaterial.countryOfOrigin
Definition | The country where the plant material is harvested or the countries where the plasma is sourced from as laid down in accordance with the Plasma Master File. For “Plasma-derived substances” the attribute country of origin provides information about the countries used for the manufacturing of the Cryopoor plama or Crioprecipitate.
Cardinality | 0..*
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.geographicalLocation
Element Id | SubstanceSourceMaterial.geographicalLocation
Definition | The place/region where the plant is harvested or the places/regions where the animal source material has its habitat.
Cardinality | 0..*
Type | string
Summary | true
SubstanceSourceMaterial.developmentStage
Element Id | SubstanceSourceMaterial.developmentStage
Definition | Stage of life for animals, plants, insects and microorganisms. This information shall be provided only when the substance is significantly different in these stages (e.g. foetal bovine serum).
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.fractionDescription
Element Id | SubstanceSourceMaterial.fractionDescription
Definition | Many complex materials are fractions of parts of plants, animals, or minerals. Fraction elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from Plants, fraction information will be captured at the Substance information level (. Oils, Juices and Exudates). Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified Substance Group 1 information level. For plasma-derived products fraction information will be captured at the Substance and the Specified Substance Group 1 levels.
Cardinality | 0..*
Summary | true
SubstanceSourceMaterial.fractionDescription.fraction
Element Id | SubstanceSourceMaterial.fractionDescription.fraction
Definition | This element is capturing information about the fraction of a plant part, or human plasma for fractionation.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.fractionDescription.materialType
Element Id | SubstanceSourceMaterial.fractionDescription.materialType
Definition | The specific type of the material constituting the component. For Herbal preparations the particulars of the extracts (liquid/dry) is described in Specified Substance Group 1.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism
Element Id | SubstanceSourceMaterial.organism
Definition | This subclause describes the organism which the substance is derived from. For vaccines, the parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described for the Substance Name: ., Leaf.
Cardinality | 0..1
Summary | true
SubstanceSourceMaterial.organism.family
Element Id | SubstanceSourceMaterial.organism.family
Definition | The family of an organism shall be specified.

Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.genus
Element Id | SubstanceSourceMaterial.organism.genus
Definition | The genus of an organism shall be specified; refers to the Latin epithet of the genus element of the plant/animal scientific name; it is present in names for genera, species and infraspecies.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.species
Element Id | SubstanceSourceMaterial.organism.species
Definition | The species of an organism shall be specified; refers to the Latin epithet of the species of the plant/animal; it is present in names for species and infraspecies.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.intraspecificType
Element Id | SubstanceSourceMaterial.organism.intraspecificType
Definition | The Intraspecific type of an organism shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.intraspecificDescription
Element Id | SubstanceSourceMaterial.organism.intraspecificDescription
Definition | The intraspecific description of an organism shall be specified based on a controlled vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the antigen in line with the WHO convention.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.author
Element Id | SubstanceSourceMaterial.organism.author
Definition | 4.9.13.6.1 Author type (Conditional).
Cardinality | 0..*
Summary | true
SubstanceSourceMaterial.organism.author.authorType
Element Id | SubstanceSourceMaterial.organism.author.authorType
Definition | The type of author of an organism species shall be specified. The parenthetical author of an organism species refers to the first author who published the plant/animal name (of any rank). The primary author of an organism species refers to the first author(s), who validly published the plant/animal name.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.author.authorDescription
Element Id | SubstanceSourceMaterial.organism.author.authorDescription
Definition | The author of an organism species shall be specified. The author year of an organism shall also be specified when applicable; refers to the year in which the first author(s) published the infraspecific plant/animal name (of any rank).
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.hybrid
Element Id | SubstanceSourceMaterial.organism.hybrid
Definition | 4.9.13.8.1 Hybrid species maternal organism ID (Optional).
Cardinality | 0..1
Summary | true
SubstanceSourceMaterial.organism.hybrid.maternalOrganismId
Element Id | SubstanceSourceMaterial.organism.hybrid.maternalOrganismId
Definition | The identifier of the maternal species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.hybrid.maternalOrganismName
Element Id | SubstanceSourceMaterial.organism.hybrid.maternalOrganismName
Definition | The name of the maternal species constituting the hybrid organism shall be specified. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.hybrid.paternalOrganismId
Element Id | SubstanceSourceMaterial.organism.hybrid.paternalOrganismId

Definition | The identifier of the paternal species constituting the hybrid organism shall be specified based on a controlled vocabulary.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.hybrid.paternalOrganismName
Element Id | SubstanceSourceMaterial.organism.hybrid.paternalOrganismName
Definition | The name of the paternal species constituting the hybrid organism shall be specified.
Cardinality | 0..1
Type | string
Summary | true
SubstanceSourceMaterial.organism.hybrid.hybridType
Element Id | SubstanceSourceMaterial.organism.hybrid.hybridType
Definition | The hybrid type of an organism shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.organismGeneral
Element Id | SubstanceSourceMaterial.organism.organismGeneral
Definition | 4.9.13.7.1 Kingdom (Conditional).
Cardinality | 0..1
Summary | true
SubstanceSourceMaterial.organism.organismGeneral.kingdom
Element Id | SubstanceSourceMaterial.organism.organismGeneral.kingdom
Definition | The kingdom of an organism shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.organismGeneral.phylum
Element Id | SubstanceSourceMaterial.organism.organismGeneral.phylum
Definition | The phylum of an organism shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.organismGeneral.class
Element Id | SubstanceSourceMaterial.organism.organismGeneral.class
Definition | The class of an organism shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.organism.organismGeneral.order
Element Id | SubstanceSourceMaterial.organism.organismGeneral.order
Definition | The order of an organism shall be specified,.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.partDescription
Element Id | SubstanceSourceMaterial.partDescription
Definition | To do.
Cardinality | 0..*
Summary | true
SubstanceSourceMaterial.partDescription.part
Element Id | SubstanceSourceMaterial.partDescription.part
Definition | Entity of anatomical origin of source material within an organism.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceSourceMaterial.partDescription.partLocation
Element Id | SubstanceSourceMaterial.partDescription.partLocation
Definition | The detailed anatomic location when the part can be extracted from different anatomical locations of the organism. Multiple alternative locations may apply.
Cardinality | 0..1
Type | CodeableConcept
Summary | true

11.26.5 Resource SubstanceSourceMaterial - Mappings

URL: https://hl7.org/fhir/R4/substancesourcematerial-mappings.html

11.26.5 Resource SubstanceSourceMaterial - Mappings

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings for the substancesourcematerial resource (see Mappings to Other Standards for further information & status).

11.26.6 Resource SubstanceSourceMaterial - Extensions & Profiles

URL: https://hl7.org/fhir/R4/substancesourcematerial-profiles.html

11.26.6 Resource SubstanceSourceMaterial - Extensions & Profiles

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SubstanceSourceMaterial resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

11.26.6.1 Profiles

No Profiles defined for this resource

11.26.6.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

11.26.6.3 Search Extensions

No Search Extensions defined for this resource

11.26.7 JSON Schema for SubstanceSourceMaterial

URL: https://hl7.org/fhir/R4/substancesourcematerial.schema.json.html

11.26.7 JSON Schema for SubstanceSourceMaterial

Biomedical Research and Regulation Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SubstanceSourceMaterial, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SubstanceSourceMaterial",
  "$ref": "#/definitions/SubstanceSourceMaterial",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SubstanceSourceMaterial": {
      "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
    fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
     used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
    described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
     starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
    defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
    Herbal preparations, the fraction information will be captured at the Substance information level and additional
    information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
    further explanation the Substance Class: Structurally Diverse and the herbal annex.",
      "properties": {
        "resourceType": {
          "description": "This is a SubstanceSourceMaterial resource",
          "const": "SubstanceSourceMaterial"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
```

```
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "sourceMaterialClass": {
      "description": "General high level classification of the source material specific to the origin of the
material.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "sourceMaterialType": {
      "description": "The type of the source material shall be specified based on a controlled vocabulary. For
vaccines, this subclause refers to the class of infectious agent.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "sourceMaterialState": {
      "description": "The state of the source material when extracted.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "organismId": {
      "description": "The unique identifier associated with the source material parent organism shall be
specified.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "organismName": {
      "description": "The organism accepted Scientific name shall be provided based on the organism taxonomy.",
      "$ref": "#/definitions/string"
    },
    "_organismName": {
      "description": "Extensions for organismName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "parentSubstanceId": {
      "description": "The parent of the herbal drug Ginkgo biloba, Leaf is the substance ID of the substance (fresh)
 of Ginkgo biloba L. or Ginkgo biloba L. (Whole plant).",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "parentSubstanceName": {
      "description": "The parent substance of the Herbal Drug, or Herbal preparation.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_parentSubstanceName": {
      "description": "Extensions for parentSubstanceName",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "countryOfOrigin": {
      "description": "The country where the plant material is harvested or the countries where the plasma is sourced
 from as laid down in accordance with the Plasma Master File. For “Plasma-derived substances” the attribute country
of origin provides information about the countries used for the manufacturing of the Cryopoor plama or
Crioprecipitate.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "geographicalLocation": {
      "description": "The place/region where the plant is harvested or the places/regions where the animal source
material has its habitat.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_geographicalLocation": {
      "description": "Extensions for geographicalLocation",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "developmentStage": {
      "description": "Stage of life for animals, plants, insects and microorganisms. This information shall be
provided only when the substance is significantly different in these stages (e.g. foetal bovine serum).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "fractionDescription": {
      "description": "Many complex materials are fractions of parts of plants, animals, or minerals. Fraction
elements are often necessary to define both Substances and Specified Group 1 Substances. For substances derived from
 Plants, fraction information will be captured at the Substance information level ( . Oils, Juices and Exudates).
Additional information for Extracts, such as extraction solvent composition, will be captured at the Specified
Substance Group 1 information level. For plasma-derived products fraction information will be captured at the
Substance and the Specified Substance Group 1 levels.",
      "items": {
        "$ref": "#/definitions/SubstanceSourceMaterial_FractionDescription"
      },
      "type": "array"
    },
    "organism": {
      "description": "This subclause describes the organism which the substance is derived from. For vaccines, the
parent organism shall be specified based on these subclause elements. As an example, full taxonomy will be described
 for the Substance Name: ., Leaf.",
      "$ref": "#/definitions/SubstanceSourceMaterial_Organism"
    },
    "partDescription": {
      "description": "To do.",
      "items": {
        "$ref": "#/definitions/SubstanceSourceMaterial_PartDescription"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"SubstanceSourceMaterial_FractionDescription": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
 used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
 starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
```

```
defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
Herbal preparations, the fraction information will be captured at the Substance information level and additional
information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "fraction": {
      "description": "This element is capturing information about the fraction of a plant part, or human plasma for
fractionation.",
      "$ref": "#/definitions/string"
    },
    "_fraction": {
      "description": "Extensions for fraction",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "materialType": {
      "description": "The specific type of the material constituting the component. For Herbal preparations the
particulars of the extracts (liquid/dry) is described in Specified Substance Group 1.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"SubstanceSourceMaterial_Organism": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
 used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
 starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
Herbal preparations, the fraction information will be captured at the Substance information level and additional
information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
      "type": "array"
    },
    "family": {
      "description": "The family of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "genus": {
      "description": "The genus of an organism shall be specified; refers to the Latin epithet of the genus element
of the plant/animal scientific name; it is present in names for genera, species and infraspecies.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "species": {
      "description": "The species of an organism shall be specified; refers to the Latin epithet of the species of
the plant/animal; it is present in names for species and infraspecies.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "intraspecificType": {
      "description": "The Intraspecific type of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "intraspecificDescription": {
      "description": "The intraspecific description of an organism shall be specified based on a controlled
vocabulary. For Influenza Vaccine, the intraspecific description shall contain the syntax of the antigen in line
with the WHO convention.",
      "$ref": "#/definitions/string"
    },
    "_intraspecificDescription": {
      "description": "Extensions for intraspecificDescription",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "4.9.13.6.1 Author type (Conditional).",
      "items": {
        "$ref": "#/definitions/SubstanceSourceMaterial_Author"
      },
      "type": "array"
    },
    "hybrid": {
      "description": "4.9.13.8.1 Hybrid species maternal organism ID (Optional).",
      "$ref": "#/definitions/SubstanceSourceMaterial_Hybrid"
    },
    "organismGeneral": {
      "description": "4.9.13.7.1 Kingdom (Conditional).",
      "$ref": "#/definitions/SubstanceSourceMaterial_OrganismGeneral"
    }
  },
  "additionalProperties": false
},
"SubstanceSourceMaterial_Author": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
 used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
 starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
Herbal preparations, the fraction information will be captured at the Substance information level and additional
information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "authorType": {
```

```
      "description": "The type of author of an organism species shall be specified. The parenthetical author of an organism species refers to the first author who published the plant/animal name (of any rank). The primary author of an organism species refers to the first author(s), who validly published the plant/animal name.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "authorDescription": {
      "description": "The author of an organism species shall be specified. The author year of an organism shall also be specified when applicable; refers to the year in which the first author(s) published the infraspecific plant/animal name (of any rank).",
      "$ref": "#/definitions/string"
    },
    "_authorDescription": {
      "description": "Extensions for authorDescription",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"SubstanceSourceMaterial_Hybrid": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( . starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For Herbal preparations, the fraction information will be captured at the Substance information level and additional information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "maternalOrganismId": {
      "description": "The identifier of the maternal species constituting the hybrid organism shall be specified based on a controlled vocabulary. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal.",
      "$ref": "#/definitions/string"
    },
    "_maternalOrganismId": {
      "description": "Extensions for maternalOrganismId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "maternalOrganismName": {
      "description": "The name of the maternal species constituting the hybrid organism shall be specified. For plants, the parents aren’t always known, and it is unlikely that it will be known which is maternal and which is paternal.",
      "$ref": "#/definitions/string"
    },
    "_maternalOrganismName": {
      "description": "Extensions for maternalOrganismName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "paternalOrganismId": {
      "description": "The identifier of the paternal species constituting the hybrid organism shall be specified based on a controlled vocabulary.",
      "$ref": "#/definitions/string"
    },
    "_paternalOrganismId": {
      "description": "Extensions for paternalOrganismId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "paternalOrganismName": {
      "description": "The name of the paternal species constituting the hybrid organism shall be specified.",
```

```
      "$ref": "#/definitions/string"
    },
    "_paternalOrganismName": {
      "description": "Extensions for paternalOrganismName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "hybridType": {
      "description": "The hybrid type of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"SubstanceSourceMaterial_OrganismGeneral": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
 used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
 starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
Herbal preparations, the fraction information will be captured at the Substance information level and additional
information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "kingdom": {
      "description": "The kingdom of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "phylum": {
      "description": "The phylum of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "class": {
      "description": "The class of an organism shall be specified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "order": {
      "description": "The order of an organism shall be specified,.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"SubstanceSourceMaterial_PartDescription": {
  "description": "Source material shall capture information on the taxonomic and anatomical origins as well as the
fraction of a material that can result in or can be modified to form a substance. This set of data elements shall be
 used to define polymer substances isolated from biological matrices. Taxonomic and anatomical origins shall be
described using a controlled vocabulary as required. This information is captured for naturally derived polymers ( .
 starch) and structurally diverse substances. For Organisms belonging to the Kingdom Plantae the Substance level
defines the fresh material of a single species or infraspecies, the Herbal Drug and the Herbal preparation. For
Herbal preparations, the fraction information will be captured at the Substance information level and additional
information for herbal extracts will be captured at the Specified Substance Group 1 information level. See for
further explanation the Substance Class: Structurally Diverse and the herbal annex.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
```

```
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "part": {
      "description": "Entity of anatomical origin of source material within an organism.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "partLocation": {
      "description": "The detailed anatomic location when the part can be extracted from different anatomical
    locations of the organism. Multiple alternative locations may apply.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
 }
}
```


11.26.8 Resource SubstanceSourceMaterial - Maturity Dependency View

URL: https://hl7.org/fhir/R4/substancesourcematerial-dependencies.html

11.26.8 Resource SubstanceSourceMaterial - Maturity Dependency View

Biomedical Research and Regulation Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SubstanceSourceMaterial FMM level 0

Name | Card. | Type | Dependency Analysis
SubstanceSourceMaterial | | DomainResource
sourceMaterialClass | 0..1 | CodeableConcept
sourceMaterialType | 0..1 | CodeableConcept
sourceMaterialState | 0..1 | CodeableConcept
organismId | 0..1 | Identifier
organismName | 0..1 | string
parentSubstanceId | 0..* | Identifier
parentSubstanceName | 0..* | string
countryOfOrigin | 0..* | CodeableConcept
geographicalLocation | 0..* | string
developmentStage | 0..1 | CodeableConcept
fractionDescription | 0..* | BackboneElement
fraction | 0..1 | string
materialType | 0..1 | CodeableConcept
organism | 0..1 | BackboneElement
family | 0..1 | CodeableConcept
genus | 0..1 | CodeableConcept
species | 0..1 | CodeableConcept
intraspecificType | 0..1 | CodeableConcept
intraspecificDescription | 0..1 | string
author | 0..* | BackboneElement
authorType | 0..1 | CodeableConcept
authorDescription | 0..1 | string
hybrid | 0..1 | BackboneElement
maternalOrganismId | 0..1 | string
maternalOrganismName | 0..1 | string
paternalOrganismId | 0..1 | string
paternalOrganismName | 0..1 | string
hybridType | 0..1 | CodeableConcept

organismGeneral | 0..1 | BackboneElement
kingdom | 0..1 | CodeableConcept
phylum | 0..1 | CodeableConcept
class | 0..1 | CodeableConcept
order | 0..1 | CodeableConcept
partDescription | 0..* | BackboneElement
part | 0..1 | CodeableConcept
partLocation | 0..1 | CodeableConcept

11.27 MarketingStatus

URL: https://hl7.org/fhir/R4/marketingstatus.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

This type is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft design not suitable for production implementation

The MarketingStatus structure defines TODO.

Note: the MarketingStatus structure allows modifier extensions.

11.27.0.1 Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
MarketingStatus | ΣD | | BackBoneElement | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available Elements defined in Ancestors: id, extension, modifierExtension
country | Σ | 1..1 | CodeableConcept | The country in which the marketing authorisation has been granted shall be specified It should be specified using the ISO 3166 - 1 alpha-2 code elements
jurisdiction | Σ | 0..1 | CodeableConcept | Where a Medicines Regulatory Agency has granted a marketing authorisation for which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate controlled terminology The controlled term and the controlled term identifier shall be specified
status | Σ | 1..1 | CodeableConcept | This attribute provides information on the status of the marketing of the medicinal product See ISO/TS 20443 for more information and examples
dateRange | Σ | 1..1 | Period | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain
restoreDate | Σ | 0..1 | dateTime | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MarketingStatus xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <country><!-- 1..1 CodeableConcept The country in which the marketing authorisation has been granted shall be specified
     It should be specified using the ISO 3166 - 1 alpha-2 code elements --></country>
 <jurisdiction><!-- 0..1 CodeableConcept Where a Medicines Regulatory Agency has granted a marketing authorisation for
    which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate
    controlled terminology The controlled term and the controlled term identifier shall be specified --></jurisdiction>
 <status><!-- 1..1 CodeableConcept This attribute provides information on the status of the marketing of the medicinal
    product See ISO/TS 20443 for more information and examples --></status>
 <dateRange><!-- 1..1 Period The date when the Medicinal Product is placed on the market by the Marketing Authorisation
    Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A
    complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on
    the market” refers to the release of the Medicinal Product into the distribution chain --></dateRange>
 <restoreDate value="[dateTime]"/><!-- 0..1 The date when the Medicinal Product is placed on the market by the Marketing
     Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    “Placed on the market” refers to the release of the Medicinal Product into the distribution chain -->
</MarketingStatus>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "country" : { CodeableConcept }, // R!  The country in which the marketing authorisation has been granted shall be
```

```
    specified It should be specified using the ISO 3166 â€‘ 1 alpha-2 code elements
  "jurisdiction" : { CodeableConcept }, // Where a Medicines Regulatory Agency has granted a marketing authorisation for
     which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate
    controlled terminology The controlled term and the controlled term identifier shall be specified
  "status" : { CodeableConcept }, // R!  This attribute provides information on the status of the marketing of the
    medicinal product See ISO/TS 20443 for more information and examples
  "dateRange" : { Period }, // R!  The date when the Medicinal Product is placed on the market by the Marketing
    Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
  "restoreDate" : "<dateTime>" // The date when the Medicinal Product is placed on the market by the Marketing
    Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:MarketingStatus.country [ CodeableConcept ]; # 1..1 The country in which the marketing authorisation has been
    granted shall be specified It should be specified using the ISO 3166 â€‘ 1 alpha-2 code elements
  fhir:MarketingStatus.jurisdiction [ CodeableConcept ]; # 0..1 Where a Medicines Regulatory Agency has granted a
    marketing authorisation for which specific provisions within a jurisdiction apply, the jurisdiction can be specified
     using an appropriate controlled terminology The controlled term and the controlled term identifier shall be
    specified
  fhir:MarketingStatus.status [ CodeableConcept ]; # 1..1 This attribute provides information on the status of the
    marketing of the medicinal product See ISO/TS 20443 for more information and examples
  fhir:MarketingStatus.dateRange [ Period ]; # 1..1 The date when the Medicinal Product is placed on the market by the
    Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction
    shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date
    format NOTE â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
  fhir:MarketingStatus.restoreDate [ dateTime ]; # 0..1 The date when the Medicinal Product is placed on the market by
    the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or
    jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO
    8601 date format NOTE â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the
    distribution chain
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

MarketingStatus | ΣD | | BackBoneElement | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available Elements defined in Ancestors: id, extension, modifierExtension

country | Σ | 1..1 | CodeableConcept | The country in which the marketing authorisation has been granted shall be specified It should be specified using the ISO 3166 - 1 alpha-2 code elements

jurisdiction | Σ | 0..1 | CodeableConcept | Where a Medicines Regulatory Agency has granted a marketing authorisation for which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate controlled terminology The controlled term and the controlled term identifier shall be specified

status | Σ | 1..1 | CodeableConcept | This attribute provides information on the status of the marketing of the medicinal product See ISO/TS 20443 for more information and examples

dateRange | Σ | 1..1 | Period | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain

restoreDate | Σ | 0..1 | dateTime | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MarketingStatus xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <country><!-- 1..1 CodeableConcept The country in which the marketing authorisation has been granted shall be specified
     It should be specified using the ISO 3166 - 1 alpha-2 code elements --></country>
 <jurisdiction><!-- 0..1 CodeableConcept Where a Medicines Regulatory Agency has granted a marketing authorisation for
    which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate
    controlled terminology The controlled term and the controlled term identifier shall be specified --></jurisdiction>
 <status><!-- 1..1 CodeableConcept This attribute provides information on the status of the marketing of the medicinal
    product See ISO/TS 20443 for more information and examples --></status>
 <dateRange><!-- 1..1 Period The date when the Medicinal Product is placed on the market by the Marketing Authorisation
    Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A
    complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on
    the market” refers to the release of the Medicinal Product into the distribution chain --></dateRange>
 <restoreDate value="[dateTime]"/><!-- 0..1 The date when the Medicinal Product is placed on the market by the Marketing
     Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    “Placed on the market” refers to the release of the Medicinal Product into the distribution chain -->
</MarketingStatus>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "country" : { CodeableConcept }, // R!  The country in which the marketing authorisation has been granted shall be
    specified It should be specified using the ISO 3166 â€‘ 1 alpha-2 code elements
  "jurisdiction" : { CodeableConcept }, // Where a Medicines Regulatory Agency has granted a marketing authorisation for
     which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate
    controlled terminology The controlled term and the controlled term identifier shall be specified
  "status" : { CodeableConcept }, // R!  This attribute provides information on the status of the marketing of the
    medicinal product See ISO/TS 20443 for more information and examples
  "dateRange" : { Period }, // R!  The date when the Medicinal Product is placed on the market by the Marketing
    Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
  "restoreDate" : "<dateTime>" // The date when the Medicinal Product is placed on the market by the Marketing
    Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be
    provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE
    â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:MarketingStatus.country [ CodeableConcept ]; # 1..1 The country in which the marketing authorisation has been
    granted shall be specified It should be specified using the ISO 3166 â€‘ 1 alpha-2 code elements
  fhir:MarketingStatus.jurisdiction [ CodeableConcept ]; # 0..1 Where a Medicines Regulatory Agency has granted a
    marketing authorisation for which specific provisions within a jurisdiction apply, the jurisdiction can be specified
     using an appropriate controlled terminology The controlled term and the controlled term identifier shall be
    specified
  fhir:MarketingStatus.status [ CodeableConcept ]; # 1..1 This attribute provides information on the status of the
    marketing of the medicinal product See ISO/TS 20443 for more information and examples
  fhir:MarketingStatus.dateRange [ Period ]; # 1..1 The date when the Medicinal Product is placed on the market by the
    Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction
    shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date
    format NOTE â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the distribution chain
  fhir:MarketingStatus.restoreDate [ dateTime ]; # 0..1 The date when the Medicinal Product is placed on the market by
    the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or
    jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO
    8601 date format NOTE â€œPlaced on the marketâ€ refers to the release of the Medicinal Product into the
    distribution chain
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

MarketingStatus is used in the following places: MedicinalProduct and MedicinalProductPackaged

11.27.0.2 Interpretation of MarketingStatus in different contexts

11.27.1 Examples

URL: https://hl7.org/fhir/R4/marketingstatus-examples.html

11.27.1 MarketingStatus Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicinalProductPackaged examples for examples using MarketingStatus.

11.27.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/marketingstatus-definitions.html

11.27.2 MarketingStatus Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.27.2.1 MarketingStatus

MarketingStatus
Standards Status | This element has a standards status of "Draft"
Element Id | MarketingStatus
Definition | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available.
Cardinality | 0..*
Type | BackboneElement
Summary | true
MarketingStatus.country
Element Id | MarketingStatus.country

Definition | The country in which the marketing authorisation has been granted shall be specified It should be specified using the ISO 3166 - 1 alpha-2 code elements.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MarketingStatus.jurisdiction
Element Id | MarketingStatus.jurisdiction
Definition | Where a Medicines Regulatory Agency has granted a marketing authorisation for which specific provisions within a jurisdiction apply, the jurisdiction can be specified using an appropriate controlled terminology The controlled term and the controlled term identifier shall be specified.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MarketingStatus.status
Element Id | MarketingStatus.status
Definition | This attribute provides information on the status of the marketing of the medicinal product See ISO/TS 20443 for more information and examples.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
MarketingStatus.dateRange
Element Id | MarketingStatus.dateRange
Definition | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain.
Cardinality | 1..1
Type | Period
Summary | true
MarketingStatus.restoreDate
Element Id | MarketingStatus.restoreDate
Definition | The date when the Medicinal Product is placed on the market by the Marketing Authorisation Holder (or where applicable, the manufacturer/distributor) in a country and/or jurisdiction shall be provided A complete date consisting of day, month and year shall be specified using the ISO 8601 date format NOTE “Placed on the market” refers to the release of the Medicinal Product into the distribution chain.
Cardinality | 0..1
Type | dateTime
Summary | true

11.27.3 Mappings

URL: https://hl7.org/fhir/R4/marketingstatus-mappings.html

11.27.3 MarketingStatus Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for MarketingStatus (see Mappings to Other Standards for further information & status).

11.28 Population

URL: https://hl7.org/fhir/R4/population.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

This type is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft design not suitable for production implementation

The Population structure defines TODO.

Note: the Population structure allows modifier extensions.

11.28.0.1 Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints

Population | ΣD | | BackBoneElement | A definition of a set of people that apply to some clinically related context, for example people contraindicated for a certain medication Elements defined in Ancestors: id, extension, modifierExtension
age[x] | Σ | 0..1 | | The age of the specific population
ageRange | | | Range
ageCodeableConcept | | | CodeableConcept
gender | Σ | 0..1 | CodeableConcept | The gender of the specific population
race | Σ | 0..1 | CodeableConcept | Race of the specific population
physiologicalCondition | Σ | 0..1 | CodeableConcept | The existing physiological conditions of the specific population to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Population xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <age[x]><!-- 0..1 Range|CodeableConcept The age of the specific population --></age[x]>
 <gender><!-- 0..1 CodeableConcept The gender of the specific population --></gender>
 <race><!-- 0..1 CodeableConcept Race of the specific population --></race>
 <physiologicalCondition><!-- 0..1 CodeableConcept The existing physiological conditions of the specific population to
    which this applies --></physiologicalCondition>
</Population>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  // age[x]: The age of the specific population. One of these 2:
  "ageRange" : { Range },
  "ageCodeableConcept" : { CodeableConcept },
  "gender" : { CodeableConcept }, // The gender of the specific population
  "race" : { CodeableConcept }, // Race of the specific population
  "physiologicalCondition" : { CodeableConcept } // The existing physiological conditions of the specific population to
    which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  # Population.age[x] : 0..1 The age of the specific population. One of these 2
    fhir:Population.ageRange [ Range ]
    fhir:Population.ageCodeableConcept [ CodeableConcept ]
  fhir:Population.gender [ CodeableConcept ]; # 0..1 The gender of the specific population
  fhir:Population.race [ CodeableConcept ]; # 0..1 Race of the specific population
  fhir:Population.physiologicalCondition [ CodeableConcept ]; # 0..1 The existing physiological conditions of the
    specific population to which this applies
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints
Population | ΣD | | BackBoneElement | A definition of a set of people that apply to some clinically related context, for example people contraindicated for a certain medication Elements defined in Ancestors: id, extension, modifierExtension
age[x] | Σ | 0..1 | | The age of the specific population
ageRange | | | Range
ageCodeableConcept | | | CodeableConcept
gender | Σ | 0..1 | CodeableConcept | The gender of the specific population
race | Σ | 0..1 | CodeableConcept | Race of the specific population
physiologicalCondition | Σ | 0..1 | CodeableConcept | The existing physiological conditions of the specific population to which this applies
Documentation for this format

UML Diagram (Legend)

XML Template


```
<Population xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <age[x]><!-- 0..1 Range|CodeableConcept The age of the specific population --></age[x]>
 <gender><!-- 0..1 CodeableConcept The gender of the specific population --></gender>
 <race><!-- 0..1 CodeableConcept Race of the specific population --></race>
 <physiologicalCondition><!-- 0..1 CodeableConcept The existing physiological conditions of the specific population to
    which this applies --></physiologicalCondition>
</Population>
```


JSON Template


```
{
  // from BackboneElement: extension, modifierExtension
  // age[x]: The age of the specific population. One of these 2:
  "ageRange" : { Range },
  "ageCodeableConcept" : { CodeableConcept },
  "gender" : { CodeableConcept }, // The gender of the specific population
```

```
  "race" : { CodeableConcept }, // Race of the specific population
  "physiologicalCondition" : { CodeableConcept } // The existing physiological conditions of the specific population to
    which this applies
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  # Population.age[x] : 0..1 The age of the specific population. One of these 2
    fhir:Population.ageRange [ Range ]
    fhir:Population.ageCodeableConcept [ CodeableConcept ]
  fhir:Population.gender [ CodeableConcept ]; # 0..1 The gender of the specific population
  fhir:Population.race [ CodeableConcept ]; # 0..1 Race of the specific population
  fhir:Population.physiologicalCondition [ CodeableConcept ]; # 0..1 The existing physiological conditions of the
    specific population to which this applies
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

Population is used in the following places: MedicinalProductContraindication, MedicinalProductIndication and MedicinalProductUndesirableEffect

11.28.0.2 Interpretation of Population in different contexts

11.28.1 Examples

URL: https://hl7.org/fhir/R4/population-examples.html

11.28.1 Population Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicinalProductPackaged examples for examples using Population.

11.28.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/population-definitions.html

11.28.2 Population Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.28.2.1 Population

Population
Standards Status | This element has a standards status of "Draft"
Element Id | Population
Definition | A populatioof people with some set of grouping criteria.
Cardinality | 0..*
Type | BackboneElement
Summary | true
Population.age[x]
Element Id | Population.age[x]
Definition | The age of the specific population.
Cardinality | 0..1
Type | Range|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Population.gender
Element Id | Population.gender
Definition | The gender of the specific population.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
Population.race
Element Id | Population.race
Definition | Race of the specific population.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
Population.physiologicalCondition
Element Id | Population.physiologicalCondition
Definition | The existing physiological conditions of the specific population to which this applies.
Cardinality | 0..1

Type | CodeableConcept
Summary | true

11.28.3 Mappings

URL: https://hl7.org/fhir/R4/population-mappings.html

11.28.3 Population Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for Population (see Mappings to Other Standards for further information & status).

11.29 ProdCharacteristic

URL: https://hl7.org/fhir/R4/prodcharacteristic.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

This type is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft design not suitable for production implementation

The ProdCharacteristic structure defines TODO.

Note: the ProdCharacteristic structure allows modifier extensions.

11.29.0.1 Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ProdCharacteristic | ΣD | | BackBoneElement | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available Elements defined in Ancestors: id, extension, modifierExtension
height | Σ | 0..1 | Quantity | Where applicable, the height can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
width | Σ | 0..1 | Quantity | Where applicable, the width can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
depth | Σ | 0..1 | Quantity | Where applicable, the depth can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
weight | Σ | 0..1 | Quantity | Where applicable, the weight can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
nominalVolume | Σ | 0..1 | Quantity | Where applicable, the nominal volume can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
externalDiameter | Σ | 0..1 | Quantity | Where applicable, the external diameter can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
shape | Σ | 0..1 | string | Where applicable, the shape can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used
color | Σ | 0..* | string | Where applicable, the color can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used
imprint | Σ | 0..* | string | Where applicable, the imprint can be specified as text
image | Σ | 0..* | Attachment | Where applicable, the image can be provided The format of the image attachment shall be specified by regional implementations
scoring | Σ | 0..1 | CodeableConcept | Where applicable, the scoring can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProdCharacteristic xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <height><!-- 0..1 Quantity Where applicable, the height can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
```

```
    The symbol and the symbol identifier shall be used --></height>
 <width><!-- 0..1 Quantity Where applicable, the width can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></width>
 <depth><!-- 0..1 Quantity Where applicable, the depth can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></depth>
 <weight><!-- 0..1 Quantity Where applicable, the weight can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></weight>
 <nominalVolume><!-- 0..1 Quantity Where applicable, the nominal volume can be specified using a numerical value and its
     unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting
    terminology The symbol and the symbol identifier shall be used --></nominalVolume>
 <externalDiameter><!-- 0..1 Quantity Where applicable, the external diameter can be specified using a numerical value
    and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used --></externalDiameter>
 <shape value="[string]"/><!-- 0..1 Where applicable, the shape can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used -->
 <color value="[string]"/><!-- 0..* Where applicable, the color can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used -->
 <imprint value="[string]"/><!-- 0..* Where applicable, the imprint can be specified as text -->
 <image><!-- 0..* Attachment Where applicable, the image can be provided The format of the image attachment shall be
    specified by regional implementations --></image>
 <scoring><!-- 0..1 CodeableConcept Where applicable, the scoring can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used --></scoring>
</ProdCharacteristic>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "height" : { Quantity }, // Where applicable, the height can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "width" : { Quantity }, // Where applicable, the width can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "depth" : { Quantity }, // Where applicable, the depth can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "weight" : { Quantity }, // Where applicable, the weight can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "nominalVolume" : { Quantity }, // Where applicable, the nominal volume can be specified using a numerical value and
    its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting
    terminology The symbol and the symbol identifier shall be used
  "externalDiameter" : { Quantity }, // Where applicable, the external diameter can be specified using a numerical value
     and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  "shape" : "<string>", // Where applicable, the shape can be specified An appropriate controlled vocabulary shall be
    used The term and the term identifier shall be used
  "color" : ["<string>"], // Where applicable, the color can be specified An appropriate controlled vocabulary shall be
    used The term and the term identifier shall be used
  "imprint" : ["<string>"], // Where applicable, the imprint can be specified as text
  "image" : [{ Attachment }], // Where applicable, the image can be provided The format of the image attachment shall be
     specified by regional implementations
  "scoring" : { CodeableConcept } // Where applicable, the scoring can be specified An appropriate controlled vocabulary
     shall be used The term and the term identifier shall be used
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ProdCharacteristic.height [ Quantity ]; # 0..1 Where applicable, the height can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.width [ Quantity ]; # 0..1 Where applicable, the width can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.depth [ Quantity ]; # 0..1 Where applicable, the depth can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.weight [ Quantity ]; # 0..1 Where applicable, the weight can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.nominalVolume [ Quantity ]; # 0..1 Where applicable, the nominal volume can be specified using
     a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO
    11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.externalDiameter [ Quantity ]; # 0..1 Where applicable, the external diameter can be specified
     using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with
    ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.shape [ string ]; # 0..1 Where applicable, the shape can be specified An appropriate
    controlled vocabulary shall be used The term and the term identifier shall be used
  fhir:ProdCharacteristic.color [ string ], ... ; # 0..* Where applicable, the color can be specified An appropriate
    controlled vocabulary shall be used The term and the term identifier shall be used
  fhir:ProdCharacteristic.imprint [ string ], ... ; # 0..* Where applicable, the imprint can be specified as text
  fhir:ProdCharacteristic.image [ Attachment ], ... ; # 0..* Where applicable, the image can be provided The format of
    the image attachment shall be specified by regional implementations
```

```
  fhir:ProdCharacteristic.scoring [ CodeableConcept ]; # 0..1 Where applicable, the scoring can be specified An
    appropriate controlled vocabulary shall be used The term and the term identifier shall be used
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

ProdCharacteristic | ΣD | | BackBoneElement | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available Elements defined in Ancestors: id, extension, modifierExtension

height | Σ | 0..1 | Quantity | Where applicable, the height can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

width | Σ | 0..1 | Quantity | Where applicable, the width can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

depth | Σ | 0..1 | Quantity | Where applicable, the depth can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

weight | Σ | 0..1 | Quantity | Where applicable, the weight can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

nominalVolume | Σ | 0..1 | Quantity | Where applicable, the nominal volume can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

externalDiameter | Σ | 0..1 | Quantity | Where applicable, the external diameter can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

shape | Σ | 0..1 | string | Where applicable, the shape can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used

color | Σ | 0..* | string | Where applicable, the color can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used

imprint | Σ | 0..* | string | Where applicable, the imprint can be specified as text

image | Σ | 0..* | Attachment | Where applicable, the image can be provided The format of the image attachment shall be specified by regional implementations

scoring | Σ | 0..1 | CodeableConcept | Where applicable, the scoring can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProdCharacteristic xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <height><!-- 0..1 Quantity Where applicable, the height can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></height>
 <width><!-- 0..1 Quantity Where applicable, the width can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></width>
 <depth><!-- 0..1 Quantity Where applicable, the depth can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></depth>
 <weight><!-- 0..1 Quantity Where applicable, the weight can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used --></weight>
 <nominalVolume><!-- 0..1 Quantity Where applicable, the nominal volume can be specified using a numerical value and its
     unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting
    terminology The symbol and the symbol identifier shall be used --></nominalVolume>
 <externalDiameter><!-- 0..1 Quantity Where applicable, the external diameter can be specified using a numerical value
    and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used --></externalDiameter>
 <shape value="[string]"/><!-- 0..1 Where applicable, the shape can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used -->
 <color value="[string]"/><!-- 0..* Where applicable, the color can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used -->
 <imprint value="[string]"/><!-- 0..* Where applicable, the imprint can be specified as text -->
 <image><!-- 0..* Attachment Where applicable, the image can be provided The format of the image attachment shall be
    specified by regional implementations --></image>
 <scoring><!-- 0..1 CodeableConcept Where applicable, the scoring can be specified An appropriate controlled vocabulary
    shall be used The term and the term identifier shall be used --></scoring>
</ProdCharacteristic>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "height" : { Quantity }, // Where applicable, the height can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "width" : { Quantity }, // Where applicable, the width can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
```

```
  "depth" : { Quantity }, // Where applicable, the depth can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "weight" : { Quantity }, // Where applicable, the weight can be specified using a numerical value and its unit of
    measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology
    The symbol and the symbol identifier shall be used
  "nominalVolume" : { Quantity }, // Where applicable, the nominal volume can be specified using a numerical value and
    its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting
    terminology The symbol and the symbol identifier shall be used
  "externalDiameter" : { Quantity }, // Where applicable, the external diameter can be specified using a numerical value
     and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  "shape" : "<string>", // Where applicable, the shape can be specified An appropriate controlled vocabulary shall be
    used The term and the term identifier shall be used
  "color" : ["<string>"], // Where applicable, the color can be specified An appropriate controlled vocabulary shall be
    used The term and the term identifier shall be used
  "imprint" : ["<string>"], // Where applicable, the imprint can be specified as text
  "image" : [{ Attachment }], // Where applicable, the image can be provided The format of the image attachment shall be
     specified by regional implementations
  "scoring" : { CodeableConcept } // Where applicable, the scoring can be specified An appropriate controlled vocabulary
     shall be used The term and the term identifier shall be used
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ProdCharacteristic.height [ Quantity ]; # 0..1 Where applicable, the height can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.width [ Quantity ]; # 0..1 Where applicable, the width can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.depth [ Quantity ]; # 0..1 Where applicable, the depth can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.weight [ Quantity ]; # 0..1 Where applicable, the weight can be specified using a numerical
    value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.nominalVolume [ Quantity ]; # 0..1 Where applicable, the nominal volume can be specified using
     a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO
    11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.externalDiameter [ Quantity ]; # 0..1 Where applicable, the external diameter can be specified
     using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with
    ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProdCharacteristic.shape [ string ]; # 0..1 Where applicable, the shape can be specified An appropriate
    controlled vocabulary shall be used The term and the term identifier shall be used
  fhir:ProdCharacteristic.color [ string ], ... ; # 0..* Where applicable, the color can be specified An appropriate
    controlled vocabulary shall be used The term and the term identifier shall be used
  fhir:ProdCharacteristic.imprint [ string ], ... ; # 0..* Where applicable, the imprint can be specified as text
  fhir:ProdCharacteristic.image [ Attachment ], ... ; # 0..* Where applicable, the image can be provided The format of
    the image attachment shall be specified by regional implementations
  fhir:ProdCharacteristic.scoring [ CodeableConcept ]; # 0..1 Where applicable, the scoring can be specified An
    appropriate controlled vocabulary shall be used The term and the term identifier shall be used
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

ProdCharacteristic is used in the following places: DeviceDefinition, MedicinalProductManufactured and MedicinalProductPackaged

11.29.0.2 Interpretation of ProdCharacteristic in different contexts

11.29.1 Examples

URL: https://hl7.org/fhir/R4/prodcharacteristic-examples.html

11.29.1 ProdCharacteristic Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicinalProductPackaged examples for examples using ProdCharacteristic.

11.29.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/prodcharacteristic-definitions.html

11.29.2 ProdCharacteristic Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.29.2.1 ProdCharacteristic

ProdCharacteristic
Standards Status | This element has a standards status of "Draft"
Element Id | ProdCharacteristic
Definition | The marketing status describes the date when a medicinal product is actually put on the market or the date as of which it is no longer available.
Cardinality | 0..*
Type | BackboneElement
Summary | true
ProdCharacteristic.height
Element Id | ProdCharacteristic.height
Definition | Where applicable, the height can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.width
Element Id | ProdCharacteristic.width
Definition | Where applicable, the width can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.depth
Element Id | ProdCharacteristic.depth
Definition | Where applicable, the depth can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.weight
Element Id | ProdCharacteristic.weight
Definition | Where applicable, the weight can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.nominalVolume
Element Id | ProdCharacteristic.nominalVolume
Definition | Where applicable, the nominal volume can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.externalDiameter
Element Id | ProdCharacteristic.externalDiameter
Definition | Where applicable, the external diameter can be specified using a numerical value and its unit of measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 0..1
Type | Quantity
Summary | true
ProdCharacteristic.shape
Element Id | ProdCharacteristic.shape
Definition | Where applicable, the shape can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used.
Cardinality | 0..1
Type | string
Summary | true
ProdCharacteristic.color
Element Id | ProdCharacteristic.color
Definition | Where applicable, the color can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used.
Cardinality | 0..*
Type | string
Summary | true
ProdCharacteristic.imprint
Element Id | ProdCharacteristic.imprint

Definition | Where applicable, the imprint can be specified as text.
Cardinality | 0..*
Type | string
Summary | true
ProdCharacteristic.image
Element Id | ProdCharacteristic.image
Definition | Where applicable, the image can be provided The format of the image attachment shall be specified by regional implementations.
Cardinality | 0..*
Type | Attachment
Summary | true
ProdCharacteristic.scoring
Element Id | ProdCharacteristic.scoring
Definition | Where applicable, the scoring can be specified An appropriate controlled vocabulary shall be used The term and the term identifier shall be used.
Cardinality | 0..1
Type | CodeableConcept
Summary | true

11.29.3 Mappings

URL: https://hl7.org/fhir/R4/prodcharacteristic-mappings.html

11.29.3 ProdCharacteristic Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for ProdCharacteristic (see Mappings to Other Standards for further information & status).

11.30 ProductShelfLife

URL: https://hl7.org/fhir/R4/productshelflife.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

This type is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft design not suitable for production implementation

The ProductShelfLife structure defines TODO.

Note: the ProductShelfLife structure allows modifier extensions.

11.30.0.1 Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ProductShelfLife | ΣD | | BackBoneElement | The shelf-life and storage information for a medicinal product item or container can be described using this class Elements defined in Ancestors: id, extension, modifierExtension
identifier | Σ | 0..1 | Identifier | Unique identifier for the packaged Medicinal Product
type | Σ | 1..1 | CodeableConcept | This describes the shelf life, taking into account various scenarios such as shelf life of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life after the first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
period | Σ | 1..1 | Quantity | The shelf life time period can be specified using a numerical value for the period of time and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
specialPrecautionsForStorage | Σ | 0..* | CodeableConcept | Special precautions for storage, if any, can be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProductShelfLife xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier for the packaged Medicinal Product --></identifier>
 <type><!-- 1..1 CodeableConcept This describes the shelf life, taking into account various scenarios such as shelf life
     of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life after the
```

```
    first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled vocabulary
    The controlled term and the controlled term identifier shall be specified --></type>
 <period><!-- 1..1 Quantity The shelf life time period can be specified using a numerical value for the period of time
    and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used --></period>
 <specialPrecautionsForStorage><!-- 0..* CodeableConcept Special precautions for storage, if any, can be specified using
     an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
    --></specialPrecautionsForStorage>
</ProductShelfLife>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "identifier" : { Identifier }, // Unique identifier for the packaged Medicinal Product
  "type" : { CodeableConcept }, // R!  This describes the shelf life, taking into account various scenarios such as
    shelf life of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life
    after the first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled
    vocabulary The controlled term and the controlled term identifier shall be specified
  "period" : { Quantity }, // R!  The shelf life time period can be specified using a numerical value for the period of
    time and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and
    the resulting terminology The symbol and the symbol identifier shall be used
  "specialPrecautionsForStorage" : [{ CodeableConcept }] // Special precautions for storage, if any, can be specified
    using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ProductShelfLife.identifier [ Identifier ]; # 0..1 Unique identifier for the packaged Medicinal Product
  fhir:ProductShelfLife.type [ CodeableConcept ]; # 1..1 This describes the shelf life, taking into account various
    scenarios such as shelf life of the packaged Medicinal Product itself, shelf life after transformation where
    necessary and shelf life after the first opening of a bottle, etc. The shelf life type shall be specified using an
    appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
  fhir:ProductShelfLife.period [ Quantity ]; # 1..1 The shelf life time period can be specified using a numerical value
    for the period of time and its unit of time measurement The unit of measurement shall be specified in accordance
    with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProductShelfLife.specialPrecautionsForStorage [ CodeableConcept ], ... ; # 0..* Special precautions for storage,
    if any, can be specified using an appropriate controlled vocabulary The controlled term and the controlled term
    identifier shall be specified
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

ProductShelfLife | ΣD | | BackBoneElement | The shelf-life and storage information for a medicinal product item or container can be described using this class Elements defined in Ancestors: id, extension, modifierExtension

identifier | Σ | 0..1 | Identifier | Unique identifier for the packaged Medicinal Product

type | Σ | 1..1 | CodeableConcept | This describes the shelf life, taking into account various scenarios such as shelf life of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life after the first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified

period | Σ | 1..1 | Quantity | The shelf life time period can be specified using a numerical value for the period of time and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used

specialPrecautionsForStorage | Σ | 0..* | CodeableConcept | Special precautions for storage, if any, can be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ProductShelfLife xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <identifier><!-- 0..1 Identifier Unique identifier for the packaged Medicinal Product --></identifier>
 <type><!-- 1..1 CodeableConcept This describes the shelf life, taking into account various scenarios such as shelf life
     of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life after the
    first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled vocabulary
    The controlled term and the controlled term identifier shall be specified --></type>
 <period><!-- 1..1 Quantity The shelf life time period can be specified using a numerical value for the period of time
    and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and the
    resulting terminology The symbol and the symbol identifier shall be used --></period>
 <specialPrecautionsForStorage><!-- 0..* CodeableConcept Special precautions for storage, if any, can be specified using
     an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
    --></specialPrecautionsForStorage>
</ProductShelfLife>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  "identifier" : { Identifier }, // Unique identifier for the packaged Medicinal Product
  "type" : { CodeableConcept }, // R!  This describes the shelf life, taking into account various scenarios such as
    shelf life of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life
```

```
    after the first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled
    vocabulary The controlled term and the controlled term identifier shall be specified
  "period" : { Quantity }, // R!  The shelf life time period can be specified using a numerical value for the period of
    time and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and
    the resulting terminology The symbol and the symbol identifier shall be used
  "specialPrecautionsForStorage" : [{ CodeableConcept }] // Special precautions for storage, if any, can be specified
    using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  fhir:ProductShelfLife.identifier [ Identifier ]; # 0..1 Unique identifier for the packaged Medicinal Product
  fhir:ProductShelfLife.type [ CodeableConcept ]; # 1..1 This describes the shelf life, taking into account various
    scenarios such as shelf life of the packaged Medicinal Product itself, shelf life after transformation where
    necessary and shelf life after the first opening of a bottle, etc. The shelf life type shall be specified using an
    appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified
  fhir:ProductShelfLife.period [ Quantity ]; # 1..1 The shelf life time period can be specified using a numerical value
    for the period of time and its unit of time measurement The unit of measurement shall be specified in accordance
    with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used
  fhir:ProductShelfLife.specialPrecautionsForStorage [ CodeableConcept ], ... ; # 0..* Special precautions for storage,
    if any, can be specified using an appropriate controlled vocabulary The controlled term and the controlled term
    identifier shall be specified
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

ProductShelfLife is used in the following places: DeviceDefinition and MedicinalProductPackaged

11.30.0.2 Interpretation of ProductShelfLife in different contexts

11.30.1 Examples

URL: https://hl7.org/fhir/R4/productshelflife-examples.html

11.30.1 ProductShelfLife Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicinalProductPackaged examples for examples using ProductShelfLife.

11.30.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/productshelflife-definitions.html

11.30.2 ProductShelfLife Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.30.2.1 ProductShelfLife

ProductShelfLife
Standards Status | This element has a standards status of "Draft"
Element Id | ProductShelfLife
Definition | The shelf-life and storage information for a medicinal product item or container can be described using this class.
Cardinality | 0..*
Type | BackboneElement
Summary | true
ProductShelfLife.identifier
Element Id | ProductShelfLife.identifier
Definition | Unique identifier for the packaged Medicinal Product.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Summary | true
ProductShelfLife.type
Element Id | ProductShelfLife.type
Definition | This describes the shelf life, taking into account various scenarios such as shelf life of the packaged Medicinal Product itself, shelf life after transformation where necessary and shelf life after the first opening of a bottle, etc. The shelf life type shall be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified.
Cardinality | 1..1
Type | CodeableConcept
Summary | true
ProductShelfLife.period
Element Id | ProductShelfLife.period

Definition | The shelf life time period can be specified using a numerical value for the period of time and its unit of time measurement The unit of measurement shall be specified in accordance with ISO 11240 and the resulting terminology The symbol and the symbol identifier shall be used.
Cardinality | 1..1
Type | Quantity
Summary | true
ProductShelfLife.specialPrecautionsForStorage
Element Id | ProductShelfLife.specialPrecautionsForStorage
Definition | Special precautions for storage, if any, can be specified using an appropriate controlled vocabulary The controlled term and the controlled term identifier shall be specified.
Cardinality | 0..*
Type | CodeableConcept
Summary | true

11.30.3 Mappings

URL: https://hl7.org/fhir/R4/productshelflife-mappings.html

11.30.3 ProductShelfLife Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for ProductShelfLife (see Mappings to Other Standards for further information & status).

11.31 SubstanceAmount

URL: https://hl7.org/fhir/R4/substanceamount.html
Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

This type is still undergoing development and review by the appropriate Workgroups. At this time, is considered only as a draft design not suitable for production implementation

The SubstanceAmount structure defines TODO.

Note: the SubstanceAmount structure allows modifier extensions.

11.31.0.1 Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
SubstanceAmount | ΣD | | BackBoneElement | Chemical substances are a single substance type whose primary defining element is the molecular structure. Chemical substances shall be defined on the basis of their complete covalent molecular structure; the presence of a salt (counter-ion) and/or solvates (water, alcohols) is also captured. Purity, grade, physical form or particle size are not taken into account in the definition of a chemical substance or in the assignment of a Substance ID Elements defined in Ancestors: id, extension, modifierExtension
amount[x] | Σ | 0..1 | | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field
amountQuantity | | | Quantity
amountRange | | | Range
amountString | | | string
amountType | Σ | 0..1 | CodeableConcept | Most elements that require a quantitative value will also have a field called amount type. Amount type should always be specified because the actual value of the amount is often dependent on it. EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use same the amount type for all related definitional elements
amountText | Σ | 0..1 | string | A textual comment on a numeric value
referenceRange | Σ | 0..1 | Element | Reference range of possible or expected values
lowLimit | Σ | 0..1 | Quantity | Lower limit possible or expected
highLimit | Σ | 0..1 | Quantity | Upper limit possible or expected
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceAmount xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <amount[x]><!-- 0..1 Quantity|Range|string Used to capture quantitative values for a variety of elements. If only
```

```
    limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is
    given, it would be captured in this field --></amount[x]>
 <amountType><!-- 0..1 CodeableConcept Most elements that require a quantitative value will also have a field called
    amount type. Amount type should always be specified because the actual value of the amount is often dependent on it.
     EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate
    whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use
    same the amount type for all related definitional elements --></amountType>
 <amountText value="[string]"/><!-- 0..1 A textual comment on a numeric value -->
 <referenceRange>  <!-- 0..1 Reference range of possible or expected values -->
  <lowLimit><!-- 0..1 Quantity Lower limit possible or expected --></lowLimit>
  <highLimit><!-- 0..1 Quantity Upper limit possible or expected --></highLimit>
 </referenceRange>
</SubstanceAmount>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  // amount[x]:
     Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would
    be the average. If only a single definite value for a given element is given, it would be captured in this field.
    One of these 3:
  "amountQuantity" : { Quantity },
  "amountRange" : { Range },
  "amountString" : "<string>",
  "amountType" : { CodeableConcept }, // Most elements that require a quantitative value will also have a field called
    amount type. Amount type should always be specified because the actual value of the amount is often dependent on it.
     EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate
    whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use
    same the amount type for all related definitional elements
  "amountText" : "<string>", // A textual comment on a numeric value
  "referenceRange" : { // Reference range of possible or expected values
    "lowLimit" : { Quantity }, // Lower limit possible or expected
    "highLimit" : { Quantity } // Upper limit possible or expected
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  # SubstanceAmount.amount[x] : 0..1 Used to capture quantitative values for a variety of elements. If only limits are
    given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it
    would be captured in this field. One of these 3
    fhir:SubstanceAmount.amountQuantity [ Quantity ]
    fhir:SubstanceAmount.amountRange [ Range ]
    fhir:SubstanceAmount.amountString [ string ]
  fhir:SubstanceAmount.amountType [ CodeableConcept ]; # 0..1 Most elements that require a quantitative value will also
    have a field called amount type. Amount type should always be specified because the actual value of the amount is
    often dependent on it. EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is
    essential to indicate whether the amount refers to a mole ratio or weight ratio. For any given element an effort
    should be made to use same the amount type for all related definitional elements
  fhir:SubstanceAmount.amountText [ string ]; # 0..1 A textual comment on a numeric value
  fhir:SubstanceAmount.referenceRange [ # 0..1 Reference range of possible or expected values
    fhir:SubstanceAmount.referenceRange.lowLimit [ Quantity ]; # 0..1 Lower limit possible or expected
    fhir:SubstanceAmount.referenceRange.highLimit [ Quantity ]; # 0..1 Upper limit possible or expected
  ];
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Structure

Name | Flags | Card. | Type | Description & Constraints

SubstanceAmount | ΣD | | BackBoneElement | Chemical substances are a single substance type whose primary defining element is the molecular structure. Chemical substances shall be defined on the basis of their complete covalent molecular structure; the presence of a salt (counter-ion) and/or solvates (water, alcohols) is also captured. Purity, grade, physical form or particle size are not taken into account in the definition of a chemical substance or in the assignment of a Substance ID Elements defined in Ancestors: id, extension, modifierExtension

amount[x] | Σ | 0..1 | | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field

amountQuantity | | | Quantity

amountRange | | | Range

amountString | | | string

amountType | Σ | 0..1 | CodeableConcept | Most elements that require a quantitative value will also have a field called amount type. Amount type should always be specified because the actual value of the amount is often dependent on it. EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use same the amount type for all related definitional elements

amountText | Σ | 0..1 | string | A textual comment on a numeric value

referenceRange | Σ | 0..1 | Element | Reference range of possible or expected values

lowLimit | Σ | 0..1 | Quantity | Lower limit possible or expected

highLimit | Σ | 0..1 | Quantity | Upper limit possible or expected

Documentation for this format

UML Diagram (Legend)

XML Template

```
<SubstanceAmount xmlns="http://hl7.org/fhir">
 <!-- from BackboneElement: extension, modifierExtension -->
 <amount[x]><!-- 0..1 Quantity|Range|string Used to capture quantitative values for a variety of elements. If only
    limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is
    given, it would be captured in this field --></amount[x]>
 <amountType><!-- 0..1 CodeableConcept Most elements that require a quantitative value will also have a field called
    amount type. Amount type should always be specified because the actual value of the amount is often dependent on it.
     EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate
    whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use
    same the amount type for all related definitional elements --></amountType>
 <amountText value="[string]"/><!-- 0..1 A textual comment on a numeric value -->
 <referenceRange>  <!-- 0..1 Reference range of possible or expected values -->
  <lowLimit><!-- 0..1 Quantity Lower limit possible or expected --></lowLimit>
  <highLimit><!-- 0..1 Quantity Upper limit possible or expected --></highLimit>
 </referenceRange>
</SubstanceAmount>
```

JSON Template

```
{
  // from BackboneElement: extension, modifierExtension
  // amount[x]:
     Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would
    be the average. If only a single definite value for a given element is given, it would be captured in this field.
    One of these 3:
  "amountQuantity" : { Quantity },
  "amountRange" : { Range },
  "amountString" : "<string>",
  "amountType" : { CodeableConcept }, // Most elements that require a quantitative value will also have a field called
    amount type. Amount type should always be specified because the actual value of the amount is often dependent on it.
     EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate
    whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use
    same the amount type for all related definitional elements
  "amountText" : "<string>", // A textual comment on a numeric value
  "referenceRange" : { // Reference range of possible or expected values
    "lowLimit" : { Quantity }, // Lower limit possible or expected
    "highLimit" : { Quantity } // Upper limit possible or expected
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[
 # from BackboneElement: Element.extension, BackboneElement.modifierextension
  # SubstanceAmount.amount[x] : 0..1 Used to capture quantitative values for a variety of elements. If only limits are
    given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it
    would be captured in this field. One of these 3
    fhir:SubstanceAmount.amountQuantity [ Quantity ]
    fhir:SubstanceAmount.amountRange [ Range ]
    fhir:SubstanceAmount.amountString [ string ]
  fhir:SubstanceAmount.amountType [ CodeableConcept ]; # 0..1 Most elements that require a quantitative value will also
    have a field called amount type. Amount type should always be specified because the actual value of the amount is
    often dependent on it. EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is
    essential to indicate whether the amount refers to a mole ratio or weight ratio. For any given element an effort
    should be made to use same the amount type for all related definitional elements
  fhir:SubstanceAmount.amountText [ string ]; # 0..1 A textual comment on a numeric value
  fhir:SubstanceAmount.referenceRange [ # 0..1 Reference range of possible or expected values
    fhir:SubstanceAmount.referenceRange.lowLimit [ Quantity ]; # 0..1 Lower limit possible or expected
    fhir:SubstanceAmount.referenceRange.highLimit [ Quantity ]; # 0..1 Upper limit possible or expected
  ];
]
```

Changes since Release 3

This complex-type did not exist in Release 2

Constraints

SubstanceAmount is used in the following places: SubstancePolymer

11.31.0.2 Interpretation of SubstanceAmount in different contexts

11.31.1 Examples

URL: https://hl7.org/fhir/R4/substanceamount-examples.html

11.31.1 SubstanceAmount Examples

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

See MedicinalProductPackaged examples for examples using SubstanceAmount.

11.31.2 Detailed Descriptions

URL: https://hl7.org/fhir/R4/substanceamount-definitions.html

11.31.2 SubstanceAmount Detailed Descriptions

Pharmacy Work Group | Maturity Level: 3 | Standards Status: Trial Use

11.31.2.1 SubstanceAmount

SubstanceAmount
Standards Status | This element has a standards status of "Draft"
Element Id | SubstanceAmount
Definition | Chemical substances are a single substance type whose primary defining element is the molecular structure. Chemical substances shall be defined on the basis of their complete covalent molecular structure; the presence of a salt (counter-ion) and/or solvates (water, alcohols) is also captured. Purity, grade, physical form or particle size are not taken into account in the definition of a chemical substance or in the assignment of a Substance ID.
Cardinality | 0..*
Type | BackboneElement
Summary | true
SubstanceAmount.amount[x]
Element Id | SubstanceAmount.amount[x]
Definition | Used to capture quantitative values for a variety of elements. If only limits are given, the arithmetic mean would be the average. If only a single definite value for a given element is given, it would be captured in this field.
Cardinality | 0..1
Type | Quantity|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
SubstanceAmount.amountType
Element Id | SubstanceAmount.amountType
Definition | Most elements that require a quantitative value will also have a field called amount type. Amount type should always be specified because the actual value of the amount is often dependent on it. EXAMPLE: In capturing the actual relative amounts of substances or molecular fragments it is essential to indicate whether the amount refers to a mole ratio or weight ratio. For any given element an effort should be made to use same the amount type for all related definitional elements.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
SubstanceAmount.amountText
Element Id | SubstanceAmount.amountText
Definition | A textual comment on a numeric value.
Cardinality | 0..1
Type | string
Summary | true
SubstanceAmount.referenceRange
Element Id | SubstanceAmount.referenceRange
Definition | Reference range of possible or expected values.
Cardinality | 0..1
Summary | true
SubstanceAmount.referenceRange.lowLimit
Element Id | SubstanceAmount.referenceRange.lowLimit
Definition | Lower limit possible or expected.
Cardinality | 0..1
Type | Quantity
Summary | true
SubstanceAmount.referenceRange.highLimit
Element Id | SubstanceAmount.referenceRange.highLimit
Definition | Upper limit possible or expected.
Cardinality | 0..1
Type | Quantity
Summary | true

11.31.3 Mappings

URL: https://hl7.org/fhir/R4/substanceamount-mappings.html

11.31.3 SubstanceAmount Mappings

Pharmacy Work Group | Maturity Level: N/A | Standards Status: Informative

This page provides mappings for SubstanceAmount (see Mappings to Other Standards for further information & status).